

Exhibit CC-109

External Memory Interfaces Agilex™ 7 M-Series FPGA IP User Guide

Updated for Quartus® Prime Design Suite: **26.1**

IP Version: **4.2.0**



Online Version



Send Feedback

772538

2026.04.06

Contents

1. About the External Memory Interfaces Agilex™ 7 M-Series FPGA IP.....	10
1.1. Release Information.....	10
2. Agilex™ 7 M-Series FPGA EMIF IP – Introduction.....	12
2.1. Agilex 7 M-Series EMIF IP Protocol and Feature Support.....	12
2.2. Agilex 7 M-Series EMIF IP Design Flow.....	13
2.3. Agilex 7 M-Series EMIF IP Design Checklist.....	14
3. Agilex 7 M-Series FPGA EMIF IP – Product Architecture.....	16
3.1. Agilex 7 M-Series EMIF Architecture: Introduction.....	16
3.1.1. Agilex 7 M-Series EMIF Architecture: I/O Subsystem.....	17
3.1.2. Agilex 7 M-Series EMIF Architecture: I/O SSM.....	17
3.1.3. Agilex 7 M-Series EMIF Architecture: I/O Bank.....	18
3.1.4. Agilex 7 M-Series EMIF Architecture: I/O Lane.....	35
3.1.5. Agilex 7 M-Series EMIF Architecture: Input DQS Clock Tree.....	37
3.1.6. Agilex 7 M-Series EMIF Architecture: PHY Clock Tree.....	38
3.1.7. Agilex 7 M-Series EMIF Architecture: PLL Reference Clock Networks.....	39
3.1.8. Agilex 7 M-Series EMIF Architecture: Clock Phase Alignment.....	40
3.1.9. User Clock in Different Core Access Modes.....	41
3.2. Agilex 7 M-Series EMIF Sequencer.....	42
3.3. Agilex 7 M-Series EMIF Controller.....	43
3.3.1. Hard Memory Controller.....	43
3.4. Agilex 7 M-Series EMIF IP Interface Protocol: Interface Width, User Access Mode, and ECC.....	45
3.5. Agilex 7 M-Series EMIF IP for Hard Processor Subsystem (HPS).....	47
3.5.1. Restrictions on I/O Bank Usage for Agilex 7 M-Series EMIF IP with HPS.....	48
4. Agilex 7 M-Series FPGA EMIF IP – End-User Signals.....	50
4.1. IP Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component	50
4.1.1. s0_axi4_clock_in for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component.....	51
4.1.2. s0_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component.....	51
4.1.3. s0_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component.....	51
4.1.4. core_init_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component.....	51
4.1.5. s0_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component.....	52
4.1.6. s0_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component.....	54
4.1.7. s0_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component.....	55
4.1.8. s0_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component.....	55
4.1.9. mem_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component.....	56

4.1.10. mem_ck_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component.....	56
4.1.11. mem_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component.....	57
4.1.12. oct_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component.....	57
4.1.13. ref_clk for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component.....	57
4.2. IP Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM	57
4.2.1. s0_axi4_clock_in for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM.....	58
4.2.2. s0_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM.....	58
4.2.3. s0_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM.....	59
4.2.4. core_init_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM.....	59
4.2.5. s0_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM.....	59
4.2.6. s0_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM.....	62
4.2.7. s0_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM.....	62
4.2.8. s0_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM.....	62
4.2.9. mem_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM.....	63
4.2.10. mem_ck_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM.....	64
4.2.11. mem_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM.....	64
4.2.12. oct_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM.....	64
4.2.13. ref_clk for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM.....	64
4.3. IP Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component	65
4.3.1. s0_axi4_clock_in for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	66
4.3.2. s0_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	66
4.3.3. s0_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	66
4.3.4. core_init_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	66
4.3.5. s0_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	67
4.3.6. s1_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	69
4.3.7. s0_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	71

4.3.8. s0_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	71
4.3.9. s0_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	71
4.3.10. mem_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	72
4.3.11. mem_ck_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	72
4.3.12. mem_reset_n_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	73
4.3.13. mem_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	73
4.3.14. mem_ck_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	73
4.3.15. mem_reset_n_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	73
4.3.16. oct_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	74
4.3.17. oct_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	74
4.3.18. ref_clk for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component.....	74
4.4. IP Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM	74
4.4.1. s0_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	76
4.4.2. s0_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	76
4.4.3. s1_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	76
4.4.4. s1_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	76
4.4.5. s0_axi4_clock_in for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	77
4.4.6. core_init_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	77
4.4.7. s0_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	77
4.4.8. s1_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	80
4.4.9. s0_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	81
4.4.10. s0_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	81
4.4.11. s0_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	82
4.4.12. s1_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	82
4.4.13. s1_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	83
4.4.14. s1_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	83

4.4.15. mem_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	84
4.4.16. mem_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	84
4.4.17. mem_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	84
4.4.18. mem_ck_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	85
4.4.19. mem_ck_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	85
4.4.20. mem_i3c for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	85
4.4.21. mem_lb_dq for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	85
4.4.22. mem_lb_dqs for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	86
4.4.23. oct_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	86
4.4.24. oct_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	86
4.4.25. ref_clk for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM.....	86
4.5. IP Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5	87
4.5.1. s0_axi4_clock_in for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	88
4.5.2. core_init_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	88
4.5.3. s0_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	89
4.5.4. s0_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	89
4.5.5. s1_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	89
4.5.6. s1_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	89
4.5.7. s0_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	90
4.5.8. s1_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	92
4.5.9. s2_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	94
4.5.10. s3_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	95
4.5.11. s0_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	96
4.5.12. s0_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	96
4.5.13. s0_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	97
4.5.14. s1_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	97

4.5.15. s1_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	98
4.5.16. s1_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	98
4.5.17. mem_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	99
4.5.18. mem_ck_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	99
4.5.19. mem_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	99
4.5.20. mem_ck_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	100
4.5.21. mem_2 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	100
4.5.22. mem_ck_2 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	101
4.5.23. mem_3 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	101
4.5.24. mem_ck_3 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	101
4.5.25. mem_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	102
4.5.26. oct_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5	102
4.5.27. oct_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5	102
4.5.28. oct_2 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5	102
4.5.29. oct_3 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5	103
4.5.30. ref_clk for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5.....	103
5. Agilex 7 M-Series FPGA EMIF IP – Simulating Memory IP.....	104
5.1. Simulation Walkthrough.....	104
5.1.1. Calibration	105
5.1.2. Simulation Scripts.....	105
5.1.3. Functional Simulation with Verilog HDL.....	105
5.1.4. Simulating the Design Example.....	106
5.1.5. Simulating with sim_lib2.....	107
6. Agilex 7 M-Series FPGA EMIF IP – DDR4 Support.....	109
6.1. Agilex 7 M-Series DDR4 Component External Memory Interfaces (EMIF) IP Parameter Descriptions.....	109
6.2. Agilex 7 M-Series DDR4 DIMM External Memory Interfaces (EMIF) IP Parameter Descriptions.....	121
6.3. Agilex 7 M-Series FPGA EMIF IP Pin and Resource Planning.....	135
6.3.1. Agilex 7 M-Series FPGA EMIF IP Interface Pins.....	135
6.3.2. Agilex 7 M-Series FPGA EMIF IP Resources.....	137
6.3.3. Pin Guidelines for Agilex 7 M-Series FPGA EMIF IP.....	138
6.3.4. Pin Placements for Agilex 7 M-Series FPGA DDR4 EMIF IP.....	138
6.4. DDR4 Board Design Guidelines.....	152
6.4.1. Terminations for DDR4 with Agilex 7 M-Series Devices.....	153
6.4.2. General Layout Routing Guidelines.....	154
6.4.3. Reference Stackup.....	158

6.4.4. Agilex 7 M-Series EMIF-Specific Routing Guidelines for Various DDR4 Topologies.....	159
6.4.5. DDR4 Routing Guidelines: Discrete (Component) Topologies.....	164
7. Agilex 7 M-Series FPGA EMIF IP – DDR5 Support.....	180
7.1. Agilex 7 M-Series DDR5 Component External Memory Interfaces (EMIF) IP Parameter Descriptions.....	180
7.2. Agilex 7 M-Series DDR5 DIMM External Memory Interfaces (EMIF) IP Parameter Descriptions.....	199
7.3. Agilex 7 M-Series FPGA EMIF IP Pin and Resource Planning.....	220
7.3.1. Agilex 7 M-Series FPGA EMIF IP Interface Pins.....	220
7.3.2. Agilex 7 M-Series FPGA EMIF IP Resources.....	223
7.3.3. Pin Guidelines for Agilex 7 M-Series FPGA EMIF IP.....	224
7.3.4. Pin Placements for Agilex 7 M-Series FPGA DDR5 EMIF IP.....	230
7.3.5. Agilex 7 M-Series EMIF Pin Swapping Guidelines.....	238
7.4. DDR5 Board Design Guidelines.....	242
7.4.1. PCB Stack-up and Design Considerations.....	242
7.4.2. General Design Considerations.....	243
7.4.3. DDR Differential Signals Routing.....	244
7.4.4. Ground Plane and Return Path.....	246
7.4.5. RDIMM, UDIMM, and SODIMM Break-in Layout Guidelines.....	246
7.4.6. DRAM Break-in Layout Guidelines.....	247
7.4.7. General Notes for EMIF Routing Guidelines Tables.....	248
7.4.8. DDR5 PCB Layout Guidelines.....	249
7.4.9. DDR5 RDIMM Power Management IC.....	255
7.4.10. DDR5 Simulation Strategy.....	255
8. Agilex 7 M-Series FPGA EMIF IP – LPDDR5 Support.....	257
8.1. Agilex 7 M-Series LPDDR5 External Memory Interfaces (EMIF) IP Parameter Descriptions.....	257
8.2. Agilex 7 M-Series FPGA EMIF IP Pin and Resource Planning.....	277
8.2.1. Agilex 7 M-Series FPGA EMIF IP Interface Pins.....	277
8.2.2. Agilex 7 M-Series FPGA EMIF IP Resources.....	281
8.2.3. Pin Guidelines for Agilex 7 M-Series FPGA EMIF IP.....	281
8.2.4. Pin Placements for Agilex 7 M-Series FPGA LPDDR5 EMIF IP.....	285
8.3. LPDDR5 Board Design Guidelines.....	289
8.3.1. PCB Stack-up and Design Considerations.....	290
8.3.2. General Design Considerations.....	290
8.3.3. DDR Differential Signals Routing.....	291
8.3.4. Ground Plane and Return Path.....	293
8.3.5. DRAM Break-in Layout Guidelines.....	293
8.3.6. General Notes for EMIF Routing Guidelines Tables.....	294
8.3.7. LPDDR5 PCB Layout Guidelines.....	295
8.3.8. LPDDR5 Simulation Strategy.....	300
9. Agilex 7 M-Series FPGA EMIF IP – Timing Closure.....	301
9.1. Timing Closure	301
9.1.1. Timing Analysis.....	301
9.2. Optimizing Timing.....	302
10. Agilex 7 M-Series FPGA EMIF IP – Controller Optimization.....	304
10.1. Optimizing Efficiency for Secondary Controller.....	304

10.2. Interface Standard.....	306
10.3. Bank Management Efficiency.....	306
10.4. Data Transfer.....	307
10.5. Improving Controller Efficiency.....	307
10.5.1. Using Multiple AXI IDs.....	307
10.5.2. Controller Performance Profile.....	307
10.5.3. Frequency of Operation.....	308
10.5.4. Controller Performance Profile.....	308
10.5.5. Series of Reads or Writes.....	308
10.5.6. Special AXI Interface Signals.....	309
10.5.7. Address Mapping — AXI Interface to DDR Memory.....	309
10.5.8. AXI Interface: Write and Read Example.....	311
11. Agilex 7 M-Series FPGA EMIF IP – Debugging.....	313
11.1. Interface Configuration Performance Issues.....	313
11.1.1. Interface Configuration Bottleneck and Efficiency Issues.....	313
11.2. Functional Issue Evaluation.....	314
11.2.1. Altera IP Memory Model.....	314
11.2.2. Vendor Memory Model.....	315
11.2.3. Transcript Window Messages.....	315
11.3. Timing Issue Characteristics.....	317
11.3.1. Evaluating FPGA Timing Issues.....	317
11.3.2. Evaluating External Memory Interface Timing Issues.....	318
11.4. Verifying Memory IP Using the Signal Tap Logic Analyzer.....	319
11.5. Debugging with the External Memory Interface Debug Toolkit.....	320
11.5.1. Prerequisites for Using the EMIF Debug Toolkit.....	320
11.5.2. Configuring Your Design to Use the EMIF Debug Toolkit.....	321
11.5.3. Launching the EMIF Debug Toolkit.....	321
11.5.4. Using the EMIF Debug Toolkit.....	322
11.6. Generating Traffic with the Test Engine IP.....	327
11.7. Guidelines for Developing HDL for Traffic Generator.....	328
11.8. Guidelines for Traffic Generator Status Check.....	329
11.8.1. Status Check Using the Signal Tap Logic Analyzer.....	329
11.8.2. Exporting the Status Interface to the Top-Level Design.....	331
11.9. Hardware Debugging Guidelines.....	333
11.9.1. Create a Simplified Design that Demonstrates the Same Issue.....	334
11.9.2. Measure Power Distribution Network.....	334
11.9.3. Measure Signal Integrity and Setup and Hold Margin.....	334
11.9.4. Vary Voltage.....	334
11.9.5. Operate at a Lower Speed.....	334
11.9.6. Determine Whether the Issue Exists in Previous Versions of Software.....	335
11.9.7. Determine Whether the Issue Exists in the Current Version of Software.....	335
11.9.8. Try A Different PCB.....	335
11.9.9. Try Other Configurations.....	336
11.9.10. Debugging Checklist.....	336
11.10. Categorizing Hardware Issues.....	337
11.10.1. Signal Integrity Issues.....	337
11.10.2. Hardware and Calibration Issues.....	340
11.11. Debugging Intermittent Issues.....	347

12. Agilex 7 FPGA EMIF IP - Mailbox Support.....	348
12.1. Agilex 7 Mailbox Structure and Register Definitions.....	348
12.1.1. Mailbox Supported Commands.....	350
12.1.2. Mailbox Command Definitions.....	356
12.1.3. Address Mapping Reference.....	368
12.1.4. VREF Calculation Reference Data.....	369
12.1.5. ECC Syndrome Codes.....	369
12.1.6. ECC Error Handling.....	370
12.1.7. Accessing Read-Only Registers.....	373
12.1.8. Sending a Mailbox Command.....	374
12.1.9. IOSSM Mailbox Access Script.....	381
13. Document Revision History for External Memory Interfaces Agilex 7 M-Series FPGA IP User Guide.....	389

1. About the External Memory Interfaces Agilex™ 7 M-Series FPGA IP

1.1. Release Information

IP versions are the same as the Quartus® Prime Design Suite software versions up to v19.1. From Quartus Prime Design Suite software version 19.2 or later, IP cores have a new IP versioning scheme.

The IP version (X.Y.Z) number may change from one Quartus Prime software version to another. A change in:

- X indicates a major revision of the IP. If you update your Quartus Prime software, you must regenerate the IP.
- Y indicates the IP includes new features. Regenerate your IP to include these new features.
- Z indicates the IP includes minor changes. Regenerate your IP to include these changes.

Release Information

IP Name	IP Version	Quartus Version	Release Date
External Memory Interfaces (EMIF) IP - DDR4 Component	4.2.0	26.1	2026.04.06
External Memory Interfaces (EMIF) IP - DDR4 DIMM	4.2.0	26.1	2026.04.06
External Memory Interfaces (EMIF) IP - DDR5 Component	4.3.0	26.1	2026.04.06
External Memory Interfaces (EMIF) IP - DDR5 DIMM	4.3.0	26.1	2026.04.06
External Memory Interfaces (EMIF) IP - LPDDR5 Component	4.2.0	26.1	2026.04.06

Note: This documentation is preliminary and subject to change.

Related Information

- [External Memory Interfaces Agilex™ 7 M-Series FPGA IP Design Example User Guide](#)
Design example and quick-start information for the Agilex™ 7 M-Series EMIF IP, which provides external memory interface support for the DDR4, DDR5, and LPDDR5 memory protocols.
- [External Memory Interfaces Agilex™ 7 M-Series FPGA IP Release Notes](#)
Release notes for the Agilex™ 7 M-Series EMIF IP.

2. Agilex™ 7 M-Series FPGA EMIF IP – Introduction

Altera's fast, efficient, and low-latency external memory interface (EMIF) intellectual property (IP) cores interface with today's higher speed memory devices.

You can implement the EMIF IP core functions through the Quartus Prime software.

The *External Memory Interfaces Agilex™ 7 M-Series FPGA IP* (referred to hereafter as the *Agilex 7 M-Series EMIF IP*) provides the following components:

- A physical layer interface (PHY) which builds the data path and manages timing transfers between the FPGA and the memory device.
- A memory controller which implements all the memory commands and protocol-level requirements.

For information on the maximum speeds supported by the external memory interface IP, refer to the *External Memory Interface Spec Estimator*, available here: <https://www.intel.com/content/www/us/en/programmable/support/support-resources/support-centers/external-memory-interfaces-support/emif.html>.

2.1. Agilex 7 M-Series EMIF IP Protocol and Feature Support

- The Agilex 7 M-Series FPGA EMIF IP supports DDR4 with hard memory controller and hard PHY.
- The Agilex 7 M-Series FPGA EMIF IP supports DDR5 with hard memory controller and hard PHY.
- The Agilex 7 M-Series FPGA EMIF IP supports LPDDR5 with hard memory controller and hard PHY.

The following table details support for memory format, rank, component and DIMM density across memory protocols:

Table 2.

Device	Memory Format	Rank	Base DRAM Width	ES	Production	Component Width	Max Component	Max Capacity
DDR4	Component *	1R	x16	Y	Y	x8 / x16	16 Gb	N/A
		1R	x8	Y	Y			
		2R	x16	Y	Y			
		2R	x8	Y	Y			
	SODIMM **	1R	x16	Y	Y	x4 / x8 / x16	N/A	64 GB
		1R	x8	Y	Y			
		2R	x8	Y	Y			

continued...

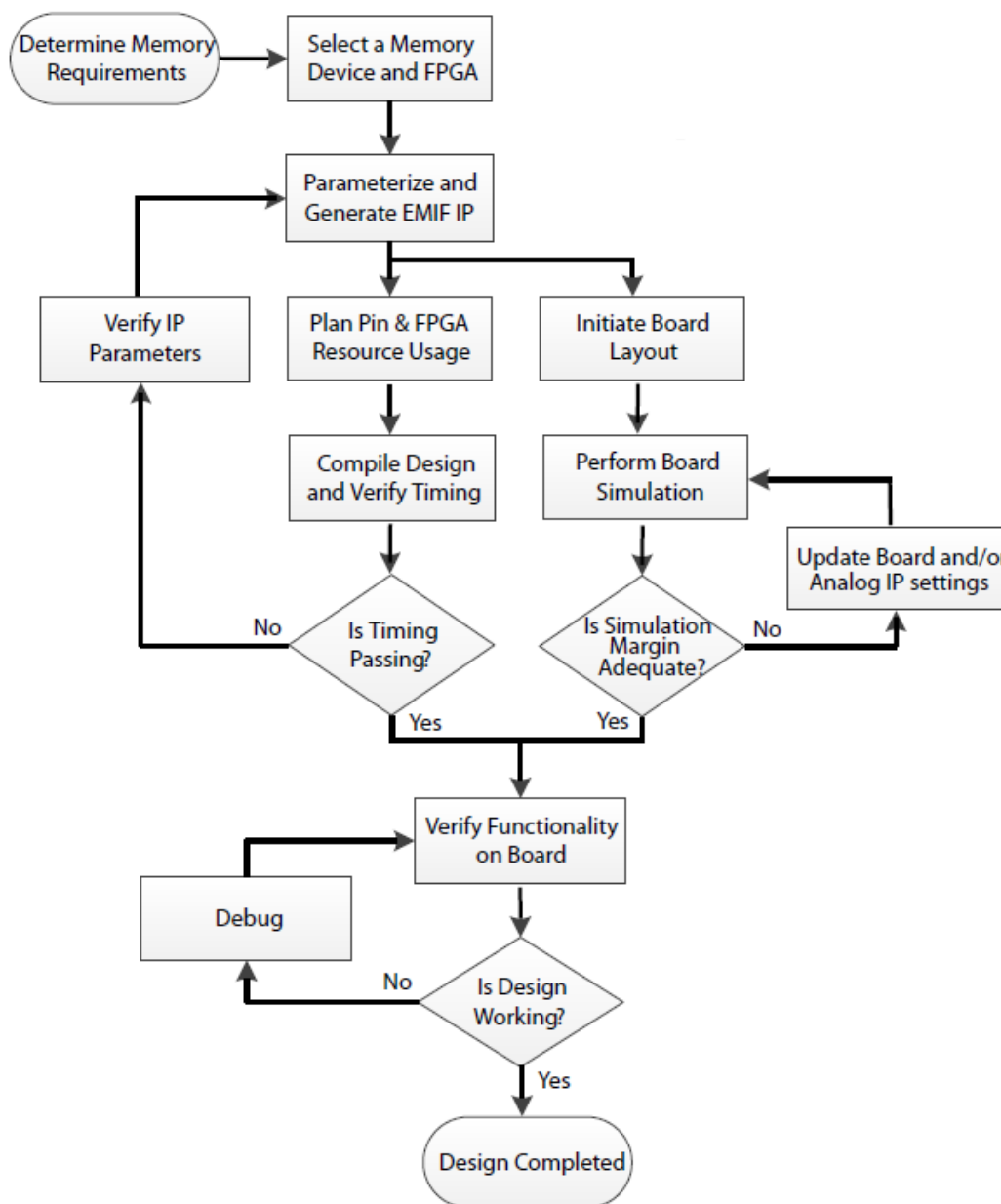
Device	Memory Format	Rank	Base DRAM Width	ES	Production	Component Width	Max Component	Max Capacity
	UDIMM **	1R	x16	Y	Y			
		1R	x8	Y	Y			
		2R	x8	Y	Y			
	RDIMM **	1R	x8	Y	Y			
		1R	x4	Y	Y			
		2R	x8	Y	Y			
		2R	x4	Y	Y			
DDR5	Component	1R	x16	Y	Y	x8 / x16	16 Gb 24 Gb 32 Gb	N/A
		1R	x8	Y	Y			
		2R	x16	Y	Y			
		2R	x8	Y	Y			
	SODIMM **	1R	x16	Y	Y	x8 / x16	N/A	64 GB
		1R	x8	Y	Y			
		2R	x8	Y	Y			
	UDIMM **	1R	x16	Y	Y	x8 / x16	N/A	64 GB
		1R	x8	Y	Y			
		2R	x8	Y	Y			
	RDIMM **	1R	x8	Y	Y	x4 / x8	N/A	128 GB (Fabric EMIF) 64 GB (HPS EMIF)
		1R	x4	Y	Y			
		2R	x8	Y	Y			
		2R	x4	Y	Y			
Note: * Clamshell support is not available. ** 3DS support is not available. ** Single sub-channel implementation is not supported; you must configure and use both sub-channels for a DIMM.								

2.2. Agilex 7 M-Series EMIF IP Design Flow

Altera recommends creating an example top-level file with the desired pin outs and all interface IPs instantiated. This enables the Quartus Prime software to validate the design and resource allocation before PCB and schematic sign off.

The following figure shows the design flow to provide the fastest out-of-the-box experience with the EMIF IP.

Figure 1. EMIF IP Design Flow



2.3. Agilex 7 M-Series EMIF IP Design Checklist

Refer to the following checklist as a quick reference for information about steps in the EMIF design flow.

Table 3. EMIF Design Checklist

Design Step	Description	Resources
Select an FPGA	Not all Altera FPGAs support all memory types and configurations. To help with the FPGA selection process, refer to the resources listed in the right column.	• External Memory Interfaces Support Center • External Memory Interface Spec Estimator
Parameterize the IP	Correct IP parameterization is important for good EMIF IP operation. The resources listed in the right column define the memory parameters during IP generation.	• DDR4 Component Parameter Descriptions • DDR4 DIMM Parameter Descriptions • DDR5 Component Parameter Descriptions • DDR5 DIMM Parameter Descriptions • LPDDR5 Parameter Descriptions
Generate initial IP and example design	After you have parameterized the EMIF IP, you can generate the IP, along with an optional example design. Refer to the Quick-Start Guide for a walkthrough of this process.	• External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide
Perform functional simulation	Simulation of the EMIF design helps to determine correct operation. The resources listed in the right column explain how to perform simulation and what differences exist between simulation and hardware implementation.	• External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide • Simulating Memory IP
Make pin assignments	For guidance on pin placement, refer to the resources listed in the right column.	• DDR4 Component Parameter Descriptions • DDR4 DIMM Parameter Descriptions • DDR5 Component Parameter Descriptions • DDR5 DIMM Parameter Descriptions • LPDDR5 Parameter Descriptions
Perform board simulation	Board simulation helps determine optimal settings for signal integrity, drive strength, as well as sufficient timing margins and eye openings. For guidance on board simulation, refer to the resources listed in the right column.	• Board Design Guidelines • Timing Closure
Verify timing closure	For information regarding compilation, system-level timing closure and timing reports refer to the Timing Closure section of this User Guide.	• Timing Closure
Run the design on hardware	For instructions on how to program a FPGA refer to the Quick-Start section of the Design Example User Guide.	• External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide
Debug issues with preceding steps	Operational problems can generally be attributed to one of the following: interface configuration, pin/resource planning, signal integrity, or timing. The resources listed in the right column contain information on typical debug procedures and available tools to help diagnose hardware issues.	• Debugging • External Memory Interfaces Support Center

3. Agilex 7 M-Series FPGA EMIF IP – Product Architecture

This chapter describes the Agilex 7 M-Series FPGA EMIF IP product architecture.

3.1. Agilex 7 M-Series EMIF Architecture: Introduction

The Agilex 7 M-Series EMIF architecture contains many new hardware features designed to meet the high-speed requirements of emerging memory protocols, while consuming the smallest amount of core logic area and power.

Note: The current version of the External Memory Interfaces Agilex 7 M-Series FPGA IP supports the DDR4, DDR5, and LPDDR5 memory protocols.

The following are key hardware features of the Agilex 7 M-Series EMIF architecture:

Hard Sequencer

The sequencer employs a hardened processor, and can perform memory calibration for a wide range of protocols. For Agilex 7 M-Series devices, the sequencer and calibration are localized to each I/O bank.

Note: You cannot use the hardened processor for any user applications after calibration is complete.

Hard PHY

The PHY circuitry in Agilex 7 M-Series devices is hardened in the silicon, which simplifies the challenges of achieving timing closure and minimizing power consumption.

Hard Memory Controller

The hard memory controller reduces latency and minimizes core logic consumption in the external memory interface. The hard memory controller supports the DDR4, DDR5, and LPDDR5 memory protocols.

High-Speed PHY Clock Tree

Dedicated high speed PHY clock networks clock the I/O buffers in Agilex 7 M-Series EMIF IP. The PHY clock trees exhibit low jitter and low duty cycle distortion, maximizing the data valid window.

Automatic Clock Phase Alignment

Automatic clock phase alignment circuitry dynamically adjusts the clock phase of core clock networks to match the clock phase of the PHY clock networks. The clock phase alignment circuitry minimizes clock skew that can complicate timing closure in transfers between the FPGA core and the periphery.

Network-on-Chip (NoC) Interface

The Agilex 7 M-Series EMIF IP supports a new Network-on-Chip (NoC) interface. Each IO96 bank contains two 256-bit AXI4 targets and one 32-bit AXI4-Lite target that connect to the NoC. NoC segments span one FPGA clock sector and consists of three AXI4 initiators on the FPGA fabric side. A network of switches transfer packets horizontally across the high-speed interconnect NoC and connect the initiators and targets. Refer to the *NoC User Guide* for additional information.

3.1.1. Agilex 7 M-Series EMIF Architecture: I/O Subsystem

In Agilex 7 M-Series devices, the I/O subsystem consists of two rows at the edge of the core.

The I/O subsystem provides the following features:

- General-purpose I/O registers and I/O buffers
- Compensation Block (Comp block)
 - ☐ On-chip termination control (OCT)
- I/O PLLs
 - ☐ I/O Bank I/O PLL for external memory interfaces and user logic
 - ☐ Fabric-feeding for non-EMIF/non-LVDS SERDES IP applications
- True differential signaling
- External memory interface components, as follows:
 - ☐ A Primary hard memory controller, which has connectivity to 8 lanes (up to 4 byte lanes for data, and optionally one additional lane for out-of-band error correction code (ECC) data)
 - ☐ A Secondary hard memory controller, which has connectivity to 4 lanes (up to 2 byte lanes for data)
 - ☐ Hard PHY
 - ☐ Hardened processor and calibration logic
 - ☐ Dynamic-link library (DLL)

3.1.2. Agilex 7 M-Series EMIF Architecture: I/O SSM

Each I/O bank includes one I/O subsystem manager (I/O SSM), which contains a hardened processor with dedicated memory. The I/O SSM is responsible for calibration of all the EMIFs in the I/O bank.

The I/O SSM includes dedicated memory which stores both the calibration algorithm and calibration run-time data. The hardened processor and the dedicated memory can be used only by an external memory interface, and cannot be employed for any other use.

The on-chip configuration network clocks the I/O SSM, and therefore the I/O SSM does not consume a PLL.

Each EMIF instance must be connected to the I/O SSM through the External Memory Interfaces Calibration IP. The Calibration IP exposes a calibration bus master port, which must be connected to the slave calibration bus port on every EMIF instance.

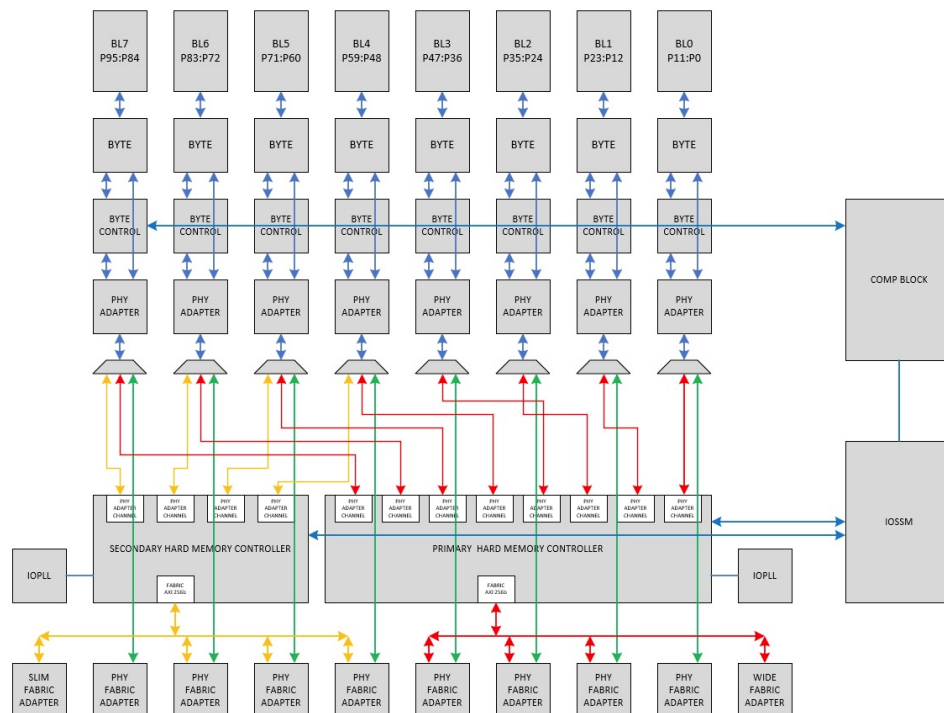
3.1.3. Agilex 7 M-Series EMIF Architecture: I/O Bank

Each I/O row contains up to four I/O banks; the exact number of banks depends on device size and pin package.

Each I/O bank consists of two sub-banks, and each sub-bank contains the following components:

- I/O PLL and PHY clock trees
- DLL
- Input DQS clock trees
- 48 pins, organized into four I/O lanes of 12 pins each

Figure 3. I/O Bank Architecture in Agilex 7 M-Series Devices



Within an I/O bank, the top sub-bank is pin indexes P95:P48, and the bottom sub-bank is pin indexes P47:P0.

Agilex 7 M-Series devices have two hard memory controllers: primary and secondary. The primary hard memory controller has access to all 96 pins in an I/O bank. The secondary hard memory controller has access only to the top sub-bank. In the above figure, the yellow signals highlight the connections for the secondary hard memory controller, while the red signals show the connections for the primary hard memory controller. The green signals show where both hard memory controllers are bypassed to provide access to the PHY from the core logic.

3.1.3.1. I/O Byte Lane Organization in Different Agilex 7 M-Series Devices

There are three different package options for Agilex 7 M-series devices.

R47A and R47B packages have the same byte lane configuration, and differ in the available transceiver tiles. R31B is a package that enables design migration from Agilex 7 I-Series to Agilex 7 M-Series devices that share this same package.

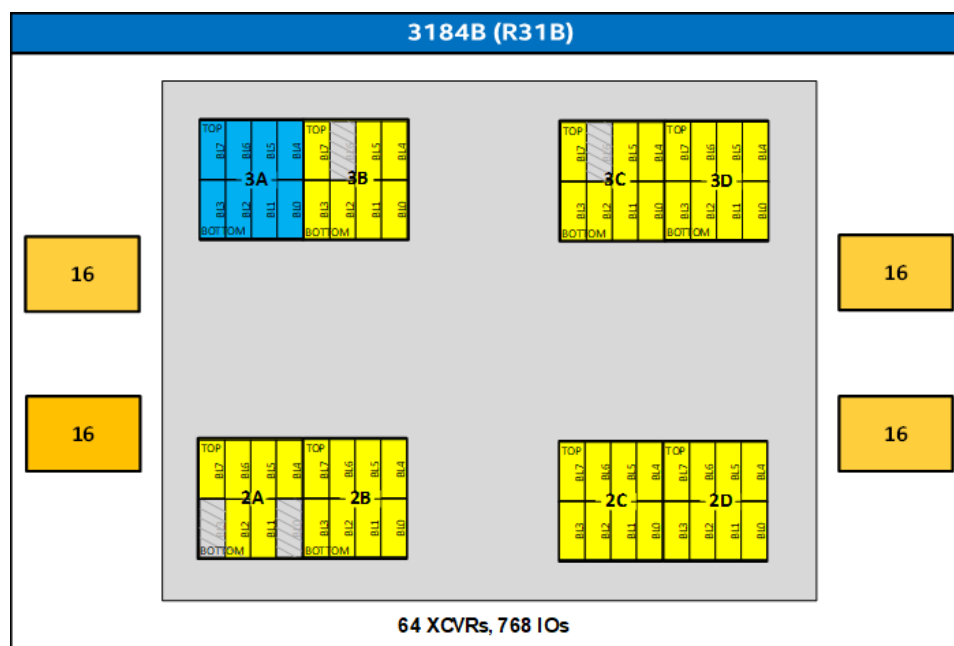
The following figures show the I/O byte lanes organization and availability across the different I/O Banks for each package.

R31B Package

In bank 3B and 3C top sub-bank, byte lane #6 is not available; in bank 2A bottom sub-bank, byte lanes #0 and #3 are not available. This affects the maximum number of external memory interfaces per device.

For more information, refer to [Maximum Number of DDR4 Interfaces](#), [Maximum Number of DDR5 Interfaces](#), and [Maximum Number of LPDDR5 Interfaces](#).

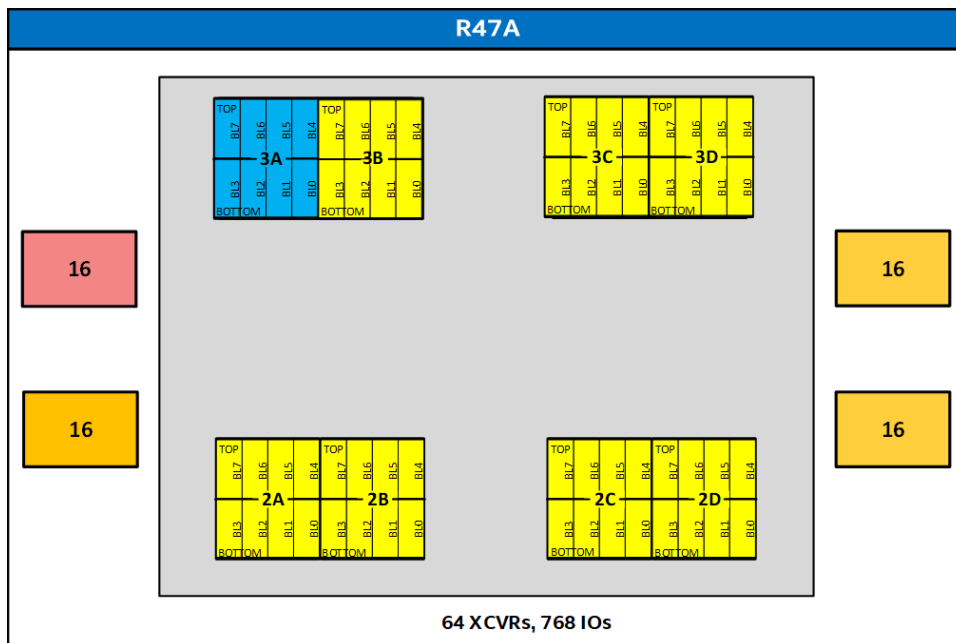
Figure 4. R31B Package



R47A Package

All byte lanes are available in all banks. For information on the maximum number of interfaces for a given protocol, refer to [Maximum Number of DDR4 Interfaces](#), [Maximum Number of DDR5 Interfaces](#), and [Maximum Number of LPDDR5 Interfaces](#).

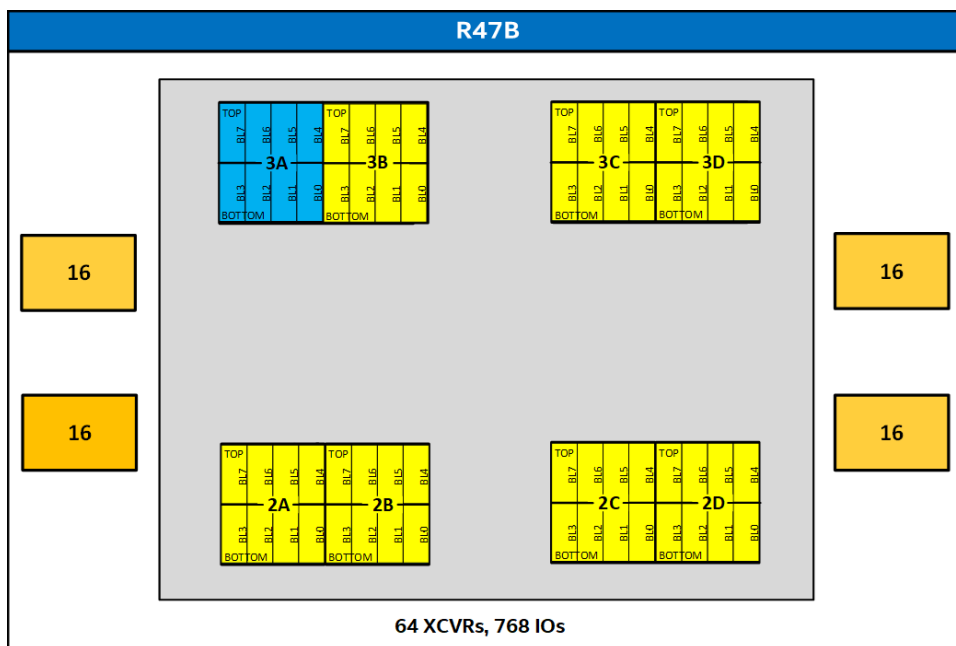
Figure 5. R47A Package



R47B Package

All byte lanes are available in all banks. For information on the maximum number of interfaces for a given protocol, refer to [Maximum Number of DDR4 Interfaces](#), [Maximum Number of DDR5 Interfaces](#), and [Maximum Number of LPDDR5 Interfaces](#).

Figure 6. R47B Package



3.1.3.2. Lockstep Configuration

To support user data widths greater than 32 bits in DDR4, the external memory interface (EMIF) IP instantiates multiple memory controllers driven in lockstep. In lockstep configurations, multiple controllers run the same set of operations simultaneously in parallel.

The primary controller drives the address and command bus and 32-bits of the DQ bus, while the other controllers drive the remainder of the DQ bus.

The following table summarizes the supported lockstep configurations:

Table 4. Supported Lockstep Controller Configurations

Memory Protocol	Configuration	DQ Width	AXI Interface
DDR4	x40	40	256 b + 64 b USER DATA
DDR4	x64	64	512 b
DDR4	x64 with ECC	72	512 b
DDR4	x72	72	512 b + 64 b USER DATA

To ensure that the controllers remain coordinated in lockstep, the following points apply:

- Lockstep configurations support only synchronous fabric clocking mode.
- Some of the controller scheduling and optimization features are disabled.
- There are limitations on the types of AXI transactions supported. Refer to the following table for details.

Table 6. Limitations for Lockstep Configuration in ES Devices

Supported Protocol	Device	Configuration	Known Limitations
DDR4	AGMX039R47AXEXVR0	x72	☐ Supports AXI transfer size=6 only
DDR4	All Agilex 7 M-Series devices except AGMX039R47AXEXVR0	x72	☐ AXI transfer size min = 3 (8 bytes + 1 byte WUSER/RUSER) ☐ Supports 8-byte aligned transfers only
DDR4	All Agilex 7 M-Series devices	x64	☐ AXI transfer size min = 1 (2 bytes) ☐ Supports 2-byte aligned transfers only
DDR4	AGMX039R47AXEXVR0	x40	☐ Supports AXI transfer size=5 only ☐ Supports 32-bit aligned transfers only
DDR4	All Agilex 7 M-Series devices except AGMX039R47AXEXVR0	x40	☐ AXI transfer size min = 2 (4 bytes + 1 byte RUSER/WUSER) ☐ Support 32-bit aligned transfers only

Narrow Read Transfer Support

Narrow read transfer happens when the AXI master generates a read data transfer that is narrower than its data bus width. For example, for a 256-bit AXI data bus, a narrow read transfer happens when ARSIZE < 5.

Configuration	Controller	OPN	Narrow Read Transfer Support?
Lockstep	Both Primary & Secondary	All Agilex 7 M-Series devices	- No. Must perform full width read data transfer. - Can support up to 6-bit of ARID/AWID for lockstep configuration that uses both Primary and Secondary Controller (x40 lockstep, x64+ECC lockstep and x72 lockstep).
	Primary only	All Agilex 7 M-Series devices	- No. Must perform full width read data transfer. - Can support up to 7-bit of ARID/AWID for lockstep configuration that uses Primary Controller only (x64 lockstep).
Non-Lockstep	Both Primary & Secondary	AGMX039R47AXEXVR0	- No. Must perform full width read data transfer. - Can support up to 6-bit of ARID/AWID.
	Primary Controller	All Agilex 7 M-Series devices except AGMX039R47AXEXVR0	Yes. Up to 7-bit of ARID/AWID.
	Secondary Controller		Yes. Up to 6-bit of ARID/AWID.

Lockstep configuration does not support narrow read transfer. For non-lockstep configuration, AGMX039R47AXEXVR0 devices do not support narrow read transfer.

Two Controllers in Lockstep Within One IO96 Bank

The EMIF IP instantiates 2 controllers within one IO96 bank to support x40 configurations. The AXI bus is configured as 256-bits wide plus 64-bits of user data (WUSER/RUSER), to generate the required 320-bits of data to transfer a burst-of-8 of 40-bit DQ. The following table illustrates how the AXI WDATA/RDATA and WUSER/RUSER can be mapped to the DQ lanes.

Table 7. Mapping of WDATA/RDATA & WUSER/RUSER in x40 Configuration

Transfer	0	1	2	3	4	5	6	7
WDATA/RDATA	31:0	63:32	95:64	127:96	159:128	191:160	223:192	255:224
DQ	[31:0]							
WUSER/RUSER	7:0	15:8	23:16	31:24	39:32	47:40	55:48	63:56
DQ	[39:32]							

This configuration can support only 3 address/command lane configurations, because there are only 8 byte lanes in one IO96 bank. The WUSER/RUSER signal is mapped to byte lane 7 (DQ lane with prefix s) in x40 configuration.

Table 8. Supported Byte Lane Placement for x40 configuration

Scheme	BL0	BL1	BL2	BL3	BL4	BL5	BL6	BL7
DDR4_AC_TOP	DQ[4]	DQ[3]	DQ[2]	DQ[1]	AC1	AC2	AC0	sDQ[0]
DDR4_AC_BOT	DQ[0]	AC0	AC1	AC2	DQ[1]	DQ[2]	DQ[3]	sDQ[4]

Two/Three Controllers in Lockstep Within One IO96 Bank

This configuration is supported only in DDR4. The EMIF IP instantiates two or three controllers across two adjacent IO96 banks for the configurations listed in the table below.

Table 9. Supported 2/3 Controller Configurations in Lockstep

Configuration	DQ Width	AXI Interface	Notes
x64	64	512b	2 controllers used.
x64 with ECC	72	512b	3 controllers used. ECC will be generated and checked by a soft IP block within the EMIF IP.
x72	72	512b + 64b USER DATA	3 controllers used.

For the x72 configuration, the AXI bus is configured as 512-bits wide, plus 64-bits of user data (WUSER/RUSER). The following example illustrates how you can map the AXI WDATA/RDATA and WUSER/RUSER to the DQ lanes. In this illustration, the WUSER/RUSER is mapped to the byte lane used for DQ [71:64].

The actual DQ lane to which the WUSER/RUSER is mapped depends on the address and command placement used. In each supported address and command placement scheme, the WUSER/RUSER is mapped to the DQ lane that has a prefix s (for example, sDQ0, sDQ4 or sDQ8). Refer to the following tables in the [DDR4 Data Width Mapping](#) topic, to identify the actual DQ lane used for WUSER/RUSER:

- Supported Lockstep configuration for DDR4 x64
- Supported Lockstep configuration for DDR4 x72 or x64 (with ECC)

Table 10. Example Mapping of WDATA/RDATA & WUSER/RUSER in x72 Configuration

Transfer	0	1	2	3	4	5	6	7
WDATA/ RDATA	63:0	127:64	191:128	255:192	319:256	383:320	447:384	511:448
DQ	[63:0]							
WUSER/ RUSER	7:0	15:8	23:16	31:24	39:32	47:40	55:48	63:56
DQ	[71:64]							

The generated IP has 2 mem DQ ports, collectively labeled as mem_DQ0 and mem_DQ1. The mapping below shows how these 2 ports are grouped into one single wide DQ port:

Figure 7.

Configuration	x64		3AC -x72 or x64+ECC		4AC -x72 or x64+ECC	
mem_DQ0	31:0		39:0		31:0	
mem_DQ1		31:0		31:0		39:0
	↓	↓	↓	↓	↓	↓
mem_DQ	31:0	63:32	39:0	71:40	31:0	63:32

AXI wstrb Signal Grouping Requirements for Lockstep Mode

When EMIF controllers operate in lockstep mode, the write strobe (wstrb) signal is subject to additional restrictions beyond those defined by the standard AXI specification. Normally, the AXI protocol allows byte-level granularity for wstrb, enabling individual bytes to be selected; however, in lockstep mode, wstrb must be set with a granularity of four-byte groups.

Specifically, within each consecutive 4-bit group of the wstrb signal, all bits must have the same value (either all zeros or all ones). This means that, for any 32-bit wstrb, each set of 4 bits (e.g., bits 0–3, 4–7, 8–11, etc.) must be uniformly set.

Table 11.

AXI wstrb	Legal in Lockstep?
1011_1001_1111_0000_1111_0111_0000_1011	No
0000_1111_1001_1111_1111_1000_1111_1111	No
1111_0000_0000_1111_0000_1111_0000_0000	Yes
0000_0000_0000_1111_1111_1111_0000_0000	Yes

It is the responsibility of the user logic to ensure this restriction is followed, as there is no hardware checking or error reporting for violations. Failure to comply may cause the controllers to fall out of lockstep, potentially leading to functional errors.

3.1.3.3. DDR4 Pin Placement

Table 12. DDR4 Pin Placement

Lane Number	Pin Index	x32+ECC *	x 32	x16 + ECC *	x16
BL7	95	MEM_DQ[39]*			
	94	MEM_DQ[38] *			
	93	MEM_DQ[37] *			
	92	MEM_DQ[36] *			
	91				
	90	MEM_DM_N[4]			
continued...					

Lane Number	Pin Index	x32+ECC *	x 32	x16 + ECC *	x16
	89	MEM_DQS_C[4]			
	88	MEM_DQS_T[4]			
	87	MEM_DQ[35] *			
	86	MEM_DQ[34] *			
	85	MEM_DQ[33] *			
	84	MEM_DQ[32] *			
BL6	83	MEM_DQ[31]	MEM_DQ[31]		
	82	MEM_DQ[30]	MEM_DQ[30]		
	81	MEM_DQ[29]	MEM_DQ[29]		
	80	MEM_DQ[28]	MEM_DQ[28]		
	79				
	78	MEM_DM_N[3]	MEM_DM_N[3]		
	77	MEM_DQS_C[3]	MEM_DQS_C[3]		
	76	MEM_DQS_T[3]	MEM_DQS_T[3]		
	75	MEM_DQ[27]	MEM_DQ[27]		
	74	MEM_DQ[26]	MEM_DQ[26]		
	73	MEM_DQ[25]	MEM_DQ[25]		
	72	MEM_DQ[24]	MEM_DQ[24]		
BL5	71	MEM_DQ[23]	MEM_DQ[23]	MEM_DQ[23] *	
	70	MEM_DQ[22]	MEM_DQ[22]	MEM_DQ[22] *	
	69	MEM_DQ[21]	MEM_DQ[21]	MEM_DQ[21] *	
	68	MEM_DQ[20]	MEM_DQ[20]	MEM_DQ[20] *	
	67				
	66	MEM_DM_N[2]	MEM_DM_N[2]	MEM_DM_N[2]	
	65	MEM_DQS_C[2]	MEM_DQS_C[2]	MEM_DQS_C[2]	
	64	MEM_DQS_T[2]	MEM_DQS_T[2]	MEM_DQS_T[2]	
	63	MEM_DQ[19]	MEM_DQ[19]	MEM_DQ[19] *	
	62	MEM_DQ[18]	MEM_DQ[18]	MEM_DQ[18] *	
	61	MEM_DQ[17]	MEM_DQ[17]	MEM_DQ[17] *	
	60	MEM_DQ[16]	MEM_DQ[16]	MEM_DQ[16] *	
BL4	59	MEM_DQ[15]	MEM_DQ[15]	MEM_DQ[15]	MEM_DQ[15]
	58	MEM_DQ[14]	MEM_DQ[14]	MEM_DQ[14]	MEM_DQ[14]
	57	MEM_DQ[13]	MEM_DQ[13]	MEM_DQ[13]	MEM_DQ[13]
	56	MEM_DQ[12]	MEM_DQ[12]	MEM_DQ[12]	MEM_DQ[12]
	55				
continued...					

Lane Number	Pin Index	x32+ECC *	x 32	x16 + ECC *	x16
	54	MEM_DM_N[1]	MEM_DM_N[1]	MEM_DM_N[1]	MEM_DM_N[1]
	53	MEM_DQS_C[1]	MEM_DQS_C[1]	MEM_DQS_C[1]	MEM_DQS_C[1]
	52	MEM_DQS_T[1]	MEM_DQS_T[1]	MEM_DQS_T[1]	MEM_DQS_T[1]
	51	MEM_DQ[11]	MEM_DQ[11]	MEM_DQ[11]	MEM_DQ[11]
	50	MEM_DQ[10]	MEM_DQ[10]	MEM_DQ[10]	MEM_DQ[10]
	49	MEM_DQ[9]	MEM_DQ[9]	MEM_DQ[9]	MEM_DQ[9]
	48	MEM_DQ[8]	MEM_DQ[8]	MEM_DQ[8]	MEM_DQ[8]
BL3	47	MEM_BG[0]	MEM_BG[0]	MEM_BG[0]	MEM_BG[0]
	46	MEM_BA[1]	MEM_BA[1]	MEM_BA[1]	MEM_BA[1]
	45	MEM_BA[0]	MEM_BA[0]	MEM_BA[0]	MEM_BA[0]
	44	MEM_ALERT_N[0]	MEM_ALERT_N[0]	MEM_ALERT_N[0]	MEM_ALERT_N[0]
	43	MEM_A[16]	MEM_A[16]	MEM_A[16]	MEM_A[16]
	42	MEM_A[15]	MEM_A[15]	MEM_A[15]	MEM_A[15]
	41	MEM_A[14]	MEM_A[14]	MEM_A[14]	MEM_A[14]
	40	MEM_A[13]	MEM_A[13]	MEM_A[13]	MEM_A[13]
	39	MEM_A[12]	MEM_A[12]	MEM_A[12]	MEM_A[12]
	38	RZQ Site	RZQ Site	RZQ Site	RZQ Site
	37	Differential "N-Side" Reference Clock Input Site	Differential "N-Side" Reference Clock Input Site	Differential "N-Side" Reference Clock Input Site	Differential "N-Side" Reference Clock Input Site
	36	Differential "P-Side" Reference Clock Input Site	Differential "P-Side" Reference Clock Input Site	Differential "P-Side" Reference Clock Input Site	Differential "P-Side" Reference Clock Input Site
BL2	35	MEM_A[11]	MEM_A[11]	MEM_A[11]	MEM_A[11]
	34	MEM_A[10]	MEM_A[10]	MEM_A[10]	MEM_A[10]
	33	MEM_A[9]	MEM_A[9]	MEM_A[9]	MEM_A[9]
	32	MEM_A[8]	MEM_A[8]	MEM_A[8]	MEM_A[8]
	31	MEM_A[7]	MEM_A[7]	MEM_A[7]	MEM_A[7]
	30	MEM_A[6]	MEM_A[6]	MEM_A[6]	MEM_A[6]
	29	MEM_A[5]	MEM_A[5]	MEM_A[5]	MEM_A[5]
	28	MEM_A[4]	MEM_A[4]	MEM_A[4]	MEM_A[4]
	27	MEM_A[3]	MEM_A[3]	MEM_A[3]	MEM_A[3]
	26	MEM_A[2]	MEM_A[2]	MEM_A[2]	MEM_A[2]
	25	MEM_A[1]	MEM_A[1]	MEM_A[1]	MEM_A[1]
	24	MEM_A[0]	MEM_A[0]	MEM_A[0]	MEM_A[0]
BL1	23	MEM_PAR[0]	MEM_PAR[0]	MEM_PAR[0]	MEM_PAR[0]
	22	MEM_CS_N[1]	MEM_CS_N[1]	MEM_CS_N[1]	MEM_CS_N[1]

continued...

Lane Number	Pin Index	x32+ECC *	x 32	x16 + ECC *	x16
	21	MEM_CK_C[0]	MEM_CK_C[0]	MEM_CK_C[0]	MEM_CK_C[0]
	20	MEM_CK_T[0]	MEM_CK_T[0]	MEM_CK_T[0]	MEM_CK_T[0]
	19	MEM_CKE[1]	MEM_CKE[1]	MEM_CKE[1]	MEM_CKE[1]
	18	MEM_CKE[0]	MEM_CKE[0]	MEM_CKE[0]	MEM_CKE[0]
	17	MEM_ODT[1]	MEM_ODT[1]	MEM_ODT[1]	MEM_ODT[1]
	16	MEM_ODT[0]	MEM_ODT[0]	MEM_ODT[0]	MEM_ODT[0]
	15	MEM_ACT_N[0]	MEM_ACT_N[0]	MEM_ACT_N[0]	MEM_ACT_N[0]
	14	MEN_CS_N[0]	MEN_CS_N[0]	MEN_CS_N[0]	MEN_CS_N[0]
	13	MEM_RESET_N[0]	MEM_RESET_N[0]	MEM_RESET_N[0]	MEM_RESET_N[0]
	12	MEM_BG[1]	MEM_BG[1]	MEM_BG[1]	MEM_BG[1]
BL0	11	MEM_DQ[7]	MEM_DQ[7]	MEM_DQ[7]	MEM_DQ[7]
	10	MEM_DQ[6]	MEM_DQ[6]	MEM_DQ[6]	MEM_DQ[6]
	9	MEM_DQ[5]	MEM_DQ[5]	MEM_DQ[5]	MEM_DQ[5]
	8	MEM_DQ[4]	MEM_DQ[4]	MEM_DQ[4]	MEM_DQ[4]
	7				
	6	MEM_DM_N[0]	MEM_DM_N[0]	MEM_DM_N[0]	MEM_DM_N[0]
	5	MEM_DQS_C[0]	MEM_DQS_C[0]	MEM_DQS_C[0]	MEM_DQS_C[0]
	4	MEM_DQS_T[0]	MEM_DQS_T[0]	MEM_DQS_T[0]	MEM_DQS_T[0]
	3	MEM_DQ[3]	MEM_DQ[3]	MEM_DQ[3]	MEM_DQ[3]
	2	MEM_DQ[2]	MEM_DQ[2]	MEM_DQ[2]	MEM_DQ[2]
	1	MEM_DQ[1]	MEM_DQ[1]	MEM_DQ[1]	MEM_DQ[1]
	0	MEM_DQ[0]	MEM_DQ[0]	MEM_DQ[0]	MEM_DQ[0]

Note: The presence of an asterisk (*) in the above table indicates an ECC byte location.

3.1.3.4. DDR5 Pin Placement

Table 13. DDR5 Pin Placement

Lane Number	Pin Index	x32+ECC *	x 32	2ch x16	x16 + ECC *	x16
BL7	95			MEM_1_MEM_DQ[15]		
	94			MEM_1_MEM_DQ[14]		
	93			MEM_1_MEM_DQ[13]		
	92			MEM_1_MEM_DQ[12]		
continued...						

Lane Number	Pin Index	x32+ECC *	x 32	2ch x16	x16 + ECC *	x16
	91					
	90			MEM_1_MEM_DM_N[1]		
	89			MEM_1_MEM_DQS_C[1]		
	88			MEM_1_MEM_DQS_T[1]		
	87			MEM_1_MEM_DQ[11]		
	86			MEM_1_MEM_DQ[10]		
	85			MEM_1_MEM_DQ[9]		
	84			MEM_1_MEM_DQ[8]		
BL6	83	MEM_DQ[39]*		MEM_1_MEM_DQ[7]		
	82	MEM_DQ[38]*		MEM_1_MEM_DQ[6]		
	81	MEM_DQ[37]*		MEM_1_MEM_DQ[5]		
	80	MEM_DQ[36]*		MEM_1_MEM_DQ[4]		
	79					
	78	MEM_DM_N[4]		MEM_1_MEM_DM_N[0]		
	77	MEM_DQS_C[4]		MEM_1_MEM_DQS_C[0]		
	76	MEM_DQS_T[4]		MEM_1_MEM_DQS_T[0]		
	75	MEM_DQ[35]*		MEM_1_MEM_DQ[3]		
	74	MEM_DQ[34]*		MEM_1_MEM_DQ[2]		
	73	MEM_DQ[33]*		MEM_1_MEM_DQ[1]		
	72	MEM_DQ[32]*		MEM_1_MEM_DQ[0]		
BL5	71	MEM_DQ[31]	MEM_DQ[31]	MEM_1_CK_C[1]		
	70	MEM_DQ[30]	MEM_DQ[30]	MEM_1_CK_T[1]		
	69	MEM_DQ[29]	MEM_DQ[29]	MEM_1_MEM_CS_N[0]		
continued...						

Lane Number	Pin Index	x32+ECC *	x 32	2ch x16	x16 + ECC *	x16
	68	MEM_DQ[28]	MEM_DQ[28]	MEM_1_MEM_CS_N[1]		
	67			MEM_1_CK_C[0]		
	66	MEM_DM_N[3]	MEM_DM_N[3]	MEM_1_CK_T[0]		
	65	MEM_DQS_C[3]	MEM_DQS_C[3]	MEM_1_MEM_CA[12]		
	64	MEM_DQS_T[3]	MEM_DQS_T[3]	MEM_1_MEM_CA[11]		
	63	MEM_DQ[27]	MEM_DQ[27]	MEM_1_RESET_N		
	62	MEM_DQ[26]	MEM_DQ[26]	OCT_1_OCT_RZQIN		
	61	MEM_DQ[25]	MEM_DQ[25]	MEM_1_ALERT_N		
	60	MEM_DQ[24]	MEM_DQ[24]	MEM_1_MEM_CA[10]		
BL4	59	MEM_DQ[23]	MEM_DQ[23]	Differential "NSide" Reference Clock Input Site	MEM_DQ[23]*	
	58	MEM_DQ[22]	MEM_DQ[22]	Differential "PSide" Reference Clock Input Site	MEM_DQ[22]*	
	57	MEM_DQ[21]	MEM_DQ[21]	MEM_1_MEM_CA[9]	MEM_DQ[21]*	
	56	MEM_DQ[20]	MEM_DQ[20]	MEM_1_MEM_CA[8]	MEM_DQ[20]*	
	55			MEM_1_MEM_CA[7]		
	54	MEM_DM_N[2]	MEM_DM_N[2]	MEM_1_MEM_CA[6]	MEM_DM_N[2]	
	53	MEM_DQS_C[2]	MEM_DQS_C[2]	MEM_1_MEM_CA[5]	MEM_DQS_C[2]	
	52	MEM_DQS_T[2]	MEM_DQS_T[2]	MEM_1_MEM_CA[4]	MEM_DQS_T[2]	
	51	MEM_DQ[19]	MEM_DQ[19]	MEM_1_MEM_CA[3]	MEM_DQ[19]*	
	50	MEM_DQ[18]	MEM_DQ[18]	MEM_1_MEM_CA[2]	MEM_DQ[18]*	
	49	MEM_DQ[17]	MEM_DQ[17]	MEM_1_MEM_CA[1]	MEM_DQ[17]*	
	48	MEM_DQ[16]	MEM_DQ[16]	MEM_1_MEM_CA[0]	MEM_DQ[16]*	
continued...						

Lane Number	Pin Index	x32+ECC *	x 32	2ch x16	x16 + ECC *	x16
BL3	47	MEM_CK_C[1]	MEM_CK_C[1]	MEM_0_CK_C[1]	MEM_CK_C[1]	MEM_CK_C[1]
	46	MEM_CK_T[1]	MEM_CK_T[1]	MEM_0_CK_T[1]	MEM_CK_T[1]	MEM_CK_T[1]
	45	MEM_CS_N[0]	MEM_CS_N[0]	MEM_0_MEM_CS_N[0]	MEM_CS_N[0]	MEM_CS_N[0]
	44	MEM_CS_N[1]	MEM_CS_N[1]	MEM_0_MEM_CS_N[1]	MEM_CS_N[1]	MEM_CS_N[1]
	43	MEM_CK_C[0]	MEM_CK_C[0]	MEM_0_CK_C[0]	MEM_CK_C[0]	MEM_CK_C[0]
	42	MEM_CK_T[0]	MEM_CK_T[0]	MEM_0_CK_T[0]	MEM_CK_T[0]	MEM_CK_T[0]
	41	MEM_CA[12]	MEM_CA[12]	MEM_0_MEM_CA[12]	MEM_CA[12]	MEM_CA[12]
	40	MEM_CA[11]	MEM_CA[11]	MEM_0_MEM_CA[11]	MEM_CA[11]	MEM_CA[11]
	39	MEM_RESET_N[0]	MEM_RESET_N[0]	MEM_0_RESET_N	MEM_RESET_N[0]	MEM_RESET_N[0]
	38	RZQ Site	RZQ Site	OCT_0_OCT_RZQIN	RZQ Site	RZQ Site
	37	MEM_ALERT_N[0]	MEM_ALERT_N[0]	MEM_0_ALERT_N	MEM_ALERT_N[0]	MEM_ALERT_N[0]
	36	MEM_CA[10]	MEM_CA[10]	MEM_0_MEM_CA[10]	MEM_CA[10]	MEM_CA[10]
BL2	35	Differential "N-Side" Reference Clock Input Site	Differential "N-Side" Reference Clock Input Site	Differential "N-Side" Reference Clock Input Site	Differential "N-Side" Reference Clock Input Site	Differential "N-Side" Reference Clock Input Site
	34	Differential "P-Side" Reference Clock Input Site	Differential "P-Side" Reference Clock Input Site	Differential "P-Side" Reference Clock Input Site	Differential "P-Side" Reference Clock Input Site	Differential "P-Side" Reference Clock Input Site
	33	MEM_CA[9]	MEM_CA[9]	MEM_0_MEM_CA[9]	MEM_CA[9]	MEM_CA[9]
	32	MEM_CA[8]	MEM_CA[8]	MEM_0_MEM_CA[8]	MEM_CA[8]	MEM_CA[8]
	31	MEM_CA[7]	MEM_CA[7]	MEM_0_MEM_CA[7]	MEM_CA[7]	MEM_CA[7]
	30	MEM_CA[6]	MEM_CA[6]	MEM_0_MEM_CA[6]	MEM_CA[6]	MEM_CA[6]
	29	MEM_CA[5]	MEM_CA[5]	MEM_0_MEM_CA[5]	MEM_CA[5]	MEM_CA[5]
	28	MEM_CA[4]	MEM_CA[4]	MEM_0_MEM_CA[4]	MEM_CA[4]	MEM_CA[4]
	27	MEM_CA[3]	MEM_CA[3]	MEM_0_MEM_CA[3]	MEM_CA[3]	MEM_CA[3]
continued...						

Lane Number	Pin Index	x32+ECC *	x 32	2ch x16	x16 + ECC *	x16
	26	MEM_CA[2]	MEM_CA[2]	MEM_0_MEM_CA[2]	MEM_CA[2]	MEM_CA[2]
	25	MEM_CA[1]	MEM_CA[1]	MEM_0_MEM_CA[1]	MEM_CA[1]	MEM_CA[1]
	24	MEM_CA[0]	MEM_CA[0]	MEM_0_MEM_CA[0]	MEM_CA[0]	MEM_CA[0]
BL1	23	MEM_DQ[7]	MEM_DQ[7]	MEM_0_MEM_DQ[7]	MEM_DQ[7]	MEM_DQ[7]
	22	MEM_DQ[6]	MEM_DQ[6]	MEM_0_MEM_DQ[6]	MEM_DQ[6]	MEM_DQ[6]
	21	MEM_DQ[5]	MEM_DQ[5]	MEM_0_MEM_DQ[5]	MEM_DQ[5]	MEM_DQ[5]
	20	MEM_DQ[4]	MEM_DQ[4]	MEM_0_MEM_DQ[4]	MEM_DQ[4]	MEM_DQ[4]
	19					
	18	MEM_DM_N[0]	MEM_DM_N[0]	MEM_0_MEM_DM_N[0]	MEM_DM_N[0]	MEM_DM_N[0]
	17	MEM_DQS_C[0]	MEM_DQS_C[0]	MEM_0_MEM_DQS_C[0]	MEM_DQS_C[0]	MEM_DQS_C[0]
	16	MEM_DQS_T[0]	MEM_DQS_T[0]	MEM_0_MEM_DQS_T[0]	MEM_DQS_T[0]	MEM_DQS_T[0]
	15	MEM_DQ[3]	MEM_DQ[3]	MEM_0_MEM_DQ[3]	MEM_DQ[3]	MEM_DQ[3]
	14	MEM_DQ[2]	MEM_DQ[2]	MEM_0_MEM_DQ[2]	MEM_DQ[2]	MEM_DQ[2]
	13	MEM_DQ[1]	MEM_DQ[1]	MEM_0_MEM_DQ[1]	MEM_DQ[1]	MEM_DQ[1]
	12	MEM_DQ[0]	MEM_DQ[0]	MEM_0_MEM_DQ[0]	MEM_DQ[0]	MEM_DQ[0]
BL0	11	MEM_DQ[15]	MEM_DQ[15]	MEM_0_MEM_DQ[15]	MEM_DQ[15]	MEM_DQ[15]
	10	MEM_DQ[14]	MEM_DQ[14]	MEM_0_MEM_DQ[14]	MEM_DQ[14]	MEM_DQ[14]
	9	MEM_DQ[13]	MEM_DQ[13]	MEM_0_MEM_DQ[13]	MEM_DQ[13]	MEM_DQ[13]
	8	MEM_DQ[12]	MEM_DQ[12]	MEM_0_MEM_DQ[12]	MEM_DQ[12]	MEM_DQ[12]
	7					
	6	MEM_DM_N[1]	MEM_DM_N[1]	MEM_0_MEM_DM_N[1]	MEM_DM_N[1]	MEM_DM_N[1]
	5	MEM_DQS_C[1]	MEM_DQS_C[1]	MEM_0_MEM_DQS_C[1]	MEM_DQS_C[1]	MEM_DQS_C[1]
	4	MEM_DQS_T[1]	MEM_DQS_T[1]	MEM_0_MEM_DQS_T[1]	MEM_DQS_T[1]	MEM_DQS_T[1]
continued...						

Lane Number	Pin Index	x32+ECC *	x 32	2ch x16	x16 + ECC *	x16
	3	MEM_DQ[11]	MEM_DQ[11]	MEM_0_MEM_DQ[11]	MEM_DQ[11]	MEM_DQ[11]
	2	MEM_DQ[10]	MEM_DQ[10]	MEM_0_MEM_DQ[10]	MEM_DQ[10]	MEM_DQ[10]
	1	MEM_DQ[9]	MEM_DQ[9]	MEM_0_MEM_DQ[9]	MEM_DQ[9]	MEM_DQ[9]
	0	MEM_DQ[8]	MEM_DQ[8]	MEM_0_MEM_DQ[8]	MEM_DQ[8]	MEM_DQ[8]

Note: The presence of an asterisk (*) in the above table indicates an ECC byte location.

3.1.3.5. LPDDR5 Pin Placement

Table 15. LPDDR5 Pin Placement

Lane Number	Pin Index	x32	2 Channel x16
BL7	95	MEM_DQ[31]	MEM_1_MEM_DQ[15]
	94	MEM_DQ[30]	MEM_1_MEM_DQ[14]
	93	MEM_DQ[29]	MEM_1_MEM_DQ[13]
	92	MEM_DQ[28]	MEM_1_MEM_DQ[12]
	91		
	90	MEM_DMI[3]	MEM_1_MEM_DMI[1]
	89	MEM_RDQS_C[3]	MEM_1_MEM_RDQS_C[1]
	88	MEM_RDQS_T[3]	MEM_1_MEM_RDQS_T[1]
	87	MEM_DQ[27]	MEM_1_MEM_DQ[11]
	86	MEM_DQ[26]	MEM_1_MEM_DQ[10]
	85	MEM_DQ[25]	MEM_1_MEM_DQ[9]
	84	MEM_DQ[24]	MEM_1_MEM_DQ[8]
BL6	83	MEM_DQ[23]	MEM_1_MEM_DQ[7]
	82	MEM_DQ[22]	MEM_1_MEM_DQ[6]
	81	MEM_DQ[21]	MEM_1_MEM_DQ[5]
	80	MEM_DQ[20]	MEM_1_MEM_DQ[4]
	79		
	78	MEM_DMI[2]	MEM_1_MEM_DMI[0]
	77	MEM_RDQS_C[2]	MEM_1_MEM_RDQS_C[0]
	76	MEM_RDQS_T[2]	MEM_1_MEM_RDQS_T[0]
	75	MEM_DQ[19]	MEM_1_MEM_DQ[3]
	74	MEM_DQ[18]	MEM_1_MEM_DQ[2]
continued...			

Lane Number	Pin Index	x32	2 Channel x16
BL5	73	MEM_DQ[17]	MEM_1_MEM_DQ[1]
	72	MEM_DQ[16]	MEM_1_MEM_DQ[0]
	71		
	70		
	69		
	68		MEM_1_MEM_CS[1]
	67		MEM_1_CK_C
	66		MEM_1_CK_T
	65		MEM_1_MEM_CS[0]
	64		MEM_1_MEM_CA[6]
	63		MEM_1_RESET_N
	62		OCT_1_OCT_RZQIN
	61		
	60		
BL4	59		Differential "NSide" Reference Clock Input Site
	58		Differential "PSide" Reference Clock Input Site
	57		MEM_1_MEM_CA[5]
	56		MEM_1_MEM_CA[4]
	55		MEM_1_MEM_WCK_C[1]
	54		MEM_1_MEM_WCK_T[1]
	53		MEM_1_MEM_WCK_C[0]
	52		MEM_1_MEM_WCK_T[0]
	51		MEM_1_MEM_CA[3]
	50		MEM_1_MEM_CA[2]
	49		MEM_1_MEM_CA[1]
	48		MEM_1_MEM_CA[0]
BL3	47		
	46		
	45		
	44	MEM_CS[1]	MEM_0_MEM_CS[1]
	43	MEM_CK_C	MEM_0_CK_C
	42	MEM_CK_T	MEM_0_CK_T
	41	MEM_CS[0]	MEM_0_MEM_CS[0]
	40	MEM_CA[6]	MEM_0_MEM_CA[6]

continued...

Lane Number	Pin Index	x32	2 Channel x16
	39	MEM_RESET_N	MEM_0_RESET_N
	38	RZQ Site	OCT_0_OCT_RZQIN
	37		
	36		
BL2	35	Differential "N-Side" Reference Clock Input Site	Differential "NSide" Reference Clock Input Site
	34	Differential "P-Side" Reference Clock Input Site	Differential "PSide" Reference Clock Input Site
	33	MEM_CA[5]	MEM_0_MEM_CA[5]
	32	MEM_CA[4]	MEM_0_MEM_CA[4]
	31	MEM_WCK_C[1]	MEM_0_MEM_WCK_C[1]
	30	MEM_WCK_T[1]	MEM_0_MEM_WCK_T[1]
	29	MEM_WCK_C[0]	MEM_0_MEM_WCK_C[0]
	28	MEM_WCK_T[0]	MEM_0_MEM_WCK_T[0]
	27	MEM_CA[3]	MEM_0_MEM_CA[3]
	26	MEM_CA[2]	MEM_0_MEM_CA[2]
	25	MEM_CA[1]	MEM_0_MEM_CA[1]
	24	MEM_CA[0]	MEM_0_MEM_CA[0]
BL1	23	MEM_DQ[15]	MEM_0_MEM_DQ[15]
	22	MEM_DQ[14]	MEM_0_MEM_DQ[14]
	21	MEM_DQ[13]	MEM_0_MEM_DQ[13]
	20	MEM_DQ[12]	MEM_0_MEM_DQ[12]
	19		
	18	MEM_DMI[1]	MEM_0_MEM_DMI[1]
	17	MEM_RDQS_C[1]	MEM_0_MEM_RDQS_C[1]
	16	MEM_RDQS_T[1]	MEM_0_MEM_RDQS_T[1]
	15	MEM_DQ[11]	MEM_0_MEM_DQ[11]
	14	MEM_DQ[10]	MEM_0_MEM_DQ[10]
	13	MEM_DQ[9]	MEM_0_MEM_DQ[9]
	12	MEM_DQ[8]	MEM_0_MEM_DQ[8]
BL0	11	MEM_DQ[7]	MEM_0_MEM_DQ[7]
	10	MEM_DQ[6]	MEM_0_MEM_DQ[6]
	9	MEM_DQ[5]	MEM_0_MEM_DQ[5]
	8	MEM_DQ[4]	MEM_0_MEM_DQ[4]
	7		
	6	MEM_DMI[0]	MEM_0_MEM_DMI[0]
continued...			

Lane Number	Pin Index	x32	2 Channel x16
	5	MEM_RDQS_C[0]	MEM_0_MEM_RDQS_C[0]
	4	MEM_RDQS_T[0]	MEM_0_MEM_RDQS_T[0]
	3	MEM_DQ[3]	MEM_0_MEM_DQ[3]
	2	MEM_DQ[2]	MEM_0_MEM_DQ[2]
	1	MEM_DQ[1]	MEM_0_MEM_DQ[1]
	0	MEM_DQ[0]	MEM_0_MEM_DQ[0]

Note: It is important to strictly follow the pin placement for a given memory topology when assigning pin locations for your EMIF IP.

The recommended approach is to manually constrain some interface signals and allow the Quartus Prime Fitter to place the pins. For this method of I/O placement, you must constrain the following signals:

- PLL reference clock
- RZQ pin
- MEM_RESET_N

Do not change the location for the EMIF pin using a .qsf assignment or the Pin Planner if you need to swap the DQ pins within a DQS group or the DQS group to simplify board design.

Refer to the [Configuring DQ Pin Swizzling](#) topic in the *External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide* for more information on how to swap the DQ pin and DQS group.

For dual-rank component interfaces, you cannot have different swizzling specifications for rank 0 and rank 1.

3.1.3.6. I/O Sub-Bank Usage

The pins in an I/O bank can serve as address and command pins, data pins, or clock and strobe pins for an external memory interface.

A given sub-bank cannot be shared between multiple EMIFs.

All the sub-banks are capable of functioning as the address and command bank.

3.1.4. Agilex 7 M-Series EMIF Architecture: I/O Lane

An I/O bank contains two sub-banks. Each sub-bank contains 48 I/O pins, organized into four I/O lanes of 12 pins each. You can identify where a pin is located within an I/O bank based on its `Index` within `I/O Bank` in the device pinout.

Table 16. Pin Index Mapping

Pin Index	Lane	Sub-bank Location
0-11	0	Bottom
12-23	1	

continued...

Pin Index	Lane	Sub-bank Location
24-35	2	Top
36-47	3	
48-59	4	
60-71	5	
72-83	6	
84-95	7	

Each I/O lane can implement one x8 read capture group (DQS group), with two pins functioning as the read capture clock/strobe pair (DQS/DQS#), and up to 10 pins functioning as data pins (DQ and DM pins). To implement an EMIF targeting a x16 interface, use multiple lanes within the same sub bank.

It is also possible to implement a pair of x4 groups in a lane. In this case, four pins function as clock/strobe pair, and 8 pins function as data pins. DM is not available for x4 groups. There must be an even number of x4 groups for each interface.

For x4 groups, you must place DQS0 and DQS1 in the same I/O lane as a pair. Similarly, DQS2 and DQS3 must be paired. In general, DQS(x) and DQS(x+1) must be paired in the same I/O lane.

For DQ and DQS pin assignments for various configurations, refer to the AgilEx 7 M-Series device pin tables.

Table 17. Lanes Used Per DQS Group

Group Size	Number of Lanes Used	Maximum Number of Data Pins per Group
x8	1	10
x16	2	22
pair of x4	1	4 per group, 8 per lane

Figure 8. x4 Group

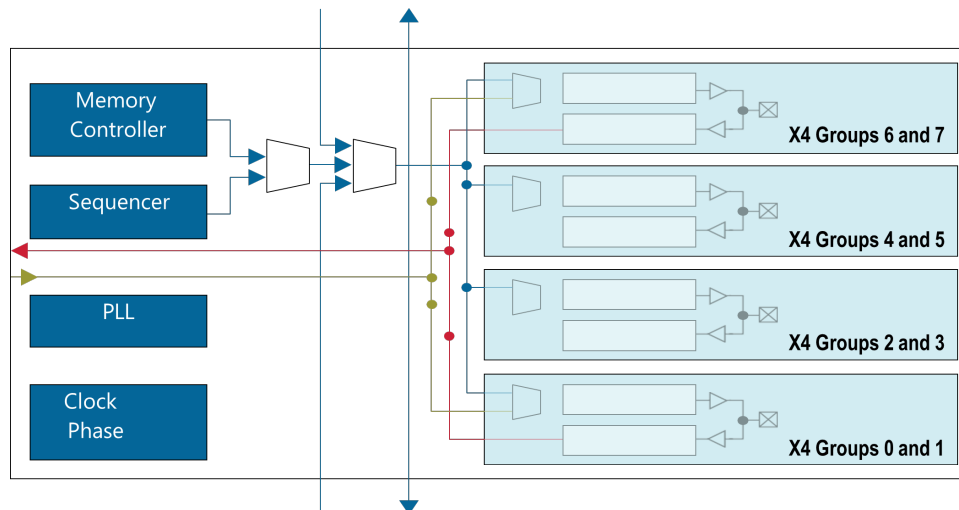


Figure 9. x8 Group

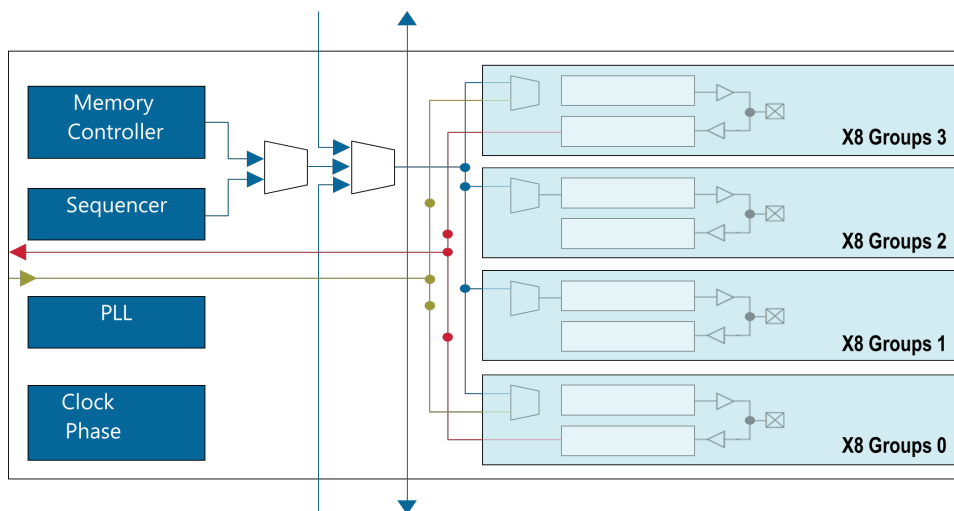
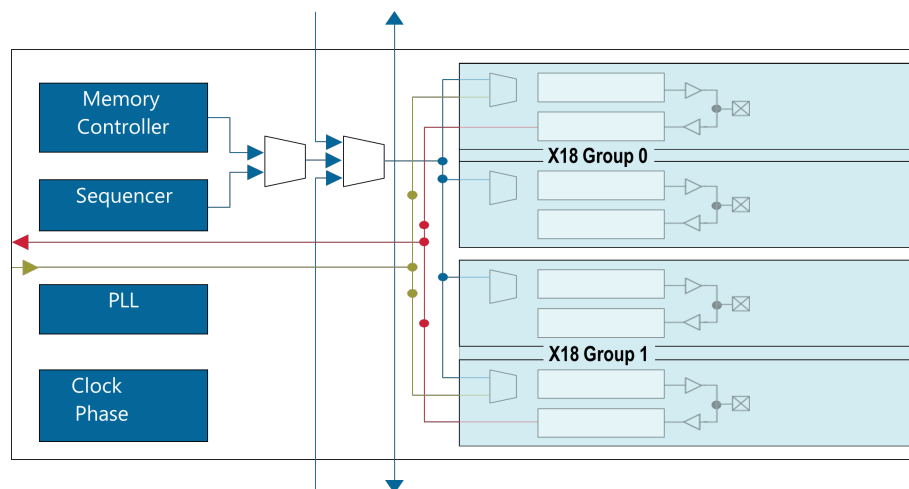


Figure 10. x16 Group



3.1.5. Agilex 7 M-Series EMIF Architecture: Input DQS Clock Tree

The input DQS clock tree is a balanced clock network that distributes the read capture clock (such as QK/QK# which are free-running read clocks) and strobe (such as DQS_T/DQS_C) from the external memory device to the read capture registers inside the I/Os.

You can configure an input DQS clock tree in x4 mode, x8 mode, or x16 mode.

Within every bank, only certain physical pins at specific locations can drive the input DQS clock trees. The pin locations that can drive the input DQS clock trees vary, depending on the size of the group.

Table 18. Pins Usable as Read Capture Clock / Strobe Pair

Group Size	Index of Lanes Spanned by Clock Tree ¹	Sub-Bank	Index of Pins Usable as Read Capture Clock / Strobe Pair	
			DQS_T	DQS_C
x4	0A	Bottom	4	5
x4	0B		6	7
x4	1A		16	17
x4	1B		18	19
x4	2A		28	29
x4	2B		30	31
x4	3A		40	41
x4	3B		42	43
x8	0		4	5
x8	1		16	17
x8	2		28	29
x8	3		40	41
x16	0, 1		4	5
x16	2, 3		28	29
x4	0A	Top	52	53
x4	0B		54	55
x4	1A		64	65
x4	1B		66	67
x4	2A		76	77
x4	2B		78	79
x4	3A		88	89
x4	3B		90	91
x8	0		52	53
x8	1		64	65
x8	2		76	77
x8	3		88	89
x16	0,1		52	53
x16	2,3		76	77

Note: ¹ A and B refer to the two nibbles within the lane.

3.1.6. Agilex 7 M-Series EMIF Architecture: PHY Clock Tree

Dedicated high-speed clock networks drive I/Os in the Agilex 7 M-Series EMIF.

The relatively short span of the PHY clock trees results in low jitter and low duty-cycle distortion, maximizing the data valid window.

The PHY clock tree in Agilex 7 M-Series devices can run as fast as 1.6 GHz. All Agilex 7 M-Series external memory interfaces use the PHY clock trees.

3.1.7. Agilex 7 M-Series EMIF Architecture: PLL Reference Clock Networks

Each sub-bank includes an I/O bank I/O PLL that can drive the PHY clock trees of that bank, through dedicated connections. In addition to supporting EMIF-specific functions, the I/O bank I/O PLLs can also serve as general-purpose PLLs for user logic.

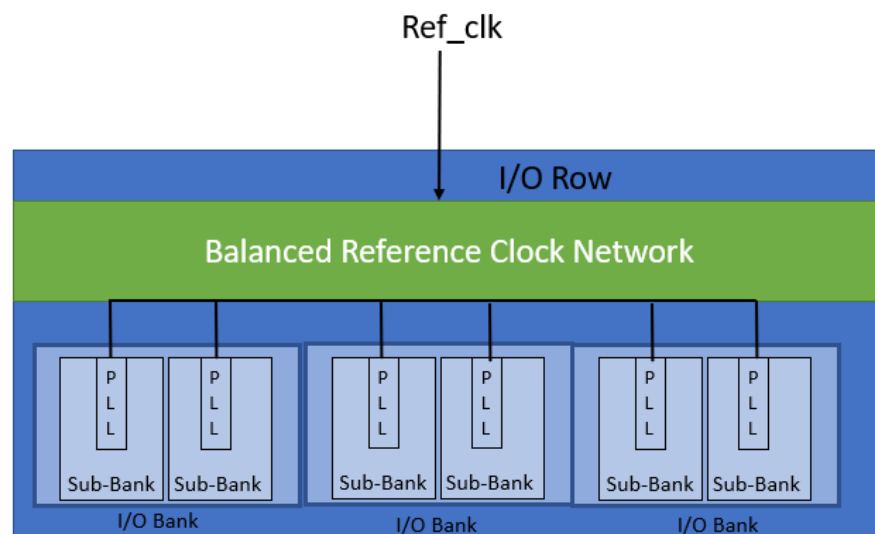
The PLL reference clock must be constrained to the address and command sub-bank only.

Agilex 7 M-Series external memory interfaces that span multiple banks use the PLL in each bank. The Agilex 7 M-Series architecture allows for relatively short PHY clock networks, reducing jitter and duty-cycle distortion.

The following mechanisms ensure that the clock outputs of individual I/O bank I/O PLLs in a multi-bank interface remain in phase:

- A single PLL reference clock source feeds all I/O bank I/O PLLs. The reference clock signal reaches the PLLs by a balanced PLL reference clock tree. The Quartus Prime software automatically configures the PLL reference clock tree so that it spans the correct number of banks. This clock must be free-running and stable prior to FPGA configuration.
- The EMIF IP sets the PLL configuration (counter settings, bandwidth settings, compensation and feedback mode setting) values appropriately to maintain synchronization among the clock dividers across the PLLs. This requirement restricts the legal PLL reference clock frequencies for a given memory interface frequency and clock rate. If you plan to use an on-board oscillator, you must ensure that its frequency matches the PLL reference clock frequency that you select from the displayed list.

Figure 11. PLL Balanced Reference Clock Tree



3.1.8. Agilex 7 M-Series EMIF Architecture: Clock Phase Alignment

In Agilex 7 M-Series external memory interfaces, a global clock network clocks registers inside the FPGA core, and the PHY clock network clocks registers inside the FPGA periphery. Clock phase alignment circuitry employs negative feedback to dynamically adjust the phase of the core clock signal to match the phase of the PHY clock signal.

The clock phase alignment feature effectively eliminates the clock skew effect in all transfers between the core and the periphery, facilitating timing closure. All Agilex 7 M-Series external memory interfaces employ clock phase alignment circuitry.

Figure 12. Clock Phase Alignment Illustration

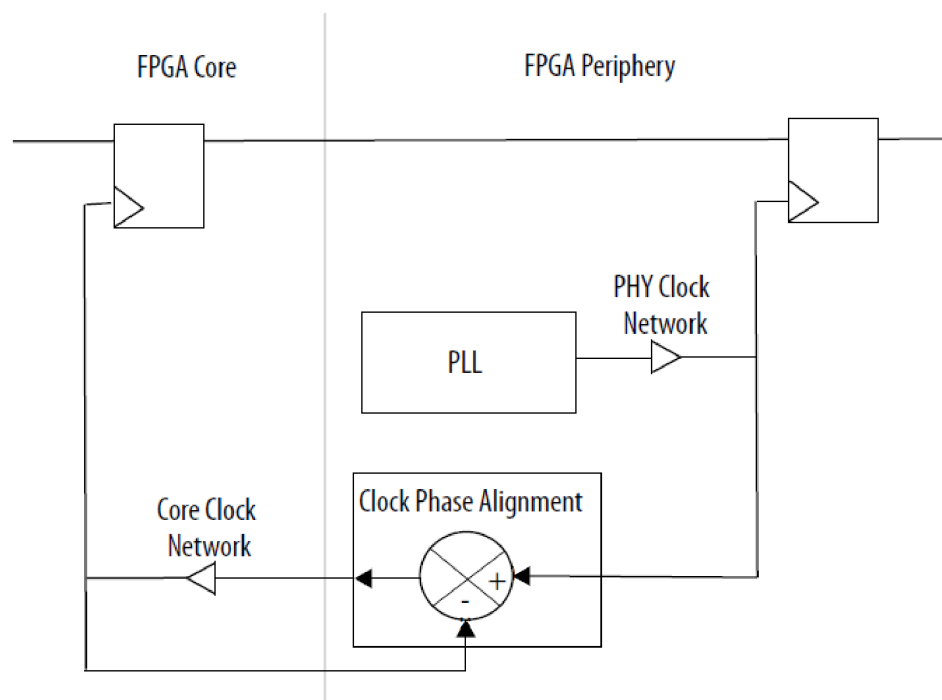
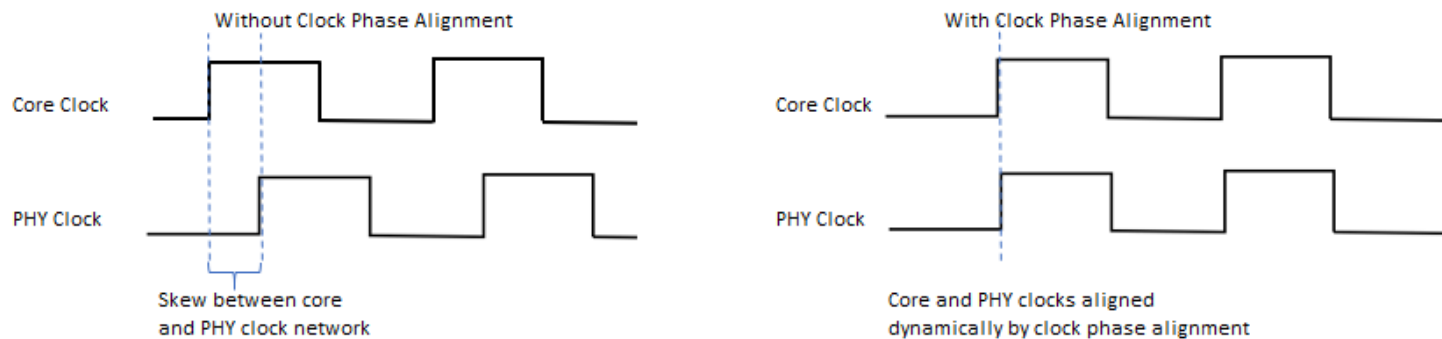


Figure 13. Effect of Clock Phase Alignment



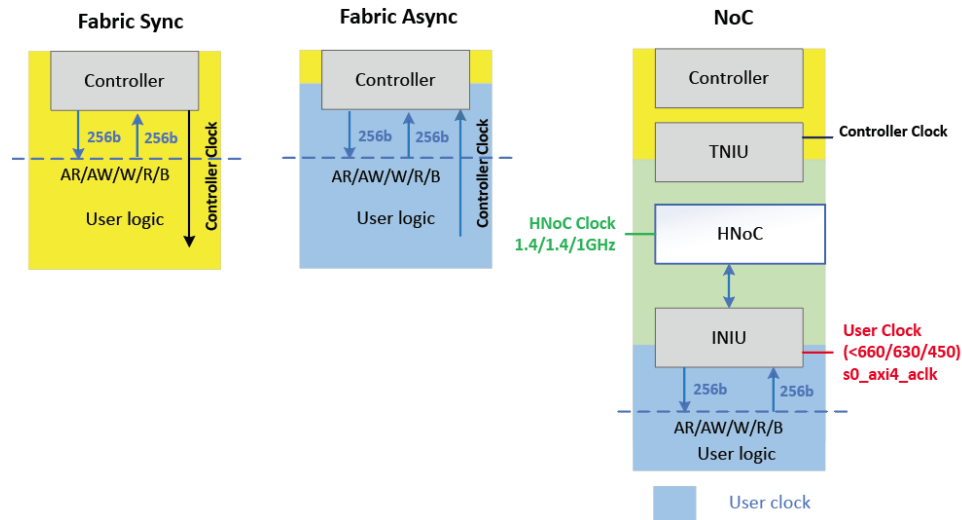
3.1.9. User Clock in Different Core Access Modes

The EMIF IP for Agilex 7 M-Series devices supports three user access modes.

- Synchronous fabric clocking, where the EMIF IP provides a user clock.
 - The user clock frequency is limited by the maximum core-to-periphery (C2P) and periphery-to-core (P2C) frequency of 400 MHz.
 - In DDR4, the user clock frequency will be one-quarter of the memory clock frequency $((\text{mem_CK})/4)$.
 - In DDR5 and LPDDR5, the user clock frequency will be one-eighth of the memory clock frequency $((\text{mem_CK})/8)$.
- Asynchronous fabric clocking, where you provide the clock to the EMIF IP.
 - The asynchronous user clock can come from any user clock source on the device.
 - The user clock frequency has no dependency on the memory clock (mem_CK).
- NoC Mode.
 - The initiator clock can come from any user clock source on the device.
 - For additional information, refer to the [Agilex 7 M-Series FPGA Network-on-Chip \(NoC\) User Guide](#).

The following figures illustrate the different clocking styles available for the Agilex 7 M-Series EMIF IP. The NoC mode shown is the simplest NoC mode.

Figure 14. Access Modes



Benefits of Each Access Mode

- Synchronous fabric clocking is required for DDR4 DIMM or DDR4/DDR5 x40 interfaces.
- Asynchronous fabric access mode has the lowest latency.
- NoC mode can achieve the highest bandwidth and efficiency, and has the lowest initiator blockage.

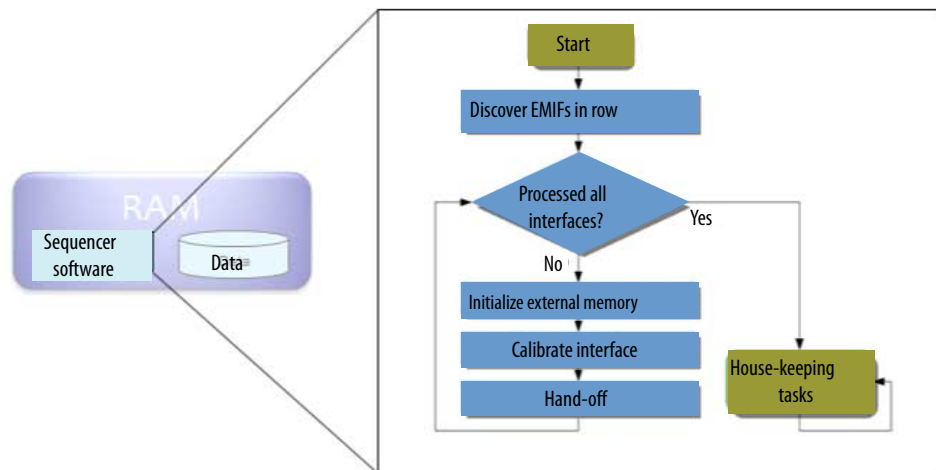
3.2. Agilex 7 M-Series EMIF Sequencer

The Agilex 7 M-Series EMIF sequencer is fully hardened in silicon, with executable code to handle protocols and topologies. Hardened RAM contains the calibration algorithm.

The Agilex 7 M-Series EMIF sequencer is responsible for the following operations:

- Initializes memory devices.
- Calibrates the external memory interface.
- Governs the hand-off of control to the memory controller.
- Handles recalibration requests and debug requests.
- Handles all supported protocols and configurations.

Figure 15. Agilex 7 M-Series EMIF Sequencer Operation



3.3. Agilex 7 M-Series EMIF Controller

3.3.1. Hard Memory Controller

The Agilex 7 M-Series hard memory controller is designed for high speed, high performance, high flexibility, and area efficiency. The Agilex 7M-Series hard memory controller supports the DDR4, DDR5, and LPDDR5 memory standards.

The hard memory controller implements efficient pipelining techniques and advanced dynamic command and data reordering algorithms to improve bandwidth usage and reduce latency, providing a high-performance solution.

The hard memory controller consists of the following logic blocks:

- Core and PHY interfaces
- Main control path
- Data buffer controller
- Read and write data buffers

The controller user interface uses the AXI4 protocol. The controller communicates to the PHY using the DDR PHY Interface (DFI).

3.3.1.1. Hard Memory Controller Features

Table 19. Features of the Agilex 7 M-Series Hard Memory Controller

Feature	Description
Memory standards support	Supports DDR4, DDR5, and LPDDR5 SDRAM.
Memory devices support	Supports the following memory devices: - Discrete (DDR4, DDR5, LPDDR5) - UDIMM (DDR5) - SODIMM (DDR5) - RDIMM (DDR5)
Interface protocols support	Supports the AXI4 interface.
Configurable memory interface width	- DDR4 supports DQ widths: 16, 32, 40 - Discrete component : 16, 16+ECC, 24, 32, 32+ECC, 40 - DIMM : 64, 64 + ECC, 72 - DDR5 supports DQ widths: 16, 16+ECC (1ch/2ch), 32, 32+ECC (1ch/2ch) - LPDDR5 supports DQ widths: 16 (1ch/2ch), 32 (1ch)
Maximum rank support	2 ranks with single slot.
Burst length support	- DDR4: BL8 - DDR5: BL16 - LPDDR5: BL16
Efficiency optimization features	- Open-page policy—by default, opens page on every access. However, the controller intelligently closes a row based on incoming traffic, which improves the efficiency of the controller especially for random traffic. - Pre-emptive bank management—the controller issues bank management commands early, which ensures that the required row is open when the read or write occurs. - Data reordering—the controller reorders read/write commands. - Additive latency—the controller can issue a READ/WRITE command after the ACTIVATE command to the memory bank prior to t_{RCD}, which increases the command efficiency.
Starvation counter	Ensures all requests are served before a predefined time-out period, which ensures that low priority access are not left behind while reordering data for efficiency.
Bank interleaving	Able to issue read or write commands continuously to "random" addresses. You must correctly cycle the bank addresses.
On-die termination	In DDR4, the controller controls the on-die termination signal for the memory. This feature improves signal integrity and simplifies your board design.
Refresh features	- Per-rank refresh—allows refresh for each individual rank. - Controller-controlled refresh.

continued...

Feature	Description
Power saving features	- Low power modes (power down and self-refresh)—optionally, you can request the controller to put the memory into one of the two low power states. - Automatic power down—puts the memory device in power down mode when the controller is idle. You can configure the idle waiting time. - Memory clock gating.
Memory features	- Bank group support—supports different timing parameters for between bank groups. - Command/Address parity—command and address bus parity check.
User ZQ calibration	Long or short ZQ calibration request for DDR4.

3.4. Agilex 7 M-Series EMIF IP Interface Protocol: Interface Width, User Access Mode, and ECC

Depending on your choice of user access mode (explained in previous sections) and memory protocol, Agilex 7 M-Series devices regulate interface data width support and handling of ECC. The following tables highlight different types of ECC support-access, memory protocol, format, and user access mode.

Table 20. ECC, Format, and Access Mode for DDR4

DDR4				
Protocol	Interface Data Width	Access Mode	Data Rate 1R	ECC
UDIMM/RDIMM/SODIMM	x72 all data	Lockstep fabric sync	3200	N/A
	x64 all data	Lockstep fabric sync	3200	N/A
	x64 data + ECC	Lockstep fabric sync	3200	Out-of-band ECC, soft logic adapter to calculate ECC on 64b data and sends ECC data on secondary hard memory controller (64b in total ECC on 512b data). x32 and another x32 data goes to two primary controllers. WUSER and RUSER bits are not used.
Component	x40	Lockstep fabric sync	3200	N/A
		Fabric sync	3200	N/A
		Fabric Async		
		NoC		
	x32 + ECC	Fabric sync	3200	Controller generates out-of-band 4-bit ECC. You must use five I/O lanes, including the
		Fabric Async		
		NoC		

continued...

DDR4				
Protocol	Interface Data Width	Access Mode	Data Rate 1R	ECC
				one for ECC — so, a x40 physical interface.)
	x16	Fabric sync	3200	N/A
		Fabric Async		
		NoC		
	x16 + ECC	Fabric sync	3200	Controller generates out-of-band 4-bit ECC. You must use three I/O lanes, including the one for ECC — so, a x24 physical interface.)
		Fabric Async		
		NoC		

Table 21. ECC, Format, and Access Mode for DDR5

DDR5				
Device/Protocol	Interface Data Width	Access Mode	Data Rate 1R	ECC
UDIMM/RDIMM/SoDIMM	x32	Fabric Sync	5600	N/A
		Fabric Async	5600	
		NoC	5600	
	x32 + ECC (4 bit)	Fabric Sync	5600	Controller generated out-of-band ECC. For RDIMM interface, this requires 2 x40 sub-channel interfaces. For SoDIMM and UDIMM, this requires 2 x36 sub-channel interfaces.
		Fabric Async	5600	
		NoC	5600	
Component	x32	Fabric Sync	3200	N/A
		Fabric Async	5600	
		NoC	5600	
	x32 + ECC	Fabric Sync	5600	Out-of-band Controller calculated ECC. You must provide an extra DRAM component to handle ECC.
		Fabric Async	5600	
		NoC	5600	
	x16	Fabric Sync	5600	N/A
		Fabric Async	5600	
		NoC	5600	
	x16 + ECC	Fabric Sync	5600	Out-of-band Controller calculated ECC. You must provide an extra DRAM component to handle ECC.
		Fabric Async	5600	
		NoC	5600	

Table 22. ECC, Format, and Access Mode for LPDDR5

LPDDR5				
Device/Protocol	Interface Data Width	Access Mode	Data Rate 1R	ECC
Component	1ch x 16 / 2ch x 16 / 4ch x 16	Fabric Sync	5500	N/A
		Fabric Async	5500	
		NoC	5500	
	1ch x 16 + Link ECC / 2ch x 16 + Link ECC / 4ch x 16 + Link ECC	Fabric Sync	5500	Link ECC can be selected for Write and Read operations.
		Fabric Async	5500	
		NoC	5500	
	1ch x 16 + In-line ECC / 2ch x 16 + In-line ECC / 4ch x 16 + In-line ECC	Fabric Sync	5500	One-eighth of the memory is used for the ECC bits. This reduces the available AXI Address range by one-eighth.
		Fabric Async	5500	
		NoC	5500	

3.5. Agilex 7 M-Series EMIF IP for Hard Processor Subsystem (HPS)

The Agilex 7 M-Series FPGA EMIF IP can access external DRAM memory devices using the External Memory Interfaces for HPS FPGA IP.

To enable connectivity between the HPS and the Agilex 7 M-Series EMIF IP, you must create and configure an instance of the EMIF for HPS IP, and connect it to the Agilex 7 FPGA hard processor subsystem instance in your system.

Table 23. IO96 Bank and Lane Usage for HPS EMIF

Data Width Usage	BL7	BL6	BL5	BL4	BL3	BL2	BL1	BL0
DDR4								
DDR4x16	—	—	—	DQ[1]	AC2	AC1	AC0	DQ[0]
DDR4x16+ECC	—	—	DQ[ECC]	DQ[1]	AC2	AC1	AC0	DQ[0]
DDR4x32	—	DQ[3]	DQ[2]	DQ[1]	AC2	AC1	AC0	DQ[0]
DDR4x32+ECC	DQ[EC C]	DQ[3]	DQ[2]	DQ[1]	AC2	AC1	AC0	DQ[0]
DDR4x64	Not Supported							
DDR4x64+ECC	Not Supported							
DDR5								
DDR5x16	—	—	—	—	AC1	AC0	DQ[0]	DQ[1]
DDR5x16	DQ[1]	DQ[0]	AC1	AC0	—	—	—	—
DDR5x16+ECC	—	—	—	DQ[ECC]	AC1	AC0	DQ[0]	DQ[1]
DDR5x32	—	—	DQ[3]	DQ[2]	AC1	AC0	DQ[0]	DQ[1]
DDR5x32+ECC	—	DQ[ECC]	DQ[3]	DQ[2]	AC1	AC0	DQ[0]	DQ[1]
continued...								

continued...

Data Width Usage	BL7	BL6	BL5	BL4	BL3	BL2	BL1	BL0
LPDDR5								
LPDDR5x16	—	—	—	—	AC1	AC0	DQ[1]	DQ[0]
LPDDR5x32	DQ[3]	DQ[2]	—	—	AC1	AC0	DQ[1]	DQ[0]

3.5.1. Restrictions on I/O Bank Usage for Agilex 7 M-Series EMIF IP with HPS

Restrictions on I/O Bank Usage for Agilex 7 M-Series EMIF IP with HPS

- Only the two IO96 banks adjacent to the HPS MPFE can be used for HPS-EMIF (bank 3C and bank 3D).
- If only one IO96 bank is to be used by HPS-EMIF, it must be the one adjacent to the HPS MPFE (bank 3D). If HPS is using only 3D, 3C can be used for non-HPS EMIF.
- No individual data channel can span multiple IO96 banks. Thus DDR5 2x32 is good because each channel is within its own IO96 bank; however, DDR4x72 is a single wide channel and therefore is not supported.
- Unused pins in an HPS-EMIF occupied IO96 bank must be left unused; you cannot use them as general-purpose I/O pins.
- Unused lanes in an HPS-EMIF occupied IO96 bank should be left unconnected; you cannot use them as general-purpose I/O pins.
- Reference clock sharing is allowed between HPS-EMIF IP and other IPs in certain cases.
- For multi-channel EMIFs or when multiple EMIFs are used inside HPS-EMIF IP, they must have identical IP parameters.
- Initiators and targets must only be connected according to the following table:

	HPS Initiator	Non-HPS Initiator	HPS Initiator-lite	Non-HPS Initiator-lite
HPS EMIF Target	Yes	No	—	—
Non-HPS EMIF Target	No	Yes	—	—
HPS EMIF Target-lite	—	—	Yes *	No
Non-HPS EMIF Target-lite	—	—	No	Yes

Note: * The Quartus Prime software may make this connection automatically.

3.5.1.1. Using the EMIF Debug Toolkit with Agilex 7 M-Series HPS EMIF Interfaces

The External Memory Interface Debug Toolkit is not directly compatible with HPS EMIF interfaces.

To debug your HPS EMIF interface using the EMIF Debug Toolkit, you should create an identically parameterized, non-HPS version of your interface (fabric EMIF), and apply the toolkit to that interface. When you finish debugging this fabric EMIF interface, you can then apply any needed changes to your HPS EMIF interface, and continue your design development.

4. Agilex 7 M-Series FPGA EMIF IP – End-User Signals

4.1. IP Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

The interfaces in the Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component each have signals that can be connected in Platform Designer. The following table lists the interfaces and corresponding interface types.

Table 24. Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

Interface Name	Interface Type	Description
s0_axi4_clock_in	clock	Input user clock for mainband; for MAINBAND_ACCESS_MODE = ASYNC only.
s0_axi4_clock_out	clock	Output user clock for mainband (from CPA of primary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.
s0_axi4_ctrl_ready	reset	Reset for mainband, from primary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.
core_init_n	reset	An input to indicate that core configuration is complete.
s0_axi4	axi4	Mainband AXI4 from fabric to controller, channel 0.
s0_axi4lite_clock	clock	Clock for sideband interface (primary I/O bank).
s0_axi4lite_reset_n	reset	Reset for sideband interface (primary I/O bank).
s0_axi4lite	axi4lite	Sideband interface (primary I/O bank) that will connect to the IOSSM, through a gearbox in the core.
mem_0	conduit	Interface to the memory (channel 0), including all CA pins, DQ pins, and DQS pins.
mem_ck_0	conduit	Clock pin to the memory (channel 0).
mem_reset_n	conduit	Reset pin to the memory. Must always be placed along with channel 0, but shared for entire interface (all channels within one EMIF).
oct_0	conduit	On-Chip Termination (OCT) interface, representing RZQ pin (channel 0).
ref_clk	clock	Reference clock used by the EMIF PLL.

4.1.1. s0_axi4_clock_in for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

Input user clock for mainband; for MAINBAND_ACCESS_MODE = ASYNC only.

Table 25. Interface: s0_axi4_clock_in

Interface type: clock

Port Name	Direction	Description
s0_axi4_clock_in	Input	User clock for mainband axi. Input clock to the EMIF IP, no relationship to PHY clock.

4.1.2. s0_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

Output user clock for mainband (from CPA of primary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.

Table 26. Interface: s0_axi4_clock_out

Interface type: clock

Port Name	Direction	Description
s0_axi4_clock_out	Output	User clock for maiband axi (primary I/O bank). Output clock from the EMIF IP (output from CPA block, synchronous to PHY clock).

4.1.3. s0_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

Reset for mainband, from primary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.

Table 27. Interface: s0_axi4_ctrl_ready

Interface type: reset

Port Name	Direction	Description
s0_axi4_reset_n	Output	Output signal from EMIF IP (primary I/O bank), indicating that Calibration of the channels in this I/O bank is complete, and controllers in this I/O bank are ready for use.

4.1.4. core_init_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

An input to indicate that core configuration is complete.

Table 28. Interface: core_init_n

Interface type: reset

Port Name	Direction	Description
core_init_n	Input	Core init signal going into EMIF. Used to generate the reset signal on the core-EMIF interface in fabric modes. When high, indicates core initialization is complete.

4.1.5. s0_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

Mainband AXI4 from fabric to controller, channel 0.

Table 29. Interface: s0_axi4

Interface type: axi4

Port Name	Direction	Description
Write Address (Command) Channel		
s0_axi4_awaddr	input	Write address
s0_axi4_awburst	input	Write burst type. 'b00 = Reserved (FIXED is not supported). 'b01 = INCR. 'b10 = WRAP. 'b11 = Reserved.
s0_axi4_awid	input	Write address ID
s0_axi4_awlen	input	Write burst length. Any value between 0 and 255 is valid, representing a transfer of 1 to 256 beats.
s0_axi4_awlock	input	Write lock type. This 2-bit signal is used to control exclusive accesses and locking. 'b0 = Normal Access. 'b1 = Exclusive Access.
s0_axi4_awqos	input	Write quality of service. Supported priority values range from 0 to 3, with 0 as the lowest priority. Only bits[3:2] are used, bits[1:0] are ignored.
s0_axi4_awsz	input	Write burst size. AWSIZE = 5 (32 bytes) is supported by the memory controller when the AXI port is 256 bits, and only AWSIZE = 4 (16 bytes) is supported by the memory controller when the AXI port is 128 bits.
s0_axi4_awvalid	input	Write address valid
s0_axi4_awuser	input	Write address user signal. [0]: Enable/Disable Auto-precharge. Drive 1 to enable Auto-precharge. An auto-precharge will be issued after the write command is completed. [1]: ALLSTRB: When all write strobes are driven (no byte enable signals not asserted), this signal can be enabled to improve controller performance. [13:2]: Not connected. Drive 0.
s0_axi4_awprot	input	Write protection type. This 2-bit signal is used to control privileged and secure accesses.
<i>continued...</i>		

Port Name	Direction	Description
		'b00 = Non-Privileged & Secure Access. 'b01 = Privileged & Secure Access. 'b10 = Non-Privileged & Non-Secure. 'b11 = Privileged & Non-Secure.
s0_axi4_awready	output	Write address ready
Write Data Channel		
s0_axi4_wdata	input	Write data
s0_axi4_wlast	input	Write last. This signal indicates the last transfer in a write burst.
s0_axi4_wready	output	Write ready. Indicates that the AXI port is ready to accept write data.
s0_axi4_wstrb	input	Write strobes
s0_axi4_wuser	input	Write user signal. Only applicable to the x40/x72 lockstep cases. The additional user bits to be written are sent on this interface. If a x36 interface is used, then only the lowest 32-bits are connected.
s0_axi4_wvalid	input	Write valid
Write Response Channel		
s0_axi4_bready	input	Response ready
s0_axi4_bid	output	Write response ID
s0_axi4_bresp	output	Write response. A response is sent for the entire burst. 'b00 = OKAY. Write command was successfully processed, or exclusive write command was not processed as exclusive. 'b01 = EXOKAY. Exclusive write command was successfully processed. 'b10 = SLVERR. Slave has received the read write command but there is an error in the transaction. 'b11 = DECERR. Slave does not exist and/or there is an error with the transaction.
s0_axi4_bvalid	output	Write response valid.
Read Address (Command) Channel		
s0_axi4_araddr	input	Read address.
s0_axi4_arburst	input	Read burst type. 'b00 = Reserved (FIXED is not supported). 'b01 = INCR. 'b10 = WRAP. 'b11 = Reserved.
s0_axi4_arid	input	Read write address ID
s0_axi4_arlen	input	Read burst length. Any value between 0 and 128 255 is valid, representing a transfer of 1 to 256 beats.
s0_axi4_arlock	input	Read lock type. This 2-bit signal is used to control exclusive accesses and locking. 'b0 = Normal Access. 'b1 = Exclusive Access.
<i>continued...</i>		

Port Name	Direction	Description
s0_axi4_arqos	input	Read quality of service Supported priority values range from 0 to 3, with 0 as the lowest priority. Only bits[3:2] are used, bit[1:0] are ignored.
s0_axi4_arsize	input	Read burst size. AWSIZE = 5 (32 bytes) is supported by the memory controller when the AXI port is 256 bits, and only AWSIZE = 4 (16 bytes) is supported by the memory controller when the AXI port is 128 bits.
s0_axi4_arvalid	input	Read address valid.
s0_axi4_aruser	inout	Read address user signal. [0]: Enable/Disable Auto-precharge. Drive 1 to enable auto-precharge. An auto-precharge will be issued after the read command is completed. [13:1]: Not connected. Drive 0.
s0_axi4_arprot	input	Read protection type. This 2-bit signal is used to control privileged and secure accesses. 'b00 = Non-Privileged & Secure Access. 'b01 = Privileged & Secure Access. 'b10 = Non-Privileged & Non-Secure. 'b11 = Privileged & Non-Secure.
s0_axi4_arready	output	Read address ready
Ready Data Channel		
s0_axi4_rdata	output	Read data
s0_axi4_rid	output	Read ID
s0_axi4_rlast	output	Read last. This signal indicates the last transfer in a read burst.
s0_axi4_rready	input	Read ready
s0_axi4_rresp	output	read response. A response is sent with each burst, indicating the status of that burst. 'b00 = OKAY. Read command was successfully processed, or exclusive read command was not processed as exclusive. 'b01 = EXOKAY. Exclusive read command was successfully processed. 'b10 = SLVERR. Slave has received the read command but there is an error in the transaction. 'b11 = DECERR. Slave does not exist and/or there is an error with the transaction.
s0_axi4_ruser	output	Read user signal. Only applicable to the x40/x72 lockstep cases. These are the additional user bits received on this interface. If a x36 interface is used, then only the lowest 32-bits are connected.
s0_axi4_rvalid	output	Read valid.

4.1.6. s0_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

Clock for sideband interface (primary I/O bank).

Table 31. Interface: s0_axi4lite_clock

Interface type: clock

Port Name	Direction	Description
s0_axi4lite_clock	Input	Axi-Lite clock, to primary IOSSM.

4.1.7. s0_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

Reset for sideband interface (primary I/O bank).

Table 32. Interface: s0_axi4lite_reset_n

Interface type: reset

Port Name	Direction	Description
s0_axi4lite_reset_n	Input	Axi-Lite reset_n, to primary IOSSM.

4.1.8. s0_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

Sideband interface (primary I/O bank) that will connect to the IOSSM, through a gearbox in the core.

Table 33. Interface: s0_axi4lite

Interface type: axi4lite

Port Name	Direction	Description
s0_axi4lite_awaddr	Input	Axi-Lite Write Address, to primary IOSSM.
s0_axi4lite_awprot	Input	Axi-Lite Write Address Protection Signal, to primary IOSSM.
s0_axi4lite_awvalid	Input	Axi-Lite Write Address Valid, to primary IOSSM.
s0_axi4lite_awready	Output	Axi-Lite Write Address Ready, to primary IOSSM.
s0_axi4lite_araddr	Input	Axi-Lite Read Address, to primary IOSSM.
s0_axi4lite_arprot	Input	Axi-Lite Read Address Protection Signal, to primary IOSSM.
s0_axi4lite_arvalid	Input	Axi-Lite Read Address Valid, to primary IOSSM.
s0_axi4lite_arready	Output	Axi-Lite Read Address Ready, to primary IOSSM.
s0_axi4lite_wdata	Input	Axi-Lite Write Data, to primary IOSSM.
s0_axi4lite_wstrb	Input	Axi-Lite Write Strobe, to primary IOSSM.
s0_axi4lite_wvalid	Input	Axi-Lite Write Valid, to primary IOSSM.
s0_axi4lite_wready	Output	Axi-Lite Write Ready, to primary IOSSM.
s0_axi4lite_bready	Input	Axi-Lite Write Response Ready, to primary IOSSM.
s0_axi4lite_bresp	Output	Axi-Lite Write Response, to primary IOSSM.
s0_axi4lite_bvalid	Output	Axi-Lite Write Response Valid, to primary IOSSM.
s0_axi4lite_rready	Input	Axi-Lite Read Ready, to primary IOSSM.

continued...

Port Name	Direction	Description
s0_axi4lite_rdata	Output	Axi-Lite Read Data, to primary IOSSM.
s0_axi4lite_rresp	Output	Axi-Lite Read Response, to primary IOSSM.
s0_axi4lite_rvalid	Output	Axi-Lite Read Valid, to primary IOSSM.

4.1.9. mem_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

Interface to the memory (channel 0), including all CA pins, DQ pins, and DQS pins.

Table 34. Interface: mem_0

Interface type: conduit

Port Name	Direction	Description
mem_0_cke	Output	Clock Enable channel 0.
mem_0_odt	Output	On-Die Termination channel 0.
mem_0_cs_n	Output	Chip Select channel 0.
mem_0_a	Output	Address channel 0.
mem_0_ba	Output	Bank Address channel 0.
mem_0_bg	Output	Bank Group channel 0.
mem_0_act_n	Output	Activation Command channel 0.
mem_0_par	Output	Command/Address Parity channel 0.
mem_0_dq	Bidir	Data (read/write) channel 0.
mem_0_dqs_t	Bidir	Data Strobe (true) channel 0.
mem_0_dqs_c	Bidir	Data Strobe (complement) channel 0.
mem_0_alert_n	Input	Indicates Write CRC Error channel 0.

4.1.10. mem_ck_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

Clock pin to the memory (channel 0).

Table 35. Interface: mem_ck_0

Interface type: conduit

Port Name	Direction	Description
mem_0_ck_t	Output	CK Clock (true) channel 0.
mem_0_ck_c	Output	CK Clock (complement) channel 0.

4.1.11. mem_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

Reset pin to the memory. Must always be placed along with channel 0, but shared for entire interface (all channels within one EMIF).

Table 36. Interface: mem_reset_n

Interface type: conduit

Port Name	Direction	Description
mem_0_reset_n	Output	Asynchronous Reset channel 0.

4.1.12. oct_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

On-Chip Termination (OCT) interface, representing RZQ pin (channel 0).

Table 37. Interface: oct_0

Interface type: conduit

Port Name	Direction	Description
oct_rzqin_0	Input	Calibrated On-Chip Termination (OCT) input pin channel 0.

4.1.13. ref_clk for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 Component

Reference clock used by the EMIF PLL.

Table 38. Interface: ref_clk

Interface type: clock

Port Name	Direction	Description
ref_clk	Input	PLL reference clock input.

4.2. IP Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

The interfaces in the Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM each have signals that can be connected in Platform Designer. The following table lists the interfaces and corresponding interface types.

Table 39. Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

Interface Name	Interface Type	Description
s0_axi4_clock_in	clock	Input user clock for mainband; for MAINBAND_ACCESS_MODE = ASYNC only.
s0_axi4_clock_out	clock	Output user clock for mainband (from CPA of primary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.
s0_axi4_ctrl_ready	reset	Reset for mainband, from primary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.
core_init_n	reset	An input to indicate that core configuration is complete.
s0_axi4	axi4	Mainband AXI4 from fabric to controller, channel 0.
s0_axi4lite_clock	clock	Clock for sideband interface (primary I/O bank).
s0_axi4lite_reset_n	reset	Reset for sideband interface (primary I/O bank).
s0_axi4lite	axi4lite	Sideband interface (primary I/O bank) that will connect to the IOSSM, through a gearbox in the core.
mem_0	conduit	Interface to the memory (channel 0), including all CA pins, DQ pins, and DQS pins.
mem_ck_0	conduit	Clock pin to the memory (channel 0).
mem_reset_n	conduit	Reset pin to the memory. Must always be placed along with channel 0, but shared for entire interface (all channels within one EMIF).
oct_0	conduit	On-Chip Termination (OCT) interface, representing RZQ pin (channel 0).
ref_clk	clock	Reference clock used by the EMIF PLL.

4.2.1. s0_axi4_clock_in for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

Input user clock for mainband; for MAINBAND_ACCESS_MODE = ASYNC only.

Table 40. Interface: s0_axi4_clock_in

Interface type: clock

Port Name	Direction	Description
s0_axi4_clock_in	Input	User clock for mainband axi. Input clock to the EMIF IP, no relationship to PHY clock.

4.2.2. s0_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

Output user clock for mainband (from CPA of primary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.

Table 41. Interface: s0_axi4_clock_out

Interface type: clock

Port Name	Direction	Description
s0_axi4_clock_out	Output	User clock for maiband axi (primary I/O bank). Output clock from the EMIF IP (output from CPA block, synchronous to PHY clock).

4.2.3. s0_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

Reset for mainband, from primary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.

Table 42. Interface: s0_axi4_ctrl_ready

Interface type: reset

Port Name	Direction	Description
s0_axi4_reset_n	Output	Output signal from EMIF IP (primary I/O bank), indicating that Calibration of the channels in this I/O bank is complete, and controllers in this I/O bank are ready for use.

4.2.4. core_init_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

An input to indicate that core configuration is complete.

Table 43. Interface: core_init_n

Interface type: reset

Port Name	Direction	Description
core_init_n	Input	Core init signal going into EMIF. Used to generate the reset signal on the core-EMIF interface in fabric modes. When high, indicates core initialization is complete.

4.2.5. s0_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

Mainband AXI4 from fabric to controller, channel 0.

Table 44. Interface: s0_axi4

Interface type: axi4

Port Name	Direction	Description
Write Address (Command) Channel		
s0_axi4_awaddr	input	Write address
s0_axi4_awburst	input	Write burst type.
<i>continued...</i>		

Port Name	Direction	Description
		'b00 = Reserved (FIXED is not supported). 'b01 = INCR. 'b10 = WRAP. 'b11 = Reserved.
s0_axi4_awid	input	Write address ID
s0_axi4_awlen	input	Write burst length. Any value between 0 and 255 is valid, representing a transfer of 1 to 256 beats.
s0_axi4_awlock	input	Write lock type. This 2-bit signal is used to control exclusive accesses and locking. 'b0 = Normal Access. 'b1 = Exclusive Access.
s0_axi4_awqos	input	Write quality of service. Supported priority values range from 0 to 3, with 0 as the lowest priority. Only bits[3:2] are used, bits[1:0] are ignored.
s0_axi4_awsz	input	Write burst size. AWSIZE = 5 (32 bytes) is supported by the memory controller when the AXI port is 256 bits, and only AWSIZE = 4 (16 bytes) is supported by the memory controller when the AXI port is 128 bits.
s0_axi4_awvalid	input	Write address valid
s0_axi4_awuser	input	Write address user signal. [0]: Enable/Disable Auto-precharge. Drive 1 to enable Auto-precharge. An auto-precharge will be issued after the write command is completed. [1]: ALLSTRB: When all write strobes are driven (no byte enable signals not asserted), this signal can be enabled to improve controller performance. [13:2]: Not connected. Drive 0.
s0_axi4_awprot	input	Write protection type. This 2-bit signal is used to control privileged and secure accesses. 'b00 = Non-Privileged & Secure Access. 'b01 = Privileged & Secure Access. 'b10 = Non-Privileged & Non-Secure. 'b11 = Privileged & Non-Secure.
s0_axi4_awready	output	Write address ready
Write Data Channel		
s0_axi4_wdata	input	Write data
s0_axi4_wlast	input	Write last. This signal indicates the last transfer in a write burst.
s0_axi4_wready	output	Write ready. Indicates that the AXI port is ready to accept write data.
s0_axi4_wstrb	input	Write strobes
s0_axi4_wuser	input	Write user signal. Only applicable to the x40/x72 lockstep cases. The additional user bits to be written are sent on this interface. If a x36 interface is used, then only the lowest 32-bits are connected.
s0_axi4_wvalid	input	Write valid
Write Response Channel		
s0_axi4_bready	input	Response ready
<i>continued...</i>		

Port Name	Direction	Description
s0_axi4_bid	output	Write response ID
s0_axi4_bresp	output	Write response. A response is sent for the entire burst. 'b00 = OKAY. Write command was successfully processed, or exclusive write command was not processed as exclusive. 'b01 = EXOKAY. Exclusive write command was successfully processed. 'b10 = SLVERR. Slave has received the read write command but there is an error in the transaction. 'b11 = DECERR. Slave does not exist and/or there is an error with the transaction.
s0_axi4_bvalid	output	Write response valid.
Read Address (Command) Channel		
s0_axi4_araddr	input	Read address.
s0_axi4_arburst	input	Read burst type. 'b00 = Reserved (FIXED is not supported). 'b01 = INCR. 'b10 = WRAP. 'b11 = Reserved.
s0_axi4_arid	input	Read write address ID
s0_axi4_arlen	input	Read burst length. Any value between 0 and 128 255 is valid, representing a transfer of 1 to 256 beats.
s0_axi4_arlock	input	Read lock type. This 2-bit signal is used to control exclusive accesses and locking. 'b0 = Normal Access. 'b1 = Exclusive Access.
s0_axi4_arqos	input	Read quality of service Supported priority values range from 0 to 3, with 0 as the lowest priority. Only bits[3:2] are used, bit[1:0] are ignored.
s0_axi4_arsize	input	Read burst size. AWSIZE = 5 (32 bytes) is supported by the memory controller when the AXI port is 256 bits, and only AWSIZE = 4 (16 bytes) is supported by the memory controller when the AXI port is 128 bits.
s0_axi4_arvalid	input	Read address valid.
s0_axi4_aruser	inout	Read address user signal. [0]: Enable/Disable Auto-precharge. Drive 1 to enable auto-precharge. An auto-precharge will be issued after the read command is completed. [13:1]: Not connected. Drive 0.
s0_axi4_arprot	input	Read protection type. This 2-bit signal is used to control privileged and secure accesses. 'b00 = Non-Privileged & Secure Access. 'b01 = Privileged & Secure Access. 'b10 = Non-Privileged & Non-Secure. 'b11 = Privileged & Non-Secure.
s0_axi4_arready	output	Read address ready
Ready Data Channel		
s0_axi4_rdata	output	Read data
<i>continued...</i>		

Port Name	Direction	Description
s0_axi4_rid	output	Read ID
s0_axi4_rlast	output	Read last. This signal indicates the last transfer in a read burst.
s0_axi4_rready	input	Read ready
s0_axi4_rresp	output	read response. A response is sent with each burst, indicating the status of that burst. 'b00 = OKAY. Read command was successfully processed, or exclusive read command was not processed as exclusive. 'b01 = EXOKAY. Exclusive read command was successfully processed. 'b10 = SLVERR. Slave has received the read command but there is an error in the transaction. 'b11 = DECERR. Slave does not exist and/or there is an error with the transaction.
s0_axi4_ruser	output	Read user signal. Only applicable to the x40/x72 lockstep cases. These are the additional user bits received on this interface. If a x36 interface is used, then only the lowest 32-bits are connected.
s0_axi4_rvalid	output	Read valid.

4.2.6. s0_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

Clock for sideband interface (primary I/O bank).

Table 46. Interface: s0_axi4lite_clock

Interface type: clock

Port Name	Direction	Description
s0_axi4lite_clock	Input	Axi-Lite clock, to primary IOSSM.

4.2.7. s0_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

Reset for sideband interface (primary I/O bank).

Table 47. Interface: s0_axi4lite_reset_n

Interface type: reset

Port Name	Direction	Description
s0_axi4lite_reset_n	Input	Axi-Lite reset_n, to primary IOSSM.

4.2.8. s0_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

Sideband interface (primary I/O bank) that will connect to the IOSSM, through a gearbox in the core.

Table 48. Interface: s0_axi4lite

Interface type: axi4lite

Port Name	Direction	Description
s0_axi4lite_awaddr	Input	Axi-Lite Write Address, to primary IOSSM.
s0_axi4lite_awprot	Input	Axi-Lite Write Address Protection Signal, to primary IOSSM.
s0_axi4lite_awvalid	Input	Axi-Lite Write Address Valid, to primary IOSSM.
s0_axi4lite_awready	Output	Axi-Lite Write Address Ready, to primary IOSSM.
s0_axi4lite_araddr	Input	Axi-Lite Read Address, to primary IOSSM.
s0_axi4lite_arprot	Input	Axi-Lite Read Address Protection Signal, to primary IOSSM.
s0_axi4lite_arvalid	Input	Axi-Lite Read Address Valid, to primary IOSSM.
s0_axi4lite_arready	Output	Axi-Lite Read Address Ready, to primary IOSSM.
s0_axi4lite_wdata	Input	Axi-Lite Write Data, to primary IOSSM.
s0_axi4lite_wstrb	Input	Axi-Lite Write Strobe, to primary IOSSM.
s0_axi4lite_wvalid	Input	Axi-Lite Write Valid, to primary IOSSM.
s0_axi4lite_wready	Output	Axi-Lite Write Ready, to primary IOSSM.
s0_axi4lite_bready	Input	Axi-Lite Write Response Ready, to primary IOSSM.
s0_axi4lite_bresp	Output	Axi-Lite Write Response, to primary IOSSM.
s0_axi4lite_bvalid	Output	Axi-Lite Write Response Valid, to primary IOSSM.
s0_axi4lite_rready	Input	Axi-Lite Read Ready, to primary IOSSM.
s0_axi4lite_rdata	Output	Axi-Lite Read Data, to primary IOSSM.
s0_axi4lite_rresp	Output	Axi-Lite Read Response, to primary IOSSM.
s0_axi4lite_rvalid	Output	Axi-Lite Read Valid, to primary IOSSM.

4.2.9. mem_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

Interface to the memory (channel 0), including all CA pins, DQ pins, and DQS pins.

Table 49. Interface: mem_0

Interface type: conduit

Port Name	Direction	Description
mem_0_cke	Output	Clock Enable channel 0.
mem_0_odt	Output	On-Die Termination channel 0.
mem_0_cs_n	Output	Chip Select channel 0.
mem_0_a	Output	Address channel 0.
mem_0_ba	Output	Bank Address channel 0.
mem_0_bg	Output	Bank Group channel 0.
mem_0_act_n	Output	Activation Command channel 0.
<i>continued...</i>		

Port Name	Direction	Description
mem_0_par	Output	Command/Address Parity channel 0.
mem_0_dq	Bidir	Data (read/write) channel 0.
mem_0_dqs_t	Bidir	Data Strobe (true) channel 0.
mem_0_dqs_c	Bidir	Data Strobe (complement) channel 0.
mem_0_alert_n	Input	Indicates Write CRC Error channel 0.

4.2.10. mem_ck_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

Clock pin to the memory (channel 0).

Table 50. Interface: mem_ck_0

Interface type: conduit

Port Name	Direction	Description
mem_0_ck_t	Output	CK Clock (true) channel 0.
mem_0_ck_c	Output	CK Clock (complement) channel 0.

4.2.11. mem_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

Reset pin to the memory. Must always be placed along with channel 0, but shared for entire interface (all channels within one EMIF).

Table 51. Interface: mem_reset_n

Interface type: conduit

Port Name	Direction	Description
mem_0_reset_n	Output	Asynchronous Reset channel 0.

4.2.12. oct_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

On-Chip Termination (OCT) interface, representing RZQ pin (channel 0).

Table 52. Interface: oct_0

Interface type: conduit

Port Name	Direction	Description
oct_rzqin_0	Input	Calibrated On-Chip Termination (OCT) input pin channel 0.

4.2.13. ref_clk for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR4 DIMM

Reference clock used by the EMIF PLL.

Table 53. Interface: ref_clk

Interface type: clock

Port Name	Direction	Description
ref_clk	Input	PLL reference clock input.

4.3. IP Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

The interfaces in the Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component each have signals that can be connected in Platform Designer. The following table lists the interfaces and corresponding interface types.

Table 54. Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Interface Name	Interface Type	Description
s0_axi4_clock_in	clock	Input user clock for mainband; for MAINBAND_ACCESS_MODE = ASYNC only.
s0_axi4_clock_out	clock	Output user clock for mainband (from CPA of primary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.
s0_axi4_ctrl_ready	reset	Reset for mainband, from primary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.
core_init_n	reset	An input to indicate that core configuration is complete.
s0_axi4	axi4	Mainband AXI4 from fabric to controller, channel 0.
s1_axi4	axi4	Mainband AXI4 from fabric to controller, channel 1.
s0_axi4lite_clock	clock	Clock for sideband interface (primary I/O bank).
s0_axi4lite_reset_n	reset	Reset for sideband interface (primary I/O bank).
s0_axi4lite	axi4lite	Sideband interface (primary I/O bank) that will connect to the IOSSM, through a gearbox in the core.
mem_0	conduit	Interface to the memory (channel 0), including all CA pins, DQ pins, and DQS pins.
mem_ck_0	conduit	Clock pin to the memory (channel 0).
mem_reset_n_0	conduit	Reset pin to the memory (channel 0).
mem_1	conduit	Interface to the memory (channel 1), including all CA pins, DQ pins, and DQS pins.
mem_ck_1	conduit	Clock pin to the memory (channel 1).
mem_reset_n_1	conduit	Reset pin to the memory (channel 1).
oct_0	conduit	On-Chip Termination (OCT) interface, representing RZQ pin (channel 0).
oct_1	conduit	On-Chip Termination (OCT) interface, representing RZQ pin (channel 1).
ref_clk	clock	Reference clock used by the EMIF PLL.

4.3.1. s0_axi4_clock_in for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Input user clock for mainband; for MAINBAND_ACCESS_MODE = ASYNC only.

Table 55. Interface: s0_axi4_clock_in

Interface type: clock

Port Name	Direction	Description
s0_axi4_clock_in	Input	User clock for mainband axi. Input clock to the EMIF IP, no relationship to PHY clock.

4.3.2. s0_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Output user clock for mainband (from CPA of primary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.

Table 56. Interface: s0_axi4_clock_out

Interface type: clock

Port Name	Direction	Description
s0_axi4_clock_out	Output	User clock for maiband axi (primary I/O bank). Output clock from the EMIF IP (output from CPA block, synchronous to PHY clock).

4.3.3. s0_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Reset for mainband, from primary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.

Table 57. Interface: s0_axi4_ctrl_ready

Interface type: reset

Port Name	Direction	Description
s0_axi4_reset_n	Output	Output signal from EMIF IP (primary I/O bank), indicating that Calibration of the channels in this I/O bank is complete, and controllers in this I/O bank are ready for use.

4.3.4. core_init_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

An input to indicate that core configuration is complete.

Table 58. Interface: core_init_n

Interface type: reset

Port Name	Direction	Description
core_init_n	Input	Core init signal going into EMIF. Used to generate the reset signal on the core-EMIF interface in fabric modes. When high, indicates core initialization is complete.

4.3.5. s0_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Mainband AXI4 from fabric to controller, channel 0.

Table 59. Interface: s0_axi4

Interface type: axi4

Port Name	Direction	Description
Write Address (Command) Channel		
s0_axi4_awaddr	input	Write address
s0_axi4_awburst	input	Write burst type. 'b00 = Reserved (FIXED is not supported). 'b01 = INCR. 'b10 = WRAP. 'b11 = Reserved.
s0_axi4_awid	input	Write address ID
s0_axi4_awlen	input	Write burst length. Any value between 0 and 255 is valid, representing a transfer of 1 to 256 beats.
s0_axi4_awlock	input	Write lock type. This 2-bit signal is used to control exclusive accesses and locking. 'b0 = Normal Access. 'b1 = Exclusive Access.
s0_axi4_awqos	input	Write quality of service. Supported priority values range from 0 to 3, with 0 as the lowest priority. Only bits[3:2] are used, bits[1:0] are ignored.
s0_axi4_awsz	input	Write burst size. AWSIZE = 5 (32 bytes) is supported by the memory controller when the AXI port is 256 bits, and only AWSIZE = 4 (16 bytes) is supported by the memory controller when the AXI port is 128 bits.
s0_axi4_awvalid	input	Write address valid
s0_axi4_awuser	input	Write address user signal. [0]: Enable/Disable Auto-precharge. Drive 1 to enable Auto-precharge. An auto-precharge will be issued after the write command is completed. [1]: ALLSTRB: When all write strobes are driven (no byte enable signals not asserted), this signal can be enabled to improve controller performance. [13:2]: Not connected. Drive 0.
s0_axi4_awprot	input	Write protection type. This 2-bit signal is used to control privileged and secure accesses.
<i>continued...</i>		

Port Name	Direction	Description
		'b00 = Non-Privileged & Secure Access. 'b01 = Privileged & Secure Access. 'b10 = Non-Privileged & Non-Secure. 'b11 = Privileged & Non-Secure.
s0_axi4_awready	output	Write address ready
Write Data Channel		
s0_axi4_wdata	input	Write data
s0_axi4_wlast	input	Write last. This signal indicates the last transfer in a write burst.
s0_axi4_wready	output	Write ready. Indicates that the AXI port is ready to accept write data.
s0_axi4_wstrb	input	Write strobes
s0_axi4_wuser	input	Write user signal. Only applicable to the x40/x72 lockstep cases. The additional user bits to be written are sent on this interface. If a x36 interface is used, then only the lowest 32-bits are connected.
s0_axi4_wvalid	input	Write valid
Write Response Channel		
s0_axi4_bready	input	Response ready
s0_axi4_bid	output	Write response ID
s0_axi4_bresp	output	Write response. A response is sent for the entire burst. 'b00 = OKAY. Write command was successfully processed, or exclusive write command was not processed as exclusive. 'b01 = EXOKAY. Exclusive write command was successfully processed. 'b10 = SLVERR. Slave has received the read write command but there is an error in the transaction. 'b11 = DECERR. Slave does not exist and/or there is an error with the transaction.
s0_axi4_bvalid	output	Write response valid.
Read Address (Command) Channel		
s0_axi4_araddr	input	Read address.
s0_axi4_arburst	input	Read burst type. 'b00 = Reserved (FIXED is not supported). 'b01 = INCR. 'b10 = WRAP. 'b11 = Reserved.
s0_axi4_arid	input	Read write address ID
s0_axi4_arlen	input	Read burst length. Any value between 0 and 128 255 is valid, representing a transfer of 1 to 256 beats.
s0_axi4_arlock	input	Read lock type. This 2-bit signal is used to control exclusive accesses and locking. 'b0 = Normal Access. 'b1 = Exclusive Access.
<i>continued...</i>		

Port Name	Direction	Description
s0_axi4_arqos	input	Read quality of service Supported priority values range from 0 to 3, with 0 as the lowest priority. Only bits[3:2] are used, bit[1:0] are ignored.
s0_axi4_arsize	input	Read burst size. AWSIZE = 5 (32 bytes) is supported by the memory controller when the AXI port is 256 bits, and only AWSIZE = 4 (16 bytes) is supported by the memory controller when the AXI port is 128 bits.
s0_axi4_arvalid	input	Read address valid.
s0_axi4_aruser	inout	Read address user signal. [0]: Enable/Disable Auto-precharge. Drive 1 to enable auto-precharge. An auto-precharge will be issued after the read command is completed. [13:1]: Not connected. Drive 0.
s0_axi4_arprot	input	Read protection type. This 2-bit signal is used to control privileged ad secure accesses. 'b00 = Non-Privileged & Secure Access. 'b01 = Privileged & Secure Access. 'b10 = Non-Privileged & Non-Secure. 'b11 = Privileged & Non-Secure.
s0_axi4_arready	output	Read address ready
Ready Data Channel		
s0_axi4_rdata	output	Read data
s0_axi4_rid	output	Read ID
s0_axi4_rlast	output	Read last. This signal indicates the last transfer in a read burst.
s0_axi4_rready	input	Read ready
s0_axi4_rresp	output	read response. A response is sent with each burst, indicating the status of that burst. 'b00 = OKAY. Read command was successfully processed, or exclusive read command was not processed as exclusive. 'b01 = EXOKAY. Exclusive read command was successfully processed. 'b10 = SLVERR. Slave has received the read command but there is an error in the transaction. 'b11 = DECERR. Slave does not exist and/or there is an error with the transaction.
s0_axi4_ruser	output	Read user signal. Only applicable to the x40/x72 lockstep cases. These are the additional user bits received on this interface. If a x36 interface is used, then only the lowest 32-bits are connected.
s0_axi4_rvalid	output	Read valid.

4.3.6. s1_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Mainband AXI4 from fabric to controller, channel 1.

Table 61. Interface: s1_axi4

Interface type: axi4

Port Name	Direction	Description
s1_axi4_awaddr	Input	Write Address , channel 1.
s1_axi4_awburst	Input	Write Burst Type, channel 1.
s1_axi4_awid	Input	Write Address ID, channel 1.
s1_axi4_awlen	Input	Write Burst Length, channel 1.
s1_axi4_awlock	Input	Write Lock Type, channel 1.
s1_axi4_awqos	Input	Write Quality of Service, channel 1.
s1_axi4_awsz	Input	Write Burst Size, channel 1.
s1_axi4_awvalid	Input	Write Address Valid, channel 1.
s1_axi4_awuser	Input	Write Address User Signal, channel 1.
s1_axi4_awprot	Input	Write Protection Type, channel 1.
s1_axi4_awready	Output	Write Address Ready, channel 1.
s1_axi4_araddr	Input	Read Address , channel 1.
s1_axi4_arburst	Input	Read Burst Type, channel 1.
s1_axi4_arid	Input	Read Address ID, channel 1.
s1_axi4_arlen	Input	Read Burst Length, channel 1.
s1_axi4_arlock	Input	Read Lock Type, channel 1.
s1_axi4_arqos	Input	Read Quality of Service, channel 1.
s1_axi4_arsz	Input	Read Burst Size, channel 1.
s1_axi4_arvalid	Input	Read Address Valid, channel 1.
s1_axi4_aruser	Input	Read Address User Signal, channel 1.
s1_axi4_arprot	Input	Read Protection Type, channel 1.
s1_axi4_arready	Output	Read Address Ready, channel 1.
s1_axi4_wdata	Input	Write Data , channel 1.
s1_axi4_wstrb	Input	Write Strobcs, channel 1.
s1_axi4_wlast	Input	Write Last, channel 1.
s1_axi4_wvalid	Input	Write Valid, channel 1.
s1_axi4_wready	Output	Write Ready, channel 1.
s1_axi4_bready	Input	Write Response Ready, channel 1.
s1_axi4_bid	Output	Write Response ID, channel 1.
s1_axi4_bresp	Output	Write Response , channel 1.
s1_axi4_bvalid	Output	Write Response Valid, channel 1.
s1_axi4_rready	Input	Read Ready, channel 1.
s1_axi4_rdata	Output	Read Data, channel 1.
s1_axi4_rid	Output	Read ID , channel 1.

continued...

Port Name	Direction	Description
s1_axi4_rlast	Output	Read Last, channel 1.
s1_axi4_rresp	Output	Read Response, channel 1.
s1_axi4_rvalid	Output	Read Valid, channel 1.

4.3.7. s0_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Clock for sideband interface (primary I/O bank).

Table 62. Interface: s0_axi4lite_clock

Interface type: clock

Port Name	Direction	Description
s0_axi4lite_clock	Input	Axi-Lite clock, to primary IOSSM.

4.3.8. s0_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Reset for sideband interface (primary I/O bank).

Table 63. Interface: s0_axi4lite_reset_n

Interface type: reset

Port Name	Direction	Description
s0_axi4lite_reset_n	Input	Axi-Lite reset_n, to primary IOSSM.

4.3.9. s0_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Sideband interface (primary I/O bank) that will connect to the IOSSM, through a gearbox in the core.

Table 64. Interface: s0_axi4lite

Interface type: axi4lite

Port Name	Direction	Description
s0_axi4lite_awaddr	Input	Axi-Lite Write Address, to primary IOSSM.
s0_axi4lite_awprot	Input	Axi-Lite Write Address Protection Signal, to primary IOSSM.
s0_axi4lite_awvalid	Input	Axi-Lite Write Address Valid, to primary IOSSM.
s0_axi4lite_awready	Output	Axi-Lite Write Address Ready, to primary IOSSM.
s0_axi4lite_araddr	Input	Axi-Lite Read Address, to primary IOSSM.
s0_axi4lite_arprot	Input	Axi-Lite Read Address Protection Signal, to primary IOSSM.
s0_axi4lite_arvalid	Input	Axi-Lite Read Address Valid, to primary IOSSM.

continued...

Port Name	Direction	Description
s0_axi4lite_arready	Output	Axi-Lite Read Address Ready, to primary IOSSM.
s0_axi4lite_wdata	Input	Axi-Lite Write Data, to primary IOSSM.
s0_axi4lite_wstrb	Input	Axi-Lite Write Strobe, to primary IOSSM.
s0_axi4lite_wvalid	Input	Axi-Lite Write Valid, to primary IOSSM.
s0_axi4lite_wready	Output	Axi-Lite Write Ready, to primary IOSSM.
s0_axi4lite_bready	Input	Axi-Lite Write Response Ready, to primary IOSSM.
s0_axi4lite_bresp	Output	Axi-Lite Write Response, to primary IOSSM.
s0_axi4lite_bvalid	Output	Axi-Lite Write Response Valid, to primary IOSSM.
s0_axi4lite_rready	Input	Axi-Lite Read Ready, to primary IOSSM.
s0_axi4lite_rdata	Output	Axi-Lite Read Data, to primary IOSSM.
s0_axi4lite_rresp	Output	Axi-Lite Read Response, to primary IOSSM.
s0_axi4lite_rvalid	Output	Axi-Lite Read Valid, to primary IOSSM.

4.3.10. mem_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Interface to the memory (channel 0), including all CA pins, DQ pins, and DQS pins.

Table 65. Interface: mem_0

Interface type: conduit

Port Name	Direction	Description
mem_0_cs_n	Output	Chip Select channel 0.
mem_0_ca	Output	Command/Address Bus channel 0.
mem_0_dq	Bidir	Data (read/write) channel 0.
mem_0_dqs_t	Bidir	Data Strobe (true) channel 0.
mem_0_dqs_c	Bidir	Data Strobe (complement) channel 0.
mem_0_alert_n	Input	Indicates Write CRC Error channel 0.

4.3.11. mem_ck_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Clock pin to the memory (channel 0).

Table 66. Interface: mem_ck_0

Interface type: conduit

Port Name	Direction	Description
mem_0_ck_t	Output	CK Clock (true) channel 0.
mem_0_ck_c	Output	CK Clock (complement) channel 0.

4.3.12. mem_reset_n_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Reset pin to the memory (channel 0).

Table 67. Interface: mem_reset_n_0

Interface type: conduit

Port Name	Direction	Description
mem_0_reset_n	Output	Asynchronous Reset channel 0.

4.3.13. mem_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Interface to the memory (channel 1), including all CA pins, DQ pins, and DQS pins.

Table 68. Interface: mem_1

Interface type: conduit

Port Name	Direction	Description
mem_1_cs_n	Output	Chip Select channel 1.
mem_1_ca	Output	Command/Address Bus channel 1.
mem_1_dq	Bidir	Data (read/write) channel 1.
mem_1_dqs_t	Bidir	Data Strobe (true) channel 1.
mem_1_dqs_c	Bidir	Data Strobe (complement) channel 1.
mem_1_alert_n	Input	Indicates Write CRC Error channel 1.

4.3.14. mem_ck_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Clock pin to the memory (channel 1).

Table 69. Interface: mem_ck_1

Interface type: conduit

Port Name	Direction	Description
mem_1_ck_t	Output	CK Clock (true) channel 1.
mem_1_ck_c	Output	CK Clock (complement) channel 1.

4.3.15. mem_reset_n_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Reset pin to the memory (channel 1).

Table 70. Interface: mem_reset_n_1

Interface type: conduit

Port Name	Direction	Description
mem_1_reset_n	Output	Asynchronous Reset channel 1.

4.3.16. oct_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

On-Chip Termination (OCT) interface, representing RZQ pin (channel 0).

Table 71. Interface: oct_0

Interface type: conduit

Port Name	Direction	Description
oct_rzqin_0	Input	Calibrated On-Chip Termination (OCT) input pin channel 0.

4.3.17. oct_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

On-Chip Termination (OCT) interface, representing RZQ pin (channel 1).

Table 72. Interface: oct_1

Interface type: conduit

Port Name	Direction	Description
oct_rzqin_1	Input	Calibrated On-Chip Termination (OCT) input pin channel 1.

4.3.18. ref_clk for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 Component

Reference clock used by the EMIF PLL.

Table 73. Interface: ref_clk

Interface type: clock

Port Name	Direction	Description
ref_clk	Input	PLL reference clock input.

4.4. IP Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

The interfaces in the Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM each have signals that can be connected in Platform Designer. The following table lists the interfaces and corresponding interface types.

Table 74. Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Interface Name	Interface Type	Description
s0_axi4_clock_out	clock	Output user clock for mainband (from CPA of primary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.
s0_axi4_ctrl_ready	reset	Reset for mainband, from primary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.
s1_axi4_clock_out	clock	Output user clock for mainband (from CPA of secondary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.
s1_axi4_ctrl_ready	reset	Reset for mainband, from secondary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.
s0_axi4_clock_in	clock	Input user clock for mainband; for MAINBAND_ACCESS_MODE = ASYNC only.
core_init_n	reset	An input to indicate that core configuration is complete.
s0_axi4	axi4	Mainband AXI4 from fabric to controller, channel 0.
s1_axi4	axi4	Mainband AXI4 from fabric to controller, channel 1.
s0_axi4lite_clock	clock	Clock for sideband interface (primary I/O bank).
s0_axi4lite_reset_n	reset	Reset for sideband interface (primary I/O bank).
s0_axi4lite	axi4lite	Sideband interface (primary I/O bank) that will connect to the IOSSM, through a gearbox in the core.
s1_axi4lite_clock	clock	Clock for sideband interface (secondary I/O bank).
s1_axi4lite_reset_n	reset	Reset for sideband interface (secondary I/O bank).
s1_axi4lite	axi4lite	Sideband interface (secondary I/O bank) that will connect to the IOSSM, through a gearbox in the core.
mem_0	conduit	Interface to the memory (channel 0), including all CA pins, DQ pins, and DQS pins.
mem_1	conduit	Interface to the memory (channel 1), including all CA pins, DQ pins, and DQS pins.
mem_reset_n	conduit	Reset pin to the memory. Must always be placed along with channel 0, but shared for entire interface (all channels within one EMIF).
mem_ck_0	conduit	Clock pin to the memory (channel 0).
mem_ck_1	conduit	Clock pin to the memory (channel 1).
mem_i3c	conduit	I3C to the DDR5 DIMM.
mem_lb_dq	conduit	LBDQ going to DDR5 DIMMs; RDIMM only. Always placed along with channel 0.
mem_lb_dqs	conduit	LBDQS going to DDR5 DIMMs; RDIMM only. Always placed along with channel 1.
oct_0	conduit	On-Chip Termination (OCT) interface, representing RZQ pin (channel 0).
oct_1	conduit	On-Chip Termination (OCT) interface, representing RZQ pin (channel 1).
ref_clk	clock	Reference clock used by the EMIF PLL.

4.4.1. s0_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Output user clock for mainband (from CPA of primary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.

Table 75. Interface: s0_axi4_clock_out

Interface type: clock

Port Name	Direction	Description
s0_axi4_clock_out	Output	User clock for maiband axi (primary I/O bank). Output clock from the EMIF IP (output from CPA block, synchronous to PHY clock).

4.4.2. s0_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Reset for mainband, from primary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.

Table 76. Interface: s0_axi4_ctrl_ready

Interface type: reset

Port Name	Direction	Description
s0_axi4_reset_n	Output	Output signal from EMIF IP (primary I/O bank), indicating that Calibration of the channels in this I/O bank is complete, and controllers in this I/O bank are ready for use.

4.4.3. s1_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Output user clock for mainband (from CPA of secondary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.

Table 77. Interface: s1_axi4_clock_out

Interface type: clock

Port Name	Direction	Description
s1_axi4_clock_out	Output	User clock for maiband axi (secondary I/O bank). Output clock from the EMIF IP (output from CPA block, synchronous to PHY clock).

4.4.4. s1_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Reset for mainband, from secondary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.

Table 78. Interface: s1_axi4_ctrl_ready

Interface type: reset

Port Name	Direction	Description
s1_axi4_reset_n	Output	Output signal from EMIF IP (secondary I/O bank), indicating that Calibration of the channels in this I/O bank is complete, and controllers in this I/O bank are ready for use.

4.4.5. s0_axi4_clock_in for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Input user clock for mainband; for MAINBAND_ACCESS_MODE = ASYNC only.

Table 79. Interface: s0_axi4_clock_in

Interface type: clock

Port Name	Direction	Description
s0_axi4_clock_in	Input	User clock for mainband axi. Input clock to the EMIF IP, no relationship to PHY clock.

4.4.6. core_init_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

An input to indicate that core configuration is complete.

Table 80. Interface: core_init_n

Interface type: reset

Port Name	Direction	Description
core_init_n	Input	Core init signal going into EMIF. Used to generate the reset signal on the core-EMIF interface in fabric modes. When high, indicates core initialization is complete.

4.4.7. s0_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Mainband AXI4 from fabric to controller, channel 0.

Table 81. Interface: s0_axi4

Interface type: axi4

Port Name	Direction	Description
Write Address (Command) Channel		
s0_axi4_awaddr	input	Write address
s0_axi4_awburst	input	Write burst type. 'b00 = Reserved (FIXED is not supported). 'b01 = INCR. 'b10 = WRAP. 'b11 = Reserved.
<i>continued...</i>		

Port Name	Direction	Description
s0_axi4_awid	input	Write address ID
s0_axi4_awlen	input	Write burst length. Any value between 0 and 255 is valid, representing a transfer of 1 to 256 beats.
s0_axi4_awlock	input	Write lock type. This 2-bit signal is used to control exclusive accesses and locking. 'b0 = Normal Access. 'b1 = Exclusive Access.
s0_axi4_awqos	input	Write quality of service. Supported priority values range from 0 to 3, with 0 as the lowest priority. Only bits[3:2] are used, bits[1:0] are ignored.
s0_axi4_awsz	input	Write burst size. AWSIZE = 5 (32 bytes) is supported by the memory controller when the AXI port is 256 bits, and only AWSIZE = 4 (16 bytes) is supported by the memory controller when the AXI port is 128 bits.
s0_axi4_awvalid	input	Write address valid
s0_axi4_awuser	input	Write address user signal. [0]: Enable/Disable Auto-precharge. Drive 1 to enable Auto-precharge. An auto-precharge will be issued after the write command is completed. [1]: ALLSTRB: When all write strobes are driven (no byte enable signals not asserted), this signal can be enabled to improve controller performance. [13:2]: Not connected. Drive 0.
s0_axi4_awprot	input	Write protection type. This 2-bit signal is used to control privileged and secure accesses. 'b00 = Non-Privileged & Secure Access. 'b01 = Privileged & Secure Access. 'b10 = Non-Privileged & Non-Secure. 'b11 = Privileged & Non-Secure.
s0_axi4_awready	output	Write address ready
Write Data Channel		
s0_axi4_wdata	input	Write data
s0_axi4_wlast	input	Write last. This signal indicates the last transfer in a write burst.
s0_axi4_wready	output	Write ready. Indicates that the AXI port is ready to accept write data.
s0_axi4_wstrb	input	Write strobes
s0_axi4_wuser	input	Write user signal. Only applicable to the x40/x72 lockstep cases. The additional user bits to be written are sent on this interface. If a x36 interface is used, then only the lowest 32-bits are connected.
s0_axi4_wvalid	input	Write valid
Write Response Channel		
s0_axi4_bready	input	Response ready
s0_axi4_bid	output	Write response ID
s0_axi4_bresp	output	Write response. A response is sent for the entire burst.
continued...		

Port Name	Direction	Description
		'b00 = OKAY. Write command was successfully processed, or exclusive write command was not processed as exclusive. 'b01 = EXOKAY. Exclusive write command was successfully processed. 'b10 = SLVERR. Slave has received the read write command but there is an error in the transaction. 'b11 = DECERR. Slave does not exist and/or there is an error with the transaction.
s0_axi4_bvalid	output	Write response valid.
Read Address (Command) Channel		
s0_axi4_araddr	input	Read address.
s0_axi4_arburst	input	Read burst type. 'b00 = Reserved (FIXED is not supported). 'b01 = INCR. 'b10 = WRAP. 'b11 = Reserved.
s0_axi4_arid	input	Read write address ID
s0_axi4_arlen	input	Read burst length. Any value between 0 and 128 255 is valid, representing a transfer of 1 to 256 beats.
s0_axi4_arlock	input	Read lock type. This 2-bit signal is used to control exclusive accesses and locking. 'b0 = Normal Access. 'b1 = Exclusive Access.
s0_axi4_arqos	input	Read quality of service Supported priority values range from 0 to 3, with 0 as the lowest priority. Only bits[3:2] are used, bit[1:0] are ignored.
s0_axi4_arsize	input	Read burst size. AWSIZE = 5 (32 bytes) is supported by the memory controller when the AXI port is 256 bits, and only AWSIZE = 4 (16 bytes) is supported by the memory controller when the AXI port is 128 bits.
s0_axi4_arvalid	input	Read address valid.
s0_axi4_aruser	inout	Read address user signal. [0]: Enable/Disable Auto-precharge. Drive 1 to enable auto-precharge. An auto-precharge will be issued after the read command is completed. [13:1]: Not connected. Drive 0.
s0_axi4_arprot	input	Read protection type. This 2-bit signal is used to control privileged and secure accesses. 'b00 = Non-Privileged & Secure Access. 'b01 = Privileged & Secure Access. 'b10 = Non-Privileged & Non-Secure. 'b11 = Privileged & Non-Secure.
s0_axi4_arready	output	Read address ready
Ready Data Channel		
s0_axi4_rdata	output	Read data
s0_axi4_rid	output	Read ID
<i>continued...</i>		

Port Name	Direction	Description
s0_axi4_rlast	output	Read last. This signal indicates the last transfer in a read burst.
s0_axi4_rready	input	Read ready
s0_axi4_rresp	output	read response. A response is sent with each burst, indicating the status of that burst. 'b00 = OKAY. Read command was successfully processed, or exclusive read command was not processed as exclusive. 'b01 = EXOKAY. Exclusive read command was successfully processed. 'b10 = SLVERR. Slave has received the read command but there is an error in the transaction. 'b11 = DECERR. Slave does not exist and/or there is an error with the transaction.
s0_axi4_ruser	output	Read user signal. Only applicable to the x40/x72 lockstep cases. These are the additional user bits received on this interface. If a x36 interface is used, then only the lowest 32-bits are connected.
s0_axi4_rvalid	output	Read valid.

4.4.8. s1_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Mainband AXI4 from fabric to controller, channel 1.

Table 83. Interface: s1_axi4

Interface type: axi4

Port Name	Direction	Description
s1_axi4_awaddr	Input	Write Address , channel 1.
s1_axi4_awburst	Input	Write Burst Type, channel 1.
s1_axi4_awid	Input	Write Address ID, channel 1.
s1_axi4_awlen	Input	Write Burst Length, channel 1.
s1_axi4_awlock	Input	Write Lock Type, channel 1.
s1_axi4_awqos	Input	Write Quality of Service, channel 1.
s1_axi4_awsiz	Input	Write Burst Size, channel 1.
s1_axi4_awvalid	Input	Write Address Valid, channel 1.
s1_axi4_awuser	Input	Write Address User Signal, channel 1.
s1_axi4_awprot	Input	Write Protection Type, channel 1.
s1_axi4_awready	Output	Write Address Ready, channel 1.
s1_axi4_araddr	Input	Read Address , channel 1.
s1_axi4_arburst	Input	Read Burst Type, channel 1.
s1_axi4_arid	Input	Read Address ID, channel 1.
s1_axi4_arlen	Input	Read Burst Length, channel 1.
continued...		

Port Name	Direction	Description
s1_axi4_arlock	Input	Read Lock Type, channel 1.
s1_axi4_arqos	Input	Read Quality of Service, channel 1.
s1_axi4_arsize	Input	Read Burst Size, channel 1.
s1_axi4_arvalid	Input	Read Address Valid, channel 1.
s1_axi4_aruser	Input	Read Address User Signal, channel 1.
s1_axi4_arprot	Input	Read Protection Type, channel 1.
s1_axi4_arready	Output	Read Address Ready, channel 1.
s1_axi4_wdata	Input	Write Data , channel 1.
s1_axi4_wstrb	Input	Write Strobes, channel 1.
s1_axi4_wlast	Input	Write Last, channel 1.
s1_axi4_wvalid	Input	Write Valid, channel 1.
s1_axi4_wready	Output	Write Ready, channel 1.
s1_axi4_bready	Input	Write Response Ready, channel 1.
s1_axi4_bid	Output	Write Response ID, channel 1.
s1_axi4_bresp	Output	Write Response , channel 1.
s1_axi4_bvalid	Output	Write Response Valid, channel 1.
s1_axi4_rready	Input	Read Ready, channel 1.
s1_axi4_rdata	Output	Read Data, channel 1.
s1_axi4_rid	Output	Read ID , channel 1.
s1_axi4_rlast	Output	Read Last, channel 1.
s1_axi4_rresp	Output	Read Response, channel 1.
s1_axi4_rvalid	Output	Read Valid, channel 1.

4.4.9. s0_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Clock for sideband interface (primary I/O bank).

Table 84. Interface: s0_axi4lite_clock

Interface type: clock

Port Name	Direction	Description
s0_axi4lite_clock	Input	Axi-Lite clock, to primary IOSSM.

4.4.10. s0_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Reset for sideband interface (primary I/O bank).

Table 85. Interface: s0_axi4lite_reset_n

Interface type: reset

Port Name	Direction	Description
s0_axi4lite_reset_n	Input	Axi-Lite reset_n, to primary IOSSM.

4.4.11. s0_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Sideband interface (primary I/O bank) that will connect to the IOSSM, through a gearbox in the core.

Table 86. Interface: s0_axi4lite

Interface type: axi4lite

Port Name	Direction	Description
s0_axi4lite_awaddr	Input	Axi-Lite Write Address, to primary IOSSM.
s0_axi4lite_awprot	Input	Axi-Lite Write Address Protection Signal, to primary IOSSM.
s0_axi4lite_awvalid	Input	Axi-Lite Write Address Valid, to primary IOSSM.
s0_axi4lite_awready	Output	Axi-Lite Write Address Ready, to primary IOSSM.
s0_axi4lite_araddr	Input	Axi-Lite Read Address, to primary IOSSM.
s0_axi4lite_arprot	Input	Axi-Lite Read Address Protection Signal, to primary IOSSM.
s0_axi4lite_arvalid	Input	Axi-Lite Read Address Valid, to primary IOSSM.
s0_axi4lite_arready	Output	Axi-Lite Read Address Ready, to primary IOSSM.
s0_axi4lite_wdata	Input	Axi-Lite Write Data, to primary IOSSM.
s0_axi4lite_wstrb	Input	Axi-Lite Write Strobe, to primary IOSSM.
s0_axi4lite_wvalid	Input	Axi-Lite Write Valid, to primary IOSSM.
s0_axi4lite_wready	Output	Axi-Lite Write Ready, to primary IOSSM.
s0_axi4lite_bready	Input	Axi-Lite Write Response Ready, to primary IOSSM.
s0_axi4lite_bresp	Output	Axi-Lite Write Response, to primary IOSSM.
s0_axi4lite_bvalid	Output	Axi-Lite Write Response Valid, to primary IOSSM.
s0_axi4lite_rready	Input	Axi-Lite Read Ready, to primary IOSSM.
s0_axi4lite_rdata	Output	Axi-Lite Read Data, to primary IOSSM.
s0_axi4lite_rresp	Output	Axi-Lite Read Response, to primary IOSSM.
s0_axi4lite_rvalid	Output	Axi-Lite Read Valid, to primary IOSSM.

4.4.12. s1_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Clock for sideband interface (secondary I/O bank).

Table 87. Interface: s1_axi4lite_clock

Interface type: clock

Port Name	Direction	Description
s1_axi4lite_clock	Input	Axi-Lite clock, to secondary IOSSM.

4.4.13. s1_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Reset for sideband interface (secondary I/O bank).

Table 88. Interface: s1_axi4lite_reset_n

Interface type: reset

Port Name	Direction	Description
s1_axi4lite_reset_n	Input	Axi-Lite reset_n, to secondary IOSSM.

4.4.14. s1_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Sideband interface (secondary I/O bank) that will connect to the IOSSM, through a gearbox in the core.

Table 89. Interface: s1_axi4lite

Interface type: axi4lite

Port Name	Direction	Description
s1_axi4lite_awaddr	Input	Axi-Lite Write Address, to secondary IOSSM.
s1_axi4lite_awprot	Input	Axi-Lite Write Address Protection Signal, to secondary IOSSM.
s1_axi4lite_awvalid	Input	Axi-Lite Write Address Valid, to secondary IOSSM.
s1_axi4lite_awready	Output	Axi-Lite Write Address Ready, to secondary IOSSM.
s1_axi4lite_araddr	Input	Axi-Lite Read Address, to secondary IOSSM.
s1_axi4lite_arprot	Input	Axi-Lite Read Address Protection Signal, to secondary IOSSM.
s1_axi4lite_arvalid	Input	Axi-Lite Read Address Valid, to secondary IOSSM.
s1_axi4lite_arready	Output	Axi-Lite Read Address Ready, to secondary IOSSM.
s1_axi4lite_wdata	Input	Axi-Lite Write Data, to secondary IOSSM.
s1_axi4lite_wstrb	Input	Axi-Lite Write Strobe, to secondary IOSSM.
s1_axi4lite_wvalid	Input	Axi-Lite Write Valid, to secondary IOSSM.
s1_axi4lite_wready	Output	Axi-Lite Write Ready, to secondary IOSSM.
s1_axi4lite_bready	Input	Axi-Lite Write Response Ready, to secondary IOSSM.
s1_axi4lite_bresp	Output	Axi-Lite Write Response, to secondary IOSSM.
s1_axi4lite_bvalid	Output	Axi-Lite Write Response Valid, to secondary IOSSM.

continued...

Port Name	Direction	Description
s1_axi4lite_rready	Input	Axi-Lite Read Ready, to secondary IOSSM.
s1_axi4lite_rdata	Output	Axi-Lite Read Data, to secondary IOSSM.
s1_axi4lite_rresp	Output	Axi-Lite Read Response, to secondary IOSSM.
s1_axi4lite_rvalid	Output	Axi-Lite Read Valid, to secondary IOSSM.

4.4.15. mem_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Interface to the memory (channel 0), including all CA pins, DQ pins, and DQS pins.

Table 90. Interface: mem_0

Interface type: conduit

Port Name	Direction	Description
mem_0_cs_n	Output	Chip Select channel 0.
mem_0_ca	Output	Command/Address Bus channel 0.
mem_0_dq	Bidir	Data (read/write) channel 0.
mem_0_dqs_t	Bidir	Data Strobe (true) channel 0.
mem_0_dqs_c	Bidir	Data Strobe (complement) channel 0.
mem_0_alert_n	Input	Indicates Write CRC Error channel 0.

4.4.16. mem_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Interface to the memory (channel 1), including all CA pins, DQ pins, and DQS pins.

Table 91. Interface: mem_1

Interface type: conduit

Port Name	Direction	Description
mem_1_cs_n	Output	Chip Select channel 1.
mem_1_ca	Output	Command/Address Bus channel 1.
mem_1_dq	Bidir	Data (read/write) channel 1.
mem_1_dqs_t	Bidir	Data Strobe (true) channel 1.
mem_1_dqs_c	Bidir	Data Strobe (complement) channel 1.
mem_1_alert_n	Input	Indicates Write CRC Error channel 1.

4.4.17. mem_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Reset pin to the memory. Must always be placed along with channel 0, but shared for entire interface (all channels within one EMIF).

Table 92. Interface: mem_reset_n

Interface type: conduit

Port Name	Direction	Description
mem_0_reset_n	Output	Asynchronous Reset channel 0.

4.4.18. mem_ck_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Clock pin to the memory (channel 0).

Table 93. Interface: mem_ck_0

Interface type: conduit

Port Name	Direction	Description
mem_0_ck_t	Output	CK Clock (true) channel 0.
mem_0_ck_c	Output	CK Clock (complement) channel 0.

4.4.19. mem_ck_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Clock pin to the memory (channel 1).

Table 94. Interface: mem_ck_1

Interface type: conduit

Port Name	Direction	Description
mem_1_ck_t	Output	CK Clock (true) channel 1.
mem_1_ck_c	Output	CK Clock (complement) channel 1.

4.4.20. mem_i3c for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

I3C to the DDR5 DIMM.

Table 95. Interface: mem_i3c

Interface type: conduit

Port Name	Direction	Description
i3c_scl_0	Output	I3C Serial Clock - to DDR5 DIMM.
i3c_sda_0	Bidir	I3C Serial Data - to DDR5 DIMM.

4.4.21. mem_lb_dq for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

LBDQ going to DDR5 DIMMs; RDIMM only. Always placed along with channel 0.

Table 96. Interface: mem_lb_dq

Interface type: conduit

Port Name	Direction	Description
mem_lbd_0	Input	Loopback data input pin.

4.4.22. mem_lb_dqs for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

LBDQS going to DDR5 DIMMs; RDIMM only. Always placed along with channel 1.

Table 97. Interface: mem_lb_dqs

Interface type: conduit

Port Name	Direction	Description
mem_lbs_0	Input	Loopback strobe input pin.

4.4.23. oct_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

On-Chip Termination (OCT) interface, representing RZQ pin (channel 0).

Table 98. Interface: oct_0

Interface type: conduit

Port Name	Direction	Description
oct_rzqin_0	Input	Calibrated On-Chip Termination (OCT) input pin channel 0.

4.4.24. oct_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

On-Chip Termination (OCT) interface, representing RZQ pin (channel 1).

Table 99. Interface: oct_1

Interface type: conduit

Port Name	Direction	Description
oct_rzqin_1	Input	Calibrated On-Chip Termination (OCT) input pin channel 1.

4.4.25. ref_clk for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - DDR5 DIMM

Reference clock used by the EMIF PLL.

Table 100. Interface: ref_clk

Interface type: clock

Port Name	Direction	Description
ref_clk	Input	PLL reference clock input.

4.5. IP Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

The interfaces in the Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5 each have signals that can be connected in Platform Designer. The following table lists the interfaces and corresponding interface types.

Table 101. Interfaces for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Interface Name	Interface Type	Description
s0_axi4_clock_in	clock	Input user clock for mainband; for MAINBAND_ACCESS_MODE = ASYNC only.
core_init_n	reset	An input to indicate that core configuration is complete.
s0_axi4_ctrl_ready	reset	Reset for mainband, from primary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.
s0_axi4_clock_out	clock	Output user clock for mainband (from CPA of primary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.
s1_axi4_ctrl_ready	reset	Reset for mainband, from secondary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.
s1_axi4_clock_out	clock	Output user clock for mainband (from CPA of secondary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.
s0_axi4	axi4	Mainband AXI4 from fabric to controller, channel 0.
s1_axi4	axi4	Mainband AXI4 from fabric to controller, channel 1.
s2_axi4	axi4	Mainband AXI4 from fabric to controller, channel 2.
s3_axi4	axi4	Mainband AXI4 from fabric to controller, channel 3.
s0_axi4lite_clock	clock	Clock for sideband interface (primary I/O bank).
s0_axi4lite_reset_n	reset	Reset for sideband interface (primary I/O bank).
s0_axi4lite	axi4lite	Sideband interface (primary I/O bank) that will connect to the IOSSM, through a gearbox in the core.
s1_axi4lite_clock	clock	Clock for sideband interface (secondary I/O bank).
s1_axi4lite_reset_n	reset	Reset for sideband interface (secondary I/O bank).
s1_axi4lite	axi4lite	Sideband interface (secondary I/O bank) that will connect to the IOSSM, through a gearbox in the core.
mem_0	conduit	Interface to the memory (channel 0), including all CA pins, DQ pins, and DQS pins.
mem_ck_0	conduit	Clock pin to the memory (channel 0).

continued...

Interface Name	Interface Type	Description
mem_1	conduit	Interface to the memory (channel 1), including all CA pins, DQ pins, and DQS pins.
mem_ck_1	conduit	Clock pin to the memory (channel 1).
mem_2	conduit	Interface to the memory (channel 2), including all CA pins, DQ pins, and DQS pins.
mem_ck_2	conduit	Clock pin to the memory (channel 2).
mem_3	conduit	Interface to the memory (channel 3), including all CA pins, DQ pins, and DQS pins.
mem_ck_3	conduit	Clock pin to the memory (channel 3).
mem_reset_n	conduit	Reset pin to the memory. Must always be placed along with channel 0, but shared for entire interface (all channels within one EMIF).
oct_0	conduit	On-Chip Termination (OCT) interface, representing RZQ pin (channel 0).
oct_1	conduit	On-Chip Termination (OCT) interface, representing RZQ pin (channel 1).
oct_2	conduit	On-Chip Termination (OCT) interface, representing RZQ pin (channel 2).
oct_3	conduit	On-Chip Termination (OCT) interface, representing RZQ pin (channel 3).
ref_clk	clock	Reference clock used by the EMIF PLL.

4.5.1. s0_axi4_clock_in for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Input user clock for mainband; for MAINBAND_ACCESS_MODE = ASYNC only.

Table 102. Interface: s0_axi4_clock_in

Interface type: clock

Port Name	Direction	Description
s0_axi4_clock_in	Input	User clock for mainband axi. Input clock to the EMIF IP, no relationship to PHY clock.

4.5.2. core_init_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

An input to indicate that core configuration is complete.

Table 103. Interface: core_init_n

Interface type: reset

Port Name	Direction	Description
core_init_n	Input	Core init signal going into EMIF. Used to generate the reset signal on the core-EMIF interface in fabric modes. When high, indicates core initialization is complete.

4.5.3. s0_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Reset for mainband, from primary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.

Table 104. Interface: s0_axi4_ctrl_ready

Interface type: reset

Port Name	Direction	Description
s0_axi4_reset_n	Output	Output signal from EMIF IP (primary I/O bank), indicating that Calibration of the channels in this I/O bank is complete, and controllers in this I/O bank are ready for use.

4.5.4. s0_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Output user clock for mainband (from CPA of primary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.

Table 105. Interface: s0_axi4_clock_out

Interface type: clock

Port Name	Direction	Description
s0_axi4_clock_out	Output	User clock for maiband axi (primary I/O bank). Output clock from the EMIF IP (output from CPA block, synchronous to PHY clock).

4.5.5. s1_axi4_ctrl_ready for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Reset for mainband, from secondary I/O bank, indicating the calibration is complete. Only available if mainband is accessed through fabric.

Table 106. Interface: s1_axi4_ctrl_ready

Interface type: reset

Port Name	Direction	Description
s1_axi4_reset_n	Output	Output signal from EMIF IP (secondary I/O bank), indicating that Calibration of the channels in this I/O bank is complete, and controllers in this I/O bank are ready for use.

4.5.6. s1_axi4_clock_out for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Output user clock for mainband (from CPA of secondary I/O bank); for MAINBAND_ACCESS_MODE = SYNC only.

Table 107. Interface: s1_axi4_clock_out

Interface type: clock

Port Name	Direction	Description
s1_axi4_clock_out	Output	User clock for maiband axi (secondary I/O bank). Output clock from the EMIF IP (output from CPA block, synchronous to PHY clock).

4.5.7. s0_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Mainband AXI4 from fabric to controller, channel 0.

Table 108. Interface: s0_axi4

Interface type: axi4

Port Name	Direction	Description
Write Address (Command) Channel		
s0_axi4_awaddr	input	Write address
s0_axi4_awburst	input	Write burst type. 'b00 = Reserved (FIXED is not supported). 'b01 = INCR. 'b10 = WRAP. 'b11 = Reserved.
s0_axi4_awid	input	Write address ID
s0_axi4_awlen	input	Write burst length. Any value between 0 and 255 is valid, representing a transfer of 1 to 256 beats.
s0_axi4_awlock	input	Write lock type. This 2-bit signal is used to control exclusive accesses and locking. 'b0 = Normal Access. 'b1 = Exclusive Access.
s0_axi4_awqos	input	Write quality of service. Supported priority values range from 0 to 3, with 0 as the lowest priority. Only bits[3:2] are used, bits[1:0] are ignored.
s0_axi4_awsz	input	Write burst size. AWSIZE = 5 (32 bytes) is supported by the memory controller when the AXI port is 256 bits, and only AWSIZE = 4 (16 bytes) is supported by the memory controller when the AXI port is 128 bits.
s0_axi4_awvalid	input	Write address valid
s0_axi4_awuser	input	Write address user signal. [0]: Enable/Disable Auto-precharge. Drive 1 to enable Auto-precharge. An auto-precharge will be issued after the write command is completed. [1]: ALLSTRB: When all write strobes are driven (no byte enable signals not asserted), this signal can be enabled to improve controller performance. [13:2]: Not connected. Drive 0.
s0_axi4_awprot	input	Write protection type. This 2-bit signal is used to control privileged and secure accesses.
<i>continued...</i>		

Port Name	Direction	Description
		'b00 = Non-Privileged & Secure Access. 'b01 = Privileged & Secure Access. 'b10 = Non-Privileged & Non-Secure. 'b11 = Privileged & Non-Secure.
s0_axi4_awready	output	Write address ready
Write Data Channel		
s0_axi4_wdata	input	Write data
s0_axi4_wlast	input	Write last. This signal indicates the last transfer in a write burst.
s0_axi4_wready	output	Write ready. Indicates that the AXI port is ready to accept write data.
s0_axi4_wstrb	input	Write strobes
s0_axi4_wuser	input	Write user signal. Only applicable to the x40/x72 lockstep cases. The additional user bits to be written are sent on this interface. If a x36 interface is used, then only the lowest 32-bits are connected.
s0_axi4_wvalid	input	Write valid
Write Response Channel		
s0_axi4_bready	input	Response ready
s0_axi4_bid	output	Write response ID
s0_axi4_bresp	output	Write response. A response is sent for the entire burst. 'b00 = OKAY. Write command was successfully processed, or exclusive write command was not processed as exclusive. 'b01 = EXOKAY. Exclusive write command was successfully processed. 'b10 = SLVERR. Slave has received the read write command but there is an error in the transaction. 'b11 = DECERR. Slave does not exist and/or there is an error with the transaction.
s0_axi4_bvalid	output	Write response valid.
Read Address (Command) Channel		
s0_axi4_araddr	input	Read address.
s0_axi4_arburst	input	Read burst type. 'b00 = Reserved (FIXED is not supported). 'b01 = INCR. 'b10 = WRAP. 'b11 = Reserved.
s0_axi4_arid	input	Read write address ID
s0_axi4_arlen	input	Read burst length. Any value between 0 and 128 255 is valid, representing a transfer of 1 to 256 beats.
s0_axi4_arlock	input	Read lock type. This 2-bit signal is used to control exclusive accesses and locking. 'b0 = Normal Access. 'b1 = Exclusive Access.
<i>continued...</i>		

Port Name	Direction	Description
s0_axi4_arqos	input	Read quality of service Supported priority values range from 0 to 3, with 0 as the lowest priority. Only bits[3:2] are used, bit[1:0] are ignored.
s0_axi4_arsize	input	Read burst size. AWSIZE = 5 (32 bytes) is supported by the memory controller when the AXI port is 256 bits, and only AWSIZE = 4 (16 bytes) is supported by the memory controller when the AXI port is 128 bits.
s0_axi4_arvalid	input	Read address valid.
s0_axi4_aruser	inout	Read address user signal. [0]: Enable/Disable Auto-precharge. Drive 1 to enable auto-precharge. An auto-precharge will be issued after the read command is completed. [13:1]: Not connected. Drive 0.
s0_axi4_arprot	input	Read protection type. This 2-bit signal is used to control privileged and secure accesses. 'b00 = Non-Privileged & Secure Access. 'b01 = Privileged & Secure Access. 'b10 = Non-Privileged & Non-Secure. 'b11 = Privileged & Non-Secure.
s0_axi4_arready	output	Read address ready
Ready Data Channel		
s0_axi4_rdata	output	Read data
s0_axi4_rid	output	Read ID
s0_axi4_rlast	output	Read last. This signal indicates the last transfer in a read burst.
s0_axi4_rready	input	Read ready
s0_axi4_rresp	output	read response. A response is sent with each burst, indicating the status of that burst. 'b00 = OKAY. Read command was successfully processed, or exclusive read command was not processed as exclusive. 'b01 = EXOKAY. Exclusive read command was successfully processed. 'b10 = SLVERR. Slave has received the read command but there is an error in the transaction. 'b11 = DECERR. Slave does not exist and/or there is an error with the transaction.
s0_axi4_ruser	output	Read user signal. Only applicable to the x40/x72 lockstep cases. These are the additional user bits received on this interface. If a x36 interface is used, then only the lowest 32-bits are connected.
s0_axi4_rvalid	output	Read valid.

4.5.8. s1_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Mainband AXI4 from fabric to controller, channel 1.

Table 110. Interface: s1_axi4

Interface type: axi4

Port Name	Direction	Description
s1_axi4_awaddr	Input	Write Address , channel 1.
s1_axi4_awburst	Input	Write Burst Type, channel 1.
s1_axi4_awid	Input	Write Address ID, channel 1.
s1_axi4_awlen	Input	Write Burst Length, channel 1.
s1_axi4_awlock	Input	Write Lock Type, channel 1.
s1_axi4_awqos	Input	Write Quality of Service, channel 1.
s1_axi4_awsz	Input	Write Burst Size, channel 1.
s1_axi4_awvalid	Input	Write Address Valid, channel 1.
s1_axi4_awuser	Input	Write Address User Signal, channel 1.
s1_axi4_awprot	Input	Write Protection Type, channel 1.
s1_axi4_awready	Output	Write Address Ready, channel 1.
s1_axi4_araddr	Input	Read Address , channel 1.
s1_axi4_arburst	Input	Read Burst Type, channel 1.
s1_axi4_arid	Input	Read Address ID, channel 1.
s1_axi4_arlen	Input	Read Burst Length, channel 1.
s1_axi4_arlock	Input	Read Lock Type, channel 1.
s1_axi4_arqos	Input	Read Quality of Service, channel 1.
s1_axi4_arsz	Input	Read Burst Size, channel 1.
s1_axi4_arvalid	Input	Read Address Valid, channel 1.
s1_axi4_aruser	Input	Read Address User Signal, channel 1.
s1_axi4_arprot	Input	Read Protection Type, channel 1.
s1_axi4_arready	Output	Read Address Ready, channel 1.
s1_axi4_wdata	Input	Write Data , channel 1.
s1_axi4_wstrb	Input	Write Strobes, channel 1.
s1_axi4_wlast	Input	Write Last, channel 1.
s1_axi4_wvalid	Input	Write Valid, channel 1.
s1_axi4_wready	Output	Write Ready, channel 1.
s1_axi4_bready	Input	Write Response Ready, channel 1.
s1_axi4_bid	Output	Write Response ID, channel 1.
s1_axi4_bresp	Output	Write Response , channel 1.
s1_axi4_bvalid	Output	Write Response Valid, channel 1.
s1_axi4_rready	Input	Read Ready, channel 1.
s1_axi4_rdata	Output	Read Data, channel 1.
s1_axi4_rid	Output	Read ID , channel 1.

continued...

Port Name	Direction	Description
s1_axi4_rlast	Output	Read Last, channel 1.
s1_axi4_rresp	Output	Read Response, channel 1.
s1_axi4_rvalid	Output	Read Valid, channel 1.

4.5.9. s2_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Mainband AXI4 from fabric to controller, channel 2.

Table 111. Interface: s2_axi4

Interface type: axi4

Port Name	Direction	Description
s2_axi4_awaddr	Input	Write Address , channel 2.
s2_axi4_awburst	Input	Write Burst Type, channel 2.
s2_axi4_awid	Input	Write Address ID, channel 2.
s2_axi4_awlen	Input	Write Burst Length, channel 2.
s2_axi4_awlock	Input	Write Lock Type, channel 2.
s2_axi4_awqos	Input	Write Quality of Service, channel 2.
s2_axi4_awsiz	Input	Write Burst Size, channel 2.
s2_axi4_awvalid	Input	Write Address Valid, channel 2.
s2_axi4_awuser	Input	Write Address User Signal, channel 2.
s2_axi4_awprot	Input	Write Protection Type, channel 2.
s2_axi4_awready	Output	Write Address Ready, channel 2.
s2_axi4_araddr	Input	Read Address , channel 2.
s2_axi4_arburst	Input	Read Burst Type, channel 2.
s2_axi4_arid	Input	Read Address ID, channel 2.
s2_axi4_arlen	Input	Read Burst Length, channel 2.
s2_axi4_arlock	Input	Read Lock Type, channel 2.
s2_axi4_arqos	Input	Read Quality of Service, channel 2.
s2_axi4_arsiz	Input	Read Burst Size, channel 2.
s2_axi4_arvalid	Input	Read Address Valid, channel 2.
s2_axi4_aruser	Input	Read Address User Signal, channel 2.
s2_axi4_arprot	Input	Read Protection Type, channel 2.
s2_axi4_arready	Output	Read Address Ready, channel 2.
s2_axi4_wdata	Input	Write Data , channel 2.
s2_axi4_wstrb	Input	Write Stobes, channel 2.
s2_axi4_wlast	Input	Write Last, channel 2.
continued...		

Port Name	Direction	Description
s2_axi4_wvalid	Input	Write Valid, channel 2.
s2_axi4_wready	Output	Write Ready, channel 2.
s2_axi4_bready	Input	Write Response Ready, channel 2.
s2_axi4_bid	Output	Write Response ID, channel 2.
s2_axi4_bresp	Output	Write Response , channel 2.
s2_axi4_bvalid	Output	Write Response Valid, channel 2.
s2_axi4_rready	Input	Read Ready, channel 2.
s2_axi4_rdata	Output	Read Data, channel 2.
s2_axi4_rid	Output	Read ID , channel 2.
s2_axi4_rlast	Output	Read Last, channel 2.
s2_axi4_rresp	Output	Read Response, channel 2.
s2_axi4_rvalid	Output	Read Valid, channel 2.

4.5.10. s3_axi4 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Mainband AXI4 from fabric to controller, channel 3.

Table 112. Interface: s3_axi4

Interface type: axi4

Port Name	Direction	Description
s3_axi4_awaddr	Input	Write Address , channel 3.
s3_axi4_awburst	Input	Write Burst Type, channel 3.
s3_axi4_awid	Input	Write Address ID, channel 3.
s3_axi4_awlen	Input	Write Burst Length, channel 3.
s3_axi4_awlock	Input	Write Lock Type, channel 3.
s3_axi4_awqos	Input	Write Quality of Service, channel 3.
s3_axi4_awsz	Input	Write Burst Size, channel 3.
s3_axi4_awvalid	Input	Write Address Valid, channel 3.
s3_axi4_awuser	Input	Write Address User Signal, channel 3.
s3_axi4_awprot	Input	Write Protection Type, channel 3.
s3_axi4_awready	Output	Write Address Ready, channel 3.
s3_axi4_araddr	Input	Read Address , channel 3.
s3_axi4_arburst	Input	Read Burst Type, channel 3.
s3_axi4_arid	Input	Read Address ID, channel 3.
s3_axi4_arlen	Input	Read Burst Length, channel 3.
s3_axi4_arlock	Input	Read Lock Type, channel 3.
continued...		

Port Name	Direction	Description
s3_axi4_arqos	Input	Read Quality of Service, channel 3.
s3_axi4_arsize	Input	Read Burst Size, channel 3.
s3_axi4_arvalid	Input	Read Address Valid, channel 3.
s3_axi4_aruser	Input	Read Address User Signal, channel 3.
s3_axi4_arprot	Input	Read Protection Type, channel 3.
s3_axi4_arready	Output	Read Address Ready, channel 3.
s3_axi4_wdata	Input	Write Data , channel 3.
s3_axi4_wstrb	Input	Write Strobes, channel 3.
s3_axi4_wlast	Input	Write Last, channel 3.
s3_axi4_wvalid	Input	Write Valid, channel 3.
s3_axi4_wready	Output	Write Ready, channel 3.
s3_axi4_bready	Input	Write Response Ready, channel 3.
s3_axi4_bid	Output	Write Response ID, channel 3.
s3_axi4_bresp	Output	Write Response , channel 3.
s3_axi4_bvalid	Output	Write Response Valid, channel 3.
s3_axi4_rready	Input	Read Ready, channel 3.
s3_axi4_rdata	Output	Read Data, channel 3.
s3_axi4_rid	Output	Read ID , channel 3.
s3_axi4_rlast	Output	Read Last, channel 3.
s3_axi4_rresp	Output	Read Response, channel 3.
s3_axi4_rvalid	Output	Read Valid, channel 3.

4.5.11. s0_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Clock for sideband interface (primary I/O bank).

Table 113. Interface: s0_axi4lite_clock

Interface type: clock

Port Name	Direction	Description
s0_axi4lite_clock	Input	Axi-Lite clock, to primary IOSSM.

4.5.12. s0_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Reset for sideband interface (primary I/O bank).

Table 114. Interface: s0_axi4lite_reset_n

Interface type: reset

Port Name	Direction	Description
s0_axi4lite_reset_n	Input	Axi-Lite reset_n, to primary IOSSM.

4.5.13. s0_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Sideband interface (primary I/O bank) that will connect to the IOSSM, through a gearbox in the core.

Table 115. Interface: s0_axi4lite

Interface type: axi4lite

Port Name	Direction	Description
s0_axi4lite_awaddr	Input	Axi-Lite Write Address, to primary IOSSM.
s0_axi4lite_awprot	Input	Axi-Lite Write Address Protection Signal, to primary IOSSM.
s0_axi4lite_awvalid	Input	Axi-Lite Write Address Valid, to primary IOSSM.
s0_axi4lite_awready	Output	Axi-Lite Write Address Ready, to primary IOSSM.
s0_axi4lite_araddr	Input	Axi-Lite Read Address, to primary IOSSM.
s0_axi4lite_arprot	Input	Axi-Lite Read Address Protection Signal, to primary IOSSM.
s0_axi4lite_arvalid	Input	Axi-Lite Read Address Valid, to primary IOSSM.
s0_axi4lite_arready	Output	Axi-Lite Read Address Ready, to primary IOSSM.
s0_axi4lite_wdata	Input	Axi-Lite Write Data, to primary IOSSM.
s0_axi4lite_wstrb	Input	Axi-Lite Write Strobe, to primary IOSSM.
s0_axi4lite_wvalid	Input	Axi-Lite Write Valid, to primary IOSSM.
s0_axi4lite_wready	Output	Axi-Lite Write Ready, to primary IOSSM.
s0_axi4lite_bready	Input	Axi-Lite Write Response Ready, to primary IOSSM.
s0_axi4lite_bresp	Output	Axi-Lite Write Response, to primary IOSSM.
s0_axi4lite_bvalid	Output	Axi-Lite Write Response Valid, to primary IOSSM.
s0_axi4lite_rready	Input	Axi-Lite Read Ready, to primary IOSSM.
s0_axi4lite_rdata	Output	Axi-Lite Read Data, to primary IOSSM.
s0_axi4lite_rresp	Output	Axi-Lite Read Response, to primary IOSSM.
s0_axi4lite_rvalid	Output	Axi-Lite Read Valid, to primary IOSSM.

4.5.14. s1_axi4lite_clock for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Clock for sideband interface (secondary I/O bank).

Table 116. Interface: s1_axi4lite_clock

Interface type: clock

Port Name	Direction	Description
s1_axi4lite_clock	Input	Axi-Lite clock, to secondary IOSSM.

4.5.15. s1_axi4lite_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Reset for sideband interface (secondary I/O bank).

Table 117. Interface: s1_axi4lite_reset_n

Interface type: reset

Port Name	Direction	Description
s1_axi4lite_reset_n	Input	Axi-Lite reset_n, to secondary IOSSM.

4.5.16. s1_axi4lite for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Sideband interface (secondary I/O bank) that will connect to the IOSSM, through a gearbox in the core.

Table 118. Interface: s1_axi4lite

Interface type: axi4lite

Port Name	Direction	Description
s1_axi4lite_awaddr	Input	Axi-Lite Write Address, to secondary IOSSM.
s1_axi4lite_awprot	Input	Axi-Lite Write Address Protection Signal, to secondary IOSSM.
s1_axi4lite_awvalid	Input	Axi-Lite Write Address Valid, to secondary IOSSM.
s1_axi4lite_awready	Output	Axi-Lite Write Address Ready, to secondary IOSSM.
s1_axi4lite_araddr	Input	Axi-Lite Read Address, to secondary IOSSM.
s1_axi4lite_arprot	Input	Axi-Lite Read Address Protection Signal, to secondary IOSSM.
s1_axi4lite_arvalid	Input	Axi-Lite Read Address Valid, to secondary IOSSM.
s1_axi4lite_arready	Output	Axi-Lite Read Address Ready, to secondary IOSSM.
s1_axi4lite_wdata	Input	Axi-Lite Write Data, to secondary IOSSM.
s1_axi4lite_wstrb	Input	Axi-Lite Write Strobe, to secondary IOSSM.
s1_axi4lite_wvalid	Input	Axi-Lite Write Valid, to secondary IOSSM.
s1_axi4lite_wready	Output	Axi-Lite Write Ready, to secondary IOSSM.
s1_axi4lite_bready	Input	Axi-Lite Write Response Ready, to secondary IOSSM.
s1_axi4lite_bresp	Output	Axi-Lite Write Response, to secondary IOSSM.
s1_axi4lite_bvalid	Output	Axi-Lite Write Response Valid, to secondary IOSSM.

continued...

Port Name	Direction	Description
s1_axi4lite_rready	Input	Axi-Lite Read Ready, to secondary IOSSM.
s1_axi4lite_rdata	Output	Axi-Lite Read Data, to secondary IOSSM.
s1_axi4lite_rresp	Output	Axi-Lite Read Response, to secondary IOSSM.
s1_axi4lite_rvalid	Output	Axi-Lite Read Valid, to secondary IOSSM.

4.5.17. mem_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Interface to the memory (channel 0), including all CA pins, DQ pins, and DQS pins.

Table 119. Interface: mem_0

Interface type: conduit

Port Name	Direction	Description
mem_0_cs	Output	Chip Select channel 0.
mem_0_ca	Output	Command/Address Bus channel 0.
mem_0_dq	Bidir	Data (read/write) channel 0.
mem_0_rdqs_t	Bidir	Read Data Strobe (true) channel 0.
mem_0_rdqs_c	Bidir	Read Data Strobe (complement) channel 0.
mem_0_dmi	Bidir	Data Mask/Data Inversion channel 0.
mem_0_wck_t	Output	Write Clock (true) channel 0.
mem_0_wck_c	Output	Write Clock (complement) channel 0.

4.5.18. mem_ck_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Clock pin to the memory (channel 0).

Table 120. Interface: mem_ck_0

Interface type: conduit

Port Name	Direction	Description
mem_0_ck_t	Output	CK Clock (true) channel 0.
mem_0_ck_c	Output	CK Clock (complement) channel 0.

4.5.19. mem_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Interface to the memory (channel 1), including all CA pins, DQ pins, and DQS pins.

Table 121. Interface: mem_1

Interface type: conduit

Port Name	Direction	Description
mem_1_cs	Output	Chip Select channel 1.
mem_1_ca	Output	Command/Address Bus channel 1.
mem_1_dq	Bidir	Data (read/write) channel 1.
mem_1_rdqs_t	Bidir	Read Data Strobe (true) channel 1.
mem_1_rdqs_c	Bidir	Read Data Strobe (complement) channel 1.
mem_1_dmi	Bidir	Data Mask/Data Inversion channel 1.
mem_1_wck_t	Output	Write Clock (true) channel 1.
mem_1_wck_c	Output	Write Clock (complement) channel 1.

4.5.20. mem_ck_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Clock pin to the memory (channel 1).

Table 122. Interface: mem_ck_1

Interface type: conduit

Port Name	Direction	Description
mem_1_ck_t	Output	CK Clock (true) channel 1.
mem_1_ck_c	Output	CK Clock (complement) channel 1.

4.5.21. mem_2 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Interface to the memory (channel 2), including all CA pins, DQ pins, and DQS pins.

Table 123. Interface: mem_2

Interface type: conduit

Port Name	Direction	Description
mem_2_cs	Output	Chip Select channel 2.
mem_2_ca	Output	Command/Address Bus channel 2.
mem_2_dq	Bidir	Data (read/write) channel 2.
mem_2_rdqs_t	Bidir	Read Data Strobe (true) channel 2.
mem_2_rdqs_c	Bidir	Read Data Strobe (complement) channel 2.
mem_2_dmi	Bidir	Data Mask/Data Inversion channel 2.
mem_2_wck_t	Output	Write Clock (true) channel 2.
mem_2_wck_c	Output	Write Clock (complement) channel 2.

4.5.22. mem_ck_2 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Clock pin to the memory (channel 2).

Table 124. Interface: mem_ck_2

Interface type: conduit

Port Name	Direction	Description
mem_2_ck_t	Output	CK Clock (true) channel 2.
mem_2_ck_c	Output	CK Clock (complement) channel 2.

4.5.23. mem_3 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Interface to the memory (channel 3), including all CA pins, DQ pins, and DQS pins.

Table 125. Interface: mem_3

Interface type: conduit

Port Name	Direction	Description
mem_3_cs	Output	Chip Select channel 3.
mem_3_ca	Output	Command/Address Bus channel 3.
mem_3_dq	Bidir	Data (read/write) channel 3.
mem_3_rdqs_t	Bidir	Read Data Strobe (true) channel 3.
mem_3_rdqs_c	Bidir	Read Data Strobe (complement) channel 3.
mem_3_dmi	Bidir	Data Mask/Data Inversion channel 3.
mem_3_wck_t	Output	Write Clock (true) channel 3.
mem_3_wck_c	Output	Write Clock (complement) channel 3.

4.5.24. mem_ck_3 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Clock pin to the memory (channel 3).

Table 126. Interface: mem_ck_3

Interface type: conduit

Port Name	Direction	Description
mem_3_ck_t	Output	CK Clock (true) channel 3.
mem_3_ck_c	Output	CK Clock (complement) channel 3.

4.5.25. mem_reset_n for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Reset pin to the memory. Must always be placed along with channel 0, but shared for entire interface (all channels within one EMIF).

Table 127. Interface: mem_reset_n

Interface type: conduit

Port Name	Direction	Description
mem_0_reset_n	Output	Asynchronous Reset channel 0.

4.5.26. oct_0 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

On-Chip Termination (OCT) interface, representing RZQ pin (channel 0).

Table 128. Interface: oct_0

Interface type: conduit

Port Name	Direction	Description
oct_rzqin_0	Input	Calibrated On-Chip Termination (OCT) input pin channel 0.

4.5.27. oct_1 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

On-Chip Termination (OCT) interface, representing RZQ pin (channel 1).

Table 129. Interface: oct_1

Interface type: conduit

Port Name	Direction	Description
oct_rzqin_1	Input	Calibrated On-Chip Termination (OCT) input pin channel 1.

4.5.28. oct_2 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

On-Chip Termination (OCT) interface, representing RZQ pin (channel 2).

Table 130. Interface: oct_2

Interface type: conduit

Port Name	Direction	Description
oct_rzqin_2	Input	Calibrated On-Chip Termination (OCT) input pin channel 2.

4.5.29. oct_3 for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

On-Chip Termination (OCT) interface, representing RZQ pin (channel 3).

Table 131. Interface: oct_3

Interface type: conduit

Port Name	Direction	Description
oct_rzqin_3	Input	Calibrated On-Chip Termination (OCT) input pin channel 3.

4.5.30. ref_clk for Agilex 7 M-Series External Memory Interfaces (EMIF) IP - LPDDR5

Reference clock used by the EMIF PLL.

Table 132. Interface: ref_clk

Interface type: clock

Port Name	Direction	Description
ref_clk	Input	PLL reference clock input.

5. Agilex 7 M-Series FPGA EMIF IP – Simulating Memory IP

To simulate your design you require the following components:

- A simulator—The simulator must be an Altera-supported Verilog HDL simulator:
 - ☐ Siemens EDA* ModelSim
 - ☐ Siemens EDA QuestaSIM*
 - ☐ Synopsys* VCS/VCS-MX
- A design using Altera's External Memory Interface (EMIF) IP
- An example driver or traffic generator (to initiate read and write transactions)
- A testbench and a suitable memory simulation model

The Altera External Memory Interface IP is not compatible with the Platform Designer Testbench System. Instead, use the simulation design example from your generated IP to validate memory interface operation, or as a reference for creating a full simulatable design. The provided simulation design example contains the generated memory interface, a memory model, and a traffic generator. For more information about the EMIF simulation design example, refer to the *External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide*.

Memory Simulation Models

There are two types of memory simulation models that you can use:

- Altera-provided generic memory model
- Vendor-specific memory model

The Quartus Prime software generates the generic memory simulation model with the simulation design example. The model adheres to all the memory protocol specifications, and can be parameterized.

Vendor-specific memory models are simulation models for specific memory components from memory vendors such as Micron and Samsung. You can obtain these simulation models from the memory vendor's website.

Note: Altera does not provide support for vendor-specific memory models.

5.1. Simulation Walkthrough

Simulation is a good way to determine the latency of your system. However, the latency reflected in simulation may be different than the latency found on the board because functional simulation does not take into account board trace delays and different process, voltage, and temperature scenarios.

A given design may display different latency values on different boards, due to differences in board implementation.

The Agilex 7 M-Series EMIF IP supports functional simulation through the design example using the traffic generator IP.

To perform functional simulation for an Agilex 7 M-Series EMIF IP design example, locate the design example files in the design example directory.

You can use the IP functional simulation model with any supported VHDL or Verilog HDL simulator.

After you have generated the memory IP, you can locate multiple file sets for various supported simulations in the `sim/ed_sim` subdirectory. For more information about the EMIF simulation design example, refer to the *External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide*.

5.1.1. Calibration

Calibration occurs shortly after the memory device is initialized, to compensate for uncertainties in the hardware system, including silicon PVT variation, circuit board trace delays, and skewed arrival times. The Agilex 7 M-Series FPGA EMIF IP provides skip calibration mode for simulating the design example.

Skip Calibration Mode

In Skip Calibration mode, the calibration processor assumes an ideal hardware environment, where PVT variations, board delays, and trace skews are all zero. Instead of running the actual calibration routine, the calibration processor calculates the expected arrival time of read data based on the memory latency values entered during EMIF IP generation, resulting in reduced simulation time. Skip calibration mode is recommended for use during system development, because it allows you to focus on interacting with the controller and optimizing your memory access patterns, thus facilitating rapid RTL development.

If you enable Skip Calibration Mode, the interface still performs some memory initialization, sending DRAM Mode Register Set (MRS) commands, or commands to program register code words for RDIMM/LRDIMM, before starting normal operation. These initialization commands are necessary to set up the memory model operation and latencies.

5.1.2. Simulation Scripts

The Quartus Prime software generates simulation scripts during project generation for several different third party simulation tools—Cadence, Synopsys, and Siemens EDA.

The simulation scripts are located under the `sim/ed_sim` directory, in separate folders named after each supported simulator.

5.1.3. Functional Simulation with Verilog HDL

Simulation scripts for the Synopsys and Siemens EDA simulators are provided for you to run the design example.

The simulation scripts are located in the following main folder locations:

Simulation scripts in the simulation folders are located as follows:

- `sim\ed_sim\mentor\msim_setup.tcl`
- `sim\ed_sim\synopsys\vcs\vcs_setup.sh`
- `sim\ed_sim\synopsys\vcsmx\vcsmx_setup.sh`

For more information about simulating Verilog HDL or VHDL designs using command lines, refer to the *Questa - Intel FPGA Edition, ModelSim, and QuestaSim Simulator Support* chapter in the [Quartus Prime Pro Edition User Guide, Third-party Simulation](#).

5.1.4. Simulating the Design Example

This topic describes how to simulate the design example in Synopsys, and Siemens EDA simulators.

To simulate the example design in the Quartus Prime software using the Synopsys simulator, follow these steps:

1. At the Linux* shell command prompt, change directory to `sim\ed_sim\synopsys\vcsmx`
2. Run the simulation by typing the following command at the command prompt:

```
sh vcsmx_setup.sh
```

To simulate the example design in the Quartus Prime software using the Siemens EDA simulator, follow these steps:

1. At the Linux or Windows shell command prompt, change directory to `sim\ed_sim\mentor`
2. Execute the **msim_setup.tcl** script that automatically compiles and runs the simulation by typing the following command at the Linux or Windows command prompt:

```
vsim -do msim_setup.tcl
```

or

Type the following command at the ModelSim* command prompt:

```
source msim_setup.tcl
```

This command compiles all the design files, elaborates the top-level design, and runs the simulation.

If you require the `-voptargs=+acc` option, follow these steps:

- a. Open the `msim_setup.tcl` script in an editor.
- b. Go to the end of the file, and search for the `ld` command.
- c. Change the `ld` command to `ld_debug`, and run the command again. The end of the `cfile` should look like this:

```
ld_debug
run -all
```

For more information about simulating the external memory interface using the Siemens EDA simulator, refer to the *Simulating External Memory Interface IP With ModelSim* chapter in the *External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide*.

Note: Altera does not provide the `run.do` file for the example design with the EMIF interface.

For more information about simulation, refer to the *Quartus Prime Pro Edition User Guide, Third-party Simulation*.

If your Quartus Prime project appears to be configured correctly but the example testbench still fails, check the known issues on the Altera FPGA Knowledge Base before filing a service request.

5.1.5. Simulating with sim_lib2

The Quartus Prime software v4.0.0 (and later) includes a new simulation models library called `sim_lib2`, which provides improved simulation models.

All generated Agilex 7 M-Series EMIF IP design examples automatically select and use this library.

Designs created from scratch do not use the `sim_lib2` library. If you want to simulate, after generating the simulator setup scripts, perform one of the following steps, appropriate for your simulator:

ModelSim and QuestaSim

Locate the generated `../mentor/` directory in your project folder structure, then open the `run_msim_setup.tcl` file in an editor and add the following lines before the line that sources the `msim_setup.tcl` script:

```
set QUARTUS_INSTALL_DIR "<Quartus installation directory>"
set QUARTUS_SIM_LIB_DIR "$QUARTUS_INSTALL_DIR/eda/sim_lib2/"
set DEVICES_SIM_LIB_DIR "$QUARTUS_INSTALL_DIR/../devices/sim_lib2/"
```

The resulting file should look like the following example:

```
if ![info exists QSYS_SIMDIR] {
    set QSYS_SIMDIR "../.."
}

set QUARTUS_INSTALL_DIR "<Quartus installation directory>"
set QUARTUS_SIM_LIB_DIR "$QUARTUS_INSTALL_DIR/eda/sim_lib2/"
set DEVICES_SIM_LIB_DIR "$QUARTUS_INSTALL_DIR/../devices/sim_lib2/"

source $QSYS_SIMDIR/mentor/msim_setup.tcl
ld
run -all
quit
```

VCS and VCSMX

Pass the variables `DEVICES_SIM_LIB_DIR` and `QUARTUS_SIM_LIB_DIR` as command-line arguments when sourcing the generated simulation script `.../synopsys/vcs/vcs_setup.sh` or `.../synopsys/vcsmx/vcsmx_setup.sh`:

```
sh vcs_setup.sh <current arguments> DEVICES_SIM_LIB_DIR=<QPDS Installation>/
devices/sim_lib2
QUARTUS_SIM_LIB_DIR=<QPDS Installation>/quartus/eda/sim_lib2
```

Xcelium

Pass the variables `DEVICES_SIM_LIB_DIR` and `QUARTUS_SIM_LIB_DIR` as command-line arguments when sourcing Platform Designer generated simulation script `.../xcelium/xcelium_setup.sh`:

```
sh xcelium_setup.sh <current arguments> DEVICES_SIM_LIB_DIR=<QPDS
Installation>/devices/sim_lib2  QUARTUS_SIM_LIB_DIR=<QPDS Installation>/
quartus/eda/sim_lib2
```

6. Agilex 7 M-Series FPGA EMIF IP – DDR4 Support

This chapter contains IP parameter descriptions, pin planning information, and board design guidelines for Agilex 7 M-Series FPGA external memory interface IP - DDR4 Component, and Agilex 7 M-Series FPGA external memory interface IP - DDR4 DIMM.

6.1. Agilex 7 M-Series DDR4 Component External Memory Interfaces (EMIF) IP Parameter Descriptions

The following tables describe the parameters available on each tab of the IP parameter editor, which you can use to configure your IP.

Table 133. Group: High-level Configuration / Memory Device

Parameter Name	Description
Data DQ Width	Number of DQ pins per memory channel, used for data. Default value is 32 Legal values are: 16, 32, 40 (Identifier: MEM_CHANNEL_DATA_DQ_WIDTH)
ECC DQ Width	Number of additional DQ pins per memory channel, used for out-of-band ECC. If bigger than 0, controller will enable out-of-band ECC. Otherwise, out-of-band ECC will be disabled. Default value is 0 Legal values are: 0, 8 (Identifier: MEM_CHANNEL_ECC_DQ_WIDTH)
Die DQ Width	Number of DQ pins in each die that makes up the interface. For dual-die packages, this is the width of the die, not the width of full the package. Default value is 16 Legal values are: 8, 16 (Identifier: MEM_DIE_DQ_WIDTH)
Die Density	Capacity of each memory die (in Gbits), per channel per die. For dual-die packages, this is the density of each die, not the density of the full package. Default value is 8 Legal values are: • 2Gbits (2) • 4Gbits (4) • 8Gbits (8) • 16Gbits (16) (Identifier: MEM_DIE_DENSITY_GBITS)
CS Width	Specifies the total number of CS pins used by each channel. Default value is 1 Legal values are: 1, 2 (Identifier: MEM_CHANNEL_CS_WIDTH)
<i>continued...</i>	

Parameter Name	Description
Memory Speedbin	Specifies the speedbin of the memory device(s) of which the interface consists. Default value is 3200W Legal values are: • 1600J (CL10) (1600J) • 1600K (CL11) (1600K) • 1600L (CL12) (1600L) • 1866L (CL12) (1866L) • 1866M (CL13) (1866M) • 1866N (CL14) (1866N) • 2133N (CL14) (2133N) • 2133P (CL15) (2133P) • 2133R (CL16) (2133R) • 2400P (CL15) (2400P) • 2400R (CL16) (2400R) • 2400T (CL17) (2400T) • 2400U (CL18) (2400U) • 2666T (CL17) (2666T) • 2666U (CL18) (2666U) • 2666V (CL19) (2666V) • 2666W (CL20) (2666W) • 2933V (CL19) (2933V) • 2933W (CL20) (2933W) • 2933Y (CL21) (2933Y) • 2933AA (CL22) (2933AA) • 3200W (CL20) (3200W) • 3200AA (CL22) (3200AA) • 3200AC (CL24) (3200AC) (Identifier: MEM_SPEEDBIN)
Use AC Mirroring	Enable command-address mirroring for multi-rank DDR4 interfaces per JEDEC Standard. Default value is false (Identifier: MEM_AC_MIRRORING_EN)
Share CK Pins Between Ranks	Specifies whether all the ranks in the same channel should share one pair of memory interface differential clock. Default value is false (Identifier: MEM_RANKS_SHARE_CK_EN)
Use AC Parity	Specifies whether address-command parity is enabled. If enabled then command latency is increased by the value of parameter "Address-Command Latency Mode". Default value is false (Identifier: MEM_AC_PARITY_EN)
Auto-set Memory Operating Frequency	If true, let IP select max frequency that this configuration can support for the current device speedgrade. If false, user can set custom value for operating frequency. Default value is true (Identifier: MEM_OPERATING_FREQ_MHZ_AUTOSSET_EN)
Memory Operating Frequency	Specifies the frequency at which the memory interface will run. Value is specified in megahertz Legal values are: 666.667, 800, 933.333, 1066.667, 1200, 1333.333, 1466.667, 1600 (Identifier: MEM_OPERATING_FREQ_MHZ)

Table 134. Group: High-level Configuration / PHY

Parameter Name	Description
Auto-set PLL Reference Clock Frequency	If true, let IP select max PLL refclk frequency that this configuration can support. If false, user can set custom value for PLL refclk frequency. Default value is true (Identifier: PHY_REFCLK_FREQ_MHZ_AUTOSET_EN)
Enable Advanced List of PLL Reference Clock Frequencies	If true, provide extended list of possible refclk values. Otherwise, prune possible list of refclk values to a more reasonable length. Default value is false (Identifier: PHY_REFCLK_ADVANCED_SELECT_EN)
Reference Clock Frequency	Specifies the reference clock frequency for the EMIF IOPLL. Value is specified in megahertz (Identifier: PHY_REFCLK_FREQ_MHZ)
AC Placement	Indicates location on the device where the interface will reside (specifically, the location of the AC lanes in terms I/O BANK and TOP vs BOT part of the I/O BANK). Legal ranges are derived from device floorplan. Default value is BOT Legal values are: • AC Bottom Sub-bank (lanes 0-3) (BOT) • AC Top Sub-bank (lanes 4-7) (TOP) (Identifier: PHY_AC_PLACEMENT)
Alert_n AC-Lane Index	Specifies the AC lane index in which to place the ALERT_N pin. Default value is AC2 Legal values are: AC2, AC3 (Identifier: PHY_ALERT_N_PLACEMENT)
Force Using 4 AC Lanes	Specifies if the minimum number of AC lanes for the memory interface should be forced to 4. Default value is false (Identifier: PHY_FORCE_MIN_4_AC_LANES_EN)
Auto-set Mainband Access Mode	If true, let IP select most likely usecase for the PHY_MAINBAND_ACCESS_MODE; if false, let user set a custom value for sideband access mode. Default value is true (Identifier: PHY_MAINBAND_ACCESS_MODE_AUTOSET_EN)
Mainband Access Mode	Specifies the path through which the EMIF QHIP mainband interface is exposed to the user. The mainband interface is the AXI4 interface to the memory controller. Legal values are: • NoC (NOC) • Fabric Direct - User Clock Asynchronous to PHY (ASYNC) • Fabric Direct - User Clock Synchronous to PHY (SYNC) (Identifier: PHY_MAINBAND_ACCESS_MODE)
Auto-set Sideband Access Mode	If true, let IP select most likely usecase for the PHY_SIDEHAND_ACCESS_MODE; if false, let user set a custom value for sideband access mode. Default value is true (Identifier: PHY_SIDEHAND_ACCESS_MODE_AUTOSET_EN)
Sideband Access Mode	Specifies the path through which the EMIF QHIP sideband interface is exposed to the user. The sideband interface is the AXI4-Lite interface to the IOSSM.
<i>continued...</i>	

Parameter Name	Description
	Legal values are: • NoC (NOC) • Fabric Direct (FABRIC) (Identifier: PHY_SIDEHAND_ACCESS_MODE)
Pin Swizzle Map	Specifies the swizzle map for the data lanes and pins. (Identifier: PHY_SWIZZLE_MAP)
Use Debug Toolkit	If enabled, the AXI-L port will be connected to SLD nodes, allowing for a system-console avalon manager interface to interact with this AXI-L subordinate interface. Default value is false (Identifier: DEBUG_TOOLS_EN)
Instance ID	Instance ID of the EMIF IP. This is useful when using a discovery mechanism over the side-band interface, to identify which EMIF instance's mailbox is at which offset. If expecting to use a discovery mechanism in hardware, this parameter must be set uniquely for all EMIFs that share a sideband. Otherwise, this parameter can be ignored / kept at the default value. Default value is 0 Legal values are: from 0 to 6 (Identifier: INSTANCE_ID)

Table 135. Group: High-level Configuration / Controller / Controller Options

Parameter Name	Description
Use ECC Autocorrection	If ECC is enabled, specifies whether single-bit-errors (SBEs) should be corrected or just reported. Default value is true (Identifier: CTRL_ECC_AUTOCORRECT_EN)
Use Data Masking	Specifies whether Data Masking is enabled by the controller. RMW will occur under these conditions: AXI transactions using a subset of write strobe, or one beat long AXI transaction in DDR5, LPDDR5, or LPDDR4 interfaces. Altera recommends enabling the DM option to improve efficiency for the following cases: non-ECC design with AXI transaction listed above, and LPDDR4 or LPDDR5 designs with In-Line ECC enabled. Default value is false (Identifier: CTRL_DM_EN)
Use WDBI	Specifies whether write Data-bus-inversion is enabled by the controller. Default value is false (Identifier: CTRL_WR_DBI_EN)
Use RDBI	Specifies whether read Data-bus-inversion is enabled by the controller. Default value is false (Identifier: CTRL_RD_DBI_EN)
Controller Performance Profile	Group of Controller settings to optimize performance based on expected traffic patterns. Choose "Custom" for fine-granular control Default value is SEQ Legal values are: • Sequential Access Optimized (SEQ) • Random Access Optimized (RAND) • Custom (CUSTOM) (Identifier: CTRL_PERFORMANCE_PROFILE)
<i>continued...</i>	

Parameter Name	Description
Force Auto-Precharge	Controller will terminate all transactions with an auto-precharge when enabled. For per-transaction granularity, leave this option disabled and use the AxUSER control signals instead. Default value is false (Identifier: CTRL_AUTO_PRECHARGE_EN)
Bank Group Rotation	Moves a number of Bank Group pins lower in the address order Default value is 1 Legal values are: • No BGs (disabled) (0) • Two BGs (BG[0]) (1) • Four BGs (BG[1:0]) (2) (Identifier: CTRL_BG_ROTATE_EN)
Enable Row/CS Swap	Specifies whether the highest order address bits are the CS bit (false) or Row bits (true) Default value is false (Identifier: DIAG_HMC_ADDR_SWAP_EN)

Table 136. Group: High-level Configuration / Controller / Data Bus Turnaround Times

Parameter Name	Description
Additional read-to-write turnaround time (same rank)	Additional memory clock cycles of delay between read and write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2W_SAMECS_CYC)
Additional read-to-read turnaround time (same rank)	Additional memory clock cycles of delay between read operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2R_SAMECS_CYC)
Additional write-to-write turnaround time (same rank)	Additional memory clock cycles of delay between write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2W_SAMECS_CYC)
Additional write-to-read turnaround time (same rank)	Additional memory clock cycles of delay between read and write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2R_SAMECS_CYC)
Additional read-to-write turnaround time (different ranks)	Additional controller cycles of delay between read and write operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2W_DIFFCS_CYC)
<i>continued...</i>	

Parameter Name	Description
Additional read-to-read turnaround time (different ranks)	Additional controller cycles of delay between read operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2R_DIFFCS_CYC)
Additional write-to-write turnaround time (different ranks)	Additional controller cycles of delay between write operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2W_DIFFCS_CYC)
Additional write-to-read turnaround time (different ranks)	Additional controller cycles of delay between write and read operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2R_DIFFCS_CYC)

Table 137. Group: High-level Configuration / Advanced Calibration Settings

Parameter Name	Description
Enable Margining during Calibration	Enable margining stages of calibration for use with the EMIF Debug Toolkit. This will increase calibration time Default value is false (Identifier: ADV_CAL_ENABLE_MARGIN)
Enable REQ Calibration	Enable the Read DQ Equalization stage as part of calibration. This will increase calibration time but will also improve top-line performance Default value is false (Identifier: ADV_CAL_ENABLE_REQ)
Enable WEQ Calibration	Run the Write DQ Equalization stage as part of calibration. This will increase calibration time but will also improve top-line performance Default value is false (Identifier: ADV_CAL_ENABLE_WEQ)

Table 138. Group: Advanced: Memory Timing / Overrides / JEDEC_TABLE

Parameter Name	Description
JEDEC Parameter	Name of JEDEC Parameter to explicitly override; the values will be applied and appear in the list below. Default value is Legal values are: • Write Preamble Length (MEM_WR_PREAMBLE_MODE) • Read Preamble Length (MEM_RD_PREAMBLE_MODE) • Read Latency (MEM_CL_CYC) • Write Latency (MEM_CWL_CYC) • tREFI (MEM_TREFI_NS) • tRAS (MEM_TRAS_NS) • tRCD (MEM_TRCD_NS) • tRP (MEM_TRP_NS) • tRC (MEM_TRC_NS)

Parameter Name	Description
	• tCCD_L (MEM_TCCD_L_NS) • tCCD_S (MEM_TCCD_S_NS) • tRRD_L (MEM_TRRD_L_NS) • tRRD_S (MEM_TRRD_S_NS) • tFAW (MEM_TFAW_NS) • tWTR_L (MEM_TWTR_L_NS) • tWTR_S (MEM_TWTR_S_NS) • tWR (MEM_TWR_NS) • tMRD (MEM_TMRD_NS) • tCKSRE (MEM_TCKSRE_NS) • tCKSRX (MEM_TCKSRX_NS) • tCKE (MEM_TCKE_NS) • tCKESR (MEM_TCKESR_CYC) • tMPRR (MEM_TMPRR_NS) • tRFC (MEM_TRFC_NS) • tDQSCK (MEM_TDQSCK_NS) • tXP (MEM_TXP_NS) • tXS (MEM_TXS_NS) • tXSDLL (MEM_TXS_DLL_NS) • tCPDED (MEM_TCPDED_NS) • tMOD (MEM_TMOD_NS) • tZQCS (MEM_TZQCS_NS) • tZQINIT (MEM_TZQINIT_CYC) • tZQOPER (MEM_TZQOPER_CYC) (Identifier: JEDEC_OVERRIDE_TABLE_PARAM_NAME)

Table 139. Group: Advanced: Memory Timing / Values

Parameter Name	Description
Write Preamble Length	Specifies the write preamble length in cycles. (Identifier: MEM_WR_PREAMBLE_MODE)
Read Preamble Length	Specifies the read preamble length in cycles. (Identifier: MEM_RD_PREAMBLE_MODE)
Read Latency	Read Latency of the memory device in clock cycles. (Identifier: MEM_CL_CYC)
Write Latency	Write Latency in clock cycles. (Identifier: MEM_CWL_CYC)
tREFI	Specifies the average refresh interval in nanoseconds. (Identifier: MEM_TREFI_NS)
tRAS	Specifies the activation-to-precharge command period in nanoseconds. (Identifier: MEM_TRAS_NS)
tRCD	Specifies the activation to interval read or write delay interval in nanoseconds. (Identifier: MEM_TRCD_NS)
tRP	Specifies the precharge command period in nanoseconds. (Identifier: MEM_TRP_NS)
tRC	Specifies the activate-to-activate or activate-to-refresh command period in nanoseconds.
<i>continued...</i>	

Parameter Name	Description
	(Identifier: MEM_TRC_NS)
tCCD_L	Specifies the CAS-to-CAS command delay for the same bank group in nanoseconds. (Identifier: MEM_TCCD_L_NS)
tCCD_S	Specifies the CAS-to-CAS command delay for different bank groups in nanoseconds. (Identifier: MEM_TCCD_S_NS)
tRRD_L	Specifies the activation-to-activation command delay for the same bank group in nanoseconds. (Identifier: MEM_TRRD_L_NS)
tRRD_S	Specifies the activation-to-activation command delay for different bank groups in nanoseconds. (Identifier: MEM_TRRD_S_NS)
tFAW	Specifies the four-activate-window in nanoseconds. (Identifier: MEM_TFAW_NS)
tWTR_L	Specifies the minimum delay from the start of an internal write transaction to the immediately next internal read command for the same bank group in nanoseconds. (Identifier: MEM_TWTR_L_NS)
tWTR_S	Specifies the minimum delay from the start of an internal write transaction to the immediately next internal read command for different bank groups in nanoseconds. (Identifier: MEM_TWTR_S_NS)
tWR	Specifies the write recovery time in nanoseconds. (Identifier: MEM_TWR_NS)
tMRD	Specifies the mode-register command cycle time in nanoseconds. (Identifier: MEM_TMRD_NS)
tCKSRE	Specifies the amount of time, in nanoseconds, required after self-refresh entry or power-down entry. (Identifier: MEM_TCKSRE_NS)
tCKSRX	Specifies the amount of time, in nanoseconds, required before self-refresh exit, power-down exit, or reset exit. (Identifier: MEM_TCKSRX_NS)
tCKE	Specifies the minimum CKE low pulse width from self-refresh entry to self-refresh exit in nanoseconds. (Identifier: MEM_TCKE_NS)
tCKESR	Specifies the minimum CKE low pulse width from self-refresh entry to self-refresh exit in memory clock cycles. (Identifier: MEM_TCKESR_CYC)
tMPRR	Specifies the multi-purpose register recovery time measured in nanoseconds. (Identifier: MEM_TMPRR_NS)
tRFC	Specifies the refresh-to-activate or refresh-to-refresh command period in nanoseconds. (Identifier: MEM_TRFC_NS)
<i>continued...</i>	

Parameter Name	Description
tDQSCK	Specifies the minimum DQS_t, DQS_c rising edge output timing location from rising CK_t, CK_c in nanoseconds. (Identifier: MEM_TDQSCK_NS)
tXP	Specifies the delay from power down exit with DLL on to any valid command, or from precharge power down with DLL frozen to commands not requiring a locked DLL. Measured in nanoseconds. (Identifier: MEM_TXP_NS)
tXS	Specifies the delay from self refresh exit to commands not requiring a locked DLL in nanoseconds. (Identifier: MEM_TXS_NS)
tXSDLL	Specifies the delay from self refresh exit to commands requiring a locked DLL in nanoseconds. (Identifier: MEM_TXS_DLL_NS)
tCPDED	Specifies the command pass disable delay measured in nanoseconds. (Identifier: MEM_TCPDED_NS)
tMOD	Specifies the mode register set command update delay in nanoseconds. (Identifier: MEM_TMOD_NS)
tZQCS	Specifies the normal operation short calibration time in nanoseconds. (Identifier: MEM_TZQCS_NS)
tZQINIT	Specifies the power-up and reset calibration time in cycles. (Identifier: MEM_TZQINIT_CYC)
tZQOPER	Specifies the normal operation full calibration time in cycles. (Identifier: MEM_TZQOPER_CYC)

Table 140. Group: Advanced: Analog Overrides / Overrides / ANALOG_TABLE

Parameter Name	Description
Analog Parameter	Name of Analog Parameter to explicitly override; the values will be applied and appear in the list below. Default value is Legal values are: • Phy AC Drive Strength (PHY_TERM_X_R_S_AC_OUTPUT_OHM) • Phy CK Drive Strength (PHY_TERM_X_R_S_CK_OUTPUT_OHM) • Phy DQ Drive Strength (PHY_TERM_X_R_S_DQ_OUTPUT_OHM) • Phy DQ Slew Rate (PHY_TERM_X_DQ_SLEW_RATE) • Phy DQ Input Termination (PHY_TERM_X_R_T_DQ_INPUT_OHM) • Phy DQ Initial Vrefin (PHY_TERM_X_DQ_VREF) • Phy PLL Reference Clock Input Termination (PHY_TERM_X_R_T_REFCLK_INPUT_OHM) • Mem Target Write Termination (MEM_ODT_DQ_X_TGT_WR) • Mem Non-Target Write Termination (MEM_ODT_DQ_X_NON_TGT_WR) • Mem Non-Target Read Termination (MEM_ODT_DQ_X_NON_TGT_RD) • Mem DQ Drive Strength (MEM_ODT_DQ_X_RON) • Mem VrefDQ Range (MEM_VREF_DQ_X_RANGE) • Mem VrefDQ Value (MEM_VREF_DQ_X_VALUE) (Identifier: ANALOG_PARAM_DERIVATION_PARAM_NAME)

Table 141. Group: Advanced: Analog Overrides / Values

Parameter Name	Description
Phy AC Drive Strength	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the refclk input pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 34 Ohm (with calibration) (SERIES_34_OHM_CAL) • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_AC_OUTPUT_OHM)
Phy CK Drive Strength	This parameter allows you to change the output on chip termination settings for the selected I/O standard on the CK Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 34 Ohm (with calibration) (SERIES_34_OHM_CAL) • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_CK_OUTPUT_OHM)
Phy DQ Drive Strength	This parameter allows you to change the output on chip termination settings for the selected I/O standard on the DQ Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 34 Ohm (with calibration) (SERIES_34_OHM_CAL) • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_DQ_OUTPUT_OHM)
Phy DQ Slew Rate	Specifies the slew rate of the data bus pins. The slew rate (or edge rate) describes how quickly the signal can transition, measured in voltage per unit time. <i>Perform board simulations to determine the slew rate that provides the best eye opening for the data bus signals.</i> Legal values are: • Slow (SLOW) • Medium (MEDIUM) • Fast (FAST) • Fastest (FASTEST) (Identifier: PHY_TERM_X_DQ_SLEW_RATE)
Phy DQ Input Termination	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the DQ Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 40 Ohm (with calibration) (RT_40_OHM_CAL) • 50 Ohm (with calibration) (RT_50_OHM_CAL) • 60 Ohm (with calibration) (RT_60_OHM_CAL) (Identifier: PHY_TERM_X_R_T_DQ_INPUT_OHM)
Phy DQ Initial Vrefin	Specifies the initial value for the reference voltage on the data pins(Vrefin). The specified value serves as a starting point and may be overridden by calibration to provide better timing margins. Legal values are: from 0 to 100 (Identifier: PHY_TERM_X_DQ_VREF)
Phy PLL Reference Clock Input Termination	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the refclk input pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • No Termination (RT_OFF) • Differential Termination (RT_DIFF) (Identifier: PHY_TERM_X_R_T_REFCLK_INPUT_OHM)

continued...

Parameter Name	Description
Mem Target Write Termination	Specifies the target termination to be used during a write. The value of this parameter represents X, where: termination = RZQ/X = (240 Ohm)/X. Legal values are: • Dynamic ODT Off (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_TGT_WR)
Mem Non-Target Write Termination	Specifies the termination to be used for the non-target rank in a multi-rank configuration during a write. The value of this parameter represents X, where: termination = RZQ/X = (240 Ohm)/X. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_NON_TGT_WR)
Mem Non-Target Read Termination	Specifies the termination to be used for the non-target rank in a multi-rank configuration during a read. The value of this parameter represents X, where: termination = RZQ/X = (240 Ohm)/X. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_NON_TGT_RD)
Mem DQ Drive Strength	Specifies the termination to be used when driving read data from memory. The value of this parameter represents X, where: termination = RZQ/X = (240 Ohm)/X. Legal values are: • 34 Ohm (RZQ/7) (7) • 48 Ohm (RZQ/5) (5) (Identifier: MEM_ODT_DQ_X_RON)
Mem VrefDQ Range	Specifies which of the memory protocol defined ranges will be used. Legal values are: • Range 1: 60%-92.5% (1) • Range 2: 45%-77.5% (2) (Identifier: MEM_VREF_DQ_X_RANGE)
Mem VrefDQ Value	Specifies the initial VrefDQ value to be used. Legal values are: from 60.00 to 92.50, from 45.00 to 75.00

Parameter Name	Description
	(Identifier: MEM_VREF_DQ_X_VALUE)

Table 142. Group: Example Design / Fileset Types

Parameter Name	Description
HDL Selection	This option lets you choose the format of HDL in which generated simulation and synthesis files are created. You can select either Verilog or VHDL. Default value is VERILOG Legal values are: • Verilog (VERILOG) • VHDL (VHDL) (Identifier: EX_DESIGN_HDL_FORMAT)
Generate Synthesis Fileset	Generate Synthesis Example Design. Default value is true (Identifier: EX_DESIGN_GEN_SYNTH)
Generate Simulation Fileset	Generate Simulation Example Design. Default value is true (Identifier: EX_DESIGN_GEN_SIM)

Table 143. Group: Example Design / User PLL

Parameter Name	Description
Auto-set User PLL Output Clock Frequency	If true, let IP select a reference clock frequency for the user PLL in the example design; if false, let user set a custom value for this parameter. Default value is true (Identifier: EX_DESIGN_USER_PLL_OUTPUT_FREQ_MHZ_AUTOSSET_EN)
User PLL Output Clock Frequency	Frequency of the core clock in MHz. This clock drives the traffic generator and NoC initiator (If in NoC mode). Value is specified in megahertz Default value is 570 (Identifier: EX_DESIGN_USER_PLL_OUTPUT_FREQ_MHZ)
User PLL Reference Clock Frequency	PLL reference clock frequency in MHz for PLL supplying the core clock. Value is specified in megahertz Default value is 100 (Identifier: EX_DESIGN_USER_PLL_REFCLK_FREQ_MHZ)
NOC Reference Clock Frequency	Reference Clock Frequency for the NOC control IP. Value is specified in megahertz Default value is 100 Legal values are: 25, 100, 125 (Identifier: EX_DESIGN_NOC_PLL_REFCLK_FREQ_MHZ)

Table 144. Group: Example Design / Traffic Generator

Parameter Name	Description
Traffic Generator Remote Access	Specifies whether the Traffic Generator control and status registers are accessible via JTAG, exported to the fabric, or just disabled. Default value is JTAG
continued...	

Parameter Name	Description
	Legal values are: • Exported for Onchip control (EXPORT) • Remote through JTAG (JTAG) (Identifier: EX_DESIGN_TG_CSR_ACCESS_MODE)
Traffic Generator Program	Specifies the traffic pattern to be run. Default value is MEDIUM Legal values are: • Short Traffic Pattern (SHORT) • Medium-length Traffic Pattern (MEDIUM) • Long Traffic Pattern (LONG) • Infinite Traffic Pattern (INFINITE) (Identifier: EX_DESIGN_TG_PROGRAM)

Table 145. Group: Example Design / Performance Monitor

Parameter Name	Description
Enable Performance Monitor for Channel 0	If true, example design will include a Performance Monitor instance connected to Channel 0. Default value is false (Identifier: EX_DESIGN_PMON_CH0_EN)

6.2. Agilex 7 M-Series DDR4 DIMM External Memory Interfaces (EMIF) IP Parameter Descriptions

The following tables describe the parameters available on each tab of the IP parameter editor, which you can use to configure your IP.

Table 146. Group: High-level Configuration / Memory Device

Parameter Name	Description
Data DQ Width	Number of DQ pins per memory channel, used for data. Default value is 64 Legal values are: 64, 72 (Identifier: MEM_CHANNEL_DATA_DQ_WIDTH)
ECC DQ Width	Number of additional DQ pins per memory channel, used for out-of-band ECC. If bigger than 0, controller will enable out-of-band ECC. Otherwise, out-of-band ECC will be disabled. Default value is 0 Legal values are: 0, 8 (Identifier: MEM_CHANNEL_ECC_DQ_WIDTH)
DIMM Type	Specifies the type of DIMM that is used with this interface. Default value is UDIMM Legal values are: UDIMM, SODIMM, RDIMM (Identifier: MEM_DIMM_TYPE)
Die DQ Width	Number of DQ pins in each die that makes up the interface. For dual-die packages, this is the width of the die, not the width of full the package. Default value is 8 Legal values are: 4, 8, 16 (Identifier: MEM_DIE_DQ_WIDTH)
<i>continued...</i>	

Parameter Name	Description
Die Density	Capacity of each memory die (in Gbits), per channel per die. For dual-die packages, this is the density of each die, not the density of the full package. Default value is 8 Legal values are: • 2Gbits (2) • 4Gbits (4) • 8Gbits (8) • 16Gbits (16) • 32Gbits (32) (Identifier: MEM_DIE_DENSITY_GBITS)
CS Width	Specifies the total number of CS pins used by each channel. Default value is 1 Legal values are: 1, 2 (Identifier: MEM_CHANNEL_CS_WIDTH)
Memory Speedbin	Specifies the speedbin of the memory device(s) of which the interface consists. Default value is 3200W Legal values are: • 1600J (CL10) (1600J) • 1600K (CL11) (1600K) • 1600L (CL12) (1600L) • 1866L (CL12) (1866L) • 1866M (CL13) (1866M) • 1866N (CL14) (1866N) • 2133N (CL14) (2133N) • 2133P (CL15) (2133P) • 2133R (CL16) (2133R) • 2400P (CL15) (2400P) • 2400R (CL16) (2400R) • 2400T (CL17) (2400T) • 2400U (CL18) (2400U) • 2666T (CL17) (2666T) • 2666U (CL18) (2666U) • 2666V (CL19) (2666V) • 2666W (CL20) (2666W) • 2933V (CL19) (2933V) • 2933W (CL20) (2933W) • 2933Y (CL21) (2933Y) • 2933AA (CL22) (2933AA) • 3200W (CL20) (3200W) • 3200AA (CL22) (3200AA) • 3200AC (CL24) (3200AC) (Identifier: MEM_SPEEDBIN)
Use AC Mirroring	Enable command-address mirroring for multi-rank DDR4 interfaces per JEDEC Standard. Default value is false (Identifier: MEM_AC_MIRRORING_EN)
Share CK Pins Between Ranks	Specifies whether all the ranks in the same channel should share one pair of memory interface differential clock. Default value is false (Identifier: MEM_RANKS_SHARE_CK_EN)
<i>continued...</i>	

Parameter Name	Description
Use AC Parity	Specifies whether address-command parity is enabled. If enabled then command latency is increased by the value of parameter "Address-Command Latency Mode". Default value is false (Identifier: MEM_AC_PARITY_EN)
Auto-set Memory Operating Frequency	If true, let IP select max frequency that this configuration can support for the current device speedgrade. If false, user can set custom value for operating frequency. Default value is true (Identifier: MEM_OPERATING_FREQ_MHZ_AUTOSET_EN)
Memory Operating Frequency	Specifies the frequency at which the memory interface will run. Value is specified in megahertz Legal values are: 666.667, 800, 933.333, 1066.667, 1200, 1333.333, 1466.667, 1600 (Identifier: MEM_OPERATING_FREQ_MHZ)

Table 147. Group: High-level Configuration / PHY

Parameter Name	Description
Auto-set PLL Reference Clock Frequency	If true, let IP select max PLL refclk frequency that this configuration can support. If false, user can set custom value for PLL refclk frequency. Default value is true (Identifier: PHY_REFCLK_FREQ_MHZ_AUTOSET_EN)
Enable Advanced List of PLL Reference Clock Frequencies	If true, provide extended list of possible refclk values. Otherwise, prune possible list of refclk values to a more reasonable length. Default value is false (Identifier: PHY_REFCLK_ADVANCED_SELECT_EN)
Reference Clock Frequency	Specifies the reference clock frequency for the EMIF IOPLL. Value is specified in megahertz (Identifier: PHY_REFCLK_FREQ_MHZ)
Alert_n AC-Lane Index	Specifies the AC lane index in which to place the ALERT_N pin. Default value is AC2 Legal values are: AC2, AC3 (Identifier: PHY_ALERT_N_PLACEMENT)
Force Using 4 AC Lanes	Specifies if the minimum number of AC lanes for the memory interface should be forced to 4. Default value is false (Identifier: PHY_FORCE_MIN_4_AC_LANES_EN)
AC Placement	Indicates location on the device where the interface will reside (specifically, the location of the AC lanes and the DQ lanes in terms of I/O BANK and TOP vs BOT part of the I/O BANK). Legal ranges are derived from device floorplan. Default value is TOP_TOP

continued...

Parameter Name	Description
	Legal values are: AC Primary Bot Sub-Bank / Secondary DQ Top Sub-Bank (BOT_BOT) AC Primary Top Sub-Bank / Secondary DQ Bot Sub-Bank (TOP_TOP) AC Primary Bot Sub-Bank / Secondary DQ Bot Sub-Bank (BOT_TOP) AC Primary Top Sub-Bank / Secondary DQ Top Sub-Bank (TOP_BOT) (Identifier: PHY_AC_PLACEMENT)
Mainband Access Mode	Specifies the path through which the EMIF QHIP mainband interface is exposed to the user. The mainband interface is the AXI4 interface to the memory controller. Legal values are: Fabric Direct - User Clock Synchronous to PHY (SYNC) (Identifier: PHY_MAINBAND_ACCESS_MODE)
Auto-set Sideband Access Mode	If true, let IP select most likely usecase for the PHY_SIDEHAND_ACCESS_MODE; if false, let user set a custom value for sideband access mode. Default value is true (Identifier: PHY_SIDEHAND_ACCESS_MODE_AUTOSET_EN)
Sideband Access Mode	Specifies the path through which the EMIF QHIP sideband interface is exposed to the user. The sideband interface is the AXI4-Lite interface to the IOSSM. Legal values are: Fabric Direct (FABRIC) (Identifier: PHY_SIDEHAND_ACCESS_MODE)
Pin Swizzle Map	Specifies the swizzle map for the data lanes and pins. (Identifier: PHY_SWIZZLE_MAP)
Use Debug Toolkit	If enabled, the AXI-L port will be connected to SLD nodes, allowing for a system-console avalon manager interface to interact with this AXI-L subordinate interface. Default value is false (Identifier: DEBUG_TOOLS_EN)
Instance ID	Instance ID of the EMIF IP. This is useful when using a discovery mechanism over the side-band interface, to identify which EMIF instance's mailbox is at which offset. If expecting to use a discovery mechanism in hardware, this parameter must be set uniquely for all EMIFs that share a sideband. Otherwise, this parameter can be ignored / kept at the default value. Default value is 0 Legal values are: from 0 to 6 (Identifier: INSTANCE_ID)

Table 148. Group: High-level Configuration / Controller / Controller Options

Parameter Name	Description
Use ECC Autocorrection	If ECC is enabled, specifies whether single-bit-errors (SBEs) should be corrected or just reported. Default value is true
<i>continued...</i>	

Parameter Name	Description
	(Identifier: CTRL_ECC_AUTOCORRECT_EN)
Use Data Masking	Specifies whether Data Masking is enabled by the controller. RMW will occur under these conditions: AXI transactions using a subset of write strobe, or one beat long AXI transaction in DDR5, LPDDR5, or LPDDR4 interfaces. Altera recommends enabling the DM option to improve efficiency for the following cases: non-ECC design with AXI transaction listed above, and LPDDR4 or LPDDR5 designs with In-Line ECC enabled. Default value is false (Identifier: CTRL_DM_EN)
Use WDBI	Specifies whether write Data-bus-inversion is enabled by the controller. Default value is false (Identifier: CTRL_WR_DBI_EN)
Use RDBI	Specifies whether read Data-bus-inversion is enabled by the controller. Default value is false (Identifier: CTRL_RD_DBI_EN)
Controller Performance Profile	Group of Controller settings to optimize performance based on expected traffic patterns. Choose "Custom" for fine-granular control Default value is SEQ Legal values are: • Sequential Access Optimized (SEQ) • Random Access Optimized (RAND) • Custom (CUSTOM) (Identifier: CTRL_PERFORMANCE_PROFILE)
Force Auto-Precharge	Controller will terminate all transactions with an auto-precharge when enabled. For per-transaction granularity, leave this option disabled and use the AxUSER control signals instead. Default value is false (Identifier: CTRL_AUTO_PRECHARGE_EN)
Bank Group Rotation	Moves a number of Bank Group pins lower in the address order Default value is 1 Legal values are: • No BGs (disabled) (0) • Two BGs (BG[0]) (1) • Four BGs (BG[1:0]) (2) (Identifier: CTRL_BG_ROTATE_EN)
Enable Row/CS Swap	Specifies whether the highest order address bits are the CS bit (false) or Row bits (true) Default value is false (Identifier: DIAG_HMC_ADDR_SWAP_EN)

Table 149. Group: High-level Configuration / Controller / Data Bus Turnaround Times

Parameter Name	Description
Additional read-to-write turnaround time (same rank)	Additional memory clock cycles of delay between read and write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 <i>continued...</i>

Parameter Name	Description
	(Identifier: TURNAROUND_R2W_SAMECS_CYC)
Additional read-to-read turnaround time (same rank)	Additional memory clock cycles of delay between read operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2R_SAMECS_CYC)
Additional write-to-write turnaround time (same rank)	Additional memory clock cycles of delay between write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2W_SAMECS_CYC)
Additional write-to-read turnaround time (same rank)	Additional memory clock cycles of delay between read and write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2R_SAMECS_CYC)
Additional read-to-write turnaround time (different ranks)	Additional controller cycles of delay between read and write operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2W_DIFFCS_CYC)
Additional read-to-read turnaround time (different ranks)	Additional controller cycles of delay between read operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2R_DIFFCS_CYC)
Additional write-to-write turnaround time (different ranks)	Additional controller cycles of delay between write operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2W_DIFFCS_CYC)
Additional write-to-read turnaround time (different ranks)	Additional controller cycles of delay between write and read operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2R_DIFFCS_CYC)

Table 150. Group: High-level Configuration / Advanced Calibration Settings

Parameter Name	Description
Enable Margining during Calibration	Enable margining stages of calibration for use with the EMIF Debug Toolkit. This will increase calibration time Default value is false
<i>continued...</i>	

Parameter Name	Description
	(Identifier: ADV_CAL_ENABLE_MARGIN)
Enable REQ Calibration	Enable the Read DQ Equalization stage as part of calibration. This will increase calibration time but will also improve top-line performance Default value is false (Identifier: ADV_CAL_ENABLE_REQ)
Enable WEQ Calibration	Run the Write DQ Equalization stage as part of calibration. This will increase calibration time but will also improve top-line performance Default value is false (Identifier: ADV_CAL_ENABLE_WEQ)

Table 151. Group: Advanced: Memory Timing / Overrides / JEDEC_TABLE

Parameter Name	Description
JEDEC Parameter	Name of JEDEC Parameter to explicitly override; the values will be applied and appear in the list below. Default value is Legal values are: • Write Preamble Length (MEM_WR_PREAMBLE_MODE) • Read Preamble Length (MEM_RD_PREAMBLE_MODE) • Read Latency (MEM_CL_CYC) • Write Latency (MEM_CWL_CYC) • tREFI (MEM_TREFI_NS) • tRAS (MEM_TRAS_NS) • tRCD (MEM_TRCD_NS) • tRP (MEM_TRP_NS) • tRC (MEM_TRC_NS) • tCCD_L (MEM_TCCD_L_NS) • tCCD_S (MEM_TCCD_S_NS) • tRRD_L (MEM_TRRD_L_NS) • tRRD_S (MEM_TRRD_S_NS) • tFAW (MEM_TFAW_NS) • tWTR_L (MEM_TWTR_L_NS) • tWTR_S (MEM_TWTR_S_NS) • tWR (MEM_TWR_NS) • tMRD (MEM_TMRD_NS) • tCKSRE (MEM_TCKSRE_NS) • tCKSRX (MEM_TCKSRX_NS) • tCKE (MEM_TCKE_NS) • tCKESR (MEM_TCKESR_CYC) • tMPRR (MEM_TMPRR_NS) • tRFC (MEM_TRFC_NS) • tDQSCK (MEM_TDQSCK_NS) • tXP (MEM_TXP_NS) • tXS (MEM_TXS_NS) • tXSDLL (MEM_TXS_DLL_NS) • tCPDED (MEM_TCPDED_NS) • tMOD (MEM_TMOD_NS) • tZQCS (MEM_TZQCS_NS) • tZQINIT (MEM_TZQINIT_CYC) • tZQOPER (MEM_TZQOPER_CYC) • RDIMM Serial Presence Detect (SPD) Byte 137 (MEM_SPD137_RCD_CA_DRV) • RDIMM Serial Presence Detect (SPD) Byte 138 (MEM_SPD138_RCD_CK_DRV)

Parameter Name	Description
	(Identifier: JEDEC_OVERRIDE_TABLE_PARAM_NAME)

Table 152. Group: Advanced: Memory Timing / Values

Parameter Name	Description
Write Preamble Length	Specifies the write preamble length in cycles. (Identifier: MEM_WR_PREAMBLE_MODE)
Read Preamble Length	Specifies the read preamble length in cycles. (Identifier: MEM_RD_PREAMBLE_MODE)
Read Latency	Read Latency of the memory device in clock cycles. (Identifier: MEM_CL_CYC)
Write Latency	Write Latency in clock cycles. (Identifier: MEM_CWL_CYC)
tREFI	Specifies the average refresh interval in nanoseconds. (Identifier: MEM_TREFI_NS)
tRAS	Specifies the activation-to-precharge command period in nanoseconds. (Identifier: MEM_TRAS_NS)
tRCD	Specifies the activation to interval read or write delay interval in nanoseconds. (Identifier: MEM_TRCD_NS)
tRP	Specifies the precharge command period in nanoseconds. (Identifier: MEM_TRP_NS)
tRC	Specifies the activate-to-activate or activate-to-refresh command period in nanoseconds. (Identifier: MEM_TRC_NS)
tCCD_L	Specifies the CAS-to-CAS command delay for the same bank group in nanoseconds. (Identifier: MEM_TCCD_L_NS)
tCCD_S	Specifies the CAS-to-CAS command delay for different bank groups in nanoseconds. (Identifier: MEM_TCCD_S_NS)
tRRD_L	Specifies the activation-to-activation command delay for the same bank group in nanoseconds. (Identifier: MEM_TRRD_L_NS)
tRRD_S	Specifies the activation-to-activation command delay for different bank groups in nanoseconds. (Identifier: MEM_TRRD_S_NS)
tFAW	Specifies the four-activate-window in nanoseconds. (Identifier: MEM_TFAW_NS)
tWTR_L	Specifies the minimum delay from the start of an internal write transaction to the immediately next internal read command for the same bank group in nanoseconds. (Identifier: MEM_TWTR_L_NS)
<i>continued...</i>	

Parameter Name	Description
tWTR_S	Specifies the minimum delay from the start of an internal write transaction to the immediately next internal read command for different bank groups in nanoseconds. (Identifier: MEM_TWTR_S_NS)
tWR	Specifies the write recovery time in nanoseconds. (Identifier: MEM_TWR_NS)
tMRD	Specifies the mode-register command cycle time in nanoseconds. (Identifier: MEM_TMRD_NS)
tCKSRE	Specifies the amount of time, in nanoseconds, required after self-refresh entry or power-down entry. (Identifier: MEM_TCKSRE_NS)
tCKSRX	Specifies the amount of time, in nanoseconds, required before self-refresh exit, power-down exit, or reset exit. (Identifier: MEM_TCKSRX_NS)
tCKE	Specifies the minimum CKE low pulse width from self-refresh entry to self-refresh exit in nanoseconds. (Identifier: MEM_TCKE_NS)
tCKESR	Specifies the minimum CKE low pulse width from self-refresh entry to self-refresh exit in memory clock cycles. (Identifier: MEM_TCKESR_CYC)
tMPRR	Specifies the multi-purpose register recovery time measured in nanoseconds. (Identifier: MEM_TMPRR_NS)
tRFC	Specifies the refresh-to-activate or refresh-to-refresh command period in nanoseconds. (Identifier: MEM_TRFC_NS)
tDQCK	Specifies the minimum DQS_t, DQS_c rising edge output timing location from rising CK_t, CK_c in nanoseconds. (Identifier: MEM_TDQCK_NS)
tXP	Specifies the delay from power down exit with DLL on to any valid command, or from precharge power down with with DLL frozen to commands not requiring a locked DLL. Measured in nanoseconds. (Identifier: MEM_TXP_NS)
tXS	Specifies the delay from self refresh exit to commands not requiring a locked DLL in nanoseconds. (Identifier: MEM_TXS_NS)
tXSDLL	Specifies the delay from self refresh exit to commands requiring a locked DLL in nanoseconds. (Identifier: MEM_TXS_DLL_NS)
tCPDED	Specifies the command pass disable delay measured in nanoseconds. (Identifier: MEM_TCPDED_NS)
tMOD	Specifies the mode register set command update delay in nanoseconds. (Identifier: MEM_TMOD_NS)
tZQCS	Specifies the normal operation short calibration time in nanoseconds. (Identifier: MEM_TZQCS_NS)
continued...	

Parameter Name	Description
tZQINIT	Specifies the power-up and reset calibration time in cycles. (Identifier: MEM_TZQINIT_CYC)
tZQOPER	Specifies the normal operation full calibration time in cycles. (Identifier: MEM_TZQOPER_CYC)
RDIMM Serial Presence Detect (SPD) Byte 137	Specifies the value of SPD Byte 137 as an integer in decimal. (Identifier: MEM_SPD137_RCD_CA_DRV)
RDIMM Serial Presence Detect (SPD) Byte 138	Specifies the value of SPD Byte 138 as an integer in decimal. (Identifier: MEM_SPD138_RCD_CK_DRV)
DDR4 RDIMM RCD F0RC0E Value	Specifies the value of DDR4 RDIMM RCD control word F0RC0E - Parity, NV Mode Enable, and Alert_N Configuration. Internal use only, and value is default to 13. (Identifier: MEM_RCD_PARITY_LATENCY_CONTROL_WORD)

Table 153. Group: Advanced: Analog Overrides / Overrides / ANALOG_TABLE

Parameter Name	Description
Analog Parameter	Name of Analog Parameter to explicitly override; the values will be applied and appear in the list below. Default value is Legal values are: • Phy AC Drive Strength (PHY_TERM_X_R_S_AC_OUTPUT_OHM) • Phy CK Drive Strength (PHY_TERM_X_R_S_CK_OUTPUT_OHM) • Phy DQ Drive Strength (PHY_TERM_X_R_S_DQ_OUTPUT_OHM) • Phy DQ Slew Rate (PHY_TERM_X_DQ_SLEW_RATE) • Phy DQ Input Termination (PHY_TERM_X_R_T_DQ_INPUT_OHM) • Phy DQ Initial Vrefin (PHY_TERM_X_DQ_VREF) • Phy PLL Reference Clock Input Termination (PHY_TERM_X_R_T_REFCLK_INPUT_OHM) • Mem Target Write Termination (MEM_ODT_DQ_X_TGT_WR) • Mem Non-Target Write Termination (MEM_ODT_DQ_X_NON_TGT_WR) • Mem Non-Target Read Termination (MEM_ODT_DQ_X_NON_TGT_RD) • Mem DQ Drive Strength (MEM_ODT_DQ_X_RON) • Mem VrefDQ Range (MEM_VREF_DQ_X_RANGE) • Mem VrefDQ Value (MEM_VREF_DQ_X_VALUE) • Mem RCD CA Input Termination (MEM_RCD_DCA_IBT) • Mem RCD CS Input Termination (MEM_RCD_DCS_IBT) • Mem RCD CKE Input Termination (MEM_RCD_DCKE_IBT) • Mem RCD ODT Input Termination (MEM_RCD_DODT_IBT) (Identifier: ANALOG_PARAM_DERIVATION_PARAM_NAME)

Table 154. Group: Advanced: Analog Overrides / Values

Parameter Name	Description
Phy AC Drive Strength	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the refclk input pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 34 Ohm (with calibration) (SERIES_34_OHM_CAL) • 40 Ohm (with calibration) (SERIES_40_OHM_CAL)

continued...

Parameter Name	Description
	(Identifier: PHY_TERM_X_R_S_AC_OUTPUT_OHM)
Phy CK Drive Strength	This parameter allows you to change the output on chip termination settings for the selected I/O standard on the CK Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 34 Ohm (with calibration) (SERIES_34_OHM_CAL) • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_CK_OUTPUT_OHM)
Phy DQ Drive Strength	This parameter allows you to change the output on chip termination settings for the selected I/O standard on the DQ Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 34 Ohm (with calibration) (SERIES_34_OHM_CAL) • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_DQ_OUTPUT_OHM)
Phy DQ Slew Rate	Specifies the slew rate of the data bus pins. The slew rate (or edge rate) describes how quickly the signal can transition, measured in voltage per unit time. <i>Perform board simulations to determine the slew rate that provides the best eye opening for the data bus signals.</i> Legal values are: • Slow (SLOW) • Medium (MEDIUM) • Fast (FAST) • Fastest (FASTEST) (Identifier: PHY_TERM_X_DQ_SLEW_RATE)
Phy DQ Input Termination	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the DQ Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 40 Ohm (with calibration) (RT_40_OHM_CAL) • 50 Ohm (with calibration) (RT_50_OHM_CAL) • 60 Ohm (with calibration) (RT_60_OHM_CAL) (Identifier: PHY_TERM_X_R_T_DQ_INPUT_OHM)
Phy DQ Initial Vrefin	Specifies the initial value for the reference voltage on the data pins(Vrefin). The specified value serves as a starting point and may be overridden by calibration to provide better timing margins. Legal values are: from 0 to 100 (Identifier: PHY_TERM_X_DQ_VREF)
Phy PLL Reference Clock Input Termination	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the refclk input pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • No Termination (RT_OFF) • Differential Termination (RT_DIFF) (Identifier: PHY_TERM_X_R_T_REFCLK_INPUT_OHM)
Mem Target Write Termination	Specifies the target termination to be used during a write. The value of this parameter represents X, where: $\text{termination} = \text{RZQ}/X = (240 \text{ Ohm})/X$. Legal values are: • Dynamic ODT Off (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2)

continued...

Parameter Name	Description
	• 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_TGT_WR)
Mem Non-Target Write Termination	Specifies the termination to be used for the non-target rank in a multi-rank configuration during a write. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_NON_TGT_WR)
Mem Non-Target Read Termination	Specifies the termination to be used for the non-target rank in a multi-rank configuration during a read. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_NON_TGT_RD)
Mem DQ Drive Strength	Specifies the termination to be used when driving read data from memory. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. Legal values are: • 34 Ohm (RZQ/7) (7) • 48 Ohm (RZQ/5) (5) (Identifier: MEM_ODT_DQ_X_RON)
Mem VrefDQ Range	Specifies which of the memory protocol defined ranges will be used. Legal values are: • Range 1: 60%-92.5% (1) • Range 2: 45%-77.5% (2) (Identifier: MEM_VREF_DQ_X_RANGE)
Mem VrefDQ Value	Specifies the initial VrefDQ value to be used. Legal values are: from 60.00 to 92.50, from 45.00 to 75.00 (Identifier: MEM_VREF_DQ_X_VALUE)
Mem RCD CA Input Termination	This parameter allows you to select the input bus termination of RCD CA input bus. See JEDEC specifications on DDR4CD02 for the pins covered. The values map to specific termination strengths: 0 -> "100 Ohm", 1 -> "150 Ohm", 2 -> "300 Ohm", 3 -> "OFF".

continued...

Parameter Name	Description
	Legal values are: • 100 Ohm (0) • 150 Ohm (1) • 300 Ohm (2) • Off (3) (Identifier: MEM_RCD_DCA_IBT)
Mem RCD CS Input Termination	This parameter allows you to select the input bus termination of RCD CS input bus. See JEDEC specifications on DDR4CD02 for the pins covered. The values map to specific termination strengths: 0 -> "100 Ohm", 1 -> "150 Ohm", 2 -> "300 Ohm", 3 -> "OFF". Legal values are: • 100 Ohm (0) • 150 Ohm (1) • 300 Ohm (2) • Off (3) (Identifier: MEM_RCD_DCS_IBT)
Mem RCD CKE Input Termination	This parameter allows you to select the input bus termination of CKE input bus. See JEDEC specifications on DDR4CD02 for the pins covered. The values map to specific termination strengths: 0 -> "100 Ohm", 1 -> "150 Ohm", 2 -> "300 Ohm", 3 -> "OFF". Legal values are: • 100 Ohm (0) • 150 Ohm (1) • 300 Ohm (2) • Off (3) (Identifier: MEM_RCD_DCKE_IBT)
Mem RCD ODT Input Termination	This parameter allows you to select the input bus termination of ODT input bus. See JEDEC specifications on DDR4CD02 for the pins covered. The values map to specific termination strengths: 0 -> "100 Ohm", 1 -> "150 Ohm", 2 -> "300 Ohm", 3 -> "OFF". Legal values are: • 100 Ohm (0) • 150 Ohm (1) • 300 Ohm (2) • Off (3) (Identifier: MEM_RCD_DODT_IBT)

Table 155. Group: Example Design / Fileset Types

Parameter Name	Description
HDL Selection	This option lets you choose the format of HDL in which generated simulation and synthesis files are created. You can select either Verilog or VHDL. Default value is VERILOG Legal values are: • Verilog (VERILOG) • VHDL (VHDL) (Identifier: EX_DESIGN_HDL_FORMAT)
Generate Synthesis Fileset	Generate Synthesis Example Design. Default value is true (Identifier: EX_DESIGN_GEN_SYNTH)
Generate Simulation Fileset	Generate Simulation Example Design.

continued...

Parameter Name	Description
	Default value is true (Identifier: EX_DESIGN_GEN_SIM)

Table 156. Group: Example Design / User PLL

Parameter Name	Description
Auto-set User PLL Output Clock Frequency	If true, let IP select a reference clock frequency for the user PLL in the example design; if false, let user set a custom value for this parameter. Default value is true (Identifier: EX_DESIGN_USER_PLL_OUTPUT_FREQ_MHZ_AUTOSET_EN)
User PLL Output Clock Frequency	Frequency of the core clock in MHz. This clock drives the traffic generator and NoC initiator (If in NoC mode). Value is specified in megahertz Default value is 570 (Identifier: EX_DESIGN_USER_PLL_OUTPUT_FREQ_MHZ)
User PLL Reference Clock Frequency	PLL reference clock frequency in MHz for PLL supplying the core clock. Value is specified in megahertz Default value is 100 (Identifier: EX_DESIGN_USER_PLL_REFCLK_FREQ_MHZ)
NOC Reference Clock Frequency	Reference Clock Frequency for the NOC control IP. Value is specified in megahertz Default value is 100 Legal values are: 25, 100, 125 (Identifier: EX_DESIGN_NOC_PLL_REFCLK_FREQ_MHZ)

Table 157. Group: Example Design / Traffic Generator

Parameter Name	Description
Traffic Generator Remote Access	Specifies whether the Traffic Generator control and status registers are accessible via JTAG, exported to the fabric, or just disabled. Default value is JTAG Legal values are: Exported for Onchip control (EXPORT) Remote through JTAG (JTAG) (Identifier: EX_DESIGN_TG_CSR_ACCESS_MODE)
Traffic Generator Program	Specifies the traffic pattern to be run. Default value is MEDIUM Legal values are: Short Traffic Pattern (SHORT) Medium-length Traffic Pattern (MEDIUM) Long Traffic Pattern (LONG) Infinite Traffic Pattern (INFINITE) (Identifier: EX_DESIGN_TG_PROGRAM)

Table 158. Group: Example Design / Performance Monitor

Parameter Name	Description
Enable Performance Monitor for Channel 0	If true, example design will include a Performance Monitor instance connected to Channel 0. Default value is false (Identifier: EX_DESIGN_PMON_CH0_EN)

6.3. Agilex 7 M-Series FPGA EMIF IP Pin and Resource Planning

The following topics provide guidelines on pin placement for external memory interfaces.

Typically, all external memory interfaces require the following FPGA resources:

- Interface pins
- PLL and clock network
- Other FPGA resources—for example, core fabric logic and debug interfaces

Once all the requirements are known for your external memory interface, you can begin planning your system.

6.3.1. Agilex 7 M-Series FPGA EMIF IP Interface Pins

Any I/O banks that do not support transceiver operations in Agilex 7 M-Series FPGAs support external memory interfaces. However, DQS (data strobe or data clock) and DQ (data) pins are listed in the device pin tables and are fixed at specific locations in the device. You must adhere to these pin locations to optimize routing, minimize skew, and maximize margins. Always check the pin table for the actual locations of the DQS and DQ pins.

Note: Maximum interface width varies from device to device depending on the number of I/O pins and DQS or DQ groups available. Achievable interface width also depends on the number of address and command pins that the design requires. To ensure adequate PLL, clock, and device routing resources are available, you should always test fit any IP in the Quartus Prime software before PCB sign-off.

Note: The greater the number of banks, the greater the skew, hence Altera recommends that you always generate a test project of your desired configuration and confirm that it meets timing.

6.3.1.1. Estimating Pin Requirements

You should use the Quartus Prime software for final pin fitting. However, you can estimate whether you have enough pins for your memory interface by performing the following steps:

1. Determine how many read/write data pins are associated per data strobe or clock pair.
2. Calculate the number of other memory interface pins needed, including any other clocks (write clock or memory system clock), address, command, and RZQ. Refer to the External Memory Interface Pin Table to determine necessary Address/Command/Clock pins based on your desired configuration.
3. Calculate the total number of I/O banks required to implement the memory interface, given that an I/O bank supports up to 96 pins.

You should test the proposed pin-outs with the rest of your design in the Quartus Prime software (with the correct I/O standard and OCT connections) before finalizing the pin-outs. There can be interactions between modules that are illegal in the Quartus Prime software that you might not know about unless you compile the design and use the Quartus Prime Pin Planner.

6.3.1.2. DIMM Options

Unbuffered DIMMs (UDIMMs) require one set of chip-select (CS#), on-die termination (ODT), clock-enable (CKE), and clock pair (CK/CKn) for every physical rank on the DIMM. Many registered DIMMs use only one pair of clocks; however, this is not a universal rule, so you should check your memory vendor's data sheet to be sure. DDR4 registered DIMMs require a minimum of one chip-select signal.

Table 159. UDIMM and RDIMM Pin Options for DDR4

Pins	UDIMM Pins (Single Rank)	UDIMM Pins (Dual Rank)	RDIMM Pins (Single Rank)	RDIMM Pins (Dual Rank)
Data	72 bit DQ[71:0] = {CB[7:0], DQ[63:0]}	72 bit DQ[71:0] = {CB[7:0], DQ[63:0]}	72 bit DQ[71:0] = {CB[7:0], DQ[63:0]}	72 bit DQ[71:0] = {CB[7:0], DQ[63:0]}
Data Mask	DM#/DBI#[8:0] ⁽¹⁾	DM#/DBI#[8:0] ⁽¹⁾	DM#/DBI#[8:0] ⁽¹⁾	DM#/DBI#[8:0] ⁽¹⁾
Data Strobe	x8: DQS[8:0] and DQS#[8:0]	x8: DQS[8:0] and DQS#[8:0]	x8: DQS[8:0] and DQS#[8:0] x4: DQS[17:0] and DQS#[17:0]	x8: DQS[8:0] and DQS#[8:0] x4: DQS[17:0] and DQS#[17:0]
Address	BA[1:0], BG[1:0], A[16:0] - 4GB: A[14:0] 8GB: A[15:0] 16GB: A[16:0] ⁽²⁾	BA[1:0], BG[1:0], A[16:0] - 8GB: A[14:0] 16GB: A[15:0] 32GB: A[16:0] ⁽²⁾	BA[1:0], BG[1:0], x8: A[16:0] - 4GB: A[14:0] 8GB: A[15:0] 16GB: A[16:0] ⁽²⁾ 32GB: A[17:0] ⁽³⁾	BA[1:0], BG[1:0], x8: A[16:0] x4: A[17:0] - 8GB: A[14:0] 16GB: A[15:0] 32GB: A[16:0] ⁽²⁾ 64GB: A[17:0] ⁽³⁾
Clock	CK0/CK0#	CK0/CK0#, CK1/CK1#	CK0/CK0#	CK0/CK0#, CK1/CK1#
Command	ODT, CS#, CKE, ACT#, RAS#/A16, CAS#/A15, WE#/A14	ODT[1:0], CS#[1:0], CKE[1:0], ACT#, RAS#/A16, CAS#/A15, WE#/A14	ODT, CS#, CKE, ACT#, RAS#/A16, CAS#/A15, WE#/A14	ODT[1:0], CS#[1:0], CKE, ACT#, RAS#/A16, CAS#/A15, WE#/A14
Parity	PAR, ALERT#	PAR, ALERT#	PAR, ALERT#	PAR, ALERT#
Other Pins	SA[2:0], SDA, SCL, EVENT#, RESET#	SA[2:0], SDA, SCL, EVENT#, RESET#	SA[2:0], SDA, SCL, EVENT#, RESET#	SA[2:0], SDA, SCL, EVENT#, RESET#
Notes to Table: 1. DM/DBI pins are available only for DIMMs constructed using x8 or greater components. 2. This density requires 4Gb x4 or 2Gb x8 DRAM components. 3. This density requires 8Gb x4 DRAM components. 4. The Agilex 7 M-Series memory controller can support up to two ranks per channel. Agilex 7 M-Series supports only one DIMM per channel (1DPC).				

6.3.1.3. Maximum Number of Interfaces

The maximum number of interfaces supported for a given memory protocol varies, depending on the FPGA in use.

Unless otherwise noted, the calculation for the maximum number of interfaces is based on independent interfaces where the address or command pins are not shared.

Note: You may need to share PLL clock outputs depending on your clock network usage.

Table 160. Maximum Number of DDR4 Interfaces

Device	Package	Component Interface	DIMM Interface
AGMA032 / AGMA039 / AGMB032 / AGMB039	R31B	7 - x16 7 - x16 + ECC 3AC 5 - x16 + ECC 4AC 5 - x32 5 - x32 + ECC	2
AGME032 / AGME039 / AGMF032 / AGMF039 / AGMG032 / AGMG039 / AGMH032 / AGMH039	R47A	8	4
AGMF032 / AGMF039 / AGMH032 / AGMH039	R47B	8	4

- *Component Interface* refers to x16, x16 + ECC, x32 or x32+ ECC which can be implemented within a single IO96B bank.
- One DIMM interface requires two adjacent IO96B banks located on the edge of the device.

Timing closure depends on device resource and routing utilization. For more information about timing closure, refer to the *Area and Timing Optimization Techniques* chapter in the *Quartus Prime Handbook*.

6.3.2. Agilex 7 M-Series FPGA EMIF IP Resources

The Agilex 7 M-Series FPGA memory interface IP uses several FPGA resources to implement the memory interface.

6.3.2.1. OCT

You require an OCT calibration block if you are using an Agilex 7 M-Series FPGA OCT calibrated series, parallel, or dynamic termination for any I/O in your design. There are two OCT blocks in an I/O bank, one for each sub-bank.

- You must observe the following requirements when using OCT blocks:
The I/O bank where you place the OCT calibration block must use the same V_{CCIO_PIO} voltage as the memory interface.
- The OCT calibration block uses a single fixed R_{ZQ} . You must ensure that an external termination resistor is connected to the correct pin for a given OCT block.

6.3.2.2. PLL

When using PLL for external memory interfaces, you must consider the following guidelines:

- For the clock source, use the clock input pin specifically dedicated to the PLL that you want to use with your external memory interface. The input and output pins are only fully compensated when you use the dedicated PLL clock input pin. Agilex 7 M-Series devices support only differential I/O standard on dedicated PLL clock input pin for EMIF IP.
- Altera recommends using the fastest possible PLL reference clock frequency available in the drop-down list in the EMIF IP Platform Designer, because doing so provides the best jitter performance.

6.3.3. Pin Guidelines for Agilex 7 M-Series FPGA EMIF IP

The Agilex 7 M-Series FPGA contains I/O banks on the top and bottom edges of the device, which can be used by external memory interfaces.

Agilex 7 M-Series FPGA I/O banks contain 96 I/O pins. Each bank is divided into two sub-banks with 48 I/O pins in each. Sub-banks are further divided into four byte-lanes, where each byte-lane is a group of twelve I/O ports.

Agilex 7 M-Series FPGAs do not support flexible DQ group assignments. Only specific byte-lanes can be used as Address/Command lanes or data lanes. As you increase the interface width, only specific byte-lanes can be used. Refer to [Pin Placement for Agilex M-Series FPGA DDR4 IP](#) for more information.

The I/O bank, byte-lane, and pairing pin for every physical I/O pin can be uniquely identified by the following naming convention in the device pin table:

- The I/O pins in a bank are represented as P#, where:
 - P# represents the pin number in a bank. It ranges from P0 to P95, for 96 pins in a bank. Because an IO96 bank comprises two IO48 sub-banks, all pins with P# value less than 48 (P# <48) belong to the bottom I/O sub-bank. All other pins belong to the top IO48 sub-bank.
- The *Index Within I/O Bank* value falls within one of the following ranges: 0 to 11, 12 to 23, 24 to 35, or 36 to 47, and represents one of byte-lanes 0, 1, 2, or 3, respectively.
- To determine whether I/O banks are adjacent, you can refer to [Architecture: I/O Bank](#) in the *Product Architecture* chapter. In general, the two sub-banks within an I/O bank are adjacent to each other when there is at least one byte-lane in each sub-bank that is bonded out and available for EMIF use.
- The pairing pin for an I/O pin is in the same I/O bank. You can identify the pairing pin by adding 1 to its *Index Within I/O Bank* number (if it is an even number), or by subtracting 1 from its *Index Within I/O Bank* number (if it is an odd number).

6.3.4. Pin Placements for Agilex 7 M-Series FPGA DDR4 EMIF IP

6.3.4.1. Address and Command Pin Placement for DDR4

Table 161. Address and Command Pin Placement for DDR4 IP

Address/ Command Lane	Index Within Byte Lane	DDR4				
		Scheme 1	Scheme 1A	Scheme 2	Scheme 3	Scheme 3A
AC3	11	CK_C[1]	CK_C[1]	Not used by Address/ Command pins in this scheme.	CK_C[1]	CK_C[1]
	10	CK_T[1]	CK_T[1]		CK_T[1]	CK_T[1]
	9					
	8		ALERT_N			ALERT_N
	7					
	6					
	5					
	4					
	3					
	2					
	1					
	0				C[0]	C[0]
AC2	11	BG[0]	BG[0]	BG[0]	BG[0]	BG[0]
	10	BA[1]	BA[1]	BA[1]	BA[1]	BA[1]
	9	BA[0]	BA[0]	BA[0]	BA[0]	BA[0]
	8	ALERT_N	A[17]	ALERT_N	ALERT_N	A[17]
	7	A[16]	A[16]	A[16]	A[16]	A[16]
	6	A[15]	A[15]	A[15]	A[15]	A[15]
	5	A[14]	A[14]	A[14]	A[14]	A[14]
	4	A[13]	A[13]	A[13]	A[13]	A[13]
	3	A[12]	A[12]	A[12]	A[12]	A[12]
	2	RZQ site				
	1	Differential "N-side" reference clock input site.				
	0	Differential "P-side" reference clock input site.				
AC1	11	A[11]	A[11]	A[11]	A[11]	A[11]
	10	A[10]	A[10]	A[10]	A[10]	A[10]
	9	A[9]	A[9]	A[9]	A[9]	A[9]
	8	A[8]	A[8]	A[8]	A[8]	A[8]
	7	A[7]	A[7]	A[7]	A[7]	A[7]
	6	A[6]	A[6]	A[6]	A[6]	A[6]
	5	A[5]	A[5]	A[5]	A[5]	A[5]
	4	A[4]	A[4]	A[4]	A[4]	A[4]
	3	A[3]	A[3]	A[3]	A[3]	A[3]
	2	A[2]	A[2]	A[2]	A[2]	A[2]

continued...

Address/ Command Lane	Index Within Byte Lane	DDR4				
		Scheme 1	Scheme 1A	Scheme 2	Scheme 3	Scheme 3A
	1	A[1]	A[1]	A[1]	A[1]	A[1]
	0	A[0]	A[0]	A[0]	A[0]	A[0]
AC0	11	PAR[0]	PAR[0]	PAR[0]	PAR[0]	PAR[0]
	10	CS_N[1]	CS_N[1]	CS_N[1]	CS_N[1]	CS_N[1]
	9	CK_C[0]	CK_C[0]	CK_C[0]	CK_C[0]	CK_C[0]
	8	CK_T[0]	CK_T[0]	CK_T[0]	CK_T[0]	CK_T[0]
	7	CKE[1]	CKE[1]	CKE[1]	CKE[1]	CKE[1]
	6	CKE[0]	CKE[0]	CKE[0]	CKE[0]	CKE[0]
	5	ODT[1]	ODT[1]	ODT[1]	ODT[1]	ODT[1]
	4	ODT[0]	ODT[0]	ODT[0]	ODT[0]	ODT[0]
	3	ACT_N[0]	ACT_N[0]	ACT_N[0]	ACT_N[0]	ACT_N[0]
	2	CS_N[0]	CS_N[0]	CS_N[0]	CS_N[0]	CS_N[0]
	1	RESET_N[0]	RESET_N[0]	RESET_N[0]	RESET_N[0]	RESET_N[0]
	0	BG[1]	BG[1]	BG[1]	BG[1]	BG[1]

Agilex 7 M-Series FPGA DDR4 IP supports fixed Address and Command pin placement as shown in the preceding table. The IP supports up to 2 ranks for the following schemes:

- Scheme 1 supports component, UDIMM, RDIMM, and SODIMM.
- Scheme 1A supports x4 component and RDIMM with A[17] (that is, with 16Gb, x4 DQ/DQS group base component).
- Scheme 2 supports component, UDIMM, RDIMM, and SODIMM. Scheme 2 is the only scheme for HPS DDR4 EMIF, available for fabric EMIF as well.
- Schemes 3 and 3A are similar to schemes 1 and 1A. Schemes 3 and 3A do not support DIMM.
- The Agilex 7 M-Series FPGA EMIF IP for DDR4 does not support 3DS.

6.3.4.2. DDR4 Data Width Mapping

Agilex 7 M-Series devices do not support flexible data lanes placement. Only fixed byte lanes within the I/O bank can be used as data lanes. The following table lists the supported address and command and data lane placements in an I/O bank.

Table 162. DDR4 Data Width Mapping

Controller	Address / Command Scheme	Data Width Usage	BL7 [P95:P84]	BL6 [P83:P72]	BL5 [P71:P60]	BL4 [P59:P48]	BL3 [P47:P36]	BL2 [P35:P24]	BL1 [P23:P12]	BL0 [P11:P0]
Primary	Scheme 2	DDR4 x16	GPIO ²	GPIO ²	GPIO ²	DQ[1]	AC2	AC1	AC0	DQ[0]
	Scheme 1		GPIO ²	GPIO ²	DQ[1]	DQ[0]	AC2	AC1	AC0	AC3
	Scheme 1a		GPIO ²	GPIO ²	DQ[1]	DQ[0]	AC2	AC1	AC0	AC3
	Scheme 3		GPIO ²	GPIO ²	DQ[1]	DQ[0]	AC2	AC1	AC0	AC3
	Scheme 3a		GPIO ²	GPIO ²	DQ[1]	DQ[0]	AC2	AC1	AC0	AC3
	Scheme 2	DDR4 x16 + ECC	GPIO ²	GPIO ²	DQ[ECC]	DQ[1]	AC2	AC1	AC0	DQ[0]
	Scheme 1		GPIO ²	DQ[ECC]	DQ[1]	DQ[0]	AC2	AC1	AC0	AC3
	Scheme 1a		GPIO ²	DQ[ECC]	DQ[1]	DQ[0]	AC2	AC1	AC0	AC3
	Scheme 3		GPIO ²	DQ[ECC]	DQ[1]	DQ[0]	AC2	AC1	AC0	AC3
	Scheme 3a		GPIO ²	DQ[ECC]	DQ[1]	DQ[0]	AC2	AC1	AC0	AC3
	Scheme 2	DDR4 x32	GPIO ²	DQ[3]	DQ[2]	DQ[1]	AC2	AC1	AC0	DQ[0]
	Scheme 1		DQ[3]	DQ[2]	DQ[1]	DQ[0]	AC2	AC1	AC0	AC3
	Scheme 1a		DQ[3]	DQ[2]	DQ[1]	DQ[0]	AC2	AC1	AC0	AC3
	Scheme 2	DDR4 x32 + ECC	DQ[ECC]	DQ[3]	DQ[2]	DQ[1]	AC2	AC1	AC0	DQ[0]
Primary + Secondary	Scheme 2	DDR4 x40 ¹	sDQ[0]	wDQ[3]	wDQ[2]	wDQ[1]	AC2	AC1	AC0	wDQ[0]
	Scheme 2 Flip	DDR4 x40	sDQ[0]	AC0	AC2	AC1	wDQ[1]	wDQ[2]	wDW[3]	wDQ[4]
Note: 1. DDR4 x40 requires both controllers within an I/O bank in a lockstep configuration, and AXI user data. 2. GPIO – available for GPIO/PHYLite. 3. DQ[ECC] – DQ/DQS group used as ECC.										

Table 163. Supported DDR4 Mapping for Lockstep Configuration for x64

A/C	A/C Placement Option	BL0	BL1	BL2	BL3	BL4	BL5	BL6	BL7	BL0	BL1	BL2	BL3	BL4	BL5	BL6	BL7
		Primary								Secondary							
3	AC Pri Top/Sec DQ Bot	DQ[0]	DQ[3]	DQ[2]	DQ[1]	AC1	AC2	AC0	GPIO	DQ[7]	DQ[6]	DQ[5]	DQ[4]	GPIO	GPIO	GPIO	GPIO
	AC Pri Bot/Sec DQ Bot	DQ[0]	AC0	AC1	AC2	DQ[1]	DQ[2]	DQ[3]	GPIO	DQ[7]	DQ[6]	DQ[5]	DQ[4]	GPIO	GPIO	GPIO	GPIO
	AC Pri Top/Sec DQ Top(m)	DQ[0]	DQ[3]	DQ[2]	DQ[1]	AC1	AC2	AC0	GPIO	GPIO	X	X	X	DQ[4]	DQ[5]	DQ[6]	DQ[7]
4	AC Pri Top/Sec DQ Bot(m)	DQ[3]	DQ[2]	DQ[1]	DQ[0]	AC1	AC2	AC0	AC3	DQ[7]	DQ[6]	DQ[5]	DQ[4]	GPIO	GPIO	GPIO	GPIO
	AC Pri Top/Sec DQ Top(m)	DQ[3]	DQ[2]	DQ[1]	DQ[0]	AC1	AC2	AC0	AC3	GPIO	X	X	X	DQ[4]	DQ[5]	DQ[6]	DQ[7]
	AC Pri Bot/Sec DQ Top(m)	AC3	AC0	AC1	AC2	DQ[0]	DQ[1]	DQ[2]	DQ[3]	GPIO	X	X	X	DQ[4]	DQ[5]	DQ[6]	DQ[7]

Note:

- GPIO = For DDR4 lockstep, unused GPIOs can be used only for GPIO, not for PHYLite or for LVDS.
- X = not available for GPIO/PHYLite.
- AC_PRI_TOP/SEC_DQ_BOT = AC Primary Top Sub-Bank/Secondary DQ Bot Sub-Bank.
- AC_PRI_BOT/SEC_DQ_BOT = AC Primary Bot Sub-Bank/Secondary DQ Bot Sub-Bank.
- AC_PRI_TOP/SEC_DQ_TOP = AC Primary Top Sub-Bank/Secondary DQ Top Sub-Bank.
- AC_PRI_BOT/SEC_DQ_TOP = AC Primary Bot Sub-Bank/Secondary DQ Top Sub-Bank.

Table 164. Supported Lockstep configuration for DDR4 x72 or x64 (with ECC)

A/C	A/C Placement Option	BL0	BL1	BL2	BL3	BL4	BL5	BL6	BL7	BL0	BL1	BL2	BL3	BL4	BL5	BL6	BL7
		Primary								Secondary							
3	AC Pri Top/Sec DQ Bot	DQ[4]	DQ[3]	DQ[2]	DQ[1]	AC1	AC2	AC0	sDQ[0] _J	DQ[8]	DQ[7]	DQ[6]	DQ[5]	GPIO	GPIO	GPIO	GPIO
	AC Pri Bot/Sec DQ Bot	DQ[0]	AC0	AC1	AC2	DQ[1]	DQ[2]	DQ[3]	sDQ[4] _J	DQ[8]	DQ[7]	DQ[6]	DQ[5]	GPIO	GPIO	GPIO	GPIO
	AC Pri Top/Sec DQ Top(m)	DQ[4]	DQ[3]	DQ[2]	DQ[1]	AC1	AC2	AC0	sDQ[0] _J	GPIO	X	X	X	DQ[5]	DQ[6]	DQ[7]	DQ[8]
4	AC Pri Top/Sec DQ Bot(m)	DQ[3]	DQ[2]	DQ[1]	DQ[0]	AC1	AC2	AC0	AC3	DQ[8]	DQ[7]	DQ[6]	DQ[5]	X	X	GPIO	sDQ[4] _J
	AC Pri Top/Sec DQ Top(m)	DQ[3]	DQ[2]	DQ[1]	DQ[0]	AC1	AC2	AC0	AC3	DQ[4]	X	X	X	DQ[5]	DQ[6]	DQ[7]	sDQ[8] _J
	AC Pri Top/Sec DQ Top(m)	AC3	AC0	AC1	AC2	DQ[0]	DQ[1]	DQ[2]	DQ[3]	DQ[4]	X	X	X	DQ[5]	DQ[6]	DQ[7]	sDQ[8] _J

- GPIO = available for GPIO/PHYLite.
- X = not available for GPIO/PHYLite.
- DQ lane with an s prefix is the lane used for the following:
 - WUSER/RUSER in x72 configuration or
 - ECC Lane for x64 + ECC configuration
- AC_PRI_TOP/SEC_DQ_BOT = AC Primary Top Sub-Bank/Secondary DQ Bot Sub-Bank.
- AC_PRI_BOT/SEC_DQ_BOT = AC Primary Bot Sub-Bank/Secondary DQ Bot Sub-Bank.
- AC_PRI_TOP/SEC_DQ_TOP = AC Primary Top Sub-Bank/Secondary DQ Top Sub-Bank.
- AC_PRI_BOT/SEC_DQ_TOP = AC Primary Bot Sub-Bank/Secondary DQ Top Sub-Bank.
- x64 with ECC is not supported in the current version of the Quartus Prime software.

Figure 16. x16 + ECC DDR4, Single Rank Using x8 Memory Component, Scheme 2

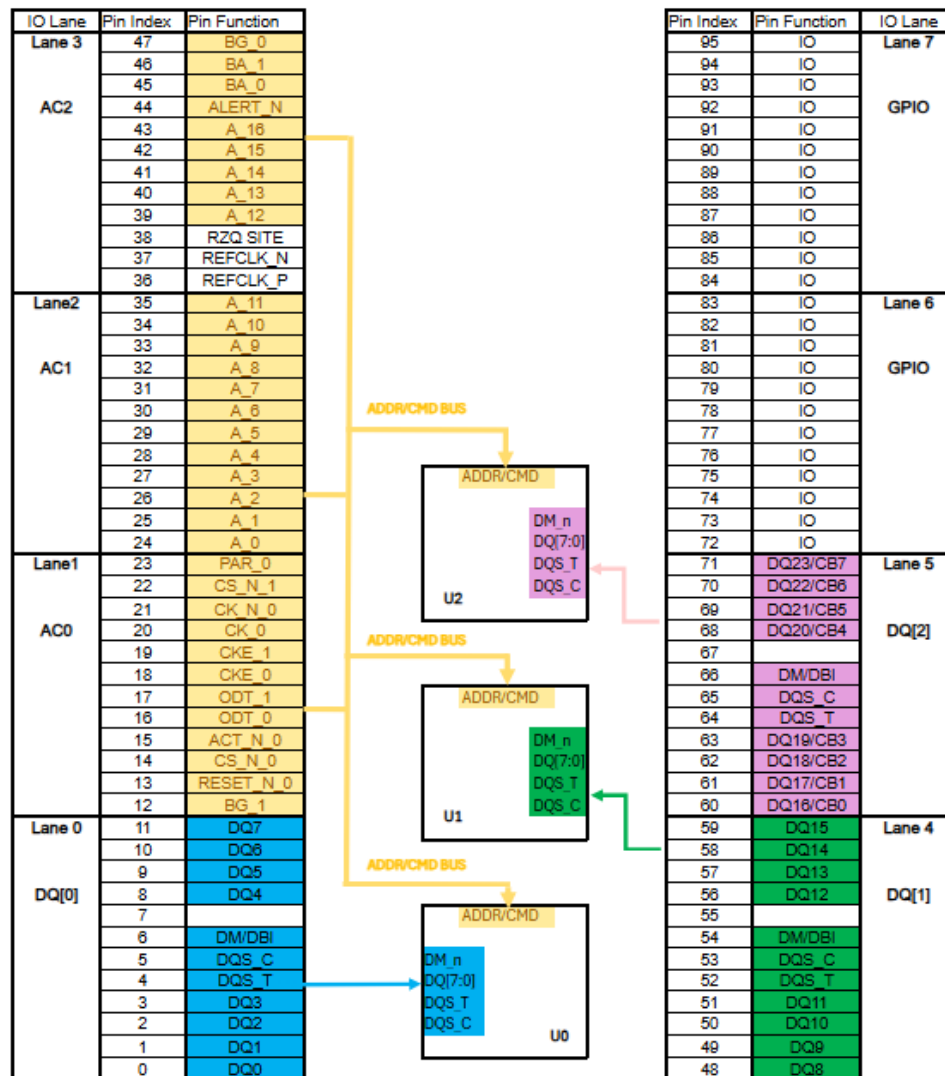


Figure 17. x32 + ECC DDR4, Single Rank Using x8 Memory Component, Scheme 2

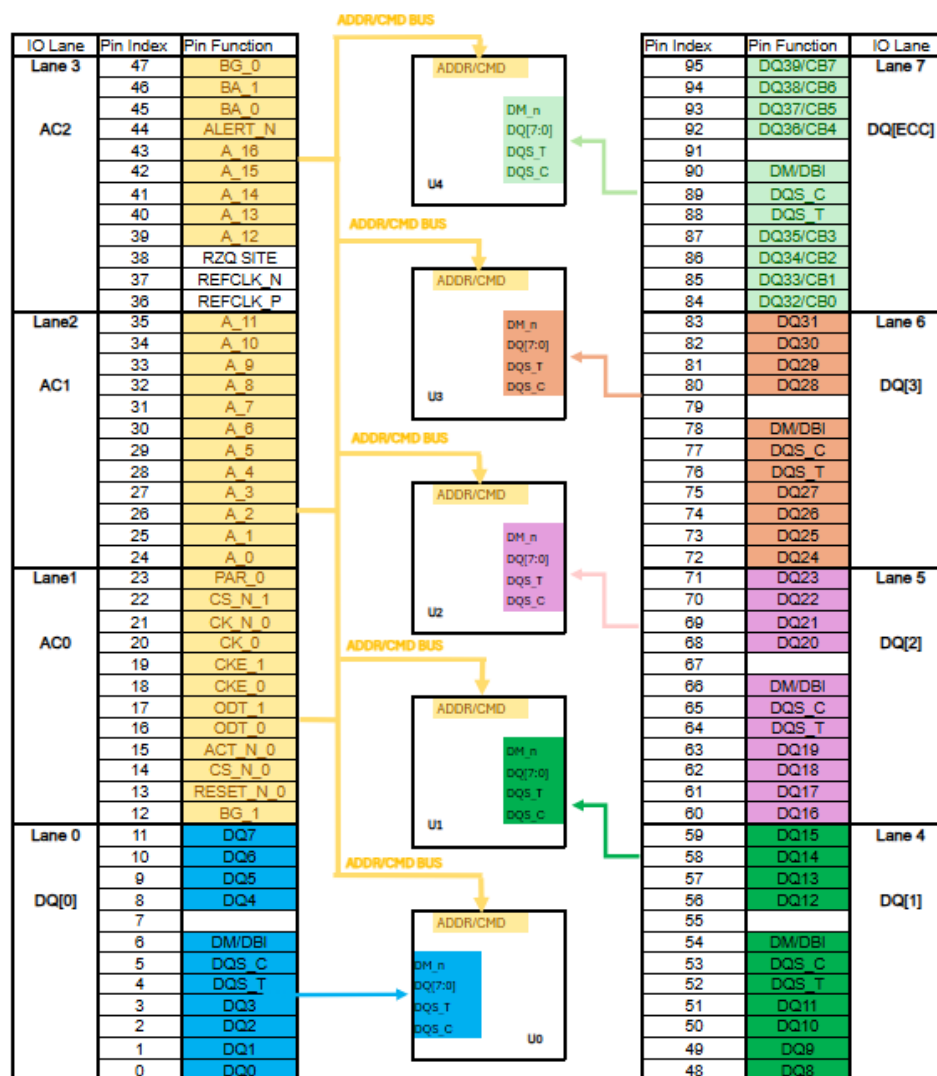
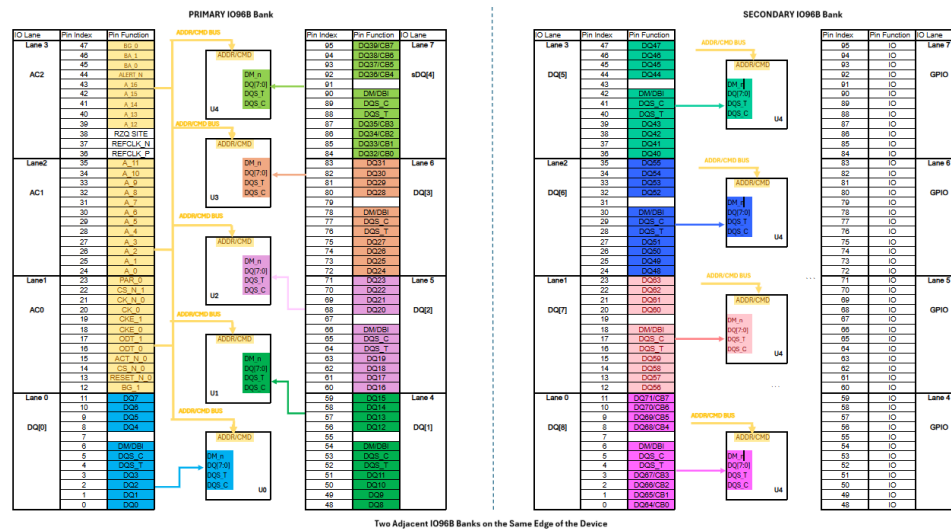


Figure 18. x64 + ECC DDR4 DIMM Interface (3AC - AC_PRI_BOT/SEC_DQ_BOT), Single Rank Using x8 Memory Component



6.3.4.3. General Guidelines - DDR4

Observe the following general guidelines when placing pins for your Agilex 7 M-Series external memory interface.

Note:

- EMIF IP pin-out requirements for the Agilex 7 M-Series Hard Processor Subsystem (HPS) are more restrictive than for a non-HPS memory interface. The HPS EMIF IP defines a fixed pin-out in the Quartus Prime Pro Edition IP file (.qip), based on the IP configuration.
- Ensure that the pins of a single external memory interface reside on the same edge I/O.
 - The address and command pins and their associated clock pins in the address and command bank must follow a fixed pin-out scheme, as defined in the table in the [Address and Command Pin Placement for DDR4](#) topic.
 - Not every byte lane can function as an address and command lane or a data lane. The pin assignment must adhere to the DDR4 data width mapping defined in [DDR4 Data Width Mapping](#).
 - A byte lane must not be used by both address and command pins and data pins.
 - An I/O 96 bank cannot be used for more than one interface – meaning that two sub-banks belonging to two different EMIF interfaces are not permitted.
 - Sharing of byte lanes within a sub-bank for two different interfaces is not permitted; you can assign byte lanes within a sub-bank to one EMIF interface only.
 - Any pin in the same bank that is not used by an external memory interface may not be available for use as a general purpose I/O pin.

- For fabric EMIF, unused pins in an I/O lane assigned to an EMIF interface cannot be used as general-purpose I/O pins. In the same sub-bank, pins in an I/O lane that is not assigned to an EMIF interface, can be used as general purpose I/O pins.
 - For HPS EMIF, unused pins in an I/O lane assigned to an EMIF interface cannot be used as general-purpose I/O pins. In the same bank, pins in an I/O lane that is not assigned to an EMIF interface cannot be used as general purpose I/O pins either.
 - When the network-on-a-chip (NoC) is used, the use of initiator may block the byte lane in the I/O bank adjacent to the sector where the initiator is located:
 - ☐ initiator 0 blocks the access for byte lane (or I/O lane) BL4, BL5, BL6, and partial BL7 (pin index 48 to 88 in the IO96 bank).
 - ☐ initiator 1 blocks the access to the fabric AXI command/control port for the primary/secondary controller in the same I/O bank, making the hard controller unavailable for EMIF purposes.
 - ☐ initiator 2 blocks the access for BL0,1,2,3 (P0 to P47).
8. All address and command pins and their associated clock pins (CK and CK#) must reside within a single sub-bank. Refer to the table in [DDR4 Data Width Mapping](#) for the supported address and command and data lane placements for DDR4.
 9. An external memory interface can occupy one or more banks on the same edge. When an interface must occupy multiple banks, ensure the following:
 - That the banks are adjacent to one another.
 - That you used only the supported data width mapping as defined in the [DDR4 Data Width Mapping](#) table in the [DDR4 Data Width Mapping](#) topic. Be aware that not every byte lane can be used as an address and command lane or a data lane.
 - That for lockstep configuration, you used only supported configuration as defined in the [Supported DDR4 Mapping for Lockstep Configuration for x64](#) and [Supported Lockstep configuration for DDR4 x72 or x64 \(with ECC\)](#) tables in the [DDR4 Data Width Mapping](#) topic.
 10. Place read data groups according to the DQS grouping in the pin table and the Pin Planner. Read data strobes (such as DQS and DQS#) must reside at physical pins capable of functioning as DQS/CQ and DQSn/CQn for a specific read data group size. You must place the associated read data pins (such as DQ and Q), within the same group.

Note: For DDR4 interfaces with x4 components, you can use the strobe pins with either of the upper or lower DQ nibbles that are placed within a x8 DQS group in an I/O lane. You must place the DQ pins and associated strobes entirely in either the upper or lower half of a 12-bit bank sub-group.

Consult the pin table for your device to identify the association between DQ pins and DQS pins for x4 mode operation. Additional restrictions apply for x4/x8 DIMM interoperability.
 11. One of the sub-banks in the device (typically the left sub-bank within corner Bank 3A) may not be available if you use certain device configuration schemes. For some schemes, there may be one byte lane available for EMIF data group:

- AVST-8 – This is contained entirely within the SDM, therefore all lanes of sub-bank 3A can be used by the external memory interface.
- AVST-32 – Byte lanes 4, 5, 6, and 7 are all effectively occupied and are not usable by the external memory interface.
- AVST-16 – Byte lanes 6 contains SDM_DATA[25:16], and is not used by AVSTx16. However, the external memory interface cannot use byte lane 6 when byte lanes 4 and 5 are not usable for EMIF purposes.

6.3.4.4. x4 DIMM Implementation

DIMMS using a x4 DQS configuration require remapping of the DQS signals to achieve compatibility between the EMIF IP and the JEDEC standard DIMM socket connections.

The necessary remapping is shown in the table below. You can implement this DQS remapping in either RTL logic or in your schematic wiring connections.

Table 165. Mapping of DQS Signals Between DIMM and the EMIF IP

DIMM		Quartus Prime EMIF IP	
DQS0_A	DQ[3:0]_A	DQS0	DQ[3:0]_A
DQS5_A	DQ[7:4]_A	DQS1	DQ[7:4]_A
DQS1_A	DQ[11:8]_A	DQS2	DQ[11:8]_A
DQS6_A	DQ[15:12]_A	DQS3	DQ[15:12]_A
DQS2_A	DQ[19:16]_A	DQS4	DQ[19:16]_A
DQS7_A	DQ[23:20]_A	DQS5	DQ[23:20]_A
DQS3_A	DQ[27:24]_A	DQS6	DQ[27:24]_A
DQS8_A	DQ[31:28]_A	DQS7	DQ[31:28]_A
DQS4_A	CB[3:0]_A	DQS8	CB[3:0]_A
DQS9_A	CB[7:4]_A	DQS9	CB[7:4]_A
DQS0_B	DQ[3:0]_B	DQS10	DQ[3:0]_B
DQS5_B	DQ[7:4]_B	DQS11	DQ[7:4]_B
DQS1_B	DQ[11:8]_B	DQS12	DQ[11:8]_B
DQS6_B	DQ[15:12]_B	DQS13	DQ[15:12]_B
DQS2_B	DQ[19:16]_B	DQS14	DQ[19:16]_B
DQS7_B	DQ[23:20]_B	DQS15	DQ[23:20]_B
DQS3_B	DQ[27:24]_B	DQS16	DQ[27:24]_B
DQS8_B	DQ[31:28]_B	DQS17	DQ[31:28]_B
DQS4_B	CB[3:0]_B	DQS18	CB[3:0]_B
DQS9_B	CB[7:4]_B	DQS19	CB[7:4]_B

Data Bus Connection Mapping Flow

1. Connect all FPGA DQ pins accordingly to DIMM DQ pins. No remapping is required.
2. DQS/DQSn remapping is required either on the board schematics or in the RTL code.

When designing a board to support x4 DQS groups, Altera recommends that you make it compatible for x8 mode, for the following reasons:

- Provides the flexibility of x4 and x8 DIMM support.
- Allows use of x8 DQS group connectivity rules.
- Allows use of x8 timing rules for matching. Adhere to x4/x8 interoperability rules when designing a DIMM interface, even if the primary use case is to support x4 DIMMs only, because doing so facilitates debug and future migration capabilities. Regardless, the rules for length matching for two nibbles in a x4 interface must match those of the signals for a corresponding x8 interface, as the data terminations are turned on and off at the same time for both x4 DQS groups in an I/O lane. If the two x4 DQS groups were to have significantly different trace delays, it could adversely affect signal integrity. Trace delays for two nibbles packed within the IO12 lanes are matched using the same guidelines as a single x8 byte lane.

6.3.4.5. Specific Pin Connection Requirements

To compile a design that instantiates an Agilex 7 M-Series EMIF IP, you must explicitly define a legal placement and I/O standard for all EMIF pins.

This topic discusses the required constraints per pin type. To simplify this process, you can leverage the **Back Annotate Assignments** tool. For more information, refer to *Back Annotating Pin Placement and I/O Standard Assignments in the External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide*.

PLL

You must constrain the PLL reference clock to the address and command sub-bank only.

- You must constrain differential reference clocks to pin indices 0 and 1 in lane AC2.
- The sharing of PLL reference clocks across multiple interfaces is permitted; however, pin indices 0 and 1 of lane 2 of the address and command sub-bank for all slave EMIF interfaces can be used only for supplying reference clocks. Altera recommends that you consider connecting these clock input pins to a reference clock source to facilitate greater system implementation flexibility.

Note: Agilex 7 M-Series FPGAs do not support single-ended I/O PLL reference clocks for EMIF IP.

OCT

For DDR4, you must constrain the RZQ pin to pin index 2 in lane AC2.

- Every EMIF instance requires its own dedicated RZQ pin.
- The sharing of RZQ pins is not permitted.

Address and Command

For DDR4, you must constrain the ALERT_N pin to the address and command lanes only.

- In three-lane address and command schemes, you can place the ALERT_N pin at pin index 8 in lane AC2 only.
- In four-lane address and command schemes, you can place the ALERT_N pin at pin index 8 in lane AC2 or at pin index 8 in lane AC3. When you generate the IP, the resulting RTL specifies which connection to use.

DQS/DQ/DM

For DDR4 x8 DQS grouping, the following rules apply:

- You may use pin indices 0, 1, 2, 3, 8, 9, 10, and 11 within a lane for DQ mode pins only.
- You must use pin index 4 for the DQS_t pin only.
- You must use pin index 5 for the DQS_c pin only.
- You must ensure that pin index 7 remains unused. Pin index 7 is not available for use as a general purpose I/O.
- You must use pin index 6 for the DM/DBI_N pin only.

For DDR4 x4 DQS grouping, the following rules apply:

- You may use pin indices 0, 1, 2, and 3 within a lane for DQ mode pins for the lower nibble only. Pin rotation within this group is permitted.
- You must use pin index 4 for the DQS_t pin only of the lower nibble.
- You must use pin index 5 for the DQS_c pin only of the lower nibble.
- You may use pin indices 8, 9, 10, and 11 within a lane for the DQ mode pins only for the upper nibble.
- Pin rotation within this group is permitted.
- You must use pin index 6 for the DQS_t pin only of the upper nibble.
- You must use pin index 7 for the DQS_c pin only of the upper nibble.

6.3.4.6. Command and Address Signals

Command and address signals in SDRAM devices are clocked into the memory device using the CK or CK# signal. These pins operate at single data rate (SDR) using only one clock edge. The number of address pins depends on the SDRAM device capacity. The address pins are multiplexed, so two clock cycles are required to send the row, column, and bank address.

Although DDR4 operates in fundamentally the same way as other SDRAM, there are no dedicated pins for RAS#, CAS#, and WE#, as those are shared with higher-order address pins. DDR4 has CS#, CKE, ODT, and RESET# pins, similar to DDR3. DDR4 also has some additional pins, including the ACT# (activate) pin and BG (bank group) pins.

6.3.4.7. Clock Signals

DDR4 SDRAM devices use CK and CK# signals to clock the address and command signals into the memory.

The memory uses these clock signals to generate the DQS signal during a read through the DLL inside the memory. The SDRAM data sheet specifies the following timings:

- t_{DQSCK} is the skew between the CK or CK# signals and the SDRAM-generated DQS signal.
- t_{DSH} is the DQS falling edge from CK rising edge hold time.
- t_{DSS} is the DQS falling edge from CK rising edge setup time.
- t_{DQSS} is the positive DQS latching edge to CK rising edge.

SDRAM devices have a write requirement (t_{DQSS}) that states the positive edge of the DQS signal on writes must be within $\pm 25\%$ ($\pm 90^\circ$) of the positive edge of the SDRAM clock input. Therefore, you should generate the CK and CK# signals using the DDR registers in the IOE to match with the DQS signal and reduce any variations across process, voltage, and temperature. The positive edge of the SDRAM clock, CK, is aligned with the DQS write to satisfy t_{DQSS} .

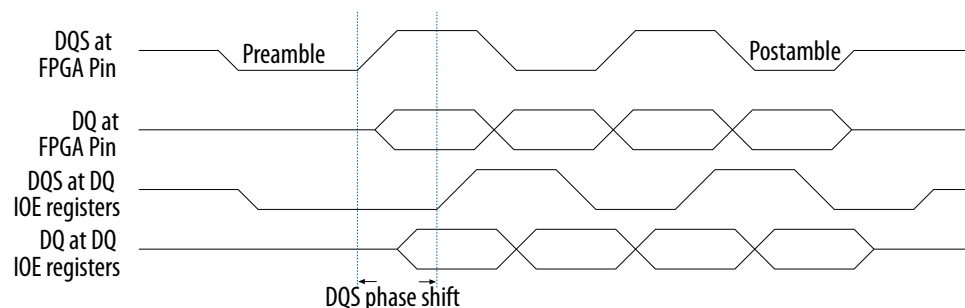
6.3.4.8. Data, Data Strobes, DM/DBI, and Optional ECC Signals

DDR4 SDRAM devices use bidirectional differential data strobes. Differential DQS operation enables improved system timing due to reduced crosstalk and less simultaneous switching noise on the strobe output drivers. The DQ pins are also bidirectional.

DQ pins in DDR4 SDRAM interfaces can operate in either $\times 4$ or $\times 8$ mode DQS groups, depending on your chosen memory device or DIMM, regardless of interface width. The $\times 4$ and $\times 8$ configurations use one pair of bidirectional data strobe signals, DQS and DQSn, to capture input data. However, two pairs of data strobes, UDQS and UDQS# (upper byte) and LDQS and LDQS# (lower byte), are required by $\times 16$ configurations. A group of DQ pins must remain associated with its respective DQS and DQSn pins.

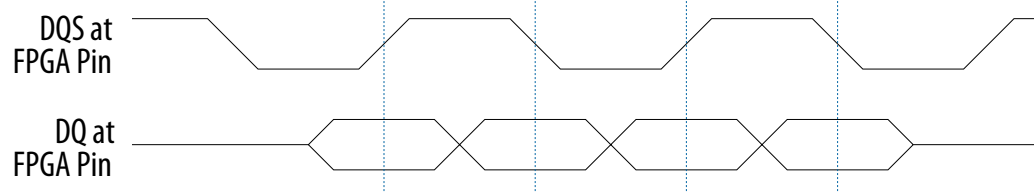
The DQ signals are edge-aligned with the DQS signal during a read from the memory and are center-aligned with the DQS signal during a write to the memory. The memory controller shifts the DQ signals by -90 degrees during a write operation to center align the DQ and DQS signals. The PHY IP delays the DQS signal during a read, so that the DQ and DQS signals are center aligned at the capture register. Altera devices use a phase-locked loop (PLL) to center-align the DQS signal with respect to the DQ signals during writes and use dedicated DQS phase-shift circuitry to shift the incoming DQS signal during reads. The following figure shows an example where the DQS signal is shifted by 90 degrees for a read from the SDRAM.

Figure 19. Edge-aligned DQ and DQS Relationship During a SDRAM Read in Burst-of-Four Mode



The following figure shows an example of the relationship between the data and data strobe during a burst-of-four write.

Figure 20. DQ and DQS Relationship During a SDRAM Write in Burst-of-Four Mode



The memory device's setup (t_{DS}) and hold times (t_{DH}) for the DQ and DM pins during writes are relative to the edges of DQS write signals and not the CK or CK# clock. Setup and hold requirements are not necessarily balanced.

The DQS signal is generated on the positive edge of the system clock to meet the t_{DQSS} requirement. DQ and DM signals use a clock shifted -90 degrees from the system clock, so that the DQS edges are centered on the DQ or DM signals when they arrive at the SDRAM. The DQS, DQ, and DM board trace lengths need to be tightly matched (within 20 ps).

The SDRAM uses the DM pins during a write operation. Driving the DM pins low shows that the write is valid. The memory masks the DQ signals if the DM pins are driven high. To generate the DM signal, Altera recommends that you use the spare DQ pin within the same DQS group as the respective data, to minimize skew.

The DM signal's timing requirements at the SDRAM input are identical to those for DQ data. The DDR registers, clocked by the -90 degree shifted clock, create the DM signals.

DDR4 supports DM similarly to other SDRAM, except that in DDR4 DM is active LOW and bidirectional, because it supports Data Bus Inversion (DBI) through the same pin. DM is multiplexed with DBI by a Mode Register setting whereby only one function can be enabled at a time. DBI is an input/output identifying whether to store/output the true or inverted data. When enabled, if DBI is LOW, during a write operation the data is inverted and stored inside the DDR4 SDRAM; during a read operation, the data is inverted and output. The data is not inverted if DBI is HIGH. For Agilex 7 interfaces, the DM/DBI pins do not need to be paired with a DQ pin.

Some SDRAM modules support error correction coding (ECC) to allow the controller to detect and automatically correct error in data transmission. The 72-bit SDRAM modules contain eight extra data pins in addition to 64 data pins. The eight extra ECC pins should be connected to a single DQS or DQ group on the FPGA.

6.4. DDR4 Board Design Guidelines

The following topics provide guidelines for improving the signal integrity of your system and for successfully implementing a DDR4 SDRAM interface on your system.

The following areas are discussed:

- comparison of various types of termination schemes, and their effects on the signal quality on the receiver
- proper drive strength setting on the FPGA to optimize the signal integrity at the receiver
- effects of different loading types, such as components versus DIMM configuration, on signal quality

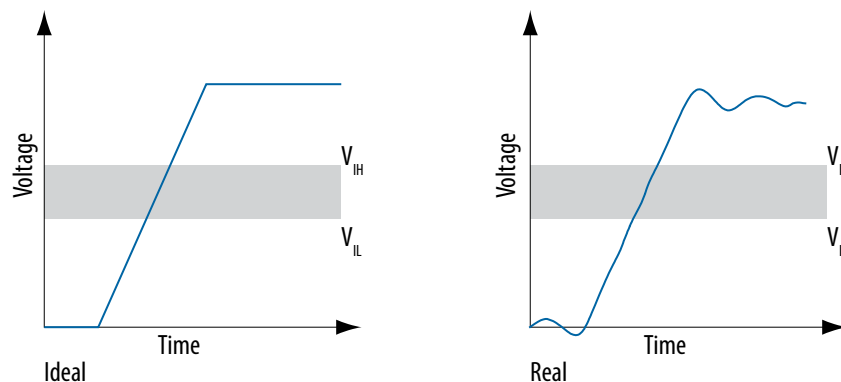
It is important to understand the trade-offs between different types of termination schemes, the effects of output drive strengths, and different loading types, so that you can swiftly navigate through the multiple combinations and choose the best possible settings for your designs.

The following key factors affect signal quality at the receiver:

- Leveling and dynamic ODT
- Proper use of termination
- Layout guidelines

As memory interface performance increases, board designers must pay closer attention to the quality of the signal seen at the receiver because poorly transmitted signals can dramatically reduce the overall data-valid margin at the receiver. The following figure shows the differences between an ideal and real signal seen by the receiver.

Figure 21. Ideal and Real Signal at the Receiver



6.4.1. Terminations for DDR4 with Agilex 7 M-Series Devices

The following topics describe considerations specific to DDR4 external memory interface protocols on Agilex 7 M-Series devices.

6.4.1.1. Dynamic On-Chip Termination (OCT)

Depending upon the R_s (series) and R_t (parallel) OCT values that you want, you should choose appropriate values for the RZQ resistor and connect this resistor to the RZQ pin of the FPGA.

Refer to the *External Memory Interfaces Agilex 7 M-Series FPGA IP* parameter editor to determine the supported termination values.

6.4.1.2. Dynamic On-Die Termination (ODT) in DDR4

In DDR4, in addition to the `Rtt_nom` and `Rtt_wr` values, which are applied during read and write respectively, a third option called `Rtt_park` is available. When `Rtt_park` is enabled, a selected termination value is set in the DRAM when ODT is driven low.

Refer to the DDR4 JEDEC* specification or your memory vendor data sheet for details about available termination values and functional description for dynamic ODT in DDR4 devices.

6.4.1.3. Choosing Terminations on Agilex 7 M-Series FPGA Devices

To determine optimal on-chip termination (OCT) and on-die termination (ODT) values for best signal integrity, you should simulate your memory interface in HyperLynx or a similar tool, using a simulation model extracted from the PCB of your memory interface channel.

If the optimal OCT and ODT values as determined by simulation are not available in the list of available values in the parameter editor, select the closest available termination values for OCT and ODT.

For information about available ODT choices, refer to your memory vendor data sheet.

6.4.1.4. On-Chip Termination Recommendations for Agilex 7 M-Series FPGA Devices

In the EMIF IP parameter editor you can select values from drop-down lists for each of the following:

- output mode drive strength for the address/command bus.
- output mode drive strength for the memory clock.
- output mode drive strength for the data bus.
- input mode termination strength for the data bus.

The range of available values may vary, depending on your memory protocol and silicon revision.

You can use the default values as starting points; however, for best results, you should sweep the entire range of legal values and generate multiple hardware designs to determine the optimal settings for your board and memory device.

Once you have found the optimal settings for your design, uncheck the **Use Default I/O settings** checkbox and use your optimal settings for all future compilations, even if those settings align with the default settings. This ensures that your settings are preserved if the IP is upgraded to a future version.

6.4.2. General Layout Routing Guidelines

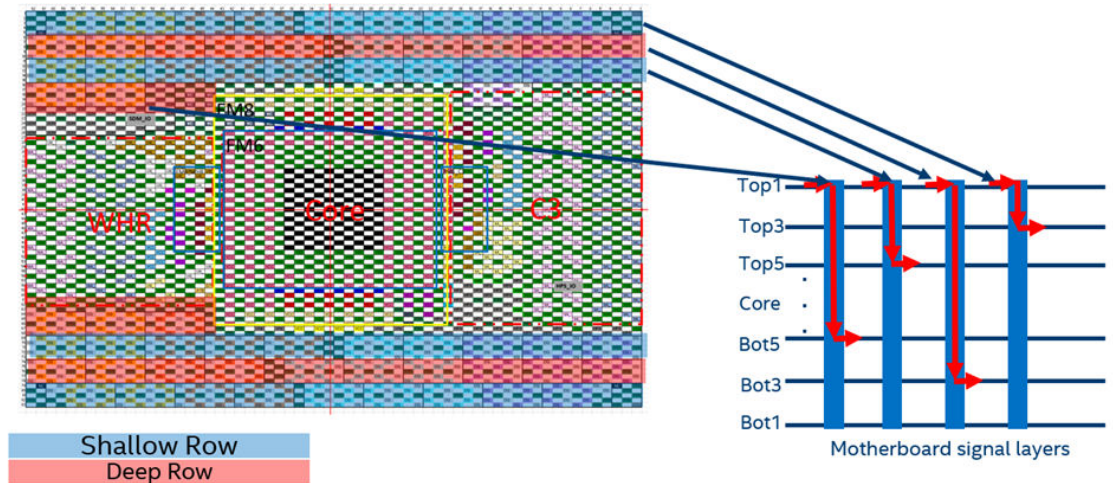
Follow the guidelines in this section for routing from the FPGA to memory for Agilex 7 M-Series devices.

For maximum channel margin, you should consider the following general routing optimizations during the layout design phase:

- The Agilex 7 M-Series FPGA EMIF IP for DDR4 does not support a clamshell layout.
- When routing the memory interface, ensure that there are solid ground reference planes without any plane splits or voids, to ensure an uninterrupted current return path.
- For signal vias in layer transitions, you must place ground stitching vias close by, within 80 mil in distance (closer is better), and in between signal vias, to minimize crosstalk among signal vias. Avoid any unnecessary signal layer transitions to minimize crosstalk, loss, and skews.
- Trace impedance plays an important role in signal integrity; board designers must follow impedance recommendations for each signal group and configuration according to the guidelines in this document. If you use a different stackup than the reference stackup in the PCB design, you must tune the trace width and geometries to achieve the impedance recommendations.
- Altera recommends using 45-degree angles (not sharp 90-degree corners) when routing signal turns. Use $3 \times h$ spacing for serpentine routing, where h is the height or distance from the trace to the nearest GND reference plane.
- Avoid referencing a signal to both power and ground planes at the same time (dual referencing), for signal return paths. When this cannot be avoided, ensure that the closer reference plane is solid ground, and the far side power plane is not noisy.
- Avoid routing two internal signal layers adjacent to each other (dual stripline routing). When this cannot be avoided, use angled routing between two signal layers to minimize crosstalk and coupling between the layers.
- Follow time-domain length and skew matching rules to ensure that your interface meets timing requirements. You should route signals from the same byte or group together on the same layer to avoid any out-of-phase crosstalk caused by varying layer transition lengths.
- To optimize memory interface margins, Altera recommends the following routing strategies:
 - ☐ For DIMM configurations, route DQ and DQS signals on shallow layers with short via transition lengths, because they have tighter timing margins than address, command, and control signals. (Shallow layers are those above the PCB core where via transition lengths are short.)
 - ☐ For discrete device configurations, route DQ and DQS on shallow layers.

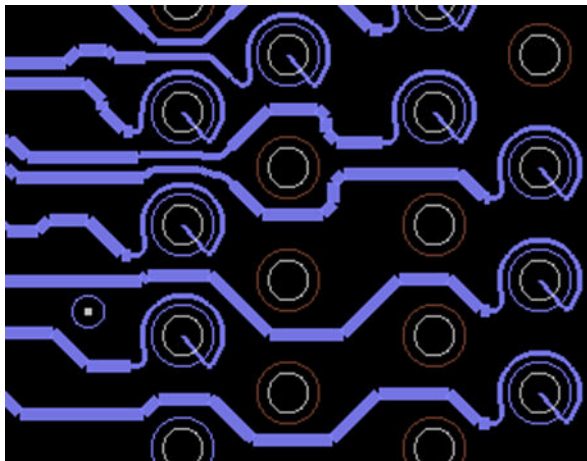
- For boards thicker than 65 mil, Altera recommends alternating adjacent FPGA EMIF BGA/ball rows with deep and shallow board via transitions to minimize crosstalk between adjacent bytes. This method is illustrated in the following figure:

Figure 22. Recommended alternate adjacent via transitions to avoid crosstalk between adjacent bytes



- For boards thicker than 65 mil, using the pin-through-hole (PTH) type of DIMM connector, Altera recommends implementing a loop-routing-around-DIMM-pin structure (Lcomp) to improve impedance matching between signal routing and the DIMM connector. Refer to the following figure.

Figure 23. Recommended Lcomp structure for better impedance matching



- For PCB designs using a surface mount technology (SMT) type of DIMM connector, Altera recommends placing a cutout (void) in the ground reference plane underneath the connector pads for DDR4 signals to minimize connector pad capacitance. Refer to the following figure for the recommended cutout on ground reference plane underneath the connector pad on surface layer.

Figure 24. Recommended Cutout on Ground Reference Plane

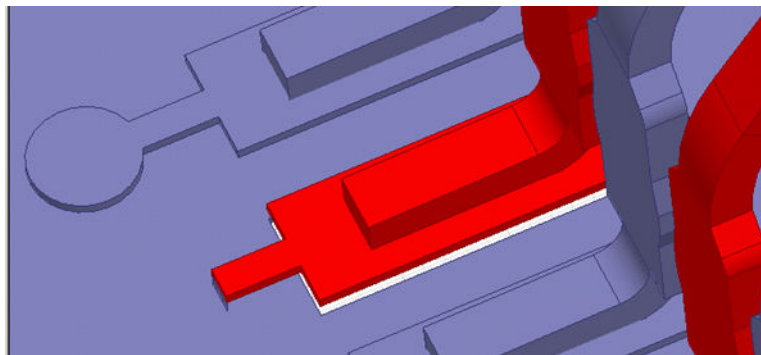


Figure 25. Closeup View of Connections

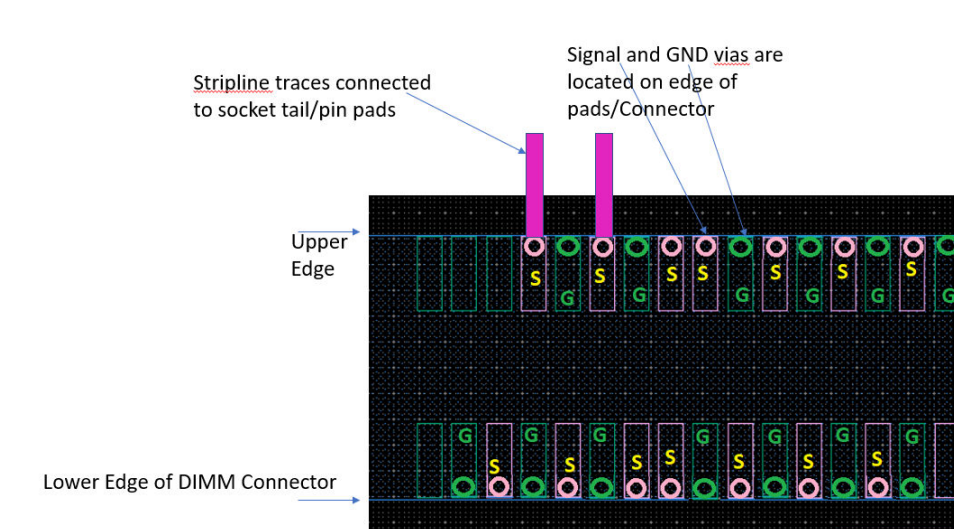
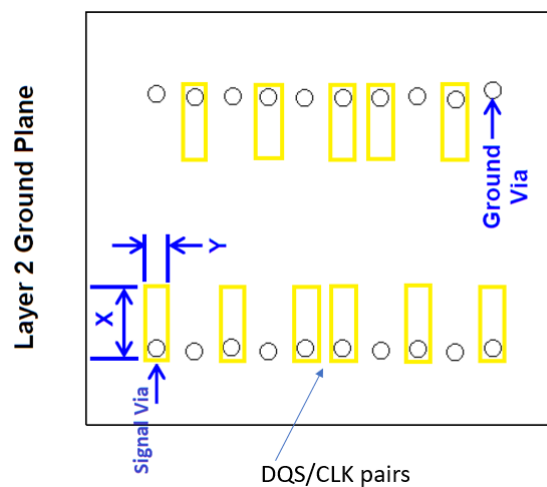


Figure 26. Closeup View of Connections



6.4.3. Reference Stackup

This topic illustrates the reference stackup on which EMIF routing design guidelines are based.

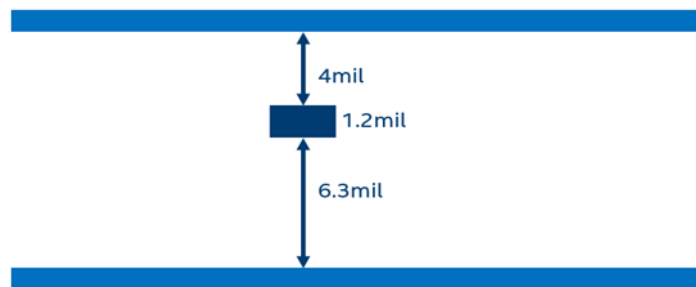
It is important to understand that trace geometry such as width, thickness, and edge-to-edge spacing, and the distance to reference planes, all impact trace impedance and crosstalk levels.

Table 166. Reference stackup details

Layer	Type	Thickness
SM TOP		0.5
L1	signal	1.8
D1	prepreg	2.7
L2	gnd/power	1.2
D2	core	4.0
L3	signal	1.2
D3	prepreg	6.3
L4	gnd/power	1.2
D4	core	4.0
L5	signal	1.2
D5	prepreg	6.3
L6	gnd/power	1.2
D6	core	4.0
L7	signal	1.2
D7	prepreg	6.3
L8	gnd	1.2
D8	core	4
	Power	1.2
	prepreg	6.3
	power	1.2
	core	4
	gnd	1.2
	prepreg	6.3
	power	1.2
	core	4
L9	gnd	1.2
D9	prepreg	6.3
L10	signal	1.2
<i>continued...</i>		

Layer	Type	Thickness
D10	core	4.0
L11	gnd/power	1.2
D11	prepreg	6.3
L12	signal	1.2
D12	core	4.0
L13	gnd/power	1.2
D13	prepreg	6.3
L14	signal	1.2
D14	core	4.0
L15	gnd/power	1.2
D15	prepreg	2.7
L16	signal	1.8
SM BOT		0.5
	Total	120.1

Figure 27. Reference trace geometries



The reference stackup height is selected to be 120 mil to cover maximum signal via coupling (110mil) in simulation while extracting EMIF design guideline. Altera recommends that board designers do not exceed 110mil signal via coupling (stripline routing on inner layers) in the EMIF layout PCB design for DDR4 interfaces.

If the PCB stackup exceeds 120 mil in height, Altera recommends routing EMIF signals on upper layers, not to exceed more than 110 mil of signal via coupling.

The reference stackup materials in the above figure are selected as FR4, to represent worst-case signal loss in design phase simulation. In case of low-loss materials, the maximum end-to-end routing length shall be larger than the recommended end-to-end routing length in the design guidelines; however, you must perform time-domain channel simulation to ensure that timing requirements are met.

6.4.4. Agilex 7 M-Series EMIF-Specific Routing Guidelines for Various DDR4 Topologies

This section discusses EMIF-related layout guidelines for Agilex 7 M-Series devices.

The Agilex 7 M-Series family pin floorplan is a HEX pattern with 0.92mm pitch. The following figure shows an example of DDR routing for an IO12 (one-byte data) on PCB within FPGA fan-out region.

Figure 28. Agilex 7 1mm HEX pin pattern/floorplan and recommended routing for one byte of data (IO12)



The following general notes apply to the EMIF routing guidelines tables in subsequent topics:

- All spacing requirements are the minimum requirement to be met on PCB in EMIF routing guideline table.
- Breakout (BO1/BO2) spacings have two different values in guideline tables. The first value represents minimum spacing between two signals routed as a pair (tightly coupled signals); this value is marked as A (5 mil) in the above figure. The second value represents minimum spacing between two pairs, and this value is marked as B (17 mil) in the above figure.
- Main route (M) spacings have both value in mil and formula. In formula, h represents the trace-to-nearest-reference-plane height or distance. In cases using a stackup different than the reference stackup, board designers shall use formula to calculate the correct spacing requirements.
- There is no differential impedance target for CLK nor DQS. Board designers shall follow single-ended impedance target and keep the signals within the pair closely coupled, within 3-4 mil spacing. For information on DQS/DQSB and CLK/CLKB, refer to the [Skew Matching Guidelines for DDR4 DIMM Topologies](#) and [Skew Matching Guidelines for DDR4 Discrete Topologies](#) tables, for DIMM and discrete device implementations, respectively.

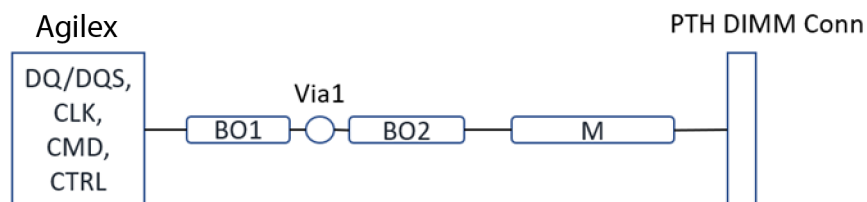
- In guideline tables, *SL* stands for stripline routing recommendation and *US* stands for upper surface (Microstrip) routing recommendation.
- The trace width value/geometry in guideline tables stands for trace designed for target impedance based on the reference stackup. This trace geometry shall be designed based on actual stackup and target impedance in guideline table.
- In guideline tables, *BO1* and *BO2* represent fan-out routing lengths. *M* stands for out of fan-out (PCB main) routing lengths

6.4.4.1. One DIMM per Channel (1DPC) for UDIMM, RDIMM, and SODIMM DDR4 Topologies

The interface covers data bytes (DQ/DQS), address signals, command signals (BA, BG, RAS, CAS, WE, ACT, PAR), control signals (CKE, CS, ODT) and clocks (CLK).

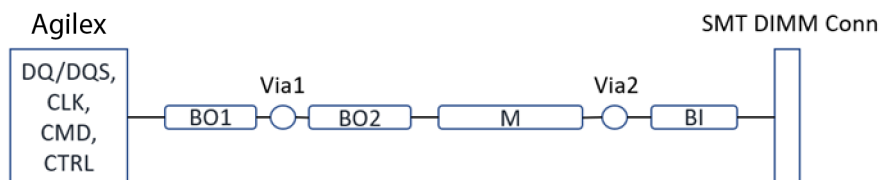
The following figure illustrates the signal connection topology for a PTH type of connector for UDIMM and RDIMM topologies.

Figure 29. Signal connections for DDR4 1DPC DIMM configuration using PTH DIMM connector



The following figure illustrates the signal connection topology for an SMT type of connector for UDIMM, RDIMM, and SODIMM topologies.

Figure 30. Signal connections for DDR4 1DPC DIMM configuration using SMT DIMM connector



The following table provides specific routing guidelines for one DIMM per channel in UDIMM, RDIMM, and SODIMM topologies for all supported signals in the interface.

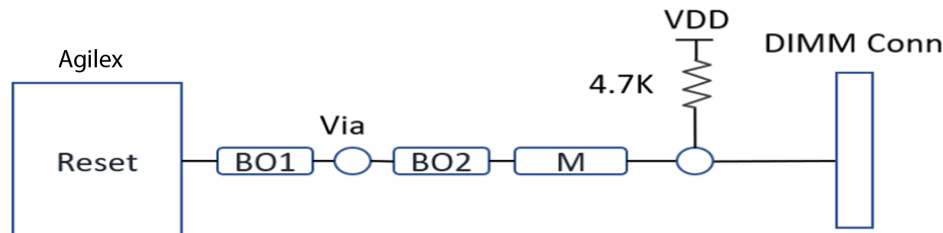
Table 167. Specific DDR4 1DPC routing guidelines for UDIMM, RDIMM, and SODIMM configurations

Signal Group	Segment	Routing Layer	Max Length (mil)		Target Zse (ohms)	Target Width, W (mil)	Trace Spacing, S1 (mil): Within Group	Trace Spacing, S2 (mil): CMD/CTRL/CLK to DQ/DQS	Trace Spacing, S3 (mil): DQ Nibble to Nibble	Trace Spacing (mil): Within DIFF pair	Trace Spacing (mil): DQS pair to DQ	Trace Spacing (mil): CLK pair to CMD/CTRL/CKE	Channel to Channel Spacing (DQ to DQ, between two channels)
			Segment	Total MB									
CLK	BO1	US	50	4500		4	5, 17	5, 17		4		17	
	BO2	SL	1000			4	5, 17	5, 17		4		17	
	M	SL			45	4.5		3h		4		3h	
	BI	US	50			4		3h		4		3h	
CMD, CTRL, ALERT	BO1	US	50	4500		4	5, 17	5, 17					
	BO2	SL	1000			4	5, 17	5, 17					
	M	SL			45	4.5	2h	3h					
	BI	US	100			4	2h	3h					
DQ	BO1	US	50	4500		3	5, 17		17				17
	BO2	SL	1000			3	5, 17		17				17
	M	SL			50	3.5	2h		3h				4h
	BI	US	50			3.5	2h		3h				4h
DQS	BO1	US	50	4500		3	5, 17			4	17		
	BO2	SL	1000			3	5, 17			4	17		
	M	SL			50	3.5				4	3h		
	BI	US	50			3.5							

For an explanation of the guidelines represented in this table, refer to the bullet points immediately following Agilex 7 M-Series EMIF-Specific Routing Guidelines for Various DDR4 Topologies on page 159.

The following figure shows the RESET signal scheme and routing guideline for one DIMM per channel topologies.

Figure 31. Reset scheme for 1DPC DIMM topologies



The target impedance for the RESET signal is 50 ohms. The RESET signal shall have at least $3 \times h$ (where h stands for trace to nearest reference plane height or distance) spacing to other nearby signals on the same layer. The end-to-end RESET trace length is not limited but shall not exceed 5 inches.

6.4.4.2. Skew Matching Guidelines for DIMM Configurations

The guidelines in this topic apply to any DIMM topology, regardless of DIMM type or number of ranks.

Board designers must observe the following guidelines for DDR4 DIMM skew matching:

- Perform skew matching in time (picoseconds) rather than in actual trace length, to better account for via delays when signals are routed on different layers.
- Include both package per-pin skew and PCB delay when performing skew matching.
- Skew (length) matching for the alert signal is not required.

The following table provides skew matching guidelines for DDR4 DIMM topologies.

Table 168. Skew Matching Guidelines for DDR4 DIMM Topologies

DIMM Skew Matching Rule	Length in Time (ps)
Length matching between DQS and CLK	$-255\text{ps} < \text{CLK} - \text{DQS} < 425\text{ps}$
Length matching between DQ and DQS within byte	$-3.5\text{ps} < \text{DQ} - \text{DQS} < 3.5\text{ps}$
Length matching between DQS and DQS#	$< 1\text{ps}$
Length matching between CLK and CLK#	$< 1\text{ps}$
Length matching between CLK0 and CLK1	$< 8\text{ps}$
Length matching between CMD/ADDR/CTRL and CLK	$-20\text{ps} < \text{CLK} - \text{CMD/ADDR/CTRL} < 20\text{ps}$
Length matching among CMD/ADDR/CTRL within each channel	$< 20\text{ps}$
<i>continued...</i>	

DIMM Skew Matching Rule	Length in Time (ps)
Include package length in skew matching for FPGA device with no migration	Required
Include package length in skew matching for FPGA device with migration when all package net length are available	It is recommended to use the final migrated package net length
Include package length in skew matching for FPGA device with migration when all package net length are not available	Not recommended

6.4.4.3. Power Delivery Recommendations for the Memory / DIMM Side

This topic describes power distribution network (PDN) design guidelines for one DIMM per channel (1DPC) memory interfaces at the DIMM/memory side.

Note: For information on power distribution network design at the FPGA to meet timing margins, refer to the AG014 PDN design guideline.

In the following table, the number of decoupling capacitors is based on a single channel. If multiple channels are sharing the same power rail at the DIMM, the number of decoupling capacitors at the DIMM must be scaled accordingly.

Physically small decoupling capacitors are recommended to minimize area, inductance, and resistance on the PDN path on the printed circuit board.

Table 169. Required Decoupling Capacitors on the PCB for the Memory/DIMM Side

Memory Configuration	Power Domain	Decoupling Location	Quantity × Value (size)
DDR4 1DPC	VDDQ	4 near each side of DIMM connector close to VDDQ pins	8 × 47uF (0805)
		4 near each side of DIMM connector close to VDDQ pins	8 × 1uF (0402)
	VTT	Place capacitor on VTT plane close to DIMM	1 × 47uF (0805)
		Place capacitor on VTT plane close to DIMM	2 × 1uF (0402)
	VPP	Place capacitor close to DIMM	1 × 47uF (0805)
		Place capacitor close to DIMM	1 × 1uF (0402)
	VDDSPD	Place capacitor close to DIMM	1 × 0.1uF (0402)
		Place capacitor close to DIMM	1 × 2.2uF (0402)

6.4.5. DDR4 Routing Guidelines: Discrete (Component) Topologies

This section discusses the DDR4 x8 and DDR4 x16 memory down topologies. The Agilex 7 M-Series FPGA EMIF IP for DDR4 does not support a clamshell layout.

Altera strongly recommends that you perform simulations using extracted PCB models to ensure that component topologies remain robust under all PCB manufacturing tolerances. Also, carefully consider the number of components on the flyby chain, because every additional component on the flyby chain reduces timing margin on the address/command bus. Take care to provide a proper VTT termination voltage network with a reference voltage that feeds back to the V_{REFCA} input of every component on the flyby chain. Agilex 7 M-Series FPGA circuitry cannot compensate for discontinuities or trace length mismatches along the flyby chain, or for crosstalk between address/command or DQ signals.

6.4.5.1. Single Rank and Dual Rank x 8 Discrete (Component) Topology

This section discusses the single rank x8 and dual rank x8 memory down topology design guideline.

The following design example demonstrates a single rank x 8 memory down topology. You can adjust the design topology based on the actual PCB design (single rank x 8 or dual rank x 8).

The interface covers data bytes (DQ/DQS), address signals, command signals (BA, BG, RAS, CAS, WE, ACT, PAR), control signals (CKE, CS, ODT) and clocks (CLK).

Figure 32. Signal connections for DDR4 Single Rank x 8 Discrete Topology

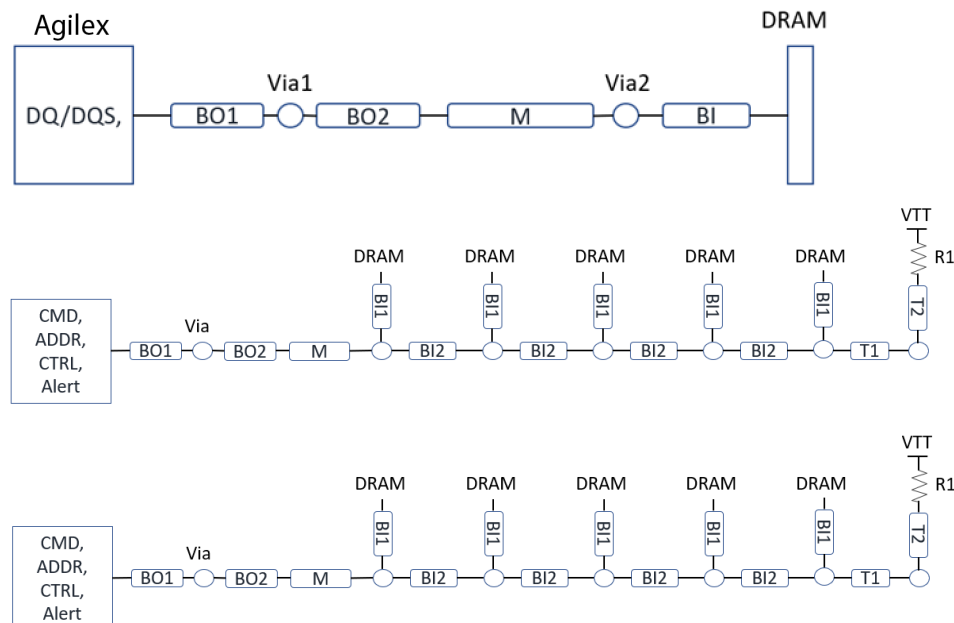


Table 170. Specific Routing Guidelines for Single Rank x8 Discrete Memory Topology for All Supported Signals in the Interface

Signal Group	Segment	Routing Layer	Max Length (mil)		Target Zse (ohm)	Trace Width, W (mil)	Trace Spacing, S1 (mil): Within Group	Trace Spacing, S2 (mil): CMD/CTRL/CLK to DQ/DQS	Trace Spacing, S3 (mil): DQ Nibble to Nibble	Trace Spacing (mil), Within DIFF pair	Trace Spacing (mil), DQS pair to DQ	Trace Spacing (mil), CLK pair to CMD/CTRL/CKE	Rtt / Ctt
			Segment	Total MB									
CLK	BO1	US	50	To first DRAM: 4000 To last DRAM: 9600		4	5, 17	17		4		17	R1=36Ω C1=10nF
	BO2	SL	1000			4	5, 17	17		4		17	
	M	SL			40	5.5		3h		4		3h	
	BI1	US	50			3		3h		4		3h	
	BI2	SL	700		50	3		3h		4		3h	
	T1	SL	300			3		3h		4		3h	
	T2	US	50			3		3h		4		3h	
CMD, CTRL, Alert	BO1	US	50	To first DRAM: 4000 To last DRAM: 9600		4	5, 17	17					R1=36Ω alert_n requires an external pullup resistor to VDD (1.2V) of approximately 1KΩ.
	BO2	SL	1000			4	5, 17	17					
	M	SL			40	5.5	2h	3h					
	BI1	US	50			3	2h	3h					
	BI2	SL	700		50	3	2h	3h					
	T1	SL	300			3	2h	3h					
	T2	US	50			3	2h	3h					
DQ	BO1	US	50	5000		4	5, 17		17				
	BO2	SL	1000			4	5, 17		17				
	M	SL			45	4.5	2h		3h				
	BI	US	50			4	2h		3h				
DQS	BO1	US	50	5000		4				4	17		
	BO2	SL	1000			4				4	17		
continued...													

Signal Group	Segment	Routing Layer	Max Length (mil)		Target Zse (ohm)	Trace Width, W (mil)	Trace Spacing, S1 (mil): Within Group	Trace Spacing, S2 (mil): CMD/CTRL/CLK to DQ/DQS	Trace Spacing, S3 (mil): DQ Nibble to Nibble	Trace Spacing (mil), Within DIFF pair	Trace Spacing (mil), DQS pair to DQ	Trace Spacing (mil), CLK pair to CMD/CTRL/CKE	Rtt / Ctt
			Segment	Total MB									
	M	SL			45	4.5				4	3h		
	BI	US	50			4				4	3h		

For an explanation of the guidelines represented in this table, refer to the bullet points immediately following Figure 28 on page 160.

6.4.5.2. Single Rank x16 and Dual Rank x16 Discrete (Component) Topology

This section discuss the single rank x16 and dual rank x16 memory down topology design guidelines.

The following design example demonstrates a single rank x 16 memory down topology. You can adjust the design topology based on the actual PCB design (single rank x 16 or dual rank x 16)

The interface covers data bytes (DQ/DQS), address signals, command signals (BA, BG, RAS, CAS, WE, ACT, PAR), control signals (CKE, CS, ODT) and clocks (CLK).

Figure 33. Signal connections for DDR4 Single Rank × 16 Discrete Topology

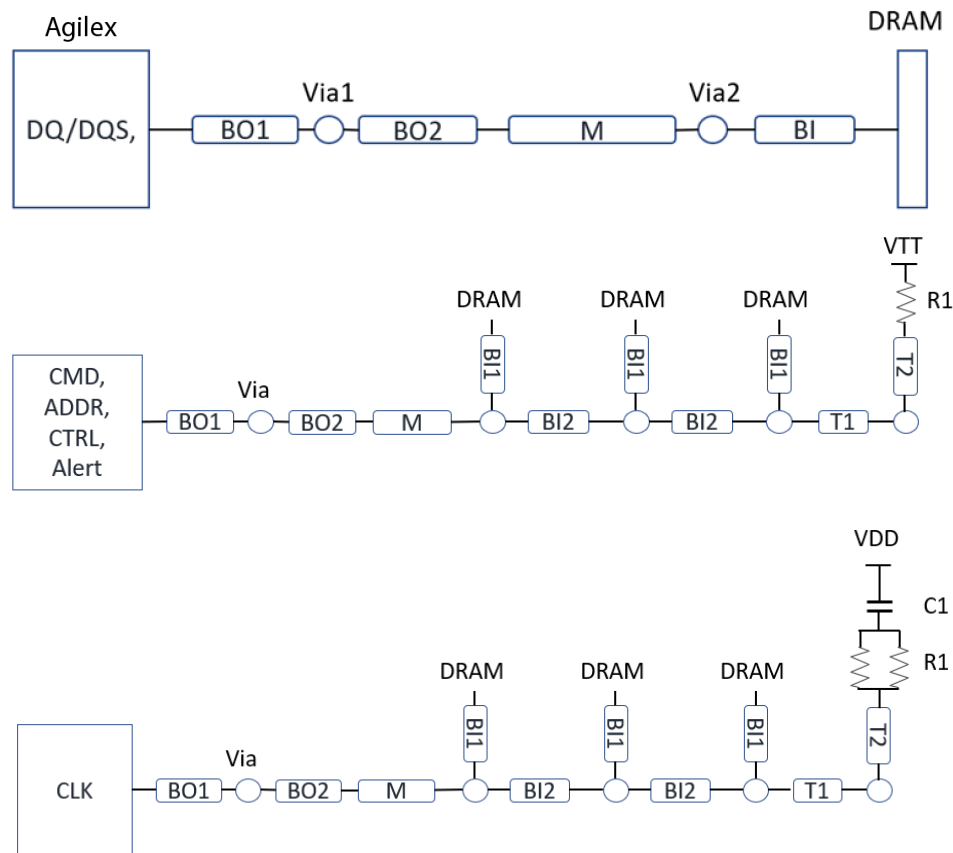


Table 171. Specific Routing Guidelines for Single Rank x16 Discrete Memory Topology for All Supported Signals in the Interface

Signal Group	Segment	Routing Layer	Max Length (mil)		Target Zse (ohm)	Trace Width, W (mil)	Trace Spacing, S1 (mil): Within Group	Trace Spacing, S2 (mil): CMD/CTRL/CLK to DQ/DQS	Trace Spacing, S3 (mil): DQ Nibble to Nibble	Trace Spacing (mil): Within DIFF pair	Trace Spacing (mil): DQS pair to DQ	Trace Spacing (mil), CLK pair to CMD/CTRL/CKE	Rtt / Ctt
			Segment	Total MB									
CLK	BO1	US	50	To first DRAM: 4000. To last DRAM: 6800.		4	5, 17	17		4		17	R1=36Ω. C1=10nF
	BO2	SL	1000			4	5, 17	17		4		17	
	M	SL			40	5.5		3h		4		3h	
	BI1	US	50			3		3h		4		3h	
	BI2	SL	700		50	3		3h		4		3h	
	T1	SL	300			3		3h		4		3h	
	T2	US	50			3		3h		4		3h	
CMD, CTRL, Alert	BO1	US	50	To first DRAM: 4000. To last DRAM: 6800.		4	5, 17	17					R1=36Ω alert_n requires an external pullup resistor to VDD (1.2V) of approximately 1KΩ.
	BO2	SL	1000			4	5, 17	17					
	M	SL			40	5.5	2H	3h					
	BI1	US	50			3	2H	3h					
	BI2	SL	700		50	3	2H	3h					
	T1	SL	300			3	2H	3h					
	T2	US	50			3	2H	3h					
DQ	BO1	US	50	5000		4	5, 17		17				
	BO2	SL	1000			4	5, 17		17				
	M	SL			45	4.5	2H		3h				
	BI	US	50			4	2H		3h				
DQS	BO1	US	50	5000		4				4	17		
	BO2	SL	1000			4				4	17		

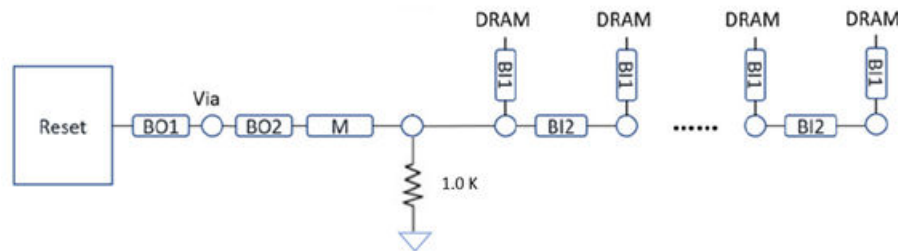
Signal Group	Segment	Routing Layer	Max Length (mil)		Target Zse (ohm)	Trace Width, W (mil)	Trace Spacing, S1 (mil): Within Group	Trace Spacing, S2 (mil): CMD/CTRL/CLK to DQ/DQS	Trace Spacing, S3 (mil): DQ Nibble to Nibble	Trace Spacing (mil), Within DIFF pair	Trace Spacing (mil), DQS pair to DQ	Trace Spacing (mil), CLK pair to CMD/CTRL/CKE	Rtt / Ctt
			Segment	Total MB									
	M	SL			45	4.5				4	3h		
	BI	US	50			4				4	3h		
For an explanation of the guidelines represented in this table, refer to the bullet points immediately following Figure 28 on page 160.													

6.4.5.3. ADDR/CMD Reference Voltage/RESET Signal Routing Guidelines for Single Rank x 8 and Single Rank x 16 Discrete (Component) Topologies

The target impedance for the RESET signal is 50 ohms. The RESET signal must have at least $3 \times h$ (where h is the distance from the trace to the nearest reference plane) spacing to other nearby signals on the same layer. The end-to-end RESET trace length is not limited but must not exceed 5 inches to the first DRAM. Altera recommends using a 1K-ohm pull-down resistor for reset signal termination.

The following figure shows the RESET routing scheme, which you can apply to both single rank x 8 and single rank x16 topologies.

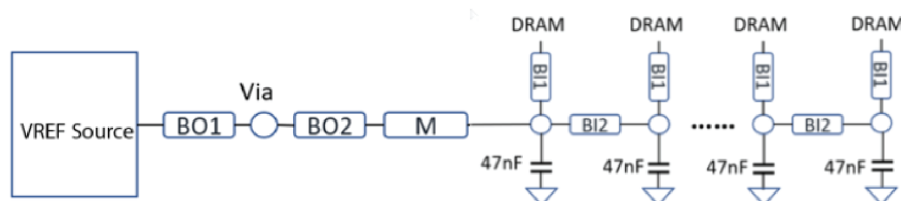
Figure 34. RESET Scheme for Single Rank x8 and Single Rank x16 Topologies



The Address/Command reference voltage input (V_{REF_CA}) must track the V_{TT} regulator output as closely as possible. There are two methods to achieve this:

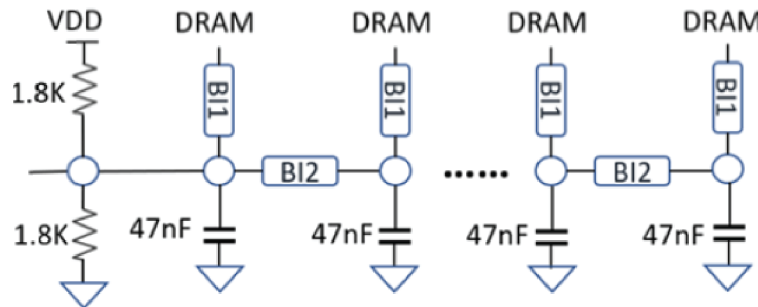
One method is to use a regulator that provides a dedicated tracking voltage reference output that can be connected directly to memory component V_{REF_CA} inputs, as shown in the figure below.

Figure 35. VTT Regulator Supplies V_{REF_CA} Output



A second method is to create a voltage divider using precision resistors. Place the resistor network in a location that is likely to track IR losses on the V_{DD} supply due to memory loading (that is, close to the V_{TT} regulator or memory components, rather than next to the V_{DD} regulator output). The following figure illustrates this configuration.

Figure 36. Resistor Divider Provides the V_{REF_CA} Signal



Altera recommends using a PCB trace width of at least 10 mils for V_{REF_CA} routing. The V_{REF_CA} signal must have at least $3 \times h$ spacing (where h is the distance or height from the trace to the nearest reference plane) to other nearby signals on the same layer.

6.4.5.4. Skew Matching Guidelines for DDR4 Discrete Configurations

This topic describes skew matching guidelines for single rank x 8 and single rank x 16 topologies.

Observe the following rules when skew matching DDR4 discrete configurations:

- Perform skew matching in time (picoseconds) rather than in actual trace length, to better account for via delays when signals are routed on different layers.
- Include both package per-pin skew and PCB delay when performing skew matching.
- Skew (length) matching for the alert signal is not required.

The following table provides skew matching guidelines for DDR4 down-memory topologies.

Table 172. Skew Matching Guidelines for DDR4 Discrete Topologies

DDR4 Device-down Length Matching Rules	Length Matching in Time (ps)
Skew matching between DQS and CLK	$-85\text{ps} < \text{CLK} - \text{DQS} < 935\text{ps}$
Skew matching between DQ and DQS within byte	$-3.5\text{ps} < \text{DQ} - \text{DQS} < 3.5\text{ps}$
Skew matching between DQS and DQS#	$< 1\text{ps}$
Skew matching between CLK and CLK#	$< 1\text{ps}$
Skew matching between CMD/ADDR/CTRL and Clock	$-20\text{ps} < \text{CLK} - \text{CMD/ADDR/CTRL} < 20\text{ps}$
Skew matching among CMD/ADDR/CTRL within each channel	$< 20\text{ps}$
Include package length in skew matching for FPGA device with no migration	Required
Include package length in skew matching for FPGA device with migration when all package net length are available	It is recommended to use the final migrated package net length
Include package length in skew matching for FPGA device with migration when all package net length are not available	Not recommended

6.4.5.5. Power Delivery Recommendations for DDR4 Discrete Configurations

This topic describes power distribution network (PDN) design guidelines for the memory side in discrete topologies.

Note: For information on power distribution network design at the FPGA to meet timing margins, refer to the Agilex 7 M-Series PDN design guidelines.

In the following table, the number of decoupling capacitors is based on a single channel. If multiple channels are sharing the same power rail, the number of decoupling capacitors at the memories must be scaled accordingly for all channels.

Physically small decoupling capacitors are recommended to minimize area, inductance, and resistance on the PDN path on the printed circuit board.

Table 173. Required Decoupling Capacitors on the PCB for the Memory Side

Memory Configuration	Power Domain	Decoupling Location	Quantity x Value (size)
Discrete (Component) Single Rank x8	VDDQ/VDD shorted	4 near each x8 DRAM device	36 x 1uF (0402)
		Distribute around DRAM devices	9 x 10uF (0603)
	VPP	2 near each x8 DRAM device	18 x 1uF (0402)
		Distribute around DRAM devices	5 x 10uF (0603)
	VTT	Place near Rtt (termination resistors)	16 x 1uF (0402)
		Place near Rtt (termination resistors)	4 x 10uF (0603)
Discrete (Component) Single Rank x16	VDDQ/VDD shorted	4 near each x16 DRAM device	18 x 1uF (0402)
		Distribute around DRAM devices	5 x 10uF (0603)
	VPP	2 near each x16 DRAM device	10 x 1uF (0402)
		Distribute around DRAM devices	3 x 10uF (0603)
	VTT	Place near Rtt (termination resistors)	8 x 1uF (0402)
		Place near Rtt (termination resistors)	2 x 10uF (0603)

6.4.5.6. Agilex 7 M-Series EMIF Pin Swapping Guidelines

In Agilex 7 M-Series devices, EMIF pin swapping is allowed under certain conditions.

A byte lane in an EMIF data byte includes 12 signal pins (pins 0,1,2,3,4,5,6,7,8,9,10,11) at the package level. These 12 x I/O pins are arranged into 6 groups of 2 pins each, called *pairs* (pair 0 for pins 0/1, pair 1 for pins 2/3, pair 2 for pins 4/5, pair 3 for pins 6/7, pair 4 for pins 8/9, and pair 5 for pins 10/11).

6.4.5.6.1. DDR4 Byte Lane Swapping

The data lane can be swapped when the byte-lanes are utilized as DQ/DQS pins. Byte lane swapping on utilized lanes is allowed when you swap all the DQ/DQS/DM/DBI pins in the same byte lane with the other utilized byte lane.

The rules for swapping DQ byte lane are as follows:

- You can only swap between utilized DQ lanes.
- You cannot swap a DQ lane with an AC lane.
- You cannot swap a DQ lane with an ECC lane when out-of-band ECC is enabled. For x40 interfaces, the highest-indexed DQ byte lane cannot be swapped.
- Additional restrictions apply when you use a x16 memory component:
 - ☐ You must place DQ group 0 and DQ group 1 on adjacent byte lanes, unless they are separated by AC Lanes. These 2 groups must be connected to the same x16 memory component.
 - ☐ You must place DQ group 2 and DQ group 3 on adjacent byte lanes, unless they are separated by AC Lanes. These 2 groups must be connected to the same x16 memory component.
 - ☐ If you use only one byte of the x16 memory component, you must use only the lower byte of the memory component.
- Additional restrictions apply in lockstep configuration implemented with 2 adjacent IO96 banks:
 - ☐ Any DQ lane with a letter s prefix must remain in its designated byte lane. You cannot swap an s-prefixed DQ lane with any other byte-lane.
 - ☐ DQ lane swapping between IO96 banks is not allowed. You can only swap between utilized DQ lanes within the same IO96 bank.

Table 174. Byte Lane Swapping

Address/ Command Scheme	Data Width usage	BL7 [P95:P84]	BL6 [P83:P72]	BL5 [P71:P60]	BL4 [P59:P48]	BL3 [P47:P36]	BL2 [P35:P24]	BL1 [P23:P12]	BL0 [P11:P0]
Scheme 2	DDR4 x32 + ECC	DQ[ECC]	DQ[3]	DQ[2]	DQ[1]	AC2	AC1	AC0	DQ[0]
Scheme 2	DDR4 x40	DQ[4]	DQ[3]	DQ[2]	DQ[1]	AC2	AC1	AC0	DQ[0]

Example: DDR4 x 32 +ECC implemented with AC Scheme 2

BL7 is used as ECC DQ lane, while Lane 0, 4, 5 and 6 are used DQ lanes. Byte lane swapping between BL0,4,5,6 is allowed.

Example: DDR4 x 40 implemented with AC Scheme 2

BL0,4,5,6,7 are used as DQ lanes. Byte lane swapping between BL0,4,5,6 is allowed. The highest-index DQ byte lane (that is, DQ[4]), cannot be swapped and must be placed at BL7.

Table 175. Byte Lane Swapping for Lockstep Configuration

A/C Placem ent Option	BL0	BL1	BL2	BL3	BL4	BL5	BL6	BL7	BL0	BL1	BL2	BL3	BL4	BL5	BL6	BL7
	Primary								Secondary							
AC Pri Top Sub-Bank / Sec DQ Bot	DQ[4]	DQ[3]	DQ[2]	DQ[1]	AC1	AC2	AC0	sDQ[0]	DQ[8]	DQ[7]	DQ[6]	DQ[5]	GPIO	GPIO	GPIO	GPIO
AC Pri Bot Sub-Bank / Sec DQ Bot	DQ[0]	AC0	AC1	AC2	DQ[1]	DQ[2]	DQ[3]	sDQ[4]	DQ[8]	DQ[7]	DQ[6]	DQ[5]	GPIO	GPIO	GPIO	GPIO
AC Pri Top Sub-Bank / Sec DQ Top(m)	DQ[4]	DQ[3]	DQ[2]	DQ[1]	AC1	AC2	AC0	sDQ[0]	GPIO	X	X	X	DQ[5]	DQ[6]	DQ[7]	DQ[8]

Example: x72 or x64+ECC Configuration with AC Pri Top Sub-Bank/Sec DQ Bot Placement

In Primary IO96 Bank: sDQ[0] is used for ECC or RUSER/WUSER. It cannot be swapped to other byte lanes.
Byte lane swapping between BL0, 1,2 and 3 is allowed.

In Secondary IO96 Bank: BL0,1,2,3 are used as DQ lanes. Byte lane swapping between BL0,1,2,3 is allowed.

Example: x72 or x64+ECC Configuration with AC Pri Bot Sub-Bank/Sec DQ Bot Placement

In Primary IO96 Bank: sDQ[4] is used for ECC or RUSER/WUSER. It cannot be swapped to other byte lanes.
Byte lane swapping between BL0, 4,5 and 6 is allowed.

In Secondary IO96 Bank: BL0,1,2,3 are used as DQ lanes. Byte lane swapping between BL0,1,2,3 is allowed.

Example: x72 or x64+ECC Configuration with AC Pri Top Sub-Bank / Sec DQ Top(m)

In Primary IO96 Bank: sDQ[0] is used for ECC or RUSER/WUSER. It cannot be swapped to other byte lanes.
Byte lane swapping between BL0, 1,2 and 3 is allowed.

In Secondary IO96 Bank: BL4,5,6,7 are used as DQ lanes. Byte lane swapping between BL4,5,6,7 is allowed.

6.4.5.6.2. DDR4 Address and Command and CLK Lane

Address and command and control signals in a bank cannot be swapped.

Pin mapping must adhere to the requirements defined in the table in the [Address and Command Pin Placement for DDR4](#) topic.

You cannot swap address and command lanes. You cannot swap among AC1/AC2/AC3/AC4 lanes. The address and command lane placement must adhere to the specific placement defined in the table in the [DDR4 Data Width Mapping](#) topic.

The T and C lanes for the CLKt/c cannot be swapped with each other, nor can the T and C lanes for the DQS-T/DQS-N be swapped with each other.

6.4.5.6.3. DDR4 Interface x8 Data Lane

A byte lane in an external memory interface consists of 12 signal pins, denoted 0-11.

For DDR4 interfaces composed of x8 devices, two pins are reserved for DQS-T and DQS-C signals, one pin is reserved for the optional DM/DBI signal, one pin must be reserved, and the remaining eight pins are for DQ signals. One-byte data lane must be

assigned for each byte lane, where the byte lane covers DQ [0:7], DQS_T/DQS_C and DBI_N. The following are EMIF I/O pin swapping restrictions applicable to a DDR4 interface with a x8 data lane:

- DQS_T must go to pin 4 in IO12 pins.
- DQS_C must go to pin 5 in IO12 pins.
- DBI_N must go to pin 6 in IO12 pins. If the interface does not use the DBI_N pin, this pin 6 in IO12 lane must remain unconnected.
- Pin 7 in IO12 lane remains unconnected. Altera recommends that you connect this pin 7 to the T_{DQS} dummy load of the memory component and route it as a differential trace along with DBI_N (pin 6). This facilitates x4 or x8 data interoperability in DIMMs configuration.
- You can connect data byte (DQ [0:7]) to any pins [0,1,2,3,8,9,10,11] in the byte lane. Any permutation within selected pins is permitted.

Table 176. Pin Swapping Rules for DDR4 x8 Interfaces

Pin Index Within Byte Lane	DDR4 x8 Data Lane Function	Swap Consideration
0	DQ Pin	Swap group A
1	DQ Pin	Swap group A
2	DQ Pin	Swap group A
3	DQ Pin	Swap group A
4	DQS-T Pin	Fixed location (not swappable)
5	DQS-C Pin	Fixed location (not swappable)
6	DM/DBI Pin	Fixed location (not swappable)
7	Unused	Fixed location (not swappable)
8	DQ Pin	Swap group A
9	DQ Pin	Swap group A
10	DQ Pin	Swap group A
11	DQ Pin	Swap group A

6.4.5.6.4. DDR4 Interface x4 Data Lane

For DDR4 x4 interfaces, two nibbles must be packed into the same IO12 lane.

Four pins are reserved for DQS_T and DQS_C signals and the remaining eight pins implement the DQ signals. The IO12 lane is divided into upper and lower halves to accommodate each nibble. You cannot swap signals belonging to one nibble with signals belonging to the other nibble. DQ signals within a nibble swap group may be swapped with each other. You may also swap entire nibbles—that is, nibble 0 and nibble 1—with each other provided the DQS pin functionality transfers to the correct pin locations. However, this process is not recommended for JEDEC-compliant DIMM interfaces, as it prohibits the interoperability between DIMMs constructed with x4 components and DIMMs constructed with x8 components.

The following table lists the supported pin functionality in x4 mode and the pins that may be swapped with each other. Pins belonging to the same swap group may be freely interchanged with each other.

Table 177. Pin Swapping Rules for DDR4 x4

Pin Index Within Byte Lane	DDR4 x4 Data Lane Function	Swap Consideration	
0	DQ Pin (lower nibble)	Swap group A	Nibble 0
1	DQ Pin (lower nibble)	Swap group A	
2	DQ Pin (lower nibble)	Swap group A	
3	DQ Pin (lower nibble)	Swap group A	
4	DQS_T Pin (lower nibble)	Fixed location (not swappable)	
5	DQS_C Pin (lower nibble)	Fixed location (not swappable)	
6	DQS_T Pin (upper nibble)	Fixed location (not swappable)	Nibble 1
7	DQS_C Pin (upper nibble)	Fixed location (not swappable)	
8	DQ Pin (upper nibble)	Swap group B	
9	DQ Pin (upper nibble)	Swap group B	
10	DQ Pin (upper nibble)	Swap group B	
11	DQ Pin (upper nibble)	Swap group B	

- Nibble 1 must correspond to DQS[17:9] on a physical JEDEC-compliant DIMM for x4/x8 interoperability.
- Nibbles 0 and 1 must follow the same skew matching rules among all 12 signals in the IO12 lane as are specified for a x8-based DQS group.

Note:

- Although the current version of the Quartus Prime software may not enforce all of the rules listed in the above table, be aware that all of these rules may be enforced in later versions of the software.
- At present, the Quartus Prime software checks the following:
 - ☐ Address and command pin placement, per the table in the [Address and Command Pin Placement for DDR4](#) topic, or the *Agilex 7 External Memory Interface Pin Information* file, which is available here: [Pin-Out Files for Altera FPGA Devices](#).
 - ☐ For x8, the Quartus Prime software checks the following:
 - DQS T/C are on pin index 4 and pin index 5 in a byte lane.
 - DM/DBI is on pin index 6.
 - DQ[x] are on pin indices [11:8] and [3:0].
 - ☐ For x4, the Quartus Prime software checks the following:
 - DQS T/C on pin index 4 and pin index 5 and associated DQs are within the corresponding byte lane.
 - DQS T/C on pin index 6 and pin index 7 and associated DQs are within the corresponding byte lane.

You are responsible for ensuring that these conditions are met.
- The Quartus Prime software does not currently check whether DQ pins associated with the lower nibble DQS are actually placed in pin[3:0] or whether DQ pins associated with the upper nibble DQS are actually placed in pin[11:8].

6.4.5.6.5. Pin Swizzling

For information on pin swizzling, refer to [Configuring DQ Pin Swizzling](#) in the *External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide*.

7. Agilex 7 M-Series FPGA EMIF IP – DDR5 Support

This chapter contains IP parameter descriptions and pin planning information for Agilex 7 M-Series FPGA external memory interface - DDR5 Component, and Agilex 7 M-Series FPGA external memory interface IP - DDR5 DIMM.

7.1. Agilex 7 M-Series DDR5 Component External Memory Interfaces (EMIF) IP Parameter Descriptions

The following tables describe the parameters available on each tab of the IP parameter editor, which you can use to configure your IP.

Table 178. Group: High-level Configuration / Memory Device

Parameter Name	Description
Number of Channels	Specifies the number of channels that the interface should implement. For multi-channel devices, this should always match the number of channels on the device. Default value is 1 Legal values are: 1, 2 (Identifier: MEM_NUM_CHANNELS)
Data DQ Width	Number of DQ pins per memory channel, used for data. Default value is 32 Legal values are: 16, 32, 40 (Identifier: MEM_CHANNEL_DATA_DQ_WIDTH)
ECC DQ Width	Number of additional DQ pins per memory channel, used for out-of-band ECC. If bigger than 0, controller will enable out-of-band ECC. Otherwise, out-of-band ECC will be disabled. Default value is 0 Legal values are: 0, 8 (Identifier: MEM_CHANNEL_ECC_DQ_WIDTH)
Die DQ Width	Number of DQ pins in each die that makes up the interface. For dual-die packages, this is the width of the die, not the width of full the package. Default value is 16 Legal values are: 8, 16 (Identifier: MEM_DIE_DQ_WIDTH)
Die Density	Capacity of each memory die (in Gbits), per channel per die. For dual-die packages, this is the density of each die, not the density of the full package. Default value is 8 Legal values are: • 8Gbits (8) • 16Gbits (16) • 24Gbits (24) • 32Gbits (32) (Identifier: MEM_DIE_DENSITY_GBITS)
<i>continued...</i>	

Parameter Name	Description
CS Width	Specifies the total number of CS pins used by each channel. Default value is 1 Legal values are: 1, 2 (Identifier: MEM_CHANNEL_CS_WIDTH)
Memory Speedbin	Specifies the speedbin of the memory device(s) of which the interface consists. Default value is 5600AN Legal values are: • 3200AN (CL24) (3200AN) • 3200B (CL26) (3200B) • 3200BN (CL26) (3200BN) • 3200C (CL28) (3200C) • 3600AN (CL26) (3600AN) • 3600B (CL30) (3600B) • 3600BN (CL30) (3600BN) • 3600C (CL32) (3600C) • 4000AN (CL28) (4000AN) • 4000B (CL32) (4000B) • 4000BN (CL32) (4000BN) • 4000C (CL36) (4000C) • 4400AN (CL32) (4400AN) • 4400B (CL36) (4400B) • 4400BN (CL36) (4400BN) • 4400C (CL40) (4400C) • 4800AN (CL34) (4800AN) • 4800B (CL40) (4800B) • 4800BN (CL40) (4800BN) • 4800C (CL42) (4800C) • 5200AN (CL38) (5200AN) • 5200B (CL42) (5200B) • 5200BN (CL42) (5200BN) • 5200C (CL46) (5200C) • 5600AN (CL40) (5600AN) • 5600B (CL46) (5600B) • 5600BN (CL46) (5600BN) • 5600C (CL50) (5600C) (Identifier: MEM_SPEEDBIN)
Auto-set Memory Operating Frequency	If true, let IP select max frequency that this configuration can support for the current device speedgrade. If false, user can set custom value for operating frequency. Default value is true (Identifier: MEM_OPERATING_FREQ_MHZ_AUTOSET_EN)
Memory Operating Frequency	Specifies the frequency at which the memory interface will run. Value is specified in megahertz Legal values are: 1600, 1800, 2000, 2200, 2400, 2600, 2800 (Identifier: MEM_OPERATING_FREQ_MHZ)

Table 179. Group: High-level Configuration / PHY

Parameter Name	Description
Auto-set PLL Reference Clock Frequency	If true, let IP select max PLL refclk frequency that this configuration can support. If false, user can set custom value for PLL refclk frequency.
<i>continued...</i>	

Parameter Name	Description
	Default value is true (Identifier: PHY_REFCLK_FREQ_MHZ_AUTOSET_EN)
Enable Advanced List of PLL Reference Clock Frequencies	If true, provide extended list of possible refclk values. Otherwise, prune possible list of refclk values to a more reasonable length. Default value is false (Identifier: PHY_REFCLK_ADVANCED_SELECT_EN)
Reference Clock Frequency	Specifies the reference clock frequency for the EMIF IOPLL. Value is specified in megahertz (Identifier: PHY_REFCLK_FREQ_MHZ)
AC Placement	Indicates location on the device where the interface will reside (specifically, the location of the AC lanes in terms I/O BANK and TOP vs BOT part of the I/O BANK). Legal ranges are derived from device floorplan. Default value is BOT Legal values are: • AC Bottom Sub-bank (lanes 0-3) (BOT) • AC Top Sub-bank (lanes 4-7) (TOP) • Ch0 Bot Sub-Bank / Ch1 Top Sub-Bank (FULL) (Identifier: PHY_AC_PLACEMENT)
Auto-set Mainband Access Mode	If true, let IP select most likely usecase for the PHY_MAINBAND_ACCESS_MODE; if false, let user set a custom value for sideband access mode. Default value is true (Identifier: PHY_MAINBAND_ACCESS_MODE_AUTOSET_EN)
Mainband Access Mode	Specifies the path through which the EMIF QHIP mainband interface is exposed to the user. The mainband interface is the AXI4 interface to the memory controller. Legal values are: • NoC (NOC) • Fabric Direct - User Clock Asynchronous to PHY (ASYNC) • Fabric Direct - User Clock Synchronous to PHY (SYNC) (Identifier: PHY_MAINBAND_ACCESS_MODE)
Auto-set Sideband Access Mode	If true, let IP select most likely usecase for the PHY_SIDEHAND_ACCESS_MODE; if false, let user set a custom value for sideband access mode. Default value is true (Identifier: PHY_SIDEHAND_ACCESS_MODE_AUTOSET_EN)
Sideband Access Mode	Specifies the path through which the EMIF QHIP sideband interface is exposed to the user. The sideband interface is the AXI4-Lite interface to the IOSSM. Legal values are: • NoC (NOC) • Fabric Direct (FABRIC) (Identifier: PHY_SIDEHAND_ACCESS_MODE)
Pin Swizzle Map	Specifies the swizzle map for the data lanes and pins. (Identifier: PHY_SWIZZLE_MAP)
Use Debug Toolkit	If enabled, the AXI-L port will be connected to SLD nodes, allowing for a system-console avalon manager interface to interact with this AXI-L subordinate interface. Default value is false
continued...	

Parameter Name	Description
	(Identifier: DEBUG_TOOLS_EN)
Instance ID	Instance ID of the EMIF IP. This is useful when using a discovery mechanism over the side-band interface, to identify which EMIF instance's mailbox is at which offset. If expecting to use a discovery mechanism in hardware, this parameter must be set uniquely for all EMIFs that share a sideband. Otherwise, this parameter can be ignored / kept at the default value. Default value is 0 Legal values are: from 0 to 6 (Identifier: INSTANCE_ID)

Table 180. Group: High-level Configuration / Controller / Controller Options

Parameter Name	Description
Use Data Masking	Specifies whether Data Masking is enabled by the controller. RMW will occur under these conditions: AXI transactions using a subset of write strobe, or one beat long AXI transaction in DDR5, LPDDR5, or LPDDR4 interfaces. Altera recommends enabling the DM option to improve efficiency for the following cases: non-ECC design with AXI transaction listed above, and LPDDR4 or LPDDR5 designs with In-Line ECC enabled. Default value is false (Identifier: CTRL_DM_EN)
Controller Performance Profile	Group of Controller settings to optimize performance based on expected traffic patterns. Choose "Custom" for fine-granular control Default value is SEQ Legal values are: • Sequential Access Optimized (SEQ) • Random Access Optimized (RAND) • Custom (CUSTOM) • Sequential Access with Simultaneous Read/Write Optimized (SEQ_SIMU) (Identifier: CTRL_PERFORMANCE_PROFILE)
Force Auto-Precharge	Controller will terminate all transactions with an auto-precharge when enabled. For per-transaction granularity, leave this option disabled and use the AxUSER control signals instead. Default value is false (Identifier: CTRL_AUTO_PRECHARGE_EN)
Bank Group Rotation	Moves a number of Bank Group pins lower in the address order Default value is 2 Legal values are: • No BGs (disabled) (0) • Two BGs (BG[0]) (1) • Four BGs (BG[1:0]) (2) • Eight BGs (BG[2:0]) (3) (Identifier: CTRL_BG_ROTATE_EN)
Enable Row/CS Swap	Specifies whether the highest order address bits are the CS bit (false) or Row bits (true) Default value is false (Identifier: DIAG_HMC_ADDR_SWAP_EN)

Table 181. Group: High-level Configuration / Controller / Data Bus Turnaround Times

Parameter Name	Description
Additional read-to-write turnaround time (same rank)	Additional memory clock cycles of delay between read and write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2W_SAMECS_CYC)
Additional read-to-read turnaround time (same rank)	Additional memory clock cycles of delay between read operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2R_SAMECS_CYC)
Additional write-to-write turnaround time (same rank)	Additional memory clock cycles of delay between write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2W_SAMECS_CYC)
Additional write-to-read turnaround time (same rank)	Additional memory clock cycles of delay between read and write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2R_SAMECS_CYC)
Additional read-to-write turnaround time (different ranks)	Additional controller cycles of delay between read and write operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2W_DIFFCS_CYC)
Additional read-to-read turnaround time (different ranks)	Additional controller cycles of delay between read operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2R_DIFFCS_CYC)
Additional write-to-write turnaround time (different ranks)	Additional controller cycles of delay between write operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2W_DIFFCS_CYC)
Additional write-to-read turnaround time (different ranks)	Additional controller cycles of delay between write and read operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2R_DIFFCS_CYC)

Table 182. Group: High-level Configuration / Advanced Calibration Settings

Parameter Name	Description
Enable Margining during Calibration	Enable margining stages of calibration for use with the EMIF Debug Toolkit. This will increase calibration time Default value is false (Identifier: ADV_CAL_ENABLE_MARGIN)
Enable REQ Calibration	Enable the Read DQ Equalization stage as part of calibration. This will increase calibration time but will also improve top-line performance Default value is false (Identifier: ADV_CAL_ENABLE_REQ)
Enable WEQ Calibration	Run the Write DQ Equalization stage as part of calibration. This will increase calibration time but will also improve top-line performance Default value is false (Identifier: ADV_CAL_ENABLE_WEQ)
Enable Read DFE Training	Enable DFE training as part of calibration Default value is false (Identifier: ADV_CAL_ENABLE_RD_DFE)
Enable Write DFE Training	Enable DFE training as part of calibration Default value is false (Identifier: ADV_CAL_ENABLE_WR_DFE)

Table 183. Group: Advanced: Memory Timing / Overrides / JEDEC_TABLE

Parameter Name	Description
JEDEC Parameter	Name of JEDEC Parameter to explicitly override; the values will be applied and appear in the list below. Default value is Legal values are: • Operating Speedbin (MEM_OPERATING_SPEEDBIN) • Read Latency (MEM_CL_CYC) • Write Latency (MEM_CWL_CYC) • Write Preamble Mode (MEM_WR_PREAMBLE_MODE) • Read Preamble Mode (MEM_RD_PREAMBLE_MODE) • Write Postamble Mode (MEM_WR_POSTAMBLE_MODE) • Read Postamble Mode (MEM_RD_POSTAMBLE_MODE) • Memory Fine Granularity Refresh Mode (MEM_FINE_GRANULARITY_REFRESH_MODE) • tREFI1 (MEM_TREFI1_NS) • tREFI2 (MEM_TREFI2_NS) • tREFISB (MEM_TREFISB_NS) • tCCD_S (MEM_TCCD_S_CYC) • tCCD_L (MEM_TCCD_L_NS) • tCCD_L_WR (MEM_TCCD_L_WR_NS) • tCCD_L_WR2 (MEM_TCCD_L_WR2_NS) • tRRD_S (MEM_TRRD_S_CYC) • tRRD_L (MEM_TRRD_L_NS) • tFAW (MEM_TFAW_NS) • tRFC1 (MEM_TRFC1_NS) • tRFC2 (MEM_TRFC2_NS) • tRFCSB (MEM_TRFCSB_NS) • tRCD (MEM_TRCD_NS) • tRP (MEM_TRP_NS)

Parameter Name	Description
	• tRAS (MEM_TRAS_NS) • tRC (tRAS+tRP) (MEM_TRC_NS) • tREFSBRD (MEM_TREFSBRD_NS) • tWR (MEM_TWR_NS) • tZQLAT (MEM_TZQLAT_NS) • tZQCAL (MEM_TZQCAL_NS) • tMRR (MEM_TMRR_NS) • tMRR_P (MEM_TMRR_P_NS) • tMRW (MEM_TMRW_NS) • tMRD (MEM_TMRD_NS) • tDFE (MEM_TDFE_NS) • tDLLK (MEM_TDLLK_NS) • tWTR_S (MEM_TWTR_S_NS) • tWTR_L (MEM_TWTR_L_NS) • tRTP (MEM_TRTP_NS) • tPPD (MEM_TPPD_CYC) • tPD (MEM_TPD_NS) • tACTPDEN (MEM_TACTPDEN_CYC) • tPRPDEN (MEM_TPRPDEN_CYC) • tREFPDEN (MEM_TREFPDEN_CYC) • tXP (MEM_TXP_NS) • tCPDED (MEM_TCPDED_CYC) • tCSL (MEM_TCSL_NS) • tCKSRX (MEM_TCKSRX_NS) • tCSH_SREXIT (MEM_TCSH_SREXIT_NS) • tDQSCK_MIN (MEM_TDQSCK_MIN_CYC) • tDQSCK_MAX (MEM_TDQSCK_MAX_CYC) • tDQSCK (MEM_TDQSCK_CYC) • tWPRE_EN (MEM_TWPRE_EN_CYC) • tDQSS (MEM_TDQSS_CYC) • tCKLCS (MEM_TCKLCS_CYC) • tWTRA (MEM_TWTRA_NS) (Identifier: JEDEC_OVERRIDE_TABLE_PARAM_NAME)

Table 184. Group: Advanced: Memory Timing / Values

Parameter Name	Description
Operating Speedbin	Specifies the operating speedbin of the memory device(s) for the current operating frequency and device speedbin. (Identifier: MEM_OPERATING_SPEEDBIN)
Read Latency	Read Latency of the memory device in clock cycles. (Identifier: MEM_CL_CYC)
Write Latency	Write Latency in clock cycles. (Identifier: MEM_CWL_CYC)
Write Preamble Mode	Specifies the write preamble mode of the memory interface (0: not supported, 1: 2-cycle preamble, 2: 3-cycle preamble, 3: 4-cycle preamble). (Identifier: MEM_WR_PREAMBLE_MODE)
Read Preamble Mode	Specifies the read preamble mode of the memory interface (0: 1-cycle preamble, 1: 2-cycle preamble, 2: 2-cycle DDR4-style preamble, 3: 3-cycle preamble, 4: 4-cycle preamble). (Identifier: MEM_RD_PREAMBLE_MODE)
<i>continued...</i>	

Parameter Name	Description
Write Postamble Mode	Specifies the write postamble mode of the memory interface (0: 0.5-cycle postamble, 1: 1.5-cycle postamble). (Identifier: MEM_WR_POSTAMBLE_MODE)
Read Postamble Mode	Specifies the read postamble mode of the memory interface (0: 0.5-tCK postamble, 1: 1.5-tCK postamble). (Identifier: MEM_RD_POSTAMBLE_MODE)
Memory Fine Granularity Refresh Mode	Specifies the Fine Granularity Refresh (FGR) mode of the memory interface. (Identifier: MEM_FINE_GRANULARITY_REFRESH_MODE)
tREFI1	Specifies the maximum average refresh interval in normal refresh mode in nanoseconds. (Identifier: MEM_TREFI1_NS)
tREFI2	Specifies the maximum average refresh interval in fine granularity refresh mode in nanoseconds. (Identifier: MEM_TREFI2_NS)
tREFISB	Specifies the maximum average refresh interval in fine granularity and same bank refresh mode in nanoseconds. (Identifier: MEM_TREFISB_NS)
tCCD_S	Specifies the CAS _n to CAS _n command delay for different bank group in cycles. (Identifier: MEM_TCCD_S_CYC)
tCCD_L	Specifies the CAS _n to CAS _n command delay for same bank group in nanoseconds. (Identifier: MEM_TCCD_L_NS)
tCCD_L_WR	Specifies the write CAS _n to write CAS _n command delay for same bank group in nanoseconds. (Identifier: MEM_TCCD_L_WR_NS)
tCCD_L_WR2	Specifies the write CAS _n to write CAS _n command delay for same bank group and the second write is not RMW, in nanoseconds. (Identifier: MEM_TCCD_L_WR2_NS)
tRRD_S	Specifies the Activate-to-Activate command delay to different bank group for 1KB page size in cycles. (Identifier: MEM_TRRD_S_CYC)
tRRD_L	Specifies the Activate-to-Activate command delay to same bank group for 1KB page size in nanoseconds. (Identifier: MEM_TRRD_L_NS)
tFAW	Specifies the four activate window for 1KB page size in nanoseconds. (Identifier: MEM_TFAW_NS)
tRFC1	Specifies the refresh operation delay in normal refresh mode in nanoseconds. (Identifier: MEM_TRFC1_NS)
tRFC2	Specifies the refresh operation delay in fine granularity refresh mode in nanoseconds. (Identifier: MEM_TRFC2_NS)
tRFCSB	Specifies the refresh operation delay in fine granularity and same bank refresh mode in nanoseconds. (Identifier: MEM_TRFCSB_NS)
<i>continued...</i>	

Parameter Name	Description
tRCD	Specifies the Activate-to-internal-Read-or-Write delay in nanoseconds. (Identifier: MEM_TRCD_NS)
tRP	Specifies the row precharge time in nanoseconds. (Identifier: MEM_TRP_NS)
tRAS	Specifies the Activate-to-Precharge command period in nanoseconds. (Identifier: MEM_TRAS_NS)
tRC (tRAS+tRP)	Specifies the Activate-to-Activate or Refresh command period in nanoseconds. (Identifier: MEM_TRC_NS)
tREFSBRD	Specifies the same bank refresh to activate delay in nanoseconds. (Identifier: MEM_TREFSBRD_NS)
tWR	Specifies the write recovery time in nanoseconds. (Identifier: MEM_TWR_NS)
tZQLAT	Specifies the ZQ calibration latch time in nanoseconds. (Identifier: MEM_TZQLAT_NS)
tZQCAL	Specifies the ZQ calibration time in nanoseconds. (Identifier: MEM_TZQCAL_NS)
tMRR	Specifies the Mode Register Read (MRR) command period in nanoseconds. (Identifier: MEM_TMRR_NS)
tMRR_P	Specifies the Mode Register Read (MRR) pattern to mode register read pattern command spacing in nanoseconds. (Identifier: MEM_TMRR_P_NS)
tMRW	Specifies the Mode Register Write (MRW) command period in nanoseconds. (Identifier: MEM_TMRW_NS)
tMRD	Specifies the Mode Register Set (MRS) command delay in nanoseconds. (Identifier: MEM_TMRD_NS)
tDFE	Specifies the Decision Feedback Equalization (DFE) Mode Register Write update delay time in nanoseconds. (Identifier: MEM_TDFE_NS)
tDLLK	Specifies the timing of DLLK in nanoseconds. (Identifier: MEM_TDLLK_NS)
tWTR_S	Specifies the delay from start of internal write transaction to internal read command for different bank group in nanoseconds. (Identifier: MEM_TWTR_S_NS)
tWTR_L	Specifies the delay from start of internal write transaction to internal read command for same bank group in nanoseconds. (Identifier: MEM_TWTR_L_NS)
tRTP	Specifies the internal read command to precharge command delay in nanoseconds. (Identifier: MEM_TRTP_NS)
tPPD	Specifies the Precharge-to-Precharge delay in cycles. (Identifier: MEM_TPPD_CYC)
<i>continued...</i>	

Parameter Name	Description
tPD	Specifies the minimum power down time in nanoseconds. (Identifier: MEM_TPD_NS)
tACTPDEN	Specifies the timing of Activate command to power down entry command in cycles. (Identifier: MEM_TACTPDEN_CYC)
tPRPDEN	Specifies the timing of Precharge All Banks (PREab), Precharge Same Bank (PREsb), or Normal Precharge (PREpb) to power down entry command in cycles. (Identifier: MEM_TPRPDEN_CYC)
tREFPDEN	Specifies the timing of Refresh All Banks (REFab) or Refresh Same Bank (REFsb) command to power down entry command in cycles. (Identifier: MEM_TREFPDEN_CYC)
tXP	Specifies the exit power down to next valid command in nanoseconds. (Identifier: MEM_TXP_NS)
tCPDED	Specifies the command pass disable delay in nanoseconds. (Identifier: MEM_TCPDED_CYC)
tCSL	Specifies the Self-Refresh CS_n low pulse width in nanoseconds. (Identifier: MEM_TCSL_NS)
tCKSRX	Specifies the valid clock requirement before SRX in nanoseconds. (Identifier: MEM_TCKSRX_NS)
tCSH_SREXIT	Specifies the self-refresh exit CS_n high pulse width in nanoseconds. (Identifier: MEM_TCSH_SREXIT_NS)
tDQSCK_MIN	Specifies the minimum DQS_t, DQS_c rising edge output timing location from rising CK_t, CK_c in cycles. (Identifier: MEM_TDQSCK_MIN_CYC)
tDQSCK_MAX	Specifies the maximum DQS_t, DQS_c rising edge output timing location from rising CK_t, CK_c in cycles. (Identifier: MEM_TDQSCK_MAX_CYC)
tDQSCK	Specifies the DQS_t, DQS_c rising edge output timing location from rising CK_t, CK_c in cycles. (Identifier: MEM_TDQSCK_CYC)
tWPRE_EN	Specifies the write preamble enable window in cycles. The window size depends on the write preamble mode. (Identifier: MEM_TWPRE_EN_CYC)
tDQSS	Specifies the host and system voltage/temperature drift window of first rising DQS_t preamble edge relative to CAS Write Latency (CWL) CK_t-CK_c edge in cycles. (Identifier: MEM_TDQSS_CYC)
tCKLCS	Specifies the valid clock requirement after SRE in cycles. (Identifier: MEM_TCKLCS_CYC)
tWTRA	Specifies the delay from start of internal write transaction to internal read with auto precharge command for same bank in nanoseconds. (Identifier: MEM_TWTRA_NS)

Table 185. Group: Advanced: Analog Overrides / Overrides / ANALOG_TABLE

Parameter Name	Description
Analog Parameter	Name of Analog Parameter to explicitly override; the values will be applied and appear in the list below. Default value is Legal values are: • Phy AC Drive Strength (PHY_TERM_X_R_S_AC_OUTPUT_OHM) • Phy CK Drive Strength (PHY_TERM_X_R_S_CK_OUTPUT_OHM) • Phy DQ Drive Strength (PHY_TERM_X_R_S_DQ_OUTPUT_OHM) • Phy DQ Slew Rate (PHY_TERM_X_DQ_SLEW_RATE) • Phy DQ Input Termination (PHY_TERM_X_R_T_DQ_INPUT_OHM) • Phy DQ Initial Vrefin (PHY_TERM_X_DQ_VREF) • Phy PLL Reference Clock Input Termination (PHY_TERM_X_R_T_REFCLK_INPUT_OHM) • Phy DFE Tap 1 (PHY_DFE_X_TAP_1) • Phy DFE Tap 2 (PHY_DFE_X_TAP_2) • Phy DFE Tap 3 (PHY_DFE_X_TAP_3) • Phy DFE Tap 4 (PHY_DFE_X_TAP_4) • Mem Target Write Termination (MEM_ODT_DQ_X_TGT_WR) • Mem Non-Target Write Termination (MEM_ODT_DQ_X_NON_TGT_WR) • Mem Non-Target Read Termination (MEM_ODT_DQ_X_NON_TGT_RD) • Mem DQ Idle Termination (MEM_ODT_DQ_X_IDLE) • Mem DQ Drive Strength (MEM_ODT_DQ_X_VALUE) • Mem VrefDQ Value (MEM_VREF_DQ_X_VALUE) • Mem CA Termination (MEM_ODT_CA_X_CA) • Mem CS Termination (MEM_ODT_CA_X_CS) • Mem CK Termination (MEM_ODT_CA_X_CK) • Mem VrefCA Value (MEM_VREF_CA_X_CA_VALUE) • Mem VrefCS Value (MEM_VREF_CA_X_CS_VALUE) • Mem DFE Tap 1 (MEM_DFE_X_TAP_1) • Mem DFE Tap 2 (MEM_DFE_X_TAP_2) • Mem DFE Tap 3 (MEM_DFE_X_TAP_3) • Mem DFE Tap 4 (MEM_DFE_X_TAP_4) (Identifier: ANALOG_PARAM_DERIVATION_PARAM_NAME)

Table 186. Group: Advanced: Analog Overrides / Values

Parameter Name	Description
Phy AC Drive Strength	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the refclk input pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 34 Ohm (with calibration) (SERIES_34_OHM_CAL) • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_AC_OUTPUT_OHM)
Phy CK Drive Strength	This parameter allows you to change the output on chip termination settings for the selected I/O standard on the CK Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 34 Ohm (with calibration) (SERIES_34_OHM_CAL) • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_CK_OUTPUT_OHM)
<i>continued...</i>	

Parameter Name	Description
Phy DQ Drive Strength	This parameter allows you to change the output on chip termination settings for the selected I/O standard on the DQ Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 34 Ohm (with calibration) (SERIES_34_OHM_CAL) • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_DQ_OUTPUT_OHM)
Phy DQ Slew Rate	Specifies the slew rate of the data bus pins. The slew rate (or edge rate) describes how quickly the signal can transition, measured in voltage per unit time. <i>Perform board simulations to determine the slew rate that provides the best eye opening for the data bus signals.</i> Legal values are: • Slow (SLOW) • Medium (MEDIUM) • Fast (FAST) • Fastest (FASTEST) (Identifier: PHY_TERM_X_DQ_SLEW_RATE)
Phy DQ Input Termination	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the DQ Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 40 Ohm (with calibration) (RT_40_OHM_CAL) • 50 Ohm (with calibration) (RT_50_OHM_CAL) • 60 Ohm (with calibration) (RT_60_OHM_CAL) (Identifier: PHY_TERM_X_R_T_DQ_INPUT_OHM)
Phy DQ Initial Vrefin	Specifies the initial value for the reference voltage on the data pins(Vrefin). The specified value serves as a starting point and may be overridden by calibration to provide better timing margins. Legal values are: from 0 to 100 (Identifier: PHY_TERM_X_DQ_VREF)
Phy PLL Reference Clock Input Termination	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the refclk input pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • No Termination (RT_OFF) • Differential Termination (RT_DIFF) (Identifier: PHY_TERM_X_R_T_REFCLK_INPUT_OHM)
Phy DFE Tap 1	This parameter allows you to select the amount of bias used on tap 1 of the FPGA DFE. Legal values are: • No Bias (0) • -77.00mV (n1) • -154.00mV (n2) • -231.00mV (n3) • -308.00mV (n4) • -385.00mV (n5) • -462.00mV (n6) • -539.00mV (n7) • -616.00mV (n8) • -693.00mV (n9) • -770.00mV (n10) • -847.00mV (n11)

continued...

Parameter Name	Description
	• -924.00mV (n12) • -1001.00mV (n13) • -1078.00mV (n14) • -1155.00mV (n15) • -1232.00mV (n16) • -1309.00mV (n17) • -1386.00mV (n18) • -1463.00mV (n19) • -1540.00mV (n20) • -1617.00mV (n21) • -1694.00mV (n22) • -1771.00mV (n23) • -1848.00mV (n24) • -1925.00mV (n25) • -2002.00mV (n26) • -2079.00mV (n27) • -2156.00mV (n28) • -2233.00mV (n29) • -2310.00mV (n30) • -2387.00mV (n31) (Identifier: PHY_DFE_X_TAP_1)
Phy DFE Tap 2	This parameter allows you to select the amount of bias used on tap 2 of the FPGA DFE. Legal values are: • +269.50mV (p7) • +231.00mV (p6) • +192.50mV (p5) • +154.00mV (p4) • +115.50mV (p3) • +77.00mV (p2) • +38.50mV (p1) • No Bias (0) • -38.50mV (n1) • -77.00mV (n2) • -115.50mV (n3) • -154.00mV (n4) • -192.50mV (n5) • -231.00mV (n6) • -269.50mV (n7) • -308.00mV (n8) (Identifier: PHY_DFE_X_TAP_2)
Phy DFE Tap 3	This parameter allows you to select the amount of bias used on tap 3 of the FPGA DFE. Legal values are: • +269.50mV (p7) • +231.00mV (p6) • +192.50mV (p5) • +154.00mV (p4) • +115.50mV (p3) • +77.00mV (p2) • +38.50mV (p1) • No Bias (0) • -38.50mV (n1)

continued...

Parameter Name	Description
	• -77.00mV (n2) • -115.50mV (n3) • -154.00mV (n4) • -192.50mV (n5) • -231.00mV (n6) • -269.50mV (n7) • -308.00mV (n8) (Identifier: PHY_DFE_X_TAP_3)
Phy DFE Tap 4	This parameter allows you to select the amount of bias used on tap 4 of the FPGA DFE. Legal values are: • +115.50mV (p3) • +77.00mV (p2) • +38.50mV (p1) • No Bias (0) • -38.50mV (n1) • -77.00mV (n2) • -115.50mV (n3) • -154.00mV (n4) (Identifier: PHY_DFE_X_TAP_4)
Mem Target Write Termination	Specifies the target termination to be used during a write. The value of this parameter represents X, where: $\text{termination} = \text{RZQ}/X = (240 \text{ Ohm})/X$. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_TGT_WR)
Mem Non-Target Write Termination	Specifies the termination to be used for the non-target rank in a multi-rank configuration during a write. The value of this parameter represents X, where: $\text{termination} = \text{RZQ}/X = (240 \text{ Ohm})/X$. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_NON_TGT_WR)
Mem Non-Target Read Termination	Specifies the termination to be used for the non-target rank in a multi-rank configuration during a read. The value of this parameter represents X, where: $\text{termination} = \text{RZQ}/X = (240 \text{ Ohm})/X$. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2)

continued...

Parameter Name	Description
	• 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_NON_TGT_RD)
Mem DQ Idle Termination	Specifies the termination to be used for RTT_PARK and DQS_RTT_PARK. For power savings it is recommended to leave this as disabled. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_IDLE)
Mem DQ Drive Strength	Specifies the termination to be used when driving read data from memory. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. Legal values are: • 34 Ohm (RZQ/7) (7) • 40 Ohm (RZQ/6) (6) • 48 Ohm (RZQ/5) (5) (Identifier: MEM_ODT_DQ_X_RON)
Mem VrefDQ Value	Specifies the initial VrefDQ value to be used. Legal values are: from 35.00 to 97.50 (Identifier: MEM_VREF_DQ_X_VALUE)
Mem CA Termination	Specifies the termination to be used for the CA bus. This setting only applies to Group B, Group A will always be unterminated. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. "off" means this termination is disabled. Legal values are: • Disable (off) • 480 Ohm (RZQ/0.5) (0p5) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 40 Ohm (RZQ/6) (6) (Identifier: MEM_ODT_CA_X_CA)
Mem CS Termination	Specifies the termination to be used for the CS bus. This setting only applies to Group B, Group A will always be unterminated. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. "off" means this termination is disabled. Legal values are: • Disable (off) • 480 Ohm (RZQ/0.5) (0p5) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2)

continued...

Parameter Name	Description
	• 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 40 Ohm (RZQ/6) (6) (Identifier: MEM_ODT_CA_X_CS)
Mem CK Termination	Specifies the termination to be used for the CK bus. This setting only applies to Group B, Group A will always be unterminated. The value of this parameter represents X, where: termination = RZQ/X = (240 Ohm)/X. "off" means this termination is disabled. Legal values are: • Disable (off) • 480 Ohm (RZQ/0.5) (0p5) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 40 Ohm (RZQ/6) (6) (Identifier: MEM_ODT_CA_X_CK)
Mem VrefCA Value	Specifies the initial VrefCA value to be used. Legal values are: from 35.00 to 97.50 (Identifier: MEM_VREF_CA_X_CA_VALUE)
Mem VrefCS Value	Specifies the initial VrefCS value to be used. Legal values are: from 35.00 to 97.50 (Identifier: MEM_VREF_CA_X_CS_VALUE)
Mem DFE Tap 1	This parameter allows you to select the amount of bias used on tap 1 of the memory DFE. Legal values are: • +50mV (p10) • +45mV (p9) • +40mV (p8) • +35mV (p7) • +30mV (p6) • +25mV (p5) • +20mV (p4) • +15mV (p3) • +10mV (p2) • +5mV (p1) • No Bias (0) • -5mV (n1) • -10mV (n2) • -15mV (n3) • -20mV (n4) • -25mV (n5) • -30mV (n6) • -35mV (n7) • -40mV (n8) • -45mV (n9) • -50mV (n10) • -55mV (n11) • -60mV (n12) • -65mV (n13) • -70mV (n14)

continued...

Parameter Name	Description
	• -75mV (n15) • -80mV (n16) • -85mV (n17) • -90mV (n18) • -95mV (n19) • -100mV (n20) • -105mV (n21) • -110mV (n22) • -115mV (n23) • -120mV (n24) • -125mV (n25) • -130mV (n26) • -135mV (n27) • -140mV (n28) • -145mV (n29) • -150mV (n30) • -155mV (n31) • -160mV (n32) • -165mV (n33) • -170mV (n34) • -175mV (n35) • -180mV (n36) • -185mV (n37) • -190mV (n38) • -195mV (n39) • -200mV (n40) (Identifier: MEM_DFE_X_TAP_1)
Mem DFE Tap 2	This parameter allows you to select the amount of bias used on tap 2 of the memory DFE. Legal values are: • +75mV (p15) • +70mV (p14) • +65mV (p13) • +60mV (p12) • +55mV (p11) • +50mV (p10) • +45mV (p9) • +40mV (p8) • +35mV (p7) • +30mV (p6) • +25mV (p5) • +20mV (p4) • +15mV (p3) • +10mV (p2) • +5mV (p1) • No Bias (0) • -5mV (n1) • -10mV (n2) • -15mV (n3) • -20mV (n4) • -25mV (n5) • -30mV (n6) • -35mV (n7)

continued...

Parameter Name	Description
	• -40mV (n8) • -45mV (n9) • -50mV (n10) • -55mV (n11) • -60mV (n12) • -65mV (n13) • -70mV (n14) • -75mV (n15) (Identifier: MEM_DFE_X_TAP_2)
Mem DFE Tap 3	This parameter allows you to select the amount of bias used on tap 3 of the memory DFE. Legal values are: • +60mV (p12) • +55mV (p11) • +50mV (p10) • +45mV (p9) • +40mV (p8) • +35mV (p7) • +30mV (p6) • +25mV (p5) • +20mV (p4) • +15mV (p3) • +10mV (p2) • +5mV (p1) • No Bias (0) • -5mV (n1) • -10mV (n2) • -15mV (n3) • -20mV (n4) • -25mV (n5) • -30mV (n6) • -35mV (n7) • -40mV (n8) • -45mV (n9) • -50mV (n10) • -55mV (n11) • -60mV (n12) (Identifier: MEM_DFE_X_TAP_3)
Mem DFE Tap 4	This parameter allows you to select the amount of bias used on tap 4 of the memory DFE. Legal values are: • +45mV (p9) • +40mV (p8) • +35mV (p7) • +30mV (p6) • +25mV (p5) • +20mV (p4) • +15mV (p3) • +10mV (p2) • +5mV (p1) • No Bias (0) • -5mV (n1) • -10mV (n2)

Parameter Name	Description
	• -15mV (n3) • -20mV (n4) • -25mV (n5) • -30mV (n6) • -35mV (n7) • -40mV (n8) • -45mV (n9) (Identifier: MEM_DFE_X_TAP_4)

Table 187. Group: Example Design / Fileset Types

Parameter Name	Description
HDL Selection	This option lets you choose the format of HDL in which generated simulation and synthesis files are created. You can select either Verilog or VHDL. Default value is VERILOG Legal values are: • Verilog (VERILOG) • VHDL (VHDL) (Identifier: EX_DESIGN_HDL_FORMAT)
Generate Synthesis Fileset	Generate Synthesis Example Design. Default value is true (Identifier: EX_DESIGN_GEN_SYNTH)
Generate Simulation Fileset	Generate Simulation Example Design. Default value is true (Identifier: EX_DESIGN_GEN_SIM)

Table 188. Group: Example Design / User PLL

Parameter Name	Description
Auto-set User PLL Output Clock Frequency	If true, let IP select a reference clock frequency for the user PLL in the example design; if false, let user set a custom value for this parameter. Default value is true (Identifier: EX_DESIGN_USER_PLL_OUTPUT_FREQ_MHZ_AUTOSSET_EN)
User PLL Output Clock Frequency	Frequency of the core clock in MHz. This clock drives the traffic generator and NoC initiator (If in NoC mode). Value is specified in megahertz Default value is 570 (Identifier: EX_DESIGN_USER_PLL_OUTPUT_FREQ_MHZ)
User PLL Reference Clock Frequency	PLL reference clock frequency in MHz for PLL supplying the core clock. Value is specified in megahertz Default value is 100 (Identifier: EX_DESIGN_USER_PLL_REFCLK_FREQ_MHZ)
NOC Reference Clock Frequency	Reference Clock Frequency for the NOC control IP. Value is specified in megahertz Default value is 100 Legal values are: 25, 100, 125 (Identifier: EX_DESIGN_NOC_PLL_REFCLK_FREQ_MHZ)

Table 189. Group: Example Design / Traffic Generator

Parameter Name	Description
Traffic Generator Remote Access	Specifies whether the Traffic Generator control and status registers are accessible via JTAG, exported to the fabric, or just disabled. Default value is JTAG Legal values are: • Exported for Onchip control (EXPORT) • Remote through JTAG (JTAG) (Identifier: EX_DESIGN_TG_CSR_ACCESS_MODE)
Traffic Generator Program	Specifies the traffic pattern to be run. Default value is MEDIUM Legal values are: • Short Traffic Pattern (SHORT) • Medium-length Traffic Pattern (MEDIUM) • Long Traffic Pattern (LONG) • Infinite Traffic Pattern (INFINITE) (Identifier: EX_DESIGN_TG_PROGRAM)

Table 190. Group: Example Design / Performance Monitor

Parameter Name	Description
Enable Performance Monitor for Channel 0	If true, example design will include a Performance Monitor instance connected to Channel 0. Default value is false (Identifier: EX_DESIGN_PMON_CH0_EN)
Enable Performance Monitor for Channel 1	If true, example design will include a Performance Monitor instance connected to Channel 1. Default value is false (Identifier: EX_DESIGN_PMON_CH1_EN)

7.2. Agilex 7 M-Series DDR5 DIMM External Memory Interfaces (EMIF) IP Parameter Descriptions

The following tables describe the parameters available on each tab of the IP parameter editor, which you can use to configure your IP.

Table 191. Group: High-level Configuration / Memory Device

Parameter Name	Description
DIMM Type	Specifies the type of DIMM that is used with this interface. Default value is UDIMM Legal values are: UDIMM, SODIMM, RDIMM (Identifier: MEM_DIMM_TYPE)
Data DQ Width	Number of DQ pins per memory channel, used for data. Default value is 32 Legal values are: 32, 40 (Identifier: MEM_CHANNEL_DATA_DQ_WIDTH)
ECC DQ Width	Number of additional DQ pins per memory channel, used for out-of-band ECC. If bigger than 0, controller will enable out-of-band ECC. Otherwise, out-of-band ECC will be disabled.
<i>continued...</i>	

Parameter Name	Description
	Default value is 4 Legal values are: 0, 4, 8 (Identifier: MEM_CHANNEL_ECC_DQ_WIDTH)
Die DQ Width	Number of DQ pins in each die that makes up the interface. For dual-die packages, this is the width of the die, not the width of full the package. Default value is 8 Legal values are: 4, 8, 16 (Identifier: MEM_DIE_DQ_WIDTH)
Die Density	Capacity of each memory die (in Gbits), per channel per die. For dual-die packages, this is the density of each die, not the density of the full package. Default value is 8 Legal values are: • 8Gbits (8) • 16Gbits (16) • 24Gbits (24) • 32Gbits (32) (Identifier: MEM_DIE_DENSITY_GBITS)
CS Width	Specifies the total number of CS pins used by each channel. Default value is 1 Legal values are: 1, 2 (Identifier: MEM_CHANNEL_CS_WIDTH)
Memory Speedbin	Specifies the speedbin of the memory device(s) of which the interface consists. Default value is 5600AN Legal values are: • 3200AN (CL24) (3200AN) • 3200B (CL26) (3200B) • 3200BN (CL26) (3200BN) • 3200C (CL28) (3200C) • 3600AN (CL26) (3600AN) • 3600B (CL30) (3600B) • 3600BN (CL30) (3600BN) • 3600C (CL32) (3600C) • 4000AN (CL28) (4000AN) • 4000B (CL32) (4000B) • 4000BN (CL32) (4000BN) • 4000C (CL36) (4000C) • 4400AN (CL32) (4400AN) • 4400B (CL36) (4400B) • 4400BN (CL36) (4400BN) • 4400C (CL40) (4400C) • 4800AN (CL34) (4800AN) • 4800B (CL40) (4800B) • 4800BN (CL40) (4800BN) • 4800C (CL42) (4800C) • 5200AN (CL38) (5200AN) • 5200B (CL42) (5200B) • 5200BN (CL42) (5200BN) • 5200C (CL46) (5200C) • 5600AN (CL40) (5600AN)

continued...

Parameter Name	Description
	• 5600B (CL46) (5600B) • 5600BN (CL46) (5600BN) • 5600C (CL50) (5600C) (Identifier: MEM_SPEEDBIN)
Auto-set Memory Operating Frequency	If true, let IP select max frequency that this configuration can support for the current device speedgrade. If false, user can set custom value for operating frequency. Default value is true (Identifier: MEM_OPERATING_FREQ_MHZ_AUTOSET_EN)
Memory Operating Frequency	Specifies the frequency at which the memory interface will run. Value is specified in megahertz Legal values are: 1600, 1800, 2000, 2200, 2400, 2600, 2800 (Identifier: MEM_OPERATING_FREQ_MHZ)

Table 192. Group: High-level Configuration / PHY

Parameter Name	Description
Auto-set PLL Reference Clock Frequency	If true, let IP select max PLL refclk frequency that this configuration can support. If false, user can set custom value for PLL refclk frequency. Default value is true (Identifier: PHY_REFCLK_FREQ_MHZ_AUTOSET_EN)
Enable Advanced List of PLL Reference Clock Frequencies	If true, provide extended list of possible refclk values. Otherwise, prune possible list of refclk values to a more reasonable length. Default value is false (Identifier: PHY_REFCLK_ADVANCED_SELECT_EN)
Reference Clock Frequency	Specifies the reference clock frequency for the EMIF IOPLL. Value is specified in megahertz (Identifier: PHY_REFCLK_FREQ_MHZ)
AC Placement	Indicates location on the device where the interface will reside (specifically, the location of the AC lanes in terms I/O BANK and TOP vs BOT part of the I/O BANK). Legal ranges are derived from device floorplan. Default value is BOT_BOT Legal values are: • Ch0 Bot Sub-Bank / Ch1 Top Sub-Bank (BOT_TOP) • Ch0 Top Sub-Bank / Ch1 Bot Sub-Bank (TOP_BOT) • Ch0 Bot Sub-Bank / Ch1 Bot Sub-Bank (BOT_BOT) (Identifier: PHY_AC_PLACEMENT)
Auto-set Mainband Access Mode	If true, let IP select most likely usecase for the PHY_MAINBAND_ACCESS_MODE; if false, let user set a custom value for sideband access mode. Default value is true (Identifier: PHY_MAINBAND_ACCESS_MODE_AUTOSET_EN)
Mainband Access Mode	Specifies the path through which the EMIF QHIP mainband interface is exposed to the user. The mainband interface is the AXI4 interface to the memory controller. Legal values are: • NoC (NOC) • Fabric Direct - User Clock Asynchronous to PHY (ASYNC) • Fabric Direct - User Clock Synchronous to PHY (SYNC) (Identifier: PHY_MAINBAND_ACCESS_MODE)

continued...

Parameter Name	Description
Auto-set Sideband Access Mode	If true, let IP select most likely usecase for the PHY_SIDEHAND_ACCESS_MODE; if false, let user set a custom value for sideband access mode. Default value is true (Identifier: PHY_SIDEHAND_ACCESS_MODE_AUTOSET_EN)
Sideband Access Mode	Specifies the path through which the EMIF QHIP sideband interface is exposed to the user. The sideband interface is the AXI4-Lite interface to the IOSSM. Legal values are: NoC (NOC) Fabric Direct (FABRIC) (Identifier: PHY_SIDEHAND_ACCESS_MODE)
Pin Swizzle Map	Specifies the swizzle map for the data lanes and pins. (Identifier: PHY_SWIZZLE_MAP)
Use Debug Toolkit	If enabled, the AXI-L port will be connected to SLD nodes, allowing for a system-console avalon manager interface to interact with this AXI-L subordinate interface. Default value is false (Identifier: DEBUG_TOOLS_EN)
Instance ID	Instance ID of the EMIF IP. This is useful when using a discovery mechanism over the side-band interface, to identify which EMIF instance's mailbox is at which offset. If expecting to use a discovery mechanism in hardware, this parameter must be set uniquely for all EMIFs that share a sideband. Otherwise, this parameter can be ignored / kept at the default value. Default value is 0 Legal values are: from 0 to 6 (Identifier: INSTANCE_ID)

Table 193. Group: High-level Configuration / Controller / Controller Options

Parameter Name	Description
Use ECC Autocorrection	If ECC is enabled, specifies whether single-bit-errors (SBEs) should be corrected or just reported. Default value is true (Identifier: CTRL_ECC_AUTOCORRECT_EN)
Use Data Masking	Specifies whether Data Masking is enabled by the controller. RMW will occur under these conditions: AXI transactions using a subset of write strobe, or one beat long AXI transaction in DDR5, LPDDR5, or LPDDR4 interfaces. Altera recommends enabling the DM option to improve efficiency for the following cases: non-ECC design with AXI transaction listed above, and LPDDR4 or LPDDR5 designs with In-Line ECC enabled. Default value is false (Identifier: CTRL_DM_EN)
Controller Performance Profile	Group of Controller settings to optimize performance based on expected traffic patterns. Choose "Custom" for fine-granular control Default value is SEQ Legal values are: Sequential Access Optimized (SEQ) Random Access Optimized (RAND) Custom (CUSTOM) (Identifier: CTRL_PERFORMANCE_PROFILE)
<i>continued...</i>	

Parameter Name	Description
Force Auto-Precharge	Controller will terminate all transactions with an auto-precharge when enabled. For per-transaction granularity, leave this option disabled and use the AxUSER control signals instead. Default value is false (Identifier: CTRL_AUTO_PRECHARGE_EN)
Bank Group Rotation	Moves a number of Bank Group pins lower in the address order Default value is 2 Legal values are: • No BGs (disabled) (0) • Two BGs (BG[0]) (1) • Four BGs (BG[1:0]) (2) • Eight BGs (BG[2:0]) (3) (Identifier: CTRL_BG_ROTATE_EN)
Enable Row/CS Swap	Specifies whether the highest order address bits are the CS bit (false) or Row bits (true) Default value is false (Identifier: DIAG_HMC_ADDR_SWAP_EN)

Table 194. Group: High-level Configuration / Controller / Data Bus Turnaround Times

Parameter Name	Description
Additional read-to-write turnaround time (same rank)	Additional memory clock cycles of delay between read and write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2W_SAMECS_CYC)
Additional read-to-read turnaround time (same rank)	Additional memory clock cycles of delay between read operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2R_SAMECS_CYC)
Additional write-to-write turnaround time (same rank)	Additional memory clock cycles of delay between write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2W_SAMECS_CYC)
Additional write-to-read turnaround time (same rank)	Additional memory clock cycles of delay between read and write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2R_SAMECS_CYC)
Additional read-to-write turnaround time (different ranks)	Additional controller cycles of delay between read and write operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16
<i>continued...</i>	

Parameter Name	Description
	(Identifier: TURNAROUND_R2W_DIFFCS_CYC)
Additional read-to-read turnaround time (different ranks)	Additional controller cycles of delay between read operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2R_DIFFCS_CYC)
Additional write-to-write turnaround time (different ranks)	Additional controller cycles of delay between write operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2W_DIFFCS_CYC)
Additional write-to-read turnaround time (different ranks)	Additional controller cycles of delay between write and read operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2R_DIFFCS_CYC)

Table 195. Group: High-level Configuration / Advanced Calibration Settings

Parameter Name	Description
Enable Margining during Calibration	Enable margining stages of calibration for use with the EMIF Debug Toolkit. This will increase calibration time Default value is false (Identifier: ADV_CAL_ENABLE_MARGIN)
Enable REQ Calibration	Enable the Read DQ Equalization stage as part of calibration. This will increase calibration time but will also improve top-line performance Default value is false (Identifier: ADV_CAL_ENABLE_REQ)
Enable WEQ Calibration	Run the Write DQ Equalization stage as part of calibration. This will increase calibration time but will also improve top-line performance Default value is false (Identifier: ADV_CAL_ENABLE_WEQ)
Enable Read DFE Training	Enable DFE training as part of calibration Default value is false (Identifier: ADV_CAL_ENABLE_RD_DFE)
Enable Write DFE Training	Enable DFE training as part of calibration Default value is false (Identifier: ADV_CAL_ENABLE_WR_DFE)

Table 196. Group: Advanced: Memory Timing / Overrides / JEDEC_TABLE

Parameter Name	Description
JEDEC Parameter	Name of JEDEC Parameter to explicitly override; the values will be applied and appear in the list below. Default value is

Parameter Name	Description
	Legal values are: • Operating Speedbin (MEM_OPERATING_SPEEDBIN) • Read Latency (MEM_CL_CYC) • Write Latency (MEM_CWL_CYC) • Write Preamble Mode (MEM_WR_PREAMBLE_MODE) • Read Preamble Mode (MEM_RD_PREAMBLE_MODE) • Write Postamble Mode (MEM_WR_POSTAMBLE_MODE) • Read Postamble Mode (MEM_RD_POSTAMBLE_MODE) • Memory Fine Granularity Refresh Mode (MEM_FINE_GANULARITY_REFRESH_MODE) • tREFI1 (MEM_TREFI1_NS) • tREFI2 (MEM_TREFI2_NS) • tREFISB (MEM_TREFISB_NS) • tCCD_S (MEM_TCCD_S_CYC) • tCCD_L (MEM_TCCD_L_NS) • tCCD_L_WR (MEM_TCCD_L_WR_NS) • tCCD_L_WR2 (MEM_TCCD_L_WR2_NS) • tRRD_S (MEM_TRRD_S_CYC) • tRRD_L (MEM_TRRD_L_NS) • tFAW (MEM_TFAW_NS) • tRFC1 (MEM_TRFC1_NS) • tRFC2 (MEM_TRFC2_NS) • tRFCSB (MEM_TRFCSB_NS) • tRCD (MEM_TRCD_NS) • tRP (MEM_TRP_NS) • tRAS (MEM_TRAS_NS) • tRC (tRAS+tRP) (MEM_TRC_NS) • tREFSBRD (MEM_TREFSBRD_NS) • tWR (MEM_TWR_NS) • tZQLAT (MEM_TZQLAT_NS) • tZQCAL (MEM_TZQCAL_NS) • tMRR (MEM_TMRR_NS) • tMRR_P (MEM_TMRR_P_NS) • tMRW (MEM_TMRW_NS) • tMRD (MEM_TMRD_NS) • tDFE (MEM_TDFE_NS) • tDLLK (MEM_TDLLK_NS) • tWTR_S (MEM_TWTR_S_NS) • tWTR_L (MEM_TWTR_L_NS) • tRTP (MEM_TRTP_NS) • tPPD (MEM_TPPD_CYC) • tPD (MEM_TPD_NS) • tACTPDEN (MEM_TACTPDEN_CYC) • tPRPDEN (MEM_TPRPDEN_CYC) • tREFPDEN (MEM_TREFPDEN_CYC) • tXP (MEM_TXP_NS) • tCPDED (MEM_TCPDED_CYC) • tCSL (MEM_TCSL_NS) • tCKSRX (MEM_TCKSRX_NS) • tCSH_SREXIT (MEM_TCSH_SREXIT_NS) • tDQSCK_MIN (MEM_TDQSCK_MIN_CYC) • tDQSCK_MAX (MEM_TDQSCK_MAX_CYC) • tDQSCK (MEM_TDQSCK_CYC) • tWPRE_EN (MEM_TWPRE_EN_CYC) • tDQSS (MEM_TDQSS_CYC)

Parameter Name	Description
	• tCKLCS (MEM_TCKLCS_CYC) • tWTRA (MEM_TWTRA_NS) • RDIMM Serial Presence Detect (SPD) Byte 248 (MEM_SPD248_CK_CONTROL_ENABLE) • RDIMM Serial Presence Detect (SPD) Byte 249 (MEM_SPD249_QCA_CS_ENABLE) • RDIMM Serial Presence Detect (SPD) Byte 250 (MEM_SPD250_QCK_SIGNAL_DRIVER_STRENGTH) • RDIMM Serial Presence Detect (SPD) Byte 252 (MEM_SPD252_QCA_QCS_SIGNAL_DRIVER_STRENGTH) • RDIMM Serial Presence Detect (SPD) Byte 254 (MEM_SPD254_CK_CA_CS_SLEW_RATE) • tRRD_DLR (MEM_TRRD_DLR_NS) • tFAW_DLR (MEM_TFAW_DLR_NS) • tCCD_DLR (MEM_TCCD_DLR_NS) (Identifier: JEDEC_OVERRIDE_TABLE_PARAM_NAME)

Table 197. Group: Advanced: Memory Timing / Values

Parameter Name	Description
Operating Speedbin	Specifies the operating speedbin of the memory device(s) for the current operating frequency and device speedbin. (Identifier: MEM_OPERATING_SPEEDBIN)
Read Latency	Read Latency of the memory device in clock cycles. (Identifier: MEM_CL_CYC)
Write Latency	Write Latency in clock cycles. (Identifier: MEM_CWL_CYC)
Write Preamble Mode	Specifies the write preamble mode of the memory interface (0: not supported, 1: 2-cycle preamble, 2: 3-cycle preamble, 3: 4-cycle preamble). (Identifier: MEM_WR_PREAMBLE_MODE)
Read Preamble Mode	Specifies the read preamble mode of the memory interface (0: 1-cycle preamble, 1: 2-cycle preamble, 2: 2-cycle DDR4-style preamble, 3: 3-cycle preamble, 4: 4-cycle preamble). (Identifier: MEM_RD_PREAMBLE_MODE)
Write Postamble Mode	Specifies the write postamble mode of the memory interface (0: 0.5-cycle postamble, 1: 1.5-cycle postamble). (Identifier: MEM_WR_POSTAMBLE_MODE)
Read Postamble Mode	Specifies the read postamble mode of the memory interface (0: 0.5-tCK postamble, 1: 1.5-tCK postamble). (Identifier: MEM_RD_POSTAMBLE_MODE)
Memory Fine Granularity Refresh Mode	Specifies the Fine Granularity Refresh (FGR) mode of the memory interface. (Identifier: MEM_FINE GRANULARITY_REFRESH_MODE)
tREFI1	Specifies the maximum average refresh interval in normal refresh mode in nanoseconds. (Identifier: MEM_TREFI1_NS)
tREFI2	Specifies the maximum average refresh interval in fine granularity refresh mode in nanoseconds. (Identifier: MEM_TREFI2_NS)
<i>continued...</i>	

Parameter Name	Description
tREFISB	Specifies the maximum average refresh interval in fine granularity and same bank refresh mode in nanoseconds. (Identifier: MEM_TREFISB_NS)
tCCD_S	Specifies the CAS_n to CAS_n command delay for different bank group in cycles. (Identifier: MEM_TCCD_S_CYC)
tCCD_L	Specifies the CAS_n to CAS_n command delay for same bank group in nanoseconds. (Identifier: MEM_TCCD_L_NS)
tCCD_L_WR	Specifies the write CAS_n to write CAS_n command delay for same bank group in nanoseconds. (Identifier: MEM_TCCD_L_WR_NS)
tCCD_L_WR2	Specifies the write CAS_n to write CAS_n command delay for same bank group and the second write is not RMW, in nanoseconds. (Identifier: MEM_TCCD_L_WR2_NS)
tRRD_S	Specifies the Activate-to-Activate command delay to different bank group for 1KB page size in cycles. (Identifier: MEM_TRRD_S_CYC)
tRRD_L	Specifies the Activate-to-Activate command delay to same bank group for 1KB page size in nanoseconds. (Identifier: MEM_TRRD_L_NS)
tFAW	Specifies the four activate window for 1KB page size in nanoseconds. (Identifier: MEM_TFAW_NS)
tRFC1	Specifies the refresh operation delay in normal refresh mode in nanoseconds. (Identifier: MEM_TRFC1_NS)
tRFC2	Specifies the refresh operation delay in fine granularity refresh mode in nanoseconds. (Identifier: MEM_TRFC2_NS)
tRFCSB	Specifies the refresh operation delay in fine granularity and same bank refresh mode in nanoseconds. (Identifier: MEM_TRFCSB_NS)
tRCD	Specifies the Activate-to-internal-Read-or-Write delay in nanoseconds. (Identifier: MEM_TRCD_NS)
tRP	Specifies the row precharge time in nanoseconds. (Identifier: MEM_TRP_NS)
tRAS	Specifies the Activate-to-Precharge command period in nanoseconds. (Identifier: MEM_TRAS_NS)
tRC (tRAS+tRP)	Specifies the Activate-to-Activate or Refresh command period in nanoseconds. (Identifier: MEM_TRC_NS)
tREFSBRD	Specifies the same bank refresh to activate delay in nanoseconds. (Identifier: MEM_TREFSBRD_NS)
tWR	Specifies the write recovery time in nanoseconds.
continued...	

Parameter Name	Description
	(Identifier: MEM_TWR_NS)
tZQLAT	Specifies the ZQ calibration latch time in nanoseconds. (Identifier: MEM_TZQLAT_NS)
tZQCAL	Specifies the ZQ calibration time in nanoseconds. (Identifier: MEM_TZQCAL_NS)
tMRR	Specifies the Mode Register Read (MRR) command period in nanoseconds. (Identifier: MEM_TMRR_NS)
tMRR_P	Specifies the Mode Register Read (MRR) pattern to mode register read pattern command spacing in nanoseconds. (Identifier: MEM_TMRR_P_NS)
tMRW	Specifies the Mode Register Write (MRW) command period in nanoseconds. (Identifier: MEM_TMRW_NS)
tMRD	Specifies the Mode Register Set (MRS) command delay in nanoseconds. (Identifier: MEM_TMRD_NS)
tDFE	Specifies the Decision Feedback Equalization (DFE) Mode Register Write update delay time in nanoseconds. (Identifier: MEM_TDFE_NS)
tDLLK	Specifies the timing of DLLK in nanoseconds. (Identifier: MEM_TDLLK_NS)
tWTR_S	Specifies the delay from start of internal write transaction to internal read command for different bank group in nanoseconds. (Identifier: MEM_TWTR_S_NS)
tWTR_L	Specifies the delay from start of internal write transaction to internal read command for same bank group in nanoseconds. (Identifier: MEM_TWTR_L_NS)
tRTP	Specifies the internal read command to precharge command delay in nanoseconds. (Identifier: MEM_TRTP_NS)
tPPD	Specifies the Precharge-to-Precharge delay in cycles. (Identifier: MEM_TPPD_CYC)
tPD	Specifies the minimum power down time in nanoseconds. (Identifier: MEM_TPD_NS)
tACTPDEN	Specifies the timing of Activate command to power down entry command in cycles. (Identifier: MEM_TACTPDEN_CYC)
tPRPDEN	Specifies the timing of Precharge All Banks (PREab), Precharge Same Bank (PREsb), or Normal Precharge (PREpb) to power down entry command in cycles. (Identifier: MEM_TPRPDEN_CYC)
tREFPDEN	Specifies the timing of Refresh All Banks (REFab) or Refresh Same Bank (REFsb) command to power down entry command in cycles. (Identifier: MEM_TREFPDEN_CYC)
tXP	Specifies the exit power down to next valid command in nanoseconds.
continued...	

Parameter Name	Description
	(Identifier: MEM_TXP_NS)
tCPDED	Specifies the command pass disable delay in nanoseconds. (Identifier: MEM_TCPDED_CYC)
tCSL	Specifies the Self-Refresh CS_n low pulse width in nanoseconds. (Identifier: MEM_TCSL_NS)
tCKSRX	Specifies the valid clock requirement before SRX in nanoseconds. (Identifier: MEM_TCKSRX_NS)
tCSH_SREXIT	Specifies the self-refresh exit CS_n high pulse width in nanoseconds. (Identifier: MEM_TCSH_SREXIT_NS)
tDQCK_MIN	Specifies the minimum DQS_t, DQS_c rising edge output timing location from rising CK_t, CK_c in cycles. (Identifier: MEM_TDQCK_MIN_CYC)
tDQCK_MAX	Specifies the maximum DQS_t, DQS_c rising edge output timing location from rising CK_t, CK_c in cycles. (Identifier: MEM_TDQCK_MAX_CYC)
tDQCK	Specifies the DQS_t, DQS_c rising edge output timing location from rising CK_t, CK_c in cycles. (Identifier: MEM_TDQCK_CYC)
tWPRE_EN	Specifies the write preamble enable window in cycles. The window size depends on the write preamble mode. (Identifier: MEM_TWPRE_EN_CYC)
tDQSS	Specifies the host and system voltage/temperature drift window of first rising DQS_t preamble edge relative to CAS Write Latency (CWL) CK_t-CK_c edge in cycles. (Identifier: MEM_TDQSS_CYC)
tCKLCS	Specifies the valid clock requirement after SRE in cycles. (Identifier: MEM_TCKLCS_CYC)
tWTRA	Specifies the delay from start of internal write transaction to internal read with auto precharge command for same bank in nanoseconds. (Identifier: MEM_TWTRA_NS)
RDIMM Serial Presence Detect (SPD) Byte 248	Specifies the value of SPD Byte 248 as an integer in decimal. (Identifier: MEM_SPD248_CK_CONTROL_ENABLE)
RDIMM Serial Presence Detect (SPD) Byte 249	Specifies the value of SPD Byte 249 as an integer in decimal. (Identifier: MEM_SPD249_QCA_CS_ENABLE)
RDIMM Serial Presence Detect (SPD) Byte 250	Specifies the value of SPD Byte 250 as an integer in decimal. (Identifier: MEM_SPD250_QCK_SIGNAL_DRIVER_STRENGTH)
RDIMM Serial Presence Detect (SPD) Byte 252	Specifies the value of SPD Byte 252 as an integer in decimal. (Identifier: MEM_SPD252_QCA_QCS_SIGNAL_DRIVER_STRENGTH)
RDIMM Serial Presence Detect (SPD) Byte 254	Specifies the value of SPD Byte 254 as an integer in decimal. (Identifier: MEM_SPD254_CK_CA_CS_SLEW_RATE)
tRRD_DLR	Specifies the Activate-to-Activate command delay to different logical ranks in nanoseconds. Only applicable to 3D stacked devices.
continued...	

Parameter Name	Description
	(Identifier: MEM_TTRD_DLR_NS)
tFAW_DLR	Specifies the four activate window for different logical ranks in nanoseconds. Only applicable to 3D stacked devices. (Identifier: MEM_TFAW_DLR_NS)
tCCD_DLR	Specifies the write CAS_n to write CAS_n command delay in different logical ranks, in nanoseconds. Only applicable to 3D stacked devices device. (Identifier: MEM_TCCD_DLR_NS)

Table 198. Group: Advanced: Analog Overrides / Overrides / ANALOG_TABLE

Parameter Name	Description
Analog Parameter	Name of Analog Parameter to explicitly override; the values will be applied and appear in the list below. Default value is Legal values are: • Phy AC Drive Strength (PHY_TERM_X_R_S_AC_OUTPUT_OHM) • Phy CK Drive Strength (PHY_TERM_X_R_S_CK_OUTPUT_OHM) • Phy DQ Drive Strength (PHY_TERM_X_R_S_DQ_OUTPUT_OHM) • Phy DQ Slew Rate (PHY_TERM_X_DQ_SLEW_RATE) • Phy DQ Input Termination (PHY_TERM_X_R_T_DQ_INPUT_OHM) • Phy DQ Initial Vrefin (PHY_TERM_X_DQ_VREF) • Phy PLL Reference Clock Input Termination (PHY_TERM_X_R_T_REFCLK_INPUT_OHM) • Phy DFE Tap 1 (PHY_DFE_X_TAP_1) • Phy DFE Tap 2 (PHY_DFE_X_TAP_2) • Phy DFE Tap 3 (PHY_DFE_X_TAP_3) • Phy DFE Tap 4 (PHY_DFE_X_TAP_4) • Mem Target Write Termination (MEM_ODT_DQ_X_TGT_WR) • Mem Non-Target Write Termination (MEM_ODT_DQ_X_NON_TGT_WR) • Mem Non-Target Read Termination (MEM_ODT_DQ_X_NON_TGT_RD) • Mem DQ Idle Termination (MEM_ODT_DQ_X_IDLE) • Mem DQ Drive Strength (MEM_ODT_DQ_X_RON) • Mem VrefDQ Value (MEM_VREF_DQ_X_VALUE) • Mem CA Termination (MEM_ODT_CA_X_CA) • Mem CS Termination (MEM_ODT_CA_X_CS) • Mem CK Termination (MEM_ODT_CA_X_CK) • Mem VrefCA Value (MEM_VREF_CA_X_CA_VALUE) • Mem VrefCS Value (MEM_VREF_CA_X_CS_VALUE) • Mem DFE Tap 1 (MEM_DFE_X_TAP_1) • Mem DFE Tap 2 (MEM_DFE_X_TAP_2) • Mem DFE Tap 3 (MEM_DFE_X_TAP_3) • Mem DFE Tap 4 (MEM_DFE_X_TAP_4) • Mem RCD CA Input Termination (MEM_RCD_DCA_IBT) • Mem RCD CS Input Termination (MEM_RCD_DCS_IBT) • Mem RCD CK Input Termination (MEM_RCD_DCK_IBT) • Mem RCD Error Input Termination (MEM_RCD_DERROR_IBT) (Identifier: ANALOG_PARAM_DERIVATION_PARAM_NAME)

Table 199. Group: Advanced: Analog Overrides / Values

Parameter Name	Description
Phy AC Drive Strength	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the refclk input pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 34 Ohm (with calibration) (SERIES_34_OHM_CAL) • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_AC_OUTPUT_OHM)
Phy CK Drive Strength	This parameter allows you to change the output on chip termination settings for the selected I/O standard on the CK Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 34 Ohm (with calibration) (SERIES_34_OHM_CAL) • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_CK_OUTPUT_OHM)
Phy DQ Drive Strength	This parameter allows you to change the output on chip termination settings for the selected I/O standard on the DQ Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 34 Ohm (with calibration) (SERIES_34_OHM_CAL) • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_DQ_OUTPUT_OHM)
Phy DQ Slew Rate	Specifies the slew rate of the data bus pins. The slew rate (or edge rate) describes how quickly the signal can transition, measured in voltage per unit time. <i>Perform board simulations to determine the slew rate that provides the best eye opening for the data bus signals.</i> Legal values are: • Slow (SLOW) • Medium (MEDIUM) • Fast (FAST) • Fastest (FASTEST) (Identifier: PHY_TERM_X_DQ_SLEW_RATE)
Phy DQ Input Termination	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the DQ Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 40 Ohm (with calibration) (RT_40_OHM_CAL) • 50 Ohm (with calibration) (RT_50_OHM_CAL) • 60 Ohm (with calibration) (RT_60_OHM_CAL) (Identifier: PHY_TERM_X_R_T_DQ_INPUT_OHM)
Phy DQ Initial Vrefin	Specifies the initial value for the reference voltage on the data pins(Vrefin). The specified value serves as a starting point and may be overridden by calibration to provide better timing margins. Legal values are: from 0 to 100 (Identifier: PHY_TERM_X_DQ_VREF)
Phy PLL Reference Clock Input Termination	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the refclk input pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • No Termination (RT_OFF) • Differential Termination (RT_DIFF) (Identifier: PHY_TERM_X_R_T_REFCLK_INPUT_OHM)

continued...

Parameter Name	Description
Phy DFE Tap 1	This parameter allows you to select the amount of bias used on tap 1 of the FPGA DFE. Legal values are: • No Bias (0) • -77.00mV (n1) • -154.00mV (n2) • -231.00mV (n3) • -308.00mV (n4) • -385.00mV (n5) • -462.00mV (n6) • -539.00mV (n7) • -616.00mV (n8) • -693.00mV (n9) • -770.00mV (n10) • -847.00mV (n11) • -924.00mV (n12) • -1001.00mV (n13) • -1078.00mV (n14) • -1155.00mV (n15) • -1232.00mV (n16) • -1309.00mV (n17) • -1386.00mV (n18) • -1463.00mV (n19) • -1540.00mV (n20) • -1617.00mV (n21) • -1694.00mV (n22) • -1771.00mV (n23) • -1848.00mV (n24) • -1925.00mV (n25) • -2002.00mV (n26) • -2079.00mV (n27) • -2156.00mV (n28) • -2233.00mV (n29) • -2310.00mV (n30) • -2387.00mV (n31) (Identifier: PHY_DFE_X_TAP_1)
Phy DFE Tap 2	This parameter allows you to select the amount of bias used on tap 2 of the FPGA DFE. Legal values are: • +269.50mV (p7) • +231.00mV (p6) • +192.50mV (p5) • +154.00mV (p4) • +115.50mV (p3) • +77.00mV (p2) • +38.50mV (p1) • No Bias (0) • -38.50mV (n1) • -77.00mV (n2) • -115.50mV (n3) • -154.00mV (n4) • -192.50mV (n5)

continued...

Parameter Name	Description
	• -231.00mV (n6) • -269.50mV (n7) • -308.00mV (n8) (Identifier: PHY_DFE_X_TAP_2)
Phy DFE Tap 3	This parameter allows you to select the amount of bias used on tap 3 of the FPGA DFE. Legal values are: • +269.50mV (p7) • +231.00mV (p6) • +192.50mV (p5) • +154.00mV (p4) • +115.50mV (p3) • +77.00mV (p2) • +38.50mV (p1) • No Bias (0) • -38.50mV (n1) • -77.00mV (n2) • -115.50mV (n3) • -154.00mV (n4) • -192.50mV (n5) • -231.00mV (n6) • -269.50mV (n7) • -308.00mV (n8) (Identifier: PHY_DFE_X_TAP_3)
Phy DFE Tap 4	This parameter allows you to select the amount of bias used on tap 4 of the FPGA DFE. Legal values are: • +115.50mV (p3) • +77.00mV (p2) • +38.50mV (p1) • No Bias (0) • -38.50mV (n1) • -77.00mV (n2) • -115.50mV (n3) • -154.00mV (n4) (Identifier: PHY_DFE_X_TAP_4)
Mem Target Write Termination	Specifies the target termination to be used during a write. The value of this parameter represents X, where: termination = RZQ/X = (240 Ohm)/X. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_TGT_WR)
Mem Non-Target Write Termination	Specifies the termination to be used for the non-target rank in a multi-rank configuration during a write. The value of this parameter represents X, where: termination = RZQ/X = (240 Ohm)/X.

continued...

Parameter Name	Description
	Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_NON_TGT_WR)
Mem Non-Target Read Termination	Specifies the termination to be used for the non-target rank in a multi-rank configuration during a read. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_NON_TGT_RD)
Mem DQ Idle Termination	Specifies the termination to be used for RTT_PARK and DQS_RTT_PARK. For power savings it is recommended to leave this as disabled. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) • 34 Ohm (RZQ/7) (7) (Identifier: MEM_ODT_DQ_X_IDLE)
Mem DQ Drive Strength	Specifies the termination to be used when driving read data from memory. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. Legal values are: • 34 Ohm (RZQ/7) (7) • 40 Ohm (RZQ/6) (6) • 48 Ohm (RZQ/5) (5) (Identifier: MEM_ODT_DQ_X_RON)
Mem VrefDQ Value	Specifies the initial VrefDQ value to be used. Legal values are: from 35.00 to 97.50 (Identifier: MEM_VREF_DQ_X_VALUE)
Mem CA Termination	Specifies the termination to be used for the CA bus. This setting only applies to Group B, Group A will always be unterminated. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. "off" means this termination is disabled.

continued...

Parameter Name	Description
	Legal values are: • Disable (off) • 480 Ohm (RZQ/0.5) (0p5) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 40 Ohm (RZQ/6) (6) (Identifier: MEM_ODT_CA_X_CA)
Mem CS Termination	Specifies the termination to be used for the CS bus. This setting only applies to Group B, Group A will always be unterminated. The value of this parameter represents X, where: termination = RZQ/X = (240 Ohm)/X. "off" means this termination is disabled. Legal values are: • Disable (off) • 480 Ohm (RZQ/0.5) (0p5) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 40 Ohm (RZQ/6) (6) (Identifier: MEM_ODT_CA_X_CS)
Mem CK Termination	Specifies the termination to be used for the CK bus. This setting only applies to Group B, Group A will always be unterminated. The value of this parameter represents X, where: termination = RZQ/X = (240 Ohm)/X. "off" means this termination is disabled. Legal values are: • Disable (off) • 480 Ohm (RZQ/0.5) (0p5) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 40 Ohm (RZQ/6) (6) (Identifier: MEM_ODT_CA_X_CK)
Mem VrefCA Value	Specifies the initial VrefCA value to be used. Legal values are: from 35.00 to 97.50 (Identifier: MEM_VREF_CA_X_CA_VALUE)
Mem VrefCS Value	Specifies the initial VrefCS value to be used. Legal values are: from 35.00 to 97.50 (Identifier: MEM_VREF_CA_X_CS_VALUE)
Mem DFE Tap 1	This parameter allows you to select the amount of bias used on tap 1 of the memory DFE. Legal values are: • +50mV (p10) • +45mV (p9) • +40mV (p8) • +35mV (p7) • +30mV (p6) • +25mV (p5) • +20mV (p4)

continued...

Parameter Name	Description
	• +15mV (p3) • +10mV (p2) • +5mV (p1) • No Bias (0) • -5mV (n1) • -10mV (n2) • -15mV (n3) • -20mV (n4) • -25mV (n5) • -30mV (n6) • -35mV (n7) • -40mV (n8) • -45mV (n9) • -50mV (n10) • -55mV (n11) • -60mV (n12) • -65mV (n13) • -70mV (n14) • -75mV (n15) • -80mV (n16) • -85mV (n17) • -90mV (n18) • -95mV (n19) • -100mV (n20) • -105mV (n21) • -110mV (n22) • -115mV (n23) • -120mV (n24) • -125mV (n25) • -130mV (n26) • -135mV (n27) • -140mV (n28) • -145mV (n29) • -150mV (n30) • -155mV (n31) • -160mV (n32) • -165mV (n33) • -170mV (n34) • -175mV (n35) • -180mV (n36) • -185mV (n37) • -190mV (n38) • -195mV (n39) • -200mV (n40) (Identifier: MEM_DFE_X_TAP_1)
Mem DFE Tap 2	This parameter allows you to select the amount of bias used on tap 2 of the memory DFE. Legal values are: • +75mV (p15) • +70mV (p14) • +65mV (p13) • +60mV (p12) • +55mV (p11)

continued...

Parameter Name	Description
	• +50mV (p10) • +45mV (p9) • +40mV (p8) • +35mV (p7) • +30mV (p6) • +25mV (p5) • +20mV (p4) • +15mV (p3) • +10mV (p2) • +5mV (p1) • No Bias (0) • -5mV (n1) • -10mV (n2) • -15mV (n3) • -20mV (n4) • -25mV (n5) • -30mV (n6) • -35mV (n7) • -40mV (n8) • -45mV (n9) • -50mV (n10) • -55mV (n11) • -60mV (n12) • -65mV (n13) • -70mV (n14) • -75mV (n15) (Identifier: MEM_DFE_X_TAP_2)
Mem DFE Tap 3	This parameter allows you to select the amount of bias used on tap 3 of the memory DFE. Legal values are: • +60mV (p12) • +55mV (p11) • +50mV (p10) • +45mV (p9) • +40mV (p8) • +35mV (p7) • +30mV (p6) • +25mV (p5) • +20mV (p4) • +15mV (p3) • +10mV (p2) • +5mV (p1) • No Bias (0) • -5mV (n1) • -10mV (n2) • -15mV (n3) • -20mV (n4) • -25mV (n5) • -30mV (n6) • -35mV (n7) • -40mV (n8) • -45mV (n9)

continued...

Parameter Name	Description
	• -50mV (n10) • -55mV (n11) • -60mV (n12) (Identifier: MEM_DFE_X_TAP_3)
Mem DFE Tap 4	This parameter allows you to select the amount of bias used on tap 4 of the memory DFE. Legal values are: • +45mV (p9) • +40mV (p8) • +35mV (p7) • +30mV (p6) • +25mV (p5) • +20mV (p4) • +15mV (p3) • +10mV (p2) • +5mV (p1) • No Bias (0) • -5mV (n1) • -10mV (n2) • -15mV (n3) • -20mV (n4) • -25mV (n5) • -30mV (n6) • -35mV (n7) • -40mV (n8) • -45mV (n9) (Identifier: MEM_DFE_X_TAP_4)
Mem RCD CA Input Termination	This parameter allows you to select the input bus termination of RCD CA input bus. See JEDEC specifications on DDR5RCD04 for the pins covered. The values map to specific termination strengths: 0 -> "60 Ohm", 1 -> "48 Ohm", 3 -> "OFF". Legal values are: • 60 Ohm (0) • 48 Ohm (1) • Off (3) (Identifier: MEM_RCD_DCA_IBT)
Mem RCD CS Input Termination	This parameter allows you to select the input bus termination of RCD CS input bus. See JEDEC specifications on DDR5RCD04 for the pins covered. The values map to specific termination strengths: 0 -> "60 Ohm", 1 -> "48 Ohm", 3 -> "OFF". Legal values are: • 60 Ohm (0) • 48 Ohm (1) • Off (3) (Identifier: MEM_RCD_DCS_IBT)
Mem RCD CK Input Termination	This parameter allows you to select the input bus termination of CK input bus. See JEDEC specifications on DDR5RCD04 for the pins covered. The values map to specific termination strengths: 0 -> "60 Ohm", 1 -> "48 Ohm", 3 -> "OFF". Legal values are: • 60 Ohm (0) • 48 Ohm (1) • Off (3)

continued...

Parameter Name	Description
	(Identifier: MEM_RCD_DCK_IBT)
Mem RCD Error Input Termination	This parameter allows you to select the input bus termination of DERROR input bus. See JEDEC specification on DDR5RCD04 for the pins covered. The values map to specific termination strengths: 0 -> "60 Ohm", 1 -> "48 Ohm", 3 -> "OFF". Legal values are: • 60 Ohm (0) • 48 Ohm (1) • Off (3) (Identifier: MEM_RCD_DERROR_IBT)

Table 200. Group: Example Design / Fileset Types

Parameter Name	Description
HDL Selection	This option lets you choose the format of HDL in which generated simulation and synthesis files are created. You can select either Verilog or VHDL. Default value is VERILOG Legal values are: • Verilog (VERILOG) • VHDL (VHDL) (Identifier: EX_DESIGN_HDL_FORMAT)
Generate Synthesis Fileset	Generate Synthesis Example Design. Default value is true (Identifier: EX_DESIGN_GEN_SYNTH)
Generate Simulation Fileset	Generate Simulation Example Design. Default value is true (Identifier: EX_DESIGN_GEN_SIM)

Table 201. Group: Example Design / User PLL

Parameter Name	Description
Auto-set User PLL Output Clock Frequency	If true, let IP select a reference clock frequency for the user PLL in the example design; if false, let user set a custom value for this parameter. Default value is true (Identifier: EX_DESIGN_USER_PLL_OUTPUT_FREQ_MHZ_AUTOSSET_EN)
User PLL Output Clock Frequency	Frequency of the core clock in MHz. This clock drives the traffic generator and NoC initiator (If in NoC mode). Value is specified in megahertz Default value is 570 (Identifier: EX_DESIGN_USER_PLL_OUTPUT_FREQ_MHZ)
User PLL Reference Clock Frequency	PLL reference clock frequency in MHz for PLL supplying the core clock. Value is specified in megahertz Default value is 100 (Identifier: EX_DESIGN_USER_PLL_REFCLK_FREQ_MHZ)
NOC Reference Clock Frequency	Reference Clock Frequency for the NOC control IP. Value is specified in megahertz Default value is 100 Legal values are: 25, 100, 125 (Identifier: EX_DESIGN_NOC_PLL_REFCLK_FREQ_MHZ)

Table 202. Group: Example Design / Traffic Generator

Parameter Name	Description
Traffic Generator Remote Access	Specifies whether the Traffic Generator control and status registers are accessible via JTAG, exported to the fabric, or just disabled. Default value is JTAG Legal values are: • Exported for Onchip control (EXPORT) • Remote through JTAG (JTAG) (Identifier: EX_DESIGN_TG_CSR_ACCESS_MODE)
Traffic Generator Program	Specifies the traffic pattern to be run. Default value is MEDIUM Legal values are: • Short Traffic Pattern (SHORT) • Medium-length Traffic Pattern (MEDIUM) • Long Traffic Pattern (LONG) • Infinite Traffic Pattern (INFINITE) (Identifier: EX_DESIGN_TG_PROGRAM)

Table 203. Group: Example Design / Performance Monitor

Parameter Name	Description
Enable Performance Monitor for Channel 0	If true, example design will include a Performance Monitor instance connected to Channel 0. Default value is false (Identifier: EX_DESIGN_PMON_CH0_EN)
Enable Performance Monitor for Channel 1	If true, example design will include a Performance Monitor instance connected to Channel 1. Default value is false (Identifier: EX_DESIGN_PMON_CH1_EN)

7.3. Agilex 7 M-Series FPGA EMIF IP Pin and Resource Planning

The following topics provide guidelines on pin placement for external memory interfaces.

Typically, all external memory interfaces require the following FPGA resources:

- Interface pins
- PLL and clock network
- RZQ pins
- Other FPGA resources—for example, core fabric logic, and debug interfaces

Once all the requirements are known for your external memory interface, you can begin planning your system.

7.3.1. Agilex 7 M-Series FPGA EMIF IP Interface Pins

Any I/O banks that do not support transceiver operations in Agilex 7 M-Series FPGAs support external memory interfaces. However, DQS (data strobe or data clock) and DQ (data) pins are listed in the device pin tables and are fixed at specific locations in

the device. You must adhere to these pin locations to optimize routing, minimize skew, and maximize margins. Always check the pin table for the actual locations of the DQS and DQ pins.

Note: Maximum interface width varies from device to device depending on the number of I/O pins and DQS or DQ groups available. Achievable interface width also depends on the number of address and command pins that the design requires. To ensure adequate PLL, clock, and device routing resources are available, you should always test fit any IP in the Quartus Prime software before PCB sign-off.

Altera devices do not limit the width of external memory interfaces beyond the following requirements:

- Maximum possible interface width in any particular device is limited by the number of DQS groups available.
- Sufficient clock networks are available to the interface PLL as required by the IP.
- Sufficient spare pins exist within the chosen bank or side of the device to include all other address and command, and clock pin placement requirements.

Note: The greater the number of banks, the greater the skew, hence Altera recommends that you always generate a test project of your desired configuration and confirm that it meets timing.

7.3.1.1. Estimating Pin Requirements

You should use the Quartus Prime software for final pin fitting. However, you can estimate whether you have enough pins for your memory interface by performing the following steps:

1. Determine how many read/write data pins are associated per data strobe or clock pair.
2. Calculate the number of other memory interface pins needed, including any other clocks (write clock or memory system clock), address, command, and RZQ. Refer to the External Memory Interface Pin Table to determine necessary Address/Command/Clock pins based on your desired configuration.
3. Calculate the total number of I/O banks required to implement the memory interface, given that an I/O bank supports up to 96 pins.

Test the proposed pin-outs with the rest of your design in the Quartus Prime software (with the correct I/O standard and OCT connections) before finalizing the pin-outs. There can be interactions between modules that are illegal in the Quartus Prime software that you might not know about unless you compile the design and use the Quartus Prime Pin Planner.

7.3.1.2. DIMM Options

DDR5 unbuffered DIMMs (UDIMMs) and small outline DIMMs (SODIMMs) command and address pins are clocked at single data rate (SDR). DDR5 registered DIMMs (RDIMMs) command and address pins are clocked at double data rate (DDR). The table below shows a pin comparison of UDIMM, SODIMM, and RDIMM modules up to dual rank. You should always check your memory vendor's data sheet to be sure.

Table 204. UDIMM, SODIMM, and RDIMM Pin Options for DDR5

Pins	UDIMM Pins	SODIMM Pins	RDIMM Pins
Data	72 bit DQ[31:0]_A DQ[31:0]_B CB[3:0]_A CB[3:0]_B	72 bit DQ[31:0]_A DQ[31:0]_B CB[3:0]_A CB[3:0]_B	80 bit DQ[31:0]_A DQ[31:0]_B CB[7:0]_A CB[7:0]_B
Data Mask	DM[3:0]_A_n ⁽¹⁾ DM[3:0]_B_n ⁽¹⁾	DM[3:0]_A_n ⁽¹⁾ DM[3:0]_B_n ⁽¹⁾	DM[4:0]_A_n ⁽¹⁾ DM[4:0]_B_n ⁽¹⁾
Data Strobe	x8: DQS[4:0]_A_t DQS[4:0]_A_c DQS[4:0]_B_t DQS[4:0]_B_c	x8: DQS[4:0]_A_t DQS[4:0]_A_c DQS[4:0]_B_t DQS[4:0]_B_c	x8: DQS[4:0]_A_t DQS[4:0]_A_c DQS[4:0]_B_t DQS[4:0]_B_c x4: DQS[9:0]_A_t DQS[9:0]_A_c DQS[9:0]_B_t DQS[9:0]_B_c
Command / Address	CA[12:0]_A CA[12:0]_B CA[1:0]_A_n CA[1:0]_B_n	CA[12:0]_A CA[12:0]_B CA[1:0]_A_n CA[1:0]_B_n	CA[6:0]_A CA[6:0]_B CS[1:0]_A_n CS[1:0]_B_n
Clock	CK[1:0]_A_t CK[1:0]_A_c CK[1:0]_B_t CK[1:0]_B_c	CK[1:0]_A_t CK[1:0]_A_c CK[1:0]_B_t CK[1:0]_B_c	CK_t CK_c
Parity	ALERT_n	ALERT_n	ALERT_n PAR_A PAR_B
Other Pins	RESET_n HSDA, HSCL, HSA	RESET_n HSDA, HSCL, HSA	RESET_n HSDA, HSCLL, HSA LBD/RSP_A_n LBS/RSP_B_n
Notes to Table: 1. DM pins are available only for DIMMs constructed using x8 or greater components. 2. The Agilex 7 M-Series memory controller supports up to 2 ranks per channel. Agilex 7 M-Series devices support only 1 DIMM per channel (1DPC).			

7.3.1.3. Maximum Number of Interfaces

The maximum number of interfaces supported for a given memory protocol varies, depending on the FPGA in use.

Unless otherwise noted, the calculation for the maximum number of interfaces is based on independent interfaces where the address or command pins are not shared.

Note: You may need to share PLL clock outputs depending on your clock network usage.

Table 205. Maximum Number of DDR5 Interfaces

Device	Package	Component Interface	DIMM Interface
AGMA032 / AGMA039 / AGMB032 / AGMB039	R31B	7 - x16 5 - 2ch x16 7 - x16 + ECC 7 - x32 5 - x32 + ECC	3 - 2ch x32 1 - 2ch x32 + ECC
AGMA032 / AGMA039 / AGMB032 / AGMB039 / AGME032 / AGME039 / AGMF032 / AGMF039 / AGMG032 / AGMG039 / AGMH032 / AGMH039	R47A	8	4
AGMF032 / AGMF039 / AGMH032 / AGMH039	R47B	8	4

- *Component Interface* refers to 2ch x16, x16, x16 + ECC, x32 and x32+ ECC which can be implemented within a single IO96B bank.
- One DIMM interface requires two adjacent IO96B banks located on the edge of the device.

Timing closure depends on device resource and routing utilization. For more information about timing closure, refer to the *Area and Timing Optimization Techniques* chapter in the *Quartus Prime Handbook*.

7.3.2. Agilex 7 M-Series FPGA EMIF IP Resources

The Agilex 7 M-Series FPGA memory interface IP uses several FPGA resources to implement the memory interface.

7.3.2.1. OCT

You require an OCT calibration block if you are using an Agilex 7 M-Series FPGA OCT calibrated series, parallel, or dynamic termination for any I/O in your design. There are two OCT blocks in an I/O bank, one for each sub-bank.

You must observe the following requirements when using OCT blocks:

- The I/O bank where you place the OCT calibration block must use the same V_{CCIO_PIO} voltage as the memory interface.
- The OCT calibration block uses a single fixed R_{ZQ} . You must ensure that an external termination resistor is connected to the correct pin for a given OCT block.

For specific pin connection requirements, refer to [Specific Pin Connection Requirements](#).

7.3.2.2. PLL

When using PLL for external memory interfaces, you must consider the following guidelines:

For the clock source, use the clock input pin specifically dedicated to the PLL that you want to use with your external memory interface. The input and output pins are only fully compensated when you use the dedicated PLL clock input pin.

For specific pin connection requirements, refer to [Specific Pin Connection Requirements](#).

7.3.3. Pin Guidelines for Agilex 7 M-Series FPGA EMIF IP

The Agilex 7 M-Series FPGA contains I/O banks on the top and bottom edges of the device, which can be used by external memory interfaces.

Agilex 7 M-Series FPGA I/O banks contain 96 I/O pins. Each bank is divided into two sub-banks with 48 I/O pins in each. Sub-banks are further divided into four I/O lanes, where each I/O lane is a group of twelve I/O ports.

The I/O bank, I/O lane, and pairing pin for every physical I/O pin can be uniquely identified by the following naming convention in the device pin table:

- The I/O pins in a bank are represented as P#X#Y#, where:
 - ☐ P# represents the pin number in a bank. It ranges from P0 to P95, for 96 pins in a bank.
 - ☐ X# represents the bank number on a given edge of the device. X0 is the farthest bank from the zipper.
 - ☐ Y# represents the top or bottom edge of the device. Y0 and Y1 refer to the I/O banks on the bottom and top edge, respectively.
- Because an IO96 bank comprises two IO48 sub-banks, all pins with P# value less than 48 (P# <48) belong to the same I/O sub-bank. All other pins belong to the second IO48 sub-bank.
- The *Index Within I/O Bank* value falls within one of the following ranges: 0 to 11, 12 to 23, 24 to 35, or 36 to 47, and represents one of I/O lanes 0, 1, 2, or 3, respectively.
- To determine whether I/O banks are adjacent, you can refer to the sub-bank-ordering figures for your device family in the [Architecture: I/O Bank](#) topic. In general, you can assume that I/O banks are adjacent within an I/O edge, unless the I/O bank is not bonded out on the package (indicated by the presence of the " - " symbol in the I/O table), or if the I/O bank does not contain 96 pins, indicating that it is only partially bonded out. If an I/O bank is not fully bonded out in a particular device, it cannot be included within the span of sub-banks for a larger external memory interface. In all cases, you should use the Quartus Prime software to verify that your usage can be implemented.
- The pairing pin for an I/O pin is in the same I/O bank. You can identify the pairing pin by adding 1 to its *Index Within I/O Bank* number (if it is an even number), or by subtracting 1 from its *Index Within I/O Bank* number (if it is an odd number).

7.3.3.1. General Guidelines - DDR5

You should follow the recommended guidelines when performing pin placement for all external memory interface pins targeting Agilex 7 M-Series devices, whether you are using the hard memory controller or your own solution.

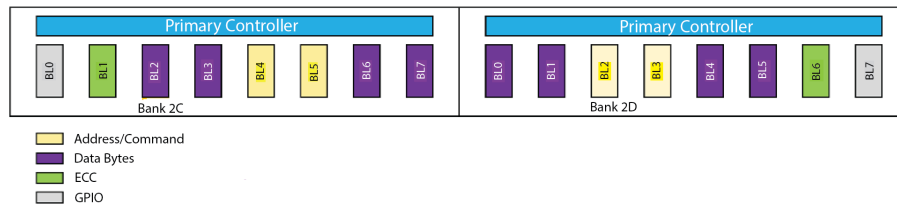
Note: QDRx is not supported with HPS.

Observe the following general guidelines when placing pins for your Agilex 7 M-Series external memory interface:

1. Ensure that the pins of a single external memory interface reside on the same edge I/O.
2. The address and command pins and their associated clock pins in the address and command bank must follow a fixed pin-out scheme, as defined in the table in the [Address and Command Pin Placement for DDR5](#) topic.
3. Not every byte lane can function as an address and command lane or a data lane. The pin assignment must adhere to the DDR5 data width mapping defined in [DDR5 Data Width Mapping](#).
4. A byte lane must not be used by both address and command pins and data pins.
5. An external memory interface can occupy one or more banks on the same edge. When an interface must occupy multiple banks, ensure that those banks are adjacent to one another.
 - If an I/O bank is shared between two interfaces—meaning that two sub-banks belong to two different EMIF interfaces—then both the interfaces must share the same voltage.
 - Sharing of I/O lanes within a sub-bank for two different EMIF interfaces is not permitted; I/O lanes within a sub-bank can be assigned to one EMIF interface only.
6. Any pin in the same bank that is not used by an external memory interface may not be available for use as a general purpose I/O pin:
 - For fabric EMIF, unused pins in an I/O lane assigned to an EMIF interface cannot be used as general-purpose I/O pins. In the same sub-bank, pins in an I/O lane that is not assigned to an EMIF interface, can be used as general-purpose I/O pins.
 - For HPS EMIF, unused pins in an I/O lane assigned to an EMIF interface cannot be used as general-purpose I/O pins. In the same bank, pins in an I/O lane that is not assigned to an EMIF interface cannot be used as general-purpose I/O pins either.
7. All address and command pins and their associated clock pins (CK_t and CK_c) must reside within a single sub-bank. The sub-bank containing the address and command pins is identified as the address and command sub-bank. Refer to the table in [DDR5 Data Width Mapping](#) for the supported address and command and data lane placements for DDR5.
8. The address and command pins and their associated clock pins in the address and command bank must follow a fixed pin-out scheme, as defined in the *Agilex 7 M-Series External Memory Interface Pin Information* file.
9. An external memory interface can occupy one or more banks on the same edge. When an interface must occupy multiple banks, ensure the following:
 - That the banks are adjacent to one another.
 - That you used only the supported data width mapping as defined in the table in [DDR5 Data Width Mapping](#). Be aware that not every byte lane can be used as an address and command lane or a data lane.

The following figure shows one possible pin placement for a DDR5 2ch x32 + ECC interface on Bank 2C and Bank 2D.

Figure 37. x72 DDR5 Pin Placement using Bank 2C and 2D



10. An unused I/O lane in the address and command sub-bank can serve to implement a data group, such as a x8 DQS group. The data group must be from the same controller as the address and command signals.
11. An I/O lane must not be used by both address and command pins and data pins.
12. Place read data groups according to the DQS grouping in the pin table and Pin Planner. Read data strobes (such as DQS_t and DQS_c) must reside at physical pins capable of functioning as DQS_t and DQS_c for a specific read data group size. You must place the associated read data pins (DQ), within the same group.

Note: For DDR5 interfaces with x4 components, place DQ pins and DQS entirely in either the upper or lower half of a 12-bit bank sub-group. Consult the pin table for your device to identify the association between DQ pins and DQS pins for x4 mode operation. Additional restrictions apply for x4/x8 DIMM interoperability.

13. One of the sub-banks in the device (typically the sub-bank within corner bank 3A) may not be available if you use certain device configuration schemes. For some schemes, there may be an I/O lane available for EMIF data group.
 - AVST-8 – This is contained entirely within the SDM, therefore all lanes of sub-bank 3A can be used by the external memory interface.
 - AVST-16/AVST-32– Lanes 4, 5, 6, and 7 are all effectively occupied and are not usable by the external memory interface.
14. Two memory interfaces cannot share an I/O 48 sub-bank.

7.3.3.2. x4 DIMM Implementation

DIMMS using a x4 DQS configuration require remapping of the DQS signals to achieve compatibility between the EMIF IP and the JEDEC standard DIMM socket connections.

The necessary remapping is shown in the table below. You can implement this DQS remapping in either RTL logic or in your schematic wiring connections.

Table 206. Mapping of DQS Signals Between DIMM and the EMIF IP

DIMM			Quartus Prime EMIF IP	
DQS0_A	DQ[3:0]_A		DQS0	DQ[3:0]_A
DQS5_A	DQ[7:4]_A		DQS1	DQ[7:4]_A
DQS1_A	DQ[11:8]_A		DQS2	DQ[11:8]_A
DQS6_A	DQ[15:12]_A		DQS3	DQ[15:12]_A
DQS2_A	DQ[19:16]_A		DQS4	DQ[19:16]_A
continued...				

DIMM			Quartus Prime EMIF IP	
DQS7_A	DQ[23:20]_A		DQS5	DQ[23:20]_A
DQS3_A	DQ[27:24]_A		DQS6	DQ[27:24]_A
DQS8_A	DQ[31:28]_A		DQS7	DQ[31:28]_A
DQS4_A	CB[3:0]_A		DQS8	CB[3:0]_A
DQS9_A	CB[7:4]_A		DQS9	CB[7:4]_A
DQS0_B	DQ[3:0]_B		DQS10	DQ[3:0]_B
DQS5_B	DQ[7:4]_B		DQS11	DQ[7:4]_B
DQS1_B	DQ[11:8]_B		DQS12	DQ[11:8]_B
DQS6_B	DQ[15:12]_B		DQS13	DQ[15:12]_B
DQS2_B	DQ[19:16]_B		DQS14	DQ[19:16]_B
DQS7_B	DQ[23:20]_B		DQS15	DQ[23:20]_B
DQS3_B	DQ[27:24]_B		DQS16	DQ[27:24]_B
DQS8_B	DQ[31:28]_B		DQS17	DQ[31:28]_B
DQS4_B	CB[3:0]_B		DQS18	CB[3:0]_B
DQS9_B	CB[7:4]_B		DQS19	CB[7:4]_B

Data Bus Connection Mapping Flow

1. Connect all FPGA DQ pins accordingly to DIMM DQ pins. No remapping is required.
2. DQS/DQSn remapping is required either on the board schematics or in the RTL code.

When designing a board to support x4 DQS groups, Altera recommends that you make it compatible for x8 mode, for the following reasons:

- Provides the flexibility of x4 and x8 DIMM support.
- Allows use of x8 DQS group connectivity rules.
- Allows use of x8 timing rules for matching. Adhere to x4/x8 interoperability rules when designing a DIMM interface, even if the primary use case is to support x4 DIMMs only, because doing so facilitates debug and future migration capabilities. Regardless, the rules for length matching for two nibbles in a x4 interface must match those of the signals for a corresponding x8 interface, as the data terminations are turned on and off at the same time for both x4 DQS groups in an I/O lane. If the two x4 DQS groups were to have significantly different trace delays, it could adversely affect signal integrity. Trace delays for two nibbles packed within the IO12 lanes are matched using the same guidelines as a single x8 byte lane.

7.3.3.3. Specific Pin Connection Requirements

To compile a design that instantiates an Agilex 7 M-Series EMIF IP, you must explicitly define a legal placement and I/O standard for all EMIF pins.

This topic discusses the required constraints per pin type. To simplify this process, you can leverage the **Back Annotate Assignments** tool. For more information, refer to *Back Annotating Pin Placement and I/O Standard Assignments* in the *External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide*.

PLL

For DDR5, you must constrain the PLL reference clock to the address and command lanes only.

- You must constrain differential reference clocks to pin indices 10 and 11 in lane 2 when placing command address pins in lane 3 and lane 2.
- You must constrain differential reference clocks to pin indices 10 and 11 in lane 4 when placing command address pins in lane 5 and lane 4.
- The sharing of PLL reference clocks across multiple DDR5 interfaces is permitted within an I/O bank.

Note: Lane 3:0 is the bottom sub-bank and lane 7:4 is the top sub-bank.

OCT

For DDR5, you must constrain the RZQ pin to the address and command lanes only.

- You must constrain RZQ to pin index 2 in lane 3 when placing command address pins in lane 3 and lane 2.
- You must constrain RZQ to pin index 2 in lane 5 when placing command address pins in lane 5 and lane 4.
- The sharing of RZQ across multiple DDR5 interfaces is permitted within an I/O bank.

Note: Lane 3:0 is the bottom sub-bank and lane 7:4 is the top sub-bank.

Address / Command / Parity

For DDR5, you must constrain the ALERT_N pin to the address and command lanes only.

- You must constrain ALERT_N to pin index 1 in lane 3 when placing command address pins in lane 3 and lane 2.
- You must constrain ALERT_N to pin index 1 in lane 5 when placing command address pins in lane 5 and lane 4.

Note: Lane 3:0 is the bottom sub-bank and lane 7:4 is the top sub-bank.

DQS/DQ/DM

For DDR5 x8 DQS/DQ/DM grouping, the following rules apply:

- You may use pin indices 0, 1, 2, 3, 8, 9, 10, and 11 within a lane for DQ mode pins only.
- You must use pin index 4 for the DQS_t pin only.
- You must use pin index 5 for the DQS_c pin only.
- You must ensure that pin index 7 remains unused. Pin index 7 is not available for use as a general purpose I/O.
- You must use pin index 6 for the DM pin only.

For DDR5 x4 DQS/DQ/DM grouping, the following rules apply:

- You may use pin indices 0, 1, 2, and 3 within a lane for DQ mode pins for the lower nibble only. Pin rotation within this group is permitted.
- You must use pin index 4 for the DQS_t pin only of the lower nibble.
- You must use pin index 5 for the DQS_c pin only of the lower nibble.
- You may use pin indices 8, 9, 10, and 11 within a lane for the DQ mode pins only for the upper nibble. Pin rotation within this group is permitted.
- You must use pin index 6 for the DQS_t pin only of the upper nibble.
- You must use pin index 7 for the DQS_c pin only of the upper nibble.

7.3.3.4. Command and Address Signals

Command and address signals in SDRAM devices are clocked into the memory device using the CK_T or CK_C signal. These pins operate at single data rate (SDR) using only one clock edge. The number of address pins depends on the SDRAM device capacity. The address pins are multiplexed, so two clock cycles are required to send the row, column, and bank address.

Although DDR5 operates in fundamentally the same way as other SDRAM, there are no dedicated pins for RAS_N, CAS_N, and WE_N, as those are shared with higher-order address pins. DDR5 has CS_N, CKE, ODT, and RESET_N pins, similar to DDR4. DDR5 also has some additional pins, including the ACT_N (activate) pin and BG (bank group) pins.

7.3.3.5. Clock Signals

DDR5 SDRAM devices use CK_t and CK_c signals to clock the address and command signals into the memory.

The memory uses these clock signals to generate the DQS signal during a read through the DLL inside the memory.

7.3.3.6. Data, Data Strobes, DM, and Optional ECC Signals

DDR5 SDRAM devices use bidirectional differential data strobes. Differential DQS operation enables improved system timing due to reduced crosstalk and less simultaneous switching noise on the strobe output drivers. The DQ pins are also bidirectional.

DQ pins in DDR5 SDRAM interfaces can operate in either ×4 or ×8 mode DQS groups, depending on your chosen memory device or DIMM, regardless of interface width. The ×4 and ×8 configurations use one pair of bidirectional data strobe signals, DQS and DQSn, to capture input data.

The DQ signals are edge-aligned with the DQS signal during a read from the memory and are center-aligned with the DQS signal during a write to the memory. The memory controller shifts the DQ signals by –90 degrees during a write operation to center align the DQ and DQS signals. The PHY IP delays the DQS signal during a read, so that the DQ and DQS signals are center aligned at the capture register. Altera devices use a phase-locked loop (PLL) to center-align the DQS signal with respect to the DQ signals during writes and use dedicated DQS phase-shift circuitry to shift the incoming DQS signal during reads.

The memory device's setup (t_{DS}) and hold times (t_{DH}) for the DQ and DM pins during writes are relative to the edges of DQS write signals and not the CK or CK# clock. Setup and hold requirements are not necessarily balanced.

The DQS signal is generated on the positive edge of the system clock to meet the t_{DQSS} requirement. DQ and DM signals use a clock shifted –90 degrees from the system clock, so that the DQS edges are centered on the DQ or DM signals when they arrive at the SDRAM. The DQS, DQ, and DM board trace lengths need to be tightly matched.

The SDRAM uses the DM pins during a write operation. Driving the DM pins low shows that the write is valid. The memory masks the DQ signals if the DM pins are driven high. To generate the DM signal, Altera recommends that you use the spare DQ pin within the same DQS group as the respective data, to minimize skew.

The DM signal's timing requirements at the SDRAM input are identical to those for DQ data. The DDR registers, clocked by the –90 degree shifted clock, create the DM signals.

Some SDRAM modules support error correction coding (ECC) to allow the controller to detect and automatically correct errors in data transmission. UDIMMs or SODIMMs with ECC will have CB[3:0] bits per sub channel. Depending on the RDIMM module you can have CB[7:0] or CB[3:0] bits per sub channel.

7.3.4. Pin Placements for Agilex 7 M-Series FPGA DDR5 EMIF IP

7.3.4.1. Address and Command Pin Placement for DDR5

Table 207. Address and Command Pin Placement

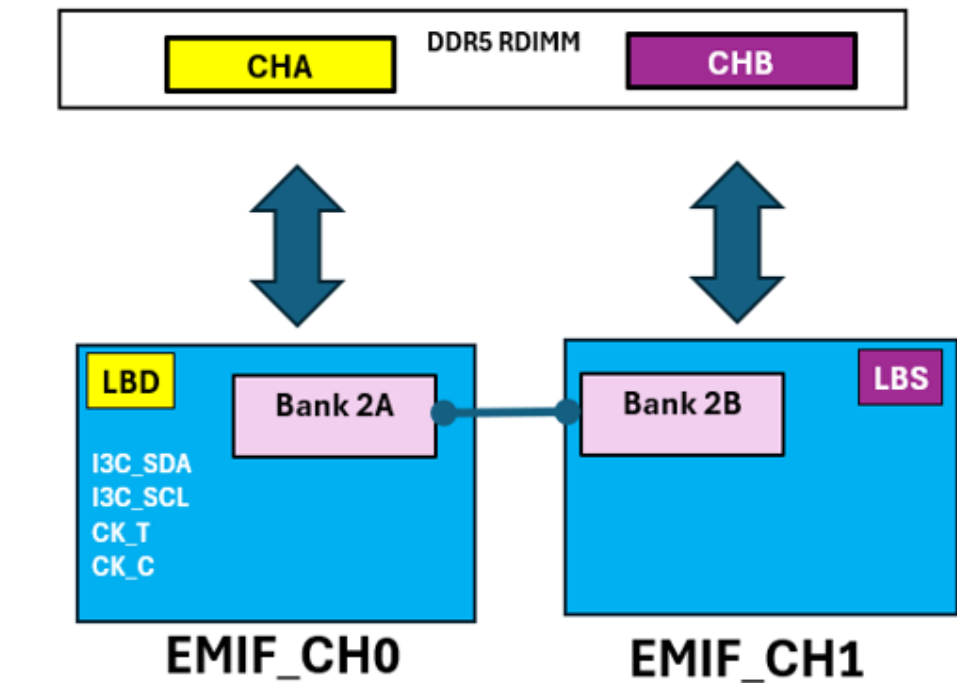
Address/Command Lane	Index Within Byte Lane	Scheme 1 UDIMM/SODIMM/ Component	Scheme 2 RDIMM
AC1	11	CK_C[1]	SCL (i3c) ¹
	10	CK_T[1]	SDA (i3c) ¹
continued...			

Address/Command Lane	Index Within Byte Lane	Scheme 1 UDIMM/SODIMM/ Component	Scheme 2 RDIMM
	9	CS_N[0]	CS_N[0] ³
	8	CS_N[1]	CS_N[1] ³
	7	CK_C[0]	CK_C[0] ¹
	6	CK_T[0]	CK_T[0] ¹
	5	CA[12]	
	4	CA[11]	
	3	RESET_N	RESET_N ¹
	2	RZQ Site	
	1	ALERT_N	ALERT_N
	0	CA[10]	
AC0	11	Differential "N-Side" reference clock input site	
	10	Differential "P-Side" reference clock input site	
	9	CA[9]	LBD, RSP_A_n ¹
	8	CA[8]	LBS, RSP_B_n ²
	7	CA[7]	PAR_A
	6	CA[6]	CA[6]
	5	CA[5]	CA[5]
	4	CA[4]	CA[4]
	3	CA[3]	CA[3]
	2	CA[2]	CA[2]
	1	CA[1]	CA[1]
	0	CA[0]	CA[0]
Note: 1. Place the pin on AC Lane for CHA/CH0 only. You cannot place it on AC Lane for CHB/CH1. 2. Place the pin on AC Lane for CHB/CH1 only. You cannot place it on AC Lane for CHA/CH0. 3. Both CS_N[0] and CS_N[1] pins are used for calibration in DDR5 RDIMM 1 rank/2 rank module. It is required to connect and route the CS_N[1] pin even when using DDR5 RDIMM 1 rank module.			

The Agilex 7 M-Series FPGA DDR5 IP supports fixed Address and Command pin placement as shown in the above table. The IP supports up to 2 ranks for the following schemes:

- Scheme 1 supports component, UDIMM, and SODIMM.
- Scheme 2 supports RDIMM.

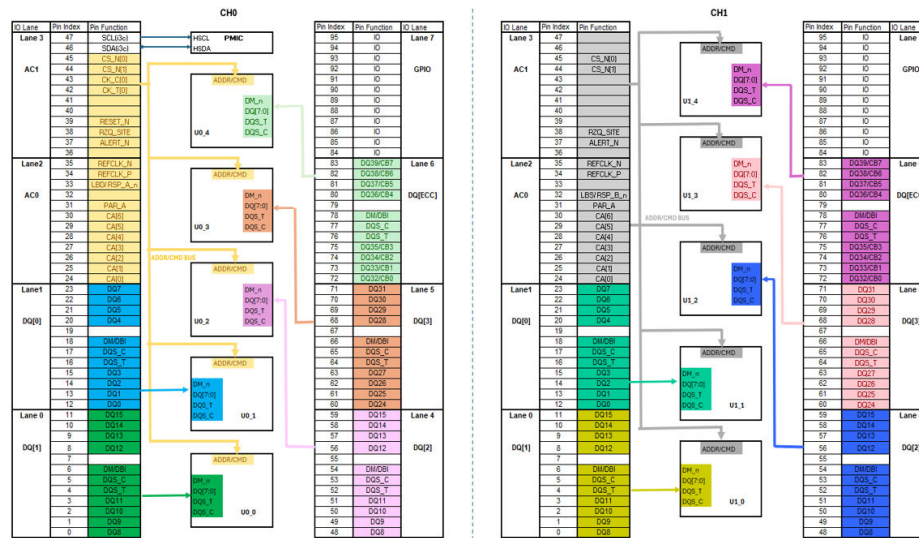
Figure 38.



The above figure shows only I3C_SDA, I3C_SCL, CK_T, CK_C, LBD and LBS. To implement one DDR5 RDIMM (2 channel) interface, two adjacent IO 96B banks on the same side of the device are required.

Connect EMIF_CH0 signals to CHA on the RDIMM, and the signals for EMIF_CH1 to CHB on the RDIMM respectively.

You cannot swap the pin location for LBD with LBS. Place LBD on the I/O bank where EMIF_CH0 resides and connect LBD to pin number 86 on the DIMM connector. Place LBS on the I/O bank where EMIF_CH1 resides and connect LBS to pin number 87 on the DIMM connector.



Two Adjacent IO96B Banks on the Same Side of the Device

7.3.4.2. DDR5 Data Width Mapping

The EMIF IP for Agilex 7 M-Series devices does not support flexible data lane placement.

Only fixed byte lanes within the I/O bank can be used as data lanes. The following table lists the supported address and command and data lane placements in an I/O bank.

Table 208. Component

Controller	Address Command Scheme	Data Width Usage	BL7 [P95:P8 4]	BL6 [P83:P7 2]	BL5 [P71:P6 0]	BL4 [P59:P4 8]	BL3 [P47:P3 6]	BL2 [P35:P2 4]	BL1 [P23:P1 2]	BL0 [P11:P0]
Primary	Scheme 1	DDR5x16	GPIO	GPIO	GPIO	GPIO	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Primary 1	Scheme 1	DDR5x16	DQ[1] ^P	DQ[0] ^P	AC1 ^P	AC0 ^P	X	X	X	GPIO
Primary	Scheme 1	DDR5x16 + ECC	GPIO	GPIO	GPIO	DQ[ECC] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P

continued...

Controller	Address Command Scheme	Data Width Usage	BL7 [P95:P84]	BL6 [P83:P72]	BL5 [P71:P60]	BL4 [P59:P48]	BL3 [P47:P36]	BL2 [P35:P24]	BL1 [P23:P12]	BL0 [P11:P0]
Primary	Scheme 1	DDR5x32	GPIO	GPIO	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Primary	Scheme 1	DDR5x32 + ECC	GPIO	DQ[ECC] ^P	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P

Note:

- ^P Primary controller.
- ^S Secondary controller.
- ¹ ES0 silicon supports this scheme using the primary controller; future revisions will enable the secondary controller, unblocking BL0-3 for GPIO use.
- X = Not available as GPIO.

DIMM Support

Table 209. Bank 3A, 3D, 2D

Address Command Scheme	Data Width per Channel	BL7 [P95:P84]	BL6 [P83:P72]	BL5 [P71:P60]	BL4 [P59:P48]	BL3 [P47:P36]	BL2 [P35:P24]	BL1 [P23:P12]	BL0 [P11:P0]
Scheme 1	DDR5 x32	GPIO	GPIO	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Scheme 1	DDR5 x32 + ECC	GPIO	DQ[ECC] ^P	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Scheme 2	DDR5 x32	GPIO	GPIO	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Scheme 2	DDR5 x32 + ECC	GPIO	DQ[ECC] ^P	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P

Note: ^P Primary controller.

Table 210. Bank 3B, 3C, 2C

Address Command Scheme	Data Width per Channel	BL7 [P95:P84]	BL6 [P83:P72]	BL5 [P71:P60]	BL4 [P59:P48]	BL3 [P47:P36]	BL2 [P35:P24]	BL1 [P23:P12]	BL0 [P11:P0]
Scheme 1	DDR5 x32	DQ[1] ^P	DQ[0] ^P	AC1 ^P	AC0 ^P	DQ[2] ^P	DQ[3] ^P	GPIO ¹	GPIO ¹
Scheme 1	DDR5 x32 + ECC	DQ[1] ^P	DQ[0] ^P	AC1 ^P	AC0 ^P	DQ[2] ^P	DQ[3] ^P	DQ[ECC] ^P	GPIO ¹
Scheme 2	DDR5 x32	DQ[1] ^P	DQ[0] ^P	AC1 ^P	AC0 ^P	DQ[2] ^P	DQ[3] ^P	GPIO ¹	GPIO ¹
Scheme 2	DDR5 x32 + ECC	DQ[1] ^P	DQ[0] ^P	AC1 ^P	AC0 ^P	DQ[2] ^P	DQ[3] ^P	DQ[ECC] ^P	GPIO ¹

Note:

- ¹ GPIO is available if using NoC access mode.
- ^P Primary controller.
- For banks 3A/3B, 3C/3D, or 2C/2D there are two options available for the address and command placement parameter (**AC Placement**):
 - ☐ **Ch0 Top Sub-Bank/Ch1 Bot Sub-Bank**, recommended to use this selection when placing Ch0 A/C pins in the top sub-bank.
 - ☐ **Ch0 Bot Sub-Bank/Ch1 Top Sub-Bank**, recommended to use this selection when placing Ch0 A/C pins in the bottom sub-bank.

Table 211. Bank 2A

Address Command Scheme	Data Width per Channel	BL7 [P95:P84]	BL6 [P83:P72]	BL5 [P71:P60]	BL4 [P59:P48]	BL3 [P47:P36]	BL2 [P35:P24]	BL1 [P23:P12]	BL0 [P11:P0]
Scheme 1	DDR5 x32	GPIO	GPIO	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Scheme 1	DDR5 x32 + ECC	GPIO	DQ[ECC] ^P	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Scheme 2	DDR5 x32	GPIO	GPIO	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Scheme 2	DDR5 x32 + ECC	GPIO	DQ[ECC] ^P	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Note: ^P Primary controller.									

Table 212. Bank 2B

Address Command Scheme	Data Width per Channel	BL7 [P95:P84]	BL6 [P83:P72]	BL5 [P71:P60]	BL4 [P59:P48]	BL3 [P47:P36]	BL2 [P35:P24]	BL1 [P23:P12]	BL0 [P11:P0]
Scheme 1	DDR5 x32	GPIO	GPIO	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Scheme 1	DDR5 x32 + ECC	GPIO	DQ[ECC] ^P	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Scheme 2	DDR5 x32	GPIO	GPIO	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Scheme 2	DDR5 x32 + ECC	GPIO	DQ[ECC] ^P	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Note: • ^P Primary controller. • For banks 2A and 2B, it is recommended to select Ch0 Bot Sub-Bank/Ch1 Bot Sub-Bank for the address and command placement parameter (AC Placement).									

Figure 39. DDR5 2chx16, Single Rank Using x8 Memory Component

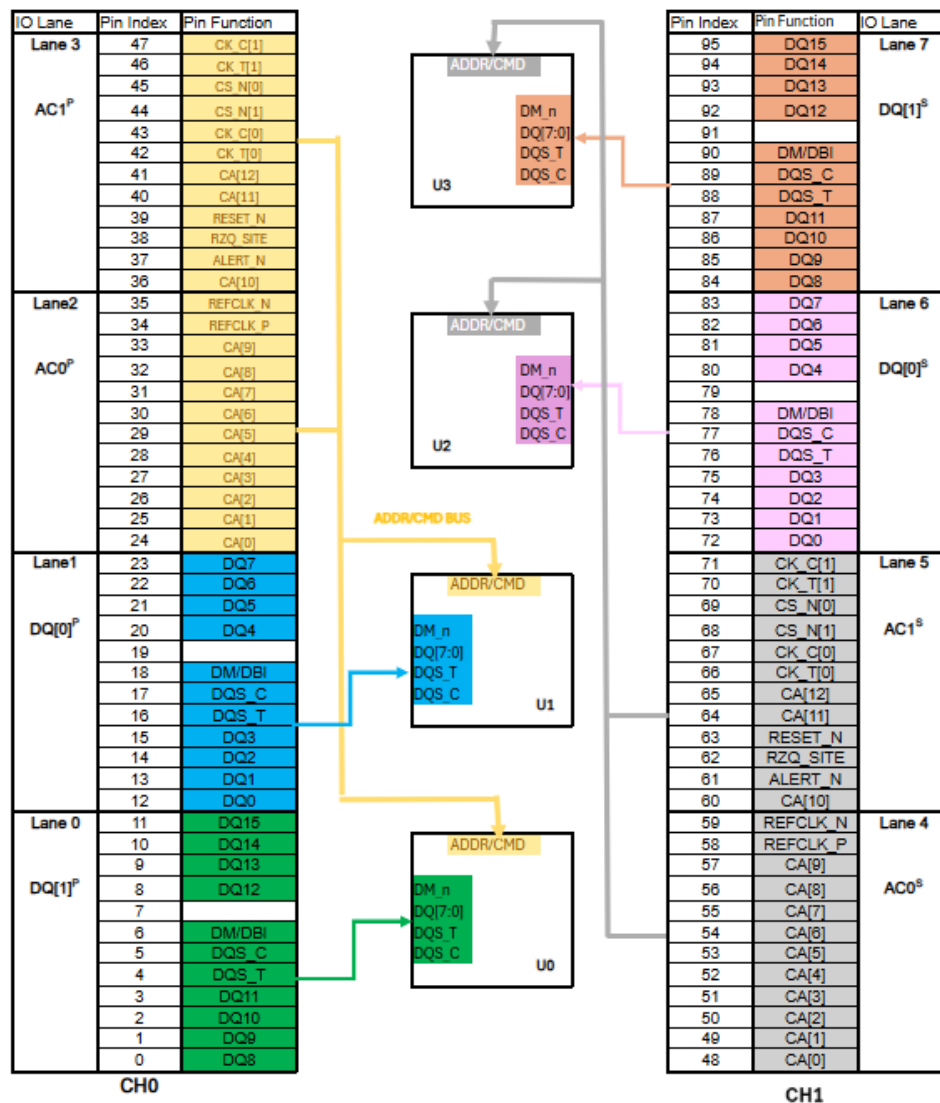
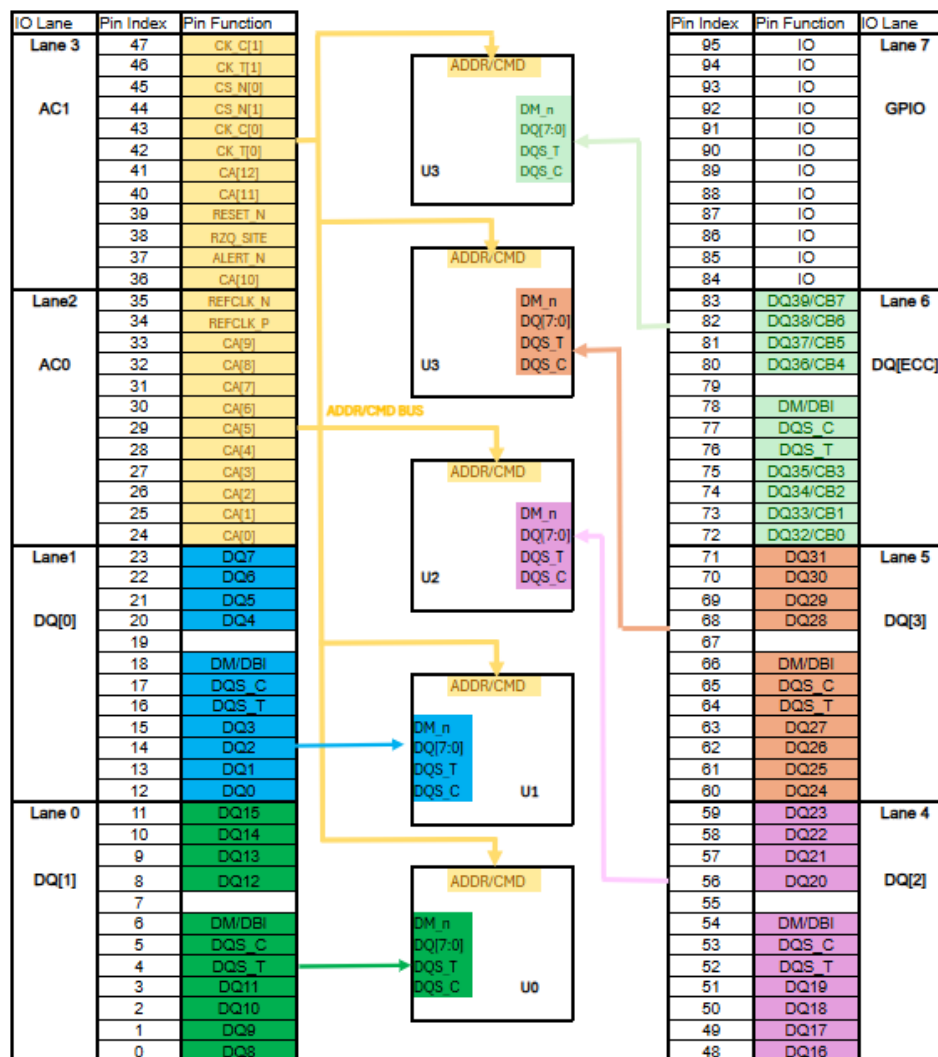


Figure 40. DDR5 x32 + ECC, Single Rank Using x8 Memory Component



CH0

Lane	Pin Index	Pin Function	Component	Pin Index	Pin Function	Lane
Lane 3	47	SCLB3	HEC	95	ID	Lane 7
AC1	46	CS_N0		94	ID	Lane 6
	45	CS_N01		93	ID	
	44	CS_N1		92	ID	
	43	CS_N10		91	ID	
42	OK_T0	90	ID	GPO	Lane 5	
41		89	ID			
40		88	ID			
39	RESET_A1	87	ID			
38	RESET_A0	86	ID			
37	RESET_A1	85	ID	GPO	Lane 4	
36	RESET_A0	84	ID			
35	RESET_A1	83	ID			
34	RESET_A0	82	ID			
33	RESET_A1	81	ID			
32	RESET_A0	80	ID	GPO	Lane 3	
31	RESET_A1	79	ID			
30	RESET_A0	78	ID			
29	RESET_A1	77	ID			
28	RESET_A0	76	ID			
27	RESET_A1	75	ID	GPO	Lane 2	
26	RESET_A0	74	ID			
25	RESET_A1	73	ID			
24	RESET_A0	72	ID			
23	RESET_A1	71	ID			
22	RESET_A0	70	ID	GPO	Lane 1	
21	RESET_A1	69	ID			
20	RESET_A0	68	ID			
19	RESET_A1	67	ID			
18	RESET_A0	66	ID			
17	RESET_A1	65	ID	GPO	Lane 0	
16	RESET_A0	64	ID			
15	RESET_A1	63	ID			
14	RESET_A0	62	ID			
13	RESET_A1	61	ID			
12	RESET_A0	60	ID	GPO	Lane 0	
11	RESET_A1	59	ID			
10	RESET_A0	58	ID			
9	RESET_A1	57	ID			
8	RESET_A0	56	ID			
7	RESET_A1	55	ID	GPO	Lane 0	
6	RESET_A0	54	ID			
5	RESET_A1	53	ID			
4	RESET_A0	52	ID			
3	RESET_A1	51	ID			
2	RESET_A0	50	ID	GPO	Lane 0	
1	RESET_A1	49	ID			
0	RESET_A0	48	ID			
	RESET_A1	47	ID			
	RESET_A0	46	ID			

CH1

Lane	Pin Index	Pin Function	Component	Pin Index	Pin Function	Lane
Lane 3	47	SCLB3	ADDRESS	95	ID	Lane 7
AC1	46	CS_N0		94	ID	Lane 6
	45	CS_N01		93	ID	
	44	CS_N1		92	ID	
	43	CS_N10		91	ID	
42	OK_T0	90	ID	GPO	Lane 5	
41		89	ID			
40		88	ID			
39	RESET_A1	87	ID			
38	RESET_A0	86	ID			
37	RESET_A1	85	ID	GPO	Lane 4	
36	RESET_A0	84	ID			
35	RESET_A1	83	ID			
34	RESET_A0	82	ID			
33	RESET_A1	81	ID			
32	RESET_A0	80	ID	GPO	Lane 3	
31	RESET_A1	79	ID			
30	RESET_A0	78	ID			
29	RESET_A1	77	ID			
28	RESET_A0	76	ID			
27	RESET_A1	75	ID	GPO	Lane 2	
26	RESET_A0	74	ID			
25	RESET_A1	73	ID			
24	RESET_A0	72	ID			
23	RESET_A1	71	ID			
22	RESET_A0	70	ID	GPO	Lane 1	
21	RESET_A1	69	ID			
20	RESET_A0	68	ID			
19	RESET_A1	67	ID			
18	RESET_A0	66	ID			
17	RESET_A1	65	ID	GPO	Lane 0	
16	RESET_A0	64	ID			
15	RESET_A1	63	ID			
14	RESET_A0	62	ID			
13	RESET_A1	61	ID			
12	RESET_A0	60	ID	GPO	Lane 0	
11	RESET_A1	59	ID			
10	RESET_A0	58	ID			

Two Adjacent IO96B Banks on the Same Side of the Device

A byte lane in an EMIF data byte includes 12 signal pins (pins 0,1,2,3,4,5,6,7,8,9,10,11) at the package level. These 12 x I/O pins are arranged into 6 groups of 2 pins each, called *pairs* (pair 0 for pins 0/1, pair 1 for pins 2/3, pair 2 for pins 4/5, pair 3 for pins 6/7, pair 4 for pins 8/9, and pair 5 for pins 10/11).

The data lane can be swapped when the byte-lanes are utilized as DQ/DQS pins. Byte lane swapping on utilized lanes is allowed when you swap all the DQ/DQS/DM pins in the same byte lane with the other utilized byte lane.

The rules for swapping DQ byte lane are as follows:

- You can only swap between utilized DQ lanes.
- You cannot swap a DQ lane with an AC lane.
- You cannot swap a DQ lane with an ECC lane when out-of-band ECC is enabled. For x40 interfaces, the highest-indexed DQ byte lane cannot be swapped.
- Additional restrictions apply when you use a x16 memory component:
 - ☐ You must place DQ group 0 and DQ group 1 on adjacent byte lanes, unless they are separated by AC Lanes. These 2 groups must be connected to the same x16 memory component.
 - ☐ You must place DQ group 2 and DQ group 3 on adjacent byte lanes, unless they are separated by AC Lanes. These 2 groups must be connected to the same x16 memory component.
 - ☐ If you use only one byte of the x16 memory component, you must use only the lower byte of the memory component.

Table 213. Byte Lane Swapping

Address/ Command Scheme	Data Width per Channel	BL7 [P95:P84]	BL6 [P83:P72]	BL5 [P71:P60]	BL4 [P59:P48]	BL3 [P47:P36]	BL2 [P35:P24]	BL1 [P23:P12]	BL0 [P11:P0]
Scheme 1	DDR5 x32	GPIO	GPIO	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P
Scheme 1	DDR5 x32+ ECC	GPIO	DQ[ECC] ^P	DQ[3] ^P	DQ[2] ^P	AC1 ^P	AC0 ^P	DQ[0] ^P	DQ[1] ^P

Note: ^P = Primary controller.

Example 1: DDR5 x32

BL0, 1, 4, 5 are used as DQ lanes. Byte lane swapping is allowed.

Example 2: DDR5 x32 + ECC

BL6 is used as ECC DQ lane, while BL0, 1, 4, and 5 are used as DQ lanes. Byte lane swapping is allowed on BL0, 1, 4, and 5 only.

7.3.5.2. DDR5 Address and Command and CLK Lane

Address and command and control signals in a bank cannot be swapped.

Pin mapping must adhere to the requirements defined in the table in the [Address and Command Pin Placement for DDR5](#) topic.

You cannot swap address and command lanes. You cannot swap among AC0/AC1 lanes. The address and command lane placement must adhere to the specific placement defined in the table in the [DDR5 Data Width Mapping](#) topic.

The T and C lanes for the CK_T/_C cannot be swapped with each other, nor can the T and C lanes for the DQS_T/DQS_C be swapped with each other.

7.3.5.3. DDR5 Interface x8 Data Lane

A byte lane in an external memory interface consists of 12 signal pins, denoted 0-11.

For DDR5 interfaces composed of ×8 devices, two pins are reserved for DQS-T and DQS-C signals, one pin is reserved for the optional DM signal, one pin must be reserved, and the remaining eight pins are for DQ signals. One-byte data lane must be assigned for each byte lane, where the byte lane covers DQ [0:7], DQS_T/DQS_C and DM_N. The following are EMIF I/O pin swapping restrictions applicable to a DDR5 interface with a ×8 data lane:

- DQS_T must go to pin 4 in IO12 pins.
- DQS_C must go to pin 5 in IO12 pins.
- DM_N must go to pin 6 in IO12 pins. If the interface does not use the DM_N pin, this pin 6 in IO12 lane must remain unconnected.
- Pin 7 in IO12 lane remains unconnected. Altera recommends that you connect this pin 7 to the T_{DQS} dummy load of the memory component and route it as a differential trace along with DM_N (pin 6). This facilitates ×4 or ×8 data interoperability in DIMMs configuration.
- You can connect data byte (DQ [0:7]) to any pins [0,1,2,3,8,9,10,11] in the byte lane. Any permutation within selected pins is permitted.

Table 214. Pin Swapping Rules for DDR5 x8 Interfaces

Pin Index Within Byte Lane	DDR5 x8 Data Lane Function	Swap Consideration
0	DQ Pin	Swap group A
1	DQ Pin	Swap group A
2	DQ Pin	Swap group A
3	DQ Pin	Swap group A
4	DQS_T Pin	Fixed location (not swappable)
5	DQS_C Pin	Fixed location (not swappable)
6	DM Pin	Fixed location (not swappable)
7	Unused	Fixed location (not swappable)
8	DQ Pin	Swap group A
9	DQ Pin	Swap group A
10	DQ Pin	Swap group A
11	DQ Pin	Swap group A

7.3.5.4. DDR5 Interface x4 Data Lane

For DDR5 x4 interfaces, two nibbles must be packed into the same IO12 lane.

Four pins are reserved for DQS_T and DQS_C signals and the remaining eight pins implement the DQ signals. The IO12 lane is divided into upper and lower halves to accommodate each nibble. You cannot swap signals belonging to one nibble with signals belonging to the other nibble. DQ signals within a nibble swap group may be swapped with each other. You may also swap entire nibbles—that is, nibble 0 and nibble 1—with each other provided the DQS pin functionality transfers to the correct pin locations. However, this process is not recommended for JEDEC-compliant DIMM interfaces, as it prohibits the interoperability between DIMMs constructed with x4 components and DIMMs constructed with x8 components.

The following table lists the supported pin functionality in x4 mode and the pins that may be swapped with each other. Pins belonging to the same swap group may be freely interchanged with each other.

Table 215. Pin Swapping Rules for DDR5 x4

Pin Index Within Byte Lane	DDR5 x4 Data Lane Function	Swap Consideration	
0	DQ Pin (lower nibble)	Swap group A	Nibble 0
1	DQ Pin (lower nibble)	Swap group A	
2	DQ Pin (lower nibble)	Swap group A	
3	DQ Pin (lower nibble)	Swap group A	
4	DQS_T Pin (lower nibble)	Fixed location (not swappable)	
5	DQS_C Pin (lower nibble)	Fixed location (not swappable)	
6	DQS_T Pin (upper nibble)	Fixed location (not swappable)	Nibble 1
7	DQS_C Pin (upper nibble)	Fixed location (not swappable)	
8	DQ Pin (upper nibble)	Swap group B	
9	DQ Pin (upper nibble)	Swap group B	
10	DQ Pin (upper nibble)	Swap group B	
11	DQ Pin (upper nibble)	Swap group B	

- Nibble 1 must correspond to DQS[17:9] on a physical JEDEC-compliant DIMM for x4/x8 interoperability.
- Nibbles 0 and 1 must follow the same skew matching rules among all 12 signals in the IO12 lane as are specified for a x8-based DQS group.

Note:

- Although the current version of the Quartus Prime software may not enforce all of the rules listed in the above table, be aware that all of these rules may be enforced in later versions of the software.
- At present, the Quartus Prime software checks the following:
 - ☐ Address and command pin placement, per the table in the [Address and Command Pin Placement for DDR5](#) topic, or the *Agilex 7 External Memory Interface Pin Information* file, which is available here: [Pin-Out Files for Altera FPGA Devices](#).
 - ☐ For x8, the Quartus Prime software checks the following:
 - DQS T/C are on pin index 4 and pin index 5 in a byte lane.
 - DM is on pin index 6.
 - DQ[x] are on pin indices [11:8] and [3:0].
 - ☐ For x4, the Quartus Prime software checks the following:
 - DQS T/C on pin index 4 and pin index 5 and associated DQs are within the corresponding byte lane.
 - DQS T/C on pin index 6 and pin index 7 and associated DQs are within the corresponding byte lane.

You are responsible for ensuring that these conditions are met.

- The Quartus Prime software does not currently check whether DQ pins associated with the lower nibble DQS are actually placed in pin[3:0] or whether DQ pins associated with the upper nibble DQS are actually placed in pin[11:8].

7.3.5.5. Pin Swizzling

For information on pin swizzling, refer to [Configuring DQ Pin Swizzling](#) in the *External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide*.

7.4. DDR5 Board Design Guidelines

This section provides board layout design recommendations and guidelines for Agilex 7 M-Series FPGAs, with GPIO-B (input/output) silicon implementation to support DDR5.

This PCB layout guideline covers various supported DDR5 topologies along with maximum supported data rate that you can use for a successful PCB design.

A successful PCB design requires not only following the topology and routing guidelines, but also meet PDN design requirements.

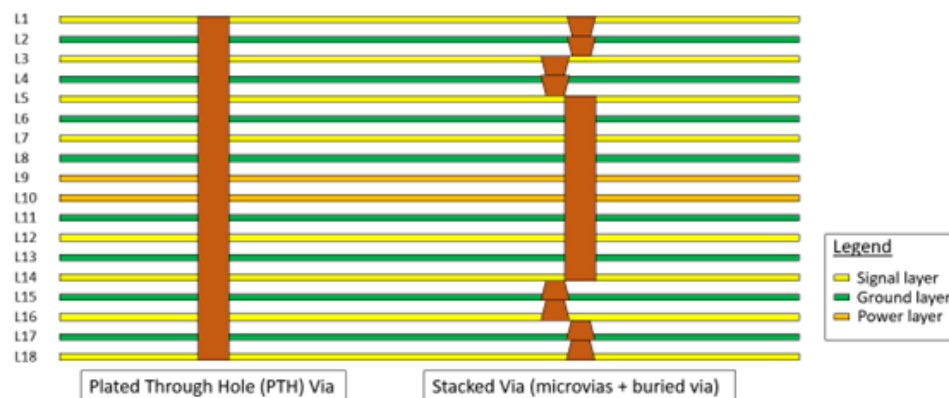
For related information, refer also to the Agilex 7 PDN design guidelines and the Agilex 7 high speed transceiver PCB design guidelines, available on the Intel website.

7.4.1. PCB Stack-up and Design Considerations

You can implement DDR5 interfaces with Type-III or Type-IV PCB stack-ups. A Type-III PCB with zero-built up layers and PTH vias which is used to implement a DDR4 design can also be used for DDR5 designs. Backdrill can be considered to reach higher supported DDR5 data rate. You can run simulation to determine whether backdrill is necessary.

A high-quality type-IV PCB with higher cost, uses not only plated-through-hole (PTH) vias to connect from the top to bottom layer, but also stacked vias, micro vias, and buried vias to connect between layers. The following figure shows a cross-sectional comparison of a PTH and stacked via.

Figure 42. Cross-sectional Comparison of Plated-Through Hole Via and Stacked Via



To support maximum data rate operation, DDR5 board design requires a high-quality PCB stackup using backdrill, micro vias, buried vias, or stacked vias to reduce crosstalk for high performance. Reducing the length of signal via is essential to minimizing the crosstalk between signals. You can perform simulation to determine whether to add backdrill or not based on your own stack-up.

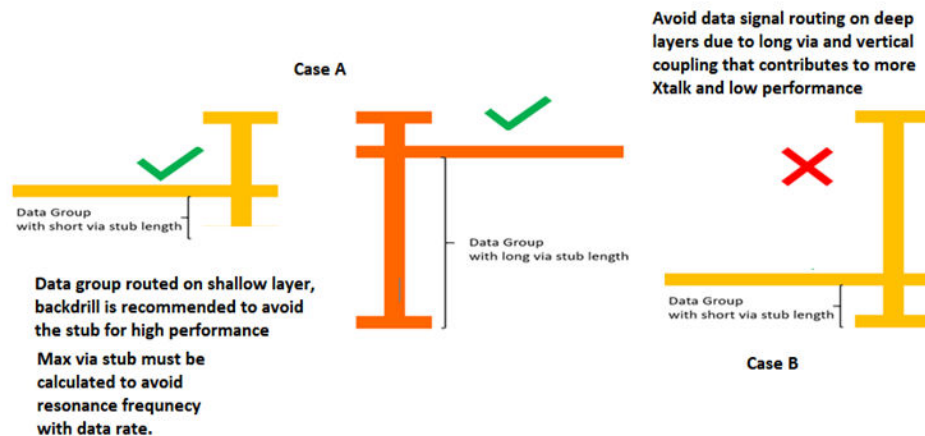
7.4.2. General Design Considerations

Altera recommends that you route all data signals within a specific group on the same layer.

The following figure illustrates a routing example for a type-III PCB for a DDR5 design. Altera recommends that you route Data Group signals such as DQ, DM and DQS on shallow layers as stripline, with the least Z-height via transition to avoid vertical crosstalk for high performance.

Minimal stub effect or back drill is recommended but not mandatory to avoid high reflection for maximum data rate performance for a DDR5 interface. You can run simulation to determine whether to use backdrill. Long via stubs will affect the intersymbol interference (ISI) of the channel, but the impact of ISI is less than the impact of crosstalk for maximum performance.

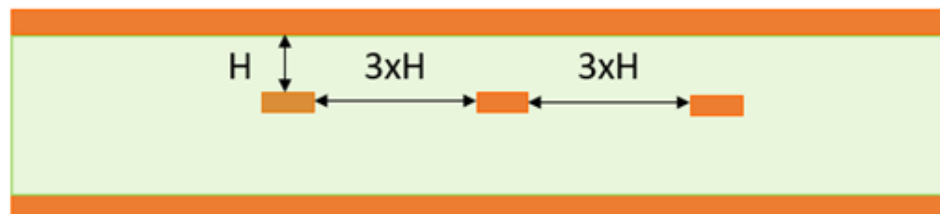
Figure 43. Suggested Routings



In the above figure, case A routing is suggested for DDR5 Data Group signals over case B, to support maximum data rate. If data signals are routed on deeper layers (as in case B, with long via and short stub), the impact of crosstalk is significant and causes reduced data rate and performance.

To minimize crosstalk horizontally between the signals on the same layer, PCB designers must maintain adequate signal trace-to-trace (edge to edge) space with a minimum spacing of $3 \times H$ separation distance, where H is the dielectric thickness to the closest reference plane, as illustrated below. Some signals within the same group can have smaller spacing, such as DQ to DQ. Refer to the routing guideline sections for detailed spacing requirements for each signal group.

Figure 44. Minimum Trace-to-Trace Separation Distance



7.4.3. DDR Differential Signals Routing

DQS and CLK signals in the DDR interface are differential signals and must be routed on PCB as differential signals unless there is a limitation for PCB routing, such as having a very small pitch at DRAM.

You should have a symmetrical fan-out routing at the FPGA pin field. Non-symmetrical routing for differential signals causes shifting on common-mode voltage and contributes to reduced timing margins at the receiver. The following figures show the recommended differential routing at the FPGA pin field for DQS/CLK signals.

Figure 45. Symmetrical Routing of Differential Signals (DQS/CLK) at FPGA Pin Field, with Length/Skew Matching Between P/N Lanes After FPGA Device Edge

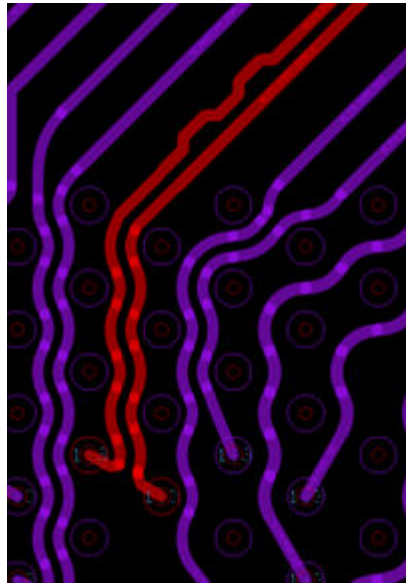
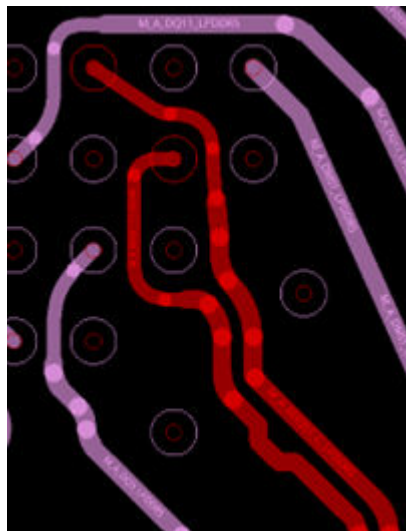


Figure 46. Single-Ended Routing for Differential Signals (DQS/CLK) at DRAM Pin Field with Very Small Pitch and Skew Matching at Edge of DRAM Pin Field



Altera recommends implementing length and skew matching for differential signals immediately after the FPGA device to avoid additional shifting on differential signals common mode voltage.

In cases where very small DRAM device pitch limits the implementation of symmetrical routing at the DRAM pin field for differential signals, it is recommended to route the differential signals as single-ended signals within the DRAM pin field, ensuring to maintain the same impedance while changing from differential to single-ended

configuration. Designers must also keep the same length of routing for each P and N single-ended lane within the DRAM pin field. The skew matching between P and N lanes must be applied before reaching the DRAM pin field.

7.4.4. Ground Plane and Return Path

A continuous and solid ground-reference plane is crucial for data lines, to ensure good signal integrity.

It is important to provide a low-impedance ground return path between the FPGA and DRAM devices, and to keep ground stitching vias within 80 mils from signal transition, for best return path on signal vias and improved signal integrity.

7.4.5. RDIMM, UDIMM, and SODIMM Break-in Layout Guidelines

The following figures show recommended signal and ground cutout and via transition, and placement patterns for RDIMMs, UDIMMs, and SODIMMs.

Altera recommends placing the vias on the connector pad, or with a short trace connected to connector pad. The placement of transition vias is critical to avoid stubs during transition to connector pads. It is also recommended to void under connector pads to avoid large capacitance and reflection in this area, and to control the impedance at the DIMM pads.

Figure 47. Recommended Signal and Ground Cutouts

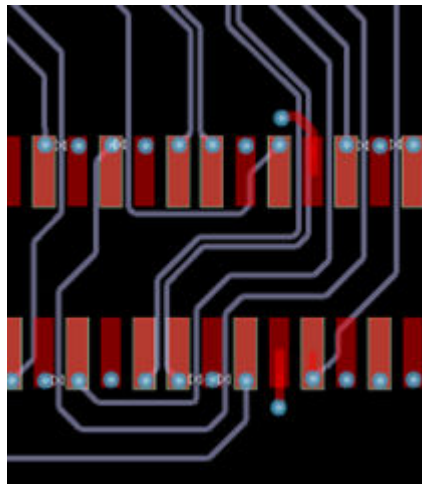
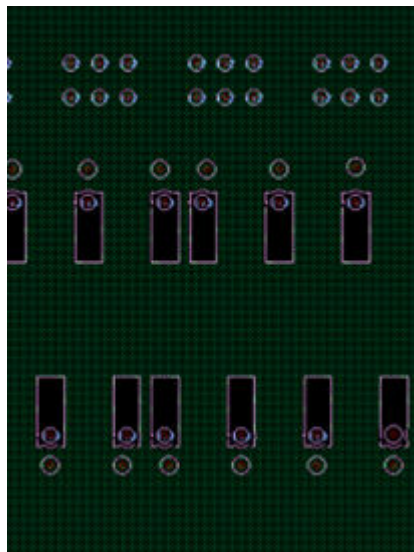


Figure 48. Recommended Signal and Ground Via Transitions

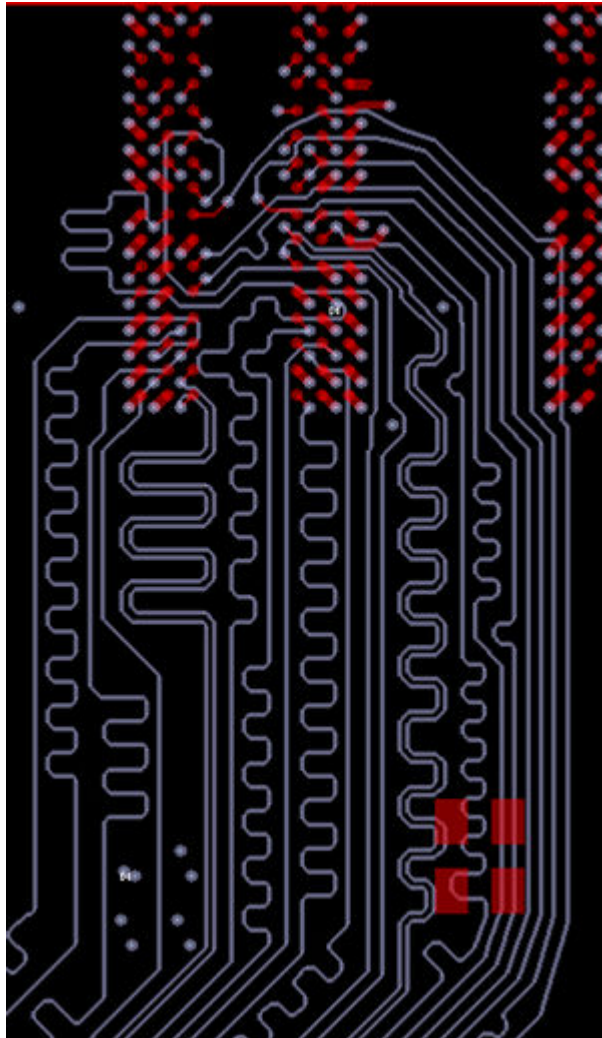


7.4.6. DRAM Break-in Layout Guidelines

For discrete DRAM components on PCB, you can use either the dog-bone or via in pad at the DRAM for the signal transition from inner layer to DRAM.

If you use dog-bone via transitions, you should separate them with larger pitch, to avoid crosstalk between the signal vias.

Figure 49. Data Signal Group Routing on PCB to Memory Down Configuration



7.4.7. General Notes for EMIF Routing Guidelines Tables

The following general notes apply to the EMIF routing guideline tables in subsequent topics.

- All spacing requirements are the minimum requirement to be met on PCB in EMIF routing guidelines table.
- In the routing guideline table, h represents the trace-to-nearest-reference-plane height or distance. Board designers should use the formula in the routing guidelines table to calculate the correct spacing requirements.
- There is no differential impedance target for CLK and DQS. Board designers should follow the EMIF routing table and keep traces to be closely coupled.

- The trace length and spacing in the guideline tables are based on FR-4 level PCB material and the PCB layout routing with the worst-case crosstalk. You should perform simulation if not following the requirements in the routing guidelines table.
- In the routing guidelines tables, *SL* stands for strip line routing recommendations and *US* stands for upper surface (Microstrip) routing recommendations.
- You need to design trace widths of BO, BO1, BO2 and BI, BI1, BI2 based on the actual PCB stack-up and PCB design. It would be better if trace widths of BO, BO1, BO2 and BI, BI1 can be designed closer to the target impedance of the M segment.
- Include the FPGA package per-pin skew and PCB delay for skew matching.
- Length matching on the Alert and Reset signals are not required.
- Altera recommends that you perform skew matching in time (picoseconds).

7.4.8. DDR5 PCB Layout Guidelines

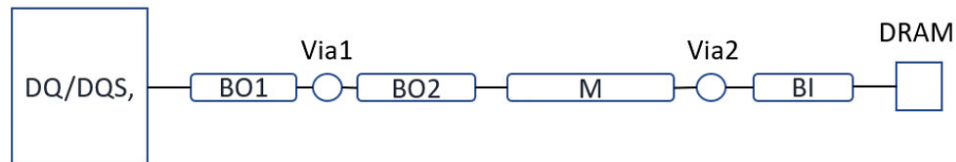
This section describes PCB layout guidelines for a DDR5 interface.

Agilex 7 M-Series devices support DDR5 interfaces for both discrete components and DIMMs, RDIMMs, SODIMMs, and LRDIMMs, with both thin and thick PCB stackups. The maximum supported data rates vary depending on the selected topology and thickness of circuit board.

7.4.8.1. DDR5 Discrete Component/Memory Down Topology: Single Rank x8 or x16, Dual Rank x8 or x16

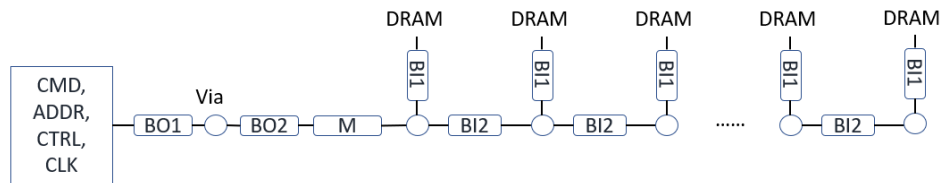
Data Group includes Data Strobe and its complement (DQS and DQS#), Data (DQ), and Data Mask (DM). The connection from the FPGA to DRAM is point-to-point topology as shown in the figure below, for single rank.

Figure 50. DRAM x8 or x16 (Single Rank)



Double rank topology has clamshell/fly-by configuration. For address, command, control and clock signals, a fly-by or clamshell topology is recommended to meet signal-integrity performance and for easier routing. The termination approach for DDR5 is through programmable ondie-termination (ODT). You can adjust the design topology based on the actual PCB design (single rank or dual rank).

Figure 51. Single Rank, DRAM x 8 bits



7.4.8.2. Routing Guidelines for DDR5 Memory Down: Single Rank or Dual Rank (x8 bit or x16 bit) Configurations

The following table provides recommended trace impedance and length for each of the DDR5 signals based on a memory down topology.

For example, the maximum length of the main trace routing can be derived from total trace length by subtracting the break-out and break-in trace segment lengths routed.

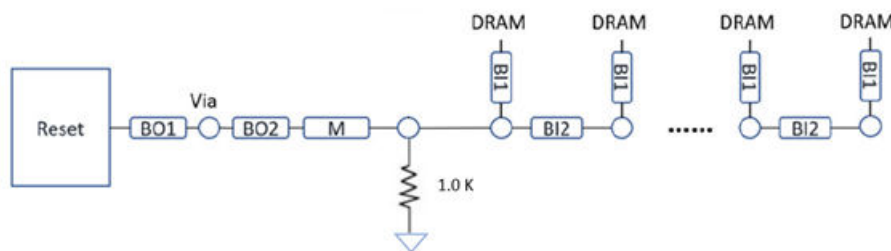
The following table shows physical trace segment routing guidelines, including the target impedance of routing for each signal and the minimum space between signal traces on the same layer. The PCB designer must meet the impedance target criteria. The h in the table indicates the minimum substrate height from signal layer to reference layer. Ensure that you follow the trace-to-trace-edge gap/space criteria in the guideline, based on h in various stackups.

Table 216. Routing Guideline for DDR5 Memory Down: Single Rank and Dual Rank (x8 bit or x16 bit)

Signal Group	Segment	Routing Layer	Max Length (mil)		Target Zse (ohm)	Trace Spacing, S1 (mil): Within Group	Trace Spacing, S2 (mil): CMD/CTRL/CLK to DQ/DQS	Trace Spacing, S3 (mil): DQ Nibble to Nibble	Trace Spacing (mil): Within DIFF pair	Trace Spacing (mil): DQS pair to DQ	Trace Spacing (mil): CLK pair to CMD/CTRL/CKE
			Segment	Total MB							
CLK	BO1	US	500	4000 to first DRAM, 6600 to the last DRAM	40	5, 17	17		4		17
	BO2	SL	1000			5, 17	17		4		17
	M	SL					3h		4		3h
	BI1	US	50				3h		4		3h
	BI2	SL	760			50+	3h		4		3h
CMD/ADDR/CTRL	BO1	US	500	4000 to first DRAM, 6600 to the last DRAM	40	5, 17	17				
	BO2	SL	1000			5, 17	17				
	M	SL				2h	3h				
	BI1	US	50			2h	3h				
	BI2	SL	760			50+	3h				
DQ	BO1	US	50	4500	45	5, 17		17			
	BO2	SL	1000			5, 17		17			
	M	SL				2h		3h			
	BI	US	50			2h		3h			
DQS	BO1	US	50	4500	45				4	17	
	BO2	SL	1000						4	17	
	M	SL							4	3h	
	BI	US	50						4	3h	

Reset signal routing also follows the CMD/ADD/CTRL routing design. Maintain an edge-to-edge space between the Reset signal and other signals on the same layer of at least $3 \times h$. There is no requirement to have skew matching between the Reset signal and CLK signal. Altera recommends using a 1K-ohm pull-down resistor for Reset signal termination, as illustrated below.

Figure 52. Reset Scheme and Setting for Memory Down Topology



The following table provides a detailed skew matching guideline. Ensure that you include both PCB physical routing skew and package routing skew in your skew matching criteria. The physical length matching criteria in the table reflects the default stackup in our platform PCB design.

Table 217. Skew Matching Requirement for DDR5 Memory Down, Single Rank and Dual Rank Configuration

Length Matching Rules	Time (assuming 170ps/in delay)
Length Matching between DQS and CLK	-255ps < CLK - DQS < 425ps
Length Matching between DQ and DQS within byte	-3.5ps < DQ - DQS < 3.5ps
Length matching between DQS and DQS#	< 1ps
Length matching between CLK and CLK#	< 1ps
Length matching between CLK0 and CLK1	< 7ps
Length matching between CS and Clock	0 < CLK - CS < 17ps
Length matching among CS within each channel	< 17ps
Length matching between CMD/ADDR and Clock	-17ps < CLK - CMD/ADDR < 17ps
Length matching among CMD/ADDR within each	< 17ps
Include package length in length matching	Required

Board thickness in the design can vary from thin to thick. In this design guideline, a standard thin board is 65 mils and 120 mils thickness is a standard thick board. You can perform simulation to determine whether to use back drill.

Table 218. DQ Routing Summary for DDR5 Memory Down

Memory Interface	DDR5	
Signal Group	DQ	
Board Thickness (mil)	65 mil or 120 mil (Data Routing must be on upper layers to avoid long via/ vertical Xtalk)	65 mil or 120 mil (Data Routing must be on upper layers to avoid long via/ vertical Xtalk)
PCB Stripline Trace Impedance (ohms)	45	45
Memory Configuration	Memory Down	Memory Down
# of Rank	1 (40bits total), x8 or x16	2 (40 bits total), x8 or x16
Maximum PCB Length Total (inches) to the DRAM	4.5	4.5
Note	Maximum package length in FPGA design is shorter than 34mm	

CA_ODT Connection Guidelines in DDR5 Component Down Fly-By Topology

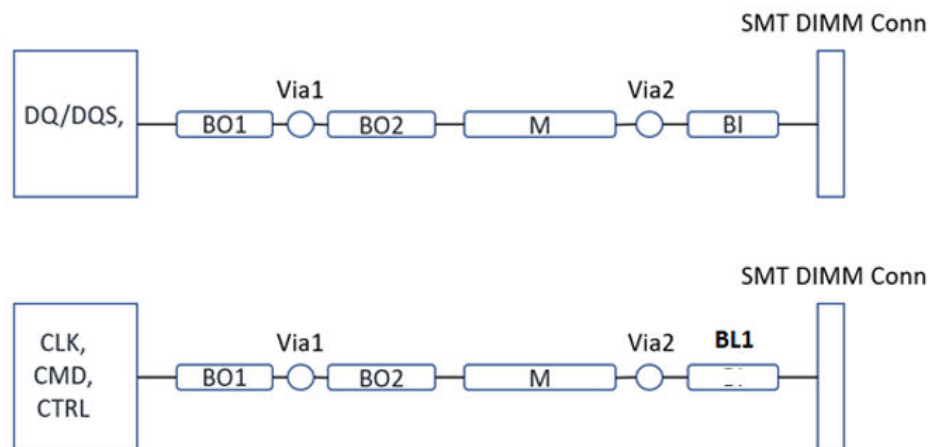
For a DDR5 Component Down interface with multiple DRAM components connected in a fly-by chain, follow these guidelines to connect the **CA_ODT** pin on each DRAM component:

1. **All DRAM Components Except the Last in the Chain:** Connect the CA_ODT pin to **GND**.
2. **Last DRAM Component in the Chain:** Connect the CA_ODT pin to **VDDQ**.

7.4.8.3. Routing Guidelines for DDR5 RDIMM, UDIMM, and SODIMM Configurations

The following figure shows DDR5 1 x DIMM per channel topology. A maximum of two transition vias are allowed for high performance.

Figure 53. DDR5 RDIMM, UDIMM, and SODIMM, 1 DIMM Per Channel Topology



The following table shows physical trace segment routing guidelines, including the target impedance of routing for each signal and the minimum space between signal traces on the same layer. The PCB designer must meet the impedance target criteria. The h in the routing guideline stands for the minimum substrate height between the signal layer and reference plane. Ensure that you follow the trace-to-trace-edge gap/space criteria in the guideline, based on “ h ” in various stackups.

Table 219. Routing Guidelines for DDR5 UDIMM, RDIMM, and SODIMM, 1 DIMM per Channel

Signal Group	Segment	Routing Layer	Max Length (mil)		Total MIB	Target Zse (ohm)	Trace Spacing, S1 (mil): Within Group	Trace Spacing, S2 (mil): CMD/CTRL/CLK to DQ/DQS	Trace Spacing, S3 (mil): DQ Nibble to Nibble	Trace Spacing (mil), Within DIFF pair	Trace Spacing (mil), DQS pair to DQ	Trace Spacing (mil), CLK pair to CMD/CTRL/CKE	Channel to Channel spacing (DQ to DQ between two channels)
			Segment										
DQ	BO1	US	50		4500		5, 17		17				17
	BO2	SL	1000				5, 17		17				17
	M	SL				50	2h		3h				4h
	BI	US	50				2h		3h				4h
DQS	BO1	US	50		4500		5, 17			4	17		
	BO2	SL	1000				5, 17			4	17		
	M	SL				50				4	3h		
	BI	US	50								3h		
CLK	BO1	US	500		4500		5, 17	17		4		17	
	BO2	SL	1000				5, 17	17		4		17	
	M	SL				40	2h	3h		4		3h	
	BI1	US	862				2h	3h		4		3h	
CMD/ADDR/CTRL	BO1	US	500		4500		5, 17	17					
	BO2	SL	1000				5, 17	17					
	M	SL				40		3h					
	BI1	US	862					3h					

Reset signal routing design also follows the command, address, and control (CMD/ADD/CTRL) routing design. Maintain the space from the Reset signal to other signals on the same layer (edge to edge) at least 3x h. There is no requirement to have skew matching between Reset and CLK signals.

Skew matching for a DDR interface consists of both package routing skew and PCB physical routing skew. You must maintain skew matching of CA and CTRL with respect to the clock signals to ensure that signals at the receiver are correctly sampled. There is also a skew matching requirement for DQ and DQS within a byte group, DQS and CLK.

The following table provides a detailed skew matching guideline to facilitate PCB trace routing. The length matching criteria in this table represents a default PCB on an Altera platform board design. Skew matching criteria must be always followed in any other stackup.

Table 220. Skew Matching Requirements for DDR5 UDIMM, RDIMM, and SODIMM, 1 DIMM per Channel. Length Matching Criteria in this Table Represents a Default Stackup.

Length Matching Rules	Time (assuming 170ps/in)
Length Matching between DQS and CLK	-255ps < CLK - DQS <
Length Matching between DQ and DQS within byte	-3.5ps < DQ - DQS < 3.5ps
Length matching between DQS and DQS#	< 1ps
Length matching between CLK and CLK#	< 1ps
Length matching between CLK0 and CLK1	< 7ps
Length matching between CS and Clock	0 < CLK - CS < 17ps
Length matching among CS within each channel	<
Length matching between CMD/ADDR and Clock	-17ps < CLK - CMD/ADDR <
Length matching among CMD/ADDR within each channel	< 17ps
Include package length in length matching	Required

The maximum DDR5 supported data rate, refer to the appropriate Agilex 7 FPGA device data sheet.

Board thickness in the design can vary from thin to thick. The 65 mil thickness is a standard thin PCB and 120 mils is a standard thick PCB in this design guideline. You can perform simulation to determine whether to use back drill.

Table 221. DQ Routing Summary for DDR5 DIMM

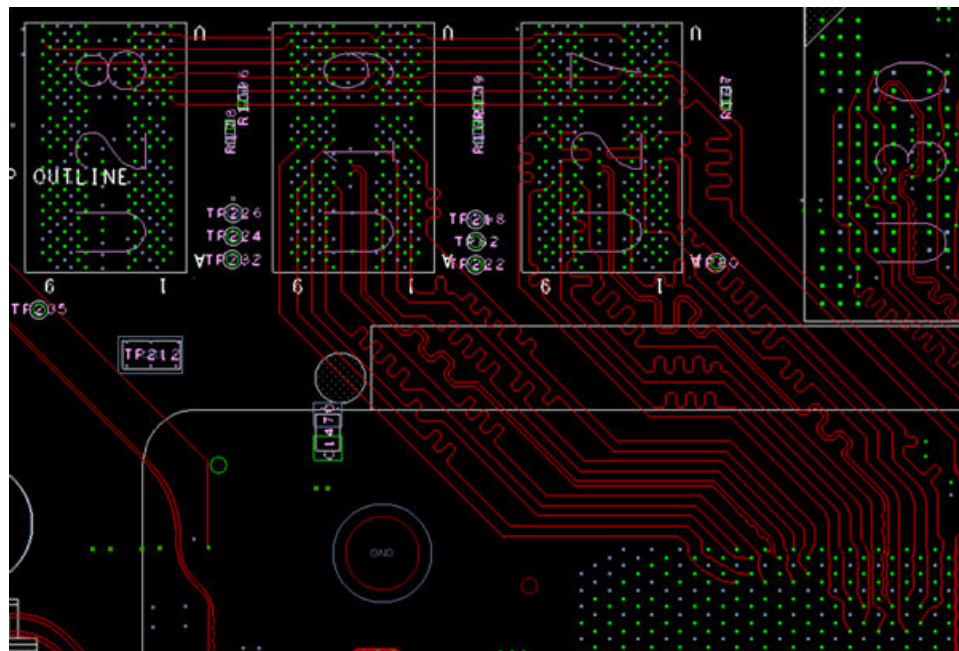
Memory Interface	DDR5 UDIMM	DDR5 RDIMM	DDR5 SODIMM
Signal group	DQ		
Board Thickness (Thin or Thick)	65 mil or 120mil (Data Routing must be on upper layers to avoid long via/ vertical Xtalk)	65 mil or 120mil (Data Routing must be on upper layers to avoid long via/ vertical Xtalk)	65mil or 120mil (Data Routing must be on upper layers to avoid long via/ vertical Xtalk)
PCB Stripline Trace Impedance (ohms)	50	50	50
Memory Configuration	1 xDIMM per Channel (40 Bits total), x8 or x16	1x DIMM per Channel (40 Bits total), x8 or x16	1x DIMM per Channel (40 bits total), x8 or x16
# of Rank	2 per DIMM	2 per DIMM	2 per DIMM
Max. Length Total (Inch)	4.5	4.5	4.5
Notes	Max. package length in FPGA design is shorter than 34mm.		

7.4.8.4. Example of a DDR5 layout on an Altera FPGA Platform Board

The following figure shows the layout example of a DDR5 x32 plus 8x ECC (3x DRAM Dual-Die Memory Down).

The DDR5 Data signal routing is on upper layers to avoid vertical crosstalk and achieve high performance; the CS/CTRL signals can be routed on deeper layers.

Figure 55. DDR5 32bits+ 8x bits ECC (3 x DRAMs Dual Die) PCB routing on an Altera FPGA Platform Board with thick stackup



7.4.9. DDR5 RDIMM Power Management IC

DDR5 RDIMM modules contain their own on-module power management IC (PMIC), which conserves board space and eliminates a need for voltage regulators on the motherboard.

The PMIC has two supply inputs, a VIN_Mgmt supply and a VIN_Bulk supply. The PMIC features four step-down switching regulators and three LDO regulators. The PMIC is powered from the VIN_Bulk input for switching regulators and VIN_Mgmt for the rest of the PMIC, including LDO voltages.

The Agilex 7 M-Series calibration algorithm does not support PMIC configuration and bring-up; you are required to perform these activities yourself. The following outlines the power-on sequence that the host must perform:

1. Ramp up VIN_Mgmt supply and VIN_Bulk supply; these may come up in either order, but VIN_Mgmt must be present for the PMIC to work. The recommendation is to ramp up VIN_Mgmt first. (The *JEDEC PMIC50x0 Power Management IC Standard* defines ramp-up and ramp-down rate limits and threshold for voltage delivered to PMIC VIN_Bulk and VIN_Mgmt).
2. Hold VIN_Mgmt supply stable for the minimum of time defined in the *JEDEC PMIC50x0 Power Management IC Standard*.
3. Hold VIN_Bulk supply stable for the minimum of time defined in the *JEDEC PMIC50x0 Power Management IC Standard*.
4. Query the status of the PMIC status register to determine whether it is safe to enable VR.
5. If it is safe to enable, send the VR Enable command.

For detail information on PMIC features, input/output electrical characteristics, register set and complete power-up sequence, refer to the *JEDEC PMIC50x0 Power Management IC Standard*.

7.4.10. DDR5 Simulation Strategy

The simulation strategy has two parts:

- Data Signal signal integrity simulation with respect to their DQS on the worse signal integrity of a data group (considering the longest routing and maximum vertical crosstalk between signals).
- CS/CTRL/CMD signal integrity simulation with respect to their CLK signals on the worst signal integrity of those signals (considering the longest routing and maximum vertical crosstalk between signals).

Take DQ simulation as an example, Altera recommends that a signal integrity engineer reviews the layout and picks the worst data group (select a victim and surrounded aggressors and DQS in the group) that has the worst signal integrity on the layout (that is, the worst cross talk coupling between deep vertical vias), long trace/PCB routing and maximum reflection on routing path due to long via stubs if backdrilling is not applied.

Altera recommends that you perform signal integrity simulation of the board layout for the selected victim surrounded by aggressor signals.

Ensure that the channel analysis is performed in the time domain (using PRBS pattern for I/O signal generator) while the channel is built, by using the actual per-pin package model at both ends, and PCB model in the format of scattering parameter along with I/O buffer model at both ends. DDR5 requires an IBIS AMI buffer model (due to equalizations/FFE/DFE at RX) at both ends to recover the data. Evaluate the eye diagram after the simulation to ensure that the design meets eye specification at both ends.

Note: Altera recommends that designers strictly follow the PCB routing design guidelines in this chapter, to achieve the maximum supported data rate for the selected configuration.

8. Agilex 7 M-Series FPGA EMIF IP – LPDDR5 Support

This chapter contains IP pin planning information for Agilex 7 M-Series FPGA external memory interface IP for LPDDR5.

8.1. Agilex 7 M-Series LPDDR5 External Memory Interfaces (EMIF) IP Parameter Descriptions

The following tables describe the parameters available on each tab of the IP parameter editor, which you can use to configure your IP.

Table 222. Group: High-level Configuration / Memory Device

Parameter Name	Description
Number of Channels	Specifies the number of channels that the interface should implement. For multi-channel devices, this should always match the number of channels on the device. Default value is 2 Legal values are: 1, 2, 4 (Identifier: MEM_NUM_CHANNELS)
Data DQ Width	Number of DQ pins per memory channel, used for data. Default value is 16 Legal values are: 16, 32 (Identifier: MEM_CHANNEL_DATA_DQ_WIDTH)
Die Density	Capacity of each memory die (in Gbits), per channel per die. For dual-die packages, this is the density of each die, not the density of the full package. Default value is 32 Legal values are: • 2Gbits (2) • 3Gbits (3) • 4Gbits (4) • 6Gbits (6) • 8Gbits (8) • 12Gbits (12) • 16Gbits (16) • 24Gbits (24) • 32Gbits (32) (Identifier: MEM_DIE_DENSITY_GBITS)
CS Width	Specifies the total number of CS pins used by each channel. Default value is 1 Legal values are: 1, 2 (Identifier: MEM_CHANNEL_CS_WIDTH)
Auto-set Memory Operating Frequency - FSP0	If true, let IP select max frequency that this configuration can support for the current device speedgrade. If false, user can set custom value for operating frequency.
<i>continued...</i>	

Parameter Name	Description
	Default value is true (Identifier: MEM_FSP0_OPERATING_FREQ_MHZ_AUTOSET_EN)
Memory Operating Frequency - FSP0	Specifies the FSP0 operating frequency of the memory interface in MHz. This is not the same as boot-frequency (boot frequency is not a user parameter). Value is specified in megahertz Legal values are: 800, 1066.667, 1375, 1600, 1866.667, 2133.333, 2400, 2750 (Identifier: MEM_FSP0_OPERATING_FREQ_MHZ)
Enable Frequency Set Point (FSP) 1	If true, users can enable and set values for a second frequency set point. Default value is false (Identifier: MEM_FSP1_EN)
Auto-set Memory Operating Frequency - FSP1	If true, let IP select max frequency that this configuration can support for the current device speedgrade. If false, user can set custom value for operating frequency. Default value is true (Identifier: MEM_FSP1_OPERATING_FREQ_MHZ_AUTOSET_EN)
Memory Operating Frequency - FSP1	Specifies the FSP1 operating frequency of the memory interface in MHz. Value is specified in megahertz Legal values are: 800, 1066.667, 1375, 1600, 1866.667, 2133.333, 2400, 2750 (Identifier: MEM_FSP1_OPERATING_FREQ_MHZ)
Enable Frequency Set Point (FSP) 2	If true, users can enable and set values for a third frequency set point. Default value is false (Identifier: MEM_FSP2_EN)
Auto-set Memory Operating Frequency - FSP2	If true, let IP select max frequency that this configuration can support for the current device speedgrade. If false, user can set custom value for operating frequency. Default value is true (Identifier: MEM_FSP2_OPERATING_FREQ_MHZ_AUTOSET_EN)
Memory Operating Frequency - FSP2	Specifies the FSP2 operating frequency of the memory interface in MHz. Value is specified in megahertz Legal values are: 800, 1066.667, 1375, 1600, 1866.667, 2133.333, 2400, 2750 (Identifier: MEM_FSP2_OPERATING_FREQ_MHZ)
Memory Operating Frequency	Specifies the frequency at which the memory interface will run. (Identifier: MEM_OPERATING_FREQ_MHZ)

Table 223. Group: High-level Configuration / PHY

Parameter Name	Description
Auto-set PLL Reference Clock Frequency	If true, let IP select max PLL refclk frequency that this configuration can support. If false, user can set custom value for PLL refclk frequency. Default value is true (Identifier: PHY_REFCLK_FREQ_MHZ_AUTOSET_EN)
Enable Advanced List of PLL Reference Clock Frequencies	If true, provide extended list of possible refclk values. Otherwise, prune possible list of refclk values to a more reasonable length. Default value is false (Identifier: PHY_REFCLK_ADVANCED_SELECT_EN)
<i>continued...</i>	

Parameter Name	Description
Reference Clock Frequency	Specifies the reference clock frequency for the EMIF IOPLL. Value is specified in megahertz (Identifier: PHY_REFCLK_FREQ_MHZ)
AC Placement	Indicates location on the device where the interface will reside (specifically, the location of the AC lanes in terms I/O BANK and TOP vs BOT part of the I/O BANK). Legal ranges are derived from device floorplan. Default value is BOT Legal values are: • AC Bottom Sub-bank (lanes 0-3) (BOT) • AC Top Sub-bank (lanes 4-7) (TOP) • Ch0 Bot Sub-Bank / Ch1 Top Sub-Bank (FULL) (Identifier: PHY_AC_PLACEMENT)
Auto-set Mainband Access Mode	If true, let IP select most likely usecase for the PHY_MAINBAND_ACCESS_MODE; if false, let user set a custom value for sideband access mode. Default value is true (Identifier: PHY_MAINBAND_ACCESS_MODE_AUTOSSET_EN)
Mainband Access Mode	Specifies the path through which the EMIF QHIP mainband interface is exposed to the user. The mainband interface is the AXI4 interface to the memory controller. Legal values are: • NoC (NOC) • Fabric Direct - User Clock Asynchronous to PHY (ASYNC) • Fabric Direct - User Clock Synchronous to PHY (SYNC) (Identifier: PHY_MAINBAND_ACCESS_MODE)
Auto-set Sideband Access Mode	If true, let IP select most likely usecase for the PHY_SIDEHAND_ACCESS_MODE; if false, let user set a custom value for sideband access mode. Default value is true (Identifier: PHY_SIDEHAND_ACCESS_MODE_AUTOSSET_EN)
Sideband Access Mode	Specifies the path through which the EMIF QHIP sideband interface is exposed to the user. The sideband interface is the AXI4-Lite interface to the IOSSM. Legal values are: • NoC (NOC) • Fabric Direct (FABRIC) (Identifier: PHY_SIDEHAND_ACCESS_MODE)
Pin Swizzle Map	Specifies the swizzle map for the data lanes and pins. (Identifier: PHY_SWIZZLE_MAP)
Use Debug Toolkit	If enabled, the AXI-L port will be connected to SLD nodes, allowing for a system-console avalon manager interface to interact with this AXI-L subordinate interface. Default value is false (Identifier: DEBUG_TOOLS_EN)
Instance ID	Instance ID of the EMIF IP. This is useful when using a discovery mechanism over the side-band interface, to identify which EMIF instance's mailbox is at which offset. If expecting to use a discovery mechanism in hardware, this parameter must be set uniquely for all EMIFs that share a sideband. Otherwise, this parameter can be ignored / kept at the default value. Default value is 0 Legal values are: from 0 to 6

Parameter Name	Description
	(Identifier: INSTANCE_ID)

Table 224. Group: High-level Configuration / Controller / Controller Options

Parameter Name	Description
Use In-Line ECC	Specifies whether in-line ECC is enabled in the controller. If this option is selected, one-eighth of the memory is used for the ECC bits; this reduces the available AXI Address range by one-eighth. Default value is false (Identifier: CTRL_ECC_INLINE_EN)
Use WR Link ECC for FSP0	Specifies whether write link ECC is enabled in the controller for Frequency Set Point 0. Default value is false (Identifier: CTRL_ECC_WR_LINK_EN_FSP0)
Use WR Link ECC for FSP1	Specifies whether write link ECC is enabled in the controller for Frequency Set Point 1. Default value is false (Identifier: CTRL_ECC_WR_LINK_EN_FSP1)
Use WR Link ECC for FSP2	Specifies whether write link ECC is enabled in the controller for Frequency Set Point 2. Default value is false (Identifier: CTRL_ECC_WR_LINK_EN_FSP2)
Use RD Link ECC for FSP0	Specifies whether read link ECC is enabled in the controller for Frequency Set Point 0. Default value is false (Identifier: CTRL_ECC_RD_LINK_EN_FSP0)
Use RD Link ECC for FSP1	Specifies whether read link ECC is enabled in the controller for Frequency Set Point 1. Default value is false (Identifier: CTRL_ECC_RD_LINK_EN_FSP1)
Use RD Link ECC for FSP2	Specifies whether read link ECC is enabled in the controller for Frequency Set Point 2. Default value is false (Identifier: CTRL_ECC_RD_LINK_EN_FSP2)
Use ECC Autocorrection	If ECC is enabled, specifies whether single-bit-errors (SBEs) should be corrected or just reported. Default value is true (Identifier: CTRL_ECC_AUTOCORRECT_EN)
Use Data Masking	Specifies whether Data Masking is enabled by the controller. RMW will occur under these conditions: AXI transactions using a subset of write strobe, or one beat long AXI transaction in DDR5, LPDDR5, or LPDDR4 interfaces. Altera recommends enabling the DM option to improve efficiency for the following cases: non-ECC design with AXI transaction listed above, and LPDDR4 or LPDDR5 designs with In-Line ECC enabled. Default value is false (Identifier: CTRL_DM_EN)
Use WDBI	Specifies whether write Data-bus-inversion is enabled by the controller. Default value is false (Identifier: CTRL_WR_DBI_EN)
<i>continued...</i>	

Parameter Name	Description
Use RDBI	Specifies whether read Data-bus-inversion is enabled by the controller. Default value is false (Identifier: CTRL_RD_DBI_EN)
Controller Performance Profile	Group of Controller settings to optimize performance based on expected traffic patterns. Choose "Custom" for fine-granular control Default value is SEQ Legal values are: • Sequential Access Optimized (SEQ) • Random Access Optimized (RAND) • Custom (CUSTOM) (Identifier: CTRL_PERFORMANCE_PROFILE)
Force Auto-Precharge	Controller will terminate all transactions with an auto-precharge when enabled. For per-transaction granularity, leave this option disabled and use the AxUSER control signals instead. Default value is false (Identifier: CTRL_AUTO_PRECHARGE_EN)
Bank Group Rotation	Moves a number of Bank Group pins lower in the address order Default value is 1 Legal values are: • No BGs (disabled) (0) • Two BGs (BG[0]) (1) • Four BGs (BG[1:0]) (2) (Identifier: CTRL_BG_ROTATE_EN)
Enable Row/CS Swap	Specifies whether the highest order address bits are the CS bit (false) or Row bits (true) Default value is false (Identifier: DIAG_HMC_ADDR_SWAP_EN)

Table 225. Group: High-level Configuration / Controller / Data Bus Turnaround Times

Parameter Name	Description
Additional read-to-write turnaround time (same rank)	Additional memory clock cycles of delay between read and write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2W_SAMECS_CYC)
Additional read-to-read turnaround time (same rank)	Additional memory clock cycles of delay between read operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2R_SAMECS_CYC)
Additional write-to-write turnaround time (same rank)	Additional memory clock cycles of delay between write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2W_SAMECS_CYC)
<i>continued...</i>	

Parameter Name	Description
Additional write-to-read turnaround time (same rank)	Additional memory clock cycles of delay between read and write operations within the same rank Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2R_SAMECS_CYC)
Additional read-to-write turnaround time (different ranks)	Additional controller cycles of delay between read and write operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2W_DIFFCS_CYC)
Additional read-to-read turnaround time (different ranks)	Additional controller cycles of delay between read operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_R2R_DIFFCS_CYC)
Additional write-to-write turnaround time (different ranks)	Additional controller cycles of delay between write operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2W_DIFFCS_CYC)
Additional write-to-read turnaround time (different ranks)	Additional controller cycles of delay between write and read operations across different ranks Value is specified in Cycles Default value is 0 Legal values are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16 (Identifier: TURNAROUND_W2R_DIFFCS_CYC)

Table 226. Group: High-level Configuration / Advanced Calibration Settings

Parameter Name	Description
Enable Margining during Calibration	Enable margining stages of calibration for use with the EMIF Debug Toolkit. This will increase calibration time Default value is false (Identifier: ADV_CAL_ENABLE_MARGIN)
Enable REQ Calibration	Enable the Read DQ Equalization stage as part of calibration. This will increase calibration time but will also improve top-line performance Default value is false (Identifier: ADV_CAL_ENABLE_REQ)
Enable WEQ Calibration	Run the Write DQ Equalization stage as part of calibration. This will increase calibration time but will also improve top-line performance Default value is false (Identifier: ADV_CAL_ENABLE_WEQ)
Enable Read DFE Training	Enable DFE training as part of calibration Default value is false
<i>continued...</i>	

Parameter Name	Description
	(Identifier: ADV_CAL_ENABLE_RD_DFE)
Enable Write DFE Training	Enable DFE training as part of calibration Default value is false (Identifier: ADV_CAL_ENABLE_WR_DFE)

Table 227. Group: Advanced: Memory Timing / Overrides / JEDEC_TABLE

Parameter Name	Description
JEDEC Parameter	Name of JEDEC Parameter to explicitly override; the values will be applied and appear in the list below. Default value is Legal values are: • [FSP0] Read Latency (MEM_FSP0_CL_CYC) • [FSP1] Read Latency (MEM_FSP1_CL_CYC) • [FSP2] Read Latency (MEM_FSP2_CL_CYC) • [FSP0] Write Latency (MEM_FSP0_CWL_CYC) • [FSP1] Write Latency (MEM_FSP1_CWL_CYC) • [FSP2] Write Latency (MEM_FSP2_CWL_CYC) • Read Postamble Mode (MEM_RDQS_POSTAMBLE_MODE) • Read Preamble Cycles (MEM_RD_PREAMBLE_CYC) • Read Postamble Cycles (MEM_RD_POSTAMBLE_CYC) • Write Postamble Cycles (MEM_WR_POSTAMBLE_CYC) • Min Number of Refs Req'd (MEM_MINNUMREFSREQ) • tRCD (MEM_TRCD_NS) • tRPab (MEM_TRPAB_NS) • tRPpb (MEM_TRPPB_NS) • tRAS (MEM_TRAS_NS) • tRAS_MAX (MEM_TRAS_MAX_NS) • tWR (MEM_TWR_NS) • [FSP0] tRRD_L (MEM_FSP0_TRRD_L_NS) • [FSP1] tRRD_L (MEM_FSP1_TRRD_L_NS) • [FSP2] tRRD_L (MEM_FSP2_TRRD_L_NS) • [FSP0] tRRD_S (MEM_FSP0_TRRD_S_NS) • [FSP1] tRRD_S (MEM_FSP1_TRRD_S_NS) • [FSP2] tRRD_S (MEM_FSP2_TRRD_S_NS) • tFAW (MEM_TFAW_NS) • [FSP0] tRBTP (MEM_FSP0_TRBTP_NS) • [FSP1] tRBTP (MEM_FSP1_TRBTP_NS) • [FSP2] tRBTP (MEM_FSP2_TRBTP_NS) • [FSP0] tWTR_S (MEM_FSP0_TWTR_S_NS) • [FSP1] tWTR_S (MEM_FSP1_TWTR_S_NS) • [FSP2] tWTR_S (MEM_FSP2_TWTR_S_NS) • [FSP0] tWTR_L (MEM_FSP0_TWTR_L_NS) • [FSP1] tWTR_L (MEM_FSP1_TWTR_L_NS) • [FSP2] tWTR_L (MEM_FSP2_TWTR_L_NS) • [FSP0] tPPD (MEM_FSP0_TPPD_NS) • [FSP1] tPPD (MEM_FSP1_TPPD_NS) • [FSP2] tPPD (MEM_FSP2_TPPD_NS) • tRC (MEM_TRC_NS) • tZQLAT (MEM_TZQLAT_NS) • tpw_RESET (MEM_TPW_RESET_NS) • tERQE (MEM_TERQE_NS) • tERQX (MEM_TERQX_NS)

Parameter Name	Description
	• tRDQE_OD (MEM_TRDQE_OD_NS) • tRDQX_OD (MEM_TRDQX_OD_NS) • tRDQSTFE (MEM_TRDQSTFE_NS) • tRDQSTFX (MEM_TRDQSTFX_NS) • tCCDMW (MEM_TCCDMW_NS) • tREFW (MEM_TREFW_NS) • tREFI (MEM_TREFI_NS) • [FSP0] tRFCab (MEM_FSP0_TRFCAB_NS) • [FSP1] tRFCab (MEM_FSP1_TRFCAB_NS) • [FSP2] tRFCab (MEM_FSP2_TRFCAB_NS) • [FSP0] tRFCpb (MEM_FSP0_TRFCPB_NS) • [FSP1] tRFCpb (MEM_FSP1_TRFCPB_NS) • [FSP2] tRFCpb (MEM_FSP2_TRFCPB_NS) • [FSP0] tpbR2pbR (MEM_FSP0_TPBR2PBR_NS) • [FSP1] tpbR2pbR (MEM_FSP1_TPBR2PBR_NS) • [FSP2] tpbR2pbR (MEM_FSP2_TPBR2PBR_NS) • tpbR2ACT (MEM_TPBR2ACT_NS) • [FSP0] tCKCSH (MEM_FSP0_TCKCSH_NS) • [FSP1] tCKCSH (MEM_FSP1_TCKCSH_NS) • [FSP2] tCKCSH (MEM_FSP2_TCKCSH_NS) • [FSP0] tCMDPD (MEM_FSP0_TCMDPD_NS) • [FSP1] tCMDPD (MEM_FSP1_TCMDPD_NS) • [FSP2] tCMDPD (MEM_FSP2_TCMDPD_NS) • [FSP0] tXP (MEM_FSP0_TXP_NS) • [FSP1] tXP (MEM_FSP1_TXP_NS) • [FSP2] tXP (MEM_FSP2_TXP_NS) • tCSH (MEM_TCSH_NS) • [FSP0] tCSLCK (MEM_FSP0_TCSLCK_NS) • [FSP1] tCSLCK (MEM_FSP1_TCSLCK_NS) • [FSP2] tCSLCK (MEM_FSP2_TCSLCK_NS) • [FSP0] tCSPD (MEM_FSP0_TCSPD_NS) • [FSP1] tCSPD (MEM_FSP1_TCSPD_NS) • [FSP2] tCSPD (MEM_FSP2_TCSPD_NS) • [FSP0] tMRWPD (MEM_FSP0_TMRWPD_NS) • [FSP1] tMRWPD (MEM_FSP1_TMRWPD_NS) • [FSP2] tMRWPD (MEM_FSP2_TMRWPD_NS) • [FSP0] tZQPD (MEM_FSP0_TZQPD_NS) • [FSP1] tZQPD (MEM_FSP1_TZQPD_NS) • [FSP2] tZQPD (MEM_FSP2_TZQPD_NS) • [FSP0] tESPD (MEM_FSP0_TESPD_NS) • [FSP1] tESPD (MEM_FSP1_TESPD_NS) • [FSP2] tESPD (MEM_FSP2_TESPD_NS) • tSR (MEM_TSR_NS) • [FSP0] tXSR (MEM_FSP0_TXSR_NS) • [FSP1] tXSR (MEM_FSP1_TXSR_NS) • [FSP2] tXSR (MEM_FSP2_TXSR_NS) • [FSP0] tMRR (MEM_FSP0_TMRR_NS) • [FSP1] tMRR (MEM_FSP1_TMRR_NS) • [FSP2] tMRR (MEM_FSP2_TMRR_NS) • [FSP0] tMRW (MEM_FSP0_TMRW_NS) • [FSP1] tMRW (MEM_FSP1_TMRW_NS) • [FSP2] tMRW (MEM_FSP2_TMRW_NS) • [FSP0] tMRD (MEM_FSP0_TMRD_NS) • [FSP1] tMRD (MEM_FSP1_TMRD_NS)

Parameter Name	Description
	• [FSP2] tMRD (MEM_FSP2_TMRD_NS) • tOSCO (MEM_TOSCO_NS) • tDQSCK (MEM_TDQSCK_NS) (Identifier: JEDEC_OVERRIDE_TABLE_PARAM_NAME)

Table 228. Group: Advanced: Memory Timing / Values

Parameter Name	Description
[FSP0] Read Latency	[FSP0] Read Latency of the memory device in clock cycles. (Identifier: MEM_FSP0_CL_CYC)
[FSP1] Read Latency	[FSP1] Read Latency of the memory device in clock cycles. (Identifier: MEM_FSP1_CL_CYC)
[FSP2] Read Latency	[FSP2] Read Latency of the memory device in clock cycles. (Identifier: MEM_FSP2_CL_CYC)
[FSP0] Write Latency	[FSP0] Write Latency in clock cycles. (Identifier: MEM_FSP0_CWL_CYC)
[FSP1] Write Latency	[FSP1] Write Latency in clock cycles. (Identifier: MEM_FSP1_CWL_CYC)
[FSP2] Write Latency	[FSP2] Write Latency in clock cycles. (Identifier: MEM_FSP2_CWL_CYC)
Read Postamble Mode	RDQS Postamble Mode. (Identifier: MEM_RDQS_POSTAMBLE_MODE)
Read Preamble Cycles	RDQS Preamble length (in cycles). (Identifier: MEM_RD_PREAMBLE_CYC)
Read Postamble Cycles	RDQS Postamble length (in cycles). (Identifier: MEM_RD_POSTAMBLE_CYC)
Write Postamble Cycles	WCK Postamble length (in cycles). (Identifier: MEM_WR_POSTAMBLE_CYC)
Min Number of Refs Req'd	Minimum Number of Refreshes Required. (Identifier: MEM_MINNUMREFSREQ)
tRCD	RAS-to-CAS Delay in nanoseconds. (Identifier: MEM_TRCD_NS)
tRPab	All-Bank Precharge Time in nanoseconds. (Identifier: MEM_TRPAB_NS)
tRPpb	Per-Bank Precharge Time in nanoseconds. (Identifier: MEM_TRPPB_NS)
tRAS	Row Active Time in nanoseconds. (Identifier: MEM_TRAS_NS)
tRAS_MAX	Specifies the maximum Activate-to-Precharge command period in nanoseconds. (Identifier: MEM_TRAS_MAX_NS)
tWR	Write Recovery Time in nanoseconds.

continued...

Parameter Name	Description
	(Identifier: MEM_TWR_NS)
[FSP0] tRRD_L	[FSP0] RAS-to-RAS (Active Bank-A to Active Bank-B) Delay Time (Long) in nanoseconds. (Identifier: MEM_FSP0_TRRD_L_NS)
[FSP1] tRRD_L	[FSP1] RAS-to-RAS (Active Bank-A to Active Bank-B) Delay Time (Long) in nanoseconds. (Identifier: MEM_FSP1_TRRD_L_NS)
[FSP2] tRRD_L	[FSP2] RAS-to-RAS (Active Bank-A to Active Bank-B) Delay Time (Long) in nanoseconds. (Identifier: MEM_FSP2_TRRD_L_NS)
[FSP0] tRRD_S	[FSP0] RAS-to-RAS (Active Bank-A to Active Bank-B) Delay Time (Short) in nanoseconds. (Identifier: MEM_FSP0_TRRD_S_NS)
[FSP1] tRRD_S	[FSP1] RAS-to-RAS (Active Bank-A to Active Bank-B) Delay Time (Short) in nanoseconds. (Identifier: MEM_FSP1_TRRD_S_NS)
[FSP2] tRRD_S	[FSP2] RAS-to-RAS (Active Bank-A to Active Bank-B) Delay Time (Short) in nanoseconds. (Identifier: MEM_FSP2_TRRD_S_NS)
tFAW	Four-bank ACTIVE window time in nanoseconds. (Identifier: MEM_TFAW_NS)
[FSP0] tRBTP	[FSP0] Read Burst End to Precharge Command Delay in nanoseconds. (Identifier: MEM_FSP0_TRBTP_NS)
[FSP1] tRBTP	[FSP1] Read Burst End to Precharge Command Delay in nanoseconds. (Identifier: MEM_FSP1_TRBTP_NS)
[FSP2] tRBTP	[FSP2] Read Burst End to Precharge Command Delay in nanoseconds. (Identifier: MEM_FSP2_TRBTP_NS)
[FSP0] tWTR_S	[FSP0] Write-to-Read Delay (Short) in nanoseconds. (Identifier: MEM_FSP0_TWTR_S_NS)
[FSP1] tWTR_S	[FSP1] Write-to-Read Delay (Short) in nanoseconds. (Identifier: MEM_FSP1_TWTR_S_NS)
[FSP2] tWTR_S	[FSP2] Write-to-Read Delay (Short) in nanoseconds. (Identifier: MEM_FSP2_TWTR_S_NS)
[FSP0] tWTR_L	[FSP0] Write-to-Read Delay (Long) in nanoseconds. (Identifier: MEM_FSP0_TWTR_L_NS)
[FSP1] tWTR_L	[FSP1] Write-to-Read Delay (Long) in nanoseconds. (Identifier: MEM_FSP1_TWTR_L_NS)
[FSP2] tWTR_L	[FSP2] Write-to-Read Delay (Long) in nanoseconds. (Identifier: MEM_FSP2_TWTR_L_NS)
[FSP0] tPPD	[FSP0] Precharge-to-precharge delay in nanoseconds. (Identifier: MEM_FSP0_TPPD_NS)
[FSP1] tPPD	[FSP1] Precharge-to-precharge delay in nanoseconds.
continued...	

Parameter Name	Description
	(Identifier: MEM_FSP1_TPPD_NS)
[FSP2] tPPD	[FSP2] Precharge-to-precharge delay in nanoseconds. (Identifier: MEM_FSP2_TPPD_NS)
tRC	Activate-to-Activate command period (same bank) in nanoseconds. (Identifier: MEM_TRC_NS)
tZQLAT	ZQCAL Latch Quiet Time in nanoseconds. (Identifier: MEM_TZQLAT_NS)
tpw_RESET	Min RESET_n low time for Reset Initialization with Stable Power Time in nanoseconds. (Identifier: MEM_TPW_RESET_NS)
terQE	Enhanced RDQS Toggle Mode Entry Time in nanoseconds. (Identifier: MEM_TERQE_NS)
terQX	Enhanced RDQS Toggle Mode Exit Time in nanoseconds. (Identifier: MEM_TERQX_NS)
trDQE_OD	ODT-disable from Enhanced RDQS Toggle Mode Entry Time in nanoseconds. (Identifier: MEM_TRDQE_OD_NS)
trDQX_OD	ODT-enable from Enhanced RDQS Toggle Mode Exit Time in nanoseconds. (Identifier: MEM_TRDQX_OD_NS)
trDQSTFE	Read/Write-based RDQS_t Training Mode Entry Time in nanoseconds. (Identifier: MEM_TRDQSTFE_NS)
trDQSTFX	Read/Write-based RDQS_t Training Mode Exit Time in nanoseconds. (Identifier: MEM_TRDQSTFX_NS)
tCCDMW	CAS-to-CAS Delay for Masked Write in nanoseconds. (Identifier: MEM_TCCDMW_NS)
tREFW	Refresh Window in nanoseconds. (Identifier: MEM_TREFW_NS)
tREFI	Refresh Interval Time in nanoseconds. (Identifier: MEM_TREFI_NS)
[FSP0] tRFCab	[FSP0] All-Bank Refresh Cycle Time in nanoseconds. (Identifier: MEM_FSP0_TRFCAB_NS)
[FSP1] tRFCab	[FSP1] All-Bank Refresh Cycle Time in nanoseconds. (Identifier: MEM_FSP1_TRFCAB_NS)
[FSP2] tRFCab	[FSP2] All-Bank Refresh Cycle Time in nanoseconds. (Identifier: MEM_FSP2_TRFCAB_NS)
[FSP0] tRFCpb	[FSP0] Per-Bank Refresh Cycle Time in nanoseconds. (Identifier: MEM_FSP0_TRFCPB_NS)
[FSP1] tRFCpb	[FSP1] Per-Bank Refresh Cycle Time in nanoseconds. (Identifier: MEM_FSP1_TRFCPB_NS)
[FSP2] tRFCpb	[FSP2] Per-Bank Refresh Cycle Time in nanoseconds. (Identifier: MEM_FSP2_TRFCPB_NS)
<i>continued...</i>	

Parameter Name	Description
[FSP0] tpbR2pbR	[FSP0] Per-Bank Refresh to Per-Bank Refresh minimum interval time in nanoseconds. (Identifier: MEM_FSP0_TPBR2PBR_NS)
[FSP1] tpbR2pbR	[FSP1] Per-Bank Refresh to Per-Bank Refresh minimum interval time in nanoseconds. (Identifier: MEM_FSP1_TPBR2PBR_NS)
[FSP2] tpbR2pbR	[FSP2] Per-Bank Refresh to Per-Bank Refresh minimum interval time in nanoseconds. (Identifier: MEM_FSP2_TPBR2PBR_NS)
tpbR2ACT	Per-Bank Refresh to Activate minimum interval time in nanoseconds. (Identifier: MEM_TPBR2ACT_NS)
[FSP0] tCKCSH	[FSP0] Valid Clock Requirement before CS goes High (Power-Down AC Timings) in nanoseconds. (Identifier: MEM_FSP0_TCKCSH_NS)
[FSP1] tCKCSH	[FSP1] Valid Clock Requirement before CS goes High (Power-Down AC Timings) in nanoseconds. (Identifier: MEM_FSP1_TCKCSH_NS)
[FSP2] tCKCSH	[FSP2] Valid Clock Requirement before CS goes High (Power-Down AC Timings) in nanoseconds. (Identifier: MEM_FSP2_TCKCSH_NS)
[FSP0] tCMDPD	[FSP0] Delay from valid command to Power Down Entry in nanoseconds. (Identifier: MEM_FSP0_TCMDPD_NS)
[FSP1] tCMDPD	[FSP1] Delay from valid command to Power Down Entry in nanoseconds. (Identifier: MEM_FSP1_TCMDPD_NS)
[FSP2] tCMDPD	[FSP2] Delay from valid command to Power Down Entry in nanoseconds. (Identifier: MEM_FSP2_TCMDPD_NS)
[FSP0] tXP	[FSP0] Exit Power-Down to Next Valid Command Delay Time in nanoseconds. (Identifier: MEM_FSP0_TXP_NS)
[FSP1] tXP	[FSP1] Exit Power-Down to Next Valid Command Delay Time in nanoseconds. (Identifier: MEM_FSP1_TXP_NS)
[FSP2] tXP	[FSP2] Exit Power-Down to Next Valid Command Delay Time in nanoseconds. (Identifier: MEM_FSP2_TXP_NS)
tCSH	Minimum CS High Pulse Width at Power Down Exit in nanoseconds. (Identifier: MEM_TCSH_NS)
[FSP0] tCSLCK	[FSP0] Valid Clock Requirement after Power Down Entry in nanoseconds. (Identifier: MEM_FSP0_TCSLCK_NS)
[FSP1] tCSLCK	[FSP1] Valid Clock Requirement after Power Down Entry in nanoseconds. (Identifier: MEM_FSP1_TCSLCK_NS)
[FSP2] tCSLCK	[FSP2] Valid Clock Requirement after Power Down Entry in nanoseconds. (Identifier: MEM_FSP2_TCSLCK_NS)
continued...	

Parameter Name	Description
[FSP0] tCSPD	[FSP0] Delay time from Power Down Entry to CS going High in nanoseconds. (Identifier: MEM_FSP0_TCSPD_NS)
[FSP1] tCSPD	[FSP1] Delay time from Power Down Entry to CS going High in nanoseconds. (Identifier: MEM_FSP1_TCSPD_NS)
[FSP2] tCSPD	[FSP2] Delay time from Power Down Entry to CS going High in nanoseconds. (Identifier: MEM_FSP2_TCSPD_NS)
[FSP0] tMRWPD	[FSP0] Delay from MRW Command to Power Down Entry in nanoseconds. (Identifier: MEM_FSP0_TMRWPD_NS)
[FSP1] tMRWPD	[FSP1] Delay from MRW Command to Power Down Entry in nanoseconds. (Identifier: MEM_FSP1_TMRWPD_NS)
[FSP2] tMRWPD	[FSP2] Delay from MRW Command to Power Down Entry in nanoseconds. (Identifier: MEM_FSP2_TMRWPD_NS)
[FSP0] tZQPD	[FSP0] Delay from ZQ Calibration Start/Latch Command to Power Down Entry in nanoseconds. (Identifier: MEM_FSP0_TZQPD_NS)
[FSP1] tZQPD	[FSP1] Delay from ZQ Calibration Start/Latch Command to Power Down Entry in nanoseconds. (Identifier: MEM_FSP1_TZQPD_NS)
[FSP2] tZQPD	[FSP2] Delay from ZQ Calibration Start/Latch Command to Power Down Entry in nanoseconds. (Identifier: MEM_FSP2_TZQPD_NS)
[FSP0] tESPD	[FSP0] Delay time from Self-Refresh Entry command to Power Down Entry command in nanoseconds. (Identifier: MEM_FSP0_TESPD_NS)
[FSP1] tESPD	[FSP1] Delay time from Self-Refresh Entry command to Power Down Entry command in nanoseconds. (Identifier: MEM_FSP1_TESPD_NS)
[FSP2] tESPD	[FSP2] Delay time from Self-Refresh Entry command to Power Down Entry command in nanoseconds. (Identifier: MEM_FSP2_TESPD_NS)
tSR	Minimum Self-Refresh Time (Entry to Exit) in nanoseconds. (Identifier: MEM_TSR_NS)
[FSP0] tXSR	Exit Self-Refresh time in nanoseconds. (Identifier: MEM_FSP0_TXSR_NS)
[FSP1] tXSR	Exit Self-Refresh time in nanoseconds. (Identifier: MEM_FSP1_TXSR_NS)
[FSP2] tXSR	Exit Self-Refresh time in nanoseconds. (Identifier: MEM_FSP2_TXSR_NS)
[FSP0] tMRR	[FSP0] Mode Register Read Command Period Time in nanoseconds. (Identifier: MEM_FSP0_TMRR_NS)
continued...	

Parameter Name	Description
[FSP1] tMRR	[FSP1] Mode Register Read Command Period Time in nanoseconds. (Identifier: MEM_FSP1_TMRR_NS)
[FSP2] tMRR	[FSP2] Mode Register Read Command Period Time in nanoseconds. (Identifier: MEM_FSP2_TMRR_NS)
[FSP0] tMRW	[FSP0] Mode Register Write Command Period Time in nanoseconds. (Identifier: MEM_FSP0_TMRW_NS)
[FSP1] tMRW	[FSP1] Mode Register Write Command Period Time in nanoseconds. (Identifier: MEM_FSP1_TMRW_NS)
[FSP2] tMRW	[FSP2] Mode Register Write Command Period Time in nanoseconds. (Identifier: MEM_FSP2_TMRW_NS)
[FSP0] tMRD	[FSP0] Mode Register Set Command Period Time in nanoseconds. (Identifier: MEM_FSP0_TMRD_NS)
[FSP1] tMRD	[FSP1] Mode Register Set Command Period Time in nanoseconds. (Identifier: MEM_FSP1_TMRD_NS)
[FSP2] tMRD	[FSP2] Mode Register Set Command Period Time in nanoseconds. (Identifier: MEM_FSP2_TMRD_NS)
tOSCO	Delay time from Stop WCK2DQI/WCK2DQO Interval Oscillator Command to Mode Register Readout time in nanoseconds. (Identifier: MEM_TOSCO_NS)
tDQSCK	DQS output access time from CK in nanoseconds. (Identifier: MEM_TDQSCK_NS)

Table 229. Group: Advanced: Analog Overrides / Overrides / ANALOG_TABLE

Parameter Name	Description
Analog Parameter	Name of Analog Parameter to explicitly override; the values will be applied and appear in the list below. Default value is Legal values are: • Phy AC Drive Strength (PHY_TERM_X_R_S_AC_OUTPUT_OHM) • Phy CK Drive Strength (PHY_TERM_X_R_S_CK_OUTPUT_OHM) • Phy DQ Drive Strength (PHY_TERM_X_R_S_DQ_OUTPUT_OHM) • Phy DQ Slew Rate (PHY_TERM_X_DQ_SLEW_RATE) • Phy DQ Input Termination (PHY_TERM_X_R_T_DQ_INPUT_OHM) • Phy DQ Initial Vrefin (PHY_TERM_X_DQ_VREF) • Phy PLL Reference Clock Input Termination (PHY_TERM_X_R_T_REFCLK_INPUT_OHM) • Phy DFE Tap 1 (PHY_DFE_X_TAP_1) • Phy DFE Tap 2 (PHY_DFE_X_TAP_2) • Phy DFE Tap 3 (PHY_DFE_X_TAP_3) • Phy DFE Tap 4 (PHY_DFE_X_TAP_4) • Mem Target Write Termination (MEM_ODT_DQ_X_TGT_WR) • Mem DQ Non-Target Termination (MEM_ODT_DQ_X_NON_TGT) • Mem DQ Drive Strength (MEM_ODT_DQ_X_RON) • Mem Data Clock Termination (MEM_ODT_DQ_X_WCK) • Mem VrefDQ Value (MEM_VREF_DQ_X_VALUE) • Mem CA Common Termination (MEM_ODT_CA_X_CA_COMM) • Mem CA Termination Enable (MEM_ODT_CA_X_CA_ENABLE)

Parameter Name	Description
	• Mem CS Termination Enable (MEM_ODT_CA_X_CS_ENABLE) • Mem CK Termination Enable (MEM_ODT_CA_X_CK_ENABLE) • Mem VrefCA Value (MEM_VREF_CA_X_CA_VALUE) • Mem DFE Tap 1 (MEM_DFE_X_TAP_1) (Identifier: ANALOG_PARAM_DERIVATION_PARAM_NAME)

Table 230. Group: Advanced: Analog Overrides / Values

Parameter Name	Description
Phy AC Drive Strength	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the refclk input pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_AC_OUTPUT_OHM)
Phy CK Drive Strength	This parameter allows you to change the output on chip termination settings for the selected I/O standard on the CK Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_CK_OUTPUT_OHM)
Phy DQ I/O Standard	Specifies the I/O electrical standard for the data bus pins. The selected I/O standard configures the circuit within the I/O buffer to match the industry standard. (Identifier: PHY_TERM_X_DQ_IO_STD_TYPE)
Phy DQ Drive Strength	This parameter allows you to change the output on chip termination settings for the selected I/O standard on the DQ Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 40 Ohm (with calibration) (SERIES_40_OHM_CAL) (Identifier: PHY_TERM_X_R_S_DQ_OUTPUT_OHM)
Phy DQ Slew Rate	Specifies the slew rate of the data bus pins. The slew rate (or edge rate) describes how quickly the signal can transition, measured in voltage per unit time. <i>Perform board simulations to determine the slew rate that provides the best eye opening for the data bus signals.</i> Legal values are: • Slow (SLOW) • Medium (MEDIUM) • Fast (FAST) • Fastest (FASTEST) (Identifier: PHY_TERM_X_DQ_SLEW_RATE)
Phy DQ Input Termination	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the DQ Pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • 40 Ohm (with calibration) (RT_40_OHM_CAL) • 50 Ohm (with calibration) (RT_50_OHM_CAL) • 60 Ohm (with calibration) (RT_60_OHM_CAL) (Identifier: PHY_TERM_X_R_T_DQ_INPUT_OHM)
Phy DQ Initial Vrefin	Specifies the initial value for the reference voltage on the data pins(Vrefin) . The specified value serves as a starting point and may be overridden by calibration to provide better timing margins.

continued...

Parameter Name	Description
	Legal values are: from 0.0 to 100.0 (Identifier: PHY_TERM_X_DQ_VREF)
Phy PLL Reference Clock Input Termination	This parameter allows you to change the input on chip termination settings for the selected I/O standard on the refclk input pins. Perform board simulation with IBIS models to determine the best settings for your design. Legal values are: • No Termination (RT_OFF) • Differential Termination (RT_DIFF) (Identifier: PHY_TERM_X_R_T_REFCLK_INPUT_OHM)
Phy DFE Tap 1	This parameter allows you to select the amount of bias used on tap 1 of the FPGA DFE. Legal values are: • No Bias (0) • -26.25mV (n1) • -52.50mV (n2) • -78.75mV (n3) • -105.00mV (n4) • -131.25mV (n5) • -157.50mV (n6) • -183.75mV (n7) • -210.00mV (n8) • -236.25mV (n9) • -262.25mV (n10) • -288.75mV (n11) • -315.00mV (n12) • -341.25mV (n13) • -367.50mV (n14) • -393.75mV (n15) • -420.00mV (n16) • -446.25mV (n17) • -472.50mV (n18) • -498.75mV (n19) • -525.00mV (n20) • -551.25mV (n21) • -577.50mV (n22) • -603.75mV (n23) • -630.00mV (n24) • -656.25mV (n25) • -682.50mV (n26) • -708.75mV (n27) • -735.00mV (n28) • -761.25mV (n29) • -787.50mV (n30) • -813.75mV (n31) (Identifier: PHY_DFE_X_TAP_1)
Phy DFE Tap 2	This parameter allows you to select the amount of bias used on tap 2 of the FPGA DFE. Legal values are: • +183.75mV (p7) • +157.70mV (p6) • +131.25mV (p5) • +105.00mV (p4)

continued...

Parameter Name	Description
	• +78.75mV (p3) • +52.50mV (p2) • +26.25mV (p1) • No Bias (0) • -26.25mV (n1) • -52.50mV (n2) • -78.75mV (n3) • -105.00mV (n4) • -131.25mV (n5) • -157.50mV (n6) • -183.75mV (n7) • -210.00mV (n8) (Identifier: PHY_DFE_X_TAP_2)
Phy DFE Tap 3	This parameter allows you to select the amount of bias used on tap 3 of the FPGA DFE. Legal values are: • +183.75mV (p7) • +157.70mV (p6) • +131.25mV (p5) • +105.00mV (p4) • +78.75mV (p3) • +52.50mV (p2) • +26.25mV (p1) • No Bias (0) • -26.25mV (n1) • -52.50mV (n2) • -78.75mV (n3) • -105.00mV (n4) • -131.25mV (n5) • -157.50mV (n6) • -183.75mV (n7) • -210.00mV (n8) (Identifier: PHY_DFE_X_TAP_3)
Phy DFE Tap 4	This parameter allows you to select the amount of bias used on tap 4 of the FPGA DFE. Legal values are: • +78.75mV (p3) • +52.50mV (p2) • +26.25mV (p1) • No Bias (0) • -26.25mV (n1) • -52.50mV (n2) • -78.75mV (n3) • -105.00mV (n4) (Identifier: PHY_DFE_X_TAP_4)
Mem Target Write Termination	Specifies the target termination to be used during a write. The value of this parameter represents X, where: termination = RZQ/X = (240 Ohm)/X. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3)
continued...	

Parameter Name	Description
	• 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) (Identifier: MEM_ODT_DQ_X_TGT_WR)
Mem DQ Non-Target Termination	Specifies the termination to be used for the non-target rank in a multi-rank configuration. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) (Identifier: MEM_ODT_DQ_X_NON_TGT)
Mem DQ Drive Strength	Specifies the termination to be used when driving read data from memory. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. Legal values are: • 40 Ohm (RZQ/6) (6) • 48 Ohm (RZQ/5) (5) • 60 Ohm (RZQ/4) (4) • 80 Ohm (RZQ/3) (3) • 120 Ohm (RZQ/2) (2) • 240 Ohm (RZQ/1) (1) (Identifier: MEM_ODT_DQ_X_RON)
Mem Data Clock Termination	Specifies the termination to be used for the data clock (WCK). The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3) • 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) (Identifier: MEM_ODT_DQ_X_WCK)
Mem VrefDQ Value	Specifies the initial VrefDQ value to be used. Legal values are: from 10.0 to 73.5 (Identifier: MEM_VREF_DQ_X_VALUE)
Mem CA Common Termination	Common termination value that can be applied to CA/CK. The value of this parameter represents X, where: termination = $RZQ/X = (240\text{ Ohm})/X$. "off" means this termination is disabled. Legal values are: • Disable (off) • 240 Ohm (RZQ/1) (1) • 120 Ohm (RZQ/2) (2) • 80 Ohm (RZQ/3) (3)
continued...	

Parameter Name	Description
	• 60 Ohm (RZQ/4) (4) • 48 Ohm (RZQ/5) (5) • 40 Ohm (RZQ/6) (6) (Identifier: MEM_ODT_CA_X_CA_COMM)
Mem CA Termination Enable	Enable the common termination value on the CA bus. Legal values are: false, true (Identifier: MEM_ODT_CA_X_CA_ENABLE)
Mem CS Termination Enable	For LPDDR5, this enables the fixed-value 80 Ohm (RZQ/3) CS termination if it is supported by the memory. Legal values are: false, true (Identifier: MEM_ODT_CA_X_CS_ENABLE)
Mem CK Termination Enable	Enable the common termination value on the CK bus. Legal values are: false, true (Identifier: MEM_ODT_CA_X_CK_ENABLE)
Mem VrefCA Value	Specifies the initial VrefCA value to be used. Legal values are: from 10.0 to 73.5 (Identifier: MEM_VREF_CA_X_CA_VALUE)
Mem DFE Tap 1	This parameter allows you to select the amount of bias used on tap 1 of the memory DFE. Legal values are: • DFE Disabled (0) • 1-Step Negative (n1) • 2-Steps Negative (n2) • 3-Steps Negative (n3) (Identifier: MEM_DFE_X_TAP_1)

Table 231. Group: Example Design / Fileset Types

Parameter Name	Description
HDL Selection	This option lets you choose the format of HDL in which generated simulation and synthesis files are created. You can select either Verilog or VHDL. Default value is VERILOG Legal values are: • Verilog (VERILOG) • VHDL (VHDL) (Identifier: EX_DESIGN_HDL_FORMAT)
Generate Synthesis Fileset	Generate Synthesis Example Design. Default value is true (Identifier: EX_DESIGN_GEN_SYNTH)
Generate Simulation Fileset	Generate Simulation Example Design. Default value is true (Identifier: EX_DESIGN_GEN_SIM)

Table 232. Group: Example Design / User PLL

Parameter Name	Description
Auto-set User PLL Output Clock Frequency	If true, let IP select a reference clock frequency for the user PLL in the example design; if false, let user set a custom value for this parameter.
<i>continued...</i>	

Parameter Name	Description
	Default value is true (Identifier: EX_DESIGN_USER_PLL_OUTPUT_FREQ_MHZ_AUTOSSET_EN)
User PLL Output Clock Frequency	Frequency of the core clock in MHz. This clock drives the traffic generator and NoC initiator (If in NoC mode). Value is specified in megahertz Default value is 570 (Identifier: EX_DESIGN_USER_PLL_OUTPUT_FREQ_MHZ)
User PLL Reference Clock Frequency	PLL reference clock frequency in MHz for PLL supplying the core clock. Value is specified in megahertz Default value is 100 (Identifier: EX_DESIGN_USER_PLL_REFCLK_FREQ_MHZ)
NOC Reference Clock Frequency	Reference Clock Frequency for the NOC control IP. Value is specified in megahertz Default value is 100 Legal values are: 25, 100, 125 (Identifier: EX_DESIGN_NOC_PLL_REFCLK_FREQ_MHZ)

Table 233. Group: Example Design / Traffic Generator

Parameter Name	Description
Traffic Generator Remote Access	Specifies whether the Traffic Generator control and status registers are accessible via JTAG, exported to the fabric, or just disabled. Default value is JTAG Legal values are: Exported for Onchip control (EXPORT) Remote through JTAG (JTAG) (Identifier: EX_DESIGN_TG_CSR_ACCESS_MODE)
Traffic Generator Program	Specifies the traffic pattern to be run. Default value is MEDIUM Legal values are: Short Traffic Pattern (SHORT) Medium-length Traffic Pattern (MEDIUM) Long Traffic Pattern (LONG) Infinite Traffic Pattern (INFINITE) (Identifier: EX_DESIGN_TG_PROGRAM)

Table 234. Group: Example Design / Performance Monitor

Parameter Name	Description
Enable Performance Monitor for Channel 0	If true, example design will include a Performance Monitor instance connected to Channel 0. Default value is false (Identifier: EX_DESIGN_PMON_CH0_EN)
Enable Performance Monitor for Channel 1	If true, example design will include a Performance Monitor instance connected to Channel 1. Default value is false (Identifier: EX_DESIGN_PMON_CH1_EN)
Enable Performance Monitor for Channel 2	If true, example design will include a Performance Monitor instance connected to Channel 2.
continued...	

Parameter Name	Description
	Default value is false (Identifier: EX_DESIGN_PMON_CH2_EN)
Enable Performance Monitor for Channel 3	If true, example design will include a Performance Monitor instance connected to Channel 3. Default value is false (Identifier: EX_DESIGN_PMON_CH3_EN)

8.2. Agilex 7 M-Series FPGA EMIF IP Pin and Resource Planning

The following topics provide guidelines on pin placement for external memory interfaces.

Typically, all external memory interfaces require the following FPGA resources:

- Interface pins
- PLL and clock network
- RZQ pins
- Other FPGA resources—for example, core fabric logic, and debug interfaces

Once all the requirements are known for your external memory interface, you can begin planning your system.

8.2.1. Agilex 7 M-Series FPGA EMIF IP Interface Pins

Any I/O banks that do not support transceiver operations in Agilex 7 M-Series FPGAs support external memory interfaces.

However, RDQS (read data strobe), WCK (write clock), and DQ (data) pins are listed in the device pin tables and are fixed at specific locations in the device. You must adhere to these pin locations to optimize routing, minimize skew, and maximize margins. Always check the pin table for the actual locations of the DQS and DQ pins.

Note: Maximum interface width varies from device to device depending on the number of I/O pins and DQS or DQ groups available. Achievable interface width also depends on the number of address and command pins that the design requires. To ensure adequate PLL, clock, and device routing resources are available, you should always test fit any IP in the Quartus Prime software before PCB sign-off.

8.2.1.1. Estimating Pin Requirements

You should use the Quartus Prime software for final pin fitting.

However, you can estimate whether you have enough pins for your memory interface by performing the following steps:

1. Determine how many read/write data pins are associated per data strobe or clock pair.
2. Calculate the number of other memory interface pins needed, including any other clocks (write clock or memory system clock), address, command, and RZQ. Refer to the External Memory Interface Pin Table to determine necessary address/command/clock pins based on your desired configuration.
3. Calculate the total number of I/O banks required to implement the memory interface, given that an I/O bank supports up to 96 pins.

Test the proposed pin-outs with the rest of your design in the Quartus Prime software (with the correct I/O standard and OCT connections) before finalizing the pin-outs. There can be interactions between modules that are illegal in the Quartus Prime software that you might not know about unless you compile the design and use the Quartus Prime Pin Planner.

8.2.1.2. LPDDR5 Component Options

The table and figures below illustrate pin placement and routing recommendation for a single 32-bit channel, and two 16-bit channels, respectively.

Note: You should always consult your memory vendor's data sheet to verify pin placement and routing plans.

Table 235. Pin Options for LPDDR5 x32 and x16

Pins	1CH x32	2CH x16	
Data	32-bit DQ[15:0]_A DQ[15:0]_B	DQ[15:0]_A	DQ[15:0]_B
Data mask	DM[1:0]_A DM[1:0]_B	DM[1:0]_A	DM[1:0]_B
Read data strobe	RDQS[1:0]_t_A RDQS[1:0]_c_A RDQS[1:0]_t_B RDQS[1:0]_c_B	RDQS[1:0]_t_A RDQS[1:0]_c_A	RDQS[1:0]_t_B RDQS[1:0]_c_B
Write clock	WCK[1:0]_t_A WCK[1:0]_c_A WCK[1:0]_t_B WCK[1:0]_c_B	WCK[1:0]_t_A WCK[1:0]_c_A	WCK[1:0]_t_B WCK[1:0]_c_B
Command/address	CA[6:0]_A CS0_A CA[6:0]_B CS0_B	CA[6:0]_A CS0_A	CA[6:0]_B CS0_B
Clock	CK_t_A CK_c_A CK_t_B	CK_t_A CK_c_A	CK_t_B CK_c_B
continued...			

Pins	1CH x32	2CH x16
	CK_c_B	
Reset	RESET_n	RESET_n (Resistor jumper to select from mem_0 or mem_1.) Note: The LPDDR5 EMIF IP design would generate one reset pin to the memory module for both single channel x16/x32 or dual channel x16 design. For early board bring up and easy debugging on dual channel x16 designs, you can test each channel independently as single channel design. You can design the board with two reset pins connected through a resistor jumper to choose the reset from the primary channel or secondary channel. The example for dual channel x16 LPDDR5 with two reset pins is shown in the figure below. For mature designs, you can design the board with one reset pin connected out from the primary channel.

Figure 56. Pin Options for LPDDR5 2ch x16

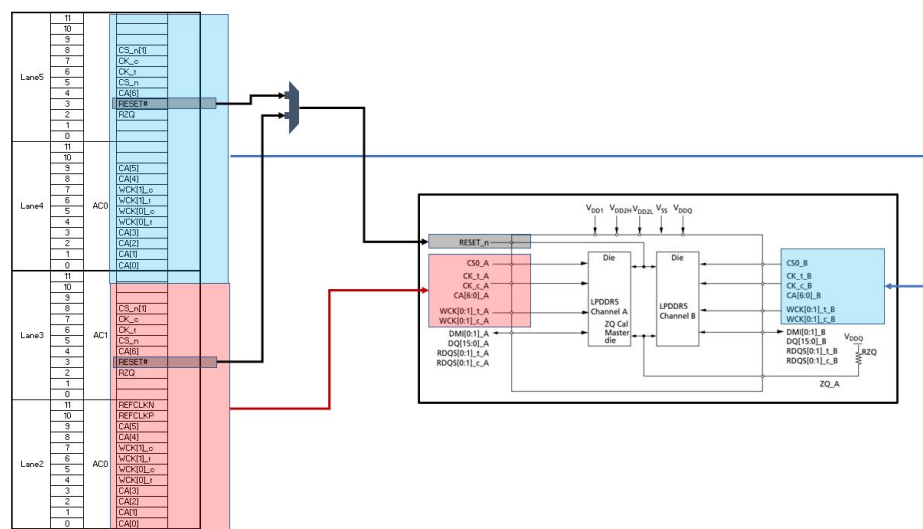
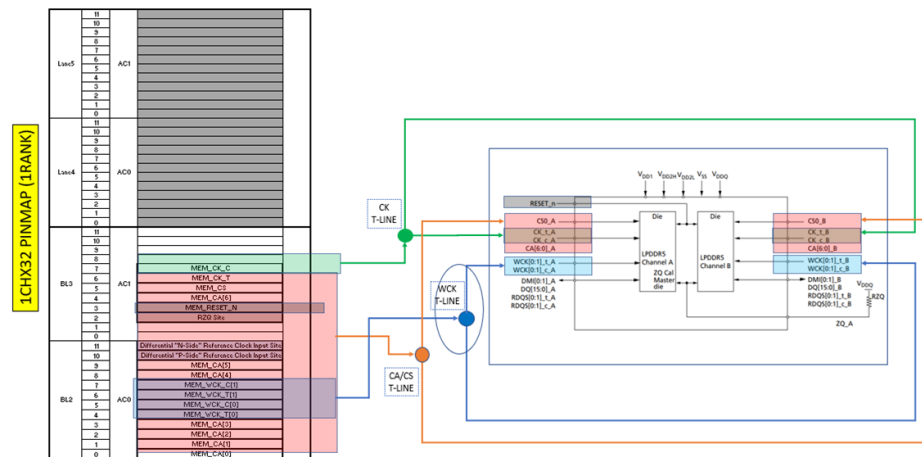


Figure 57. Pin Options for LPDDR5 x32



8.2.1.3. Maximum Number of Interfaces

The maximum number of interfaces supported for a given memory protocol varies, depending on the FPGA in use.

Unless otherwise noted, the calculation for the maximum number of interfaces is based on independent interfaces where the address or command pins are not shared.

Note: You may need to share PLL clock outputs depending on your clock network usage.

Table 236. Maximum Number of LPDDR5 Interfaces

Device	Package	1ch x32	2ch x16	4ch x16
AGMA032 / AGMA039 / AGMB032 / AGMB039	R31B	5	5	1
AGME032 / AGME039 / AGMF032 / AGMF039 / AGMG032 / AGMG039 / AGMH032 / AGMH039	R47A	8	8	4
AGMF032 / AGMF039 / AGMH032 / AGMH039	R47B	8	8	4

- 4ch x16 interface requires two adjacent IO96B banks located on the edge of the device.

Timing closure depends on device resource and routing utilization. For more information about timing closure, refer to the *Area and Timing Optimization Techniques* chapter in the *Quartus Prime Handbook*.

8.2.2. Agilex 7 M-Series FPGA EMIF IP Resources

The Agilex 7 M-Series FPGA memory interface IP uses several FPGA resources to implement the memory interface.

8.2.2.1. OCT

You require an OCT calibration block if you are using an Agilex 7 M-Series FPGA OCT calibrated series, parallel, or dynamic termination for any I/O in your design. There are two OCT blocks in an I/O bank, one for each sub-bank.

You must observe the following requirements when using OCT blocks:

- The I/O bank where you place the OCT calibration block must use the same V_{CCIO_PIO} voltage as the memory interface.
- The OCT calibration block uses a single fixed R_{ZQ} . You must ensure that an external termination resistor is connected to the correct pin for a given OCT block.

For specific pin connection requirements, refer to [Specific Pin Connection Requirements](#).

8.2.2.2. PLL

When using PLL for external memory interfaces, you must consider the following guidelines:

For the clock source, use the clock input pin specifically dedicated to the PLL that you want to use with your external memory interface. The input and output pins are only fully compensated when you use the dedicated PLL clock input pin.

For specific pin connection requirements, refer to [Specific Pin Connection Requirements](#).

8.2.3. Pin Guidelines for Agilex 7 M-Series FPGA EMIF IP

The Agilex 7 M-Series FPGA contains I/O banks on the top and bottom edges of the device, which can be used by external memory interfaces.

Agilex 7 M-Series FPGA I/O banks contain 96 I/O pins. Each bank is divided into two sub-banks with 48 I/O pins in each. Sub-banks are further divided into four I/O lanes, where each I/O lane is a group of twelve I/O ports.

The I/O bank, I/O lane, and pairing pin for every physical I/O pin can be uniquely identified by the following naming convention in the device pin table:

- The I/O pins in a bank are represented as P#X#Y#, where:
 - ☐ P# represents the pin number in a bank. It ranges from P0 to P95, for 96 pins in a bank.
 - ☐ X# represents the bank number on a given edge of the device. X0 is the farthest bank from the zipper.
 - ☐ Y# represents the top or bottom edge of the device. Y0 and Y1 refer to the I/O banks on the bottom and top edge, respectively.
- Because an IO96 bank comprises two IO48 sub-banks, all pins with P# value less than 48 (P# <48) belong to the same I/O sub-bank. All other pins belong to the second IO48 sub-bank.
- The *Index Within I/O Bank* value falls within one of the following ranges: 0 to 11, 12 to 23, 24 to 35, or 36 to 47, and represents one of I/O lanes 0, 1, 2, or 3, respectively.
- To determine whether I/O banks are adjacent, you can refer to the sub-bank-ordering figures for your device family in the [Architecture: I/O Bank](#) topic. In general, you can assume that I/O banks are adjacent within an I/O edge, unless the I/O bank is not bonded out on the package (indicated by the presence of the " - " symbol in the I/O table), or if the I/O bank does not contain 96 pins, indicating that it is only partially bonded out. If an I/O bank is not fully bonded out in a particular device, it cannot be included within the span of sub-banks for a larger external memory interface. In all cases, you should use the Quartus Prime software to verify that your usage can be implemented.
- The pairing pin for an I/O pin is in the same I/O bank. You can identify the pairing pin by adding 1 to its *Index Within I/O Bank* number (if it is an even number), or by subtracting 1 from its *Index Within I/O Bank* number (if it is an odd number).

8.2.3.1. General Guidelines - LPDDR5

You should follow the recommended guidelines when performing pin placement for all external memory interface pins targeting Agilex 7 M-Series devices, whether you are using the hard memory controller or your own solution.

Note: PHY only, RLDRAMx, and QDRx are not supported with HPS.

Observe the following general guidelines when placing pins for your Agilex 7 M-Series external memory interface:

1. Ensure that the pins of a single external memory interface reside on the same edge I/O.
2. The address and command pins and their associated clock pins in the address and command bank must follow a fixed pin-out scheme, as defined in the table in the [Address and Command Pin Placement for LPDDR5](#) topic.
3. Not every byte lane can function as an address and command lane or a data lane. The pin assignment must adhere to the LPDDR5 data width mapping defined in [LPDDR5 Data Width Mapping](#).
4. A byte lane must not be used by both address and command pins and data pins.

5. An external memory interface can occupy one or more banks on the same edge. When an interface must occupy multiple banks, ensure that those banks are adjacent to one another.
 - If an I/O bank is shared between two interfaces—meaning that two sub-banks belong to two different EMIF interfaces—then both the interfaces must share the same voltage.
 - Sharing of I/O lanes within a sub-bank for two different EMIF interfaces is not permitted; I/O lanes within a sub-bank can be assigned to one EMIF interface only.
6. Any pin in the same bank that is not used by an external memory interface may not be available for use as a general purpose I/O pin:
 - For fabric EMIF, unused pins in an I/O lane assigned to an EMIF interface cannot be used as general-purpose I/O pins. In the same sub-bank, pins in an I/O lane that is not assigned to an EMIF interface, can be used as general-purpose I/O pins.
 - For HPS EMIF, unused pins in an I/O lane assigned to an EMIF interface cannot be used as general-purpose I/O pins. In the same bank, pins in an I/O lane that is not assigned to an EMIF interface cannot be used as general-purpose I/O pins either.
7. All address and command pins and their associated clock pins (CK_t and CK_c) must reside within a single sub-bank. The sub-bank containing the address and command pins is identified as the address and command sub-bank. Refer to the table in [LPDDR5 Data Width Mapping](#) for the supported address and command and data lane placements for DDR5.
8. The address and command pins and their associated clock pins in the address and command bank must follow a fixed pin-out scheme, as defined in the *Agilex 7 M-Series External Memory Interface Pin Information* file.
9. An external memory interface can occupy one or more banks on the same edge. When an interface must occupy multiple banks, ensure the following:
 - That the banks are adjacent to one another.
 - That you used only the supported data width mapping as defined in the table in [LPDDR5 Data Width Mapping](#). Be aware that not every byte lane can be used as an address and command lane or a data lane.
10. An unused I/O lane in the address and command sub-bank can serve to implement a data group, such as a x8 DQS group. The data group must be from the same controller as the address and command signals.
11. An I/O lane must not be used by both address and command pins and data pins.
12. Place read data groups according to the DQS grouping in the pin table and Pin Planner. Read data strobes (such as RDQS_t and RDQS_c) must reside at physical pins capable of functioning as RDQS_t and RDQS_c for a specific read data group size. You must place the associated read data pins (DQ), within the same group.
13. One of the sub-banks in the device (typically the sub-bank within corner bank 3A) may not be available if you use certain device configuration schemes. For some schemes, there may be an I/O lane available for EMIF data group.

- AVST-8 – This is contained entirely within the SDM, therefore all lanes of sub-bank 3A can be used by the external memory interface.
- AVST-16/AVST-32– Lanes 4, 5, 6, and 7 are all effectively occupied and are not usable by the external memory interface.

14. Two memory interfaces cannot share an I/O 48 sub-bank.

8.2.3.2. Specific Pin Connection Requirements

To compile a design that instantiates an Agilex 7 M-Series EMIF IP, you must explicitly define a legal placement and I/O standard for all EMIF pins.

This topic discusses the required constraints per pin type. To simplify this process, you can leverage the **Back Annotate Assignments** tool. For more information, refer to *Back Annotating Pin Placement and I/O Standard Assignments* in the *External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide*.

PLL

For LPDDR5, you must constrain the PLL reference clock to the address and command lanes only.

- You must constrain differential reference clocks to pin indices 10 and 11 in lane 2 when placing command address pins in lane 3 and lane 2.
- The sharing of PLL reference clocks across multiple LPDDR5 interfaces is permitted within an I/O bank.

OCT

For LPDDR5, you must constrain the RZQ pin to the address and command lanes only.

- You must constrain RZQ to pin index 2 in lane 3 when placing command address pins in lane 3 and lane 2.
- The sharing of RZQ across multiple LPDDR5 interfaces is permitted within an I/O bank.

RDQS/DQ/DM

For LPDDR5 x8 DQS grouping, the following rules apply:

- You may use pin indices 0, 1, 2, 3, 8, 9, 10, and 11 within a lane for DQ mode pins only.
- You must use pin index 4 for the RDQS_p pin only.
- You must use pin index 5 for the RDQS_n pin only.
- You must ensure that pin index 7 remains unused. Pin index 7 is not available for use as a general purpose I/O.
- You must use pin index 6 for the DM pin only.

8.2.3.3. Command and Address Signals

Command and address signals in SDRAM devices are clocked into the memory device using the CK_t or CK_c signal.

8.2.3.4. Clock Signals

LPDDR5 SDRAM devices use CK_t and CK_c signals to clock the address and command signals into the memory.

The memory uses these clock signals to generate the DQS signal during a read through the DLL inside the memory.

8.2.4. Pin Placements for Agilex 7 M-Series FPGA LPDDR5 EMIF IP

8.2.4.1. Address and Command Pin Placement for LPDDR5

Agilex 7 M-Series FPGA LPDDR5 IP supports fixed address and command pin placement as shown in the following table. The IP supports up to 2 ranks.

Table 237. Address and Command Pin Placement

Address/Command Lane	Index Within Byte Lane	LPDDR5
AC1	11	
	10	
	9	
	8	CS_N[1]
	7	CK_C[0]
	6	CK_T[0]
	5	CS_N[0]
	4	CA[6]
	3	RESET_N
	2	RZQ Site
	1	
	0	
AC0	11	Differential "N-Side" reference clock input site
	10	Differential "P-Side" reference clock input site
	9	CA[5]
	8	CA[4]
	7	WCK_C[1]
	6	WCK_T[1]
	5	WCK_C[0]
	4	WCK_T[0]
continued...		

Address/Command Lane	Index Within Byte Lane	LPDDR5
	3	CA[3]
	2	CA[2]
	1	CA[1]
	0	CA[0]

8.2.4.2. LPDDR5 Data Width Mapping

The EMIF IP for Agilex 7 M-Series does not support flexible data lane placement.

Only fixed byte lanes within the I/O bank can be used as data lanes. The following table lists the supported address and command and data lane placements in an I/O bank.

Table 238. Component

Controller	Data Width Usage	BL7 [P95:P84]	BL6 [P83:P72]	BL5 [P71:P60]	BL4 [P59:P48]	BL3 [P47:P36]	BL2 [P35:P24]	BL1 [P23:P12]	BL0 [P11:P0]
Primary	LPDDR5 x16	GPIO	GPIO	GPIO	GPIO	AC1 P	AC0 P	DQ[1] P	DQ[0] P
Primary ¹	LPDDR5 x16	DQ[1] S	DQ[0] S	AC1 S	AC0 S	X	X	X	GPIO
Primary	LPDDR5 x32	DQ[3] P	DQ[2] P	GPIO	GPIO	AC1 P	AC0 P	DQ[1] P	DQ[0] P
Note: • P Primary controller. • S Secondary controller. • ¹ ES0 silicon supports this scheme using the primary controller; future revisions will enable the secondary controller, unblocking BL0-3 for GPIO use. • X = Not available as GPIO.									

Figure 58. LPDDR5 2chx16 , Single Rank

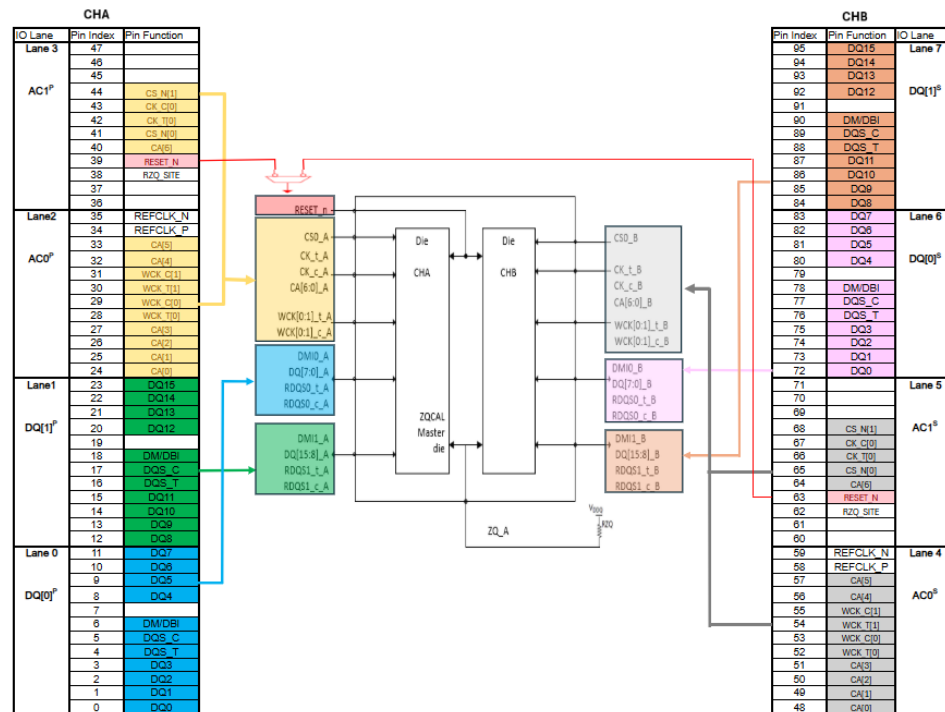
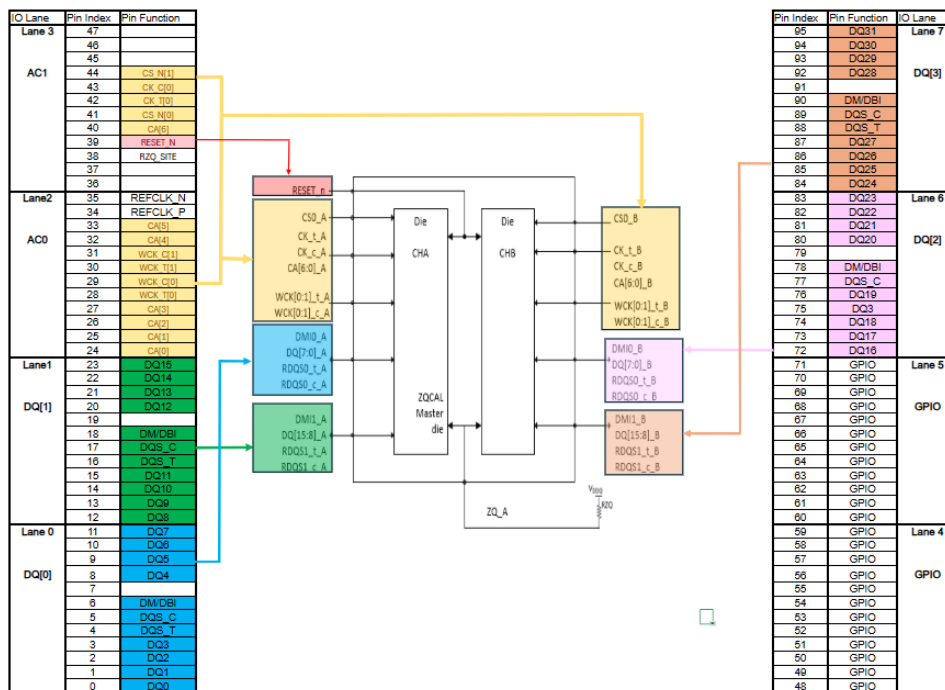


Figure 59. LPDDR5 1chx32 , Single Rank



Refer to the *Pin Options for LPDDR5 x32* figure in [LPDDR5 Component Options](#) for information on the T-line routing requirements for address and command pins.

8.2.4.3. LPDDR5 Byte Lane Swapping

The data lane can be swapped when the byte-lanes are utilized as DQ/DQS pins. Byte lane swapping on utilized lanes is allowed when you swap all the DQ/DQS/DM pins in the same byte lane with the other utilized byte lane.

The rules for swapping DQ byte lane are as follows:

- You can only swap between utilized DQ lanes.
- You cannot swap a DQ lane with an AC lane.
- Additional restrictions apply when you use a x16 memory component:
 - ☐ You must place DQ group 0 and DQ group 1 on adjacent byte lanes, unless they are separated by AC lanes. These 2 groups must be connected to the same x16 memory component.
 - ☐ You must place DQ group 2 and DQ group 3 on adjacent byte lanes, unless they are separated by AC lanes. These 2 groups must be connected to the same x16 memory component.
 - ☐ If you use only one byte of the x16 memory component, you must use only the lower byte of the memory component.

Table 239. Component

Controller	Data Width Usage	BL7 P95:P84	BL6 P83:P72	BL5 P71:P60	BL4 P59:P48	BL3 P47:P36	BL2 P35:P24	BL1 P23:P12	BL0 P11:P0
Primary & Secondary	LPDDR5 2ch x16	DQ[1] ^S	DQS[0] ^S	AC1 ^S	AC0 ^S	AC1 ^P	AC0 ^P	DQ[1] ^P	DQ[0] ^P
Primary	LPDDR5 x32	DQ[3] ^P	DQ[2] ^P	GPIO	GPIO	AC1 ^P	AC0 ^P	DQ[1] ^P	DQ[0] ^P
Note: • ^P Primary controller. • ^S Secondary controller.									

Example 1: LPDDR5 2 ch x16

DQ[0] and DQ[1] of the primary controller are can swapped with each other. DQ[0] and DQ[1] of the secondary controller can be swapped with each other.

Example 2: LPDDR5 x32

DQ[0] and DQ[1] can be swapped with each other. DQ[2] and DQ[3] can be swapped with each other.

8.3. LPDDR5 Board Design Guidelines

This section provides board layout design recommendations and guidelines for Agilex 7 M-Series FPGAs, with GPIO-B (input/output) silicon implementation to support LPDDR5.

This PCB layout guideline covers various supported LPDDR5 topologies along with maximum supported data rate that you can use for a successful PCB design.

A successful PCB design requires not only following the topology and routing guidelines, but also meet PDN design requirements.

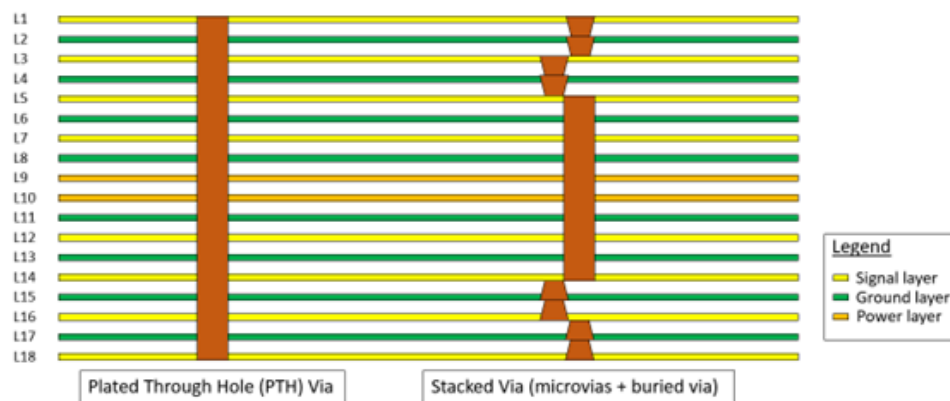
For related information, refer also to the Agilex 7 PDN design guidelines and the Agilex 7 high speed transceiver PCB design guidelines, available on the Intel website.

8.3.1. PCB Stack-up and Design Considerations

You can implement an LPDDR5 interface with a Type-III or Type-IV PCB stack-up. A Type-III PCB with zero built-up layers and PTH vias which is used to implement a DDR4 design can also be used for LPDDR5 designs. Backdrill can be considered to reach higher supported LPDDR5 data rate. You can run simulations to determine whether backdrill is necessary.

A high-quality type-IV PCB with higher cost uses not only plated-through-hole (PTH) vias to connect from the top to bottom layer, but also stacked vias, micro vias, and buried vias to connect between layers. The following figure shows a cross-sectional comparison of a PTH and stacked via.

Figure 60. Cross-sectional Comparison of Plated-Through Hole Via and Stacked Via



To support maximum data rate operation, LPDDR5 board design requires a high-quality PCB stackup using backdrill or micro vias, buried vias, or stacked vias to reduce crosstalk for high performance. Reducing the length of signal via is essential to minimize the crosstalk between signals.

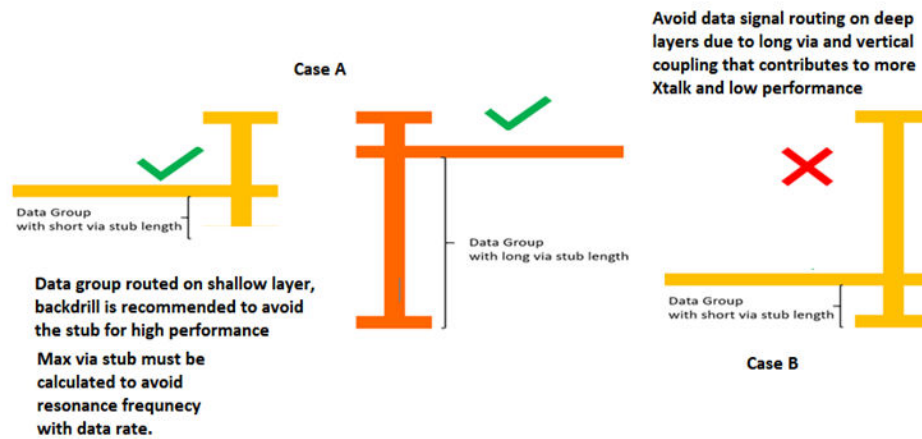
8.3.2. General Design Considerations

Altera recommends that you route all data signals within a specific group on the same layer.

The following figure illustrates a routing example for a Type-III PCB for an LPDDR5 design. Altera recommends that you route Data Group signals such as DQ, DM and DQS on shallow layers as stripline, with the least Z-height via transition to avoid vertical crosstalk for high performance.

Minimal stub effect or back drill is recommended but not mandatory to avoid high reflection for maximum data rate performance for an LPDDR5 interface. You can perform simulation to determine whether to use backdrill. Long via stubs will affect the intersymbol interference (ISI) of the channel, but the impact of ISI is less than the impact of crosstalk for maximum performance.

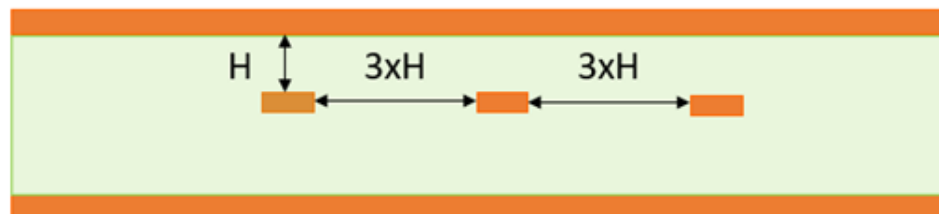
Figure 61. Suggested Routings



In the above figure, case A routing is suggested for LPDDR5 Data Group signals over case B, to support maximum data rate. If data signals are routed on deeper layers (as in case B, with long via and short stub), the impact of crosstalk is significant and causes reduced data rate and performance.

To minimize horizontal crosstalk between signals on the same layer, PCB designers must maintain adequate signal trace-to-trace (edge to edge) space with a minimum spacing of $3 \times h$ separation distance, where h is the dielectric thickness to the closest reference plane, as illustrated below. Some signals within the same group can have smaller spacing, such as DQ to DQ. Refer to the routing guidelines sections for detailed spacing requirements for each signal group.

Figure 62. Minimum Trace-to-Trace Separation Distance



8.3.3. DDR Differential Signals Routing

DQS and CLK signals in the DDR interface are differential signals and must be routed on the PCB as differential signals unless there is a limitation for PCB routing, such as having a very small pitch at DRAM.

You should have a symmetrical fan-out routing at the FPGA pin field. Non-symmetrical routing for differential signals causes shifting on common-mode voltage and contributes to reduced timing margins at the receiver. The following figures show the recommended differential routing at the FPGA pin field for DQS/CLK signals.

Figure 63. Symmetrical Routing of Differential Signals (DQS/CLK) at FPGA Pin Field, with Length/Skew Matching Between P/N Lanes After FPGA Device Edge

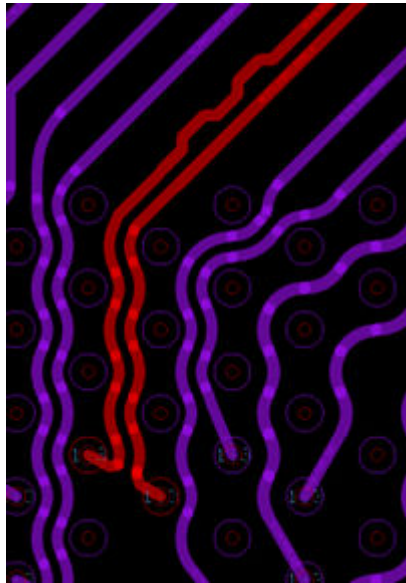
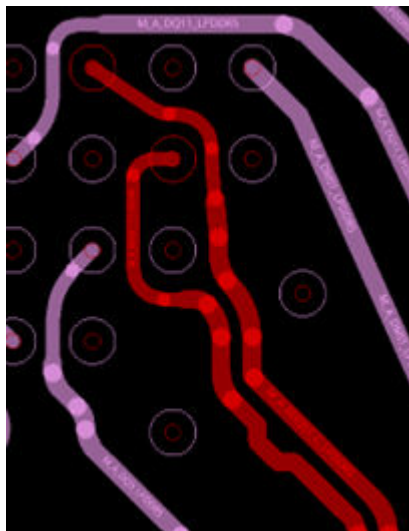


Figure 64. Single-Ended Routing for Differential Signals (DQS/CLK) at DRAM Pin Field with Very Small Pitch and Skew Matching at Edge of DRAM Pin Field



Altera recommends implementing length and skew matching for differential signals immediately after the FPGA device to avoid additional shifting on differential signals common mode voltage.

In cases where very small DRAM device pitch limits the implementation of symmetrical routing at the DRAM pin field for differential signals, it is recommended to route the differential signals as single-ended signals within the DRAM pin field, ensuring to maintain the same impedance while changing from differential to single-ended

configuration. Designers must also keep the same length of routing for each P and N single-ended lane within the DRAM pin field. The skew matching between P and N lanes must be applied before reaching the DRAM pin field.

8.3.4. Ground Plane and Return Path

A continuous and solid ground-reference plane is crucial for data lines, to ensure good signal integrity.

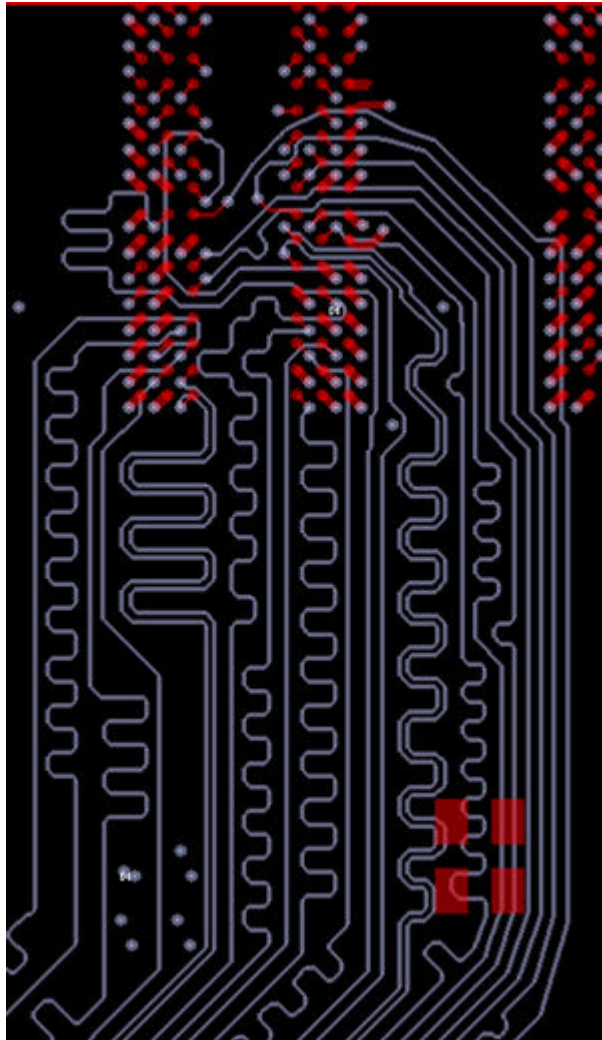
It is important to provide a low-impedance ground return path between the FPGA and DRAM devices, and to keep ground stitching vias within 80 mils from signal transition, for best return path on signal vias and improved signal integrity.

8.3.5. DRAM Break-in Layout Guidelines

For discrete DRAM components on PCB, you can use either the dog-bone or via in pad at the DRAM for the signal transition from inner layer to DRAM.

If you use dog-bone via transitions, you should separate them with larger pitch, to avoid crosstalk between the signal vias.

Figure 65. Data Signal Group Routing on PCB to Memory Down Configuration



8.3.6. General Notes for EMIF Routing Guidelines Tables

The following general notes apply to the EMIF routing guideline tables in subsequent topics.

- All spacing requirements are the minimum requirement to be met on PCB in EMIF routing guidelines table.
- In the routing guideline table, h represents the trace-to-nearest-reference-plane height or distance. Board designers should use the formula in the routing guidelines table to calculate the correct spacing requirements.
- There is no differential impedance target for CLK and DQS. Board designers should follow the EMIF routing table and keep traces to be closely coupled.

- The trace length and spacing in the guideline tables are based on FR-4 level PCB material and the PCB layout routing with the worst-case crosstalk. You should perform simulation if not following the requirements in the routing guidelines table.
- In the routing guidelines tables, *SL* stands for strip line routing recommendations and *US* stands for upper surface (Microstrip) routing recommendations.
- You need to design trace widths of BO, BO1, BO2 and BI, BI1, BI2 based on the actual PCB stack-up and PCB design. It would be better if trace widths of BO, BO1, BO2 and BI, BI1, BI2 can be designed closer to the target impedance of the M segment.
- Include the FPGA package per-pin skew and PCB delay for skew matching.
- Length matching on the Alert and Reset signals are not required.
- Altera recommends that you perform skew matching in time (picoseconds).

8.3.7. LPDDR5 PCB Layout Guidelines

This section describes PCB layout guidelines for an LPDDR5 interface.

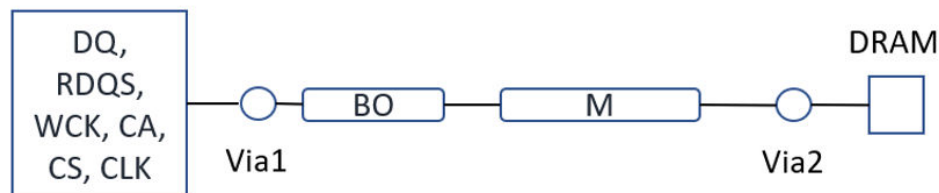
Agilex 7 M-Series devices support LPDDR5 interfaces only for memory down configuration. The LPDDR5 interface supports both thin and thick PCB stackups. The maximum supported data rates vary depending on the selected topology.

8.3.7.1. LPDDR5 Discrete Component/Memory Down Topology, Single Rank or Dual Rank

LPDDR5 memory down support is available in two configurations: single rank or dual rank.

There are four DRAM interface signal groupings: Data Group, Command-Address Group, Control Group, and Clock Group. The connection between the FPGA and DRAM uses point-to-point topology for Data, Command/Address, Control, and Clocks, as shown in the following figure.

Figure 66. Point-to-Point Connection for Data, CA, CTRL, and Clock Signals Topology for LPDDR5



The LPDDR5 interface does not support a traditional dual-directional data-strobe architecture. However, two single-directional data strobes such as Write Clock (WCK) for Write Operations and an optional Read Clock (RDQS) for Read Operations are supported.

The following two figures show the connection topology for DQ, WCK signal and CA, CLK, CTRL signals for LPDDR5.

Figure 67. WCK Signals Topology for LPDDR5 Memory Down, T-Line Connection, Depending on EMIF Topology

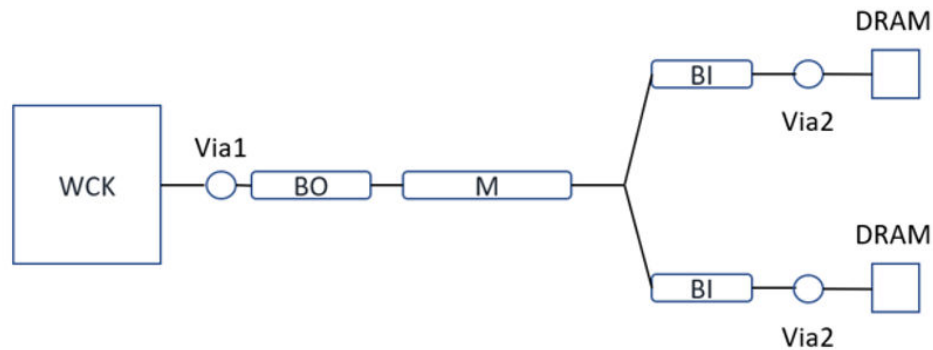
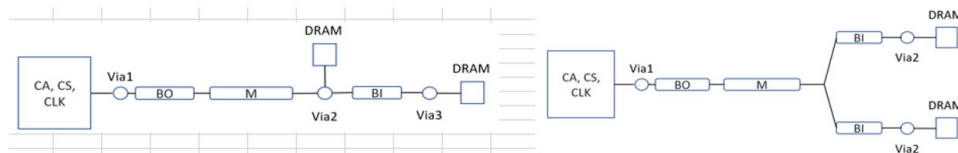


Figure 68. CS, CLK, CTRL Signals Topology for LPDDR5 Memory Down: Daisy or T-Line Connections, Depending on EMIF Topology



8.3.7.2. Routing Guidelines for LPDDR5 Memory Down Topology

The following table provides comprehensive routing guidelines (recommended trace impedance and length) for each LPDDR5 signal, based on a memory down topology. For example, the maximum length of the main trace routing can be derived from total trace length by subtracting the break-out and break-in trace segment lengths.

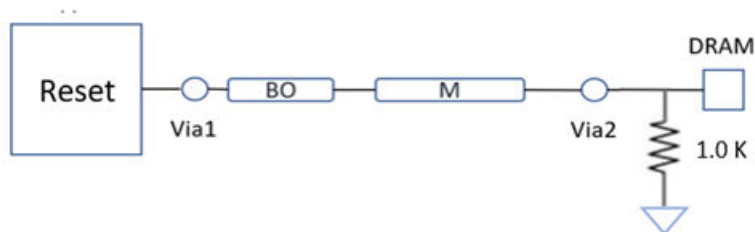
The following table shows physical trace segment routing guidelines, including the target impedance of routing for each signal and the minimum space between signal traces on the same layer. The PCB designer must meet the impedance target criteria. The h in the table indicates the minimum substrate height from signal layer to reference layer. Ensure that you follow the trace-to-trace-edge gap/space criteria in the guideline, based on h in various stackups.

Table 240. Routing Guidelines for LPDDR5 Memory Down Topology

Signal Group	Segment	Routing Layer	Max Length (mil)		Target Zse (ohm)	Trace Spacing, S1 (mil): Within Group	Trace Spacing, S2 (mil): CMD/CTRL/CLK to DQ/DQS	Trace Spacing, S3 (mil): DQ Nibble to Nibble	Trace Spacing (mil), Within DIFF pair	Trace Spacing (mil), DQS pair to DQ	Trace Spacing (mil), CLK pair to CMD/CTRL/CKE
			Segment	Total							
DQ, RDQS	BO	SL	1000	3000	40	5, 17	17	17	4		
	M	SL				2h	3h	3h	4	3h	
CA/CS/CLK (Direct Connect)	BO1/BO2	SL	450	4000	40	5, 17	17		4		17
	M	SL					3h		4		3h
	BI	SL	50				3h		4		3h
CA/CS/CLK (Daisy)	BO	SL	500	5000 (to last DRAM for Daisy)	40	5, 17	17		4		17
	M	SL				2h	3h		4		3h
	BI	SL	1000			2h	3h		4		3h
WCK	BO	SL	1000	3000	40	5, 17	17		4		
	M	SL				2h	3h		4	3h	
	BI	SL	200 ~ 500			2h	3h		4		

Reset signal routing design also follows the CMD/ADD/CTRL routing design. Maintain at least 3x h edge-to-edge spacing from the Reset signal to other signals on the same layer. There is no requirement to have skew matching between Reset signal and CLK signal. Altera recommends using a 1K-ohm pull-down resistor for Reset termination as shown in the figure below.

Figure 69. Strip line Routing for Reset Signals



Skew matching for the LPDDR5 interface consists of both package routing skew and PCB physical routing skew. You must maintain skew matching of CA and CTRL with respect to the clock signals to ensure signals at the receiver are correctly sampled. In addition, there are skew matching requirements for DQ and DQS within a byte group, DQS and CLK.

The following table provides a detailed skew matching guideline to facilitate PCB trace routing efforts. The length matching criteria in the table below represents a default PCB on an Altera platform board design. Skew matching criteria must be always followed in any other stackup.

Table 241. Skew Matching Requirement for LPDDR5 Memory Down Topology

Length Matching Rules	Time (assuming 170ps/in delay)
Length matching between DQ and WCK per x16	-68ps < DQ - WCK < 110ps
Length matching between DQ and RDQS per x8	-20ps < DQ - RDQS < 20ps
Length matching between WCK and CLK per x16	-34ps < WCK - CLK < 68ps
Length matching between DQ signals per x8	< 27ps
Length matching between channel-to-channel CLK signals	< 170ps
Length matching between CLK and CA bits per x16	-34ps < CLK - CA < 34ps
Length matching between CA bits per x16	< 51ps
Length matching between CLK and CS per x16	-85ps < CLK - CS < 85ps
Length matching between CS bits per x16	< 68ps
Length matching between RDQS_N and RDQS_P	< 1ps
Length matching between WCK_N and WCK_P	< 1ps
Length matching between CLK_N and CLK_P	< 1ps
Length matching between WCK Tee segments (BI2)	< 1ps
Include package length in length matching	Required

LPDDR5 eye margin is sensitive to crosstalk, especially when the signals are routed on deep layers in the stackup. The deep-layer vertical transition induces more vertical coupling between signals and hence more crosstalk. You can perform simulation to determine the maximum stub length, trace length, routing layer, and so forth.

Table 242. DQ Routing Summary for LPDDR5 Memory Down

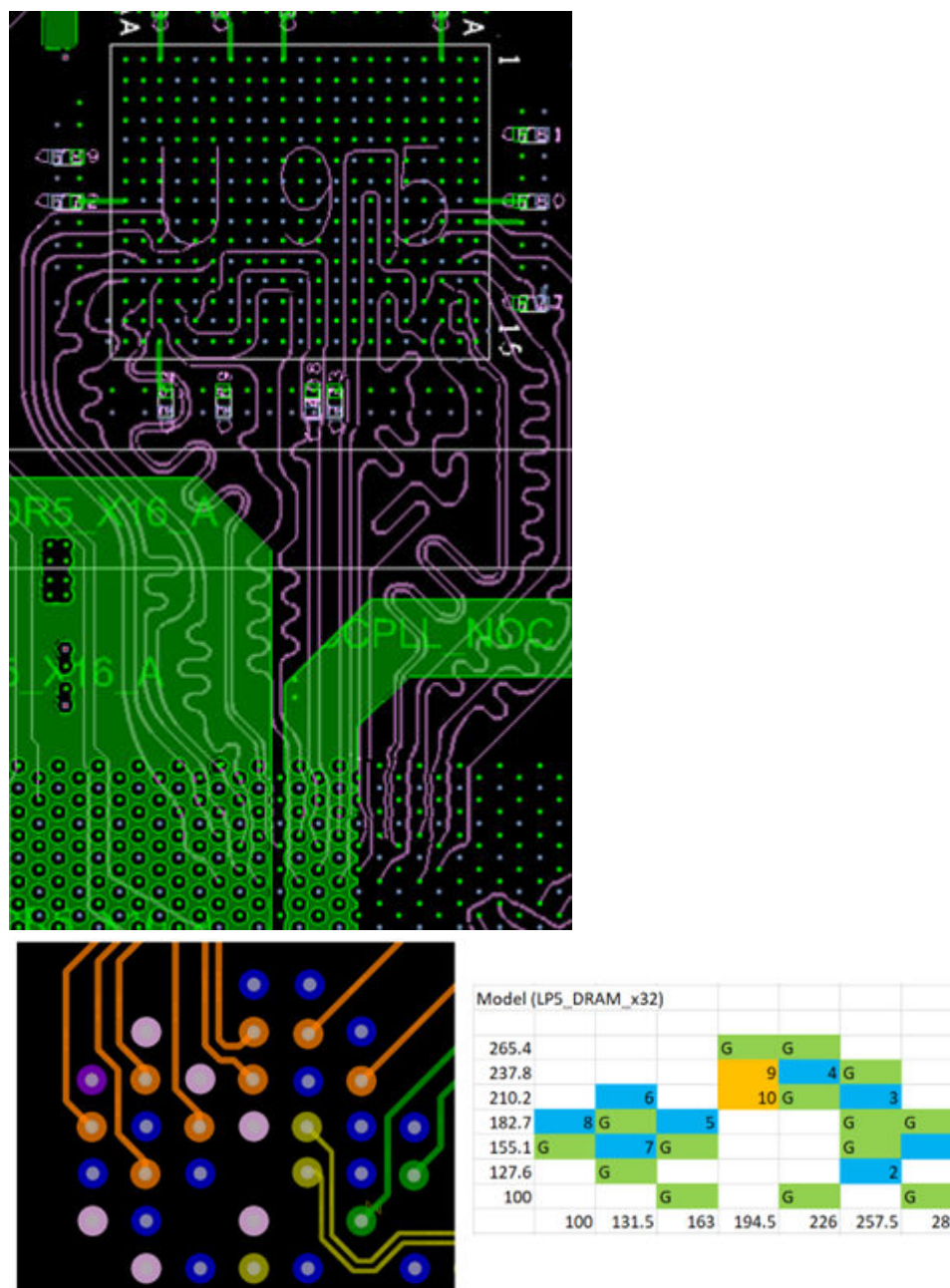
Memory	LPDDR5	
Signal; Group	DQ	
Board Thickness (mil)	65mil or 120mil (Routing must be on upper layers, max via transition depth <=50mil)	65mil or 120mil (Routing must be on upper layers, max via transition depth <=50mil)
Maximum Z-transition height (mil)	50	50
PCB Stripline Trace Impedance (Ohms)	40	40
Memory Configuration	Memory Down	Memory Down
# of Rank	1 (x16 bit or x32 bit Double Die)	2 (x 16 or x 32 bit)
Maximum Length Total (Inch)	3.0	3.0
Notes	Maximum package length in FPGA design is shorter than 34mm.	

8.3.7.3. Example of an LPDDR5 Layout on an Altera® FPGA Platform Board

The following figures show the layout example of a single rank LPDDR5 x 32-bit device with a pitch size of 0.7×0.8mm on an Altera FPGA platform design.

The LPDDR5 signal routing is on upper layers to avoid vertical crosstalk on interface and achieve high performance.

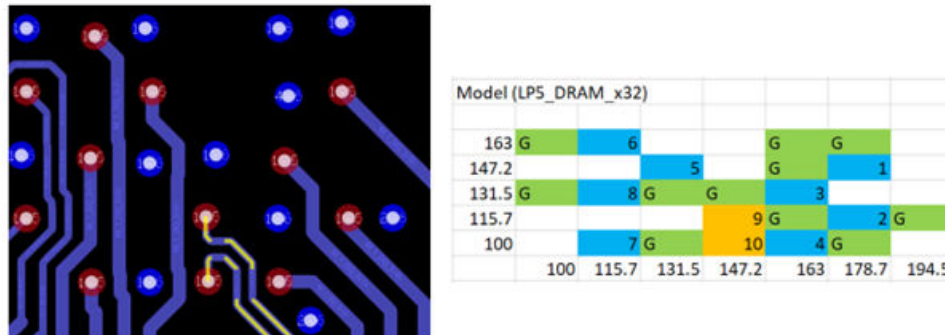
Figure 70. Board Layout and Via Pattern for Single Rank LPDDR5 x32 device on an Altera FPGA Platform Design



In addition, the following figure shows a LPDDR5 64-bit device board routing sample with pitch dimension of 0.4×0.4mm. The microvia has been used for via transitions on this interface. The microvia mentioned in the following is from top layer to layer 3, it belongs to Type-III stack-up for easier fan-out, not the same as microvia used in

Type-IV stack-up. Only one lamination cycle is required, therefore the cost is not increased as much as microvias in Type-IV. You can also use normal PTH with backdrilling to fan out.

Figure 71. Board Via Pattern for LPDDR5 64-bit Device



8.3.8. LPDDR5 Simulation Strategy

The simulation strategy has two parts:

- Data Signal signal integrity simulation with respect to their DQS on the worse signal integrity of a data group (considering the longest routing and maximum vertical crosstalk between signals).
- CS/CTRL/CMD signal integrity simulation with respect to their CLK signals on the worst signal integrity of those signals (considering the longest routing and maximum vertical crosstalk between signals).

Consider DQ simulation as an example, Altera recommends that the signal integrity engineer review the layout and pick the worst data group (select a victim and surrounded aggressors and DQS in the group) that has the worst signal integrity on the layout (that is, the worst cross talk coupling between deep vertical vias), long trace/PCB routing and maximum reflection on the routing path due to long via stubs if backdrilling is not applied.

Altera recommends that you perform signal integrity simulation of the board layout for the selected victim surrounded by aggressor signals.

The channel analysis must be performed in the time domain (using PRBS pattern for I/O signal generator) while the channel is built by using actual per-pin package model at both ends, PCB model in the format of scattering parameter along with I/O buffer model at both ends. I/O buffer IBIS model was used for DDR4 interface SI simulation; however, LPDDR5 requires an IBIS AMI buffer model (due to the equalizations/FFE/DFE at RX) at both ends to recover the data. Eye diagram is evaluated after the simulation to meet eye specification at both ends.

Note: Designers are advised to follow strictly the PCB routing design guideline in this document to meet maximum supported data rate per selected configuration.



9. Agilex 7 M-Series FPGA EMIF IP – Timing Closure

This chapter describes timing analysis and optimization techniques that you can use to achieve timing closure within the FPGA.

Note: At this time, Agilex 7 M-Series device timing models have not been verified by silicon characterization.

9.1. Timing Closure

The following sections describe the timing analysis using the respective FPGA data sheet specifications and the user-specified memory data sheet parameters.

- Core to core (C2C) transfers have timing constraints created and are analyzed by the Timing Analyzer. Core timing does not include user logic timing within core or to and from the EMIF block. The EMIF IP provides the constrained clock to the customer logic.
- Core to periphery (C2P) transfers have timing constraints created and are timing analyzed by the Timing Analyzer.
- Periphery to core (P2C) transfers have timing constraints created and are timing analyzed by the Timing Analyzer.
- Periphery to periphery (P2P) transfers are modeled entirely by a minimum pulse width violation on the hard block, and have no internal timing arc.

To account for the effects of calibration, the EMIF IP includes additional scripts that are part of the `<phy_variation_name>_report_timing.tcl` and `<phy_variation_name>_report_timing_core.tcl` files that determine the timing margin after calibration. These scripts use the setup and hold slacks of individual pins to emulate what is occurring during calibration to obtain timing margins that are representative of calibrated PHYs. The effects considered as part of the calibrated timing analysis include improvements in margin because of calibration, and quantization error and calibration uncertainty because of voltage and temperature changes after calibration.

9.1.1. Timing Analysis

Timing analysis of Agilex 7 M-Series EMIF IP is somewhat simpler than that of some earlier device families, because Agilex 7 M-Series devices have more hardened blocks and fewer soft logic registers to be analyzed, because most are user logic registers.

Your Agilex 7 M-Series EMIF IP includes a Synopsys Design Constraints File (`.sdc`) which contains timing constraints specific to your IP. The `.sdc` file also contains Tool Command Language (`.tcl`) scripts which perform various timing analyses specific to memory interfaces.

9.1.1.1. PHY or Core

Timing analysis of the PHY or core path includes the path from the last set of registers in the core to the first set of registers in the periphery (C2P), or the path from the last set of registers in the periphery to the first of registers in the core (P2C) and the ECC related path if it is enabled.

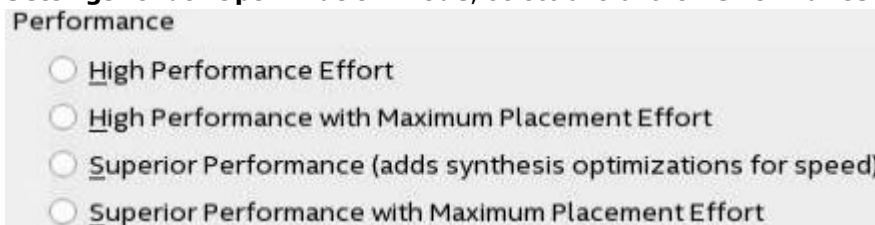
Core timing analysis excludes user logic timing to or from EMIF blocks. The EMIF IP provides a constrained clock (for example: ddr4_usr_clk) with which to clock customer logic; pll_afi_clk serves this purpose.

The PHY or core analyzes this path by calling the `report_timing` command in `<variation_name>_report_timing.tcl` and `<variation_name>_report_timing_core.tcl`.

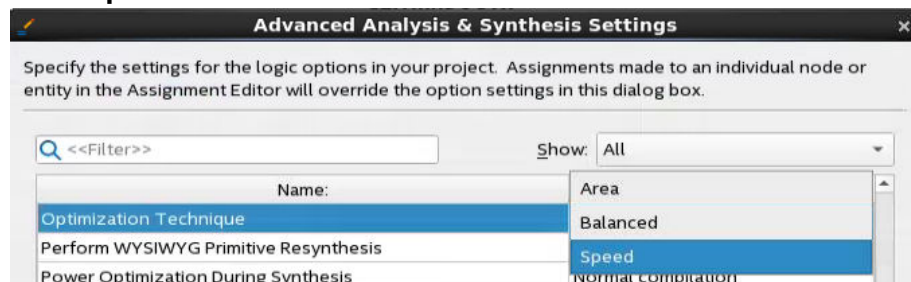
9.2. Optimizing Timing

The Quartus Prime software offers several advanced features that you can use to assist in meeting core timing requirements.

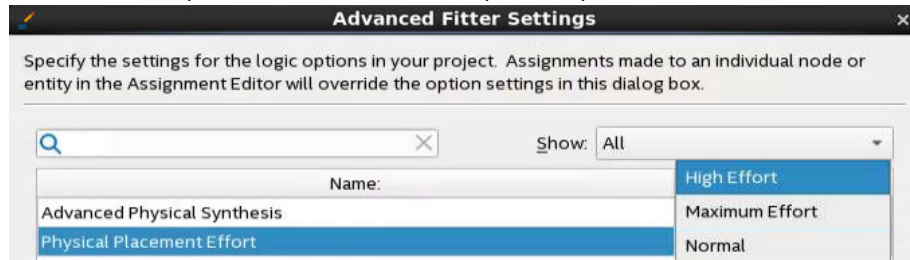
1. On the **Assignments** menu, click **Settings**. In the **Category** list, click **Compiler Settings**. Under **Optimization mode**, select one of the **Performance** options.



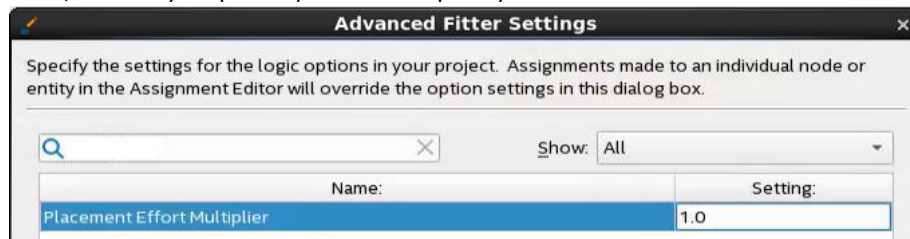
2. On the **Assignments** menu, click **Settings**. In the **Category** list, click **Compiler Settings** > **Advanced Settings (Synthesis)**. For **Optimization Technique**, select **Speed**.



3. On the **Assignments** menu, click **Settings**. In the **Category** list, click **Compiler Settings > Advanced Settings (Fitter)**. For **Physical Placement Effort**, select **High Effort** or **Maximum Effort**. The High and Maximum effort settings take additional compilation time to further optimize placement.



4. On the **Assignments** menu, click **Settings**. In the **Category** list, click **Compiler Settings > Advanced Settings (Fitter)**. For **Placement Effort Multiplier**, select a number higher than the preset value of 1.0. A higher value increases CPU time, but may improve placement quality.





10. Agilex 7 M-Series FPGA EMIF IP – Controller Optimization

When designing an external memory interface, you should understand the ways available to increase the efficiency and bandwidth of the memory controller.

The following topics discuss factors that affect controller efficiency and ways to increase the efficiency of the controller.

Controller Efficiency

Controller efficiency varies depending on data transaction. The best way to determine the efficiency of the controller is to simulate the memory controller for your specific design.

Controller efficiency is expressed as:

Efficiency = number of active cycles of data transfer/total number of cycles

The total number of cycles includes the number of cycles required to issue commands or other requests.

Note: You calculate the number of active cycles of data transfer in terms of local clock cycles.

10.1. Optimizing Efficiency for Secondary Controller

You can encounter lower than expected efficiency for sequential access patterns on the secondary memory controller in the following EMIF configurations:

- 2ch x16 LPDDR5 / DDR5
- 4ch x16 LPDDR5
- 1ch x16 of LPDDR5 / DDR5 on the top sub-bank

Asynchronous Clocking Mode (Fabric Direct - User Clock Asynchronous to PHY)

You can improve the efficiency for sequential access pattern on the secondary controller by increasing the AXI clock frequency. Recommendation is to set the AXI clock frequency to a higher frequency. You can further improve the efficiency of sequential write access pattern on the secondary controller by increasing the burst length as shown in the Table 1. The data shown in the table is obtained by using a 2ch x16 LPDDR5 design, running at 1066.667MHz.

Table 243. Controller Efficiency with Different AXI Clock Frequency and Burst Length

AXI Clock Frequency (MHz)	Pattern	Burst Length	Controller Efficiency (%)	
			Primary	Secondary
133.333 (Equivalent to Sync Mode)	Sequential write	2	~90	~40
	Sequential read		~90	~50
266.667	Sequential write	2	~90	~75
	Sequential read		~90	~90
	Sequential write	4	~90	~80
	Sequential read		~90	~90
	Sequential write	8	~90	~90
	Sequential read		~90	~90
	Sequential write	16	~90	~90
	Sequential read		~90	~90

Random access patterns or mixed traffic patterns can mitigate the differences in the efficiency for the primary controller and secondary controller.

Table 244. Controller Efficiency with Different AXI Clock Frequency for Random Traffic Pattern

AXI Clock Frequency (MHz)	Pattern	Burst Length	Controller Efficiency (%)	
			Primary	Secondary
133.333	Random write	2	~40	~39
	Random read		~38	~38
266.667	Random write	2	~41	~41
	Random read		~38	~38

With increased AXI clock frequency, the efficiency number reported by the PMON IP represents the utilization of the AXI bus instead of controller efficiency. The following equation represents the efficiency of the memory controller:

Controller efficiency = AXI transactions accepted by AXI bus / Memory Controller Bandwidth * 100 %

= (PMON efficiency * AXI Clock Freq * 256-bit) / (16-bit * Mem Clock Freq) * 100%

Synchronous Clocking Mode (Fabric Direct – User Clock Synchronous to PHY)

In LPDDR5 and DDR5 memory interface, when using x16 2-channel configuration for long burst sequential access during write or read operation, performance is limited to 50% per read/write sub-channel for the secondary controller.

If your application requires high controller efficiency for sequential access pattern, it is recommended to use asynchronous clocking mode. Random access patterns or mixed traffic patterns can mitigate the differences in the efficiency for the primary controller and secondary controller.

Note that in synchronous clocking mode, the efficiency for the primary controller used for implementing a x32 DDR5/LPDDR5 memory interface has reduced performance. This is because the bandwidth for AXI bus is only half of the bandwidth for the memory controller.

AXI Bus Bandwidth = AXI Clock Freq x 256-bit = (Memory Clock Freq / 8) * 256 = 32 * Memory Clock Frequency

Memory Controller Bandwidth = Memory Clock Freq * 2 * 32-bit = 64 * Memory Clock Frequency = 2 * AXI Bus Bandwidth

10.2. Interface Standard

Complying with certain interface standard specifications affects controller efficiency.

When interfacing the memory device to the memory controller, you must observe timing specifications and perform the following bank management operations:

- **Activate**
Before you issue any read (RD) or write (WR) commands to a bank within an SDRAM device, you must open a row in that bank using the activate (ACT) command. After you open a row, you can issue a read or write command to that row based on the t_{RCD} specification. Reading or writing to a closed row has negative impact on the efficiency as the controller has to first activate that row and then wait until t_{RCD} time to perform a read or write.
- **Precharge**
To open a different row in the same bank, you must issue a precharge command. The precharge command deactivates the open row in a particular bank or the open row in all banks. Switching a row has a negative impact on the efficiency as you must first precharge the open row, then activate the next row and wait t_{RCD} time to perform any read or write operation to the row.
- **Device CAS latency**
The memory device has its own read latency, and the higher the CAS latency, the less efficient an individual access. The higher the operating frequency, the longer the CAS latency is in number of cycles.
- **Refresh**
A refresh, in terms of cycles, consists of the precharge command and the waiting period for the auto refresh.

10.3. Bank Management Efficiency

Bank management operation affects controller efficiency.

When a read operation reads changes from a row in a bank, it has an impact on efficiency, relative to the row in the bank remaining unchanged.

When a row in the bank is unchanged, the controller does not need to issue precharge and activate commands; by not issuing precharge and activate commands, the speed of the read operation is increased, resulting in better efficiency.

Similarly, if you do not switch between read and write frequently, the efficiency of your controller improves significantly.

10.4. Data Transfer

The following methods of data transfer reduce the efficiency of your controller:

- Performing individual read or write accesses is less efficient.
- Switching between read and write operation reduces the efficiency of the controller.
- Performing read or write operations from different rows within a bank or in a different bank—if the bank and a row you are accessing is not already open—also affects the efficiency of your controller.

10.5. Improving Controller Efficiency

You can use the following methods to improve the efficiency of your controller.

- Frequency of Operation
- Series of Reads or Writes

The following sections discuss these methods in detail.

10.5.1. Using Multiple AXI IDs

The External Memory Interfaces IP memory controller can support out-of-order transactions using the ID signal in each AXI channel.

When using different IDs for write or read transactions, those sharing the same ID are completed in order, those with different IDs can be reordered by the controller to improve efficiency.

Altera recommends that you assign different IDs for write and read transactions whenever possible; combined with previous recommendations, doing so will improve controller efficiency.

10.5.2. Controller Performance Profile

You can select different controller profiles that suit your design needs under the **Controller Performance Profile** dropdown menu.

Three controller profiles are available:

- **Sequential access optimized.**
- **Random access optimized.**
- **Custom.** (This option makes available the **Force auto-precharge**, **Bank Group rotation**, and **Enable Row/CS Swap** options.)

A larger BG rotation will improve performance for sequential access and has minimal effect for random ordering. For non-continuous accesses, using different BG rotations may improve performance.

Pairing the controller profile with the right traffic type would provide better throughput for your design. You can test each of the settings with the required traffic or test data, to identify the combination that provides the best performance.

10.5.3. Frequency of Operation

Certain frequencies of operation give you the best possible latency based on the memory parameters. The memory parameters you specify through the parameter editor are converted to clock cycles and rounded up.

In most cases, the frequency and parameter combination is not optimal. If you are using a memory device that has $t_{RCD} = 15$ ns and are running the interface at 1200 MHz, you get the following results:

- For quarter-rate implementation ($t_{CK} = 3.33$ ns):
 t_{RCD} convert to clock cycle = $15/3.33 = 4.5$, rounded up to 5 clock cycles or 16.65 ns.

10.5.4. Controller Performance Profile

You can select different controller profiles to suit your design needs, under the **Controller Performance Profile** dropdown menu.

Three controller profiles are available:

- **Sequential access optimized.**
- **Random access optimized.**
- **Custom.** (This option makes available the **Force auto-precharge**, **Bank Group rotation**, **Enable Row/CS Swap**, **Enable Placement Logic**, **Enable All Strobe**, and **Enable Fixed Priority** options.)

Pairing the controller profile with the right traffic type would provide better throughput for your design. You can test each of the settings with the required traffic or test data, to identify the combination that provides the best performance.

10.5.5. Series of Reads or Writes

Performing a series of reads or writes from the same bank and row increases controller efficiency.

For best performance, minimize random reads and random writes. When you perform reads and writes to random locations, the operations require row and bank changes. To change banks, the controller must precharge the previous bank and activate the row in the new bank. Even if you change the row in the same bank, the controller has to close the bank (precharge) and reopen it again just to open a new row (activate). Because of the precharge and activate commands, efficiency can decrease by as much as 3–15%, as the controller needs more time to issue a read or write.

If you must perform a random read or write, use additive latency and bank interleaving to increase efficiency.

Controller efficiency depends on the method of data transfer between the memory device and the FPGA, the memory standards specified by the memory device vendor, and the type of memory controller.

10.5.6. Special AXI Interface Signals

The majority of the AXI interface signals presented correspond to their usage as defined in the AXI specifications.

However, the AWUSER, ARUSER, WUSER, and RUSER signals have specific functions in the Agilex 7 M-Series FPGA EMIF IP. These signals are optional and you may set them to zero if the feature is not required.

- AWUSER:
 - ☐ Bit 0: Enable/Disable Auto-precharge. (1 = Enable, 0 = Disable).
 - ☐ Bit 1: ALLSTRB – When all write strobes are asserted (that is., when no byte enable signals are deasserted), enabling this bit can improve controller performance.
 - ☐ Bits 2–13: Not connected (drive zero).
- ARUSER:
 - ☐ Bit 0: Enable/Disable Auto-precharge. (1 = Enable, 0 = Disable).
 - ☐ Bits 1–13: Not connected (drive zero).
- WUSER:
 - ☐ When ECC is enabled (in the x40/x72 lockstep cases), the additional user bits to be written are sent on this interface. If an x36 interface is used, only the lowest 32 bits are connected.
- RUSER:
 - ☐ When ECC is enabled (in the x40/x72 lockstep cases), the additional user bits are received on this interface. If an x36 interface is used, only the lowest 32 bits are connected.

10.5.7. Address Mapping — AXI Interface to DDR Memory

You can select which region of the memory (bank, bank group, row, column, and chip) to access by selecting the corresponding AXI address.

The default configuration is as follows:

Table 245. DDR4 4-Bank-Group Case

<MSB								LSB>
Chip Select	Chip ID	Row	Bank	Bank Group [1]	Column [N:3]	Bank Group [0]	Column [2:0]	Datapath

Table 246. DDR4 2-Bank-Group Case

<MSB							LSB>
Chip Select	Chip ID	Row	Bank	Column [N:3]	Bank Group [0]	Column [2:0]	Datapath

Table 247. DDR5 4-Bank-Group Case

<MSB							LSB>
Chip Select	Chip ID	Row	Bank	Column [N:4]	Bank Group [1:0]	Column [3:0]	Datapath

Table 248. DDR5 8-Bank-Group Case

<MSB								LSB>
Chip Select	Chip ID	Row	Bank	Bank Group [2]	Column [N:4]	Bank Group [1:0]	Column [3:0]	Datapath

Table 249. LPDDR5 x16 Case

<MSB								LSB>
Chip Select	Chip ID	Row	Bank	Bank Group [1]	Column [N:4]	Bank Group [0]	Column [3:0]	Datapath

Table 250. LPDDR5 x32

<MSB								LSB>
Chip Select	Chip ID	Row	Bank	Bank Group [1]	Column [N:4]	Bank Group [0]	Column [3:0]	Datapath

Datapath: For a x32 or wider interface, two bits are allocated to datapath. For a x16 interface, one bit is allocated; this bit should always be set to zero.

Row/Bank/Bank Group/Column: These are allocated a number of address bits based on the requirements of the connected memory type.

Chip Select/Chip ID: These are each one bit wide and are omitted when not using multi-rank or 3DS configurations.

Example: AXI Address mapping for a DDR4 1Gb x8 device, 2 components per rank for a total DQ Width of x16

You can obtain addressing tables from your memory device datasheet. For this example, the following information applies:

Parameter	1 Gb x 8
Bank Group	BG[1:0]
Bank Address	BA[1:0]
Rows	A[15:0]
Columns	A[9:0]

The following AXI Address to Memory Mapping table can be constructed.

Memory Address Bits	AXI Address Bits
Datapath [0]	[0] - Always 0, 1 bit for x16 interfaces
Column [2:0]	[3:1]
BG [0]	[4]
Column [9:3]	[11:5]
BG [1]	[12]
BA [1:0]	[14:13]
Row [15:0]	[30:15]

Using this information, the AXI Address to write to Column 1016 of the Row 0, Bank group 0, Bank 0 is: 0x00000FE0.

10.5.8. AXI Interface: Write and Read Example

The following diagram illustrates a Write operation through the AXI4 interface

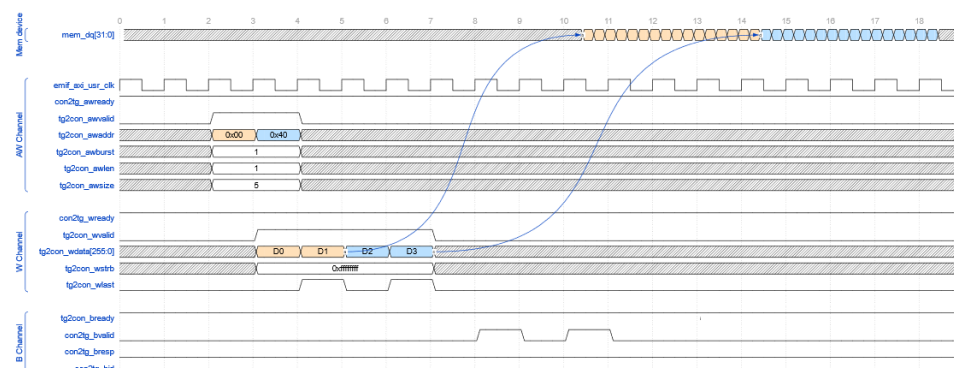
In this context, "tg" refers to any AXI4 Master, while "con" represents the EMIF IP.

In the example, two separate transactions are initiated: the first targets address 0x00, and the second targets address 0x40. Each transaction consists of two transfers (with arlen=1, indicating a burst length of 2), and each transfer is 32 bytes in size (awsize=5) for a total of 64 bytes per transaction.

On the W channel, as indicated by the color coding in the diagram, D0 and D1 represent the two 32-byte (256-bit) transfers of the first transaction. The subsequent two transfers, shown in blue, correspond to the second transaction. The wlast signal marks the end of each transaction.

In this setup, the EMIF IP is connected to DDR5 memory, which expects 16 consecutive data packets of 32 bits for each access. To meet this requirement, the EMIF IP collects the incoming AXI data and rearranges the two transfers of each transaction into two bursts of 16 data packets. These rearranged bursts are also color-coded in the diagram below.

Figure 75. EMIF IP AXI Interface: Write Example

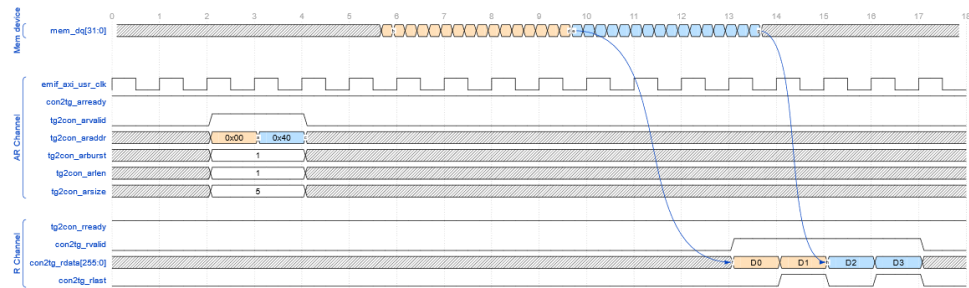


As in the previous example, the following diagram illustrates a read transaction accessing addresses 0x00 and 0x40. For each AXI4 read request, the EMIF IP issues a corresponding read command to the DDR5 memory device, which responds with a burst of 16 consecutive 32-bit data packets.

The EMIF IP collects this data and reorganizes it to be sent over the AXI4 R channel as two transactions, each composed of two 256-bit (32-byte) transfers. Each transfer is signaled on the R channel, and the end of each burst is indicated by the rlast signal. This process ensures that the data returned from DDR5 memory is properly formatted and aligned with AXI4 protocol requirements.

Figure 76. EMIF IP AXI Interface: Read Example





11. Agilex 7 M-Series FPGA EMIF IP – Debugging

This chapter discusses issues and strategies for debugging your external memory interface IP.

11.1. Interface Configuration Performance Issues

There are many interface combinations and configurations possible in an Altera design, therefore it is impractical for Altera to explicitly state the achievable f_{MAX} for every combination.

Altera seeks to provide guidance on typical performance, but this data is subject to memory component timing characteristics, interface widths, depths directly affecting timing deration requirements, and the achieved skew and timing numbers for a specific PCB.

FPGA timing issues should generally not be affected by interface loading or layout characteristics. In general, the Altera performance figures for any given device family and speed-grade combination should usually be achievable.

To resolve FPGA (PHY and PHY reset) timing issues, refer to the *Timing Closure* chapter.

Achievable interface timing (address and command, half-rate address and command, read and write capture) is directly affected by any layout issues (skew), loading issues (deration), signal integrity issues (crosstalk timing deration), and component speed grades (memory timing size and tolerance). Altera performance figures are typically stated for the default (single rank, unbuffered DIMM) case. Altera provides additional expected performance data where possible, but the f_{MAX} is not achievable in all configurations. Altera recommends that you optimize the following items whenever interface timing issues occur:

- Improve PCB layout tolerances
- Use a faster speed grade of memory component
- Ensure that the interface is fully and correctly terminated
- Reduce the loading (reduce the deration factor)

11.1.1. Interface Configuration Bottleneck and Efficiency Issues

Depending on the transaction types, efficiency issues can exist where the achieved data rate is lower than expected. Ideally, these issues should be assessed and resolved during the simulation stage because they are sometimes impossible to solve later without rearchitecting the product.

Any interface has a maximum theoretical data rate derived from the clock frequency, however, in practice this theoretical data rate can never be achieved continuously due to protocol overhead and bus turnaround times.

Simulate your desired configuration to ensure that you have specified a suitable external memory family and that your chosen controller configuration can achieve your required bandwidth.

Efficiency can be assessed in several different ways, and the primary requirement is an achievable continuous data rate. The local interface signals combined with the memory interface signals and a command decode trace should provide adequate visibility of the operation of the IP to understand whether your required data rate is sufficient and the cause of the efficiency issue.

To show if under ideal conditions the required data rate is possible in the chosen technology, follow these steps:

1. Use the memory vendor's own testbench and your own transaction engine.
2. Use either your own driver, or modify the provided example driver, to replicate the transaction types typical of your system.
3. Simulate this performance using your chosen memory controller and decide if the achieved performance is still acceptable.

Observe the following points that may cause efficiency or bottleneck issues at this stage:

- Identify the memory controller rate (full, half, or quarter) and commands, which may take two or four times longer than necessary
- Determine whether the memory controller is starved for data by observing the appropriate request signals.
- Determine whether the memory controller processor transactions at a rate sufficient to meet throughput requirements by observing appropriate signals, including the local ready signal.

Consider using either a faster interface, or a different memory type to better align your data rate requirements to the IP available directly from Altera.

11.2. Functional Issue Evaluation

Functional issues occur at all frequencies (using the same conditions) and are not altered by speed grade, temperature, or PCB changes. You should use functional simulation to evaluate functional issues.

The Altera FPGA IP includes the option to autogenerate a testbench specific to your IP configuration, which provides an easy route to functional verification.

The following issues should be considered when trying to debug functional issues in a simulation environment.

11.2.1. Altera IP Memory Model

Altera memory IP autogenerates a generic simplified memory model that works in all cases. This simple read and write model is not designed or intended to verify all entered IP parameters or transaction requirements.

The Altera-generated memory model may be suitable to evaluate some limited functional issues, but it does not provide comprehensive functional simulation.

11.2.2. Vendor Memory Model

Contact the memory vendor directly, because many additional models are available from the vendor's support system.

When using memory vendor models, ensure that the model is correctly defined for the following characteristics:

- Speed grade
- Organization
- Memory allocation
- Maximum memory usage
- Number of ranks on a DIMM
- Buffering on the DIMM
- ECC

Note: Refer to the **readme.txt** file supplied with the memory vendor model, for more information about how to define this information for your configuration. Also refer to Transcript Window messages, for more information.

Note: Altera does not provide support for vendor-specific memory models.

During simulation vendor models output a wealth of information regarding any device violations that may occur because of incorrectly parameterized IP.

11.2.3. Transcript Window Messages

When you are debugging a functional issue in simulation, vendor models typically provide much more detailed checks and feedback regarding the interface and their operational requirements than the Altera generic model.

In general, you should use a vendor-supplied model whenever one is available. Consider using second-source vendor models in preference to the Altera generic model.

Many issues can be traced to incorrectly configured IP for the specified memory components. Component data sheets usually contain settings information for several different speed grades of memory. Be aware data sheets specify parameters in fixed units of time, frequencies, or clock cycles.

The Altera generic memory model always matches the parameters specified in the IP, as it is generated using the same engine. Because vendor models are independent of the IP generation process, they offer a more robust IP parameterization check.

During simulation, review the transcript window messages and do not rely on the Simulation Passed message at the end of simulation. This message indicates only that the example driver successfully wrote and then read the correct data for a single test cycle.

Even if the interface functionally passes in simulation, the vendor model may report operational violations in the transcript window. These reported violations often explain why an interface appears to pass in simulation, but fails in hardware.

Vendor models typically perform checks to ensure that the following types of parameters are correct:

- Burst length
- Burst order
- tMRD
- tMOD
- tRFC
- tREFPDEN
- tRP
- tRAS
- tRC
- tACTPDEN
- tWR
- tWRPDEN
- tRTP
- tRDPDEN
- tINIT
- tXPDLL
- tCKE
- tRRD
- tCCD
- tWTR
- tXPR
- PRECHARGE
- CAS length
- Drive strength
- AL
- tDQS
- CAS_WL
- Refresh
- Initialization
- tIH
- tIS
- tDH
- tDS

If a vendor model can verify that all these parameters are compatible with your chosen component values and transactions, it provides a specific insight into hardware interface failures.

11.3. Timing Issue Characteristics

The PHY and controller combinations automatically generate timing constraint files to ensure that the PHY and external interface are fully constrained and that timing is analyzed during compilation. Nevertheless, timing issues can still occur. This topic discusses how to identify and resolve any timing issues that you may encounter.

Timing issues typically fall into two distinct categories:

- FPGA core timing reported issues
- External memory interface timing issues in a specific mode of operation or on a specific PCB

The Timing Analyzer reports timing issues in two categories: core-to-core and core-to-IOE transfers. These timing issues include the PHY and PHY reset sections in the Timing Analyzer Report DDR subsection of timing analysis. External memory interface timing issues are reported specifically in the Timing Analyzer Report DDR subsection, excluding the PHY and PHY reset. The Report DDR PHY and PHY reset sections only include the PHY, and specifically exclude the controller, core, PHY-to-controller and local interface. Quartus Prime timing issues should always be evaluated and corrected before proceeding to any hardware testing.

PCB timing issues are usually Quartus Prime timing issues, which are not reported in the Quartus Prime software, if incorrect or insufficient PCB topology and layout information is not supplied. PCB timing issues are typically characterized by calibration failure, or failures during user mode when the hardware is heated or cooled. Further PCB timing issues are typically hidden if the interface frequency is lowered.

11.3.1. Evaluating FPGA Timing Issues

Usually, you should not encounter timing issues with Altera-provided IP unless your design exceeds Altera's published performance range or you are using a device for which the Quartus Prime software offers only preliminary timing model support. Nevertheless, timing issues can occur in the following circumstances:

- The **.sdc** files are incorrectly added to the Quartus Prime project
- Quartus Prime analysis and synthesis settings are not correct
- Quartus Prime Fitter settings are not correct

For all of these issues, refer to the correct user guide for more information about recommended settings, and follow these steps:

1. Ensure that the IP generated **.sdc** files are listed in the Quartus Prime Timing Analyzer files to include in the project window.
2. Configure the Settings as follows, to help close timing in the design:

- a. On the **Assignments** menu click **Settings**.
- b. In the **Category** list, click **Compiler Settings**.
- c. Select **Optimization mode** ► **Performance** ► **High Performance Effort**.
- a. On the **Assignments** menu click **Settings**.
- b. In the **Category** list, click **Compiler Settings** ► **Advanced Settings (Synthesis)**.
- c. For **Optimization Technique**, select **Speed**.
- a. On the **Assignments** menu click **Settings**.
- b. In the **Category** list, click **Compiler Settings** ► **Advanced Settings (Fitter)**.
- c. For **Physical Placement Effort**, select **High Effort/Maximum Effort**.
- 3. Use **Timing Analyzer Report Ignored Constraints**, to ensure that **.sdc** files are successfully applied.
- 4. Use **Timing Analyzer Report Unconstrained Paths**, to ensure that all critical paths are correctly constrained.

More complex timing problems can occur if any of the following conditions are true:

- The design includes multiple PHY or core projects
- Devices where the resources are heavily used
- The design includes wide, distributed, maximum performance interfaces in large die sizes

Any of the above conditions can lead to suboptimal placement results when the PHY or controller are distributed around the FPGA. To evaluate such issues, simplify the design to just the autogenerated example top-level file and determine if the core meets timing and you see a working interface. Failure implies that a more fundamental timing issue exists. If the standalone design passes core timing, evaluate how this placement and fit is different than your complete design.

Use Logic Lock regions or design partitions to better define the placement of your memory controllers. When you have your interface standalone placement, repeat for additional interfaces, combine, and finally add the rest of your design.

Additionally, use fitter seeds and increase the placement and router effort multiplier.

11.3.2. Evaluating External Memory Interface Timing Issues

External memory interface timing issues usually relate to the FPGA input and output characteristics, PCB timing, and memory component characteristics.

The FPGA input and output characteristics are usually fixed values, because the IOE structure of the devices is fixed. Optimal PLL characteristics and clock routing characteristics do have an effect. Assuming the IP is correctly constrained with autogenerated assignments, and you follow implementation rules, the design should reach the stated performance figures.

Memory component characteristics are fixed for any given component or DIMM. Consider using faster components or DIMMs in marginal cases when PCB skew may be suboptimal, or your design includes multiple ranks when deration may cause read capture or write timing challenges. Using faster memory components often reduces the memory data output skew and uncertainty easing read capture, and lowering the memory's input setup and hold requirement, which eases write timing.

Increased PCB skew reduces margins on address, command, read capture and write timing. If you are narrowly failing timing on these paths, consider reducing the board skew (if possible), or using faster memory. Address and command timing typically requires you to manually balance the reported setup and hold values with the dedicated address and command phase in the IP.

Refer to the respective IP user guide for more information.

Deration because of increased loading, or suboptimal layout may result in a lower than desired operating frequency meeting timing. You should close timing in the Timing Analyzer software using your expected loading and layout rules before committing to PCB fabrication.

Ensure that any design with an Altera PHY is correctly constrained and meets timing in the Timing Analyzer software. You must address any constraint or timing failures before testing hardware.

For more information about timing constraints, refer to the Timing Analysis chapter.

11.4. Verifying Memory IP Using the Signal Tap Logic Analyzer

The Signal Tap logic analyzer shows read and write activity in the system.

For more information about using the Signal Tap logic analyzer, refer to the *Quartus Prime Pro Edition User Guide: Debug Tools*.

To add the Signal Tap logic analyzer, follow these steps:

1. On the Tools menu click **Signal Tap Logic Analyzer**.
2. In the **Signal Configuration** window next to the **Clock** box, click ... (Browse Node Finder).
3. Type the memory interface system clock in the **Named** box, for **Filter** select **Signal Tap: presynthesis** and click **Search**.
4. Select the memory interface clock that is exposed to the user logic.
5. Click **OK**.
6. Under Signal Configuration, specify the following settings:
 - For **Sample depth**, select **512**
 - For **RAM type**, select **Auto**
 - For **Trigger flow control**, select **Sequential**
 - For **Trigger position**, select **Center trigger position**
 - For **Trigger conditions**, select **1**

11.5. Debugging with the External Memory Interface Debug Toolkit

The External Memory Interface Debug Toolkit for Agilex 7 M-Series FPGAs provides access to data collected by the sequencer during memory calibration, as well as analysis tools to evaluate the stability of the calibrated interface and assess hardware conditions.

The debug toolkit provides the following types of reports:

- Interface and memory configuration, such as external memory protocol and interface width.
- Calibration results, including calibration status (pass or fail), calibration failure stage (if applicable), delay settings and margins, and VREF settings and margins..
- Traffic test results (pass or fail).
- Calibration report, providing detailed information about the margins observed during calibration.
- Debug prints, which consist of delay settings and margins, as well as VREF settings and margins in each calibration. Debug Prints are printed to a text file for further analysis.

The available task and analysis capabilities include the following:

- Ability to request recalibration of the memory interface.
- Ability to run the test engine in the design example.
- Ability to view the delay setting on any pin in the selected interface and update it if necessary.
- Ability to run VREF margining on the interface.
- Ability to run driver margining on the interface.

Note:

Because the HPS EMIF controller does not support the External Memory Interface Debug Toolkit, verify that the HPS memory interface is operational using the non-HPS memory controller first. Create a design that instantiates the FPGA memory controller, use parameters that will be used for the HPS memory interface, and route it to the same I/O that the HPS EMIF uses.

11.5.1. Prerequisites for Using the EMIF Debug Toolkit

You must complete the following prerequisites before using the EMIF Debug Toolkit.

1. You must configure your design to use the EMIF Debug Toolkit, as described in the following topics.
2. You must compile your design.
3. You must program the target device with the resulting SRAM Object File (.sof).

After completing the above steps, you are ready to run the EMIF Debug toolkit.

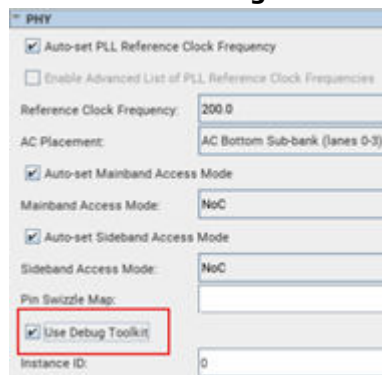
Related Information

[Launching the EMIF Debug Toolkit](#) on page 321

11.5.2. Configuring Your Design to Use the EMIF Debug Toolkit

To configure your design for use with the toolkit, you must generate a design example with the debug toolkit.

1. Navigate to the **High-Level Configuration** tab, and go to the **PHY** section.
2. Click the **Use Debug Toolkit** checkbox .



3. After you have fully parameterized the interface, click **Generate Example Design**. The system generates a design example with the debug toolkit enabled and all necessary components connected as required for a single interface.

Related Information

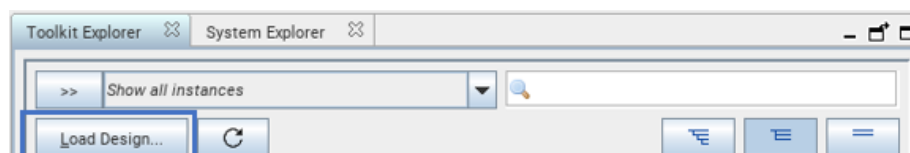
- [Design Example Quick Start Guide](#)
External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide.
- [Launching the EMIF Debug Toolkit](#) on page 321

11.5.3. Launching the EMIF Debug Toolkit

Before launching the EMIF Debug Toolkit, ensure that you have configured your device with a programming file that has the EMIF Debug Toolkit enabled.

To launch the debug toolkit, follow these steps:

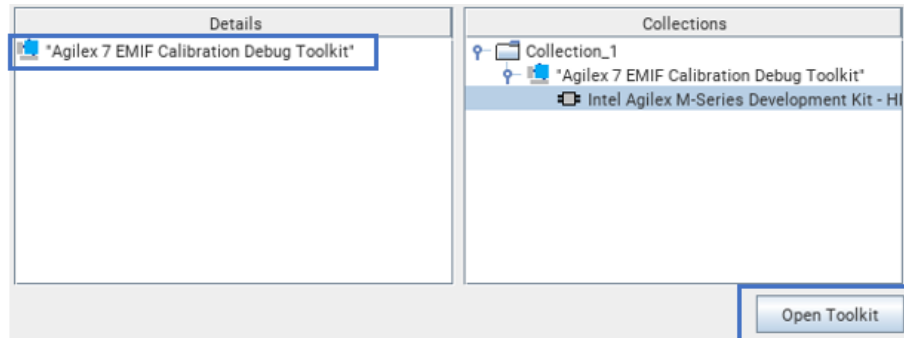
1. In the Quartus Prime software, open the System Console by clicking **Tools > System Debugging Tools > System Console**.
2. In the System Console, load the SRAM Object File (.sof) with which you have programmed the board. (Refer to the [Prerequisites for Using the EMIF Debug Toolkit](#) topic.)



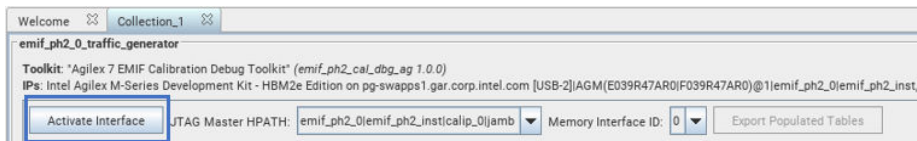
3. Select the correct instance that you want to debug.

Instances	References
ed_synth.sof	/nfs/site/disks/ceg_user_abdulaz1/EMIF/24.1/A...
Intel Agilex M-Series Development Kit - HBM2e Edition ...	
AGM(E039R47AR0 F039R47AR0)@1	
lemif_ph2_0	emif_ph2_cal_dbg_ag_1.0.0

4. Select **Agilex 7 EMIF Calibration Debug Toolkit** from the **Details** section. Click **Open Toolkit** to open the main view of the toolkit.



5. Click **Activate Interface** to allow the toolkit to read the parameters and status for the selected interface, and to perform analysis tasks.



6. Successful activation of the Debug Toolkit is indicated by a message in the Messages tab.



Related Information

- [Configuring Your Design to Use the EMIF Debug Toolkit](#) on page 321
- [Prerequisites for Using the EMIF Debug Toolkit](#) on page 320

11.5.4. Using the EMIF Debug Toolkit

The main view of the EMIF Debug Toolkit contains the **Memory Configuration**, **Calibration**, **Calibration Report**, **Driver Margining**, **VREF Margining**, and **Pin Delay Settings** tabs.

Memory Configuration Tab

The **Memory Configuration** tab shows the IP settings, which you defined when you parameterized the EMIF IP.

Figure 77. Memory Configuration Tab

Memory Configuration Calibration Calibration Report Driver Margining Pin Delay Settings				
Parameter		Value		
Technology Generation		DDR5		
Memory Format		Discrete Component		
Memory Clock Frequency		2200.0		
Memory Ranks		1		
Device DQ Width		16		
Total DQ Width		32		
Number of Channels		1		
Use NOC		false		
Burst Length		16		
DQ Width		32		
Chip Select Width		1		
Chip ID Width		0		
Memory Device Capacity in Gb		32		

Calibration Tab

The **Calibration** tab allows you to rerun calibration and the test engine.

Figure 78. Calibration Tab

Memory Configuration **Calibration**

Number of iterations: 5

☐ Store debug prints locally

Run Re-Calibration Assert TG Reset (Run TG)

Calibration run status: ●

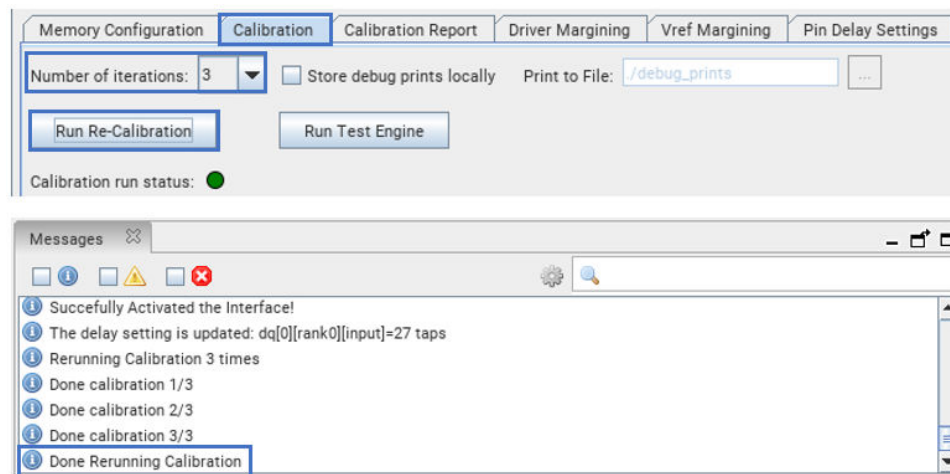
11.5.4.1. Rerunning Calibration

The **Calibration** tab lets you specify a number of iterations by which to rerun calibration.

To rerun calibration, follow these steps:

1. Select the desired number of iterations from the **Number of Iterations** pull-down menu.
2. Click **Run Re-Calibration** to repeat calibration the specified number of times. The system reports **Done Rerunning Calibration** in the **Messages** window upon completion.

A green dot in the **Calibration run status** indicator signifies that the calibration passed for all the iterations, while a red dot indicates that the calibration failed in at least one of the iterations.



Note: For multichannel interfaces, recalibrating an interface resets all channels associated with that EMIF.

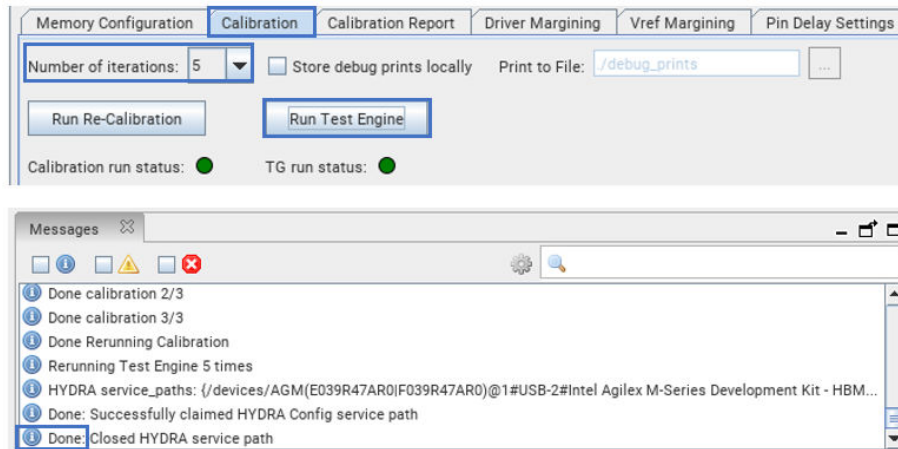
11.5.4.2. Rerunning the Test Engine

You can use the debug toolkit to rerun the test engine.

To rerun the test engine, follow these steps:

1. Select the desired number of iterations that you want to rerun the test engine from the **Number of Iterations** pulldown menu.
2. Click **Run Test Engine** to rerun the traffic generator with each iteration. A **Done** status on the **Messages** window indicates a complete run.

A green dot in the **Calibration run status** indicator signifies a pass for all iterations of the test engine, while a red dot indicates a failure in at least one iteration.

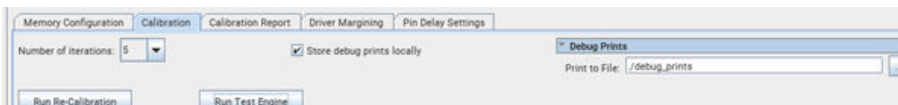


11.5.4.3. Saving Debug Print

You can save the calibration log to a Debug Print text file.

To save a Debug Print file, follow these steps:

1. On the **Calibration** tab, enable **Store debug prints locally**.
2. In the **Print to File** field, specify a path and filename for the file that you want to create.



In the example illustrated in the above diagram, clicking **Run Re-Calibration** would generate 5 debug print files, named `debug_prints_1`, `debug_prints_2`, and so forth. Each file stores the debug prints information for one iteration of calibration.

11.5.4.4. Calibration Reports

Calibration reports provide detailed information about the margins observed during calibration and the settings applied on the calibration bus during calibration.

To view the margins, click on the respective section for Address/Command, CS, DQ, DQS, DM_DBI, VREF Settings or VREF_GROUP Settings.

Figure 79. Calibration Report Tab

Memory Configuration Calibration Calibration Report Driver Margining Pin Delay Settings		
Delays and Margins (Command/Address)		
Address/Command	Margin (ps)	Delay (taps)
CA_0	-394 to 398	183
CA_1	-415 to 419	170
CA_2	-401 to 401	198
CA_3	-419 to 423	175
CA_4	-387 to 387	186
CA_5	-423 to 423	172
CA_6	-401 to 405	186
CA_7	-433 to 433	174
CA_8	-380 to 384	177
CA_9	-423 to 426	167
CA_10	-376 to 376	174
CA_11	-412 to 415	162
CA_12	-384 to 384	177
Delays and Margins (CS)		
Delays and Margins (DQ)		
Delays and Margins (DQS)		
Delays and Margins (DM,DBI)		
VREF Settings		
VREF_GROUP Settings		
Command/Address VREF Settings		

11.5.4.5. Driver Margining Tab

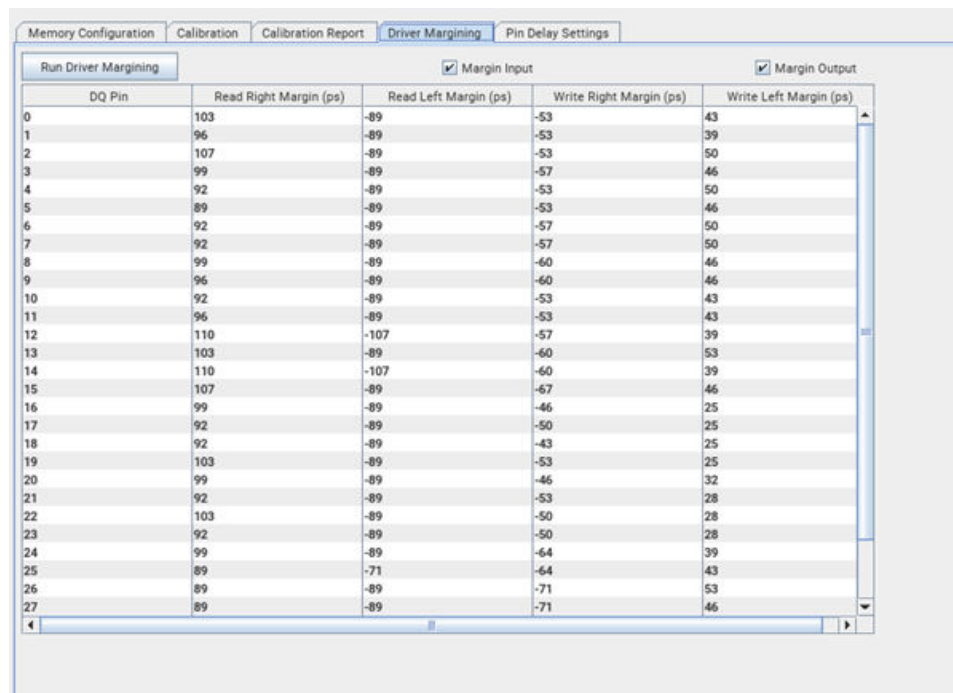
The driver margining feature lets you measure margins on your memory interface using a driver with predefined traffic patterns.

Margins measured with this feature are expected to be smaller than margins measured during calibration, because this traffic pattern is longer than those run during calibration, and is intended to stress the interface.

To use driver margining, press **Run Driver Margining**, at the top-left of the tab. The toolkit then measures margins for DQ read, DQ write, and DM. The process usually takes a few minutes, depending on the margin size, the interface size, and the duration of the driver tests.

The system displays the test results in the table when the test has completed.

Figure 80. Driver Margining Tab



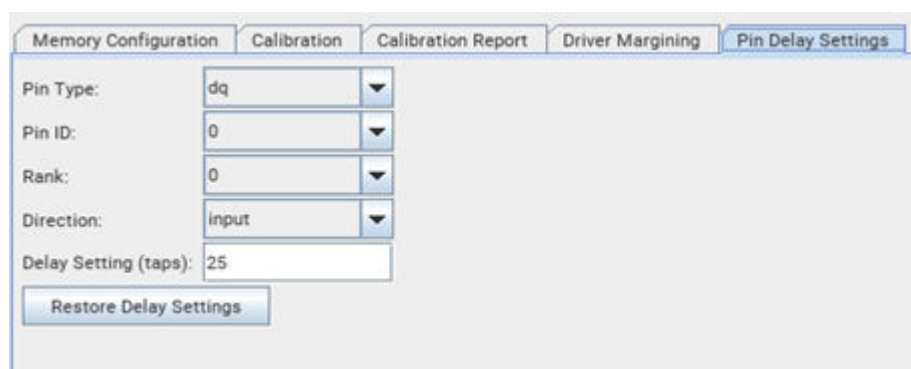
DQ Pin	Read Right Margin (ps)	Read Left Margin (ps)	Write Right Margin (ps)	Write Left Margin (ps)
0	103	-89	-53	43
1	96	-89	-53	39
2	107	-89	-53	50
3	99	-89	-57	46
4	92	-89	-53	50
5	89	-89	-53	46
6	92	-89	-57	50
7	92	-89	-57	50
8	99	-89	-60	46
9	96	-89	-60	46
10	92	-89	-53	43
11	96	-89	-53	43
12	110	-107	-57	39
13	103	-89	-60	53
14	110	-107	-60	39
15	107	-89	-67	46
16	99	-89	-46	25
17	92	-89	-50	25
18	92	-89	-43	25
19	103	-89	-53	25
20	99	-89	-46	32
21	92	-89	-53	28
22	103	-89	-50	28
23	92	-89	-50	28
24	99	-89	-64	39
25	89	-71	-64	43
26	89	-89	-71	53
27	89	-89	-71	46

11.5.4.6. Pin Delay Settings Tab

The **Pin Delay Settings** tab lets you view and change delay values on specific pins.

You can select the **Pin Type**, **Pin ID**, **Rank**, and **Direction** values of any delay that you are interested in, and the toolkit displays the delay value in the **Delay Setting (taps)** field.

Figure 81. Pin Delay Settings Tab



Memory Configuration Calibration Calibration Report Driver Margining **Pin Delay Settings**

Pin Type: dq

Pin ID: 0

Rank: 0

Direction: input

Delay Setting (taps): 25

Restore Delay Settings

11.6. Generating Traffic with the Test Engine IP

Every Agilex 7 M-Series FPGA EMIF design example includes an instance of the software-driven programmable AXI traffic generator, known as the Test Engine IP.

You can view the Test Engine IP software within the following Python scripts:

- A `main.py` file that parses the `.qsys` file and selects the traffic program to run during execution.
- A `traffic_patterns.py` file that contains many different tutorial programs and functional tests that you can refer to when writing your own traffic patterns.

For the EMIF design example, the hard-coded traffic program selected when you generate a design is the `emif_tg_emulation` traffic program, which provides these features:

- Single write and read (with `AxLEN=axlen_a1`)
- Single write and read (with `AxLEN=axlen_b2`)
- Sequential address³ block of 512 writes and 512 reads (with `AxLEN=axlen_a1`)
- Sequential address³ block of 512 writes and 512 reads (with `AxLEN=axlen_b2`)
- Random address⁴ block of 512 writes and 512 reads (with `AxLEN=axlen_a1`)

- ¹ The `axlen_a` value is dependent on the memory technology:
 - For DDR4: 0
 - For DDR5: 1
 - For LPDDR5: 3
- ² The `axlen_b` value is dependent on the memory technology:
 - For DDR4: 0
 - For DDR5: 0 (results in Read-Modify-Write or Data-Masking on the memory side).
 - For LPDDR5: 3
- ³ Sequential Address pattern starts at address=0, and increments by $(AXI_DATA_WIDTH/8)*(AxLEN + 1)$ on each transaction.
- ⁴ Random Address pattern starts at address=0, and uses pseudo-random addresses.

Running Traffic with the Test Engine IP

To run traffic through your interface with the Test Engine IP, follow these steps:

1. Open your preferred command terminal.
2. Check for the availability of the system-console command (for example, in Linux, run: `which system-console`).
3. If the system console is not available, add it to the system PATH.
4. Navigate to the example design `qii/` root directory. You should be able to see the Test Engine directory, `_sw/`.
5. From the example design root directory (`qii/`), run the following:

```
system-console --script=hydra_sw/testengine_library.tcl --sof=ed_synth.sof --
update=1 --n-loops=1
```

11.7. Guidelines for Developing HDL for Traffic Generator

If you are not getting the expected response on the AXI bus when using your own traffic generator to test your EMIF IP on hardware, ensure that your traffic generator meets the following guidelines.

1. The traffic generator issues AXI transactions only after calibration has completed successfully. You can check the calibration status by using the AXI-Lite interface. In the EMIF example design, the `cal_done_rst_n` port on the `ed_synth_axil_driver_0` corresponds to the calibration status. A value of `cal_done_rst_n=1` indicates that the calibration has completed and passed. Your traffic generator can begin to issue AXI-compliant transactions only after `cal_done_rst_n=1`.

Figure 82. `cal_done_rst_n` in `ed_synth_axil_driver_0`

```

149  ed_synth_axil_driver_0 axil_driver_0 (
150      .axil_driver_clk      (user_pll_outclk1_clk),
151      .axil_driver_rst_n    (~rst_controller_reset_out_reset),
152      .axil_driver_awaddr   (axil_driver_0_axil_driver_axi4_lite_awaddr),
153      .axil_driver_awvalid  (axil_driver_0_axil_driver_axi4_lite_awvalid),
154      .axil_driver_awready  (axil_driver_0_axil_driver_axi4_lite_awready),
155      .axil_driver_wdata    (axil_driver_0_axil_driver_axi4_lite_wdata),
156      .axil_driver_wstrb    (axil_driver_0_axil_driver_axi4_lite_wstrb),
157      .axil_driver_wvalid   (axil_driver_0_axil_driver_axi4_lite_wvalid),
158      .axil_driver_wready   (axil_driver_0_axil_driver_axi4_lite_wready),
159      .axil_driver_bresp    (axil_driver_0_axil_driver_axi4_lite_bresp),
160      .axil_driver_bvalid   (axil_driver_0_axil_driver_axi4_lite_bvalid),
161      .axil_driver_bready   (axil_driver_0_axil_driver_axi4_lite_bready),
162      .axil_driver_araddr   (axil_driver_0_axil_driver_axi4_lite_araddr),
163      .axil_driver_arvalid  (axil_driver_0_axil_driver_axi4_lite_arvalid),
164      .axil_driver_arready  (axil_driver_0_axil_driver_axi4_lite_arready),
165      .axil_driver_rdata    (axil_driver_0_axil_driver_axi4_lite_rdata),
166      .axil_driver_rresp    (axil_driver_0_axil_driver_axi4_lite_rresp),
167      .axil_driver_rvalid   (axil_driver_0_axil_driver_axi4_lite_rvalid),
168      .axil_driver_rready   (axil_driver_0_axil_driver_axi4_lite_rready),
169      .axil_driver_awprot   (axil_driver_0_axil_driver_axi4_lite_awprot),
170      .axil_driver_arprot   (axil_driver_0_axil_driver_axi4_lite_arprot),
171      .cal_done_rst_n       (axil_driver_0_cal_done_rst_n_reset)
172  );

```

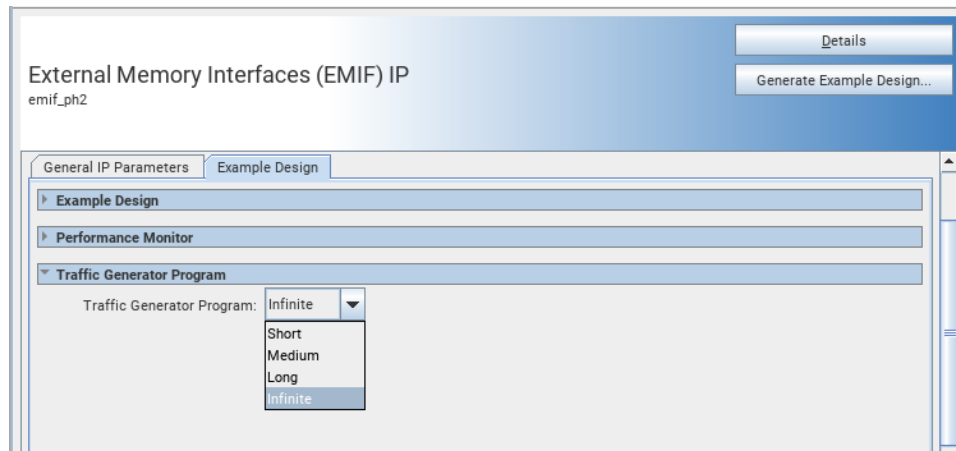
11.8. Guidelines for Traffic Generator Status Check

The Test Engine IP allows you to observe traffic generator status through the Signal Tap Logic Analyzer and allows you to export status interface so that you can observe the signals through top level design ports or access through user logic.

11.8.1. Status Check Using the Signal Tap Logic Analyzer

When you generate an Agilex 7 FPGA EMIF IP design example, you can specify the traffic pattern to run. Select the required setting, as illustrated in the figure below.

Figure 83. Selecting Traffic Generator Pattern



Refer to the *External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide* for more information.

Observing Status with the Signal Tap Logic Analyzer

Follow these steps to observe status using the Signal Tap Logic Analyzer:

1. Generate a Signal Tap Logic Analyzer file, which includes traffic generator status signals.

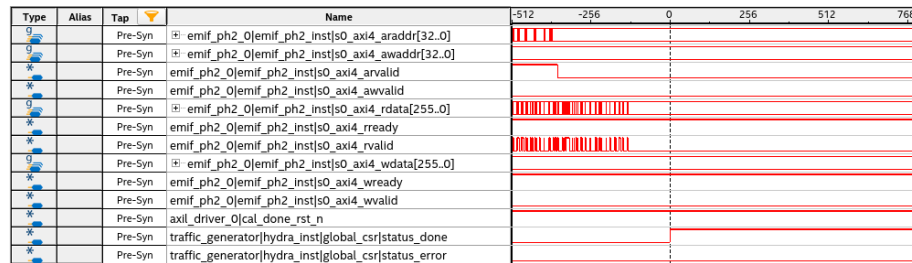
Figure 84.

Type	Alias	Tap	Node	Name	Data Enable	Trigger Enable	Trigger Conditions
Pre-Syn			emif_ph2_0 emif_ph2_inst s0_axi4_araddr[32..0]		✓	✓	1 ✓ Basic AND
Pre-Syn			emif_ph2_0 emif_ph2_inst s0_axi4_awaddr[32..0]		✓	✓	xxxxxxxxxxh
Pre-Syn			emif_ph2_0 emif_ph2_inst s0_axi4_arvalid		✓	✓	xxxxxxxxxxh
Pre-Syn			emif_ph2_0 emif_ph2_inst s0_axi4_avalid		✓	✓	
Pre-Syn			emif_ph2_0 emif_ph2_inst s0_axi4_rdata[255..0]		✓	✓	xxxxxxxxxxxxxxxxxxxx
Pre-Syn			emif_ph2_0 emif_ph2_inst s0_axi4_rready		✓	✓	
Pre-Syn			emif_ph2_0 emif_ph2_inst s0_axi4_rvalid		✓	✓	
Pre-Syn			emif_ph2_0 emif_ph2_inst s0_axi4_wdata[255..0]		✓	✓	xxxxxxxxxxxxxxxxxxxx
Pre-Syn			emif_ph2_0 emif_ph2_inst s0_axi4_wready		✓	✓	
Pre-Syn			emif_ph2_0 emif_ph2_inst s0_axi4_wvalid		✓	✓	
Pre-Syn			axil_driver_0 cal_done_rst_n		✓	✓	
Pre-Syn			traffic_generator hydra_inst global_csr status_done		✓	✓	
Pre-Syn			traffic_generator hydra_inst global_csr status_error		✓	✓	

Note: The status_done and status_error ports on the traffic_generator|hydra_inst|global_csr instance are the traffic generator status signals.

2. Save and enable the Signal Tap file in the project, and recompile the design.
3. Configure the device with your .sof file and observe the signals status through signal tap file.
4. Signal values of status_done=1 and status_error=0 indicate that the traffic test completed with no traffic errors.

Figure 85.



Note: For an infinite traffic pattern program, you will not see `status_done=1`, because the traffic continues running. Set the trigger conditions at the rising edge of the `status_error` signal and let the traffic run. Any failure in traffic is detected if the `status_error` signal goes high.

11.8.2. Exporting the Status Interface to the Top-Level Design

Do not perform this procedure if you plan to observe signal activity through the Signal Tap Logic Analyzer only.

Using the Agilex 7 FPGA EMIF IP design example, perform the following steps to export the traffic generator traffic signals:

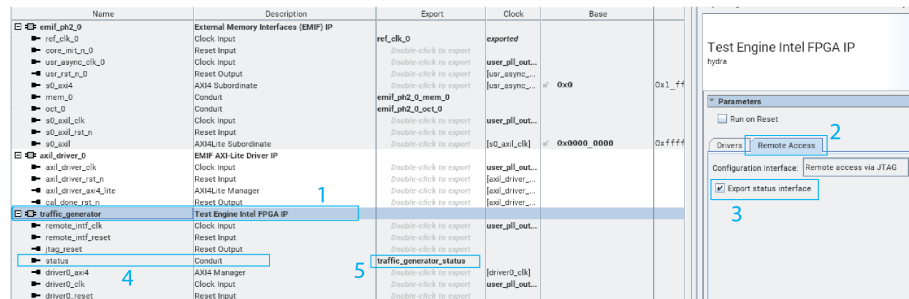
1. Navigate to the **IP Components** tab in the **Project Navigator**, and click `ed_synth.qsys` to open the design in the Platform Designer.

Figure 86.

Entity	IP Component	Version	Supported Device Families	IP File
ed_synth	<Platform Designer>		Agilex 7	ed_synth.qsys
ed_synth_axil_driver_0	<Platform Designer>		Agilex 7	ip/ed_synth/ed_synth_axil_driver_0.ip
ed_synth_emif_ph2_inst	External Memory Interfaces (EMIF) IP	6.2.0	Agilex 7	ip/ed_synth/ed_synth_emif_ph2_inst.ip
ed_synth_perf_monch0	Performance Monitor Intel FPGA IP	1.1.0	Agilex 7	ip/ed_synth/ed_synth_perf_monch0.ip
ed_synth_reset_handler	<Platform Designer>		Agilex 7	ip/ed_synth/ed_synth_reset_handler.ip
ed_synth_rrip	Reset Release Intel FPGA IP	19.4.7	Agilex 7	ip/ed_synth/ed_synth_rrip.ip
ed_synth_traffic_generator	Test Engine Intel FPGA IP	1.1.0	Agilex 7	ip/ed_synth/ed_synth_traffic_generator.ip
ed_synth_user_pll	IOPLL Intel FPGA IP	19.3.1	Agilex 7	ip/ed_synth/ed_synth_user_pll.ip

2. Click on **Test Engine Intel FPGA IP** and go to the **Remote Access** tab.
3. Check the **Export status interface** box.
4. Note the `status` port available in the IP instance. Export the status signal to the top-level design.

Figure 87.



5. Save the design.
6. Click **Sync System Infos** and **Generate HDL**.

Figure 88.



7. Compile the design. You can observe the `status_status_done` and `status_status_error` signals exported at the top level of the `traffic_generator` instance.

Figure 89.



11.9. Hardware Debugging Guidelines

Before debugging your design, confirm that it follows the recommended design flow. Refer to the [Intel Agilex 7 M-Series EMIF IP Design Flow](#) section in chapter 1 of this user guide.

Always keep a record of tests, to avoid repeating the same tests later. To start debugging the design, perform the following initial steps.

11.9.1. Create a Simplified Design that Demonstrates the Same Issue

To help debugging, create a simple design that replicates the problem.

A simple design should compile quickly and be easy to understand. The EMIF IP generates an example top-level file that is ideal for debugging. The example top-level file uses all the same parameters, pin-outs, and so on.

11.9.2. Measure Power Distribution Network

Measure voltages of the various power supplies on their hardware development platform over a suitable time base and with a suitable trigger.

Ensure that you use an appropriate probe and grounding scheme. In addition, take the measurements directly on the pins or vias of the devices in question, and with the hardware operational.

Confirm that reference voltages (V_{REF_CA}) and termination voltages are active and within specification.

11.9.3. Measure Signal Integrity and Setup and Hold Margin

Measure the signals on the PCB. When measuring any signal, consider the edge rate of the signal, not just its frequency. Modern FPGA devices have very fast edge rates, therefore you must use a suitable oscilloscope, probe, and grounding scheme when you measure the signals.

You can take measurements to capture the setup and hold time of key signal classes with respect to their clock or strobe. Ensure that the measured setup and hold margin is at least better than that reported in the Quartus Prime software. A worse margin indicates a timing discrepancy somewhere in the project; however, this issue may not be the cause of your problem.

11.9.4. Vary Voltage

Vary the voltage of your system, if you suspect a marginality issue.

Increasing the voltage usually causes devices to operate faster and also usually provides increased noise margin.

11.9.5. Operate at a Lower Speed

Test the interface at a lower speed. If the interface works at a lower speed, the interface is correctly pinned out and functional.

If the interface fails at a lower speed, determine if the test is valid. Many high-speed memory components have a minimal operating frequency or require subtly different configurations when operating at a lower speeds.

For example, DDR4 SDRAM typically requires modification to the following parameters if you want to operate the interface at lower speeds:

- t_{MRD}
- t_{WTR}
- CAS latency and CAS write latency

11.9.6. Determine Whether the Issue Exists in Previous Versions of Software

Hardware that works before an update to either the Quartus Prime software or the memory IP indicates that the development platform is not the issue.

However, the previous generation IP may be less susceptible to a PCB issue, masking the issue.

11.9.7. Determine Whether the Issue Exists in the Current Version of Software

Designs are often tested using previous generations of Intel software or IP.

Projects may not be upgraded for various reasons:

- Multiple engineers are on the same project. To ensure compatibility, a common release of Intel software is used by all engineers for the duration of the product development. The design may be several releases behind the current Quartus Prime software version.
- Many companies delay before adopting a new release of software so that they can first monitor Internet forums to get a feel for how successful other users say the software is.
- Many companies never use the latest version of any software, preferring to wait until the first service pack is released that fixes the primary issues.
- Some users may only have a license for the older version of the software and can only use that version until their company makes the financial decision to upgrade.
- The local interface specification from Intel FPGA IP to the customer's logic sometimes changes from software release to software release. If you have already spent resources designing interface logic, you may be reluctant to repeat this exercise. If a block of code is already signed off, you may be reluctant to modify it to upgrade to newer IP from Intel.

In all of the above scenarios, you must determine if the issue still exists in the latest version of the Intel software. Bug fixes and enhancements are added to the Intel FPGA IP every release. Depending on the nature of the bug or enhancement, it may not always be clearly documented in the release notes.

Finally, if the latest version of the software resolves the issue, it may be easier to debug the version of software that you are using.

11.9.8. Try A Different PCB

If you are using the same Intel FPGA IP on several different hardware platforms, determine whether the problem occurs on all platforms or just on one.

Multiple instances of the same PCB, or multiple instances of the same interface, on physically different hardware platforms may exhibit different behavior. You can determine if the configuration is fundamentally not working, or if some form of marginality is involved in the issue.

Issues are often reported on the alpha build of a development platform. These are produced in very limited numbers and often have received limited bare-board testing, or functional testing. These early boards are often more unreliable than production quality PCBs.

Additionally, if the IP is from a previous project to help save development resources, determine whether the specific IP configuration works on a previous platform.

11.9.9. Try Other Configurations

Designs are often quite large, using multiple blocks of IP in many different combinations. Determine whether any other configurations work correctly on the development platform.

The full project may have multiple external memory controllers in the same device, or may have configurations where only half the memory width or frequency is required. Find out what does and does not work to help the debugging of the issue.

11.9.10. Debugging Checklist

The following checklist is a good starting point when debugging an external memory interface.

Table 251. Checklist

Check	Item
☐	Try a different fit.
☐	Check IP parameters at the operating frequency (t_{MRD} , t_{WTR} for example).
☐	Ensure you have constrained your design with proper timing deration and have closed timing.
☐	Simulate the design. If it fails in simulation, it will fail in hardware.
☐	Analyze timing.
☐	Place and assign R_{ZQ} (OCT).
☐	Measure the power distribution network (PDN).
☐	Measure signal integrity.
☐	Measure setup and hold timing.
☐	Measure FPGA voltages.
☐	Vary voltages.
☐	Heat and cool the PCB.
continued...	

Check	Item
☐	Operate at a lower or higher frequency.
☐	Check board timing and trace Information.
☐	Check LVDS and clock sources, I/O voltages and termination.
☐	Check PLL clock source, specification, and jitter.
☐	Retarget to a smaller interface width or a single bank.

11.10. Categorizing Hardware Issues

The following topics divide issues into categories. By determining the category (or categories) in which an issue belongs, you may be able to better focus on the cause of the issue.

Hardware issues fall into three categories:

- Signal integrity issues
- Hardware and calibration issues
- Intermittent issues

11.10.1. Signal Integrity Issues

Many design issues, including some at the protocol layer, can be traced back to signal integrity problems. You should check circuit board construction, power systems, command, and data signaling to determine if they meet specifications.

If infrequent, random errors exist in the memory subsystem, product reliability suffers. Check the bare circuit board or PCB design file. Circuit board errors can cause poor signal integrity, signal loss, signal timing skew, and trace impedance mismatches. Differential traces with unbalanced lengths or signals that are routed too closely together can cause crosstalk.

11.10.1.1. Characteristics of Signal Integrity Issues

Signal integrity problems often appear when the performance of the hardware design is marginal.

The design may not always initialize and calibrate correctly, or may exhibit occasional bit errors in user mode. Severe signal integrity issues can result in total failure of an interface at certain data rates, and sporadic component failure because of electrical stress. PCB component variance and signal integrity issues often show up as failures on one PCB, but not on another identical board. Timing issues can have a similar characteristic. Multiple calibration windows or significant differences in the calibration results from one calibration to another can also indicate signal integrity issues.

11.10.1.2. Evaluating Signal Integrity Issues

Signal integrity problems can only really be evaluated in two ways:

- direct measurement using suitable test equipment like an oscilloscope and probe
- simulation using a tool like HyperLynx or Allegro PCB SI

Compare signals to the respective electrical specification. You should look for overshoot and undershoot, non-monotonicity, eye height and width, and crosstalk.

11.10.1.2.1. Skew

Ensure that all clocked signals, commands, addresses, and control signals arrive at the memory inputs at the same time.

Trace length variations cause data valid window variations between the signals, reducing margin. For example, DDR4-3200 at 1600 MHz has a data valid window that is smaller than 313 ps. Trace length skew or crosstalk can reduce this data valid window further, making it difficult to design a reliably operating memory interface. Ensure that the skew figure previously entered into the Intel FPGA IP matches that actually achieved on the PCB, otherwise Quartus Prime timing analysis of the interface is accurate.

11.10.1.2.2. Crosstalk

Crosstalk is best evaluated early in the memory design phase.

Check the clock-to-data strobes, because they are bidirectional. Measure the crosstalk at both ends of the line. Check the data strobes to clock, because the clocks are unidirectional, these only need checking at the memory end of the line.

11.10.1.2.3. Power System

Some memory interfaces draw current in spikes from their power delivery system as SDRAMs are based on capacitive memory cells.

Rows are read and refreshed one at a time, which causes dynamic currents that can stress any power distribution network (PDN). The various power rails should be checked either at or as close as possible to the SDRAM power pins. Ideally, you should use a real-time oscilloscope set to fast glitch triggering to check the power rails.

11.10.1.2.4. Clock Signals

The clock signal quality is important for any external memory system.

Measurements include frequency, digital core design (DCD), high width, low width, amplitude, jitter, rise, and fall times.

11.10.1.2.5. Address and Command Signals

Confirm that address and command signals are reaching the memory devices correctly.

For example, if you are targeting DDR4, you can probe the ALERT_N pin after the memory interface has been successfully calibrated, to determine if any memory component has encountered an address and command parity error.

11.10.1.2.6. Read Data Valid Window and Eye Diagram

The memory generates the read signals. Take measurements at the FPGA end of the line.

To ease read diagram capture, modify the example driver to mask writes or modify the PHY to include a signal that you can trigger on when performing reads.

11.10.1.2.7. Write Data Valid Window and Eye Diagram

The FPGA generates the write signals. Take measurements at the memory device end of the line.

To ease write diagram capture, modify the example driver to mask reads or modify the PHY export a signal that is asserted when performing writes.

For the FPGA, ensure that you perform the following:

- Connect the R_{ZQ} pin to the correct resistors and pull-down to ground in the schematic or PCB.
- Contain the R_{ZQ} pins within a bank of the device that is operating at the same VCCIO voltage as the interface that is terminated.
- Review the Fitter Pin-Out file for R_{ZQ} pins to ensure that they are on the correct pins, and that only the correct number of calibration blocks exists in your design.
- Check in the fitter report that the input, output, and bidirectional signals with calibrated OCT all have the termination control block applicable to the associated R_{ZQ} pins.

For the memory components, ensure that you perform the following:

- Connect the required resistor to the correct pin on each and every component, and ensure that it is pulled to the correct voltage.
- Place the required resistor close to the memory component.
- Correctly configure the IP to enable the desired termination at initialization time.
- Check that the speed grade of memory component supports the selected ODT setting.
- Check that the second source part that may have been fitted to the PCB, supports the same ODT settings as the original.

11.10.1.3. EMIF Eye Mask Margin Guidelines

You can run DRAM IBIS-based simulations for DDR4/LPDDR4 and DDR5/LPDDR5 interfaces and verify that the transmitter and receiver meet the required eye mask margins. For DDR5 designs that include receiver equalization, you can use IBIS-AMI models as a supplementary analysis, where supported.

11.10.1.3.1. Simulating FPGA Device Transmission to DDR4/LPDDR4 or DDR5/LPDDR5 Receiver

1. • For DDR4/LPDDR4 interface, configure the transmitter jitter conditions such that the resulting Total Jitter (Tj) meets the values listed in the following table:

Table 252. DDR4/LPDDR4 Transmitter (Tx) Jitter Requirements

Parameter	Tj (UI _{pk-pk})
DDR4 Tx	0.20
LPDDR4 Tx	0.23

- For DDR5 or LPDDR5 interface, configure the transmitter jitter conditions to meet the deterministic jitter (Dj) and random jitter (Rj) values listed in the following table:

Table 253. DDR5/LPDDR5 Transmitter (Tx) Jitter Requirements

Parameter	Dj (UI _{pk-pk})	Rj (UI _{pk-pk})
DDR5 Tx	0.19	0.13
LPDDR5 Tx	0.19	0.13

2. Verify that the DRAM receiver meets the eye mask margin requirements.

11.10.1.3.2. Simulating DDR4/LPDDR4 or DDR5/LPDDR5 Interface Transmission to FPGA Device

1. 1. Configure transmitter jitter conditions based on the DRAM vendor timing and signal-integrity documentation.
2. • For DDR4/LPDDR4 interface, verify that the FPGA device receiver meets the eye mask margin requirements listed in the following table:

Table 254. DDR4/LPDDR4 Receiver (Rx) Eye Mask Requirements

Parameter	Eye Mask Window (UI _{pk-pk})	Eye Mask Voltage (mV _{pk-pk})
DDR4 Rx	0.15	95
LPDDR4 Rx	0.17	95

- For DDR5/LPDDR5 interface, verify that the FPGA device receiver meets the eye mask margin requirements listed in the following table:

Table 255. DDR5/LPDDR5 Receiver (Rx) Eye Mask Requirements

Parameter	Eye Mask Window (UI _{pk-pk})	Eye Mask Voltage (mV _{pk-pk})
DDR5 Rx	0.20	75
LPDDR5 Rx	0.20	75

11.10.2. Hardware and Calibration Issues

Hardware and calibration issues have the following definitions:

- Calibration issues result in calibration failure.
- Hardware issues result in read and write failures.

Note: Ensure that functional, timing, and signal integrity issues are not the direct cause of your hardware issue, as functional, timing or signal integrity issues are usually the cause of any hardware issue.

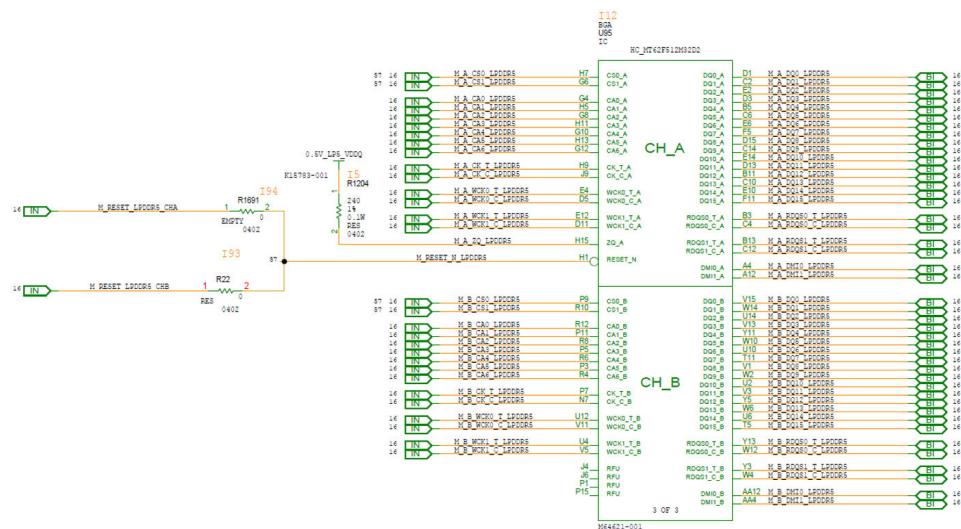
11.10.2.1. Verifying High-Level Configuration

Memory Device

Verify that you select the correct memory *Number of Channels*, *Data DQ Width*, *Die Density*, and *CS Width* for the memory configuration used. Inaccurate information can result in calibration failure or traffic failure.

The example below shows memory parameterization based on the Agilex™ 7 FPGA M-Series Development Kit targeting an MT62F512M32D2 LPDDR5 EMIF configuration.

Figure 90. MT62F512M32D2 LPDDR5 EMIF Configuration



- Number of Channels: 2 (Dedicated A/C pins for each channel.)
- Data DQ Width: 16 (DQ pins per memory channel.)
- Die Density: 8 Gbits (Select capacity of each memory die <in Gbits>, not total memory density per package.)
- CS width: 1 (Above topology is single-rank per channel. MT62F512M32D2 supports single-rank per die per channel based on the die organization in memory component datasheet.)

Figure 91. Example EMIF IP High Level Configuration for Configuration shown Above

External Memory Interfaces (EMIF) IP - LPDDR5
emif_io96b_lpddr5

High-level Configuration		Advanced: Memory Timing	Advanced: Analog Overrides	Example Design
Memory Device				
Number of Channels:	2			
Data DQ Width:	16			
Die Density:	8Gbits			
CS Width:	1			

Another example showing a configuration based on Agilex™ 7 FPGA M-Series Development Kit targeting DDR5 RDIMM is shown below.

Figure 92. DDR5 RDIMM Slot Using MTC10F1084S1RC56BG1 Module

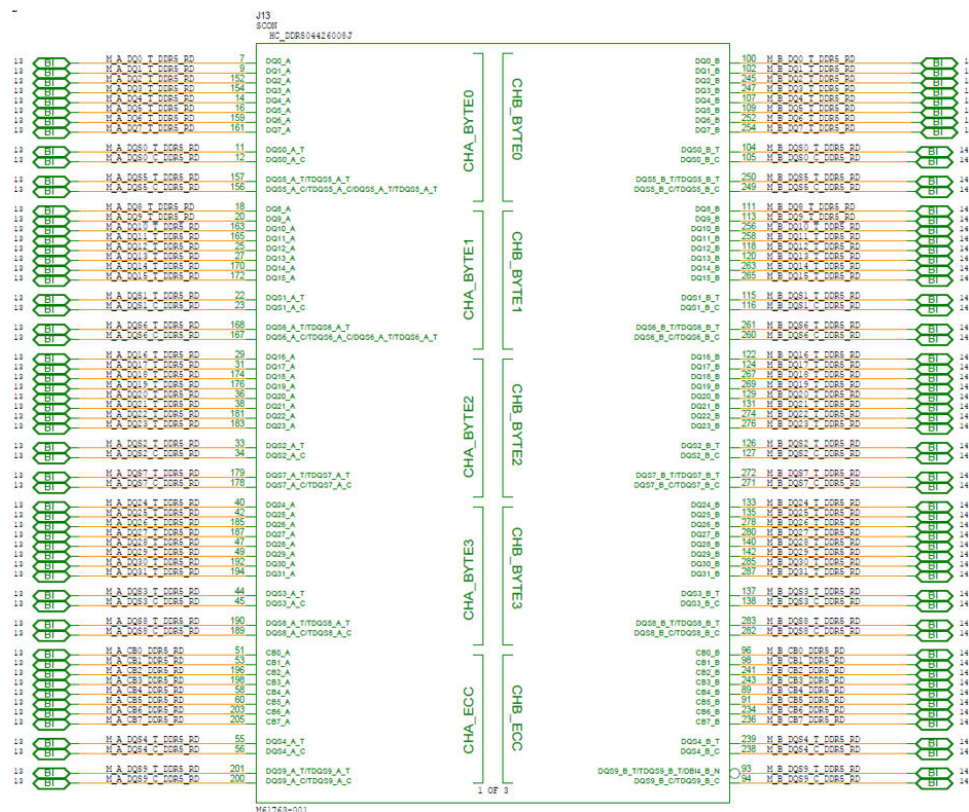
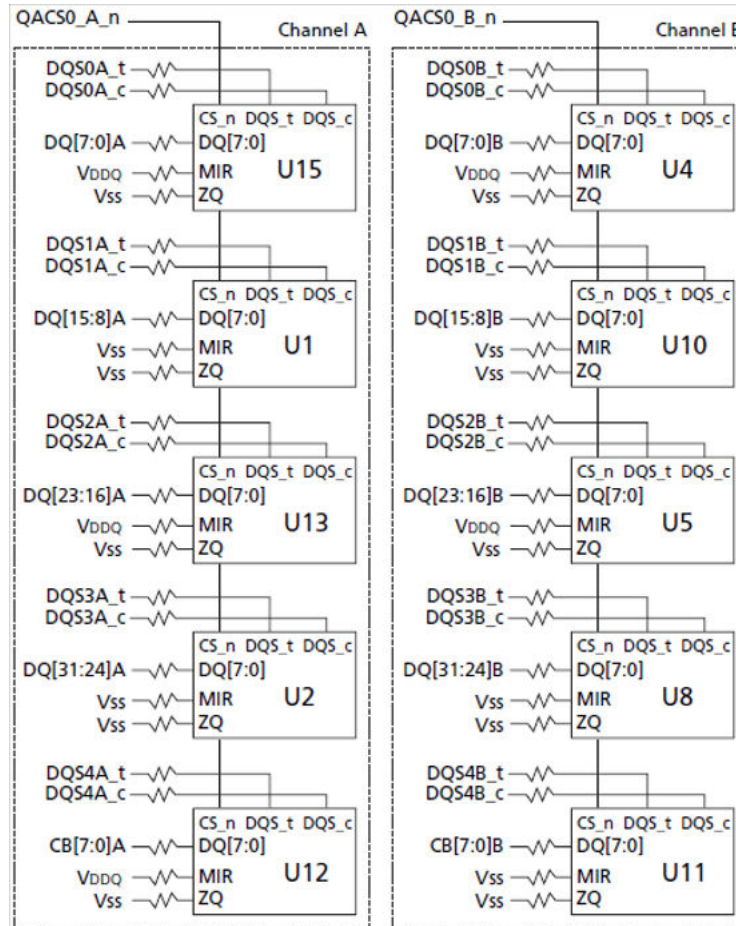


Figure 93. MTC10F1084S1RC56BG1 DDR5 RDIMM



- DIMM Type: RDIMM.
- Data DQ Width: 32 (DQ pins per memory channel, used for data.)
- ECC DQ Width: 8 (Additional DQ pins per channel, used for ECC.)
- Die DQ Width: 8 (Numbers of DQ pins in each die. For dual-die package, select width of each die, not width of full package.)
- Die Density: 16Gbits (Select capacity of each memory die <in Gbits> per channel per die. For dual-die package, this is the density of each die, not the density of the full package.)
- CS width: 1 (Above topology is Single Rank per channel.)

Figure 94. Example EMIF IP High Level Configuration for configuration shown Above

High-level Configuration | Advanced: Memory Timing | Advanced: Analog Overrides | Internal: Diagnostic Options | Example Design

Memory Device

DIMM Type: RDIMM

Data DQ Width: 32

ECC DQ Width: 8

☐ Use 3DS Device

Die DQ Width: 8

Die Density: 16Gbits

CS Width: 1

Memory Speedbin: 5600B (CL46)

Pin Swizzle map

The Quartus Prime software uses fixed pin placement for each supported memory standard, as described in the [Product Architecture](#) chapter. Do not change pin placement using a .qsf assignment or the pin planner if you need to swap the DQ pins within a DQS group or swap the DQS group within a channel or interface. You may encounter fitter errors if you enter incorrect pin locations for the EMIF interface. The example below shows fitter error for LPDDR5 protocol when an incorrect pin location is used.

Error (175001): The Fitter cannot place 1 BYTE, which is within Generic Component ed_synth_emif_io96b_lpddr5_0.

The **Pin Swizzle Map** allows you to inform the EMIF IP about the ordering of the physical connections of each DQ pin on the memory component to the FPGA. Incorrect information entered in the pin swizzle map may cause calibration errors.

Refer to the [External Memory Interfaces Agilex 7 M-Series FPGA IP Design Example User Guide](#) for details on configuring DQ pin swizzling.

Figure 95. Example of Pin Swizzle Map Entry

PHY

☐ Auto-set PLL Reference Clock Frequency

☐ Enable Advanced List of PLL Reference Clock Frequencies

Reference Clock Frequency: 100.0

☐ Auto-set Mainband Access Mode

Mainband Access Mode: Fabric Direct - User Clock Asynchronous to PHY

Sideband Access Mode: Fabric Direct

Pin Swizzle Map: BYTE_SWIZZLE_CH0=1,0,X,X,X,X,X,X;PIN_SWIZZLE_CH0_DQS1=9,11,8,10

Pin Swizzle Map for Component Interfaces

For component type interfaces, you must match the net name in the schematic to the index of default pin placements, to determine the pin swizzling specification.

Consider the example in the table below, which represents DQS group 1 for channel 1 on the LPDDR5 component interface on the Agilex™ 7 FPGA M-Series Development Kit with MT62F512M32D2 memory.

Table 256. Example of Pin Swizzle Targeting Component Type Memory Interface

Lane	Pin	Pin Index	Default Pin Placement	Net Name in Schematic	Swizzling index
BL6	K57	83	MEM_1_MEM_DQ [7]	MEM_1_MEM_DQ [14]	14
	L56	82	MEM_1_MEM_DQ [6]	MEM_1_MEM_DQ [12]	12
	H57	81	MEM_1_MEM_DQ [5]	MEM_1_MEM_DQ [13]	13
	G56	80	MEM_1_MEM_DQ [4]	MEM_1_MEM_DQ [11]	11
	K55	79			
	L54	78	MEM_1_MEM_DM_N [0]	MEM_1_MEM_DM_N [1]	
	H55	77	MEM_1_MEM_RD QS_C [0]	MEM_1_MEM_RD QS_C [1]	
	G54	76	MEM_1_MEM_RD QS_T [0]	MEM_1_MEM_RD QS_T [1]	
	K53	75	MEM_1_MEM_DQ [3]	MEM_1_MEM_DQ [9]	9
	L52	74	MEM_1_MEM_DQ [2]	MEM_1_MEM_DQ [10]	10
	H53	73	MEM_1_MEM_DQ [1]	MEM_1_MEM_DQ [8]	8
	G52	72	MEM_1_MEM_DQ [0]	MEM_1_MEM_DQ [15]	15

For this specific group, the correct pin swizzling specification is:

```
PIN_SWIZZLE_CH1_DQS1 = 15,8,10,9,11,13,12,14;
```

The sample below shows pin swizzle map parameters based on Agilex 7 M-Series Development Kit targeting LPDDR5 EMIF. All parameters must be added in the *Pin Swizzle Map* entry in a single line separated by semi-colons.

```
BYTE_SWIZZLE_CH0=1,0,X,X,X,X,X,X;
PIN_SWIZZLE_CH0_DQS1=9,11,8,10,13,15,14,12;
PIN_SWIZZLE_CH0_DQS0=6,7,4,5,0,3,2,1;
BYTE_SWIZZLE_CH1=X,X,X,X,X,X,1,0;
PIN_SWIZZLE_CH1_DQS1=15,8,10,9,11,13,12,14;
PIN_SWIZZLE_CH1_DQS0=7,6,4,5,3,2,1,0;
```

Pin Swizzle Map for DIMM interface

For DIMM topologies, you must incorporate the component-to-module DQ map (refer to the memory vendor's datasheet), to determine the pin swizzling specification for the DIMM interface.

Consider the example in the table below, which represents DQS group 0 for channel 0 on the DDR5 RDIMM interface on the Agilex™ 7 FPGA M-Series Development Kit with MTC10F1084S1RC56BG1 DDR5 RDIMM memory.

Table 257. Example of Component-to-Module DQ Map to Pin Swizzle from FPGA Schematics Targeting DIMM-type Memory Interface

Information From Dev Kit Schematic				Information from Memory Vendor Data Sheet (Component-to-Module DQ Map)		
Lane	Pin	Pin Index	Net Name in Schematic	Component Ref Number	Module DQ	Component DQ
BL6	FV23	83	M_A_DQ5_T_D DR5_RD	U4	5A	7
	FU24	82	M_A_DQ6_T_D DR5_RD		6A	6
	FY23	81	M_A_DQ7_T_D DR5_RD		7A	5
	GA24	80	M_A_DQ4_T_D DR5_RD		4A	4
	FU26	79	M_A_DQS5_C_ DDR5_RD			
	FV25	78	M_A_DQS5_T_ DDR5_RD			
	GA26	77	M_A_DQS0_C_ DDR5_RD			
	FY25	76	M_A_DQS0_T_ DDR5_RD			
	FV27	75	M_A_DQ1_T_D DR5_RD		1A	1
	FU28	74	M_A_DQ2_T_D DR5_RD		2A	0
	FY27	73	M_A_DQ3_T_D DR5_RD		3A	3
	GA28	72	M_A_DQ0_T_D DR5_RD		0A	2

For this specific group, the correct pin swizzling specification is:

```
PIN_SWIZZLE_CH0_DQS0 = 2,3,0,1,4,5,6,7;
```

(Note that the `PIN_SWIZZLE_CH0_DQS0 = 0,3,2,1,4,7,6,5;` specification would be incorrect. Pin swizzling specifications can be different when using different DIMMs.)

The example below shows the full Pin Swizzle Map parameters based on an Agilex 7 M-Series development kit targeting an MTC10F1084S1RC56BG1 DDR5 RDIMM memory interface. All parameters must be added in the *Pin Swizzle Map* entry in a

single line separated by semi-colons. Ensure that the DQ pin swizzles are based on the component DQ index and not the Module/DIMM DQ index; inaccurate swizzling will result in calibration failure.

```
BYTE_SWIZZLE_CH0=X,ECC,2,1,X,X,0,3;  
PIN_SWIZZLE_CH0_DQS0=2,3,0,1,4,5,6,7;  
PIN_SWIZZLE_CH0_DQS1=8,11,10,9,13,15,12,14;  
PIN_SWIZZLE_CH0_DQS2=18,17,16,19,20,23,21,22;  
PIN_SWIZZLE_CH0_DQS3=27,24,25,26,29,30,28,31;  
PIN_SWIZZLE_CH0_ECC=2,1,0,3,4,7,6,5;  
BYTE_SWIZZLE_CH1=0,1,X,X,3,2,ECC,X;  
PIN_SWIZZLE_CH1_DQS0=0,1,3,2,4,5,7,6;  
PIN_SWIZZLE_CH1_DQS1=8,10,9,11,12,13,15,14;  
PIN_SWIZZLE_CH1_DQS2=19,16,18,17,21,22,23,20;  
PIN_SWIZZLE_CH1_DQS3=25,26,24,27,31,28,30,29;  
PIN_SWIZZLE_CH1_ECC=6,5,7,4,0,3,2,1;
```

11.10.2.2. Verifying Memory Timing Parameters

Review and update the memory timing parameters, CAS, and Write CAS latency based on the speed bin of the targeted memory component and the operating frequency of your interface.

Incorrect memory timing parameters, CAS, or Write CAS latency can cause data corruption in the memory component.

11.10.2.3. Verifying the Correct Memory Component or DIMM is Installed

Verify that the correct memory component or DIMM is installed on the circuit board.

If an incorrect memory part is used, the memory part may not support the memory timing parameters, CAS, or Write CAS latency used in parameterizing the IP; this situation can result in data corruption. If a memory device with smaller memory capacity is installed incorrectly, data written to the higher address space overwrites the data at the lower address space.

11.11. Debugging Intermittent Issues

Intermittent failures can be challenging to isolate. In such cases, it is advisable to return to general debugging to better understand the symptoms. Use the [Debugging Checklist](#) as a starting point to narrow down the issue.

12. Agilex 7 FPGA EMIF IP - Mailbox Support

This section discusses mailbox support.

12.1. Agilex 7 Mailbox Structure and Register Definitions

The mailbox is a software structure that the calibration subsystem manager (SSM) polls periodically.

You can access the mailbox using the AXI-lite interface or through JTAG Avalon Master Bridge (JAMB) when you enable the debug toolkit option in the EMIF IP.

All accesses to the mailbox should align to 32-bit boundaries, with no byte masking support. The following tables show the mailbox structure and the calibration status register definition.

The base address for the mailbox is 0x0500_0000.

Table 258. Mailbox Structure

Register Name	Byte Offset (Decimal)	Width (bits)	Access	Description
CMD_PARAM_6	1056	32	RW	[Input] This register specifies the seventh parameter (if applicable) for the requested command.
CMD_PARAM_5	1060	32	RW	[Input] This register specifies the sixth parameter (if applicable) for the requested command.
CMD_PARAM_4	1064	32	RW	[Input] This register specifies the fifth parameter (if applicable) for the requested command.
CMD_PARAM_3	1068	32	RW	[Input] This register specifies the fourth parameter (if applicable) for the requested command.
CMD_PARAM_2	1072	32	RW	[Input] This register specifies the third parameter (if applicable) for the requested command.
CMD_PARAM_1	1076	32	RW	[Input] This register specifies the second parameter (if applicable) for the requested command.
CMD_PARAM_0	1080	32	RW	[Input] This register specifies the first parameter (if applicable) for the requested command.
CMD_REQ	1084	32	RW	[Input] This register specifies the command to be performed and the target IP type and identifier.
continued...				

Register Name	Byte Offset (Decimal)	Width (bits)	Access	Description
CMD_RESPONSE_DATA_2	1104	32	RO	[Output] For commands that return up to 3.5x 32-bit values, CMD_RESPONSE_DATA 0/1/2, and/or CMD_RESPONSE_DATA_SHORT fields contain the requested response data.
CMD_RESPONSE_DATA_1	1108	32	RO	[Output] This register can contain two types of values depending on the requested operation. For commands that return up to 3.5x 32-bit values, CMD_RESPONSE_DATA 0/1/2, and/or CMD_RESPONSE_DATA_SHORT fields contain the requested response data. For commands that return more data, this register specifies a pointer to a data buffer within the 4K User. It is a byte offset relative to the start of the 4Kbyte RAM. Do not assume that this offset value remains static as the value of this pointer offset may change depending on the requested operation.
CMD_RESPONSE_DATA_0	1112	32	RO	[Output] This register can contain two types of values depending on the requested operation. For commands that return up to 3.5x 32-bit values, CMD_RESPONSE_DATA 0/1/2, and/or CMD_RESPONSE_DATA_SHORT fields contain the requested response data. For commands that return more data, this value contains the size of the returned data structure in bytes. The location of the data buffer is specified in CMD_RESPONSE_DATA_1.
CMD_RESPONSE_STATUS	1116	32	RW	[Output] Command Interface status Captures the current state of the Mailbox's Command Interface (that is, Is the response data ready?). CMD_RESPONSE_DATA_SHORT field in this register can be used for 16-bit response data.

Table 259. Calibration Status Register

Bit	Name	Description	Access	Reset
[31:3]	Reserved			
[2]	STATUS_CAL_BUSY	Indicates calibration busy status of any external memory interfaces in the IO96: '1' - One or more EMIF instances are busy with calibration. '0' - No EMIF instances are busy with calibration.	RO	0x0
[1]	STATUS_CAL_FAIL	Indicates calibration failure status of any external memory interfaces in the IO96: '1' - One or more EMIF instances have failed to calibrate successfully. '0' - No calibration failures have been reported for any of the EMIFs.	RO	0x0
[0]	STATUS_CAL_SUCCESS	Indicates final calibration status of all interfaces in the IO96: '1' All EMIF instances within the IO96 have calibrated successfully. '0' One or more EMIF instances in the IO96 have either failed to calibrate or have not completed calibration yet.	RO	0x0

Table 260. CMD_RESPONSE_STATUS Register

Bit	Name	Description	Access	Reset
[31:16]	CMD_RESPONSE_DATA_SHORT	For commands that return up to 16-bit value, this field contains the requested response data.	RO	0x0
[15:8]	Reserved			
[7:5]	STATUS_CMD_RESPONSE_ERROR	Indicates which errors occurred while processing the command, which are specific to the command being executed. '000' - No errors. Other error codes will be outlined alongside the commands to which they are associated.	RO	0x0
[4:1]	STATUS_GENERAL_ERROR	Reports general errors that occurred while a command was being executed: '0000' - no error. '0001' - Indicates that an illegal command was received. '0010' - indicates that the interface the command is targeting doesn't exist. '0011' - indicates that the conditions for executing the command are not met. '0100' - indicates an unexpected command for 2-stage commands like BIST_MEM_INIT_START . '0110' - indicates that the specified interface is busy.	RO	0x0
[0]	STATUS_COMMAND_RESPONSE_READY	Tells whether the command response is ready. '1' - Indicates that the command response is ready.	RW	0x0

12.1.1. Mailbox Supported Commands

Read-Only Registers

The following read-only registers provide direct access to retrieve information. Note that these registers are not write-protected; you should not attempt to write to any of these registers. Additionally, you should avoid reading from registers that are not used in the target design, as doing so might result in unpredictable or invalid data being returned.

Table 261. Read-Only Registers

Register Name	Byte Offset (Hexadecimal)	Description
MAILBOX_HEADER	0x000	[Output] Mailbox Header This register specifies information about the mailbox protocol. [Fields] [2:0] MB_SPEC_VER: This field specifies the version of mailbox specification being used by the IOSSM firmware. Current version number is 1.
MEM_INTF_INFO_0	0x200	[Output] Memory Interface Information This register specifies the memory interface IP type and instance ID for the interface 0/1 in the IO96B. It will only return the primary interface of each channel. [Fields]
MEM_INTF_INFO_1	0x280	

continued...

Register Name	Byte Offset (Hexadecimal)	Description
		[31:29] IP_TYPE: Indicates the type of IP: • 0x0 – Not used. • 0x1 – Primary MC of Primary IO96B. • 0x2 – Secondary MC of Primary IO96B. • 0x3 – Primary MC of Secondary IO96B. • 0x4 – Secondary MC of Secondary IO96B. [28:24] INSTANCE_ID: IP identifier.
MEM_TECHNOLOGY_INTF0	0x210	[Output] Memory Technology This register provides the memory technology type for the memory interface 0/1. [Fields] [2:0] TECH: Reports the memory technology type: • 0x0 = DDR4 • 0x1 = DDR5 • 0x2 = DDR5_RDIMM • 0x3 = LPDDR4 • 0x4 = LPDDR5
MEM_TECHNOLOGY_INTF1	0x290	
MEMCLK_FREQ_INTF0	0x220	[Output] Memory Clock Frequency - This register reports the memory clock frequency in kilohertz (KHz) for the memory interface 0/1. For protocols with multiple FSPs, it reports the current memory clock frequency in kilohertz (KHz). [Fields] [31:0] FREQ_KHZ_FSP_CUR: Current memory clock frequency in KHz.
MEMCLK_FREQ_INTF1	0x2A0	
MEMCLK_FREQ_FSP0_INTF0	0x224	[Output] Memory Clock Frequency - Frequency Set Point 0 This register reports the memory clock frequency in kilohertz (KHz) for Frequency Set Point 0 of the memory interface 0/1. Only valid for LPDDR5 protocol. [Fields] [31:0] FREQ_KHZ_FSP0: Memory clock frequency in KHz for Frequency Set Point 0.
MEMCLK_FREQ_FSP0_INTF1	0x2A4	
MEMCLK_FREQ_FSP1_INTF0	0x228	[Output] Memory Clock Frequency - Frequency Set Point 1 This register reports the memory clock frequency in kilohertz (KHz) for Frequency Set Point 1 of the memory interface 0/1. Only valid for LPDDR5 protocol. [Fields] [31:0] FREQ_KHZ_FSP1: Memory clock frequency in KHz for Frequency Set Point 1.
MEMCLK_FREQ_FSP1_INTF1	0x2A8	
MEMCLK_FREQ_FSP2_INTF0	0x22C	MEMCLK_FREQ_INTF0_FSP2 [Output] Memory Clock Frequency - Frequency Set Point 2 This register reports the memory clock frequency in kilohertz (KHz) for Frequency Set Point 2 of the memory interface 0/1. Only valid for LPDDR5 protocol. [Fields] [31:0] FREQ_KHZ_FSP2: Memory clock frequency in KHz for Frequency Set Point 2.
MEMCLK_FREQ_FSP2_INTF1	0x2AC	
MEM_WIDTH_INFO_INTF0	0x230	[Output] Memory Width Information This register provides the memory width information for the memory interface 0/1. [Fields] [23:16] C_WIDTH: Channel width. [15:8] CS_WIDTH: Chip select width. [7:0] DQ_WIDTH: Data width
MEM_WIDTH_INFO_INTF1	0x2B0	

continued...

Register Name	Byte Offset (Hexadecimal)	Description
MEM_TOTAL_CAPACITY_INTF0	0x234	Output] Total Memory Capacity
MEM_TOTAL_CAPACITY_INTF1	0x2B4	This register reports the total memory capacity per channel in gigabits (Gb) for the memory interface 0/1. [Fields] [7:0] CAPACITY_GBITS: Total memory device capacity per channel in Gb. Total memory capacity is calculated as: $CAPACITY = (DQ_WIDTH / DEVICE_WIDTH) * NUM_RANKS * C_WIDTH * DEVICE_DENSITY$
ECC_ENABLE_INTF0	0x240	[Output] ECC Enable Status
ECC_ENABLE_INTF1	0x2C0	This register provides detailed information regarding the ECC enable status and type for the memory interface 0/1. [Fields] [18:18] RD_LINK_ECC_ENABLED: Indicates the Read Link-ECC enable status. 1'b1 – Read Link-ECC is enabled. 1'b0 – Read Link-ECC is disabled. [17:17] WR_LINK_ECC_ENABLED: Indicates the Write Link-ECC enable status. 1'b1 – Write Link-ECC is enabled. 1'b0 – Write Link-ECC is disabled. [16:16] LINK_ECC_SUPPORTED: Indicates if Link-ECC is supported by the interface. 1'b1 – Link-ECC is supported. 1'b0 – Link-ECC is not supported. [8:8] ECC_TYPE: Specifies the ECC operational mode. 1'b0 – Out-of-Band ECC. 1'b1 – In-line ECC. [1:0] ECC_ENABLE_TYPE: Reports the current ECC error reporting and correcting status. 1'b00 – ECC is disabled. Data will be written and returned without ECC verification. 1'b01 – ECC is enabled without error detection or correction. 1'b10 – ECC is enabled with error detection; errors are reported but not corrected. 1'b11 – ECC is enabled with error detection and correction; single-bit errors are automatically corrected.
ECC_SCRUB_STATUS_INTF0	0x244	[Output] ECC Scrub Status
ECC_SCRUB_STATUS_INTF1	0x2C4	This register indicates the ECC scrub operation status for the memory interface 0/1. [Fields] [1:1] ECC_SCRUB_IN_PROGRESS: Reports whether an ECC scrub operation is currently in progress. 1'b0 – Not actively performing a scrubbing operation. 1'b1 – The Controller is in the process of performing a scrubbing operation. [0:0] ECC_SCRUB_STATUS: The ECC scrub operation initiated using ECC_SCRUB_MODE_0/1_START mailbox command has completed. 1'b0 – Scrubbing operation has not completed. 1'b1 – Scrubbing operation has completed.
LP_MODE_INTF0	0x250	[Output] Low Power Mode Status
LP_MODE_INTF1	0x2D0	This register provides the current low power state and its validity for the memory interface 0/1. [Fields] [5:0] LP_STATE: Current Interface Low Power State.

continued...

Register Name	Byte Offset (Hexadecimal)	Description
		1'b000000 – Idle. 1'b000001 – Active Power Down. 1'b000010 – Active Power Down with Memory Clock Gating. 1'b000011 – Pre-Charge Power Down. 1'b000100 – Pre-Charge Power Down with Memory Clock Gating. 1'b000101 – Self-Refresh Short. 1'b000110 – Self-Refresh Short with Memory Clock Gating. 1'b001000 – Self-Refresh Long. 1'b001001 – Self-Refresh Long with Memory Clock Gating. 1'b001010 – Self-Refresh Long with Memory Clock and Controller Clock Gating. 1'b001011 – Self-Refresh Power Down Short. 1'b001100 – Self-Refresh Power Down Short with Memory Clock Gating. 1'b001101 – Self-Refresh Power Down. 1'b001110 – Self-Refresh Power Down Long with Memory Clock Gating. 1'b001111 – Self-Refresh Power Down Long with Memory and Controller Clock Gating. [6:6] LP_STATE_VALID: Indicates whether the data for the low power state is valid. 1'b1 – Valid. The low power state information is reliable. 1'b0 – Invalid. The interface is currently transitioning into or out of a low power state.
MEM_INIT_STATUS_INTF0	0x260	[Output] BIST Memory Initialization Status This register holds the status of the BIST memory initialization operation for the memory interface specified by the instance ID. The value of this register is only valid if the mailbox command BIST_MEM_INIT_START was issued previously. [Fields] [0:0] MEM_INIT_STATUS: Indicates the status of the BIST memory content initialization operation. 1'b0 – Memory content initialization operation still in progress if previously initiated. 1'b1 – Memory content initialization operation has been completed.
MEM_INIT_STATUS_INTF1	0x2E0	
BIST_STATUS_INTF0	0x264	[Output] BIST Results Status This register contains BIST results for the previously initiated BIST operation for memory interface 0/1 [Fields] [31:16] BIST_FAIL_RESULT_OFFSET: Holds the offset pointer of the BIST failure results. The data at the pointer location will be as below: - OFFSET [0] to OFFSET [8]: [287:0] BIST_EXP_DATA - Holds the expected read data for BIST data check failure. - OFFSET [9] to OFFSET [10]: [37:0] BIST_FAIL_ADDR - Holds the actual failing address for BIST data check failure. - OFFSET [11] to OFFSET [20]: [287:0] BIST_FAIL_DATA - Holds the actual failing data for BIST data check failure. [15:8] BIST_FAIL_RESULT_SIZE: Holds the size of the BIST failure results. The value of this will be 640 bits. [1:1] BIST_RESULT: Holds the result of the BIST operation. For this BIST mode, the test will end at the first failure, or completely check the specified data range if no failures were found. This value is valid when BIST_STATUS indicates that the BIST operation has completed.
BIST_STATUS_INTF1	0x2E4	

continued...

Register Name	Byte Offset (Hexadecimal)	Description
		1'b0 = Data check failed. 1'b1 = Data check passed. [0:0] BIST_STATUS: Indicates the status of the BIST operation. 1'b0 – BIST operation still in progress if previously initiated. 1'b1 – BIST operation has been completed.
ECC_ERR_STATUS	0x300	[Output] ECC Error Status This register contains the status of the ECC error buffer. [Fields] [31:16] ECC_ERR_OVERFLOW: This field indicates that buffer overflow has occurred due to a new ECC interrupt occurring after the ECC error buffer is at full capacity of 16 entries. Each new interrupt type that cannot be enqueued into the buffer are logically-OR'd to the existing contents of the overflow status field. The ECC interrupt types indicated by the different positional bits in the overflow field are as below: [0] – Single-bit error. [1] – Multiple single-bit errors. [2] – Double-bit error. [3] – Multiple double-bit errors. [8] – Single-bit error during ECC scrubbing. [9] – Write link ECC single-bit error (LPDDR5 only). [10] – Write link ECC double-bit error (LPDDR5 only). [11] – Read link ECC single-bit error (LPDDR5 only). [12] – Read link ECC double-bit error (LPDDR5 only). [13] – RMW read link ECC double-bit error (LPDDR5 only). [15:0] ECC_ERR_COUNTER: This field indicates the number of entries in ECC error buffer. The buffer has a maximum capacity of 16 entries after which overflow occurs. The mailbox command ECC_CLEAR_ERR_BUFFER clears the ECC error buffer and resets the ECC_ERR_COUNTER to 0.
ECC_ERR_DATA_START	0x310	[Output] ECC Error Data Start This register is the start of the ECC error data buffer. The buffer has a maximum capacity of 16 ECC errors. Each ECC error data entry has a size of 64 bits arranged in two 32-bit registers (Register 1 and Register 2). The mailbox command ECC_CLEAR_ERR_BUFFER clears the ECC error buffer and resets the ECC_ERR_COUNTER to 0. [Fields] R1[24:22] IP_TYPE: This field specifies the IP type of the interface that produced the ECC interrupt. IP_TYPE of 0 indicates that the buffer entry is not populated with valid data. R1[21:17] INSTANCE_ID: This field specifies the instance ID of the interface that produced the ECC interrupt. R1[16:10] ECC_ERR_SOURCE_ID: This field specifies the source ID associated with the ECC event. For AXI ports, the source ID is comprised of the Port ID (upper bit/s) and the Requestor ID, where the Requestor ID is the axi0_AWID for write commands or the axi0_ARID for read commands. R1[9:6] ECC_ERR_TYPE: This field specifies the ECC interrupt type of the ECC event. 1'b0000 – Single-bit error. 1'b0001 – Multiple single-bit errors. 1'b0010 – Double-bit error. 1'b0011 – Multiple double-bit errors. 1'b1000 – Single-bit error during ECC scrubbing. 1'b1001 – Write link ECC single-bit error (LPDDR5 only). 1'b1010 – Write link ECC double-bit error (LPDDR5 only).

continued...

Register Name	Byte Offset (Hexadecimal)	Description
		1'b1011 – Read link ECC single-bit error (LPDDR5 only) 1'b1100 – Read link ECC double-bit error (LPDDR5 only). 1'b1101 – Read link ECC double-bit error, caused by read-modify-write operation (LPDDR5 only). R1[5:0] ECC_ERR_ADDR_UPPER: This field specifies the upper 6 bits of the address of the read data that caused the ECC event. The Controller will pad this parameter with zeros for any address bits not used by the controller. R2[31:0] ECC_ERR_ADDR_LOWER: This field specifies the lower 32 bits of the address of the read data that caused the ECC event. The Controller will pad this parameter with zeros for any address bits not used by the controller. <i>Note:</i> Refer to ECC Error Handling for information on ECC Error Buffer Byte Offset.
STATUS	0x400	[Output] Status Register “At a Glance” status register. This field is automatically updated by the Calibration IOSSM and no explicit operation is required to trigger an update. 1'b001 – Calibration was successful. 1'b010 – Calibration failed. 1'b100 – Calibration ongoing. 1'b000 – Interface unused.
STATUS_CAL_INTF0	0x404	[Output] Memory Calibration Status This register provides the calibration status for memory interface 0/1. [Fields] [2:0] STATUS: Calibration Status of memory interface 1'b001 – Calibration was successful. 1'b010 – Calibration failed. 1'b100 – Calibration ongoing. 1'b000 – Interface unused.
STATUS_CAL_INTF1	0x408	

Mailbox Supported Commands

The following registers require handshaking procedures with the mailbox for successful retrieval of information. Refer to the instructions described in [Sending a Mailbox Command](#) for more information on handshaking procedures.

Table 262. Supported Commands

CMD_TYPE Enum	CMD_OPCODE Enum	Value
CMD_TRIG_CONTROLLER_OP (value=0x4)	ECC_ENABLE_SET	0x0101
	ECC_INTERRUPT_MASK	0x0105
	ECC_WRITEBACK_ENABLE	0x0106
	ECC_INJECT_ERROR	0x0109
	ECC_CLEAR_ERR_BUFFER	0x0110
	ECC_SCRUB_MODE_0_START	0x0202
	ECC_SCRUB_MODE_1_START	0x0203
	BIST_STANDARD_MODE_START	0x0301
	BIST_MEM_INIT_START	0x0303

continued...

CMD_TYPE Enum	CMD_OPCODE Enum	Value
	BIST_SET_DATA_PATTERN_UPPER	0x0305
	BIST_SET_DATA_PATTERN_LOWER	0x0306
	CHANGE_FSP_LP5	0x0c01
	LP_MODE_ENTER	0x0d01
	LP_MODE_EXIT	0x0d02
	LP_MODE_AUTO	0x0d04
CMD_TRIG_MEM_CAL_OP (value=0x5)	TRIG_MEM_CAL	0x000a
CMD_GET_MEM_CAL_INFO (value=0x3)	GET_MEM_CAL_MGN	0x000c

12.1.2. Mailbox Command Definitions

Table 264. CMD_REQ Definition

Bit	Name	Description	Access	Reset
[31:29]	CMD_TARGET_IP_TYPE	Indicates the type of IP, as follows: 0x0 – Not used. 0x1 – Primary MC of primary IO96B. 0x2 – Secondary MC of primary IO96B. 0x3 – Primary MC of secondary IO96B. 0x4 – Secondary MC of secondary IO96B.	Read-Write	0x0
[28:24]	CMD_TARGET_IP_INSTANCE_ID	IP identifier.	Read-Write	0x00
[23:16]	CMD_TYPE	The type of command that the user wants the firmware to perform.	Read-Write	0x00
[15:0]	CMD_OPCODE	The opcode of the command that the user wants the firmware to perform.	Read-Write	0x00

Table 265. CMD_TYPE Definition

CMD_TYPE	Value	Description
CMD_NOP	0x00	No operation command.
CMD_TRIG_CONTROLLER_OP	0x04	Triggering memory controller-related operations.
CMD_TRIG_MEM_CAL_OP	0x05	Triggering calibration events.
CMD_GET_MEM_CAL_INFO	0x03	Retrieving memory calibration information.

Table 266. CMD_TARGET_IP_TYPE Definition

Multi-channel/ Lockstep Configurations	CMD_TARGET_IP_TYPE			
	1 – Primary MC, Primary IO96B	2 – Secondary MC, Primary IO96B	3 – Primary MC, Secondary IO96B	4 – Secondary MC, Secondary IO96B
LPDDR4/5 2CHx16	CH1	CH2		
LPDDR4/5 4CHx16	CH1	CH2	CH3	CH4
DDR5 2CHx16	CH1	CH2		
DDR5 2CHx32	CH1		CH2	
DDR5 x40 lockstep	CH1	*		
DDR4 x40 lockstep	CH1	*		
DDR4 x64, x72 lockstep	CH1	*	*	*

Note: * These controllers are used but have no (or limited) mailbox features due to limited lockstep capabilities.

Table 267. Command Definitions

CMD_REQ	Description
<i>CMD_REQ</i> [31:29]: CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> <i>CMD_REQ</i> [28:24]: CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> <i>CMD_REQ</i> [23:16]: CMD_TYPE = CMD_TRIG_CONTROLLER_OP <i>CMD_REQ</i> [15:0]: CMD_OPCODE = ECC_ENABLE_SET	Command to enable different ECC modes for the memory interface specified using the instance ID. ECC mode can be changed only if the design was generated with ECC enabled. [Inputs] CMD_PARAM_0 [1:0]: ECC_ENABLE Set the current ECC error reporting (single-bit and double-bit errors) and correcting (single-bit errors) that is enabled. 'b00 = ECC is disabled. Data is written to the memory without ECC values, and data is returned to the user interface without being verified for accuracy. 'b01 = ECC is enabled, but without detection or correction. 'b10 = ECC is enabled with detection, but correction is not supported. When an error is found on a read operation, ECC reporting parameters are updated for read commands. Erroneous data is returned to the user on read commands and written to the memory on write commands. 'b11 = ECC is enabled with detection and correction. When an error is found on a read operation, the ECC reporting parameters are updated for read commands. Single bit errors are corrected automatically by the controller in both read and write commands. [Outputs] [Errors] CMD_RESPONSE_STATUS - STATUS_CMD_RESPONSE_ERROR: 000 - No errors 001 - ECC was not enabled during design generation. [Outputs] N/A
<i>CMD_REQ</i> [31:29]: CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> <i>CMD_REQ</i> [28:24]: CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> <i>CMD_REQ</i> [23:16]: CMD_TYPE = CMD_TRIG_CONTROLLER_OP <i>CMD_REQ</i> [15:0]: CMD_OPCODE = ECC_INTERRUPT_MASK	Command to set mask for ECC interrupts for the memory interface specified using the instance ID, in order to disable specific ECC interrupts. [Inputs] CMD_PARAM_0 [13:0]: ECC_INTERRUPT_MASK If any bit is set to 'b1 in this parameter, the corresponding interrupt does NOT trigger an interrupt on the top-level EMIF interrupt signal.

continued...

CMD_REQ	Description
	Bit [13] = A RMW Read Link ECC double-bit error has been detected Bit [12] = A Read Link ECC double-bit error has been detected. Bit [11] = A Read Link ECC single-bit error has been detected. Bit [10] = A Write Link ECC double-bit error has been detected by the periodic MRR to MR43. Bit [9] = A Write Link ECC single-bit error has been detected by the periodic MRR to MR43. Bit [8] = An ECC correctable error has been detected in a scrubbing read operation Bit [7] = The triggered scrub operation has completed. Bit [6] = One or more ECC writeback commands could not be executed. Bit [3] = Another un-correctable ECC event has been detected on a read operation, prior to the initial event being acknowledged. Bit [2] = An un-correctable ECC event has been detected on a read operation. Bit [1] = Another correctable ECC event has been detected on a read operation, prior to the initial event being acknowledged. Bit [0] = A correctable ECC event has been detected on a read operation [Outputs] N/A
CMD_REQ column is: <i>CMD_REQ</i> [31:29]: CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> <i>CMD_REQ</i> [28:24]: CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> <i>CMD_REQ</i> [23:16]: CMD_TYPE = CMD_TRIG_CONTROLLER_OP <i>CMD_REQ</i> [15:0]: CMD_OPCODE = ECC_WRITEBACK_ENABLE	Command to set automatic writing of corrected errors on read operation for the memory interface specified using the instance ID <i>CMD_PARAM_0</i> [0:0]: ECC_WRITEBACK_EN Enables automatic writing of corrected data on single bit correctable errors on read operations. This parameter is only relevant if ECC is enabled with detection and correction (ECC_ENABLE = 'b11). Note: No writebacks will be issued during BIST. 'b0 = Disable. 'b1 = Enable [Outputs] N/A [Command-Specific Errors] 'b000 – No errors 'b001 – ECC detection and correction not enabled
<i>CMD_REQ</i> [31:29]: CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> <i>CMD_REQ</i> [28:24]: CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> <i>CMD_REQ</i> [23:16]: CMD_TYPE = CMD_TRIG_CONTROLLER_OP <i>CMD_REQ</i> [15:0]: CMD_OPCODE = ECC_SCRUB_MODE_0_START	Command to start ECC scrub in mode 0 where scrub is performed at regular intervals for the memory interface specified using the instance ID. Automatic writing of corrected data should be enabled using the ECC_WRITEBACK_EN command before initiating scrub operations. [Inputs] <i>CMD_PARAM_0</i> [15:0]: ECC_SCRUB_INTERVAL Sets the minimum interval between two ECC scrubbing commands, in number of controller clock cycles. The controller clock is based on the Controller's operating frequency. Clearing this parameter to 0x0000 disables interval operation. <i>CMD_PARAM_1</i> [11:0]: ECC_SCRUB_LEN

continued...

CMD_REQ	Description
	Defines the length (in bytes) of the ECC scrubbing read command that the controller issues. This value must be an integer multiple of the memory burst length, and the lowest 3 bits of this parameter must be cleared to 'b0. The burst lengths for different protocols are: DDR4 - 8 DDR5/LPDDR4/LPDDR5 - 16 CMD_PARAM_2 [0:0]: ECC_SCRUB_FULL_MEM Defines whether to perform ECC scrub on full memory or on the specified address range. 'b0 – ECC scrub performed on address range specified using ECC_SCRUB_START_ADDR and ECC_SCRUB_END_ADDR 'b1 – ECC scrub performed on full memory address range CMD_PARAM_3 [31:0]: ECC_SCRUB_START_ADDR [31:0] CMD_PARAM_4 [5:0]: ECC_SCRUB_START_ADDR [36:32] Defines the starting address from where scrubbing operations begin. This value must be less than or equal to the value programmed into the ECC_SCRUB_END_ADDR parameter. Only used when ECC_SCRUB_FULL_MEM is 'b0. CMD_PARAM_5 [31:0]: ECC_SCRUB_END_ADDR [31:0] CMD_PARAM_6 [5:0]: ECC_SCRUB_END_ADDR [36:32] Defines the ending address at which scrubbing operations wrap around to the start address. This parameter must be programmed to a non-zero value for the scrubbing logic to operate. Only used when ECC_SCRUB_FULL_MEM is 'b0. [Outputs] CMD_RESPONSE_DATA_SHORT [0:0]: ECC_SCRUB_INITIATED 'b1 – ECC scrub initiated successfully 'b0 – ECC scrub initiation failed [Command-Specific Errors] 'b000 – No errors 'b001 – ECC not enabled
CMD_REQ [31:29]: CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> CMD_REQ [28:24]: CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> CMD_REQ [23:16]: CMD_TYPE = CMD_TRIG_CONTROLLER_OP CMD_REQ [15:0]: CMD_OPCODE = ECC_SCRUB_MODE_1_START	Command to start ECC scrub in mode 1 where scrub is performed when the controller is idle for the memory interface specified using the instance ID. Automatic writing of corrected data should be enabled using the ECC_WRITEBACK_EN command before initiating scrub operations. [Inputs] CMD_PARAM_0 [15:0]: ECC_SCRUB_IDLE_CNT Defines the number of controller clock cycles that the scrubbing engine waits in the Controller's idle state before starting scrubbing operations. The Controller is considered idle when the command queue is empty. When this condition is detected, an internal counter loads with the value programmed in this parameter and count down on each controller clock. When the counter expires, either the scrubbing operation begins or the next address is tested. The controller clock is based on the Controller's operating frequency. Clearing this parameter to 0x0000 disables idle operation. CMD_PARAM_1 [11:0]: ECC_SCRUB_LEN

continued...

CMD_REQ	Description
	Defines the length (in bytes) of the ECC scrubbing read command that the controller issues. This value must be an integer multiple of the memory burst length, and the lowest 3 bits of this parameter must be cleared to 'b0. The burst lengths for different protocols are: DDR4 - 8 DDR5/LPDDR4/LPDDR5 - 16 CMD_PARAM_2 [0:0]: ECC_SCRUB_FULL_MEM Defines whether to perform ECC scrub on full memory or on the specified address range. 'b0 – ECC scrub performed on address range specified using ECC_SCRUB_START_ADDR and ECC_SCRUB_END_ADDR 'b1 – ECC scrub performed on full memory address range CMD_PARAM_3 [31:0]: ECC_SCRUB_START_ADDR [31:0] CMD_PARAM_4 [5:0]: ECC_SCRUB_START_ADDR [36:32] Defines the starting address from where scrubbing operations begin. This value must be less than or equal to the value programmed into the ECC_SCRUB_END_ADDR parameter. Only used when ECC_SCRUB_FULL_MEM is 'b0. CMD_PARAM_5 [31:0]: ECC_SCRUB_END_ADDR [31:0] CMD_PARAM_6 [5:0]: ECC_SCRUB_END_ADDR [36:32] Defines the ending address at which scrubbing operations wrap around to the start address. This parameter must be programmed to a non-zero value for the scrubbing logic to operate. Only used when ECC_SCRUB_FULL_MEM is 'b0. [Outputs] CMD_RESPONSE_DATA_SHORT [0:0]: ECC_SCRUB_INITIATED 'b1 – ECC scrub initiated successfully 'b0 – ECC scrub initiation failed [Command-Specific Errors] 'b000 – No errors 'b001 – ECC not enabled
CMD_REQ [31:29]: CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> CMD_REQ [28:24]: CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> CMD_REQ [23:16]: CMD_TYPE = CMD_TRIG_CONTROLLER_OP CMD_REQ [15:0]: CMD_OPCODE = ECC_INJECT_ERROR	Command to force a specific check code to be written into memory interface specified using the instance ID for diagnostic purposes or for flagging a particular memory address as erroneous for future accesses. The procedure is as follows: 1. Set the ECC_ENABLE parameter to enable detection ('b1x). 2. Ensure that no writes to the controller are pending. 3. Send ECC_INJECT_ERROR command through mailbox setting the ECC_XOR_CHECK_BITS input parameter. Use the syndromes to program the ECC_XOR_CHECK_BITS parameter. Each byte of the ECC_XOR_CHECK_BITS parameter controls the ECC event forcing for a separate user-word space. For example, setting a value of 0xF4 as ECC_XOR_CHECK_BITS will result in the check bits to

continued...

CMD_REQ	Description
	be updated and written to the memory such that it flags a single-bit correctable error on bit [0] of the user-word on subsequent access of the same address. - Execute a write command for an aligned user word. The controller will XOR the ECC_XOR_CHECK_BITS parameter with the generated checksum bits from the word written to the memory. The next read command to the same address will force the ECC error. - Depending on the programming of the ECC_XOR_CHECK_BITS parameter, a single bit, double bit or multi-bit ECC error will occur. For single bit and double bit errors, the appropriate bit in the ECC_INTERRUPT_STATUS parameter will be set to 'b1 and the ECC error signature parameters will be filled with the relevant information. [Inputs] CMD_PARAM_0 [31:0]: ECC_XOR_CHECK_BITS The check bits generated by the next write operation will be XOR'ed with this parameter. The result will be written into memory as the new check value. ECC_XOR_CHECK_BITS [31:24] maps to user word [255:192] ECC_XOR_CHECK_BITS [23:16] maps to user word [191:128] ECC_XOR_CHECK_BITS [15:8] maps to user word [127:64] ECC_XOR_CHECK_BITS [7:0] maps to user word [63:0] [Outputs] N/A [Command-Specific Errors] 'b000 – No errors 'b001 – ECC detection not enabled
CMD_REQ [31:29]: CMD_TARGET_IP_TYPE = <UNUSED> CMD_REQ [28:24]: CMD_TARGET_IP_INSTANCE_ID = <UNUSED> CMD_REQ [23:16]: CMD_TYPE = CMD_TRIG_CONTROLLER_OP CMD_REQ [15:0]: CMD_OPCODE = ECC_CLEAR_ERR_BUFFER	Command to clear the ECC error buffer and reset the ECC error counter and overflow registers. [Inputs] N/A [Outputs] N/A
CMD_REQ [31:29]: CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> CMD_REQ [28:24]: CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> CMD_REQ [23:16]: CMD_TYPE = CMD_TRIG_CONTROLLER_OP CMD_REQ [15:0]: CMD_OPCODE = BIST_STANDARD_MODE_START	Command to initiate Original MOV11 3N BIST algorithm for data checking for the memory interface specified using the instance ID. Use BIST_STATUS_INTF0/ BIST_STATUS_INTF1 registers to get the status of the operation. BIST operations are supported only for designs with DBI disabled. [Inputs] CMD_PARAM_0 [5:0]: BIST_ADDR_SPACE [5:0] Used in BIST data checking to define the address space in bytes from 0 to $2^{\text{addr_space}}$ that the BIST logic will check. As an example, if the addr_space parameter was programmed to 0x1c, then the BIST logic would check 2^{28} bytes = 256 MBytes. Only used if BIST_FULL_MEM is 'b0. A BIST test must cover a minimum of 2 bursts. Therefore, the user must program this parameter to a value such that the start address and end address of the BIST test will encompass a minimum of 2 bursts. The burst length for different protocols are: DDR4 - 8 DDR5/LPDDR4/LPDDR5 - 16 CMD_PARAM_0 [6:6]: BIST_FULL_MEM

continued...

CMD_REQ	Description
	Defines whether to perform BIST on full memory or on the specified address range. 'b0 – BIST performed on address range specified using BIST_START_ADDR and BIST_ADDR_SPACE 'b1 – BIST performed on full memory address range CMD_PARAM_1 [31:0]: BIST_START_ADDR [31:0] CMD_PARAM_2 [5:0]: BIST_START_ADDR [36:32] Used in BIST data checking and memory initialization programming to define the starting address for BIST checking in bytes. Only used if BIST_FULL_MEM is 'b0. [Outputs] CMD_RESPONSE_DATA_SHORT [0:0]: BIST_INITIATED 'b1 – BIST initiated successfully 'b0 – BIST initiation failed [Command-Specific Errors] 'b00 – No errors 'b01 – A previous command's saved state not restored.
CMD_REQ [31:29]: CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> CMD_REQ [28:24]: CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> CMD_REQ [23:16]: CMD_TYPE = CMD_TRIG_CONTROLLER_OP CMD_REQ [15:0]: CMD_OPCODE = BIST_MEM_INIT_START	Command to initiate memory initialization BIST for the memory interface specified using the instance ID. Memory initialization programming allows a selectable range of memory to be initialized with a programmable data value. Use MEM_INIT_STATUS_INTF0/ MEM_INIT_STATUS_INTF1 registers to get the status of the operation. If in-line ECC is enabled and you want to complete memory initialization on the entire memory, then before running the test, you should disable in-line ECC. [Inputs] CMD_PARAM_0 [5:0]: BIST_ADDR_SPACE [5:0] Used in BIST data checking to define the address space in bytes from 0 to $2^{\text{addr_space}}$ that the BIST logic will check. As an example, if the addr_space parameter was programmed to 0x1c, then the BIST logic would check 2^{28} bytes = 256 MBytes. Only used if BIST_FULL_MEM is 'b0. Note: A BIST test must cover a minimum of 2 bursts. Therefore, the user must program this parameter to a value such that the start address and end address of the BIST test will encompass a minimum of 2 bursts. The burst length for different protocols are: DDR4 - 8 DDR5/LPDDR4/LPDDR5 - 16 CMD_PARAM_0 [6:6]: BIST_FULL_MEM Defines whether to perform BIST on full memory or on the specified address range. 'b0 – BIST performed on address range specified using BIST_START_ADDR and BIST_ADDR_SPACE 'b1 – BIST performed on full memory address range CMD_PARAM_1 [31:0]: BIST_START_ADDR [31:0] CMD_PARAM_2 [5:0]: BIST_START_ADDR [37:32] Used in BIST data checking and memory initialization programming to define the starting address for BIST checking in bytes. Only used if BIST_FULL_MEM is 'b0. CMD_PARAM_3: BIST_DATA_PATTERN Specifies the data pattern to use for the memory initialization. 'b00 – Initialize memory to all zeros.

continued...

CMD_REQ	Description
	'b10 – Use data pattern specified using the values set using commands BIST_SET_DATA_PATTERN_UPPER and BIST_SET_DATA_PATTERN_LOWER before issuing BIST_MEM_INITIAL_START command. [Outputs] CMD_RESPONSE_DATA_SHORT [0:0]: BIST_INITIATED 'b1 – BIST memory initialization initiated successfully 'b0 – BIST memory initialization initiation failed [Command-Specific Errors] 'b00 – No errors 'b01 – A previous command's saved state not restored. 'b10 – In-line ECC must be disabled before full memory initialization. Either disable in-line ECC or specify a valid address range to be initialized.
CMD_REQ [31:29]: CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> CMD_REQ [28:24]: CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> CMD_REQ [23:16]: CMD_TYPE = CMD_TRIG_CONTROLLER_OP CMD_REQ [15:0]: CMD_OPCODE = BIST_SET_DATA_PATTERN_UPPER	CMD_PARAM_3 [31:0]: BIST_DATA_PATTERN [287:256] CMD_PARAM_2 [31:0]: BIST_DATA_PATTERN [255:224] CMD_PARAM_1 [31:0]: BIST_DATA_PATTERN [223:192] CMD_PARAM_0 [31:0]: BIST_DATA_PATTERN [191:160] Defines the data pattern bits [287:160] to be used. Only data corresponding to active portion of core word is used while the inactive portion is ignored. The active portion of the core word depends on the DQ width per channel and the burst length. For example: - DDR5/LPDDR4/LPDDR5 (BL16) will use BIST_DATA_PATTERN [255:0] for x32 and BIST_DATA_PATTERN [127:0] for x16. - DDR4 (BL8) will use BIST_DATA_PATTERN [127:0] for x32 and BIST_DATA_PATTERN [63:0] for x16. If the axi4_rdata bus width is greater than the width of the active BIST_DATA_PATTERN, you will get repeated BIST_DATA_PATTERN when reading from the initialized memory locations. For example: - DDR4 x32: axi4_rdata[255:0] = { 2{BIST_DATA_PATTERN[127:0]} } - DDR4 x16: axi4_rdata[255:0] = { 4{BIST_DATA_PATTERN[63:0]} } [Outputs] N/A [Command-Specific Errors] 'b00 – No errors 'b01 – Cannot set upper data pattern for secondary hard memory interface.
CMD_REQ [31:29]: CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> CMD_REQ [28:24]: CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> CMD_REQ [23:16]: CMD_TYPE = CMD_TRIG_CONTROLLER_OP CMD_REQ [15:0]: CMD_OPCODE = BIST_SET_DATA_PATTERN_LOWER	CMD_PARAM_4 [31:0]: BIST_DATA_PATTERN [159:128] CMD_PARAM_3 [31:0]: BIST_DATA_PATTERN [127:96] CMD_PARAM_2 [31:0]: BIST_DATA_PATTERN [95:64] CMD_PARAM_1 [31:0]: BIST_DATA_PATTERN [63:32] CMD_PARAM_0 [31:0]: BIST_DATA_PATTERN [31:0] Defines the data pattern bits [223:0] to be used. Only data corresponding to active portion of the core word is used while inactive portion is ignored. The active portion of the core word depends on the DQ width per channel and the burst length. For example: - DDR5/LPDDR4/LPDDR5 (BL16) will use BIST_DATA_PATTERN [255:0] for x32 and BIST_DATA_PATTERN [127:0] for x16. - DDR4 (BL8) will use BIST_DATA_PATTERN [127:0] for x32 and BIST_DATA_PATTERN [63:0] for x16.

continued...

CMD_REQ	Description
	If the axi4_rdata bus width is greater than the width of the active BIST_DATA_PATTERN, you will get repeated BIST_DATA_PATTERN when reading from the initialized memory locations. For example: - DDR4 x32: axi4_rdata[255:0] = { 2{BIST_DATA_PATTERN[127:0]} } - DDR4 x16: axi4_rdata[255:0] = { 4{BIST_DATA_PATTERN[63:0]} }
<i>CMD_REQ [31:29]:</i> CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> <i>CMD_REQ [28:24]:</i> CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> <i>CMD_REQ [23:16]:</i> CMD_TYPE = CMD_TRIG_CONTROLLER_OP <i>CMD_REQ [15:0]:</i> CMD_OPCODE = CHANGE_FSP_LP5	Command to change the FSP when using the LPDDR5 protocol. [Inputs] <i>CMD_PARAM_0 [1:0]:</i> TARGET_FSP 'b00 – FSP 0 'b01 – FSP 1 'b10 – FSP 2 [Outputs] <i>CMD_RESPONSE_DATA_SHORT [0:0]:</i> FSP_CHANGE_STATUS 'b1 – The FSP was changed successfully. 'b0 – The FSP was not changed successfully. [Error Codes] 'b000 – No errors. 'b001 – Current Protocol is not LPDDR5. 'b010 – The current FSP is equal to the target FSP – No change is needed. 'b011 – The FSP could not be changed.
<i>CMD_REQ [31:29]:</i> CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> <i>CMD_REQ [28:24]:</i> CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> <i>CMD_REQ [23:16]:</i> CMD_TYPE = CMD_TRIG_CONTROLLER_OP <i>CMD_REQ [15:0]:</i> CMD_OPCODE = LP_MODE_ENTER	Command to cause the Interface to enter a low power state. Note that other interface operations, including recalibration and mode register reads and writes, can cause automatic exits from some low-power states. [Inputs] <i>CMD_PARAM_0[3:0]:</i> LP_STATE Specified the low power state the interface should enter: 'b1000 – Self-Refresh Long (DDR4/DDR5 Only) 'b1001 – Self-Refresh Long with Memory Clock Gating (DDR4/DDR5 Only) 'b1010 – Self-Refresh Long with Memory Clock and Controller Clock Gating (DDR4/DDR5 Only) 'b1101 – Self-Refresh Power Down Long (LPDDR4/LPDDR5 Only) 'b1110 – Self-Refresh Power Down Long with Memory Clock Gating (LPDDR4/LPDDR5 Only) 'b1111 – Self-Refresh Power Down Long with Memory and Controller Clock Gating (LPDDR4/LPDDR5 Only) [Outputs] N/A [Error Codes] 'b000 – No errors 'b001 – The Selected Low Power State is Not Available for the Current Protocol 'b010 – The Selected Low Power State is invalid/Does not Exist. 'b011 – Disable auto LP mode using the LP_MODE_AUTO command and confirm the idle state using the LP_MODE_STATUS command before using the LP_MODE_ENTER command.

continued...

CMD_REQ	Description
<i>CMD_REQ [31:29]:</i> CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> <i>CMD_REQ [28:24]:</i> CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> <i>CMD_REQ [23:16]:</i> CMD_TYPE = CMD_TRIG_CONTROLLER_OP <i>CMD_REQ [15:0]:</i> CMD_OPCODE = LP_MODE_EXIT	Command to exit any low power state. [Inputs] N/A [Outputs] N/A [Error Codes] 'b000 – No errors
<i>CMD_REQ [31:29]:</i> CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> <i>CMD_REQ [28:24]:</i> CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> <i>CMD_REQ [23:16]:</i> CMD_TYPE = CMD_TRIG_CONTROLLER_OP <i>CMD_REQ [15:0]:</i> CMD_OPCODE = LP_MODE_AUTO	Command to cause the Interface to enter certain automatic low power states. Supported low power modes for this command are listed below as valid options. Do not use LP_MODE_EXIT opcode to exit from the automatic low power states. Note that other interface operations, including recalibration and mode register reads and writes, sending read/write request to the controller can cause automatic exits from some low-power states. [Inputs] CMD_PARAM_0 [3:0]: Specified the low power state the interface should enter. 'b0000 – Disable automatic low power mode. 'b0001 – Active Power Down (All Protocols). 'b0010 – Active Power Down with Memory Clock Gating (LPDDR4/LPDDR5 only). 'b0011 – Pre-Charge Power Down (All Protocols). 'b0100 – Pre-Charge Power Down with Memory Clock Gating (LPDDR4/LPDDR5 only). 'b0101 – Self-Refresh Short (DDR4/DDR5 only). 'b0110 – Self-Refresh Short with Memory Clock Gating (DDR4/DDR5 only). 'b1011 – Self-Refresh Power Down Short (LPDDR4/LPDDR5 only). 'b1100 – Self-Refresh Power Down Short with Memory Clock Gating (LPDDR4/LPDDR5 only). CMD_PARAM_1 [11:0]: Defines the number of idle controller clocks that can elapse before the controller will automatically issue an entry into the low power state specified [Outputs] [Error Codes] 'b000 – No errors 'b001 – The selected low power state is invalid for the current protocol. 'b010 – The selected low power state is invalid/does not exist. 'b011 – The idle clock count should not be zero.
<i>CMD_REQ [31:29]:</i> CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> <i>CMD_REQ [28:24]:</i> CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> <i>CMD_REQ [23:16]:</i> CMD_TYPE = CMD_TRIG_MEM_CAL_OP <i>CMD_REQ [15:0]:</i> CMD_OPCODE = TRIG_MEM_CAL	Command to trigger a memory calibration for the memory interface specified using the instance ID. You should only recalibrate the memory interface for debug purposes. (HPS EMIF on Agilex 5 devices does not support recalibration.) For multi-channel interfaces, recalibration can be initiated by sending the TRIG_MEM_CAL command as below: • 1 channel – 1 request to the interface • 2 channel LPDDR4/LPDDR5 – 1 request to one of the interfaces • 2 channel DDR5 – 2 separate requests to the interfaces in each IO96B • 4 channel LPDDR4/LPDDR5 – 2 requests to one of the interfaces in each IO96B

continued...

CMD_REQ	Description
	[Inputs] [Outputs] CMD_RESPONSE_DATA_SHORT [0]: MEM_CAL_INITIATED 'b1 – Memory calibration triggered successfully 'b0 – Memory calibration triggering failed [Errors] CMD_RESPONSE_STATUS - STATUS_CMD_RESPONSE_ERROR: 000 - No errors 001 - Triggering memory calibration is not supported for this memory interface.
CMD_REQ [31:29]: CMD_TARGET_IP_TYPE = <TARGET_IP_TYPE> CMD_REQ [28:24]: CMD_TARGET_IP_INSTANCE_ID = <TARGET_IP_INSTANCE_ID> CMD_REQ [23:16]: CMD_TYPE = CMD_GET_MEM_CAL_INFO CMD_REQ [15:0]: CMD_OPCODE = GET_MEM_CAL_MGN	Command to retrieve the calibration margin information for each pin. [Inputs] CMD_PARAM_0 [3:0]: PIN_TYPE 'b000 – DQ delay 'b001 – DQS delay (not applicable to LPDDR5) 'b010 – DM/DBI delay 'b011 – CA delay 'b100 – CS delay 'b101 – DQ VREF CMD_PARAM_1 [6:0]: PIN_IDX Pin index for the selected PIN_TYPE CMD_PARAM_2 [0:0]: RANK Rank select [Outputs] PIN_TYPE = 'b000 (DQ delay): CMD_RESPONSE_DATA_0 [15:0]: DQ WR delay CMD_RESPONSE_DATA_0 [31:16]: DQ RD delay CMD_RESPONSE_DATA_1 [15:0]: DQ RD right margin CMD_RESPONSE_DATA_1 [31:16]: DQ RD left margin CMD_RESPONSE_DATA_2 [15:0]: DQ WR right margin CMD_RESPONSE_DATA_2 [31:16]: DQ WR left margin <i>Note:</i> DQ delays and margins are returned as absolute values. PIN_TYPE = 'b001 (DQS delay) CMD_RESPONSE_DATA_0 [15:0]: DQS delay <i>Note:</i> DQS delays are returned as absolute values. PIN_TYPE = 'b010 (DM/DBI delay) CMD_RESPONSE_DATA_0 [15:0]: DM/DBI WR delay CMD_RESPONSE_DATA_0 [31:16]: DM/DBI RD delay CMD_RESPONSE_DATA_1 [15:0]: DM/DBI RD right margin CMD_RESPONSE_DATA_1 [31:16]: DM/DBI RD left margin CMD_RESPONSE_DATA_2 [15:0]: DM/DBI WR right margin CMD_RESPONSE_DATA_2 [31:16]: DM/DBI WR left margin <i>Note:</i> DM and DBI delays and margins are returned as absolute values. PIN_TYPE = 'b011 (CA delay) CMD_RESPONSE_DATA_0 [31:16]: CA delay CMD_RESPONSE_DATA_1 [15:0]: CA right edge per-rank offset (DDR4 and DDR5 only) CMD_RESPONSE_DATA_1 [31:16]: CA left edge per-rank offset (DDR4 and DDR5 only)

CMD_REQ	Description
	CMD_RESPONSE_DATA_2 [15:0]: CA right edge offset (LPDDR4 and LPDDR5 only) CMD_RESPONSE_DATA_2 [31:16]: CA left edge offset (LPDDR4 and LPDDR5 only) Note: - CA Delay values are returned as absolute values, whereas CA right or left margin values are the offsets. Margin Calculation: right or left offset - delay. - Refer to the address mapping table for address decoding based on index. PIN_TYPE = 'b0100 (CS delay) CMD_RESPONSE_DATA_0 [31:0]: CS delay CMD_RESPONSE_DATA_1 [15:0]: CS right edge offset CMD_RESPONSE_DATA_1 [31:16]: CS left edge offset Note: CA Delay values are returned as absolute values, whereas right or left margin values are the offsets. Margin Calculation: right or left offset - delay. PIN_TYPE = 'b101 (DQ VREF) CMD_RESPONSE_DATA_0 [15:0]: DQ WR-VREF CMD_RESPONSE_DATA_0 [31:16]: DQ RD-VREF CMD_RESPONSE_DATA_1 [15:0]: DQ RD-VREF upper margin offset CMD_RESPONSE_DATA_1 [31:16]: DQ RD-VREF lower margin offset CMD_RESPONSE_DATA_2 [15:0]: DQ WR-VREF upper margin offset CMD_RESPONSE_DATA_2 [31:16]: DQ WR-VREF lower margin offset Note: VREF Delay values are returned as absolute values, whereas VREF upper or lower margin values are the offsets. RD-VREF voltage calculation: $DQ\ RD-VREF\ (V) = [DQ\ RD-VREF / 512] \times VDDQ.$ $DQ\ RD-VREF\ Upper\ (V) = [(DQ\ RD-VREF + DQ\ RD-VREF\ Upper\ Margin\ Offset) / 512] \times VDDQ.$ $DQ\ RD-VREF\ Lower\ (V) = [(DQ\ RD-VREF - DQ\ RD-VREF\ Lower\ Margin\ Offset) / 512] \times VDDQ.$ WR-VREF calculation: $Vrefout_factor = (Vrefout_max - Vrefout_min) / Vrefout_step.$ $Vrefout(\%) = (Vrefout_factor \times DQ\ WR-VREF) + Vrefout_min.$ $Vrefout_upper(\%) = [Vrefout_factor \times (DQ\ WR-VREF + DQ\ WR-VREF\ Upper)] + Vrefout_min.$ $Vrefout_lower(\%) = [Vrefout_factor \times (DQ\ WR-VREF + DQ\ WR-VREF\ Lower)] + Vrefout_min.$ $DQ\ WR-VREF\ (V) = Vrefout(\%) \times VDDQ.$ $DQ\ WR-VREF\ Upper\ (V) = Vrefout_upper(\%) \times VDDQ.$ $DQ\ WR-VREF\ Lower\ (V) = Vrefout_lower(\%) \times VDDQ.$ Note: Refer to the VREF Calculation Reference Data for computing VREF values.

CMD_REQ	Description
	All delay and margin information are represented by finite increments (taps). For delay values, one tap corresponds to 1/128 of a memory clock cycle. RD-VREF taps correspond to 1/512 of the RD-VREF range. WR-VREF tap definitions follow the protocol specification. [Error Codes] 'b000 – No errors

12.1.3. Address Mapping Reference

The following table provides address mapping by index for the GET_MEM_CAL_MGN register (CA Delay).

Table 268. Address Mapping

Index	DDR4	DDR5	LPDDR4	LPDDR5
0	ACT	CA_0	CA_0	CA_0
1	C_0	CA_1	CA_1	CA_1
2	C_1	CA_2	CA_2	CA_2
3	C_2	CA_3	CA_3	CA_3
4	BA_0	CA_4	CA_4	CA_4
5	BA_1	CA_5	CA_5	CA_5
6	BG_0	CA_6		CA_6
7	BG_1	CA_7		
8	ADD_0	CA_8		
9	ADD_1	CA_9		
10	ADD_2	CA_10		
11	ADD_3	CA_11		
12	ADD_4	CA_12		
13	ADD_5	CA_13		
14	ADD_6			
15	ADD_7			
16	ADD_8			
17	ADD_9			
18	ADD_10			
19	ADD_11			
20	ADD_12			
21	ADD_13			
22	ADD_14			
23	ADD_15			
continued...				

Index	DDR4	DDR5	LPDDR4	LPDDR5
24	ADD_16			
25	ADD_17			
26	PAR_IN			

12.1.4. VREF Calculation Reference Data

The following tables provide the necessary information for the GET_MEM_CAL_MGN register (DQ VREF).

Table 269. VDDQ Value

Protocol	VDDQ
DDR4	1.2
DDR5	1.1
LPDDR4	0.6
LPDDR5	0.5

Table 270. Vrefout Range

Protocol	Vrefout min	Vrefout max
DDR4_R1	0.6	0.925
DDR4_R2	0.45	0.775
DDR5	0.35	0.975
LPDDR4_R1	0.1	0.3
LPDDR4_R2	0.22	0.42
LPDDR5	0.15	0.735

Note: For the effective Vrefout Range setting, refer to the *Mem VrefDQ Range* in the *EMIF IP Configuration* under the *Advanced: Analog Overrides* tab.

Table 271. Vrefout Steps

Protocol	Steps
DDR4	50
DDR5	125
LPDDR4	50
LPDDR5	127

12.1.5. ECC Syndrome Codes

Table 272. ECC Syndrome Codes

Syndrome	Incorrect Bit	Syndrome	Incorrect Bit	Syndrome	Incorrect Bit
0x00	No Error	0x4f	Data [46]	0xa4	Data [22]
0x01	Check [0]	0x52	Data [45]	0xa7	Data [21]
0x02	Check [1]	0x54	Data [44]	0xa8	Data [20]
0x04	Check [2]	0x57	Data [43]	0xab	Data [19]
0x08	Check [3]	0x58	Data [42]	0xad	Data [18]
0x0b	Data [63]	0x5b	Data [41]	0xb0	Data [17]
0x0e	Data [62]	0x5e	Data [40]	0xb5	Data [16]
0x10	Check [4]	0x62	Data [39]	0xcb	Data [15]
0x13	Data [61]	0x64	Data [38]	0xce	Data [14]
0x15	Data [60]	0x67	Data [37]	0xd3	Data [13]
0x16	Data [59]	0x68	Data [36]	0xd5	Data [12]
0x19	Data [58]	0x6b	Data [35]	0xd6	Data [11]
0x1a	Data [57]	0x6d	Data [34]	0xd9	Data [10]
0x1c	Data [56]	0x70	Data [33]	0xda	Data [9]
0x20	Check [5]	0x75	Data [32]	0xdc	Data [8]
0x23	Data [55]	0x80	Check [7]	0xe3	Data [7]
0x25	Data [54]	0x8a	Data [31]	0xe5	Data [6]
0x26	Data [53]	0x8f	Data [30]	0xe6	Data [5]
0x29	Data [52]	0x92	Data [29]	0xe9	Data [4]
0x2a	Data [51]	0x94	Data [28]	0xea	Data [3]
0x2c	Data [50]	0x97	Data [27]	0xec	Data [2]
0x31	Data [49]	0x98	Data [26]	0xf1	Data [1]
0x34	Data [48]	0x9b	Data [25]	0xf4	Data [0]
0x40	Check [6]	0x9d	Data [24]		
0x4a	Data [47]	0xa2	Data [23]		
Any other Syndrome			Multi-bit error		

12.1.6. ECC Error Handling

The user RAM includes an integrated ECC error buffer, that improves ECC error management capabilities with the mailbox. The ECC error buffer is administered by the IOSSM, which routinely inspects the memory controller's interrupt status to detect ECC events. Upon identifying an ECC event, the IOSSM captures and stores relevant details from the event, including the following:

- The type of ECC error that triggered the event.
- The memory address involved in the error.
- The Source ID linked to the error-causing transaction.

- Note:**
1. For two-rank configurations, the rank bit is not reflected in the source address in the ECC buffer entries.
 2. When accessing the memory interface through NoC mode, the reported source IDs of error-causing transactions are incorrect.

Table 273. ECC Error Buffer Byte Offset

Register Name	Byte Offset (Hexadecimal)	Description
ECC Error Status	0x0300	ECC_ERROR_COUNTER, ECC_ERR_OVERFLOW
ECC Error Buffer Entry 0 – (ECC_ERR_DATA_START)	0x0310	ENTRY 0 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x0314	ENTRY 0 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 1	0x0318	ENTRY 1 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x031C	ENTRY 1 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 2	0x0320	ENTRY 2 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x0324	ENTRY 2 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 3	0x0328	ENTRY 3 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x032C	ENTRY 3 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 4	0x0330	ENTRY 4 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x0334	ENTRY 4 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 5	0x0338	ENTRY 5 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x033C	ENTRY 5 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 6	0x0340	ENTRY 6 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x0344	ENTRY 6 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 7	0x0348	ENTRY 7 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x034C	ENTRY 7 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 8	0x0350	ENTRY 8 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
continued...		

Register Name	Byte Offset (Hexadecimal)	Description
	0x0354	ENTRY 8 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 9	0x0358	ENTRY 9 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x035C	ENTRY 9 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 10	0x0360	ENTRY 10 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x0364	ENTRY 10 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 11	0x0368	ENTRY 11 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x036C	ENTRY 11 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 12	0x0370	ENTRY 12 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x0374	ENTRY 12 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 13	0x0378	ENTRY 13 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x037C	ENTRY 13 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 14	0x0380	ENTRY 14 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x0384	ENTRY 14 R2: ECC_ERR_ADDR_LOWER
ECC Error Buffer Entry 15	0x0388	ENTRY 15 R1: IP_TYPE, INSTANCE_ID, ECC_ERR_SOURCE_ID, ECC_ERR_TYPE, ECC_ERR_ADDR_UPPER
	0x038C	ENTRY 15 R2: ECC_ERR_ADDR_LOWER

When the IOSSM firmware identifies an ECC event and the buffer has not yet reached its maximum capacity (as indicated by the ECC_ERR_COUNTER not exceeding the maximum capacity of 16 entries), it then populates the next buffer entry with pertinent information in the format outlined in the table below.

Table 274. ECC Error Buffer Structure

32-Bit Register	Bit Range	Register Field	Description
Register 1 (R1)	5:0	ECC_ERR_ADDR_UPPER	Upper 6 bits of the ECC Error Address
	9:6	ECC_ERR_TYPE	Type of ECC Error
	16:10	ECC_ERR_SOURCE_ID	Source Transaction AXI ID
	21:17	INSTANCE_ID	EMIF Instance ID
	22:24	IP_TYPE	EMIF IP Type
	31:25	—	Reserved
Register 2 (R2)	31:0	ECC_ERR_ADDR_LOWER	Lower 32 bits of the ECC Error Address

ECC Error Buffer Overflow Handling

The ECC error buffer is considered full when the ECC_ERR_COUNTER reaches the buffer's maximum capacity of 16 entries. If the buffer becomes full, the IOSSM does not write new data to the ECC error buffer for any subsequent interrupts. Instead, the ECC_ERR_OVERFLOW bits corresponding to each new interrupt type that cannot be buffered are logically OR'ed with the existing overflow status field's contents.

Table 275. ECC Error Types and Overflow

ECC Error Type	ECC_ERR_TYPE	ECC_ERR_OVERFLOW
Single-bit error	'b0000	'b0000_0000_0000_0001
Multiple single-bit errors	'b0001	'b0000_0000_0000_0010
Double-bit error	'b0010	'b0000_0000_0000_0100
Multiple double-bit errors	'b0011	'b0000_0000_0000_1000
Single-bit error during ECC scrubbing	'b1000	'b0000_0000_1000_0000
Write link ECC single-bit error (LPDDR5 only)	'b1001	'b0000_0001_0000_0000
Write link ECC double-bit error (LPDDR5 only)	'b1010	'b0000_0010_0000_0000
Read link ECC single-bit error (LPDDR5 only)	'b1011	'b0000_0100_0000_0000
Read link ECC double-bit error (LPDDR5 only)	'b1100	'b0000_1000_0000_0000
Read link ECC double-bit error, caused by read-modify-write operation (LPDDR5 only)	'b1101	'b0001_0000_0000_0000

Clearing ECC Error Buffer

To clear the ECC error buffer and reset the ECC_ERR_COUNTER and the ECC_ERR_OVERFLOW to zero, you should issue the ECC_CLEAR_ERR_BUFFER mailbox command. This command ensures that the buffer is empty and allows for new entries.

12.1.7. Accessing Read-Only Registers

To access read-only registers, follow these steps:

1. Read the desired register information from base address (0x0500_0000) + Byte Offset.
2. Decode the retrieved read data by referring to the description on each register.

The following examples illustrate read-only register access.

12.1.7.1. Example 1: Reading IP_TYPE and IP_INSTANCE_ID of All the Interfaces in the IO96B Through Read-Only Registers

This example provides instructions for reading the IP_TYPE and IP_INSTANCE_ID of all the interfaces in the IO96B from read-only registers.

The values shown below are for illustrative purposes and are obtained from an EMIF example design using DDR4 x32 + ECC implemented on the Agilex 7 FPGA E-Series 065B Premium Development Kit. This configuration uses the Primary MC of the Primary IO96B.

Base address=0x500_0000

Address for each read-only register = Base address + offset of each register

Register Name	Byte Offset (Hexadecimal)	Address (Hexadecimal)
MEM_INTF_INFO_0	0x200	0x5000200
MEM_INTF_INFO_1	0x280	0x5000280

1. Read from MEM_INTF_INFO_0 (address = 0x5000200)
The expected read_data=0x2000_0000.
IP_TYPE for interface 0 = MEM_INTF_INFO_0[31:29] = 0x1, indicating interface 0 is using Primary MC of Primary IO96B.
INSTANCE_ID for interface 0 = MEM_INTF_INFO_0 [28:24] = 0x0.
2. Read from MEM_INTF_INFO_1 (address = 0x5000280)
The expected read_data=0x0000_0000.
IP_TYPE for interface 1 = MEM_INTF_INFO_1 [31:29] = 0x0, indicating interface 1 does not exist.
INSTANCE_ID for interface 1 = MEM_INTF_INFO_1 [28:24] = 0x0; not a valid data as interface 1 does not exist.

12.1.7.2. Example 2: Reading the Memory Clock Frequency for an Interface

This example illustrates the reading of the current memory clock frequency for an interface in the IO96B from read-only registers.

The values in this example are for illustrative purposes and are obtained from an EMIF example design with DDR4 x32 + ECC running at 800MHz on the Agilex 7 FPGA E-Series 065B Development Kit - Premium. This configuration uses the Primary MC of the Primary IO96B.

Base address=0x500_0000

Address for each read-only register = Base address + offset of each register

Register Name	Byte Offset (Hexadecimal)	Address (Hexadecimal)
MEMCLK_FREQ_FSP_CUR_INTF0	0x220	0x5000220

The expected read_data=0x000c_3500

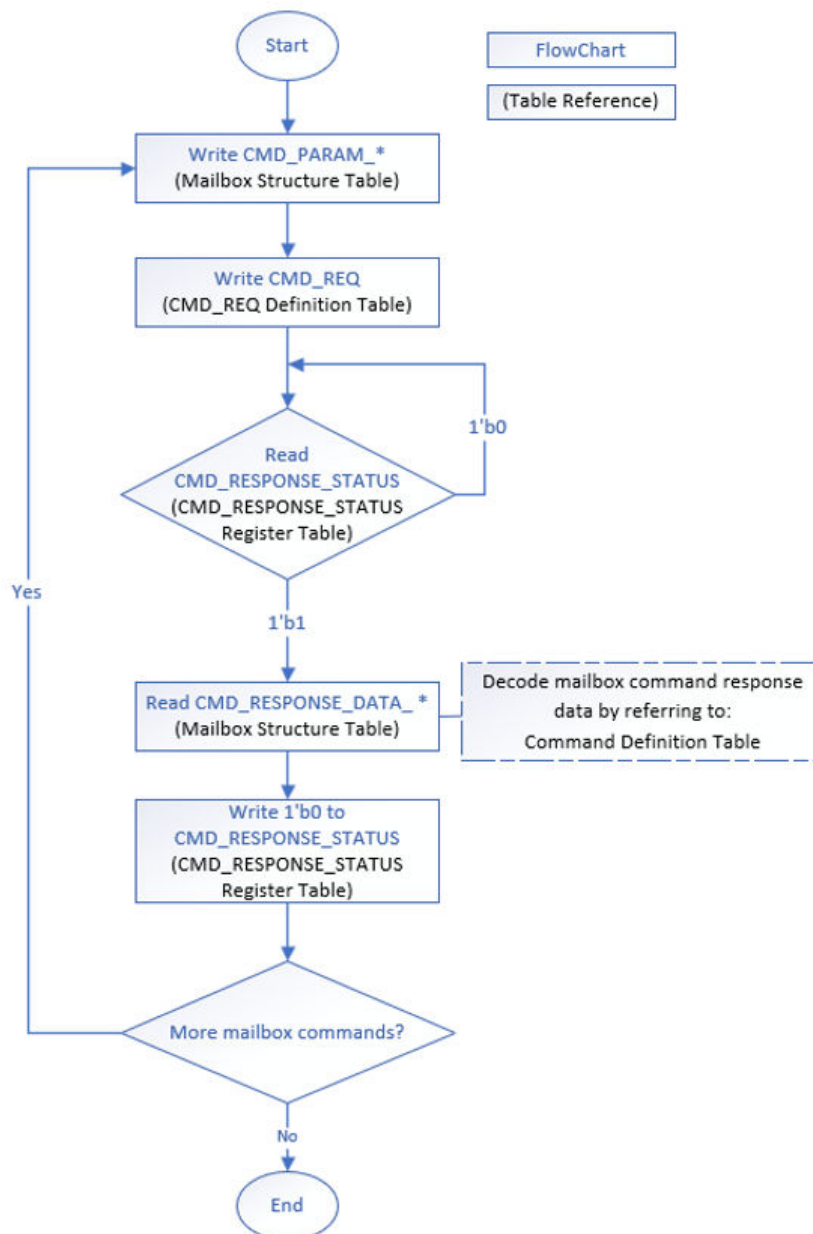
12.1.8. Sending a Mailbox Command

Follow these steps to send a mailbox command.

1. Wait for the CMD_REQ to be cleared to ensure the previous command has been dispatched.
2. Write to the CMD_PARAM_* register. (If the command does not require any CMD_PARAM – for example, TRIG_MEM_CAL – you can skip this step.)
3. Write to the CMD_REQ with the desired command (CMD_TARGET_IP_TYPE, CMD_TARGET_INSTANCE_ID, CMD_TYPE and CMD_OPCODE)

4. Wait until STATUS_CMD_RESPONSE_READY (BIT 0 of CMD_RESPONSE_STATUS) equals 1 .
5. Read the CMD_RESPONSE_DATA_* and/or CMD_RESPONSE_DATA_SHORT (as required for the CMD_REQ).
6. Clear the STATUS_COMMAND_RESPONSE_READY (Bit 0 of CMD_RESPONSE_STATUS register) . Perform a Read-Modify-Write operation:
 - a. Read from CMD_RESPONSE_STATUS register.
 - b. Write_Data = Data in (a) & 0xffff_fffe (Change only bit 0).
 - c. Write to Address = 0x500_045c Data= Write_Data from (b).

Figure 96. Sending a Mailbox Command



12.1.8.1. Example 3: Sending Recalibration Request and Reading Calibration Status Using the Mailbox

This example provides instructions for sending recalibration request and reading the calibration status using the mailbox.

The values in this example are for illustrative purposes and are obtained from an EMIF example design using DDR5 2CHx40 RDIMM, implemented on the Agilex 7 FPGA Development Kit –HBM2e Edition.

In a 2CHx40 DDR5 RDIMM design, CH1 and CH2 use the Primary MC in Primary IO96B and Secondary IO96B, respectively; hence the CMD_TARGET_IP_TYPE is 0x1 for CH1 , and 0x3 for CH2. The CMD_TARGET_IP_INSTANCE defaults to 0 in the example design generated by the Quartus Prime software.

The table below illustrates the construction of the data to be written to the CMD_REQ register for this request:

Channel	CMD_REQ [31:29]: CMD_TARGET_IP_TYPE	CMD_REQ [28:24]: CMD_TARGET_IP_INSTANCE_ID	CMD_REQ [23:16]: CMD_TYPE	CMD_REQ [15:0]: CMD_OPCODE	CMD_REQ [31:0]:
CH1	0x1	0x0	0x05	0xa	0x2005_000a
CH2	0x3	0x0	0x05	0xa	0x6005_000a

Recalibrate CH1 using the JAMB/AXI-Lite interface for CH1

1. It is not required to write to the CMD_PARAM_* register as this OPCODE does not require any cmd_param.
2. Write to the CMD_REQ(address=0x500_043c) with write_data=0x2005_000a .
3. Read from CMD_RESPONSE_STATUS(address=0x500_045c) until you get the STATUS_COMMAND_RESPONSE_READY (Bit 0 of CMD_RESPONSE_STATUS register) equals 1.
4. Clear the STATUS_COMMAND_RESPONSE_READY (Bit 0 of CMD_RESPONSE_STATUS register) Perform a Read-Modify-Write operation:
 - a. Read from CMD_RESPONSE_STATUS register.
 - b. Write_Data = Data in (a) & 0xffff_ffe (Change only bit 0).
 - c. Write to address = 0x500_045c data= write_data from (b).

Recalibrate CH2 using the JAMB/AXI-Lite interface for CH2

Repeat the same steps as listed above. For Step 2, the write_data=0x6005_000a, because the CMD_TARGET_IP_TYPE for CH2 is 0x3 in this design.

Read the CH1 Calibration Status using the JAMB/AXI-Lite interface for CH1

1. Read from read-only register STATUS(address=0x500_0400) till you get either bit_1/bit_0 is asserted.
 - bit_0 = 1 means one or more EMIF instances have calibrated successfully.
 - bit_1 = 1 means one or more EMIF instances have failed to calibrate successfully.
 - bit_2 = 1 means one or more EMIF instances calibration still ongoing.
 - bit_1 and bit_2 and bit_3 = 0 means no EMIF interface used in the design.

Read the CH2 Calibration Status using the JAMB/AXI-Lite interface for CH2

Repeat the same steps as for CH1, above.

12.1.8.1.1. Example 4: Sending ECC_INJECT_ERROR Using the Mailbox and Reading ECC Error Status, ECC Error Buffer Registers

This example provides instructions for sending ECC_INJECT_ERROR using mailbox command.

The values in this example are for illustrative purposes and are obtained from an EMIF example design using LPDDR4 1ch x32 component, implement on the Agilex 5 FPGA E-Series 065B Premium Development Kit.

For a 1ch x32 LPDDR4 component design, the interface uses the Primary MC of Primary IO96B, hence the CMD_TARGET_IP_TYPE is 0x1. The CMD_TARGET_IP_INSTANCE defaults to 0 in the design example generated by the Quartus Prime software. The design example is generated with In-Line ECC enabled.

The following table illustrates the construction of the data to be written to the CMD_REQ register for this request:

Table 277.

CMD_REQ [31:29]: CMD_TARGET_IP_TYPE	CMD_REQ [28:24]: CMD_TARGET_IP_INSTANCE_ID	CMD_REQ [23:16]: CMD_TYPE	CMD_REQ [15:0]: CMD_OPCODE	CMD_REQ [31:0]:
0x1	0x0	0x04	0x109	0x20040109

Flagging error on specific memory address using JAMB/AXI-Lite

Prerequisites:

- Ensure the design generated with ECC enabled. If you modify the ECC enable status via a mailbox command, verify ECC is enabled by reading the ECC_ENABLE_INTF0/1 read-only status register.
- Ensure the design is generated with a non-infinite Traffic Generator Program so that write traffic to a known address can be compiled and executed in user mode. Follow these steps for compiling traffic program to perform single write and read on a specific memory address:
 1. Open the *traffic_patterns.py* file from the <project directory>/qii/hydra_sw folder and navigate to a function called *tut1_block_rw*.
 2. Specify the address at "addr" parameter as shown in example below and save the file.

```
def tut1_block_rw(self):  
    return [  
        # Write to a specific address with specific data  
        *self.gencmd("wr", addr=0x2000, data=0x0123456789ABCDEF),  
        # Wait for all write responses to return  
        wait_writes(),  
        # Read from the same address and check if we get back the same data  
        *self.gencmd("rd", addr=0x2000, data=0x0123456789ABCDEF),  
        # Wait for all read responses to return  
        wait_reads(),  
  
        # Idle for 100 cycles  
        sleep(100)  
    ]
```

(Comment or remove all other commands within this function, except for the ones performing single write to a specific address and reading from the same address)

3. Execute the command to compile the traffic program:

```
quartus_py main.py --ipdir=<project directory>/qii/ --prog=tut1_block_rw
```

4. The <project directory>/qii/hydra_sw/bin folder is updated from the command execution above. Copy the generated /bin folder to <project directory>/qii. The traffic program is now ready to perform a single write to a known address and read from the same address.

Execution:

1. Write to CMD_PARAM_0(address=0x 0x5000438) register the syndrome to program the ECC_XOR_CHECK_BITS parameter(write_data=0xf4).

Note: 0xf4 will update the check bits and write to memory as single bit correctable error on bit [0]. Refer to Table 283 for ECC syndrome codes.

2. Write the CMD_REQ(address=0x500043C) with write_data=0x20040109.
3. Read from CMD_RESPONSE_STATUS(address=0x500_045c) until you get the STATUS_COMMAND_RESPONSE_READY (Bit 0 of CMD_RESPONSE_STATUS register) equals 1.
4. Clear the STATUS_COMMAND_RESPONSE_READY (Bit 0 of CMD_RESPONSE_STATUS register) Perform a Read-Modify-Write operation:
 - a. Read from CMD_RESPONSE_STATUS register.
 - b. Write_Data = Data in (a) & 0xffff_fffe (Change only bit 0).
 - c. Write to address = 0x500_045c data= write_data from (b).
5. Run traffic generator.

```
system-console --script=<path to testengine_library.tcl> --sof=<path to sof_file> --update=1 --n-loops=1
```

You should observe that the traffic has failed.

6. Read ECC_ERR_COUNTER from ECC_ERR_STATUS(address= 0x5000300) read-only register (bit 0 to bit 15). The expected read_data=0x0000_0001; indicates that 1 error encountered.
7. Read ECC Error Buffer Entry 0(R1 address= 0x5000310 , R2 address= 0x5000314).

The expected read_data from R1= 0x00400000

```
IP_TYPE [22:24]      : 1 ==> Primary MC of Primary IO96B
INSTANCE_ID [21:17]  : 0
ECC_ERR_SOURCE_ID [16:10] : 0
ECC_ERR_TYPE [9:6]   : 0 ==> Single-bit error
ECC_ERR_ADDR_UPPER [5:0] : 0 ==> 0x0
```

The expected read_data from R2= 0x00002000.

```
ECC_ERR_ADDR_LOWER[31:0]: 8192 ==> 0x2000
```

8. Repeat the above steps and observe the ECC_ERR_COUNTER increase accordingly. Read the respective ECC Error Buffer Entry (0 to 15) based on the number of ECC events.

When the ECC error buffer reaches its full capacity of 16 entries, proceed to the next steps to check ECC_ERR_OVERFLOW status.

9. Read ECC_ERR_OVERFLOW from ECC_ERR_STATUS (address= 0x5000300) read-only register (bit 16 to bit 31).

The expected read_data=0x0001; indicates that single-bit error overflow occurred.

Clear the ECC error buffer, reset the ECC error counter and reset overflow register

It is not required to write to the CMD_PARAM_* register as this OPCODE does not require any cmd_param.

The following table illustrates the construction of the data to be written to the CMD_REQ register for this request using the same design as the previous example targeting an Agilex 5 FPGA E-Series 065B Premium Development Kit:

Table 278.

CMD_REQ [31:29]: CMD_TARGET_IP_TY PE	CMD_REQ [28:24]: CMD_TARGET_IP_IN STANCE_ID	CMD_REQ [23:16]: CMD_TYPE	CMD_REQ [15:0]: CMD_OPCODE	CMD_REQ [31:0]:
0x1	0x0	0x04	0x110	0x20040110

1. Write the CMD_REQ(address=0x500043C) with write_data=0x20040110.
2. Read from CMD_RESPONSE_STATUS(address=0x500_045c) until you get the STATUS_COMMAND_RESPONSE_READY (Bit 0 of CMD_RESPONSE_STATUS register) equals 1.
3. Clear the STATUS_COMMAND_RESPONSE_READY (Bit 0 of CMD_RESPONSE_STATUS register) Perform a Read-Modify-Write operation:
 - a. Read from CMD_RESPONSE_STATUS register.
 - b. Write_Data = Data in (a) & 0xffff_fffe (Change only bit 0).
 - c. Write to address = 0x500_045c data= write_data from (b).
4. Reading ECC_ERR_STATUS (address= 0x5000300) and ECC Error Buffer Entry (0 to 15) should return 0, indicates that the previous data cleared/reset successfully.

12.1.9. IOSSM Mailbox Access Script

This section discusses mailbox scripts, which are available to perform all mailbox operations, including reading status registers through read-only registers.

12.1.9.1. Mailbox Script Overview

The mailbox script provides complete implementation of command execution as described in previous chapters. This script uses TCL code that performs read and write operations to the respective registers.

The table below lists the three TCL files that are included in the download:

Table 279. TCL Files Included in Mailbox Script Download

File Name	Description
mb_base_parameter.tcl	Parameterization of addresses for each register.
mb_iossm_command.tcl	Functions to perform mailbox access flow and reading read-only register.
mb_user_input.tcl	User input to perform desired operation.

Call the respective function with the required argument or arguments from mb_user_input.tcl to perform any mailbox access operation.

Do not modify mb_base_parameter.tcl or mb_iossm_command.tcl.

You can download the mailbox access script at this location: [Agilex 7 IOSSM Mailbox Access Script](#).

12.1.9.1.1. mb_iossm_command.tcl

This topic describes the functions available for accessing a mailbox or read-only registers.

The following table lists the mailbox access script functions. (For detailed information on the corresponding arguments for these functions, refer to the subsequent *User Input Arguments Description* table.)

Table 280. Mailbox Access Script Functions

Registers	Functions	Argumentss
MAILBOX_HEADER MEM_INTF_INFO_0 MEM_INTF_INFO_1 MEM_TECHNOLOGY_INTF0 MEM_TECHNOLOGY_INTF1 MEMCLK_FREQ_INTF0 MEMCLK_FREQ_INTF1 MEMCLK_FREQ_FSP0_INTF0 MEMCLK_FREQ_FSP0_INTF1 MEMCLK_FREQ_FSP1_INTF0 MEMCLK_FREQ_FSP1_INTF1 MEMCLK_FREQ_FSP2_INTF0 MEMCLK_FREQ_FSP2_INTF1 MEM_WIDTH_INFO_INTF0 MEM_WIDTH_INFO_INTF1 MEM_TOTAL_CAPACITY_INTF0 MEM_TOTAL_CAPACITY_INTF1	intf_info	<calibration interface>
STATUS STATUS_CAL_INTF0 STATUS_CAL_INTF1	cal_status	<calibration interface>
ECC_ENABLE_INTF0 ECC_ENABLE_INTF1 ECC_SCRUB_STATUS_INTF0 ECC_SCRUB_STATUS_INTF1	ecc_status	<calibration interface>
MEM_INIT_STATUS_INTF0 MEM_INIT_STATUS_INTF1 BIST_STATUS_INTF0 BIST_STATUS_INTF1	bist_status	<calibration interface>
LP_MODE_INTF0 LP_MODE_INTF1	lp_mode_status	<calibration interface>
ECC_ENABLE_SET	mb_ecc_enable_set	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID> <ECC_ENABLE>
ECC_INTERRUPT_MASK	mb_ecc_interrupt_mask	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID> <ECC_INTERRUPT_MASK>
continued...		

Registers	Functions	Argumentss
ECC_WRITEBACK_ENABLE	mb_ecc_writeback_enable	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID> <ECC_WRITEBACK_EN>
ECC_INJECT_ERROR	mb_ecc_inject_error	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID> <ECC_XOR_CHECK_BITS>
ECC_CLEAR_ERR_BUFFER	mb_ecc_clear_buffer	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID>
ECC_SCRUB_MODE_0_START	mb_ecc_scrub_mode_0_start	<calibration interface> <CMD_TARGET_IP_TYPE> <ECC_SCRUB_INTERVAL> <ECC_SCRUB_LEN> <ECC_SCRUB_FULL_MEM> <ECC_SCRUB_START_ADDR [31:0]> <ECC_SCRUB_START_ADDR [36:32]> <ECC_SCRUB_END_ADDR [31:0]> <ECC_SCRUB_END_ADDR [36:32]>
ECC_SCRUB_MODE_1_START	mb_ecc_scrub_mode_1_start	<calibration interface> <CMD_TARGET_IP_TYPE> <ECC_SCRUB_IDLE_CNT> <ECC_SCRUB_LEN> <ECC_SCRUB_FULL_MEM> <ECC_SCRUB_START_ADDR [31:0]> <ECC_SCRUB_START_ADDR [36:32]> <ECC_SCRUB_END_ADDR [31:0]> <ECC_SCRUB_END_ADDR [36:32]>
BIST_STANDARD_MODE_START	mb_run_bist_std_mode_start	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID> <BIST_ADDR_SPACE [5:0]>
BIST_MEM_INIT_START	mb_bist_mem_init_start	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID> <BIST_DATA_PATTERN> <BIST_START_ADDR [37:32]> <BIST_START_ADDR [31:0]> <BIST_FULL_MEM>
BIST_SET_DATA_PATTERN_UPPER	mb_bist_set_data_pattern_upper	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID> <BIST_DATA_PATTERN [287:256]> <BIST_DATA_PATTERN [255:224]> <BIST_DATA_PATTERN [223:192]> <BIST_DATA_PATTERN [191:160]>
BIST_SET_DATA_PATTERN_LOWER	mb_bist_set_data_pattern_lower	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID> <BIST_DATA_PATTERN [159:128]>

continued...

Registers	Functions	Argumentss
		<BIST_DATA_PATTERN [127:96]> <BIST_DATA_PATTERN [95:64]> <BIST_DATA_PATTERN [63:32]> <BIST_DATA_PATTERN [31:0]>
CHANGE_FSP_LP5	mb_change_fsp_lp5	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID> <TARGET_FSP>
LP_MODE_ENTER	mb_lp_mode_enter	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID> <LP_STATE>
LP_MODE_EXIT	mb_lp_mode_exit	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID>
LP_MODE_AUTO	mb_lp_mode_auto	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID> <LP_STATE_AUTO> <LP_IDLE_CNT>
TRIG_MEM_CAL	mb_reset_cal	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID>
GET_MEM_CAL_MGN	cal_data	<calibration interface> <CMD_TARGET_IP_TYPE> <CMD_TARGET_IP_INSTANCE_ID> < RANK> <(Optional) WR_VREF_RANGE>

Table 281. User Input Arguments Description

Arguments	Valid Input	Description
<calibration interface>	calip_0 – calip_7	Calibration interface instances (up to 8 total)
<CMD_TARGET_IP_TYPE>	0x1, 0x2, 0x3, 0x4	Refer to the CMD_REQ Definition table.
<CMD_TARGET_IP_INSTANCE_ID>	Unique ID defined in EMIF IP	Refer to the CMD_REQ Definition table.
<ECC_ENABLE>	0x0, 0x1, 0x2, 0x3	Refer to the ECC_ENABLE_SET description in the Command Definitions table.
<ECC_INTERRUPT_MASK>	0x0 – 0xffff	Refer to the ECC_INTERRUPT_MASK description in the Command Definitions table.
<ECC_WRITEBACK_EN>	0x0, 0x1	Refer to the ECC_WRITEBACK_EN description in the Command Definitions table.
<ECC_XOR_CHECK_BITS>	Refer table 283	Refer to the ECC_INJECT_ERROR description in the Command Definitions table.
continued...		

Arguments	Valid Input	Description
<ECC_SCRUB_INTERVAL>	User defined	Refer to the ECC_SCRUB_MODE_0_START description in the Command Definitions table.
<ECC_SCRUB_LEN>	0x8, 0x10	
<ECC_SCRUB_FULL_MEM>	0x0, 0x1	
<ECC_SCRUB_START_ADDR [31:0]> <ECC_SCRUB_START_ADDR [36:32]>	User defined	
<ECC_SCRUB_END_ADDR [31:0]> <ECC_SCRUB_END_ADDR [36:32]>	User defined	
<ECC_SCRUB_IDLE_CNT>	User defined	Refer to the ECC_SCRUB_MODE_1_START description in the Command Definitions table.
<BIST_ADDR_SPACE [5:0]>	User defined	Refer to the BIST_STANDARD_MODE_START description in the Command Definitions table.
<BIST_DATA_PATTERN>	0x0, 0x1	Refer to the BIST_MEM_INIT_START description in the Command Definitions table.
<BIST_START_ADDR [37:32]> <BIST_START_ADDR [31:0]>	User defined	Refer to the BIST_MEM_INIT_START description in the Command Definitions table.
<BIST_FULL_MEM>	0x0, 0x1	Refer to the BIST_MEM_INIT_START description in the Command Definitions table.
<BIST_DATA_PATTERN [287:256]> <BIST_DATA_PATTERN [255:224]> <BIST_DATA_PATTERN [223:192]> <BIST_DATA_PATTERN [191:160]>	User defined	Refer to the BIST_SET_DATA_PATTERN_UPPER description in the Command Definitions table.
<BIST_DATA_PATTERN [159:128]> <BIST_DATA_PATTERN [127:96]> <BIST_DATA_PATTERN [95:64]> <BIST_DATA_PATTERN [63:32]> <BIST_DATA_PATTERN [31:0]>	User defined	Refer to the BIST_SET_DATA_PATTERN_LOWER description in the Command Definitions table.
<TARGET_FSP>	0x0, 0x1, 0x2	Refer to the CHANGE_FSP_LP5 description in the Command Definitions table.
<LP_STATE>	0x8, 0x9, 0xa, 0xd, 0xe, 0xf	Refer to the LP_MODE_ENTER description in the Command Definitions table.
<LP_STATE_AUTO>	0x0, 0x1, 0x2, 0x3, 0x4, 0x5, 0x6, 0xb, 0xc	Refer to the LP_MODE_AUTO description in the Command Definitions table.
<LP_IDLE_CNT>	User defined	
< RANK>	0x1, 0x2	0x1 -> For Single Rank Configuration 0x2 -> For Dual Rank Configuration
<(Optional) WR_VREF_RANGE>	DDR4_R1, DDR4_R2, LPDDR4_R1, LPDDR4_R2	Default: DDR4_R2 / LPDDR4_R1 Specify as follows: Range 1: DDR4_R1, LPDDR4_R1 Range 2: DDR4_R2, LPDDR4_R2

Arguments	Valid Input	Description
		<i>Note:</i> - Used for WR-VREF calculation (only DDR4/LPDDR4). - Refer to the <i>Mem VrefDQ Range</i> parameter in the EMIF IP Configuration on the Advanced: Analog Overrides tab.

Calibration Interface Instances (calip_0 – calip_7)

Specifying Calibration Interface Arguments in mb_user_input.tcl

When working with EMIF designs that include multiple channels or EMIF IP instances, it is essential to understand how to specify the calibration interface in mb_user_input.tcl. Correct instantiation ensures that each EMIF interface is appropriately targeted for mailbox command operations.

Displaying Available Calibration Interface Instances

To display all available calibration interface instances (calip_[n]) in your current setup, run the mailbox script without specifying any procedure calls.

For detailed instructions, refer to *Running the Mailbox Script*. Skip step 3 if you only want to view the calibration interface print-out.

Example of Console Output 1:

Tested on the Agilex 7 M-series Development Kit – HBM2e Edition, targeting a design with multiple EMIF instances (1x32 DDR5 component and dual-channel LPDDR5 EMIF).

This setup returns the calibration instances calip_0 and calip_1.

```
Detected the following calip:
calip_0 : /devices/AGM(E039R47AR0|F039R47AR0)@1#USB-3#Intel Agilex M-Series
Development Kit - HBM2e Edition # <server_path> / (link)/JTAG/
alt_sld_fab_0 alt_sld_fab_0 sldfabric.node_0/phy_0/
emif_io96b_ddr5comp_inst_cal_0_cal_0_jamb_jamb.master
calip_1 : /devices/AGM(E039R47AR0|F039R47AR0)@1#USB-3#Intel Agilex M-Series
Development Kit - HBM2e Edition # <server_path> / (link)/JTAG/
alt_sld_fab_0 alt_sld_fab_0 sldfabric.node_1/phy_1/
emif_io96b_lpddr5_inst_cal_0_cal_0_jamb_jamb.master
```

Understanding Calibration Interface Instances

Multiple Channel Design

- Each calibration instance contains information for two channels. Example:
 - ☐ calip_n holds Channel 0 and Channel 1
 - ☐ calip_n+1 holds Channel 2 and Channel 3

Multiple EMIF Instances with Multiple Channel Design

- Each EMIF instance is associated with its own calibration instance. For multi-channel designs, each calibration instance targets two channels (as described in *Multiple Channel Design*, above.)
- Example (as shown in the console output 1):
 - calip_n0 → emif_io96b_ddr5comp_inst
 - calip_1 → emif_io96b_lpddr5_inst<CMD_TARGET_IP_TYPE>
differentiates between each channel for dual-channel LPDDR5 configuration.)

Calibration Interface Instance Usage in Mailbox Script

The examples below demonstrate how to use various calibration interface instances to target specific EMIF IP instances and channels for calibration data retrieval, corresponding to the setup shown in the *Console Output 1* example above.

Use the following command format to retrieve calibration margin data:

```
cal_data    <calibration interface>    <CMD_TARGET_IP_TYPE>
<CMD_TARGET_IP_INSTANCE_ID>    <RANK>    <(Optional)WR_VREF_RANGE>

# DDR5 Component
cal_data    calip_0    0x1    0x0    0x1

# LPDDR5 Channel 0
cal_data    calip_1    0x1    0x0    0x1

# LPDDR5 Channel 1
cal_data    calip_1    0x2    0x0    0x1
```

The above example illustrates the calibration margin retrieval function; however, this concept applies to all other functions as listed in the *Mailbox Access Script Functions* table.

For additional information, refer to the topic *mb_user_input.tcl*.

12.1.9.1.2. mb_user_input.tcl

This topic provides guidance for entering input using the *mb_user_input.tcl* script.

Follow these steps to perform function calls:

- Identify the registers you want to access. Refer to the [Read-Only Registers](#) and [Command Definitions](#) tables for details.
- Refer to the tables in the *mb_iossm_command.tcl* topic for the required arguments.
- Call the respective functions with the necessary arguments, following the format below:

```
<function name> <lists of arguments separated by spaces>
```

Example 1: Accessing MEM_INTF_INFO_0 read-only register.

Function call: *intf_info calip_0*

Example 2: Accessing ECC_INJECT_ERROR mailbox command register.

Function call: *mb_ecc_inject_error calip_0 0x1 0x0 0xf4*

Example 3: Accessing ECC_ENABLE_SET mailbox command register.

Function call: *mb_ecc_enable_set calip_0 0x1 0x0 0x2*

12.1.9.2. Running the Mailbox Script

Follow these steps to run the mailbox script:

1. Configure the device with the `.sof` design file. (Ensure that the *Use Debug Toolkit* feature is enabled in the design.)
2. Download and extract all Mailbox Access Script files into the same directory.
3. Refer to the *mb_user_input.tcl* topic and modify the function calls as needed.
4. Save the file.
5. To run the script, use the following command:

```
system-console --script=mb_user_input.tcl
--sof=<sof file directory>/<sof file name>.sof
```

(Ensure that you have configured the device with the same `.sof` design file that you specify in this command.)

13. Document Revision History for External Memory Interfaces Agilex 7 M-Series FPGA IP User Guide

Document Version	Quartus Prime Version	IP Version	Changes
2026.04.06	26.1	4.2.0	- In the <i>About</i> chapter, updated the table in the <i>Release Information</i> topic. - In the <i>Signals</i> chapter, added detail to the protocol-specific <i>s0_axi4...</i> topics. - In the <i>DDR5 Support</i> chapter, added a note to the table in the <i>Address and Command Pin Placement for DDR5</i> topic. - In the <i>Controller Optimization</i> chapter, added the <i>Controller Performance Profile</i> topic. - In the <i>Debugging</i> chapter, added the <i>EMIF Eye Mask Margin Guidelines</i> section.
2026.01.23	25.3.1	4.1.1	- In the <i>Controller Optimization</i> chapter, added the <i>Special AXI Interface Signals</i> topic. - In the <i>Mailbox Support</i> chapter: - Added content to the <i>Description</i> field of the last row in the <i>Command Definitions</i> table in the <i>Mailbox Command Definitions</i> topic. - Added <i>Address Mapping Reference</i> topic. - Added <i>VREF Calculation Reference Data</i> topic. - Updated the flowchart in the <i>Sending a Mailbox Command</i> topic. - In the <i>mb_iossm_command.tcl</i> topic: - Added <i>GET_MEM_CAL_MGN</i> register to the <i>Mailbox Access Script Functions</i> table. - Modified the <i><calibration interface></i> entry in the <i>User Input Arguments Description</i> table. - Added <i>RANK</i> and <i>WR_VREF_RANGE</i> entries to the <i>User Input Arguments Description</i> table. - Added the <i>Calibration Interface Instances (calip_0 – calip_7)</i> topic.
2025.09.29	25.3	4.1.0	- In the <i>Architecture</i> chapter: - Implemented updates to the <i>I/O Byte Lane Organization in Different Agilex 7 M-Series Devices, Lockstep Configuration, and IP Interfaces for External Memory Interfaces (EMIF) IP - DDR5 DIMM</i> topics. - Modified the <i>Using the EMIF Debug Toolkit with Agilex 7 M-Series HPS EMIF Interfaces</i> topic. - In the <i>DDR4 Support</i> chapter, modified the introductory text in the <i>Specific Pin Connection Requirements</i> topic. - In the <i>DDR5 Support</i> chapter: - Modified the introductory text in the <i>Specific Pin Connection Requirements</i> topic. - Added content to the <i>Address and Command Pin Placement for DDR5</i> topic. - Modified the figures in the <i>DDR5 Data Width Mapping</i> topic. - Added content to the <i>Routing Guidelines for DDR5 Memory Down: Single Rank or Dual Rank (x8 bit or x16 bit) Configurations</i> topic. - In the <i>LPDDR5 Support</i> chapter, modified the introductory text in the <i>Specific Pin Connection Requirements</i> topic.
continued...			

Document Version	Quartus Prime Version	IP Version	Changes
			- In the <i>Controller</i> chapter, added the <i>AXI Interface: Write and Read Example</i> topic. - In the <i>Debugging</i> chapter, applied minor updates to the <i>Guidelines for Developing HDL for Traffic Generator</i> topic. - In the <i>Mailbox Support</i> chapter: - Added the <code>CMD_GET_MEM_CAL_INFO</code> command to the <i>Supported Commands</i> table in the <i>Mailbox Supported Commands</i> topic. - Added the <code>CMD_GET_MEM_CAL_INFO</code> type to the <i>CMD_TYPE</i> Definition table in the <i>Mailbox Command Definitions</i> topic. - Added a row to the bottom of the <i>Command Definitions</i> table in the <i>Mailbox Command Definitions</i> topic.
2025.07.07	25.1.1	4.0.0	- In the <i>Introduction</i> chapter, modified some data in the <i>Protocol and Feature Support</i> topic. - In the <i>Architecture</i> chapter: - Modified the <i>Refresh features</i> description in the <i>Hard Memory Controller Features</i> topic. - In the <i>Interface Protocol: Interface Width, User Access Mode, and ECC</i> topic, added an <i>ECC, Format, and Access Mode for LPDDR5</i> table. - Modified some bullet points in the <i>Restrictions on I/O Bank Usage for Agilex 7 M-Series EMIF IP with HPS</i> topic. - In the <i>Functional Simulation</i> chapter, made some minor changes to wording in the <i>Simulating the Design Example</i> topic, and added the new <i>Simulating with sim_lib2</i> topic. - In the <i>DDR4 Support</i> chapter, updated the figures in the <i>Data Width Mapping</i> topic. - In the <i>DDR5 Support</i> chapter, updated the figures in the <i>Data Width Mapping</i> topic. - In the <i>LPDDR5 Support</i> chapter, updated the figures in the <i>Data Width Mapping</i> topic. - In the <i>Controller Optimization</i> chapter, added the new <i>Controller Performance Profile</i> topic. - In the <i>Mailbox Support</i> chapter, added notes to the <i>ECC Error Handling</i> topic.
2025.03.31	25.1	3.0.0	- In the <i>Architecture</i> chapter: - Added the <i>I/O Byte Lane Organization in Different Agilex 7 M-Series Devices</i> topic. - Reorganized and added content to the <i>EMIF IP for Hard Processor Subsystem (HPS)</i> section. - In the <i>DDR5</i> chapter: - Added a sentence to the last paragraph of the <i>General Design Considerations</i> topic. - In the <i>DDR5 Simulation Strategy</i> topic, implemented a change in the paragraph immediately preceding the note. - In the <i>LPDDR5</i> chapter: - Added a sentence to the last paragraph of the <i>General Design Considerations</i> topic. - In the <i>LPDDR5 Simulation Strategy</i> topic, implemented a change in the paragraph immediately preceding the note. - In the <i>Debugging</i> chapter, added a note to the <i>Debugging with the External Memory Interface Debug Toolkit</i> topic.
continued...			

Document Version	Quartus Prime Version	IP Version	Changes
2025.03.18	24.3.1	2.0.0	- In the <i>Introduction</i> chapter, implemented changes to the DDR5 section of the table in the <i>Protocol and Feature Support</i> topic. - In the <i>Product Architecture</i> chapter: - Removed DDR5 references from the first two tables in the <i>Lockstep Configuration</i> topic. - In the <i>EMIF IP Interface Protocol: Interface Width, User Access Mode, and ECC</i> topic, made changes to the <i>ECC, Format, and Access Mode for DDR5</i> table. - In the <i>EMIF IP for Hard Processor Subsystem (HPS)</i> topic, removed rows from the DDR5 and LPDDR5 sections of the <i>IO96 Bank and Lane Usage for HPS EMIF</i> table. - In the <i>DDR5</i> chapter, removed a row from the <i>Component</i> table in the <i>DDR5 Data Width Mapping</i> topic. - In the <i>LPDDR5</i> chapter, removed two rows from the <i>Component</i> table in the <i>LPDDR5 Data Width Mapping</i> topic. - In the <i>Controller Optimization</i> chapter, added the <i>Using Multiple AXI IDs</i> topic.
2025.01.13	24.3.1	2.0.0	- In the <i>DDR4</i> chapter, added a table to the <i>Maximum Number of Interfaces</i> topic and added figures to the <i>DDR4 Data Width Mapping</i> topic. - In the <i>DDR5</i> chapter, added a table to the <i>Maximum Number of Interfaces</i> topic and added figures to the <i>DDR5 Data Width Mapping</i> topic. Modified the first two rows of the table in the <i>Address and Command Pin Placement for DDR5</i> topic. - In the <i>LPDDR5</i> chapter, added a table to the <i>Maximum Number of Interfaces</i> topic and added figures to the <i>LPDDR5 Data Width Mapping</i> topic. - In the <i>Controller Optimization</i> chapter, modified the <i>AXI to Memory Mapping</i> topic. - In the <i>Debugging</i> chapter, added several topics to the <i>Hardware Debugging Guidelines</i> section. - Added a new <i>Mailbox Support</i> chapter.
continued...			

Document Version	Quartus Prime Version	IP Version	Changes
2024.11.18	24.3	1.0.0	- In the <i>About the External Memory Interfaces IP</i> chapter, updated the table of IPs and associated version numbers. - In the <i>Product Architecture</i> chapter, modified the <i>Lockstep Configuration</i> topic. - Moved the mailbox content from the <i>Architecture</i> chapter to a new chapter entitled <i>Agilex 7 M-Series FPGA EMIF IP - Mailbox Support</i>, later in the document. - Recast the <i>End-User Signals</i> chapter to provide signals information for DDR4 Component, DDR4 DIMM, DDR5 Component, DDR5 DIMM, and LPDDR5 interfaces. - Recast the <i>DDR4 Support</i>, <i>DDR5 Support</i>, and <i>LPDDR5 Support</i> chapters with updated parameter information. - Removed the <i>Layout Design Guidelines</i> sections from the protocol-specific chapters. Implemented various changes to the protocol-specific <i>Board Design Guidelines</i> sections. - Updated the mailbox content in the new <i>Agilex 7 M-Series FPGA EMIF IP - Mailbox Support</i> chapter.
2024.07.08	24.2	6.2.0	- In the <i>End-User Signals</i> chapter, implemented minor updates to the interface description topics. - In the <i>DDR4 Support</i>, <i>DDR5 Support</i>, and <i>LPDDR5 Support</i> chapters, implemented minor updates to the parameter description topics. - In the <i>Controller Optimization</i> chapter, added the <i>AXI to Memory Mapping</i> topic. - In the <i>Debugging</i> chapter, added the <i>Guidelines for Traffic Generator Status Check</i> section. - Implemented minor editorial improvements and branding changes throughout.
2024.04.01	24.1	6.1.0	- In the <i>Product Architecture</i> chapter: - Expanded the note after the table in the <i>LPDDR5 Pin Placement</i> topic. - Added two rows to the table in the <i>Mailbox Supported Commands</i> topic. - Added two new rows to the <i>Command Definitions</i> table in the <i>Mailbox Command Definitions</i> topic. - In the <i>Agilex 7 FPGA EMIF IP Parameter for DDR4</i> topic in the <i>DDR4 Support</i> chapter: - Repositioned the <i>Group: Example Design / Example Design</i> table. - Added a row to the <i>Group: General IP Parameters / High-Level Parameters</i> table. - Added the <i>Group: Example Design / Traffic Generator Program</i> table. - In the <i>DDR5 Support</i> chapter: - In the <i>Agilex 7 FPGA EMIF IP Parameter for DDR5</i> topic: - Repositioned the <i>Group: Example Design / Example Design</i> table. - Added the <i>Group: Example Design / Traffic Generator Program</i> table. - In the <i>Agilex 7 FPGA EMIF Memory Device Description IP (DDR5) Parameter Descriptions</i> topic, added one row to the <i>Group: Memory Timing Parameters / Timing Parameters</i> table. - In the <i>LPDDR5 Support</i> chapter: - In the <i>Agilex 7 FPGA EMIF IP Parameter for LPDDR5</i> topic: - Repositioned the <i>Group: Example Design / Example Design</i> table. - Added the <i>Group: Example Design / Traffic Generator Program</i> table. - In the <i>Agilex 7 FPGA EMIF Memory Device Description IP (LPDDR5) Parameter Descriptions</i> topic: - In the <i>Group: High-Level Parameters</i> table: - Changed the <i>Number of Channels in Memory Package</i> row to <i>Number of Channels</i>. - Changed the <i>Number of Ranks per Channel</i> row to <i>Memory Ranks</i>. - Changed the <i>Number of Individual DRAM Components per Rank</i> row to <i>Number of Components per Rank</i>.

Document Version	Quartus Prime Version	IP Version	Changes
			- In the <i>Debugging</i> chapter: - Removed the final sentence from the <i>Interface Configuration Bottleneck and Efficiency Issues</i> topic. - In the <i>Debugging with the External Memory Interface Debug Toolkit</i> section, implemented assorted updates and figure replacements in the following topics: <i>Debugging with the External Memory Interface Debug Toolkit</i>. <i>Configuring Your Design to Use the EMIF Debug Toolkit</i>. <i>Launching the EMIF Debug Toolkit</i>. <i>Using the EMIF Debug Toolkit</i>. <i>Rerunning Calibration</i>. <i>Rerunning the Test Engine</i>.
2023.12.04	23.4	6.0.0	- In the <i>Architecture</i> chapter: - In the <i>Mailbox Command Definitions</i> topic, modified the <code>CMD_TARGET_IP_TYPE</code> description in the <i>CMD_REQ Definition</i> table, and added the <i>CMD_TARGET_IP_TYPE Definition</i> table. - In the <i>End-User Signals</i> chapter, updated interfaces and signals descriptions. - In the <i>DDR4 Support</i>, <i>DDR5 Support</i>, and <i>LPDDR5 Support</i> chapters, updated the parameter description sections. - In the <i>LPDDR5 Support</i> chapter: - In the <i>LPDDR5 Data Width Mapping</i> topic, modified the <i>Component</i> table. - Added the <i>LPDDR5 Byte Lane Swapping</i> topic. - In the <i>Debugging</i> chapter: - Modified the bulleted lists in the <i>Debugging with the External Memory Interface Debug Toolkit</i> topic. - Updated figures in the <i>Rerunning the Traffic Generator</i> and <i>Saving Debug Print</i> topics. - Added <i>Calibration Reports</i>, <i>Driver Margining Tab</i>, and <i>Pin Delay Settings Tab</i> topics, to the <i>Debugging with the External Memory Interface Debug Toolkit</i> section. - Added the <i>LPDDR5 Byte Lane Swapping</i> topic.
2023.10.02	23.3	5.0.0	- In the <i>Architecture</i> chapter: - Added the <i>Lockstep Configuration</i> topic. - Added information to the table in the <i>Mailbox Supported Commands</i> topic. - Added three tables to the <i>Mailbox Command Definitions</i> topic. - In the <i>DDR4 Support</i> chapter: - Added lockstep configuration information to the <i>DDR4 Data Width Mapping</i> topic. - Added lockstep configuration information to the <i>General Guidelines - DDR4</i> topic. - Added lockstep configuration information to the <i>DDR4 Byte Lane Swapping</i> topic. - In the <i>End-User Signals</i> chapter, updated the <i>s0_axi4 for EMIF</i> topic in the <i>DDR4 Interfaces</i>, <i>DDR5 Interfaces</i>, and <i>LPDDR5 Interfaces</i> sections.
continued...			

Document Version	Quartus Prime Version	IP Version	Changes
			- In the <i>DDR5 Support</i> chapter, added the <i>Board Design Guidelines</i> section. - In the <i>LPDDR5 Support</i> chapter, added the <i>Board Design Guidelines</i> section. - In the <i>Debugging</i> chapter, added the <i>Debugging with the External Memory Interface Debug Toolkit</i> section.
2023.06.26	23.2	4.0.0	- In the <i>Architecture</i> chapter: - Added the <i>User Clock in Different Core Access Modes</i> topic. - Added the <i>Mailbox Supported Commands and Mailbox Command Definitions</i> topics. - Added the <i>Intel Agilex 7 M-Series EMIF IP for Hard Processor Subsystem (HPS)</i> topic. - In the <i>End-User Signals</i> chapter, updated the signals description topics. - In the <i>DDR4 Support</i> chapter: - Updated the parameter description topics. - Modified the <i>DDR4 Byte Lane Swapping</i> topic. - In the <i>DDR5 Support</i> chapter: - Updated the parameter description topics. - Added the <i>Address and Command Pin Placement for DDR5 and DDR5 Data Width Mapping</i> topics. - Added the <i>Pin Swapping Guidelines</i> section. - In the <i>LPDDR5 Support</i> chapter: - Updated the parameter description topics. - Added the <i>Address and Command Pin Placement for LPDDR5 and LPDDR5 Data Width Mapping</i> topics.
2023.04.03	23.1	3.0.0	Initial release.